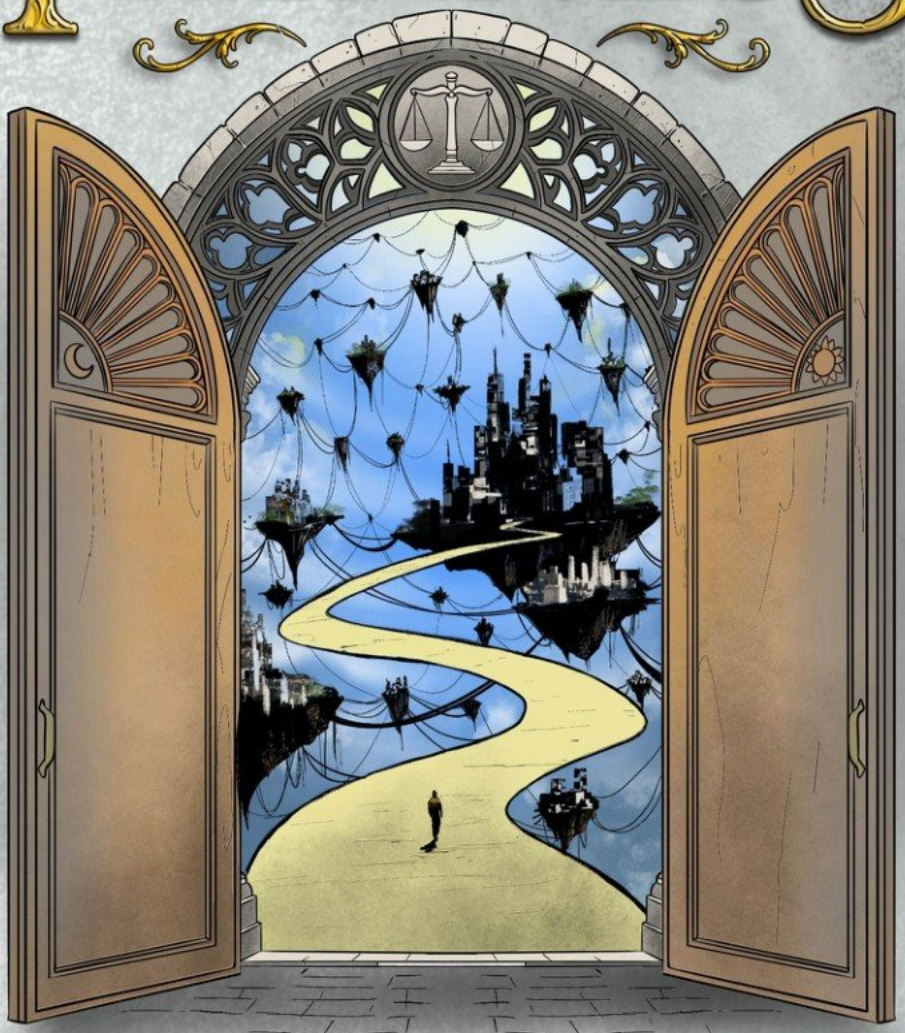
The Law of Fairness



Stephen Takowsky

The Law of Fairness

Reference Edition

Stephen Takowsky

Publishing Details

The Law of Fairness

Reference Edition

Copyright © 2026 Law of Fairness LLC. All rights reserved.

Personal, educational, and non-commercial use of this edition is welcome, including scholarly discussion, review, quotation, and public commentary on social media and elsewhere. Commercial reproduction or distribution requires prior written permission from Law of Fairness LLC.

Published in the United States of America.

Publisher: Law of Fairness LLC
Author: Stephen Takowsky
Illustrations: Diego Santos
ISBN: 979-8-953313-00-1

For permissions, corrections, publication updates, and future edition information: team@lawoffairness.com

For the current authorized edition and project updates, visit LawOfFairness.com.

Table of Contents

Table of Contents ..... *i*
Series Map ..... *vi*
How to Read This Book ..... *xi*
Illustrated Guide to Fairness ..... *xiv*
Part I — The Question and Human Stakes *1*
Chapter 1 — Why Fairness? Why Now? **4**
1.1 — The World We Can See 7
1.2 — What Fairness Usually Means 11
1.3 — The Cases That Resist Easy Answers 16
1.4 — Why Averages Cannot Settle a Whole Stream 21
Chapter 2 — The Life Only You Can Feel **25**
2.1 — Why Experience Matters 28
2.2 — How Conditions Become Lived 32
2.3 — Hidden Experience and Appearance 37
2.4 — Describing Experience Without Owning It 40
2.5 — What Affective Science Can and Cannot Clarify 47
Chapter 3 — Care Before the Claim **51**
3.1 — Care, Responsibility and Chosen Meaning 54
3.2 — One Stream Cannot Pay for Another 59
3.3 — Consent, Refusal, and No Harm 63
Part II — The Law in Plain Language *65*
Chapter 4 — The Law of Fairness **68**
4.1 — The Law in One Sentence 71
4.2 — The Parts Inside the Claim 73
4.3 — Streams, Cumulative Experience and Closure 76
Chapter 5 — A Stream's Beginning and End **79**
5.1 — Evidence for a Boundary 82
5.2 — Pauses, Returns, and Low-Unity States 85
5.3 — The Endpoint Must Be Independent 88
5.4 — Ambiguous, Branching, and Revised Endpoints 92
Chapter 6 — A Constraint Without a Judge **96**
6.1 — Constraint, Explanation and Mechanism 99
6.2 — Purpose, Meaning and Scientific Lawhood 104
6.3 — How Post Hoc Rescue Destroys the Claim 108

Part III — How Exact Closure Could Occur.....115
Chapter 7 — The Structure Closure Needs.....118
7.1 — Bearer, Experience, Time, and Boundary.....121
7.2 — Exact Zero and the Random-Endpoint Problem.....133
7.3 — Branches, Layers, and Non-Equivalent Claims.....147
7.4 — Latent Objects, Observations, and Estimates.....160
7.5 — Reachability, Reserve, and Predictions.....165
Chapter 8 — Possible Routes to Closure.....178
8.1 — Choice, Reachability, and Possibility.....181
8.2 — Repair, Reserve, and Compensatory Capacity.....187
8.3 — Candidate Routes and Their Evaluation.....190
Chapter 9 — The Shortening Horizon.....208
9.1 — The Shortening Horizon.....211
9.2 — Time, Capacity, and Opportunity.....214
9.3 — Addiction, Delay, and Future Optionality.....225
9.4 — Aging, Illness, Caregiving, and Terminal Scripts.....228
9.5 — What Horizon Patterns Could Show.....232
Part IV — From Experience to Evidence.....236
Chapter 10 — From Experience to Evidence.....239
10.1 — Why No Single Meter Is Enough.....242
10.2 — From Experience to a Measured Construct.....245
10.3 — Combining Channels Under Uncertainty.....255
10.4 — Missingness, Reactivity, and Honest Failure.....271
10.5 — When Measurement Must Stop or Abstain.....278
Chapter 11 — Comparing Experiences.....283
11.1 — The Comparison Problem.....286
11.2 — Neutral Origin, Sign, and Scale.....289
11.3 — Culture, Age, Disability, and Response Shift.....295
11.4 — Comparisons Within and Across People.....301
11.5 — Uncertainty and Partial Identification.....310
Chapter 12 — Measurement Without Scores.....317
12.1 — Consent, Burden, Minimization, and Deletion.....320
12.2 — No Human-Worth Scores.....323
12.3 — No Dossier, No Model-Training Shortcut.....326
12.4 — Measurement That Should Not Be Done.....329

Part V — Who or What Carries a Stream?.....332
Chapter 13 — What Makes One Stream?.....335
13.1 — What Unity Could Mean.....338
13.2 — Access, Memory and Continuity Across States.....342
13.3 — Disorders, Fragmentation, Fission, and Fusion.....345
13.4 — Blinded Adjudication and Honest Unknowns.....352
Chapter 14 — Animals and Machines.....355
14.1 — Animals, Insects, and Comparative Evidence.....358
14.2 — Artificial Systems and Testimony-Like Output.....371
14.3 — Protection and Research Under Uncertainty.....376
14.4 — Precaution and Uncertain Personhood.....384
Part VI — Windows into a Whole Life.....386
Chapter 15 — Dreams as a Window.....389
15.1 — Why Dreams Matter.....392
15.2 — Dream Affect, Recall, and Missingness.....398
15.3 — Dream-Affect Predictions and Their Interpretation.....406
15.4 — Safe Sampling, Coding, and Privacy.....410
Chapter 16 — At a Stream's End.....414
16.1 — End-of-Stream Evidence and Care.....417
16.2 — Sedation, Delirium, and Family Scripts.....420
16.3 — Observation and No-Estimate Outcomes.....422
16.4 — Terminal Patterns, Rivals, and Limits.....424
Chapter 17 — Questions for a Laboratory.....429
17.1 — Laboratory Questions and Process Signatures.....432
17.2 — Perturbation, Control, and Causal Limits.....445
17.3 — Negative Controls and Rival Tests.....463
17.4 — What Laboratory Failure Would Mean.....469
Chapter 18 — Following Lives with Care.....473
18.1 — What Repeated Observation Can Reveal.....476
18.2 — Life Events, Adaptation, and Context.....480
18.3 — Observation, Missingness and Model Comparison.....488
18.4 — Participation Without Surveillance.....502
Part VII — The Strongest Rivals.....504
Chapter 19 — Adaptation and Rebound.....507
19.1 — What Adaptation Explains Well.....510
19.2 — Opponent Processes, Tolerance, and Rebound.....516

19.3 — What Adaptation Can Explain.....521
Chapter 20 — Prediction and Surprise.....526
20.1 — Prediction, Surprise, and Uncertainty.....529
20.2 — Affect as Prediction Error.....537
20.3 — Viability, Prediction and Explanatory Reach.....540
Chapter 21 — Control and Homeostasis.....543
21.1 — Rewards, Set Points, and Regulation.....546
21.2 — Optimization Is Not Pathwise Closure.....553
21.3 — Constrained and Hybrid Control.....555
21.4 — Mean Reversion, Bridges, and Hybrid Models.....566
Part VIII — Evolution, Worlds, Meaning and Responsibility.....580
Chapter 22 — Evolution and Constraint.....583
22.1 — Evolution, Regulation and Candidate Constraints.....586
22.2 — Predictions and Measurement Bridges.....594
22.3 — Fitness, Feeling, and Moral Worth.....601
22.4 — What Cross-Species Failure Would Mean.....603
Chapter 23 — Could Such a World Exist?.....607
23.1 — Why Build Simulated Worlds?.....612
23.2 — Constraints, Construction and Explanation.....615
23.3 — What Simulations Can Recover.....623
23.4 — What Would Make the Model Unnecessary?.....634
Chapter 24 — Spiritual Echoes.....640
24.1 — Why Fairness Questions Recur.....643
24.2 — Karma, Providence, Fate, Grace, and Repair.....645
24.3 — Traditions Disagree.....647
24.4 — Belief, Interpretation and Evidence.....651
Chapter 25 — Limits on the Idea's Use.....654
25.1 — No Punishment, Ranking, or Eligibility.....657
25.2 — The Hidden Object Cannot Be Administered.....661
25.3 — Institutional Authority and Repair.....665
Part IX — Living with and Testing the Law.....670
Chapter 26 — Fairness in an Ordinary Day.....673
26.1 — Attention Without Accounting.....676
26.2 — Pleasure, Relief, and Moderation.....680
26.3 — Addiction, Craving, and Future Optionality.....683
26.4 — Work, Ambition, and Aspirational Dreams.....685

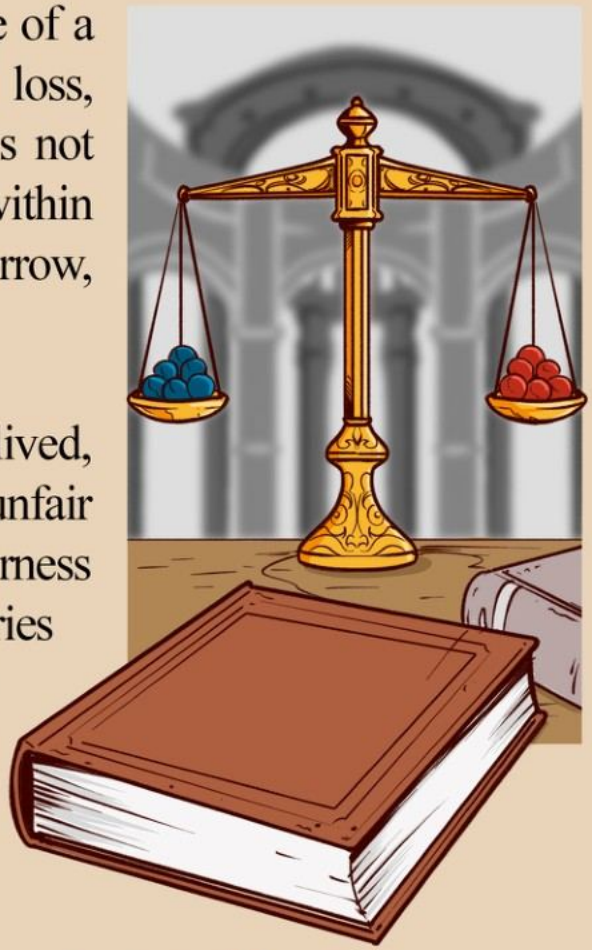
26.5 — Relationships, Grief, Rest, Joy, and Repair.....696
26.6 — Learning from Personal Records.....700
Chapter 27 — A Scientific Program.....703
27.1 — The Scientific Question and Measurement Gates.....706
27.2 — Rivals, Preregistration, and Red Teams.....710
27.3 — Replication, Recovery, and Scientific Verdicts.....717
Glossary.....723
References.....764
Index.....860
About the Author.....871

Series Map

THE LAW OF FAIRNESS

A new way of seeing every life.

We judge fairness by the visible shape of a life: health and hardship, love and loss, opportunity and misfortune. Yet life is not lived from the outside. It is lived within consciousness, through joy and sorrow, pleasure and pain, peace and fear.



Could life be fair where it is actually lived, even when it appears profoundly unfair from the outside? The Law of Fairness proposes that every felt experience carries weight according to its intensity and duration. Across the completed conscious life, total positive and negative feeling exactly balance.

Picture a scale open for a lifetime. Joy and relief add weight to one side; pain and grief to the other. It may tip heavily for years, but when the conscious life is complete, the two sides balance. Each life carries its own scale. No life balances another.

Lives may differ enormously in length, choices, pleasures, wounds, and circumstances. The scale weighs neither wealth nor virtue, status nor faith. It weighs felt experience alone. Circumstances tell the visible story of a life, but feeling is how that story is lived. The Law asks us not to mistake the visible story for the whole.

WHAT IF THE LAW IS REAL?

A new meaning of utopia.

If the Law of Fairness is real, utopia cannot mean endless pleasure. More comfort, luxury, or consumption does not make a society ideal. If conscious lives end with no lasting surplus of positive feeling, utopia must mean something deeper and steadier than limitless happiness.



"There is no free happiness" does not mean joy is wrong, pain is payment, or suffering is holy. Pleasure alone cannot define a good life, nor appearances a good society. The goal is joy without excess. A fairer world would reduce needless suffering, poverty, disease, cruelty, corruption, and abuse while protecting love, beauty, rest, play, choice, and joy. Moderation restrains excess, work builds worthy goals, and stewardship preserves the world so future generations inherit something worth protecting.

If the Law is real, its deepest equality lies beneath wealth, status, power, culture, and belief. Lives may differ while reaching the same final balance of felt experience. These ten books follow that question from science into life, spirituality, government, cities, law, transition, and peace, exploring how humanity might build a lasting utopia.

THE TEN BOOKS OF FAIRNESS

A complete path from truth to practice.

THE LAW OF FAIRNESS



Explores the core question, defines key terms, and sets the ethical bounds of the series.

THE SCIENCE OF FAIRNESS



Evaluates the Law through evidence, measurement, rival models, protocols, and falsification.

THE UTOPIA OF FAIRNESS



Brings the series into a mature view of what a peaceful world could become.

THE TRANSITION OF FAIRNESS



Asks how a move toward utopia could unfold with care, step by step, under real limits.

THE LIFE OF FAIRNESS



Brings the Law into daily life and asks what it means for how we live.

THE RELIGION OF FAIRNESS



Explores faith, world religions, and the optional path of Fairness.

THE GOVERNMENT OF FAIRNESS



Examines civic structure and the public institutions that give it form in practice.

THE CITY OF FAIRNESS



Builds a civic space for learning, service, and peace with privacy kept safe.

THE CONSTITUTION OF FAIRNESS



Sets out the rights, rules, paths, and checks a lasting utopia would need.

THE PEACE OF FAIRNESS



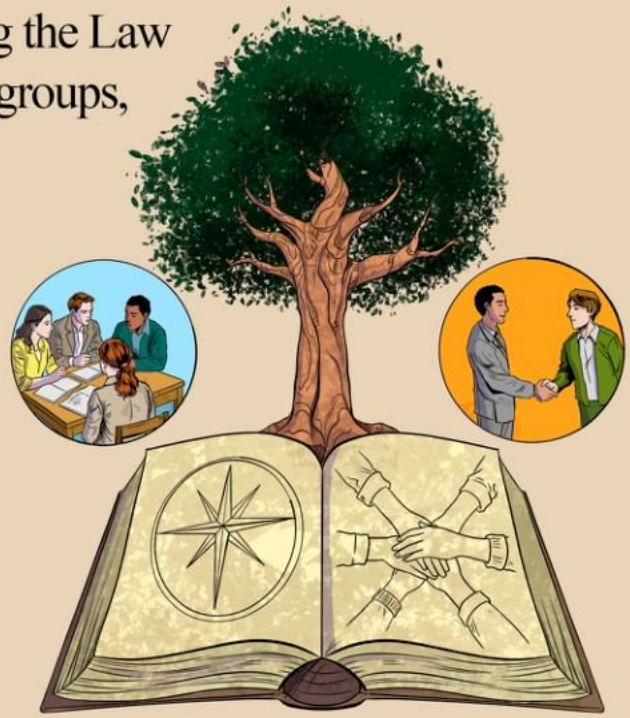
Builds the base for lasting peace and a fairer future for all humanity.

The series moves from the deepest question about reality to the structures that make a more peaceful world.

THE WORK AHEAD

A shared mission to build a better world.

Read the books, share them, and bring the Law of Fairness into classrooms, research groups, reading circles, and community conversations. Important ideas grow stronger when people examine them together, challenge assumptions, compare perspectives, and turn understanding into shared effort.



Scientists, researchers, clinicians, philosophers, and other experts can help test the Law, refine its measurements, improve its methods, and compare it with strong rivals. Serious criticism is essential. It exposes weaknesses, sharpens the questions, and helps reveal what should endure, what needs revision, and what should be rejected.

Beyond the hypothesis lies a practical challenge: building a world with less corruption and needless suffering, and more room for dignity, freedom, meaning, beauty, and opportunity. If this project speaks to you, bring your knowledge, experience, curiosity, and determination. Progress begins when people from different paths meet, share what they know, and build what survives scrutiny.

Help us make this generation
the one that brought peace to our world.

How to Read This Book

This Reference Edition explores the book's ideas in depth and develops the reasoning behind them. The companion Foundation Edition distills the essential ideas into a shorter, accessible introduction that stands on its own and requires no specialized background.

The Law of Fairness proposes that each qualifying conscious stream has finite, exactly equal total positive and negative felt magnitudes across its complete qualifying history strictly before an independently defined true endpoint. A stream is one subject's course of experience, considered separately from every other stream. In the additive account, felt magnitude reflects both intensity and duration. The primary finite branch is called Per-Stream Terminal Valence Closure, or PTVC. We explore what would follow if the Law were true and what evidence could establish it in nature.

Begin with the Series Map and the *Illustrated Guide to Fairness*, then read the chapters in order. Parts I and II introduce the human stakes and the proposed Law in plain language. Parts III–VII develop possible routes to closure, measurement, the conscious stream, evidence from a whole life, and the strongest rival explanations. Parts VIII–IX explore wider implications, responsible use, practical life, testing, objections, and possible outcomes.

Choose a route that fits your purpose. For a rapid orientation, read the Illustrated Guide and Chapters 4 and 26–27. For the human and ethical argument, follow Part I, Chapter 25, and Part IX. For the scientific architecture, follow Parts II–VII and Chapter 27. You can return to earlier definitions as the questions become more precise.

Read each example for the distinction it makes. A neutral hour can leave an earlier accumulated total unchanged. Two histories can both balance while containing very different amounts of experience. The numerical cases use invented values so that the

reasoning can be followed directly. They illustrate the proposed relation rather than measuring your life or asking you to keep a personal account of pleasure and pain.

Keep the proposed Law, a candidate explanation, and a finding distinct. The Law states a relationship across a complete qualifying history. A candidate model describes a possible way it could arise. Evidence bears on a particular claim under stated conditions. A consequence of an assumption is not a finding. Read the cited sources alongside the argument when you want to examine the evidence more closely.

Care comes before interpretation. The Law does not interpret anyone's pain as payment of a moral debt or as deserved punishment. Pain alone neither proves the Law nor conveys a private message from it. A good day need not become a prediction about pain to come. Someone's difficulty is a reason to listen and offer appropriate help. Better conditions change what people can do and experience while leaving their joys free to be enjoyed.

Use the Glossary for an unfamiliar term and the Index to return to its fuller explanation. The References identify the published work cited. In discussion, explain one example in your own words. Ask what would change if an observation were missing or the endpoint remained uncertain, and why an average across people cannot answer a question about each separate history.

This volume is self-contained. *The Science of Fairness* develops the next stage of inquiry through mathematical reasoning and research design. *The Life of Fairness* continues into private life, while *The Utopia of Fairness* considers a civilization worth inhabiting. Readers moving between the Foundation and Reference Editions should follow subjects rather than chapter numbers. Let the question that stays with you guide your next reading.

Illustrated Guide to Fairness



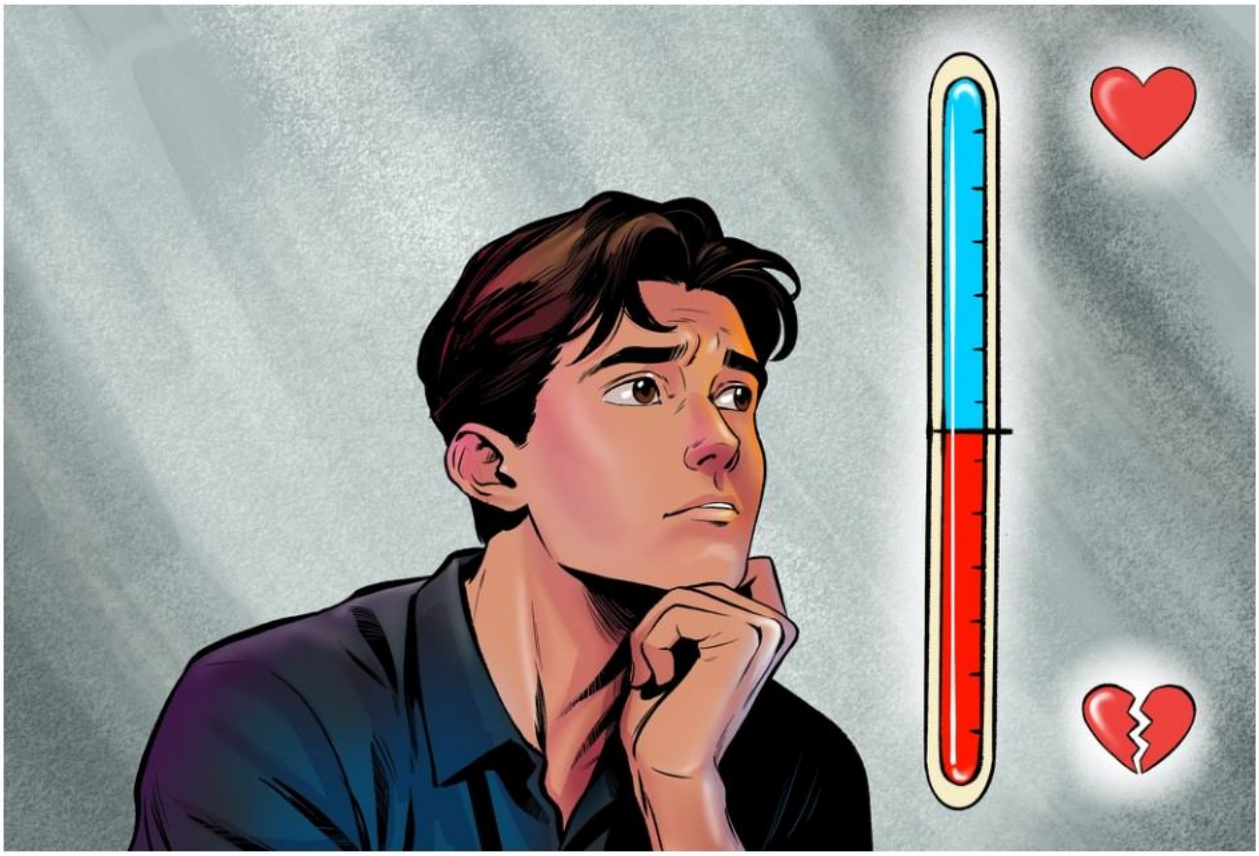
From a young age, we are told that “life is not fair.” As we look at the world, the evidence seems overwhelming. Some lives appear to overflow with unearned comfort and joy, while others are marked by relentless struggle, loss, and tragedy. On the surface, the distribution of pain and pleasure often looks unjust and even cruel. Facing this disparity, it is only natural to ask: Why is life so unfair? The Law of Fairness begins by refusing to look away from that question. It does not ask us to call visible injustice good or pretend that suffering is harmless. The Law asks one deeper question beneath the unequal world we can see: could a complete conscious life still be fair at the level of felt experience, even when its circumstances appear profoundly unequal?



Appearances can be deceiving. We do not see the entire story of another person's life. A billionaire may inhabit a private world of fear, paranoia, loneliness, or stress, while a person with little may still know genuine moments of peace, laughter, love, or relief. That does not make poverty harmless, wealth imaginary, or unequal opportunity acceptable. Circumstances matter, but they tell only the visible story. If we want to understand whether life could be fair at a deeper level, appearances alone are not enough. We have to look at experience as it is actually felt.



On social media, someone may post only bright moments, creating the illusion that joy fills nearly every day. The image may hide late-night worry, grief, coercion, exhaustion, numbness, or the quiet fear of losing what has been gained. Social comparison is profoundly uneven: we often measure selected moments from another person's life against the full private reality of our own. That imbalance can make other people seem luckier and our own lives uniquely burdened. A snapshot is not a lifetime.



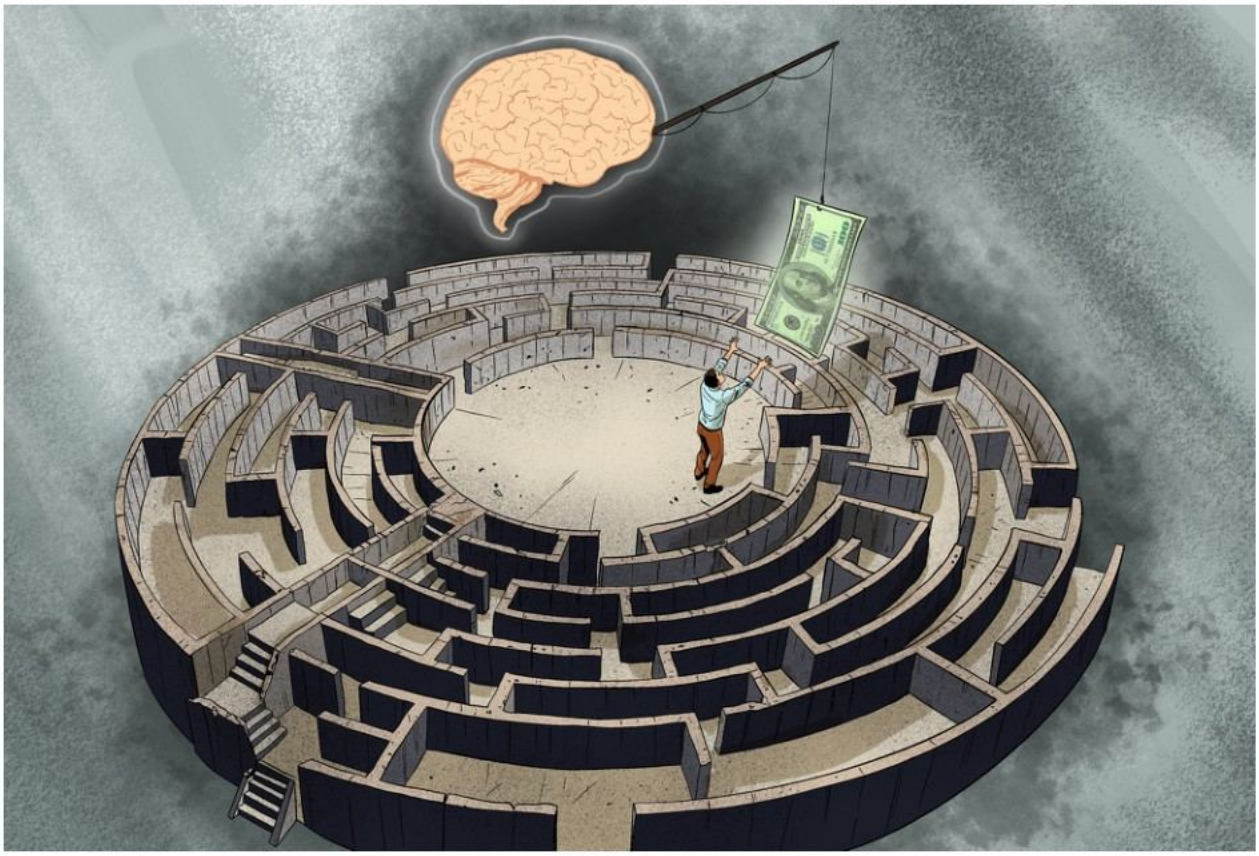
Even when life seems painfully unfair, human experience has a remarkable capacity to move. A sudden stroke of luck can bring pressures no one else sees, while people who endure terrible hardship do not always remain broken by it. Many adapt, discover strength they did not know they had, and find moments of love, purpose, or joy in places they never expected. Psychologists study part of this movement through hedonic adaptation and other forms of emotional regulation. Our feelings often shift away from their most extreme highs and lows as we adjust to what has happened. Experience changes as the person changes. It is almost as if the mind carries a quiet thermostat. And when the last visible chapter of a life is tragic, it can be easy to forget how much peace, joy, love, and satisfaction may have filled the years that came before. Do these familiar movements belong to something larger that becomes visible across the whole of a conscious life?



There is at least one remarkable way life could be fair for everyone, no matter how differently our stories unfold. Not through equal money, status, luck, health, opportunity, or length of life, but at the deeper level of felt experience. The Law of Fairness proposes that, across each qualifying conscious stream over its complete course, the total weight of positive feeling ultimately equals the total weight of negative feeling. Joy and sorrow may arrive in completely different amounts, intensities, and sequences along the way, and one life may look nothing like another. Yet when the whole journey is complete, the two sides would exactly balance. The scientific name for this proposed pattern is Per-Stream Terminal Valence Closure, or PTVC. This is the possibility at the heart of the Law of Fairness.



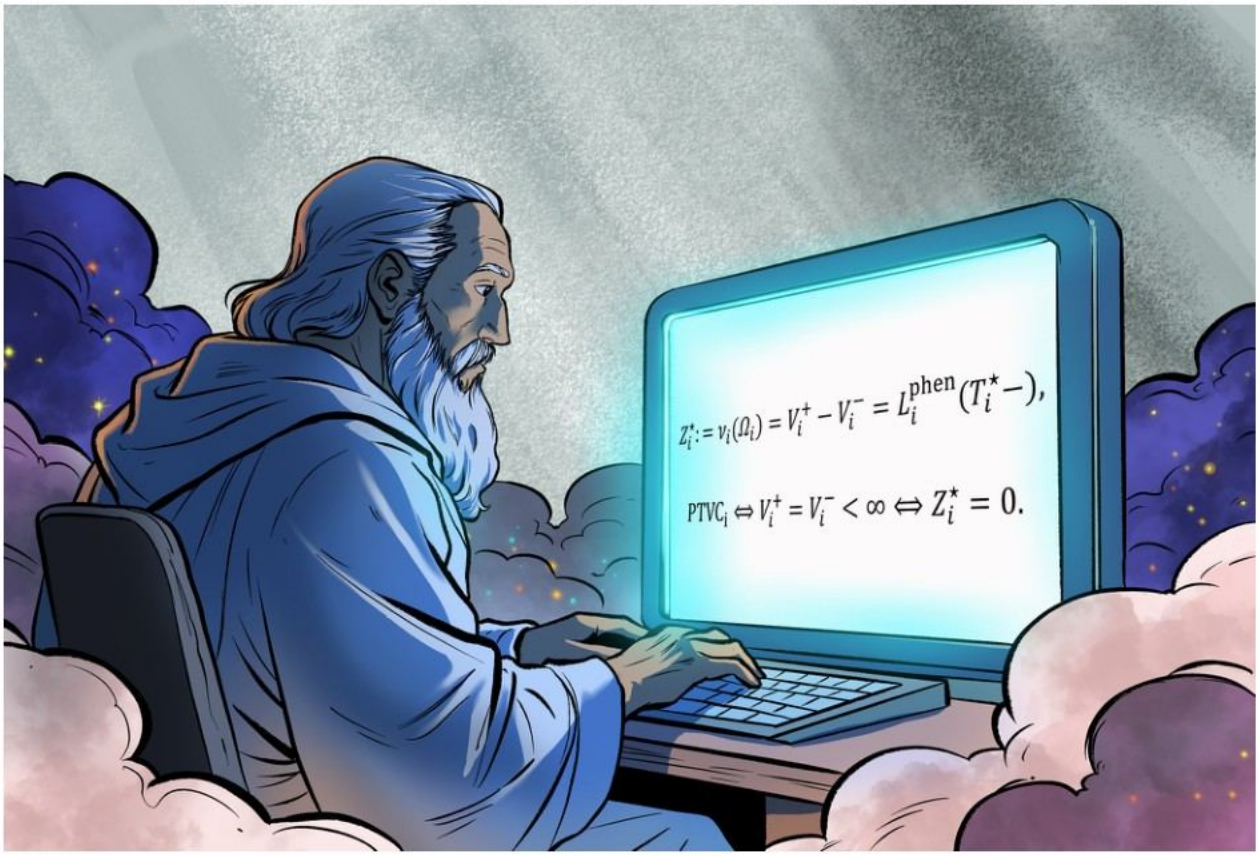
To determine whether the Law is true, we must examine the entire conscious story, not a single chapter. One joyful year cannot settle a life, and one terrible hour cannot reveal the total on its own, however morally serious that hour remains. The proposed domain begins when a qualifying stream of experience exists and ends at that stream's true endpoint. Everything felt between those boundaries belongs to the whole: joy and grief, pleasure and pain, ordinary days, forgotten moments, and experiences no outside observer may ever fully know.



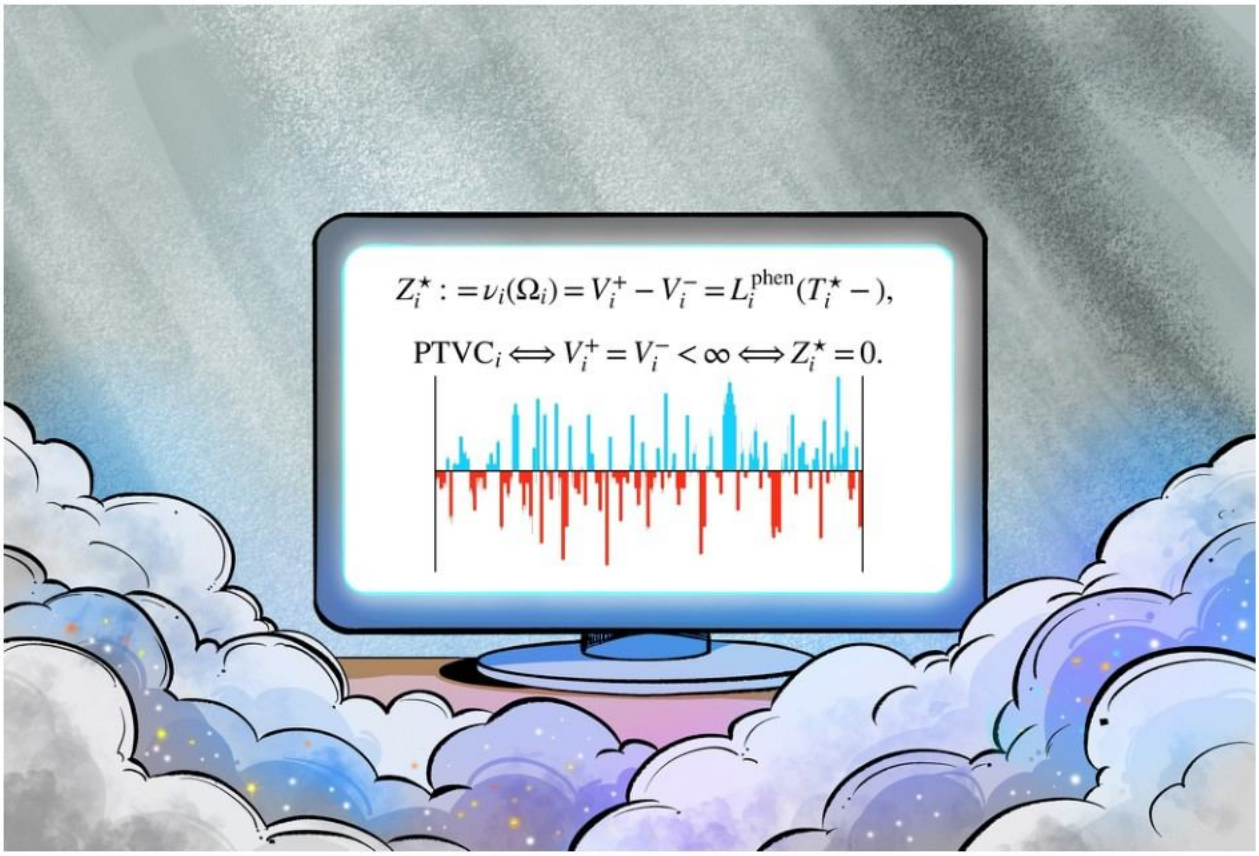
How could exact closure happen without anyone noticing? Our instincts, desires, addictions, fears, and even sudden pulls of attention help shape the experiences and possibilities ahead of us. If the Law of Fairness is true, these forces must unfold within a larger structure compatible with exact closure. Our urges and choices may sometimes function like guardrails, shaping which paths remain reachable, but they would be only one possible part of that larger structure. A conscious stream could not reach its true endpoint while its complete balance remained unresolved. Yet exact balance need not occur only at the end; the running total may return to zero at other moments along the way. How often that happens could vary enormously from one life to another, and discovering those patterns would be part of understanding how the Law works.



The numbers in this illustration are invented, not measurements of these experiences or values readers should assign to their own lives. The proposed equality does not pair one joy with one hardship or make suffering the price of pleasure. Our tears today may water the flowers of tomorrow's happiness through care or renewed understanding; at other times, the positive weight in the complete history may belong to joys already lived. The order and causes remain open. Research can ask how particular choices shape the experience that follows, then examine whether a distinctive whole-history relation also exists.



One way to picture a possible route to exact closure is what we call the Queue System. The name is a metaphor, not a claim that a literal queue, accountant, or controller exists inside the mind. Imagine instead that every conscious life unfolds through a vast field of possible futures. Some paths remain open, others narrow or disappear, and one becomes the life actually lived. Many paths could remain possible while still preserving a way to closure. Different mechanisms could produce that constraint, so the Queue leaves open how nature might accomplish it. The equation on the screen states the target precisely; the Queue pictures a possible route toward it.



Imagine felt value changing through time. Some moments rise above a neutral baseline; others fall below it. Even when an experience contains both positive and negative elements, its net felt value can be positive, negative, or zero under the chosen representation. As those moments accumulate, the running balance could climb, fall, reverse direction, or wander for years; the Law does not require it to move steadily toward zero. A measured graph would be evidence about the underlying felt history, not the history itself. Testing the claim would also require identifying the correct stream and endpoint. An average across many people would not establish a claim about each stream individually.



The Law of Fairness asks whether exact closure could be built into conscious reality as a lawful boundary, without a miracle or a judge adjusting events one by one. On one strong Queue-like route, possibilities that could no longer end in exact balance would fall outside the paths that remain available; on other routes, the constraint could lie elsewhere. This is a question about the structure through which conscious lives unfold, not a claim that an outside agent must steer each event.



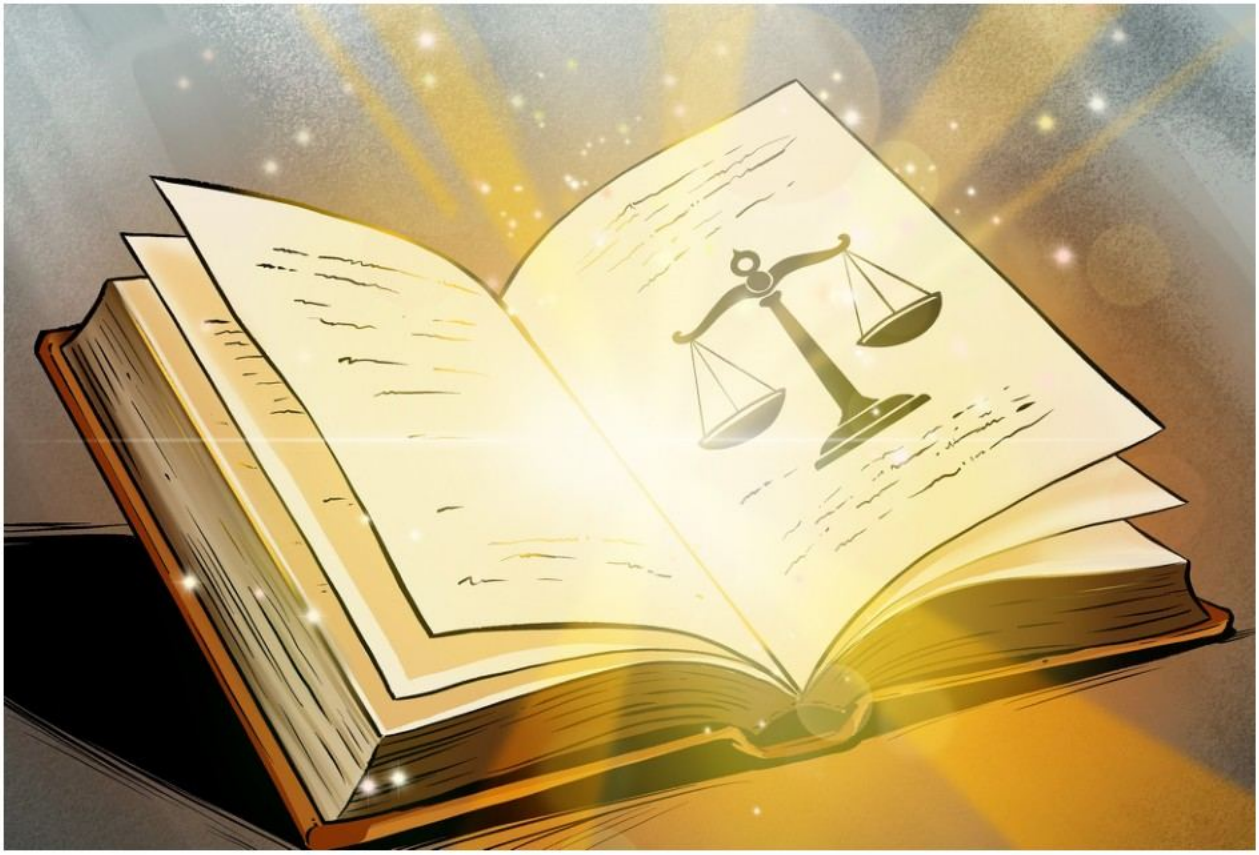
If exact closure is real, there is no way to “cheat” it at the level of the completed stream. Choosing pleasure, avoiding discomfort, changing habits, or pursuing a different path may alter what happens next, but none of those choices can place the finished conscious history outside the Law’s final condition. What we choose still matters because it helps determine the experiences, relationships, opportunities, and consequences that become part of that history. Thoughts, dreams, desires, and sudden impulses may also influence the path, but their exact role remains open to discovery.



Exact closure would connect the path of a conscious life with the point at which that life is complete. A possible route to balance is not enough if the stream can end before that route is taken. Under the hypothesis, whatever history is actually lived must satisfy the final relation. The endpoint has to be identified independently of whether the total balances; reaching zero cannot simply be our definition of completion. This opens a question about nature. Could the changing state of a stream shape when it can end? Could a broader constraint govern its complete history? The branching landscape leaves those possibilities open without fixing every step in advance.



For some readers, the Law will naturally carry spiritual meaning. A believer may imagine a Creator establishing a just structure so deep and complete that fairness need not be imposed through constant intervention, while another reader may see the same possibility as a natural regularity of conscious life. The formal claim itself remains neutral: it neither requires nor rules out God, karma, providence, reincarnation, or an afterlife. What it offers is a question believers and skeptics can approach together and test in this life: could exact closure be a real feature of conscious existence? Spiritual interpretation can enrich the meaning of that question, while evidence determines whether the underlying pattern is actually there. If exact closure were someday established, science could reveal that the pattern exists without necessarily settling why reality has such a pattern. In that sense, natural law and spiritual meaning need not be competing ideas, and the Law can be contemplated through faith without belonging to any one faith.



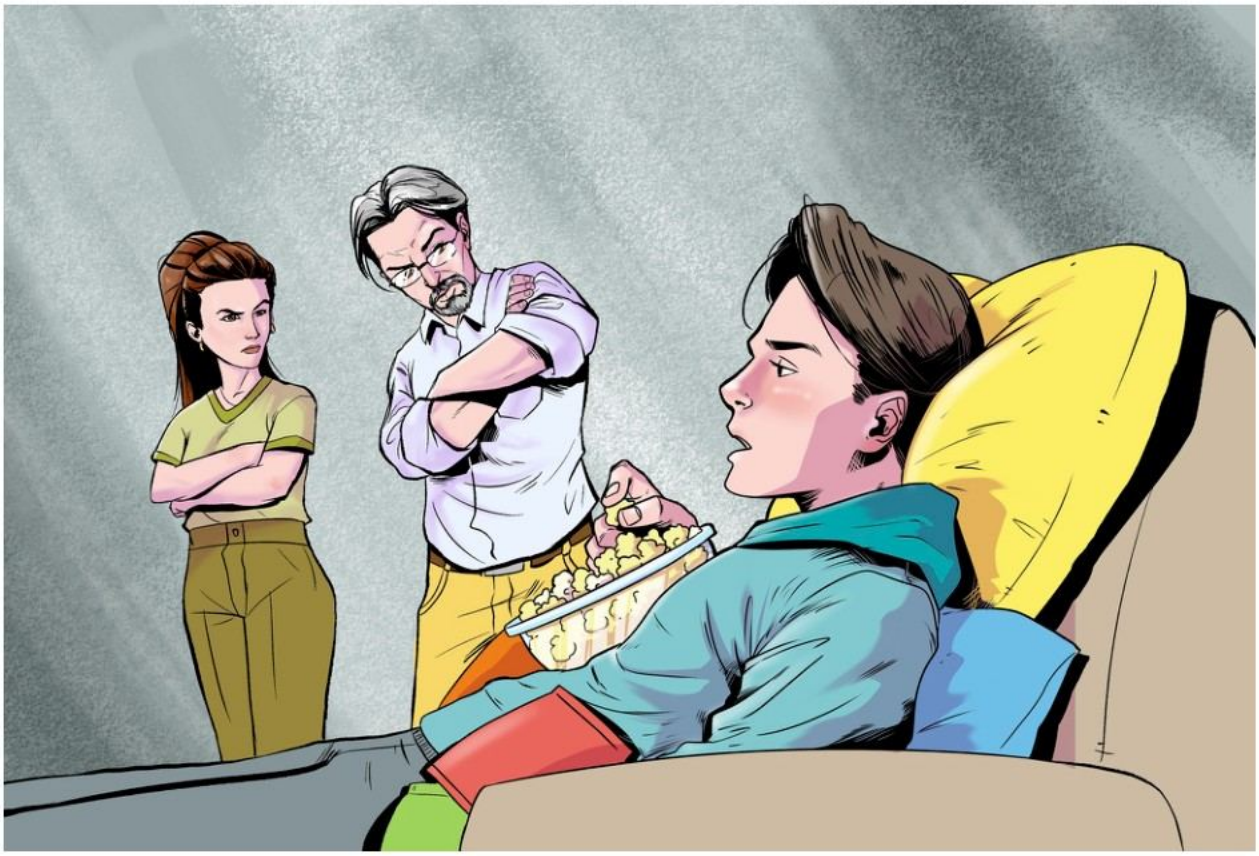
Across history, traditions have asked how to live wisely amid pleasure and suffering, desire and consequence. Long before the Law of Fairness was proposed, people built rituals, moral codes, prayers, stories, and ways of life around that question, often returning to themes of restraint, compassion, and balance. Those traditions do not prove the Law, but their persistence is difficult to ignore. If the Law is real, humanity may have been feeling its effects all along without understanding the deeper structure behind them. People may have noticed recurring patterns in how pleasure, hardship, choice, and consequence shape a life, then explained those patterns through the language available to them. What later became religion, philosophy, or moral tradition may sometimes have been an attempt to describe something real before the tools existed to test it directly. That would not make every ancient belief correct, but it raises a remarkable possibility: some of humanity's oldest ideas about how to live may be echoes of a deeper pattern in conscious experience.



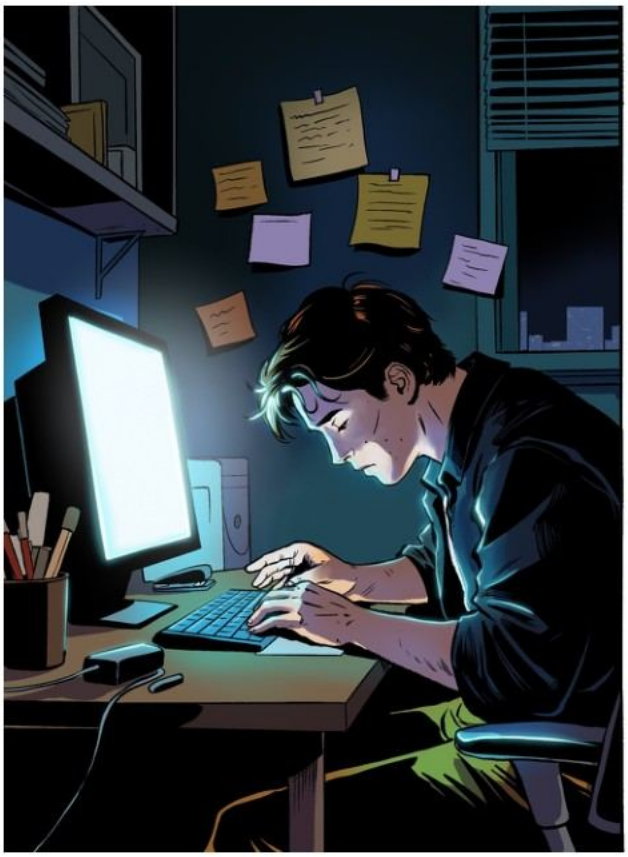
Many people sense a missing piece in our understanding of existence. Perhaps the Law feels like that piece not because it replaces science, philosophy, personal experience, or spiritual tradition, but because it gives them a new question to meet around. Science can ask whether exact closure is real and measurable. Philosophy can explore what such a pattern would mean for fairness and the human condition. Our own lives provide the felt reality the Law is actually about, while spiritual traditions may offer ways of interpreting what a deeper order would mean. These perspectives should not be collapsed into one another: adaptation is not karma, biology is not providence, and resemblance is not proof. Yet they may still be looking at different edges of the same mystery. Perhaps the missing piece is not a final answer at all, but a question precise enough to let very different ways of understanding reality finally speak to one another.



Consider the old warnings about greed, gluttony, pride, and excess. Their practical wisdom may reach deeper than simple moral condemnation. Pleasure itself is not the enemy; a good life should contain food, rest, celebration, achievement, love, and delight. The danger begins when pleasure stops enriching life and starts taking possession of it. Appetite can become compulsion, easy reward can reshape what we want, and repeated indulgence can quietly narrow the freedom to choose what comes next. That is why moderation matters: not because joy must be feared, but because freedom is worth protecting. The Law opens an even deeper question. Could some forms of excess reliably alter the later course of felt experience in ways we have not yet learned to measure? Addiction already shows that pleasure and suffering can become connected through recognizable chains of consequence. Studying the Law may reveal whether other patterns exist as well. If such patterns are found, old warnings about excess could gain a new interpretation through evidence rather than a claim of cosmic punishment.



Each of us can shape something like heaven or hell within ordinary life, not because we control everything that happens, but because our habits and choices help create the conditions we and others must live through. Comfort can become isolation, resentment can poison a home, and neglect can slowly close doors that once stood open. Honest change, treatment, forgiveness, boundaries, hard work, and care can transform what a life feels like from the inside. None of this makes hardship proof that someone is being corrected. Knowing that our choices can alter what another conscious life actually feels like, what kind of experience are we helping to create?





We pursue goals expecting satisfaction, but the effort required to reach them may be part of a deeper pattern in felt experience. Hard work can change not only what we achieve, but how strongly achievement is valued when it arrives. Delayed gratification, discipline, sacrifice, and earned success may create very different experiential trajectories from easy reward. The Law does not tell us that every obstacle was placed in our path for this purpose, and some hardships are plainly unjust. But it does give us a reason to ask whether recurring relationships exist between effort, reward, and the later balance of conscious experience. If such patterns are real, they could help explain why effortless pleasure can sometimes feel strangely empty while difficult achievements carry unusual depth and meaning. A worthy ambition may therefore be more than a destination. It may be part of a larger experiential pattern in which effort, anticipation, struggle, and fulfillment become one connected journey.



Think of daily habits as paths that can widen or narrow future freedom. A pleasure may enrich an afternoon and leave room for everything else that matters. Another pattern may gradually claim time once available for a friendship or a chosen ambition. We can notice that difference without assigning numbers to a life. A useful question is what the habit makes easier tomorrow, and what it makes harder. Understanding those consequences can suggest a practical change now. Reflection becomes worthwhile when it helps someone recover a possibility they care about.



Whenever life feels unfair, the Law should deepen compassion, never explain the wound away. Even if exact closure is real, present suffering is still part of what a person actually experiences, and we should not treat grief, illness, oppression, or danger as something that should simply be endured. A joyful event is not a warning that pain must soon arrive, and a painful one is not evidence that some hidden correction is underway. Easing another person's suffering changes the very experience the Law asks us to understand. If that deeper balance exists, compassion itself becomes part of the conscious histories through which it unfolds. Our responsibility is not to enforce the Law, but to improve the conscious experience our choices can actually touch.



We are all part of one living fabric, yet every conscious life is lived from within. No one can feel another person's joy or pain for them, and no stream can balance another. Still, our lives enter one another's stories. What we build, protect, neglect, repair, or simply say can change what someone experiences. That gives choices extraordinary weight. The Law of Fairness asks whether a deeper equality exists beneath the differences that divide us, but we do not need the answer before deciding how to treat one another. We can build a world with less needless suffering and more room for freedom, love, meaning, beauty, and joy. Perhaps humanity has been living within a deeper order all along, only now beginning to recognize its shape. Whatever nature reveals, no one can live another person's conscious life, but each of us can help change what that life feels like. The future of fairness is ours to discover and help build. If we take that responsibility seriously, we may yet create a world worthy of the lives within it and give future generations something humanity has sought for centuries: a more peaceful world.

Part I — The Question and Human Stakes

Fairness begins as a human recognition before it becomes a theory. It appears when safety gathers around one life and danger around another, when illness arrives before a child can name what is being lost, when effort and outcome part company, and when time, love, health, power, and pain arrive without visible regard for innocence or need. The question endures because human beings can see both the world’s beauty and the burdens it distributes, and because understanding that contrast may help us care more wisely.

Yet visible inequality does not exhaust the inquiry. A biography can be observed from outside, while experience is lived from within. Two people may receive similar goods and inhabit different emotional worlds. Public advantage may coexist with fear, pain, compulsion, loneliness, or shame that no photograph reveals. Severe external constraint may coexist with humor, relationship, skill, meaning, or peace, but none of those goods makes the constraint acceptable. Appearance is real evidence, not the whole field of consciousness.

The proposed Law begins at that divide. It asks whether each complete qualifying conscious stream could contain finite and exactly equal total positive and negative felt experience, under a declared representation of felt experience. This primary version is called Per-Stream Terminal Valence Closure, or PTVC. Its balance concerns one stream across its whole history before its true endpoint, with the stream, the measure, and the endpoint identified independently of the desired answer. Lives could still differ profoundly in duration, health, freedom, opportunity, memory, and everything that makes a history its own. The proposal therefore opens a separate question about experience while leaving the work of justice fully in place.

Imagine a kitchen table crowded with schoolbooks and medication schedules. A child studies tomorrow's lesson while a

parent calculates which appointment the family can reach. No photograph of that table records the child's full inward life; neither does its hardship disappear because the child sometimes laughs. This scene raises two questions that must not be traded against each other. What can be done for this family now? And what, if anything, could a proposed relation across an entire conscious history actually explain or predict? The first question calls for decisions. The second demands evidence. Their difference makes the larger inquiry worth attempting.

Three witnesses enter this inquiry. A public record tells us what happened; a person tells us how an interval was lived; an institution shows what can be done about a preventable harm. The opening chapters keep their testimony distinct. A better question becomes possible once we stop demanding that one witness tell the whole story.

There is a further pleasure in asking the question carefully: it can make familiar scenes more legible. An explanation of how a burden arose, an account of how it felt, and an idea for making the next day easier can each illuminate the same life. The inquiry begins by learning to move among those perspectives deliberately. Its ambition is a connected account in which causes, experience, and practical action retain their particular detail, and in which a larger theory earns its place by revealing a relation that those details alone had left unexplained.

Chapter 1 — Why Fairness? Why Now?

A child asks why one person has enough and another does not. An adult answers by tracing family circumstances, public decisions, and events beyond anyone's control. These causes matter. They explain how unequal conditions arise and often show where responsibility lies. Yet after the explanation, the child's question survives. Why should the burden have landed there?

A record of outward events cannot by itself settle a claim about an entire conscious life. That claim extends beyond familiar judgments about treatment, opportunity and responsibility.

A principle of just treatment can be defended by the obligations it creates. A proposed regularity of nature is different: it must risk an observation that contradicts it. The question also changes with scale. A household can be loving inside an unjust labor market; a country can improve average health while leaving one region behind. Fairness judgments become clearer when the level of analysis is named rather than allowed to slide from moment to institution to existence.

A personal case does not stop changing when it enters a book or a dataset. An interview given during a crisis may be accurate about that day while becoming misleading if quoted years later as the final account of a life. A later change of mind need not expose the first speaker as dishonest; it may record a different position in an unfolding history. Responsible comparison therefore attends to the date of an account and the conditions under which it was offered. It allows a person to revise an interpretation without surrendering ownership of the earlier words. Credibility judgments must also be checked for the prejudice that produces testimonial injustice (Fricker, 2007).

Why now? Methods that once sat apart can now be joined. Affective science distinguishes valence from arousal; intensive

longitudinal methods model within-person change; consciousness assessment and palliative-care frameworks sharpen questions about awareness and observation near the end of life; and preregistration and missing-data analysis make analytical choices inspectable (Russell, 2003; Fisher et al., 2018; Giacino et al., 2018; World Health Organization, 2020; Little & Rubin, 2019; Nosek et al., 2018). Together they divide the inquiry into problems that can be investigated in stages.

The attraction of the Law lies in what a successful explanation would connect. It would link the changes of feeling inside an ordinary day to a relation across a complete conscious history, while allowing lives to remain profoundly different. Such a finding would reach beyond a description of recovery: it would reveal a constraint on how experience can unfold. That possibility makes the question worth sustained attention.

The route into that possibility begins with ordinary judgments. We first ask what visible conditions reveal, then distinguish the standards people invoke when they call an arrangement fair. Difficult cases give those distinctions something to work on. Only after that groundwork do we turn to the difference between a group average and one complete stream. The order matters: it lets the formal question emerge from recognizable concerns, while giving the reader a way to tell which concern a proposed answer actually addresses.

1.1 — The World We Can See

The public record need not be complete to be useful. A missing meal can reveal a material shortfall. An exclusionary rule or a denied hearing can reveal a failure of procedure. Each is a specific claim that can be investigated before anyone answers the much larger question of whether a whole life is fair.

1.1.1 — The Ordinary Sentence, “Life Is Not Fair”

The ordinary sentence, life is not fair, often follows a mismatch between expectation and event. It may name an injury before the speaker knows whether the cause was chance, choice, or an institution's rule. Its imprecision is not emptiness: a disproportion has been noticed.

Consider two children entering the same school year. One arrives rested, housed, medically supported, and certain that an adult will answer the telephone. The other has moved three times, shares a room with people working opposite shifts, and has learned to hide hunger. Their test scores may later be compared as if effort entered a neutral field. The ordinary sentence protests the field before it makes any claim about inner totals.

The same sentence can also name bad luck without accusing a person. A random collision, a genetic condition, a wildfire, or the sudden loss of a caregiver may alter a future before responsibility has any meaningful role. Work on moral luck shows why judgment becomes unstable when outcomes depend heavily on factors outside control (Nagel, 1979; Williams, 1981).

Sometimes the sentence is directed at institutions rather than nature. A rule may be written in neutral terms while its cost falls predictably on people with less money, less mobility, less legal knowledge, or less time to appeal. Social conditions shape

exposure to illness, violence, pollution, and premature death, so visible unfairness is often organized rather than accidental (Marmot et al., 2008; Williams et al., 2019).

There is a difference between saying that life contains unfair arrangements and saying that life is unfair in every possible sense. The first can be supported by observable facts about rights, resources, exposure, and treatment. The second reaches toward a total verdict on existence. *The Law of Fairness* enters precisely in the space between those statements. It asks whether one deeper relation could hold across a complete conscious life without denying the injustices already in view.

1.1.2 — Why Suffering Reopens the Question

Suffering reopens the fairness question because pain is not merely information. It is experience demanding recognition. When suffering enters a household or affects someone loved, fairness becomes a protest against disproportion and a demand that the burden be seen.

Pain science reinforces the need to listen before theorizing. Pain is not identical to tissue damage and cannot be inferred perfectly from a scan, an injury, or a facial expression. It is a personal sensory and emotional experience shaped by biological, psychological, and social conditions. The person's report therefore has a privileged evidential role (Raja et al., 2020; Cassell, 1982).

The social location of suffering matters as well. The same symptom may be treated promptly in one patient and discounted in another. Differences in credibility, access, race, disability, language, gender, income, and geography can shape whose pain is believed and whose care is delayed (Bailey et al., 2017; Williams et al., 2019). These are public failures with identifiable mechanisms.

Suffering also changes time. Minutes in acute pain may feel enlarged, while months of chronic illness can blur into fatigue, appointment schedules, missed work, and repeated explanation. Anticipation can make a future procedure present long before it begins. Relief can be immediate, delayed, incomplete, or accompanied by fear that the pain will return. Any account of felt experience that ignores these temporal structures will mistake clock time for lived exposure.

The problem becomes sharper when communication fails. An infant, a person with advanced dementia, an intubated patient, or someone who does not share the clinician's language may be unable to provide the report a protocol prefers. The absence of a standard answer is not the absence of experience. Pain science therefore treats self-report as central where available while also developing behavioral, contextual, and physiological evidence for situations in which speech is constrained (Raja et al., 2020).

Disbelief can compound pain with self-doubt, financial loss, stigma, and the repeated labor of explaining what others cannot see. Epistemic injustice names the added burden when testimony is discounted because of identity, diagnosis, or social position (Fricker, 2007; Kidd & Carel, 2017).

Suffering and relief can cascade through connected pathways: pain can disrupt sleep, concentration, work, and economic security, while effective care can reverse several links at once.

1.1.3 — The Difference Between Comfort and Inquiry

Comfort responds to the person who is hurting; inquiry responds to the unresolved question.

Timing matters: a person may welcome reasons in calm times but need quiet in crisis. Medicine distinguishes disease from suffering that threatens identity, agency, relationship, or future; a correct mechanism may not address those threats (Cassell,

1982). Neither comfort nor distress, by itself, tests whether the Law is true.

Verb choice makes the boundary visible. ‘I believe that you are hurting’ recognizes testimony, and ‘we will help you find safety’ commits to action. ‘I know what your pain is for’ claims authority, while ‘everything will balance’ predicts an unseen outcome.

1.2 — What Fairness Usually Means

Fairness refers to several standards, not one settled object. Rules can be judged for procedure, distributions for outcomes, judgments for desert, and relationships for reciprocity. These standards may answer different questions about the same event. An argument becomes unreliable when it borrows authority from one sense of fairness without naming the shift.

1.2.1 — Procedural Fairness

Procedural fairness concerns rules, hearings, and decision paths. It asks whether standards were knowable and applied consistently, bias was limited, people could understand what was happening and be heard, reasons were connected to decisions, and errors could be challenged before someone with authority. These features matter because process shapes both outcomes and the experience of being treated as a participant rather than an object; research identifies voice, neutrality, respect, and trustworthy motives as important to perceived legitimacy (Tyler, 2006). Formally equal rules can still exclude through language, disability, time, access, or power.

Neutrality also requires more than an impersonal tone. Decision rules can encode old inequalities, data can omit the most burdened people, and discretion can be exercised differently across groups (Bailey et al., 2017; Williams et al., 2019). Procedure must therefore be evaluated by design, operation, and patterned consequence.

Respectful treatment matters even when the decision remains adverse. A person can be told no without being humiliated, ignored, or treated as a threat. This distinction is especially important in systems where the person depends on the decision-maker for medicine, safety, education, income, or legal status.

Dependency magnifies the experiential cost of opacity and makes retaliation protections part of fair process.

Public procedures can impose burdens even when the rule itself is formally available. Applicants may have to learn a complex process, produce repeated documentation, wait through delays, travel, disclose private facts, and absorb the anxiety of a minor error. These costs can restrict access to rights and benefits as effectively as an explicit denial, which is why procedural fairness must examine the route as lived rather than merely as written (Herd & Moynihan, 2018).

1.2.2 — Desert-Based Fairness

Desert-based fairness asks when praise, reward, blame, or sanction properly follows from what a person chose and did. The idea has a legitimate local use. Promises, contracts, offices, and relationships would become unintelligible if intention, knowledge, effort, and responsibility never mattered. The difficulty begins when a judgment about conduct expands into a judgment about the value of a person or the hidden purpose of suffering.

A useful desert judgment is narrow. It asks what duty applied, what the person understood, what alternatives were genuinely available, how freely the action was chosen, and what response is tied to the act and harm. A worker who completed difficult work may reasonably expect credit; a public official who concealed a conflict may properly lose authority; a person who damaged another's property may owe repair. None of these judgments makes distress itself the currency of justice.

Judging people only by their resources can mistake advantages in turning resources into opportunities for merit, and barriers to doing so for flaws of character (Sen, 1999; Sen, 2009).

The proposed Law concerns the net balance of experience over a complete conscious life. Its target differs from just treatment, a distribution of goods, and the tendency of mood to recover.

Equal totals would not make lives equivalent. In an invented example, one stream could accumulate ten units of positive experience and ten of negative experience; another could accumulate a thousand of each. Both net to zero, while their intensity, duration, opportunities, and histories remain profoundly different.

1.2.3 — The Experience of Receiving a Reason

A decision arrives in two parts: the outcome and the account of how it was reached. Imagine a family receiving a school placement letter. The first version states only the assigned school. The second explains the criteria, identifies the information used and gives a date for correcting an error. The placement is identical. The family's relationship to it can nevertheless change because the second letter makes the next step intelligible.

Before receiving an explanation, a parent may repeatedly reconstruct the decision. Was the address entered incorrectly? Did another family receive an exception? Is an appeal still possible? Each unanswered question creates another task. An explanation can close some of those tasks and make the remaining one specific. The effect arises through information and usable agency. It can be described without pretending that the letter makes the placement desirable.

The difference becomes clearer if the reason is accurate but unusable. A letter might cite a rule that the family cannot obtain or name an appeal office that never answers. Formal disclosure has occurred, yet the practical question remains open. A useful

reason connects an outcome to a process that a person can understand and act within. The recipient can then distinguish an error from a disagreement with the rule itself.

This example reveals something easy to miss in comparisons of outcomes. Two people can receive the same allocation while inheriting different amounts of unresolved work. One can plan the coming week. The other must keep investigating what happened. The outward result is only one moment in a longer sequence of attention and action.

The sequence also offers a better question for an institution. Instead of asking whether its decision was explained somewhere, it can ask whether the explanation reached the person in a usable form and enabled the next step. A person's experience of a procedure becomes more understandable when we follow that sequence. Fairness is then examined where the decision enters a life.

1.2.4 — Fair Shares and the Standard of Comparison

Distributive fairness asks how shared arrangements allocate resources and opportunities. A procedure tells us how a decision is made; a distributive principle tells us what the resulting arrangement should achieve. Equal shares are one possible standard. Priority to an unmet need is another. Choosing between them requires reasons about the people and the good being shared, not only an accurate division of a total (Rawls, 1999; Sen, 2009).

Consider an invented study group with twelve hours of individual mentoring available to four learners. Giving each learner three hours creates equal shares of the available time. Suppose, however, that one learner needs six hours to complete an introductory preparation that the other three have already received. Giving that learner six hours and each of the others

two hours still uses twelve hours. The second arrangement is unequal in mentoring time but aims to give everyone a workable starting point for the same course.

The arithmetic identifies what changes between the two arrangements. It cannot decide which aim should govern the allocation. That decision depends partly on what the group undertook to provide. A group organized to share a visiting tutor equally has one commitment. A group organized to prepare everyone for a common course has another. The participants can discuss those commitments directly rather than treating the word equal as a complete answer.

Now change one fact. Suppose the extra three hours are sought for an optional advanced project rather than for introductory preparation. The numerical allocation is unchanged, but the reason for priority has changed. A person who supported additional help to establish a common starting point may reasonably ask for a different justification here. The relevant comparison follows the purpose of the assistance, not merely the size of the request.

This gives the familiar fairness question a clearer shape. We can ask what is being shared, which differences are relevant, and what the arrangement is intended to make possible. Chapter 2 follows a different step: how resources become usable conditions within a particular life. Here the question concerns the standard by which resources are allocated among lives. Naming that standard makes a disagreement more specific and gives the participants something they can examine together.

1.3 — The Cases That Resist Easy Answers

Some cases expose limits that ordinary definitions conceal. A child's suffering tests any claim that hardship is earned. Early death tests what an explanation assumes about a whole life. Apparent fortune tests whether outside success can reveal inner experience.

1.3.1 — Innocent Children

Innocent children are the first hard case because they remove blame before the argument begins. A child does not choose the circumstances into which they are born or bear responsibility for harms imposed on them. Dependency gives adults and institutions greater responsibility to protect the child (United Nations, 1989).

Childhood also exposes the limits of consent and voice. Children may lack the language, authority, or practical safety to report what is happening, yet adults can still overinterpret silence, compliance, or adaptation. A child’s vulnerability is not a reason to turn the child into an emblem.

Childhood is not valuable only as preparation for adulthood. Safety, play, attachment, learning, rest, friendship, and relief are present goods in a life being lived now. Protecting them is not merely an investment in future productivity or health. The child’s current experience has standing before any long-range theory is considered (United Nations, 1989).

Children may express distress through movement, sleep, play, withdrawal, irritability, or physiological change before they can describe it (Raja et al., 2020; Trottier et al., 2022).

Development makes timing consequential. The effects of adversity depend on when it occurs, how long it lasts, how severe or persistent it is, and whether responsive relationships and other protections buffer it. Interacting pathways of stress

regulation, learning, health, and relationship may be altered without fixing a child's destiny (Shonkoff et al., 2012). These findings should direct prevention, support, and repair rather than deterministic labels.

Children reveal why social causation matters. Preventable disease, malnutrition, conflict, unsafe environments, and unequal access to essential care have institutional histories. Global child-mortality estimates show both long-term progress and a continuing concentration of deaths in regions and settings with the greatest barriers to survival (United Nations Inter-agency Group for Child Mortality Estimation [UN IGME], 2026).

Resilience should describe a process, not excuse an environment. A child may adapt through supportive relationships, skill, temperament, community, or sheer necessity, while still carrying costs that deserve attention. Quietness, vigilance, high achievement, or apparent self-sufficiency can be compatible with competence and distress at once. The evidence should direct support toward the child and the conditions, not turn endurance into proof that help was unnecessary.

1.3.2 — Early Death and Interrupted Futures

Early death brings the endpoint question forward before the reader is ready for it. A life interrupted in childhood, youth, or early adulthood seems to defeat every easy account of fairness. There was no long arc, no visible time for repair, no old age in which patterns could soften.

An interrupted future is not merely a shorter count of calendar years. It can deprive a person of projects, relationships, repair, learning, agency, and possibilities that had already begun to matter. Philosophers have long examined death as deprivation while recognizing the uncertainty of counterfactual futures

(Nagel, 1970). Those unrealized possibilities remain ethically important, but they are not phenomenal events to be entered retroactively into the ledger. The formulation developed here concerns experience within the stream; justice, medicine, and grief also track goods and futures the phenomenal target does not absorb.

Early death also makes observation difficult: fatigue, sedation, delirium, and deterioration may remove the intervals a study most needs. Loss of report cannot simply be assigned a neutral value. Chapter 5 examines the distinction between observation boundaries and the end of a stream.

Population statistics reveal patterns that individual stories cannot. UN IGME estimates that 4.9 million children died before age five in 2024, including 2.3 million newborns, with the heaviest regional burdens in sub-Saharan Africa and Southern Asia (UN IGME, 2026).

For the formal hypothesis, early death creates a reachability question. If the endpoint arrives while the cumulative object is far from zero, what lawful structure could still make exact closure possible?

Those who remain may carry medical debt, traumatic memory, altered family roles, lost income, disrupted education, and decisions that cannot be revisited. These consequences belong to their own lives and to institutions capable of reducing them. They should be named because early death is not a single private event; it reorganizes households and communities.

1.3.3 — Visible Fortune With Hidden Burden

Visible fortune can protect people by widening options and reducing exposure, but it does not reveal an entire felt history. Inquiry should follow mechanisms and repeated testimony without assuming effortless happiness or secret misery

(Kahneman & Deaton, 2010; Killingsworth et al., 2023). Longitudinal evidence can distinguish a fleeting burden from one that recurs or is structurally tied to a person's circumstances.

Social media intensifies the temptation to infer an easy inner life from a house, title, body, partner, or audience. Better questions examine control over time, exit, belonging, comparison, surveillance, and the cost of honest disclosure.

1.3.4 — Possibility and an Interrupted Future

An interrupted future can be understood through the possibilities that were open. Suppose a young person has accepted a place in an apprenticeship. The place does not determine the adult life that would follow. It creates an opportunity to learn a craft and meet people through that work. Its value begins before anyone knows whether the person would complete the training or change direction.

This distinction helps us describe loss precisely. We can know that an opportunity was available while remaining uncertain about its eventual outcome. The two judgments concern different objects. One concerns access to a path. The other concerns the particular history that might have unfolded along it. Treating an opportunity as valuable does not require selecting one imagined biography and declaring that it would certainly have occurred.

The same reasoning applies to ordinary choices within a continuing life. A place to study may be valuable partly because it opens several later directions. A reliable means of travel may make previously distant work available. A relationship can widen what someone feels able to attempt. These changes concern the shape of the available future before they produce a particular achievement.

For the whole-stream question, a further distinction follows. Anticipating a future can be an experience in the present. The hoped-for events themselves belong to a possible continuation until they occur. A person may already feel excitement about the apprenticeship, while the experiences of training remain unrealized. The present anticipation and the future possibility therefore have different places in the account.

Holding those places apart enriches the description of a life. It allows us to recognize the significance of an open future without filling the unobserved years with invented feelings. A history contains what was lived. Its human significance also includes the paths that circumstances made available and the directions a person hoped to take. The Law's formal question becomes clearer when it is situated within that larger understanding.

1.4 — Why Averages Cannot Settle a Whole Stream

A population average summarizes many lives, but the formal claim concerns each completed stream. A follow-up window captures part of a history rather than its endpoint. Regression can test a declared relation without deciding an unobserved terminal sum. Evidence must therefore keep the individual, the observation interval, and the target claim distinct.

1.4.1 — Averages Do Not Save Streams

Averages reveal patterns that anecdotes cannot, but different levels of analysis answer different questions. The ecological fallacy arises when a relation among group-level percentages is attributed to individuals. A parallel error assumes that differences between people describe change within one person (Robinson, 1950; Molenaar, 2004; Fisher et al., 2018).

The scientific program must therefore track individual trajectories and uncertainty, not only averages. Hierarchical models may borrow information across people when individual data are sparse, but partial pooling must remain visible: a person's estimate pulled toward the group is not direct knowledge of that person. A model can predict the center well while missing volatile or extreme paths, so it must report who the average fails to represent and how uncertainty changes at the tails. Section 11.4.2 works through the calculation and shows where the remaining uncertainty goes.

Aggregation can even reverse a comparison. In an invented example, Clinic A has successful outcomes in 9 of 10 lower-risk cases and 30 of 50 higher-risk cases. Clinic B succeeds in 40 of 50 lower-risk cases and 5 of 10 higher-risk cases. Clinic A does better within each group: 90% versus 80%, and 60% versus 50%. Yet its overall rate is lower, 65% versus 75%, because it treats more higher-risk cases. A per-stream question therefore

asks which units were pooled, how they were weighted, and whose paths the summary may conceal.

Population success can be distorted by attrition. If participants with worsening conditions are more likely to leave, a cohort mean may return toward baseline because some recover, some remain impaired, and the most severely affected disappear from observation. Death, institutionalization, distrust, cost, caregiving, and symptom burden can all alter the denominator. Without tracking who left and why, a reassuring average can be partly manufactured by observability rather than recovery.

1.4.2 — A Life Is Not a Follow-Up Window

Studies observe people over days, weeks, months, or years, and those windows can reveal recovery, relapse, adaptation, burden, relief, and changing physiology. Most studies begin after experience has started and end while the stream continues. Even a long follow-up represents the complete history only when justified bridges account for unobserved contributions before, between, and after the recorded intervals. The complete-stream claim retains those contributions rather than equating the recording with the life.

Timing can change the apparent conclusion. A study may end before delayed collapse, miss late repair, capture a temporary rebound, or begin after the most severe burden has already passed. The same life can support opposite stories when different windows are selected.

A finite window may test adaptation after a specified event, reserve scaling, within-person dynamics, or endpoint-proxy performance. Its length should be chosen for that question rather than treated as a miniature lifetime.

1.4.3 — The Need for an Exact Terminal Question

A simple thought experiment shows how an exact claim could face a finite challenge. On the additive branch, suppose independently justified bounds established that a stream had already accumulated a positive net total of at least eight units and could accumulate at most two further negative units before its independently established true endpoint. Its final total would have to remain positive. No exact final reading would be needed to exclude zero.

The corresponding negative case is a stream whose own evidence supports prolonged severe negative experience and whose independently established endpoint leaves no admissible complete-history bound reaching zero. Absence of observed relief alone does not supply that bound. Disability, restricted mobility and dependence on life-sustaining technology do not themselves establish negative experience.

The example identifies the information a test would need. Establishing the units, covering all relevant past and future contributions, and justifying those bounds would be the scientific work. A bound supported statistically would carry its stated uncertainty; a deductive exclusion would need bounds established to contain the true values under the declared model. This turns the measurement problem into a concrete challenge: can a proposed route supply enough independently justified information to make some outcomes impossible? Endpoint selection is examined in Section 5.3.

Several ordinary processes could imitate the appearance of closure. Regulation can pull states toward a usual range, worsening illness or medication can reduce variability, and a statistical path can be made to end near zero simply because that endpoint was built into the model. A forced ending is not evidence that nature supplied the rule.

Evidence from verbally reporting adults does not by itself settle how the Law applies to infants, people with severe disorders of consciousness, animals, or artificial systems. Each extension needs its own justification of bearer status and measurement.

Chapter 2 — The Life Only You Can Feel

A house is visible. Safety is lived. A diagnosis is recorded. Pain is felt. Status can be announced in a title, but belonging, humiliation, relief, vigilance, and loneliness unfold inside a person's experience. The distinction is simple enough to overlook and deep enough to reorganize the fairness question.

An employee may have a right to medical leave yet miss an appointment because taking time off threatens their standing at work. The written policy records authorization. Scheduling, actual uptake, and fear of retaliation reveal whether that entitlement can be used; the person's account helps clarify its felt consequences.

Ask the same person two questions: "How is your life going?" and "What did the last hour feel like?" An answer about financial security may coexist with a report of pain, worry, or delight in a particular interval. Neither answer is insincere, and neither can simply replace the other. Adjusting to disability or a changed situation may also alter the standard by which a life is evaluated. A scientific account must therefore specify which question produced its data and whether the interpretation survives a change in context. The measured constructs are different (Kahneman & Deaton, 2010; Killingsworth et al., 2023; Albrecht & Devlieger, 1999; Koch, 2000; Sprangers & Schwartz, 1999).

Time is often the hidden medium through which goods work. Money can purchase a shorter commute, childcare, a safer repair, or the ability to refuse an extra shift. These changes may reduce fear and recover attention. Yet time is distributed within households and institutions, so the same income can produce different lived conditions depending on care obligations, disability, geography, debt, and public services (Sen, 1999; Nussbaum, 2011). Goods are not automatically interchangeable. Income can purchase services, yet it may not restore health,

trust, legal standing, or lost time. Love can support a person through poverty, but it does not make poverty harmless.

Imagine two people sitting quietly in the same waiting room. One is peacefully relieved; the other is frozen with fear. Their posture may look similar while their experience differs sharply. Affective science helps explain why observation requires concepts, context, and multiple forms of evidence rather than confidence drawn from appearance alone.

A material good enters conscious life through a chain rather than a single conversion. Shelter is not only walls: stable housing can reduce exposure to weather, violence, eviction, and forced movement while creating privacy, predictable sleep, a place for medicine, and a reliable address for school or work. These consequences are partly experiential and partly practical, which is why the good matters both for what a person can do and for what daily life demands.

A change in conditions can also change what a person notices. When an urgent practical demand is resolved, attention may become available for an unfinished conversation, a remembered ambition, or an ordinary pleasure previously crowded out. That sequence makes lived improvement more intricate than replacing one sensation with another. An opportunity can matter by allowing a different part of life to come into view. Following such sequences gives the chapter its method: begin with a concrete condition, trace how it reorganizes activity and attention, and listen for the experience of that reorganization.

2.1 — Why Experience Matters

Two reductions fail in opposite directions. Material reduction treats possession or status as a complete account of life. Experiential reduction treats rights and conditions as important only through immediate feeling. The first ignores lived reality; the second makes justice hostage to reporting, awareness, adaptation, or mood.

2.1.1 — Goods Become Lived Conditions

Medicine demonstrates why benefit is multidimensional. A treatment may relieve pain, extend life, restore mobility, or reduce uncertainty. It may also cause side effects, cost, disruption, stigma, or dependence. A serious evaluation follows these consequences rather than declaring the intervention simply positive or negative. The person’s own priorities are necessary because the same trade-off can be lived differently.

Relationships are not possessions: marriage, friendship, family, and community can provide care and meaning while also involving coercion, obligation, surveillance, or loneliness. The label alone does not tell us which conditions prevail.

An opportunity often depends on conditions that work together. Imagine someone preparing a digital collection of family photographs. A scanner is available, and the person knows how to handle the prints. The scanner's controls, however, cannot be operated through the person's usual assistive interface. The practical route requires both the image-making equipment and a usable way to control it.

Suppose a second scanner arrives with the same inaccessible controls. The amount of equipment has increased, while the person remains unable to carry out the chosen task independently. Now suppose a compatible control interface becomes available. The original scanner becomes useful without

any change to its optical quality. A condition that appeared peripheral has completed the route from equipment to activity.

The reverse sequence reveals the same relationship. If the accessible interface is ready but no scanner is available, supplying the scanner completes the route. The usefulness of either improvement depends partly on which other conditions are already present. This is complementarity: together the conditions make something possible that either alone leaves unfinished.

This differs from adding two independent benefits. Two complete scanning stations might allow two people to work at once. Two incomplete stations might still leave the chosen task inaccessible. Counting devices answers a question about resources; following the whole activity reveals whether the resources can be used for the purpose that matters.

Once the route is complete, the person can choose which photographs to preserve, adjust the images, and decide how the collection will be shared. Access has become a practical capacity exercised through a project of their own. The example gives the resource-to-experience inquiry a specific structure: identify the required conditions, see how they combine, and follow what their combination enables the person to do.

2.1.2 — External Justice and the Limits of Surface Outcomes

Rights-based disability frameworks require institutions to remove barriers to participation (United Nations, 2006; World Health Organization, 2022). A person's report of well-being and an institution's accessibility duty answer different questions.

Freedom from violence, access to education, bodily integrity, legal recognition, and political participation matter even when a person cannot report their experiential effect. A sedated patient,

an infant, or a person with profound communication difficulty does not lose standing because a valence estimate is unavailable. Justice cannot wait for measurement.

A wheelchair user may report a rich and satisfying life while a staircase excludes them from a public meeting. The response should honor that testimony and remove the barrier.

External justice also protects people whose suffering is hidden by necessity. A worker may report that conditions are manageable because complaint risks employment. A child may say everything is fine because the unsafe adult is present. A patient may minimize pain because previous reports were dismissed. Public safeguards cannot depend solely on a declaration made under unequal power.

A person may accommodate to a narrow field of options and still be owed a wider one. Preference adaptation therefore cannot be the final measure of fairness (Sen, 1999; Nussbaum, 2011).

Exercising a right can itself be difficult or distressing. A person may choose an arduous life because agency and commitment matter. Welfare and liberty interact, but neither should be reduced to the other (Rawls, 1999; Sen, 2009).

Family income can rise while one member loses time, safety, or control. Public measures of success therefore need to show who carries their lived consequences.

The International Classification of Functioning, Disability and Health links body functions, activities, participation, and environment: environmental change can improve participation without altering the underlying condition (World Health Organization, 2001).

2.1.3 — The Work Hidden Inside an Outcome

Two households receive the same public service. For one, the application takes fifteen minutes at home. For the other, it

requires a translated form and a journey during working hours. The final records show two completed applications. They conceal different histories of obtaining the result.

Follow the second household through the process. Someone must first discover which document is missing. A later visit depends on arranging time away from work. If the office supplies conflicting instructions, an apparently small correction can reopen the entire sequence. The burden is partly the time spent at the desk. It is also the preparation and rearrangement that make each visit possible.

Now imagine that the office introduces a reliable interpreter and accepts the document in a form the household already possesses. The benefit may appear as a shorter process even though the service delivered at the end has not changed. An outcome comparison alone would miss much of the improvement. The intervention changed the route by which the outcome entered the person's life.

This is one reason to ask about effort before interpreting achievement. Effort can be directed toward the person's chosen aim or consumed by avoidable friction around it. An hour spent learning a useful skill and an hour spent correcting a duplicated form may occupy the same amount of clock time. Their relationships to agency can be very different.

The example also shows how external conditions become experience without passing through one simple conversion rate. A procedural change may return an afternoon to a caregiver. The afternoon can become rest or time with someone they love. Its eventual meaning depends on the life receiving it. The public improvement is concrete, while its lived consequences unfold through the person's own activities.

Following the process reveals the work behind the outcome and the possibilities a better route can open.

2.2 — How Conditions Become Lived

Changes in resources reach felt life through specific pathways. A salary increase may relieve fear of a bill while adding new demands on time. Public recognition can create opportunity or exposure, depending on circumstances. The question is which options and burdens actually change for the person involved.

2.2.1 — Wealth as Relief, Exposure, or Fear

Wealth can bring new exposure through demands for help, family pressure, intrusive scrutiny, predatory relationships, or fear of loss. A person with visible resources may begin to question whether ordinary relationships rest on affection, dependence, or an interest in their money.

Wealth should therefore be analyzed through concrete counterfactuals rather than moral caricature. What burden would disappear if income rose? What new obligation or exposure appeared after it did? Who controls the resource? Can it be converted into time, safety, treatment, mobility, or refusal? Population evidence can guide the questions, but the answers remain person- and context-specific (Kahneman & Deaton, 2010; Killingsworth et al., 2023).

At low levels of income, an additional dollar may prevent a utility shutoff or purchase medicine. At higher levels, it may change convenience, status, or future security. The experiential relation is therefore unlikely to be a single conversion rate.

Debt complicates the picture. Two people with the same income may experience very different constraint because one faces volatile obligations, medical bills, or family support commitments. Gross resources conceal timing, liquidity, and the ability to absorb shock. These mechanisms are better guides to lived condition than status labels alone.

Timing can be isolated in a small example. Imagine two households with the same monthly income of 3,000 units and the same bills. One receives the whole amount on the first day of the month. The other receives 1,500 on the first and 1,500 on the sixteenth. Suppose a payment of 2,000 is due on the fifth. The first household can make it from that month's income. The second needs an additional way to bridge the interval before its next payment.

The monthly total conceals the difference because it combines money received before the deadline with money received afterward. At the fifth day, those amounts do different work. A statement of annual income conceals the sequence more completely still. To understand the practical constraint, the relevant record places receipts and obligations on the same timeline.

One solution changes the timing of the obligation. If the second household can divide the payment into two installments that fit its receipts, the immediate gap can disappear without an increase in income. Another solution changes the receipt schedule. A third uses a reserve already available at the start of the month. These arrangements can reach the same immediate result through different changes in the household's circumstances.

For the isolated payment in this example, a starting buffer of 500 bridges the gap on the fifth. The later receipt can replenish that buffer after the household's other expenses have been accounted for. The buffer's usefulness comes from being available at the needed time. The same 500 can perform that timing role again in a later month if it is restored. Its service cannot be understood by counting it as a fresh income gain every time it is used.

Real households have several obligations, so the largest shortfall may occur on a different day. Plotting the running difference between receipts and required payments reveals the interval the buffer must cover. A change that lowers the largest gap can make the month more workable even when the final monthly balance stays exactly the same. Predictability matters alongside the amounts because a person can prepare more effectively for a known sequence.

This example identifies a pathway from money to lived conditions. The improvement may be the ability to make a commitment without repeatedly rearranging the rest of the month. That can free attention for work or family activity. The experience depends on the sequence through which resources become usable, which is why a visible total is the beginning of the inquiry into wealth rather than its complete account.

Social comparison also matters. People evaluate resources relative to needs, peers, expectations, and past conditions. Comparison can motivate, but it can also produce shame, pressure, and endless escalation.

Wealth may also shift burdens outward. Private transport, education, security, and health care can reduce the holder's dependence on public systems, while a fortune may depend on workers, renters, customers, infrastructure, or environmental costs. The holder's private distress does not erase those consequences. Their investigation follows the institutions and people affected, not only the experience of the wealthy person.

2.2.2 — Status and Social Weather

Status is social weather. It influences whose account is heard and whose mistakes are forgiven. It can protect privacy or make everyday encounters feel unsafe. These patterns of treatment can persist throughout a life. Structural racism and stigma maintain

such conditions through restricted options, unequal treatment and accumulated risk (Bailey et al., 2017; Williams et al., 2019; Young, 1990).

High status can protect. It can open doors, reduce friction, attract help, and make institutions respond. It can also trap. A public identity may become a cage of expectation. A respected person may lose the freedom to be unsure, ordinary, weak, or unseen.

When social exclusion leaves a group's experiences poorly represented in shared concepts, members can struggle to explain what happened to them. Fricker calls this hermeneutical injustice (Fricker, 2007). The gap can obstruct recognition and practical help.

Status is also context-dependent. A person respected at work may be marginalized in health care, family, or public space. A title that commands attention in one setting may offer no protection in another.

Status also affects the cost of refusal. A junior employee, patient, student, or dependent family member may agree because disagreement threatens access or belonging. A senior person may face other pressures, but their formal choice usually carries different consequences.

2.2.3 — Why Biography Cannot Replace the Stream

Biography compresses duration. Ten years of poverty, illness, study, marriage, grief, or office become one sentence, while the stream contained thousands of mornings, bodily states, interruptions, comforts, decisions, and forgotten hours.

Response shift explains why biography and report can diverge without either becoming worthless. Illness, disability, grief, or recovery may change standards, values, comparisons, and the meaning of quality of life. Those changes are information, not

automatic error, and they neither erase loss nor authorize outsiders to correct a person's testimony from afar (Sprangers & Schwartz, 1999).

Stories and reports are time-bound and audience-shaped. Memory may emphasize peaks, endings, present beliefs, or available language, while a clinical history, memoir, obituary, job interview, and private conversation select different material for legitimate reasons of purpose and privacy. Later understanding may enrich evidence without rewriting earlier experience.

Biography can exaggerate agency, reduce a person to injury, or conceal coercion. Success stories may hide luck and support; accounts of failure may moralize structural barriers; diagnoses may eclipse humor, work, relationship, or faith; polished family narratives may hide fear or denied choice. These risks require evidence and humility, not the abandonment of biography. A timeline that places housing changes beside work and caregiving demands can reveal pressures that a simple success narrative leaves unexplained.

2.3 — Hidden Experience and Appearance

Appearance does not reveal a complete felt history. Envy may overlook burdens behind a visible advantage. Pity may reduce a person with a visible limitation to that one feature. Better inquiry begins with what changed in the person's lived conditions rather than an observer's first impression.

2.3.1 — Apparent Calm, Pity, and Hidden Burdens

A familiar burden may be reported less often without becoming harmless. Chronic pain or administrative humiliation can recede from a person's descriptions while continuing to shape daily possibilities.

Trauma-related dissociation or emotional numbing can also separate outward behavior from danger. A person may sound detached while carrying an acute threat response, or report little feeling because access to the experience is disrupted. Calm appearance is evidence about presentation, not a verdict about safety or burden (SAMHSA, 2014). Pity can awaken care, but it is a poor instrument for knowing a whole life. It notices a visible burden and may let that feature stand in for the person: injury for identity, poverty for biography, disability for destiny, dependence for incapacity. The moral impulse can be sincere while the inference remains reductive (Albrecht & Devlieger, 1999; Koch, 2000).

Reductive pity asks, ‘How tragic is your life?’ before asking, ‘What do you want changed?’ The question about change leaves room for agency, privacy, and specific barriers. A verdict about tragedy recruits the person into an observer’s emotional story before the person has spoken.

Disability scholarship offers a direct correction to reductive pity because it separates impairment, environment, identity, access, and reported quality of life. Studies that found high self-

reported quality of life among some people with serious disabilities challenged outsider assumptions, while critics correctly warned against turning the finding into a paradox that centers nondisabled expectations (Albrecht & Devlieger, 1999; Koch, 2000; United Nations, 2006; World Health Organization, 2022).

Dismissing that positive account as confusion can itself be testimonial injustice (Fricker, 2007; Kidd & Carel, 2017).

Lowered expectations can be as restrictive as overt exclusion. A teacher may stop offering difficult work, a clinician may speak past the patient, or a program may offer help only in forms that remove choice. Assistance becomes respectful when it is offered, accessible, and revisable, and when declining it does not trigger punishment or disbelief.

A promotion raises an employee's income but also adds overnight travel. If disrupted sleep begins with the new schedule and eases during weeks at home, that pattern helps isolate a possible pathway. It does not establish causation by itself; workload, health, and other changes still need examination.

2.3.2 — Learning the Shape of Another Life

When people imagine living differently, they often begin with their present routines and remove one ability. They picture the stairs they usually climb or the conversation they usually hear. Everything else in the imagined world remains unchanged. This method can reveal a real obstacle. It can also miss the ways a person has organized a life around different methods and sources of meaning.

Consider an observer imagining a musician who reads music through a different sensory route. The observer may focus on the method they themselves would lose. The musician may focus on the methods they have developed, the instrument they know and

the people with whom they play. These perspectives begin from different histories of competence. Understanding the musician requires learning that history.

The useful questions become specific. How is a passage learned? Which part of rehearsal requires preparation? What arrangement of the room makes collaboration easier? An answer can reveal both an obstacle and a practiced solution. The observer gains a picture of activity rather than a single global verdict about the life.

The same approach works beyond disability. A person entering an unfamiliar culture may interpret quietness as loneliness when it expresses a comfortable rhythm of social life. Someone who values public recognition may overlook the satisfaction another person finds in skilled work done privately. In each case, the observer's own habits supply an initial guess. Conversation can replace that guess with a more faithful account.

This kind of understanding develops through revision. An early question leads to an answer that changes the next question. Attention moves from the feature that first caught the observer's eye toward the organization of the person's actual day. What once looked like one undifferentiated condition becomes a pattern of activities and relationships.

2.4 — Describing Experience Without Owning It

Describing experience does not confer ownership of it. A patient may report pain so treatment can improve. A research participant may report mood without surrendering control over the record. A friend who shares grief is seeking understanding, not granting unlimited access to inner life.

2.4.1 — Description Is Not Possession

In each case, the listener receives testimony for a purpose, not ownership of the state. The description remains connected to the person's agency, context, and right to revise or stop disclosure, while creating a responsibility to hear carefully.

Description can change the experience it describes. Asking about pain can direct more attention to it. Monitoring may prompt a person to perform for the observer, while categorization can produce shame or relief. These effects belong in both the ethical assessment and the interpretation of the data.

Third-person evidence can matter when report is unavailable or contradicted by urgent risk. Behavior, physiology, history, and context may guide care. The ethical question is what decision the evidence can support and how uncertainty is handled.

2.4.2 — Privacy of Inner Life

Inner life has a privacy that is part of dignity. A person may disclose pain, desire, grief, hope, or fear, and may choose silence or partial speech. Need can make that choice costly: someone seeking help may feel compelled to narrate trauma or surrender intimate detail. Privacy protects the ordinary space in which experience can be lived without continual interpretation.

It also shapes evidence. People report differently when they expect punishment, judgment, or uncontrolled reuse; clear limits

on recipients and consequences can support more candid accounts (O'Neill, 2002; CIOMS, 2016). Narrow safety exceptions require a specific purpose, recipient, and extent, with transparency where possible.

Disclosure is not the only route inward. Combining sleep, movement, voice, and physiological records can infer vulnerability beyond any individual channel. The inference may be mistaken and still affect opportunity. Assessing the combined profile therefore matters as much as protecting each source (National Research Council, 2013; CIOMS, 2016).

The key error analysis is human. Who is harmed by a false positive, false negative, missing report, cultural mismatch, inaccessible instrument, or model trained on the wrong population?

2.4.3 — Finding Words That Open a Conversation

Sometimes a conversation improves when a person finds a more useful word. Imagine someone returning from a gathering and saying that the evening was tiring. The listener hears a request for rest. That may be right, but the word leaves several possibilities together: the room was loud, the conversation demanded constant attention, the person felt overlooked, or they enjoyed themselves fully and simply need sleep. Each possibility gives the next response a different shape.

The speaker may initially have no better description. The evening arrived as one experience, with music, faces, effort, pleasure, and fatigue woven together. Describing it asks for a selection from that whole. A word can be accurate enough to begin and still leave something useful undiscovered. The conversation can make space for a second attempt without treating the first as a failed report.

Suppose the person says, after a pause, “I liked being there. I was working hard to keep up.” That sentence separates enjoyment from effort. A suggestion to avoid future gatherings would address a different wish from the one now expressed. A quieter seat, a smaller group, or a little more time between conversations might fit the experience better. The new description changes the available response because it distinguishes features that the first word combined.

Another evening might produce the opposite clarification: “I had energy left, but I never found a way into the conversation.” Now rest is less central than participation. The useful next step could be an invitation with a familiar person, a different activity, or a conversation about how the group welcomes someone. Similar outward quietness and the same initial word can lead to different practical questions once the person's distinction becomes clearer.

Comparison can help when a direct label is hard to find. The speaker may say that this gathering felt different from a small dinner the previous week, even though both lasted two hours. Asking what differed can bring attention to pace, turn-taking, background sound, or the ease of leaving and returning. The comparison offers a concrete reference without requiring the person to fit their experience into a complete theory of emotion.

A metaphor can do similar work. Someone describes a conversation as trying to board a moving train. The image may communicate pace and missed entry more vividly than a general rating of discomfort. The listener can then ask whether slowing the exchange or making a clear opening would help. The metaphor succeeds through the relation it makes understandable; it need not function as a literal model of the person's inner life.

Words can also acquire a local meaning between people. Two friends may learn that “full” means pleasantly saturated after a

lively day, while “crowded” means that too many demands are still competing for attention. Their vocabulary becomes useful because they have explored examples together. The same words may carry different distinctions in another relationship. Shared understanding grows through that history of use and clarification.

This kind of learning can reveal that a familiar label was too coarse for a recurring choice. A person who calls every intense state stressful may gradually distinguish excitement before a performance, effort during practice, and strain from an impossible schedule. The distinctions can support different arrangements. Practice may be welcomed, anticipation may be enjoyed, and the schedule may be changed. Greater expressive resolution creates more than one path forward.

The listener contributes by offering possibilities lightly and allowing the speaker to revise them. A proposed word is a tool for the conversation, not a verdict. A useful exchange may end with a clear description, an image, a concrete request, or recognition that the person would rather return to the subject later. Its success lies in what becomes easier to understand and respond to between the people involved.

The life only one person can feel can still become more communicable. Language does not need to reproduce that life in another mind to make a difference. It can preserve a distinction, correct an assumption, and make an offer of help more fitting. A carefully found phrase can open a possibility that a broader label concealed, allowing the next encounter to begin with a better understanding of what the person actually meant.

2.4.4 — Knowing That an Understanding Is Shared

Information can reach two people without becoming a shared understanding between them. Imagine two stage technicians

preparing to open a curtain. Each has received a message saying that the performance is ready. Each knows the fact. Yet one technician is unsure whether the other received the message, and neither wants to begin an action that requires both of them. The practical uncertainty concerns the other's information.

Suppose the first technician now receives confirmation that the second has been told. This adds another layer: the first knows that the second knows. The second may still be uncertain about the first. A further confirmation can address that uncertainty. The underlying fact has not changed throughout the exchange. What changes is the structure of the participants' knowledge about one another.

Compare a different arrangement. Both technicians stand beside a signal light, each can see the other watching it, and they have already agreed that a green light means ready. When it turns green, the event provides a public reference for the action. In the idealized setting, each sees the signal, sees that the other can see it, and understands the same rule. A shared event can perform work that two separate messages leave unfinished.

The formal idea of common knowledge takes this layering to its logical limit. Everyone knows a fact, everyone knows that everyone knows it, and the same relation continues through further levels. Ordinary cooperation rarely requires people to recite such a sequence. The concept helps identify why public announcements, mutually visible gestures, and agreed procedures can matter even when they add little new information about the external situation.

The example has a second version that separates information from agreement. Suppose both technicians know the performance is ready, but one believes the curtain should open on the conductor's gesture and the other expects it to open at a clock time. More confirmation of readiness will leave the

coordination question unresolved. They need a shared rule for action. Knowing the same fact and agreeing on what to do with it are different achievements.

A third version separates agreement from interpretation. The instruction says to open after the introduction. One technician understands the spoken welcome; the other understands the opening musical phrase. Both sincerely follow the instruction. Rehearsing the cue together can reveal the difference before the performance begins. The useful addition is a common referent for the word, established through a concrete instance.

These distinctions make conversation more precise. Sometimes a person needs to hear a fact. Sometimes they need to know that another person has heard it. Sometimes both need to settle which action follows. Sometimes they need to discover that the same expression refers to different events in their minds. Each situation calls for a different next sentence, even though all may initially look like a failure of communication.

The lived consequences can be subtle. A person can spend attention checking whether a plan is still understood, whether an earlier message arrived, or whether silence means agreement. A clear shared reference can release that attention. In the stage example, the technicians become able to watch the performance and the agreed cue instead of repeatedly reconstructing one another's information. The improvement occurs through the organization of a relationship.

A shared understanding can also be renewed when circumstances change. If the conductor alters the opening, the technicians can identify the changed cue together. They retain the parts of their agreement that still apply and revise the one that moved. The earlier coordination becomes a resource: there is already a vocabulary for explaining the change and a known moment at which everyone can check it.

For the inquiry into experience, this shows why a public description of circumstances may leave out an important condition. Two people can possess the same message while standing in different relations of confidence, uncertainty, and expected action. Understanding those relations requires asking who knows what about whom, and which agreement the participants believe they share. Felt life occurs within that informational setting as well as within the visible arrangement of objects.

Good communication can therefore create a small piece of common ground. It gives people something they can refer to together, interpret together, and revise together. The achievement is concrete: an action becomes coordinated, an ambiguity becomes visible, or a question becomes answerable by the right person. A fact has entered more than one mind, and the participants have acquired a usable way of knowing that it has.

2.5 — What Affective Science Can and Cannot Clarify

Affective science separates features that ordinary speech can blur. Feeling activated does not always mean feeling good. Remembering an event differs from the experience unfolding at the time, and mixed feelings can coexist. Those distinctions determine what a measurement may legitimately claim to represent.

2.5.1 — Valence, Arousal, and Mixed Experience

Consider a person leaving a hospital after hearing that a feared diagnosis was ruled out while learning that treatment for another condition will be difficult. Relief may be immediate, fear may remain, and the body may still be highly activated. One ordinary moment can therefore contain opposed feelings, changing expectations, and bodily intensity without collapsing into a single simple state.

Such mixed emotion can faithfully represent a situation containing opposed values (Larsen et al., 2001; Berrios et al., 2015).

Valence is the name affective science often gives to the positive or negative character of experience. In this affective context, arousal refers to felt activation or energy, from low to high. The coordinates separate questions that ordinary language often merges (Russell, 2003; Kuppens et al., 2013).

High arousal can accompany celebration, panic, anger, pain, sexual excitement, athletic effort, or awe. Low arousal can accompany serenity, boredom, fatigue, depression, sedation, or despair. Activation alone therefore cannot reveal whether an experience is welcome or burdensome. The same pulse rate can belong to very different lived situations (Russell & Carroll, 1999; Russell, 2003).

Relief has a temporal structure. In a laboratory study of 18 healthy volunteers, dread and pessimism scores correlated with reported relief pleasantness when anticipated pain was omitted. Corresponding correlations during a task involving imagined rewards were not statistically significant (Leknes et al., 2011). A rating alone can leave the history of expectation unrepresented.

Affective granularity varies across people and across time. Some distinguish irritation, dread, shame, grief, and physical discomfort precisely; others use broader labels. Greater verbal detail can improve communication, but limited vocabulary does not show limited capacity to feel. The measurement task is to separate language skill from the target of interest.

Culture and relationship shape the available language. Some contexts encourage direct reports of sadness or pain; others emphasize bodily symptoms, restraint, social harmony, or spiritual meaning. These differences can alter response patterns without providing a simple ranking of whose report is more truthful. Interpretation requires local knowledge and humility.

Interoception concerns how bodily signals such as heartbeat, breathing, temperature, gut sensation, pain, and fatigue are sensed and interpreted. It is not one transparent faculty. Sensitivity, attention, belief, and regulation can diverge, and their relationships with affect vary across tasks and conditions (Khalsa et al., 2018).

The most defensible first comparisons are often within a person. Repeated observations may reveal how that person distinguishes better from worse, how context changes reports, and which transitions predict later behavior. This does not solve the whole problem, but it avoids assuming that identical numbers already mean identical experiences across people.

Scalarization remains a scientific choice, not a fact granted by convenience. A signed one-dimensional representation may

work for some tasks. Mixed states may require separate positive and negative components, a vector, or partial ordering.

2.5.2 — No Single Channel Is the Experience Itself

Self-report has a distinctive strength because a person can describe what is felt in their own terms, including shame, relief, dread, moral conflict, and the object of fear.

Reports also have limits. Language, memory, audience, privacy, scale design, expectations, and willingness to disclose can change what is said. A person may minimize pain to leave a hospital, exaggerate urgency to be heard, or lack words for a mixed state. These possibilities call for context and follow-up, not automatic disbelief.

Experience sampling asks close to the moment and can reduce some retrospective distortion. It also interrupts life, samples only selected times, and depends on the person being able and willing to answer. The most intense episodes may be precisely when a prompt is missed. Response availability is part of the observation process, not clerical noise.

Behavior reveals approach, avoidance, persistence, withdrawal, and choice. Stillness or a smile may carry different meanings in different contexts; neither is a fixed emotional code.

Neural measures can illuminate processes associated with salience, pain, reward, attention, regulation, and report. Group-level associations may overlap too broadly for reliable individual classification, especially across development, disability, medication, and disease (National Research Council, 2013; Raja et al., 2020).

Choice tasks can reveal trade-offs and preferences. They may help locate sign, compare relative values, or examine what a person will sacrifice for relief. Choices are also shaped by risk, learning, habit, social pressure, available options, and the design

of the task. Preference is evidence about value, not a transparent unit of feeling.

Clinical observation can integrate history, examination, behavior, relationship, and change over time. Expertise adds value because patterns can be recognized across many encounters. It does not remove bias, hierarchy, or incomplete access. The interpretation remains answerable to the patient, the record, and other evidence.

Infancy, anesthesia, disorders of consciousness, severe illness, language difference, and motor impairment change which observation channels are available and require adapted measures.

2.5.3 — Combining Different Kinds of Evidence

Affective science supplies concepts and observations with which to test a proposed model of experience. It can show whether a measurement tracks its intended feature, distinguish rival explanations, or reveal where the model fails, even when the full cumulative target remains unobserved.

Mature measurement practice reinforces that separation. The updated Organisation for Economic Co-operation and Development (OECD) guidelines distinguish life evaluation, affective experience, and eudaimonic aspects of well-being, and they emphasize question design, context, comparability, respondent burden, and statistical quality rather than one universal happiness score (OECD, 2025).

Triangulation is informative when channels constrain the same latent feature through different error structures. Agreement among self-report, physiology, behavior, and neural data is less persuasive if all inherit the same hidden assumption.

Chapter 3 — Care Before the Claim

A clinician may explain distress brilliantly while overlooking the symptom that frightens the patient tonight. Decisions about relief require present needs, known risks, the person's choices, and competent help. Delay is itself lived experience: waiting for analgesia, shelter, an interpreter, or protection can deepen fear and exhaustion. A person escaping an unsafe household may need a secure place and communication channel before an account of what happened.

Care can begin provisionally while assessment continues. Temporary protection and symptom control can be revised as knowledge improves; the helper should say which assessment is pending, who will follow up, and what assistance is available now. Accountability for harm does not cancel medical need or make untreated suffering an appropriate unofficial penalty. Relief becomes part of the lived trajectory to which the hypothesis would apply. These duties do not depend on the Law's scientific fate.

Care also protects inquiry. Untreated symptoms, fear of losing support, and pressure to please researchers can alter reporting and participation. Relief, safety, competent assessment, and proportionate research burden belong together (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016; World Health Organization, 2020; SAMHSA, 2014).

A warm referral explains why another role is needed, identifies the appropriate help, and stays with the transition long enough to reduce abandonment. If the handoff is delayed, the referring service should explain the reason and remain accountable for the interim plan. Institutional convenience or reputation does not justify avoidable delay.

Access may require help with transport, insurance, language, disability accommodation, childcare, advocacy or finding a safer professional.

Policy creates a harder question than personal consolation. Hospitals allocate scarce care; cities set disaster priorities; researchers decide whom a study can safely include. Each choice requires reasons that people can challenge. Even if a terminal relation in experience were established, it could not decide these allocations by itself.

The institutional question becomes concrete when a research schedule is mistaken for a clinical obligation, or an invitation is difficult to refuse because the person asking also controls care. A protection matters at the decision itself: who can interrupt the procedure, and what follows when they do?

A worker injured at a remote site might know that reporting the injury could cost wages or a contract. A safe route therefore requires more than advising the worker to speak: medical treatment must be available, retaliation must be addressable, and the route for correction must be real.

Addiction tests the sequence without simplifying it. Treatment may involve withdrawal management, boundaries, effort, and discomfort that a person chooses within care. Those burdens differ from pain imposed or prolonged because a theory predicts later balance. The relevant questions are whether the intervention is evidence-based, proportionate, wanted, revisable, and protective of future choice.

Test the particular sentence in its setting: could a distressed listener reasonably hear a reason to wait, self-punish, or refuse help? If so, make the relevant practical limit explicit before the listener acts. Emotional timing matters because grief and acute danger can turn explanation into pressure (SAMHSA, 2014).

3.1 — Care, Responsibility and Chosen Meaning

Under exact PTVC, a completed qualifying stream with finite signed totals and a calibrated neutral origin must contain a remaining net contribution that offsets its accumulated balance. If the accumulated total were negative 1,000 units, the remainder would need to be positive 1,000; another negative 100 would raise that required remainder to positive 1,100. These invented numbers express an accounting identity, not a prediction of what will happen or an entitlement to suffering. An intervention changes the life actually lived, and its effects require causal evidence. Terminal arithmetic gives no reason to withhold help.

3.1.1 — Suffering Is Not Earned

Moral-accounting metaphors can harden into claims about what a person deserved. No equation can make a child's abuse deserved. If adults invoke terminal balance to excuse failing to protect, the failure remains theirs. The relevant questions concern who could prevent harm, who permitted it, and what repair is possible. Growth after suffering does not establish that the harm was necessary; a person can value what they became while judging what happened wrong. Nor does a calm final period vindicate a life's injuries, or a distressed ending turn them into a moral failure.

The belief that people receive what they deserve can make an unpredictable world feel orderly. It offers the observer psychological distance from misfortune: if the harmed person somehow earned the outcome, others can imagine themselves safe through virtue. That emotional function is precisely why desert language requires scrutiny rather than trust.

An account may sum to zero while recording theft and repayment; the arithmetic does not make the theft acceptable.

Likewise, a terminal equality in valence would describe a boundary relation without approving the events that produced it.

Joy creates no moral debt. PTVC specifies exact complete-stream balance, not conservation of a fixed quantity of feeling at every moment. Moderation may be wise because choices affect health, relationships, resources, and future options, not because enjoyment creates a duty to suffer.

3.1.2 — Relief and Meaning Remain Free

People may accept an overarching interpretation of experience. They may revise it when circumstances change. They may also postpone a judgment or decline the question altogether. A useful account of fairness respects each of those choices without treating uncertainty as a defect.

Meaning may grow through faith, relationship, creativity, service, or time; research and medical-humanities scholarship also show variation in whether narrative coherence is sought at all (Park, 2010; Woods, 2011). Today's refusal need not bind tomorrow's wishes.

Families may find comfort in different forms, including silence. The person most directly affected should not be overruled to complete someone else's story. 'I do not want to talk about what this means' is enough; a helper can ask what support is wanted instead. The same freedom extends to nonparticipants: a community is not committed to the project's worldview because some members find it inspiring.

Chosen vocabulary and chosen audience can differ: someone may welcome a spiritual frame privately while declining its public use, or accept ritual while doubting doctrine (Pargament, 1997).

3.1.3 — Helping Without Taking Over

A person can welcome help and still care deeply about how the work is done. Imagine a gardener preparing a small courtyard after a move. A friend arrives with tools and enthusiasm. The easiest offer is to finish the job quickly: choose the plants, arrange the pots, clear the old soil, and leave a tidy space behind. Yet the gardener may be seeking something more particular than a completed courtyard. They may want to learn the light, decide where to sit, and make a place whose details gradually become their own.

The helpful conversation begins by finding that purpose. Perhaps carrying heavy bags is difficult, while choosing plants is a pleasure. Perhaps the gardener wants advice about drainage but already has a clear idea of the colors. Perhaps there is one pot they want to fill alone because it belonged to someone they loved. These distinctions change the shape of the offer without reducing its generosity. Help becomes more exact as the friend's understanding improves.

The outcome and the method can have separate value. A level shelf is useful; learning to set it level may also be satisfying. A finished letter can solve a problem; finding the words oneself can matter independently. In the courtyard, an afternoon of arranging and rearranging pots may be part of making the place familiar. Efficiency is one possible good among several that the gardener can identify.

Assistance can therefore be layered. The friend might carry materials, demonstrate one technique, hold a planter steady, or leave a tool within easy reach. At another moment, the most useful action may be to complete an entire job that the gardener has no interest in doing. The relevant difference is not between more help and less help in the abstract. It is between forms of help and the purposes they serve in this particular activity.

A well-shaped offer also makes revision easy. After trying a task, the gardener may discover that it is tiring, more enjoyable than expected, or different from what they imagined. The friend can adjust without turning that change into a test of resolve. An initial preference was a starting point for cooperation, not a prediction that the afternoon had to fulfill. The next useful arrangement can emerge from what both people have learned.

The helper's knowledge still matters. They may know that one plant needs more light or that a container needs drainage. Sharing that knowledge gives the gardener a better basis for choice. Expertise can enlarge the person's practical understanding while leaving the desired result open to them. The conversation is strongest when the reason for a suggestion is available, so that the gardener can use it again after the friend has gone.

There can also be a deliberate handoff. The friend demonstrates how to judge soil moisture, then the gardener tries it, compares the judgment, and chooses a routine that fits the courtyard. The useful legacy is partly a finished task and partly a newly available ability. On another occasion, the preferred legacy may simply be relief: the work is done, and the person can spend their attention elsewhere. Both can be worthwhile forms of care.

This makes the recipient's experience relevant throughout the interaction. A helper may feel productive while the other person feels displaced, or may feel that little was accomplished while the other person has gained confidence and room to begin. Those differences invite a more attentive exchange about what is working. The visible output alone does not contain every reason the afternoon mattered.

The courtyard eventually acquires an arrangement. There is a chair where the late light reaches, a plant chosen for its scent,

and a practical solution to the heavy watering can. Some details came from the friend's skill; others came from the gardener's preference; several emerged through trying things together. The result can carry the contribution of both people while remaining a place the gardener recognizes as their own.

Care has an affirmative craft in such moments. It notices which part of a difficulty can be shared, which part of a choice the person wants to keep, and which new possibility an offer can make available. The achievement is more than reducing an obstacle. It is helping someone inhabit the next part of their life with a little more room, a little more capability, and a form of support they can meaningfully accept.

3.2 — One Stream Cannot Pay for Another

Influence does not turn two people into one stream. Parents and children can change each other's conditions without sharing the same inner history. Partners may respond to a common event in different ways. A community can transmit trauma or support across members, but the accounting of any one experience remains personal.

3.2.1 — Separate Streams and Responsibility

These causal connections do not fuse felt histories into one account. Each bearer’s experience remains their own unless a separate theory of stream identity justifies a different routing.

Levinas’s account of responsibility likewise resists absorbing another person’s suffering into a justification by wider harmony (Levinas, 1969). Knowledge may follow harm, but the knowledge does not travel backward and justify what happened. A discovery can be valuable while the event that made it possible remains wrong, preventable, and owed a response.

Recruitment language should explain the study’s limits without promising that participation will redeem a painful experience or help others in ways the design cannot establish (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016).

Social value is necessary but not sufficient: a scientifically invalid design cannot justify research burden by appealing to a noble purpose (Emanuel et al., 2000).

Records produced through coercion, discrimination, or inadequate consent may require restriction, contextualization, community engagement, or nonuse rather than gaining legitimacy with time.

Gratitude must attach to voluntary contribution, not to suffering itself. Researchers can thank participants for time,

candor, and chosen risk without thanking illness, abuse, or grief for becoming scientifically useful. The wording matters because it identifies what the institution believes it was entitled to receive.

3.2.2 — Social Benefit Does Not Cancel Individual Burdens

Public action can legitimately distribute burdens through vaccination, infrastructure, taxation, emergency response, or environmental rules. Some burdens cannot be individually consented to; even voluntary choices may reflect background inequality. Such decisions require lawful authority, necessity, proportionality, transparency, and appeal. Invoking the Law as a hidden phenomenal offset cannot supply those reasons.

Adults may voluntarily accept risk for family, medicine, or public service. Their consent authorizes a choice; it does not transfer the experience to its beneficiaries. Likewise, majority rule can authorize some public decisions without making a minority’s suffering unreal or morally neutral. Constitutional and human-rights protections constrain what aggregate preference can justify.

3.2.3 — What People Can Make Together

Two people can make something together that neither could make separately. Imagine a singer and an accompanist rehearsing a song. The singer lengthens a phrase. The pianist hears the change and leaves a little more space before the next chord. The singer receives that space and enters the next line differently. What emerges belongs to the encounter: a performance shaped by each person's response to the other.

The pianist's contribution is not a recording placed beside an independent vocal track. Each action becomes part of the conditions for the next action. A phrase invites a response; the response changes the phrase that follows. The finished

performance contains a history of mutual adjustment. To explain its particular shape, one would need more than a list of the notes each musician knew before entering the room.

That shared creation is fully compatible with separate experience. The singer may attend to breath, language, and the effort of reaching a note. The pianist may attend to harmony, touch, and the timing of the entrance. Both hear the performance, but each hears it from a place within the activity. They can later agree about a striking moment while describing different features of it.

The distinction gives cooperation a richer form than the transfer of a fixed good. Sometimes one person gives another an object, time, or assistance. In this rehearsal, the participants also create possibilities for one another as the activity unfolds. A pause becomes available because someone leaves it; an expressive response becomes possible because someone hears the invitation. The good being made develops through those relations.

A shared intention can itself become more definite during the work. At the beginning, the musicians may agree only that the song should feel unhurried. After trying the opening, they discover that slowing every phrase makes it heavy. They keep the tempo moving and reserve one spacious pause for a particular line. Their original aim has acquired a shape through experiment, listening, and revision.

Neither person needs to possess that final shape in advance. One proposes a change, the other notices what it makes possible, and together they retain a version neither initially imagined. This is a practical kind of discovery. The participants are learning about the activity and about what they can do together. The resulting knowledge includes a way of responding, rather than

merely an instruction that could have been written before they began.

Roles can be unequal in expertise without making participation empty. A skilled accompanist can simplify a passage so that a beginning singer has room to explore it. The beginner can suggest a phrasing that the experienced musician had not considered. What matters to the collaboration is how each contribution can affect the developing work. Being included as an author of a small decision is different from being present while someone else decides everything.

The same structure appears in a shared meal prepared from an unfamiliar recipe, a story invented with a child, or a neighborhood project whose design changes when someone notices how a place is actually used. The examples differ in their practical demands. Their common feature is a product that becomes more specific through participation. People contribute not only labor but observations, preferences, and responses that change what the group is making.

Recognition can follow those contributions accurately. One person may have conceived the project, another made the difficult joining step work, and another noticed the detail that gave it character. Naming these contributions helps participants understand the history of the result. It can also make future cooperation easier because people know whose knowledge to seek and which forms of participation they enjoyed.

A shared accomplishment can leave different lasting goods. The singer may gain confidence in a difficult passage. The pianist may discover a new way of supporting another person's timing. Both may value the relationship formed in rehearsal. These effects arise within separate lives through one connected activity. They need not be equal or interchangeable for the collaboration to matter to both people.

3.3 — Consent, Refusal, and No Harm

Even a research invitation can pressure a participant when it promises to reveal destiny or hidden meaning. No protocol in this book can discover a person's true balance. Existential claims can make an uncertain study sound like treatment or revelation. Consent requires naming that limit before recruitment, because therapeutic misconception can arise before anyone signs a form (Appelbaum et al., 1982).

3.3.1 — No Created Suffering for Data

This book prohibits deliberately inflicting serious physical or psychological suffering to test the Law. Human research must satisfy independent ethical requirements concerning scientific validity, social value, proportionality, and participant protection (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016; Emanuel et al., 2000).

The prohibition does not equate serious suffering with every inconvenience. Ethically reviewed research may use ordinary questions, brief tasks, noninvasive monitoring, or mild, reversible burdens when the objective matters, the design is credible, alternatives have been considered, and the remaining burden is proportionate. Prefer lower-burden experiments, natural variation, archival evidence, prerequisite tests, or simulation when they can answer the question.

Control conditions and observation of adversity still carry safety duties. Effective care cannot be withheld where serious harm is likely; deception needs strong justification and permissible debriefing; sensitive questions must be necessary and stoppable. A study must stop when its expected learning no longer justifies its burden. Candidate-sentience procedures for nonhuman and artificial systems are specified in Section 14.3.4.

3.3.2 — Refusal Rights

Participation requires understandable information, time to decide, a practical alternative, and freedom from retaliation. The invitation must make clear that declining research will not remove ordinary care or services; a distinct lawful requirement must be identified separately. A person may stop without supplying a persuasive explanation.

The right must be usable where the pressure arises, including when the inviter controls treatment, employment, or another needed resource. Institutions must explain what stopping changes and what happens to information already collected. Supported decision making and respect for assent and dissent matter where full consent is unavailable (World Medical Association, 2024; United Nations, 1989; Faden & Beauchamp, 1986). The operational lifecycle appears in Section 12.1.1.

Researchers should model how nonparticipation may affect representativeness rather than weakening voluntariness to improve a sample.

Part II — The Law in Plain Language

Imagine drawing a closed loop on a sheet of paper. It can be small or enormous, smooth or jagged; if it finishes where it began, its net displacement is zero. No point on the line has to know the final shape. A boundary condition concerns a whole path while leaving many different paths possible. The formulation that follows asks whether a comparable relation can be defined for experience.

The public version has to do two jobs at once: give the reader a sentence that can be remembered, and keep enough structure that the sentence can be contradicted. Notation can obscure the human question; slogans can erase the conditions that make it scientific.

The title *The Law of Fairness* names an ambition, not a completed scientific verdict. In ordinary speech, law can mean an enacted command, a moral norm, a recurrent pattern, a governing necessity, or an organizing principle. Those meanings carry different kinds of authority. The intended use here is a scientific aspiration: the proposal asks whether a general structural condition constrains completed qualifying conscious streams. That is more specific than a metaphor and less established than a scientific law.

Philosophers of science disagree about what makes a law a law. A regularity account emphasizes patterns that hold without exception in the declared domain. A best-system account asks whether a principle belongs to the strongest and simplest summary of the world. A governing account treats laws as relations or necessities that help determine what happens. Other views emphasize counterfactual stability, invariance, or the role a statement plays in successful explanation (Lewis, 1973; Armstrong, 1983; Lange, 2009; Woodward, 2003). The present project treats lawhood as a question to assess through these philosophical accounts and the evidence.

A musician creates within scale and instrument; a patient acts within illness and care; a citizen acts within law and opportunity. Constraint and authorship can coexist. A proposed boundary relation would therefore have to be understood alongside the bodies, histories, resources, institutions and chance that already shape what a person can do.

Causal explanation also depends on differences that interventions could make. If changing a condition predictably changes later possibilities or outcomes, the condition can be causally relevant even inside a lawful world. Counterfactual and interventionist approaches make room for this ordinary sense of control without requiring an exception from structure (Lewis, 1973; Woodward, 2003).

Agency has several dimensions. Identification with one's motives, reasons-responsive control, and choice at the level of persons answer different questions about freedom within causal structure (Frankfurt, 1971; Fischer and Ravizza, 1998; List, 2019). Psychological accounts add intentionality, forethought, self-regulation and reflection, while autonomy, competence and relatedness concern the conditions of self-endorsed action (Bandura, 2001; Deci and Ryan, 2000; Ryan and Deci, 2017). A worker allowed to refuse an unsafe task only at the cost of housing illustrates why formal permission can coexist with severely constrained practical choice.

Chapter 4 — The Law of Fairness

Why call a terminal relation fairness? The strongest reason is that a whole-life account should take the bearer’s own experience seriously. An improvement in other people’s circumstances cannot answer what became of this person’s burden. A relation that applied separately to each qualifying stream would give each history equal standing in the hypothesis. Social visibility, wealth, talent, and usefulness would not determine whose experience counted.

The affirmative thought is therefore one of nonabandonment: perhaps no qualifying history is left with an uncompensated net experiential burden. Such a regularity would matter even where an observer could not see its complete course. It would extend inquiry beyond whether people usually recover toward whether their histories share a deeper structure. This is a reason to investigate the proposed relation, rather than evidence that nature must satisfy it.

There are powerful objections. The two balanced histories in Chapter 1 differ by a hundredfold in their positive and negative totals. Multiplying both sides leaves the terminal arithmetic unchanged while vastly changing exposure to suffering. Duration matters too: a brief intense burden and years of recurring pain need not have the same consequences for anticipation, memory, agency, or bodily wear. Order matters as well: pain followed by relief is not lived like relief followed by pain. A final sum is deliberately insensitive to much that makes those lives different.

Distribution and opportunity pose a second objection. A person could be excluded from education, political voice, or a chosen relationship while having the same terminal total as someone with abundant opportunities. Justice concerns what people can do and become, how decisions treat them, and which inequalities require remedy. A symmetry of felt totals does not

settle those questions (Rawls, 1999; Sen, 2009; Nussbaum, 2006). Nor does an eventual positive remainder make the earlier burden freely chosen or retrospectively acceptable.

A further difficulty is that exact zero is stronger than protection against a negative total. Why should a complete history with a positive surplus be objectionable? Nothing in the ethical argument establishes that it should. The scientific symmetry must earn its place through explanation and evidence; its negative and positive sides cannot both be derived from a wish to prevent suffering.

For these reasons, terminal valence closure is the scientific name used here. *The Law of Fairness* is a philosophical interpretation of the proposed nonabandonment and equal application, open to criticism. The interpretation claims one possible relation among complete histories, not a sufficient standard for evaluating lives. A reader can reject the name while examining the same mathematical target.

Hidden quantities can nevertheless be studied. Two sealed parcels weighing ten kilograms together permit many individual weights. Greater precision in the combined weighing will not choose between them; a separate weighing or an independent constraint could. Likewise, indirect evidence about experience becomes informative when it distinguishes the alternatives relevant to the claim. The next useful measurement is one that separates those alternatives.

4.1 — The Law in One Sentence

Exactness is where the Law becomes scientifically interesting. Adaptation, resilience, mean reversion, and homeostasis can move experience toward a local baseline, but none by itself requires a complete stream to satisfy an exact terminal boundary. A successful route would have to explain why the whole history closes, not merely why local states recover. That difference creates concrete questions about remaining opportunity, dispersion, perturbation, and endpoint timing. Each question gives researchers a way to compare a Law-linked explanation with capable ordinary mechanisms.

4.1.1 — The Law, Its Endpoint, and Exact Wording

On the primary additive branch, the Law of Fairness proposes that each maximal qualifying conscious stream in the declared reference class, with its experience separately assigned under independently fixed routing rules, accumulates finite and exactly equal positive and negative magnitudes of felt experience across its complete history before an independently defined true endpoint, so that its latent cumulative residual is exactly zero immediately before that endpoint. The relevant endpoint is the irreversible end of the qualifying conscious stream on the declared formal branch, not merely the last moment for which an observer has a record.

The Illustrated Guide shows the defining equation:

$$Z_i^\star := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\mathrm{phen}}(T_i^{\star-})$$

$$\mathrm{PTVC}_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^\star = 0$$

4.1.2 — Three Boundaries and the Pre-Endpoint Domain

Three boundaries must remain separate. The truth-level endpoint belongs to the hypothesis. A protocol endpoint is the prospectively defined event a study uses to locate a candidate

stopping region. The end of observation is simply the last time usable data were obtained. These boundaries may coincide in a well-adjudicated case, but no notation makes them identical.

The phrase immediately before is deliberate. The primary public sentence concerns the complete strict-pre-endpoint domain. Any phenomenal contribution assigned exactly at the endpoint belongs to a separately declared endpoint-inclusive branch and cannot be added informally after the result is known. This small grammatical choice protects a large scientific distinction. Paraphrase can quietly create another theory. “Life is fair” turns a narrow terminal hypothesis into a verdict on visible life; “everything balances” removes bearer, currency, endpoint, and exactness; moralized versions import desert or purpose. Exact latent zero likewise cannot be softened into a near-zero estimate, trend, or comforting impression without changing the claim (Lakens, 2017; Wellek, 2010).

4.2 — The Parts Inside the Claim

PTVC contains distinct claims that must be evaluated separately. Identifying a bearer does not establish a defensible neutral origin. A well-calibrated measure cannot identify an endpoint that has not been independently specified. Evidence at one gate cannot compensate for failure at another.

4.2.1 — A Stream Exists

The first assertion is that there is a bearer for whom experience occurs. A qualifying stream can be brief; its positive and negative experiences must still be assignable to one history without pooling them with another. If no such bearer can be specified, there is no coherent intrastream object to accumulate and no terminal claim to test.

Unity can concern experience at one moment or continuity across time. Philosophy of consciousness does not offer one uncontested test that settles all of these dimensions (Bayne, 2010). The Law therefore needs an operational bearer rule while remaining open about deeper metaphysical accounts of the self.

Maximality prevents favorable slicing. A painful year cannot be declared one stream and a joyful year another merely because each selected interval produces a preferred total. Separate routing prevents favorable pooling. Two people who share a household, body of records, or common cause do not thereby become one phenomenal bearer. These rules make the stream a scientific constraint rather than a storytelling device. Every admissible contribution is assigned once.

Qualifying is an eligibility rule, not a favorable verdict. Consciousness, unity, routing, domain, and reference-class criteria must be fixed before terminal outcomes are inspected. An empty declared class supplies no empirical support. If

qualifying streams exist but none can be adequately assessed, the result is No adequate estimate, not confirmation.

4.2.2 — Each Assertion Can Fail

Bearer, ownership, and routing can fail before terminal arithmetic begins. A study may be unable to identify one qualifying stream, distinguish one stream from several, or decide whether a pause, split, fusion, or altered state preserves continuity. Experiences may also be incompletely assigned or blended across people.

A defensible neutral origin must be identified independently of the desired terminal result.

The completeness assertion can fail because observation starts late, ends early, or misses precisely the intervals that matter most. Missing prefixes and suffixes need explicit bridges, bounds, or withholding a complete-stream estimate. Unseen experience is not zero experience (Little & Rubin, 2019).

Once bearer, representation, origin, domain, endpoint, observation, and recovery gates are adequate, a terminal residual credibly bounded away from neutrality counts against the declared exact branch. Latency creates an inferential burden, not an unlimited shelter. Unfavorable evidence cannot be renamed measurement error without an independently supported account of what failed.

A universal claim applies to every member of its declared truth-domain class. Finitely many compatible cases cannot establish that every member satisfies it, especially when cases remain unobserved. Measurement and observation criteria govern whether evidence can adjudicate a stream; they do not remove an otherwise qualifying stream from that class.

For a strict exact branch, fix the class, stream and routing ontology, representation, neutral origin, domain, endpoint and

measured-to-latent bridge before inspecting the case. One qualifying stream with a nonempty deterministic truth-containing residual set that excludes zero refutes exact PTVC for that declared class under those assumptions. Confidence in the gate assessments remains an empirical question; a probabilistic interval alone is not a deductive counterexample. Reclassifying the stream, moving the endpoint, changing the origin, switching branches, or adding continuation after the adverse result creates a revised hypothesis; it does not erase the loss.

Independent replication, transport and review strengthen confidence that a purported counterexample meets the declared conditions; they do not change the logical force of a verified counterexample (Popper, 1959; Mayo, 2018).

4.3 — Streams, Cumulative Experience and Closure

A human stream is embodied rather than floating free of physical life. Pain and hunger can narrow attention before any reflective choice occurs. Sleep, illness, disability, and nervous-system state can change both experience and its accessibility to others. A stream model must distinguish those changes from the identity of the bearer itself.

4.3.1 — Stream, Body, and Archive

Yet the body alone does not settle stream identity or endpoint status.

Surface appearance can understate experience in unusual embodiment and overstate it in mere biological persistence. A person who communicates, moves, senses, or requires care differently may still bear a full stream.

An archive can extend cognition and social memory without becoming the subject that once felt. The extended-mind literature shows how notebooks, devices, and environments can participate in cognitive activity, but participation in a cognitive system does not by itself establish that the record has phenomenal experience (Clark & Chalmers, 1998). The distinction allows archives to support accountability without treating preservation of a record as evidence that the experiencing subject survives in it.

A person’s self-understanding can change through lost autobiographical memory or rejection of an earlier narrative without settling whether a stream continues. Conversely, similar records need not share a subject. Neither an indivisible ego nor resemblance between biographies determines the routing of experience (Parfit, 1984; Dennett, 1987).

4.3.2 — The Hidden Cumulative Object

The cumulative object is a latent target, not a number visible from outside a person. Reports and physiological measures supply observations rather than direct readings of it. A model must justify how those observations connect to the proposed quantity. Disagreement between measures can expose a weak bridge without deciding the underlying claim.

Construct validity asks whether different measures behave as a coherent theory predicts, whether the construct generalizes across relevant settings, and whether competing explanations have been challenged (Cronbach & Meehl, 1955; Borsboom, 2005).

The chain starts with lived experience. A truth-level formal target fixes the bearer, valence representation, accrual, and the strict-pre-endpoint domain; a candidate route, when proposed, describes a process that could realize it. An observation model connects reports, behavior, and other channels to the quantities studied. An estimand states the quantity a study seeks; an estimator maps observations to an estimate using explicit calibration and uncertainty rules. A decision rule states what the estimate and comparison license (Campbell & Fiske, 1959; Mauss & Robinson, 2009; Michell, 1999; Borsboom, 2005).

The observation model should explain expected channel disagreement and preserve unresolved conflict as uncertainty or evidence against the representation. Cumulative means that the represented value includes contributions across the path, not merely the mood visible at its end. A sufficiently informative physical state might retain the relevant history; the state-based route in Section 8.3.2 examines that possibility.

4.3.3 — Exact Closure and Unequal Lives

Terminal neutrality is a proposed property of the completed cumulative object. It does not mean that a person looks calm at the final observation. Nor does a difficult ending alone determine the sum of an entire life. A test must state the aggregation rule and endpoint before it treats any observed sign as support or challenge.

Measured compatibility is a different proposition. An estimate and its uncertainty may fall within a protocol-fixed equivalence region around an operational neutral point. That can support compatibility at the declared resolution and under the declared observation model. It does not convert a finite band into exact latent equality or a confidence interval containing zero into proof that the true residual is exactly zero (Schuirmann, 1987; Lakens, 2017; Wellek, 2010). Evaluate all admissible endpoints rather than selecting the one nearest zero. When observation is lost through withdrawal, treatment, fatigue, delirium, transfer, device failure, or communication loss, use a justified bridge or sensitivity analysis, or withhold the estimate (Little & Rubin, 2019).

The Law does not forecast the sequence of anyone's future pleasures and pains. Even an exact terminal boundary, if real, would leave countless paths to the same residual. It contains no schedule pairing hardship with reward, no happiness budget, and no rule telling a reader when relief must arrive.

Chapter 5 — A Stream’s Beginning and End

The first moment of a conscious stream is not automatically identical to birth, conception, legal recognition, first memory, or the first measurement available to science.

Evidence about that beginning may be gradual and uncertain. Development can present changing capacities, incomplete access, and disagreement about what signs are sufficient. The Law can require a prospectively stated bearer rule without pretending that one public event settles every developmental case.

Whether a return continues the same stream or begins another must be decided under a prospectively declared identity rule, not chosen to reset the cumulative object (Section 5.2).

Disorders of consciousness make repeated, standardized, multimodal assessment indispensable. Diagnostic error has been documented when one observation is treated as final, and practice guidelines emphasize reassessment, confounder control, and converging evidence (Schnakers et al., 2009; Giacino et al., 2018; Kondziella et al., 2020). Covert command following adds an especially important contribution. Functional imaging and electroencephalography have revealed task-responsive activity in some patients who show no reliable behavioral response. The finding does not identify the Law's endpoint, but it demonstrates why behavior, neural response, signal quality, confounders, and later recovery must remain distinct channels (Owen et al., 2006; Monti et al., 2010; Claassen et al., 2019; Bodien et al., 2024).

Reversibility separates a pause from an endpoint asymmetrically. If reliable unified access returns and the predeclared routing rule identifies continuity of the same stream, the earlier state was not that stream's truth-level endpoint, although this may have been unknown at the time. A protocol must prospectively define signs of return and their effect on classification, using reliable communication, command

following, validated covert command following, or another domain-appropriate marker. Failure to return, however, cannot by itself prove absence of experience or identify the exact ending (Brown et al., 2010; Giacino et al., 2018; Bodien et al., 2024).

Beginnings and endings create different missing-history problems. At the beginning, experience may precede the first reliable report, memory, or measurement, leaving an unobserved prefix. At the end, observation may stop while experience continues, leaving an unobserved suffix; in time-to-endpoint analysis, this is right censoring. A complete-domain claim must face both. Counting only the observable middle can produce a precise estimate of an incomplete stream.

Family and caregiver testimony can help interpret ordinary communication, preferences, and changes from baseline. That testimony deserves respect without giving an informant sole authority to determine the endpoint. Love, hope, exhaustion, guilt, culture, and grief can shape interpretation, so the record should preserve both the value and the limits of relational evidence.

Each biological and clinical variable needs a declared role: context, proxy, confounder, competing event, or input to endpoint adjudication. An observed signal can have different implications depending on that role.

5.1 — Evidence for a Boundary

Evidence about onset comes from records with different strengths. Family reports can describe behavior, while medical records document clinical states. Milestones or observations may narrow a time window without identifying a precise consciousness switch. The study must state whether each clue concerns arousal, responsiveness, learning, memory, neural organization, or social recognition rather than collapsing them into one category.

5.1.1 — Death Determination and the Stream Endpoint

Legal death is a public status with consequences for rights, duties, records, succession, disposition, and authorized decision-making. Clinical determination of death is governed by professional and jurisdictional standards that society cannot leave to private interpretation. The truth-level endpoint in the formulation developed here serves a different purpose: it marks the end of the relevant conscious stream for a scientific hypothesis.

Death by neurologic criteria is not declared from one sign or one intuition. Contemporary guidance requires prerequisites, exclusion of confounders, a structured neurologic examination, apnea testing, and ancillary testing when the clinical evaluation cannot be completed or interpreted as required (Greer et al., 2020; Greer et al., 2023).

Determinations using circulatory criteria and those using neurologic criteria assess death through different clinical pathways. A research protocol must identify which criteria were used, under which authority, with what timing uncertainty, and what that event can support about the proposed stream endpoint.

Confounders matter because apparent absence can be produced by reversible conditions. Sedatives, neuromuscular

blockade, metabolic disturbance, temperature, intoxication, shock, seizure, and technical failure can alter examination and monitoring. Clinical standards address these concerns for clinical determination; a research protocol must not assume that a secondary dataset preserved every condition needed for a consciousness inference.

Even a legally and clinically established death may provide an outer boundary without identifying the last phenomenal moment. A study must state what the recorded event can support.

5.1.2 — Unknown Cases Remain Unknown

Unknown applies when available evidence does not responsibly settle whether the relevant stream began, continued, paused, returned, fragmented, or ended.

Distinguish absent evidence from conflicting evidence, identify the observation that could resolve the case, and assign responsibility for obtaining it where feasible. Reclassification requires the stated resolution rule, not pressure for a cleaner denominator.

Several neighboring statuses should remain distinct. Not interpretable means the record cannot bear the intended inference because of signal failure, confounding, timing, preprocessing, or an invalid observation model. Ethically unavailable means the information should not be collected, linked, retained, or analyzed because doing so would interfere with care, exceed consent, exploit dependency, or violate privacy. Outside the declared class means the case was prospectively excluded for a stated reason rather than removed after the outcome became known.

A high rate of unknown outcomes can pressure the research program. It may show that the endpoint proxy is fragile, the

intended population too heterogeneous, the observation window unrealistic, or humane data collection unable to answer the question acceptably. Redesign, reference-class narrowing, or suspension may be the scientifically honest response.

A reference class narrowed for legitimate reasons should be named with equal prominence. Results from well-instrumented acute-care adults cannot silently become results about infants, dementia, nonhuman animals, artificial systems, or every possible conscious bearer. Unknown cases mark the study’s reach.

The route into the unknown category also matters. Missingness after emergency transfer differs from selective exclusion of difficult cases; a failed sensor differs from unresolved confounding; ethical nonuse differs from absent monitoring. These mechanisms can leave a deceptively clean group of well-instrumented cases. Denominator flow should therefore report usable, unknown, uninterpretable, ethically unavailable, returned, and excluded cases rather than hiding them behind one final sample size.

5.2 — Pauses, Returns, and Low-Unity States

An outward pause does not by itself identify a break in experience. Ordinary sleep, anesthesia, coma, and severe injury can all reduce responsiveness. They differ in mechanism and reversibility, and their implications for awareness are not interchangeable. A continuity claim must therefore identify the bearer across the low-access interval rather than infer identity from one surface sign.

5.2.1 — Sleep, Coma, and Recovery

Sleep introduces an exposure problem: some states may contain vivid experience while others yield little report, so biological time cannot be added to a phenomenal history without an exposure and unity rule.

Sleep is not one blank state. Dream reports and neural studies indicate that conscious experience can occur during rapid-eye-movement and non-rapid-eye-movement sleep, with changing content and level across the night (Siclari et al., 2017; Tononi et al., 2024).

A waking dream report is reconstructive and shaped by memory, language, culture, medication, timing, and narrative habit.

Coma is a clinical state with many causes, courses, and outcomes, not a synonym for the truth-level end of a conscious stream. Some patients recover wakefulness or awareness, some transition into other disorders of consciousness, some die, and some remain difficult to classify. If recovery meets the predeclared criteria for continuity of the same stream, it defeats that case’s earlier terminal classification. A threshold designated in advance as a nonterminal low-unity crossing makes no such terminal claim.

Chapter 13 treats bearer evidence in detail. General anesthesia is designed to be reversible; drugs, paralysis, metabolic state, temperature, and delayed emergence therefore belong in prospectively specified confounder and return rules.

Follow-up rules must accommodate delayed recovery without imposing one universal clinical duration.

Loss of autobiographical continuity does not establish loss of experience. A person may no longer narrate a stable self and still inhabit the present. Dementia and delirium can fluctuate, interact with illness and medication, and produce thresholds that cross and recross. Persistence, hysteresis, confounder control, and longitudinal evidence are therefore more informative than a convenient first crossing.

5.2.2 — Reversibility as the Key Distinction

Signal quality, persistence, and return rules protect against false stopping. Artifact, device loss, preprocessing failure, missing channels, brief fluctuations, or rescue interventions cannot define a terminal event; the proxy must meet protocol-fixed duration and evidence criteria, and a return meeting the declared continuity criteria counts against a terminal classification. This makes the rule a research safeguard rather than a prognosis or legal definition of death.

A laboratory instrument provides a simple way to separate two ideas used in these rules. Suppose its display reports values near one half. A classifier using one threshold changes to high at or above one half and to low below it. The sequence 0.49, 0.51, 0.49, 0.51 makes the classifier change state three times between four readings, even though the readings differ only slightly.

A hysteresis rule uses different thresholds for entering and leaving a state. Begin in low, switch to high only when the reading reaches 0.60, and return to low only when it falls to

0.40. The same four readings now leave the classification low throughout. If a reading of 0.65 arrives, the classification becomes high; a subsequent reading of 0.50 leaves it high. A reading of 0.35 returns it to low. Values between the two thresholds inherit the existing classification.

Persistence addresses a different feature. A rule might require three consecutive readings above 0.60 before entering high. Then one reading of 0.65 followed by 0.50 would not trigger entry, whereas three successive readings of 0.65 would. Hysteresis separates entry and exit thresholds. Persistence requires a specified duration or sequence of evidence. A classifier can use either or both, with different consequences for brief fluctuations and sustained changes.

Every classification of this instrument can be reproduced from the numbers and the rule. The same record produces different labeled transitions under different rules, so a study can compare those rules on independently classified test cases before choosing one for its declared purpose. The classification then has an inspectable history: the current reading, the previous state, and any accumulated persistence evidence each play an explicit part.

5.3 — The Endpoint Must Be Independent

An endpoint must be specified without inspecting the cumulative outcome it is meant to test. A camera can be programmed before a throw to stop when a ball touches the ground, although contact time depends on the motion. Likewise, a prospectively justified biological or consciousness endpoint may depend on the observed process without being selected to force closure. This operational separation is weaker than stochastic independence: the stopping time can still depend on the experienced path. The theorem in Section 7.2.2 assumes the stronger property, and Chapter 7 examines the consequences.

5.3.1 — No Endpoint After Cumulative Object Inspection

Operational separation requires advance rules for endpoint channels, preprocessing, signal quality, persistence, anti-chatter logic, adjudication, and return. Blinding helps, but shared sensors, physiological drivers, or preprocessing can still entangle the endpoint proxy with the cumulative proxy. If outcome information shaped the boundary, the result becomes exploratory or unusable for the confirmatory claim.

Recovery cannot be dismissed after it destabilizes an endpoint; resuscitation, medication changes, seizures, transfer, device loss, and absent channels cannot be coded automatically as terminality or neutrality.

The record should preserve why a classification was reached, which confounders remained, whether return rules were triggered, and what information each adjudicator could see. Plural expertise may be necessary, but cumulative outcome information should not enter merely because consensus is difficult. Role separation, access control, time-stamped decisions, and a concise audit trail make independence inspectable while protecting clinical privacy.

5.3.2 — Ambiguity and Endpoint Discipline

Adjudicators can also disagree for principled reasons. One may judge a confounder unresolved; another may accept the endpoint. The record should preserve the evidence, the rule applied, the disagreement, and the effect of alternative classifications. Consensus reached only after seeing which choice favors neutrality is not independent adjudication.

Branch analysis can make uncertainty visible without letting it choose the answer. Investigators should evaluate every endpoint window allowed by the protocol while keeping the decision rule fixed. A conclusion that survives the full plausible set is robust to those endpoint alternatives; one that appears only on a favorable branch is branch-dependent.

Partial identification can replace a forced point estimate. Data may justify a set of possible terminal values. If the entire set lies away from neutrality after every prerequisite passes, the case can pressure the strong claim. If neutral and non-neutral values both remain possible, the result stays unresolved; the width of the set is information about the bridge, not permission to select its most reassuring point.

An uncertain stopping window can be handled by evaluating a whole family of possible totals. Consider a program whose cumulative quantity, while it is running, follows $L(t) = (t - 2)^2 - 1/4$ during the interval from time one through time three. Its earlier history supplies the value with which this window begins. The program's stopping time is known to lie somewhere in the window, but its exact location is unavailable.

At both ends of the window, the cumulative value is three quarters. In the middle, at time two, it is minus one quarter. It crosses zero at times one and a half and two and a half. The set of possible stopping totals is therefore the complete interval from minus one quarter to three quarters.

Looking only at the earliest and latest allowed stopping times would produce the same positive answer twice. The interior of the window contains additional possibilities because the cumulative path first falls and then rises. To propagate endpoint uncertainty, the relevant operation is to map every allowed time through the cumulative function. The resulting range need not be determined by the two boundary times.

If the cumulative function is monotone across a window, the two boundary evaluations do determine its extreme values. If it turns, interior extrema also matter. With several disjoint possible stopping windows, the resulting set can itself be disconnected. Preserving that structure can be more informative than replacing every possible total by one broad interval.

A different calculation is available when the path is not known exactly but its rate is bounded. Suppose the cumulative value at a reference time t_0 is known, the endpoint differs from t_0 by at most δ time units, and the admitted rate has magnitude at most M throughout the relevant interval. The additional signed contribution then has magnitude at most $M\delta$. This follows by bounding the magnitude of an integral by the integral of the magnitude of its rate.

For example, a known reference total of 1.2, an endpoint uncertainty of at most one quarter of a time unit, and a rate bound of two units per time unit give a possible terminal range from 0.7 to 1.7. The calculation says which timing uncertainty and rate information were used. If the reference total itself has an error bound of 0.1, the combined range expands to 0.6 through 1.8.

Sharper information can produce a sharper set. A known one-sided rate, a measured turning point, or a narrower independently determined stopping window can each remove possible totals. These improvements address different parts of

the calculation. More accurate timing helps when the rate can change the total substantially over the remaining window; a better bound on the rate can help even when the timing interval stays the same.

The pattern of ambiguity is itself diagnostic. If uncertainty clusters where residuals would be least favorable, the method may be selecting, censoring, or failing at precisely the critical cases. Comparing ambiguity across predicted residuals, clinical pathways, sites, medications, and observation quality can reveal more about the design than the classified cases alone.

Rivals can benefit from ambiguity as readily as PTVC can. Endpoint-conditioned sampling, selection on observability, floor effects, and flexible latent models may all concentrate the cases that remain classifiable. Equal-budget rival comparison must therefore include the ambiguous and missing cases rather than score only the cleanest survivors.

Across sites, different endpoint definitions require harmonization or planned heterogeneity analysis; local medical authority does not automatically create one common research endpoint. Material drift can leave a comparison exploratory.

Simulation and model-recovery tests should include reversible crossings, shared confounders, abrupt dropout, and non-neutral endings. Before enrollment, use these cases to check whether the planned rule distinguishes endpoints from interruptions and whether it systematically favors easy-to-observe cases.

5.4 — Ambiguous, Branching, and Revised Endpoints

Endpoint classifications can be ambiguous for different reasons. A clinical record may be incomplete, while a dissociative case raises a different bearer question. Later evidence can also require revising an earlier classification. Each problem calls for its own adjudication rule rather than a retrospective choice made to favor closure.

5.4.1 — Clinical Ambiguity

Clinical ambiguity may arise from the condition, treatment, limits of observation, or disagreement among methods that are each valid for narrower purposes.

A protocol may use clinical information as part of an endpoint bridge, but the formal vocabulary cannot overrule a bedside team, reinterpret a chart, or create a second route for declaring death.

The confounder rule in Section 5.1.1 remains active here. Clinical ambiguity must also account for analgesia, infection, organ failure, sensory impairment, and language barriers, because a weak or absent response may reflect the channel rather than the absence of a stream.

Task-based tests for covert command following require the person to understand the instructions and sustain the requested activity (Bodien et al., 2024). Neural data also require adequate signal quality, controlled preprocessing, and checks for artifacts. Bedside behavior and family testimony contribute different information. A discordant result may therefore identify an unsuitable task or unreliable observation channel without settling consciousness status.

Because arousal and access can fluctuate, a protocol should specify the number, timing, and independence of assessments and state what later return does to an earlier classification.

During active resuscitation, provisional signals cannot establish terminality while circulation, breathing, and recoverable function may still be restored. The resulting observation window may be incomplete.

Sudden death presents a different limitation. The clinical event may be clear while no prospective unity record or complete cumulative proxy exists. The truth-level endpoint may have occurred, but its timing and the terminal residual may remain inaccessible. A well-documented ending alone cannot adjudicate complete-stream closure when the required antecedent record is missing.

A multidisciplinary adjudication panel may include neurology, critical care, rehabilitation, ethics, measurement, and statistics, with additional expertise when the protocol requires it. Each discipline should identify the particular unresolved question it can help answer.

5.4.2 — Dissociative Ambiguity and the Unknown Category

Dissociation is a clinical and experiential domain, not a dramatic metaphor for theoretical uncertainty. The American Psychiatric Association describes dissociative disorders as involving disruptions in memory, identity, emotion, perception, behavior, and sense of self (American Psychiatric Association, 2022). Those disruptions can affect how experience is accessed and narrated without automatically establishing that conscious experience ended or that several formal streams exist.

The topic requires anti-stigma discipline. Dissociative symptoms and disorders are often associated with trauma, and the literature has had to develop amid controversy, diagnostic disagreement, and public sensationalism (Dorahy et al., 2014). The model must carry the uncertainty rather than asking trauma survivors to carry the burden of making the theory interesting.

Memory gaps do not by themselves divide a stream: experience can occur without later autobiographical recall, and later narratives do not create or erase earlier experience.

Identity language also needs care. Distinct identity states or discontinuities in agency may pressure a model of access, ownership, and routing, but they do not by themselves supply a formal count of conscious streams. The relevant questions concern simultaneous or sequential experience, access relations, continuity, mutual influence, and whether ownership can be assigned prospectively without erasing any lived interval.

The project may therefore prospectively specify more than one plausible routing model for a declared class. One may treat changing states as modes of one continuing stream; another may remain agnostic where evidence cannot distinguish alternatives.

Dissociative ambiguity must also be distinguished from cognitive motor dissociation. The latter term is used in disorders-of-consciousness research for patients who are behaviorally unresponsive to commands yet show task-responsive brain activity on functional imaging or electroencephalography (Claassen et al., 2019; Bodien et al., 2024). Similar words do not make psychiatric dissociation and motor-output dissociation the same phenomenon.

A participant may describe discontinuity, co-consciousness, amnesia, conflict, or continuity without being asked to supply a definitive metaphysical interpretation of their diagnosis. Record their description separately from the study's routing judgment.

Language and culture shape how dissociative experience is recognized and described. Terms of identity, possession, detachment, memory, and agency do not travel unchanged across communities. A measurement instrument validated in one setting may misclassify people in another. Translation, cultural

consultation, and measurement-invariance review are therefore scientific requirements, not decorative sensitivity.

Review should include clinicians with relevant expertise and, where appropriate, people with lived experience who can identify stigmatizing language and unrealistic procedures.

When evidence locates an ending only between two events, carry that interval into the admissible-set analysis in Subsection 5.3.2 rather than inventing a time stamp (Little & Rubin, 2019).

When classification changes, preserve the earlier record and identify which analyses must be rerun under the new evidence (Claassen et al., 2019; Bodien et al., 2024).

5.4.3 — No Personal Endpoint Testing

No person should try to test the Law by moving themselves or anyone else toward danger, loss of consciousness, treatment refusal, self-harm, violence, deprivation, or death. Such an episode creates risk without satisfying the prerequisites of a valid test. Stop the activity and seek immediate safety and appropriate emergency, clinical, safeguarding, or legal help when danger is present.

A moderator encountering acute danger should suspend philosophical discussion and route the matter through the appropriate emergency, safeguarding or platform process.

Chapter 6 — A Constraint Without a Judge

The proposal becomes more rigorous, not less wondrous, when explanation begins with impersonal structure rather than a hidden observer arranging each life.

PTVC is a boundary hypothesis before it is a mechanism story. Reserve coupling, bridge processes, feedback regulation, adaptation, opponent dynamics, predictive control, and selection can become candidate routes or rivals only when they generate discriminating consequences. The target remains fixed while those explanations compete.

The intentional stance can help people interpret complex systems, but its usefulness does not establish that a system literally has beliefs, desires, or moral aims (Dennett, 1987).

Practice can improve competence, generosity can deepen friendship, honesty can build trust, and recklessness can produce injury. These causal relations support responsibility and learning in ordinary life.

A best-system perspective makes the apparent simplicity of the terminal statement especially demanding. The sentence may be short, yet its real complexity includes the definition of valence, neutral origin, exposure, stream ownership, complete domain, endpoint, missingness, measured-to-latent bridge, branch, and reference class. If those elements can be adjusted after outcomes are known, the one-line rule is not simple. Its complexity has merely been hidden in movable definitions (Lewis, 1973; van Fraassen, 1989).

A governing or necessitarian interpretation would be stronger still. It would suggest that the relation is not only a good summary but part of what makes admissible histories possible. The formulation developed here explores structural consequences of such a target, but the public manuscript does not claim to know whether any supported regularity would govern, emerge, supervene, or summarize. That ontological

question remains separate from the empirical question of whether the terminal relation holds.

Counterfactual support supplies another test of lawlike status. A relation is more informative when it constrains what would happen under changes that were not used to fit it. A pattern that appears only in the exact data and endpoint window from which it was constructed has little lawlike force, however elegant its retrospective fit (Lange, 2009; Woodward, 2003).

Gravity illustrates impersonality: objects are not rewarded for falling or punished for resisting. Its explanatory power comes from specified variables and tested relations, not the moral connotations of the analogy.

Symmetry and invariance become scientifically useful when they generate explicit questions. Which transformations preserve the terminal relation? Which shifts destroy it? Does the same zero survive a new instrument, culture, or temporal representation? The analogy earns value only by generating a prospectively specified invariance test rather than gesturing toward mathematical beauty (Lange, 2009).

6.1 — Constraint, Explanation and Mechanism

PTVC is framed as a boundary condition on cumulative felt experience. That makes it a constraint claim rather than a story about a deciding agent. Whether experience supports the needed signed quantity remains open. So does whether a defensible endpoint exists at which the proposed condition could be tested.

6.1.1 — Constraint, Freedom, and Formal Shape

Initial, terminal, and other boundary conditions describe relations at their respective edges; constitutive, control, functional, and metaphysical constraints ask different questions.

A descriptive regularity, a dynamical generating law, a boundary restriction and a scale-limited effective model play different explanatory roles (Armstrong, 1983; Cartwright, 1983).

Constitutive constraints help define a system, control constraints limit admissible actions or trajectories, and biological functions describe contributions under a defensible account of function. PTVC does not currently claim that closure constitutes consciousness, that a hidden controller optimizes a life, or that terminal balance is an evolved function. These categories organize candidate explanations without preselecting one.

A mechanism names entities, activities, or couplings that could produce the target and earns status through discriminating evidence. Section 6.2.3 develops the philosophical alternatives.

A terminal condition can restrict a complete path without specifying every action along it. Responsibility need not rest on metaphysical exemption from every cause; it can turn on a person’s reasons, capacities, knowledge, available alternatives, exposure to coercion, and ability to respond to consequences (Dennett, 1987).

6.1.2 — What Two Boundaries Can Determine

A boundary condition and a rule of motion perform different jobs. A simple oscillator makes the difference visible. Let a position $x(t)$ obey $x''(t) = -x(t)$, so acceleration always points toward zero with magnitude proportional to the displacement. The complete family of solutions is $x(t) = A \sin t + B \cos t$. The two constants identify a particular motion within that family.

Specify position and velocity at the beginning, and those constants are determined. If $x(0) = 0$ and $x'(0) = 1$, then $B = 0$ and $A = 1$, giving $x(t) = \sin t$. The dynamical rule carries the initial information forward. This is an initial-value problem with a unique solution for every pair of finite initial values.

Now specify positions at two different times instead. Require $x(0) = 0$ and $x(T) = 0$. The first condition again sets $B = 0$. The second requires $A \sin T = 0$. If $\sin T$ is nonzero, A must be zero. The only motion satisfying both boundary conditions is the stationary one, $x(t) = 0$ throughout.

At $T = \pi$, the situation changes. Since $\sin \pi = 0$, every amplitude A satisfies the two zero-position boundaries. The path can remain still, rise gently and return, or make a much larger excursion before returning. All those histories obey the same dynamics and the same two positions. The boundary data leave the amplitude undetermined.

Change only the final position, requiring $x(\pi) = 1$ while retaining $x(0) = 0$. No solution exists under these dynamics. Every motion starting at position zero has the form $A \sin t$ and is back at zero at time π. The final condition asks for a value outside that family at that particular time. The calculation establishes incompatibility rather than selecting a special motion.

For a nearby time T with $\sin T$ nonzero, the final condition $x(T) = 1$ does have a solution: $x(t) = \sin t / \sin T$. Yet its

amplitude becomes large as T approaches π. A small change in the specified horizon can therefore require a large change in the motion satisfying the two boundaries. Existence and uniqueness at a particular T do not by themselves establish stable dependence on the boundary data.

The sensitivity can be stated quantitatively. At $T = \pi/2$, sin T is one and the required amplitude is one. At $T = \pi - 0.1$, sin T is about 0.0998 and the amplitude is about ten. At $T = \pi - 0.01$, it is about one hundred. The dynamics have stayed fixed; the imposed final position and shrinking distance from a special horizon create the large response.

Add a physical restriction such as an amplitude limit of two, and some previously available mathematical paths cease to be admissible. Reaching final position one requires $|\sin T|$ to be at least one half under that limit. The boundary question now depends on the dynamics, the time, the requested final value, and the allowed family of motions. Each restriction has an identifiable role.

This provides a precise way to discuss a proposed terminal relation. One can ask whether any complete history satisfies it, whether the allowed history is unique, and how the family changes when initial or terminal information changes. Those questions are different from asking what process selects or generates an actual history. The oscillator's equation supplies a dynamical family; the boundaries select within it when a compatible selection exists.

Several paths sharing the same endpoint can consequently preserve substantial freedom in their interiors. At the special horizon π, the amplitude remains available to distinguish infinitely many histories that return to zero. A common terminal value does not erase what happened along the way. In this

example, the full time course and the total distance traveled vary with A even though the endpoint positions agree.

The example also makes a scientific proposal easier to articulate. A candidate account can state its dynamics, its boundary condition, and the independently justified restrictions that connect them. It can then establish whether its solutions exist, how many remain possible, and whether small changes produce stable predictions. Those are substantial achievements in understanding the formal shape of a claim before any observation is used to decide whether nature realizes it.

6.1.3 — Structure Instead of Miracles

An ad hoc patch enters a theory after a troubling result appears. Its chief job is to protect the preferred conclusion from that result. A useful auxiliary assumption instead brings independent reasons and consequences that can be tested. The distinction matters when explanations of terminal closure introduce new mechanisms only after a prediction fails.

It may protect hope as well as theory, but it makes the claim less answerable to the world. If a difficult case can always be assigned an unseen episode, a hidden dimension, an endpoint not defined in advance, or a special exception, no observation can matter.

Popper emphasized exposure to refutation, while Lakatos distinguished progressive revisions that predict new facts from degenerating adjustments that merely absorb anomalies (Popper, 1959; Lakatos, 1978). A revision can therefore strengthen the program when it narrows a mechanism and creates a new test. It weakens the program when it preserves every old conclusion and changes only what was needed to explain away the latest failure.

Tests also depend on auxiliary assumptions about instruments, samples, endpoints and observation. A failed prediction need not identify which assumption failed. The Duhem-Quine problem makes an independently supported diagnosis necessary: an auxiliary cannot be discarded solely because the outcome disappointed the theory (Quine, 1951; Lakatos, 1978).

Bearer rules can also rescue universality. A stream that violates the claim can be declared fragmented, nonmaximal, or outside the intended class after the outcome is known. A universal proposition cannot remain universal by shrinking the class one adverse case at a time.

Miracle, in this methodological sense, is not a judgment about religion. It names an explanation whose only role is to arrive after difficulty and save the desired result. Religious traditions may speak of miracles within their own meanings and warrants. Scientific inquiry asks a different question: what observation or independently motivated mechanism would distinguish this account from ordinary uncertainty?

For example, does dispersion in an estimated cumulative residual change near a prospectively defined endpoint after accounting for sedation, medication, selective observation, cultural reporting patterns, exhaustion, social expectation, and measurement error?

6.2 — Purpose, Meaning and Scientific Lawhood

Purpose language can move unnoticed from description to judgment. A model might propose a causal route by which suffering affects later behavior. That does not show the suffering was necessary, deserved, or chosen for the person's benefit. Scientific claims must remain separable from the interpretations people freely give their experiences.

6.2.1 — Purpose Language as Risk

Purpose language can cross several categories in a single sentence. A process that regulates a variable becomes an intention, the intention becomes a moral plan, and the plan becomes an explanation of suffering. Scientific description must stop that slide at the first step. A biological process may contribute to viability, learning, repair, or regulation without being good for every person or designed for a moral end.

Biology often uses teleological shorthand, such as saying that a trait is for a function, while scientific explanation traces selection, development, mechanism, and current causal role. Mayr’s distinction between proximate and evolutionary explanation helped clarify why purpose-like language need not imply a conscious planner (Mayr, 1961). The Law should be even more careful because purpose claims about suffering carry immediate moral weight.

Monod used the language of teleonomy for organized, goal-like features that can be explained without foresight or cosmic intention (Monod, 1971). The distinction is useful, but it does not grant a general license to speak as though the universe has aims. A system can maintain temperature, seek food, avoid threats, or minimize a modeled error while remaining vulnerable to failure and without assigning moral meaning to any event.

End-directed behavior is not destiny. An organism can regulate toward a range and still fail, overshoot, collapse, or be overwhelmed. Control language therefore increases empirical burden. The model must specify what is regulated, what resources are required, what happens when control fails, and which rival process could mimic the same pattern.

For the research program, every purpose word should pass an ownership test. If the owner is a person, tradition, organism, or explicit model, say so. If no owner can be specified, replace the purpose claim with the mechanism, boundary, or observation actually meant. A method preserves counterfactual alternatives and the possibility of correction. Invoking fate to treat every outcome as confirmation removes that possibility. A scientific account must state what another measure, horizon, endpoint rule, or rival would have led us to expect before the path is complete.

6.2.2 — Scientific Boundary

The scientific boundary starts by identifying the sentence type. A definition states meaning; an assumption declares what a model takes as given; a mathematical consequence follows under conditions; a candidate route proposes how a pattern might arise; an estimator maps observed data to an estimate; and a verdict states what a governed comparison licenses. Confusing these roles lets one sentence borrow authority from another.

A failed predicted consequence may implicate the target, the proposed route, or the observation bridge. The evidence map identifies the level each test can presently address.

Exact PTVC. Evidence: The formal statement and conditional models. Establishes: A defined target, not demonstrated complete-stream closure. Open bridge: Cumulative

representation and an independently identified endpoint. Next test: Measurement gates (Section 27.1).

State-change route. Evidence: Telescoping identity and closing/nonclosing countermodels (Section 8.3.2). Establishes: Cancellation under potential-difference assumptions. Open bridge: Felt contributions must track an independently identified state potential. Next test: Held-out transition-cycle study (Section 17.2.4).

Affective measurement. Evidence: Construct and calibration literature (Chapters 10–11). Establishes: Local observation relations and their limits. Open bridge: Neutral origin, cross-sign accumulation and sufficient coverage. Next test: Independent calibration pilot with bounded error and a fixed failure rule.

Observable return patterns. Evidence: Adaptation, prediction and control accounts (Chapters 19–21). Establishes: Several mechanisms generate recovery or apparent closure. Open bridge: Which mechanism predicts new paths under shared information? Next test: Common-budget held-out comparison (Section 21.4.4).

Simulator performance. Evidence: Constructed worlds, recovery and calibration checks (Chapter 23). Establishes: Behavior of a specified model and analysis pipeline. Open bridge: Transfer to an empirical system. Next test: Held-out generator tests followed by the independently calibrated local study.

The boundary routes questions without closing them. Science tests causes, functions, measurements, and predictions; philosophy examines lawhood and implication; ethics governs conduct; spiritual and narrative life may explore meaning voluntarily. When a dispute crosses domains, naming the warrant in question (empirical, philosophical, ethical, or

interpretive) keeps disagreement focused on the kind of support actually needed.

6.2.3 — Why Calling It a Law Is Still a Question

Physical conservation laws do more than assert that something balances. They identify the systems and quantities being conserved. Their symmetry and transformation rules produce independent predictions. Any analogy to conserved felt experience must earn comparable definitions and tests before it can be called a law.

Their status does not come from the loose idea that something eventually balances. A conservation analogy therefore cannot supply a neutral origin, unit, bearer, endpoint, or empirical invariance for valence. Those identification burdens remain exactly where they were before the comparison.

Thermodynamic imagery introduces another family of errors. Entropy, equilibrium, heat, and closed systems have technical meanings. A conscious organism exchanges matter, energy, information, and care with its environment, and an affective history is not made thermodynamic by calling its endpoint an equilibrium. The analogy may motivate a question about regulation, but it cannot replace a biological or phenomenal model.

Bridge processes are mathematically useful precisely because they warn against overinterpretation. A stochastic path conditioned to end at a chosen value can display terminal concentration without any special fairness law. That makes the bridge a serious comparator. It demonstrates that endpoint shape may arise from conditioning or selection and that conditional drift alone cannot identify the proposed target.

6.3 — How Post Hoc Rescue Destroys the Claim

Post hoc rescue can happen at more than one level. An analyst may change a stopping rule after examining the results. A theorist may add an untested exception, or a public account may quietly redefine failure as support. These moves need separate safeguards because they conceal different kinds of revision.

6.3.1 — Optional Stopping as Theory Risk

Optional stopping allows the decision to end observation or analysis to depend on accumulating data. Repeated, unplanned use of a fixed-sample test can inflate false-positive risk. In a terminal theory, the researcher can also change the observation window or the set of cases being evaluated, or abandon one route for another after seeing the result (Simmons et al., 2011; Gelman & Loken, 2013).

Repeatedly applying an ordinary fixed-sample equivalence test and stopping when it becomes favorable can change its error rate. Valid alternatives include planned group-sequential tests and confidence sequences with simultaneous coverage across repeated looks. The latter need not impose a statistical maximum sample size, although participant burden and resources still require governed limits (Howard et al., 2021).

For a fixed local contrast, declare measured-scale equivalence only when the current confidence-sequence interval lies wholly inside the prespecified equivalence band. A sequence with 95% simultaneous coverage limits the probability of ever making that declaration when the true contrast is at or beyond an equivalence boundary to at most 5%, under its sampling and dependence assumptions. The target and measurement model stay fixed; looking again does not redefine the stream’s endpoint. This allows evidence to accumulate flexibly around a stable scientific question.

A simple construction shows how a guarantee can cover an unlimited sequence of planned looks. Consider independent simulated trials with outcomes zero or one and an unchanging success probability p. Look at the sample average after 2, 4, 8 and successive powers of two trials. Number these looks k = 1, 2, 3 and so on. The sample size at look k is 2^k.

Set aside an overall error budget α. Give look k a budget α/[k(k + 1)]. The first look receives half of α, the second receives one sixth, the third one twelfth. These fractions sum to one across all positive integers because 1/[k(k + 1)] equals 1/k − 1/(k + 1). The partial sum through look K is 1 − 1/(K + 1), which approaches one.

Each sample average has variance p(1 − p)/2^k, at most 1/(4 × 2^k). A direct probability bound now supplies an interval. For any random quantity, its mean squared deviation is at least b^2 times the probability that the absolute deviation is at least b. Rearranging gives the familiar variance bound on that probability. Choose the half-width at look k to be the square root of k(k + 1)/(4 × 2^k × α). The chance that this interval misses p is at most that look's assigned budget.

The union bound states that the probability that any interval in the entire sequence misses p is no greater than the sum of the individual miss probabilities. That sum is at most α. The intervals reuse earlier observations, so the looks are dependent, but the sum bound does not require independence between those events. Independence was used in calculating the variance of each sample average, where the assumption belongs.

On the event that every interval contains p, a declaration made at any of these looks retains the same protection. For example, if an interval lies wholly inside a fixed equivalence band, p lies in the band too. An analyst may stop at the first qualifying look

without spending a fresh α on each inspection. All the scheduled looks were included in the original guarantee.

This elementary construction is deliberately conservative. With α = 0.05, the twentieth look occurs after 1,048,576 trials and has a half-width of about 0.0448. An average of 0.50 then gives an interval contained within the band from 0.45 to 0.55. This elementary simulator calculation favors a short proof over a narrow interval. Sharper concentration arguments produce much narrower sequential intervals, as in the methods discussed by Howard and colleagues.

The construction's intellectual value is its visible accounting. It names the unchanged parameter, the sampling assumptions, the times at which decisions may occur and the error allocated to the whole sequence. If a process changes its success probability, the original fixed-p statement no longer describes the same target. If decisions are also made between the planned looks, those additional opportunities need a corresponding guarantee. The freedom to stop is earned by defining which evidence paths the statistical statement covers.

Stopping can also occur by window selection. A study may present the terminal week rather than the terminal month, the calm interval rather than the fluctuating one, or the last completed survey rather than the period in which missingness became severe. The chosen window must follow a protocol-fixed clinical and scientific rationale. It cannot be the interval that yields the cleanest picture.

Route switching can play a similar role at the theoretical level. A reserve model may be emphasized while it fits, then replaced by adaptation, a bridge, a global constraint, or subjective-time correction when predictions fail. Candidate routes may evolve, but each successor begins a new test cycle. The failed route

remains failed for the version that was actually prospectively specified.

The reference law for the random-endpoint problem cannot be selected by the answer either. If independence produces an uncomfortable implication, the project may investigate state-coupled timing or hitting-time branches. It may not claim that the endpoint was always state-coupled solely because the observed path would otherwise count against closure. Branch choice needs independent motivation and new discriminators.

Publication creates a second stopping layer. A program can keep running variants until one produces a striking graph while null, rival-favored, and unusable results disappear. Complete study inventories, prospective specification, and public preregistration reduce the power to stop the public record at success (Nosek et al., 2018; Munafò et al., 2017). Empirical work documents selective publication of results (Franco et al., 2014).

6.3.2 — Public Rhetoric as Rescue

A claim can survive a failed test by changing language after the numbers are fixed. The target remains exact in the technical source, while public summaries quietly say tendency, deep balance, hidden fairness, or evidence of convergence. That shift may feel modest, but it changes the proposition and can turn an adverse result into an inspirational success.

Qualifier deletion is the simplest form. Per stream becomes across people; terminal becomes eventually; latent becomes measured; candidate becomes established; compatibility becomes confirmation; a route-specific pattern becomes evidence for the whole Law. Public brevity does not excuse these substitutions because each omitted word removes a failure condition.

Testimony can rescue a claim through selection. A moving account of unexpected relief, terminal calm, a fulfilled dream, or a life that appears balanced may illuminate a human question. Choosing only stories that fit and treating counterstories as incomplete turns narrative power into an undeclared sampling rule.

Specify the permissible public wording for each scientific outcome before results arrive.

6.3.3 — Change-Control Rather Than Improvisation

A living research program should change in ways readers can follow. Name whether the target, route, endpoint proxy, instrument, analysis, citation, or public explanation changed.

For each revision, preserve the earlier version and record its successor, rationale, date, responsible person, and the evidence available at the time. Identify which target, bridge, and boundary assumptions remain unchanged. Archive the prospective forecast beside the final report and version history. A pre-data correction, a safety change during collection, and a post-result revision have different evidential status.

Triggers need names: theorem correction, failed replication, clinical guidance, instrument defect, consent or privacy failure, expert review, accessibility defect, legal change, public misreading, or a burden requiring suspension. Record whether evidence, communication, safety, or target changed.

Changes to bearer, ownership, origin, complete domain, or endpoint can create a new target. A new unity rule can change which experiences accrue; a new origin can change the residual; a new endpoint can move the boundary. Test the successor prospectively, never to validate the predecessor retroactively.

Bridge-validity criteria must apply symmetrically to favorable and adverse results. Once those bounds and bridge criteria

establish adequacy, an unnamed additional channel cannot be invoked solely to rescue an adverse result.

Match a revision to the failed condition: a scalarity failure may require a bivariate or partially identified branch rather than excluding a population; endpoint uncertainty does not redefine valence. For an excluded population or endpoint, specify a prospective scientific re-entry criterion, such as validated translation, an adequate observation model or a target-preserving branch argument. This scope criterion is separate from permission to restart an experiment.

Version every change to measurement and analysis, including channel selection, normalization, weighting, smoothing, imputation, calibration, preprocessing, estimands, exclusions, stopping rules, multiplicity corrections, code, and decision thresholds. Preserve the original procedure and results, identify deviations, and state the successor's evidential status (Nosek et al., 2018; Munafò et al., 2017).

Audit the combined and directional flexibility across a sequence of adjustments, including whether successive revisions systematically preserve the favored conclusion while independently supported progressive revisions remain possible. Several small choices can jointly accommodate almost any result (Simmons et al., 2011; Gelman & Loken, 2013).

Advance control should use exact language. Protocol-fixed means written into the study protocol before outcome inspection. Prospectively specified means fixed before outcome inspection when no public-record claim is asserted. Publicly preregistered requires a timestamped external registry record with a persistent identifier. An intention, private draft, or internal timestamp does not meet that standard.

Narrowing and defeat should state what remains. A universal claim may become reference-class specific; a scalar model may

yield to a multidimensional question; a route may fail, leaving the semantic target open.

Part III — How Exact Closure Could Occur

A terminal relation does not explain what could make it occur. The next step is to examine how an unfolding path, a representation of that path and the options still available can constrain one another.

On the scalar signed-measure branch, temporal resolution changes what can be recovered. In an invented example, successive contributions of +3 and −3 become one zero total when combined into a single time bin. The signed total is preserved, but treating that bin as neutral hides three positive and three negative units. Refining exact bin totals cannot decrease the recovered positive and negative sums, which remain bounded by their true full values. The overall signed sum stays unchanged. This matters for the primary finite branch. Exact coarse-bin totals can preserve net closure while understating gross experience and supplying only lower bounds on positive and negative magnitudes. Finer noisy sampling can inflate apparent variation, so resolution and observation error must be examined together. A displayed average multiplied by duration equals a bin's phenomenal contribution only under a justified measurement and exposure model. Net equality, gross magnitude, and evidence that the full magnitudes are finite are related questions, not interchangeable measurements.

Time makes that architecture human. Calendar duration, perceived horizon, and usable opportunity can diverge, changing which choices feel available and which repairs remain possible.

Imagine a bead able to wander sideways inside a channel that tapers continuously to a point. As the bead approaches the tip, its greatest possible sideways distance from the centre shrinks to zero. No foresight is required: the available width already limits where the bead can be. A route could therefore produce terminal convergence through its changing geometry, provided the

physical endpoint and the narrowing boundary are independently connected.

Thinkable, felt-selectable, practical, and model-admissible options need not form a simple funnel. A model can allow an action the person has never imagined; a person can imagine an action that money, safety, or time makes unavailable. Locate the particular obstruction and what would reopen that path (Woodward, 2003).

When researchers estimate an unobserved state from a recorded history, their uncertainty differs from a person’s uncertainty about their own life. The model’s probability distribution describes what the data and model allow them to infer; it does not report the person’s private beliefs. A person may feel uncertain while the model assigns a narrow distribution, or feel certain while the model remains uncertain. Neither result replaces first-person testimony (Bollen, 1989; Box, 1976).

A good route explanation should become easier to follow when its pieces are taken apart. One can first ask what a state remembers, then what it permits next, then how it changes as a history unfolds. Those questions suggest different observations: histories that reach a similar present, actions that remain available from it, and transitions under a changed input. The chapters that follow use this division of labor to turn the image of a closing path into a set of concrete representational and dynamical problems.

Chapter 7 — The Structure Closure Needs

The proposed Law is compact enough to say in one sentence yet demanding enough to require an architecture. Readers must see the represented object, the stream that owns it, the boundary where the question is asked, the routes that might realize it, and the fallible observations that could bear on it.

The represented domain includes positive, negative, mixed, and neutral tones of experience. Pain and pleasure are central; dread, relief, safety, humiliation, belonging, fatigue, grief, awe, loneliness, and bodily ease give the domain its variety (Barrett, 2006; Cacioppo & Berntson, 1994).

The model must also say when experience enters the domain. Sleep, dreams, anesthesia, delirium, fragmented access, memory loss, and low responsiveness show why biological time cannot simply be copied into a phenomenal record. Accrual conditions belong to the claim and must be fixed independently of the desired terminal result.

Endpoint rules must likewise be specified independently of the desired result. State-coupled or other dependent regimes remain possible as separately declared branches with their own testable consequences.

Remaining capacity is causal, not moral. Illness, lost time, broken trust, narrowed attention, poverty, coercion, and absent support can close paths that once existed. Treatment, education, reconciliation, protection, and material help can reopen some of them. The geometry concerns possibility.

Processing can change a waveform while leaving the quantity a comparison needs recoverable. Filters occur throughout neural recording and analysis, so their responses belong in the observation model (de Cheveigné and Nelken, 2019). In a stipulated finite sequence with equal sampling intervals, let x contain signed contributions and h be a known linear filter. For the complete zero-padded convolution y, the output total equals

the input total multiplied by the sum of the filter coefficients. A unit-sum smoothing filter preserves that total; a known nonzero coefficient sum permits its recovery without reconstructing every input sample.

An invented example shows a more surprising possibility. The input (1, 2, 1) has total four. The difference filter (1, −1) produces (1, 1, −1, −1), whose total is zero. Yet the full output has not lost the input total: number its positions 0 through 3, multiply each output by its position, and negate the sum. The result is four. The filter has moved information about the total into a different observable contrast; the zero output sum is not evidence that the input balanced.

These calculations require a known filter, boundary convention, complete response, and uncertainty propagation appropriate to where noise enters. Here the target is a quantity in the unfiltered input; Chapter 19’s biological dynamic has a different role. Truncation can prevent recovery. Filter delays are not post-endpoint experience, and neural totals need an independent bridge to phenomenal valence.

7.1 — Bearer, Experience, Time, and Boundary

The bearer in this formal program is the unified conscious stream defined in Section 4.2.1. A body can support that stream without its physical boundaries alone settling the question. Experience is indexed to a time interval, and the endpoint must be defined separately. This combination fixes whose cumulative object a proposed boundary condition concerns.

7.1.1 — Stream, Accrual, and Units

At a pause or transition, the assignment rule must specify whether contributions belong to the same qualifying stream or to different streams. Accrual means that a felt valence contribution is counted within the formal path under the model's specified conditions. Elapsed biological time cannot decide this: conscious access may be suspended, degraded, fragmented, or uncertain while the clock continues.

The unity gate specifies when a felt contribution enters the modeled stream. On a rate-based branch, valence density names the signed contribution per unit of the adopted measure of experienced exposure.

Changing the clock need not change the accumulated quantity. On a rate-based branch, write the contribution as $\int v(t)\,dt$. If $s=\phi(t)$ is a smooth increasing coordinate with $\phi'(t)>0$, the same quantity is represented by $\tilde{v}(s)=\frac{v(t)}{\phi'(t)}$. Then $\int \tilde{v}(s)\,ds=\int v(t)\,dt$, and the positive and negative totals are preserved separately. For the invented rate $v(t)=2t-1$ on [0,1], the total is zero. Relabeling time as $s=t+t^2$ but carrying over the values as $\tilde{v}(s)=v(\phi^{-1}(s))$ instead gives 1/3. The difference is altered weighting, not altered experience. This consistency check does not identify phenomenal time; it separates changing coordinates from changing the measure itself.

The phenomenological relation belongs to the definition layer only. It does not imply that the candidate stochastic process has the same moment-by-moment behavior as the experienced contribution admitted by the unity condition. That bridge remains open and is treated through the observation and reachability layers in Sections 7.4 and 7.5 (Campbell & Fiske, 1959; Borsboom, 2005).

Dimensional discipline belongs inside the map. Intensity, duration, probability, capacity, and confidence are not interchangeable merely because each can be represented numerically. Units and transformation rules determine which comparisons survive rescaling, which depend on an origin, and which should never be combined (Krantz et al., 1971; Michell, 1999).

7.1.2 — What a Carrier Has to Distinguish

Imagine a model with ten steps. At each step an invented contribution is either plus one or minus one. After ten steps, the accumulated value can be minus ten, minus eight, minus six, and so on up to plus ten. There are eleven possible totals. A variable that records only the latest contribution has two possible values. It cannot, by itself, distinguish all eleven accumulated totals.

The difference is visible in two histories ending with the same move. Ten positive contributions end with plus one and total plus ten. Five negative contributions followed by five positive contributions also end with plus one, yet total zero. An observer who sees only the final move receives the same observation in both cases. An exact carrier of the accumulated value must preserve some distinction between them.

This is a requirement on a proposed representation. If the carrier uses a finite collection of discrete states, eleven different totals require at least eleven distinguishable states. Three binary

switches provide eight configurations, while four provide sixteen. Four switches are therefore enough to label the eleven totals in this particular toy model. The count describes a capacity requirement for this discrete representation. A continuous variable or a distributed pattern would organize the representation differently.

Notice how much compression is already possible. Ten binary contributions generate 1,024 ordered histories. A carrier of their sum needs to distinguish only eleven totals. Many histories can share one cumulative value because their order has been discarded. A carrier of the whole sequence would need far more distinctions. Before searching for a mechanism, the investigator can therefore ask exactly which features of history the proposed state must retain.

The count also depends on when the carrier is read. At the tenth step, only eleven totals are possible because the number of positive and negative steps must add to ten. Across all prefixes from the beginning through step ten, every integer from minus ten through plus ten is possible. A single direct code that represents the total at any of those times therefore needs twenty-one distinguishable labels. Four binary switches provide only sixteen; five provide thirty-two.

A different representation can use a clock. If the step number n is available independently, the carrier can store the number k of positive moves. The accumulated total is twice k minus n. Since k lies between zero and ten, eleven labels suffice for this part of the representation. The step count supplies the other information needed to interpret them. Four switches can encode k, while the clock has its own state and physical implementation.

This is a useful way to locate information rather than merely count it. A proposal may store the running total directly, combine a compact count with an independent clock, or

distribute both across interacting processes. What looks like a smaller carrier can be achieved by making another state do part of the work. The full explanation identifies where each required distinction is retained and how the states are read together.

7.1.3 — A Carrier That Can Correct a Mistake

Having enough states to label a quantity is one achievement. Preserving that label through disturbance is another. A four-switch register has sixteen configurations and can label eleven possible totals. If a switch changes accidentally, however, the altered configuration may be another valid label. A reader of the register could then receive a different total with no indication that anything had changed.

Redundant representation can give the disturbance a recognizable signature. Begin with the simplest case: a message contains one binary digit. Encode zero as 000 and one as 111. A decoder chooses the value held by a majority of the three positions. If at most one position flips, 000 can become 100, 010, or 001, and each still decodes to zero. The corresponding three single-flip versions of 111 still decode to one.

The unused configurations do useful work. They provide a neighborhood around each valid codeword in which a one-position disturbance can be interpreted as an error rather than a new message. The two valid codewords differ in all three positions. Their one-error neighborhoods therefore do not overlap. The decoder can identify the original message because the allowed disturbance is too small to carry one valid codeword into the other's neighborhood.

A more efficient arrangement can protect several data positions together. The following construction is a seven-bit Hamming code (Hamming, 1950). Place four data bits in positions 3, 5, 6, and 7 of a seven-bit word. Use positions 1, 2,

and 4 for parity checks. The first check covers positions 1, 3, 5, and 7. The second covers 2, 3, 6, and 7. The third covers 4, 5, 6, and 7. Choose each check bit so that the number of ones in its group is even.

These groups give each position a unique membership pattern. Position 1 belongs only to the first group; position 2 only to the second; position 3 to the first and second. Position 4 belongs only to the third; position 5 to the first and third; position 6 to the second and third; position 7 to all three. A single changed bit reverses the parity of exactly the groups containing its position.

The failed checks therefore identify where the change occurred. If the first and third checks fail while the second passes, the changed position is 5. If all three fail, it is 7. Flipping the identified position restores the encoded word under the stipulated one-error condition. The three check results are often called a syndrome: a compact description of the discrepancy that directs the correction.

There are sixteen possible four-bit messages, so this seven-bit construction can protect labels for all eleven totals in the earlier example. The extra positions are not storing a longer history of contributions. They are organizing the representation so that a specified kind of corruption can be repaired. Storage capacity and error-correction capacity are thus different uses of the available physical distinctions.

The number of available configurations also gives a lower bound on the storage required. For a binary word of length n, a valid codeword has n distinct one-bit neighbors, plus itself. Correcting any one-bit error requires these neighborhoods to be disjoint for different messages. If there are M messages, they occupy at least M(n + 1) configurations among the 2 raised to n possible binary words. Thus M(n + 1) must be no greater than 2 raised to n.

For eleven messages in six positions, the required neighborhood count is eleven times seven, or seventy-seven. Only sixty-four six-bit words exist. No assignment of six-bit codewords can therefore correct every single-bit error for all eleven messages. Five or fewer positions cannot solve the problem either, since the number of available words per required neighborhood is smaller. The obstacle is a count of distinguishable possibilities, independent of the decoder's ingenuity.

Seven positions are enough because the construction above protects sixteen messages, more than the eleven needed. For all sixteen messages, sixteen neighborhoods of eight words each exactly fill the 128 possible received words. This connects necessity with construction: six positions cannot provide the stated protection for eleven labels, while seven can. The earlier four-position register was sufficient for raw labeling; the stronger requirement of correcting a one-position change has increased the minimum to seven.

The correction rule has a definite disturbance model. Two simultaneous changes can produce a syndrome that looks like a single change at another position. A code designed for one error therefore has a different guarantee from a code designed for several errors or for detecting rather than correcting corruption. Describing the permitted disturbance is part of describing what the representation accomplishes.

For a proposed physical carrier, this suggests concrete questions beyond the number of labels. Which distinctions are stable? What disturbances can occur? Does the dynamics restore a represented state, merely signal that it has become uncertain, or move to another valid state? These questions can be investigated in an engineered model without assuming that conscious systems use binary switches or this particular code.

The larger idea is that apparent duplication can carry distinct information about reliability. Three copies of one bit do not describe three different messages, but their agreement makes certain errors recoverable. A theory of a cumulative carrier therefore needs to say both what is represented and how the representation persists. A compact code answers the first question efficiently; a protected code devotes additional structure to keeping that answer available over time.

7.1.4 — Finite Totals Across Infinitely Many Contributions

A complete total can be finite even when its mathematical representation contains infinitely many contributions. The decisive question is how their magnitudes accumulate. Consider a sequence of pairs. The first pair contributes plus one half and minus one half. The next contributes plus one quarter and minus one quarter. The next contributes plus one eighth and minus one eighth, and the pattern continues. Every completed pair has a signed total of zero.

The positive contributions form a geometric series. After n pairs their sum is one minus two to the power minus n. The negative magnitudes have the same sum. As the sequence continues, both approach one. Their difference approaches zero, and the total magnitude of all the contributions approaches two. The infinitely many terms fit inside finite positive and negative totals because each new pair is sufficiently smaller than its predecessor.

Change the sizes while retaining the pairing. Let the contributions now be plus one and minus one, plus one half and minus one half, plus one third and minus one third, and so on. At the end of every pair the signed total is again zero. Between successive pair endings, the positive excursion becomes smaller and tends to zero. The signed partial sums therefore converge to

zero. Yet the positive contributions form the divergent harmonic series, and the negative magnitudes do too. Group the terms from one half onward into blocks ending at one half, one quarter, one eighth, and successive reciprocal powers of two. Each block contains twice as many terms as the preceding block, and each adds at least one half. Arbitrarily many such blocks produce an arbitrarily large total. This sequence has a limiting signed total but no finite positive or negative total.

The two constructions answer the finiteness question differently even though their completed-pair totals look identical. In the second, writing infinity minus infinity would not define the primary branch's residual. The finite branch asks for two actual finite quantities whose equality has a definite meaning. It does not merely ask whether a selected sequence of increasingly complete differences settles near zero. That requirement determines which mathematical histories belong to the claim.

For a finite signed measure, the Jordan decomposition expresses the measure as the difference of a positive measure and a negative-magnitude measure that occupy disjoint measurable parts. Their complete masses are the positive and negative Jordan masses. They supply the positive and negative phenomenal magnitudes used by the primary formal representation. Adding those masses gives the measure's total variation; subtracting them gives its signed total, or net valence. In the geometric example the two masses are one and one, the total variation is two, and the signed total is zero. The measure-theoretic construction is developed by Folland (1999).

The disjointness matters. Suppose one wrote the same signed measure as the difference of two larger positive measures by adding an identical extra measure to both. Their difference would remain unchanged, while both totals would increase. The

Jordan decomposition removes this arbitrary common addition. It separates the positive and negative parts of the signed representation itself. A bivariate account of simultaneously positive and negative experience may retain different information; Section 7.3.1 specifies that separate representational choice.

Finiteness also supplies a useful approximation rule. In the geometric construction, after n complete pairs the uncounted positive mass is two to the power minus n, and so is the uncounted negative mass. The total absolute amount still uncounted is twice that value. After ten pairs it is one five-hundred-twelfth. Any partial collection from that remaining tail has a signed magnitude no greater than this absolute bound. The assurance comes from controlling what remains, rather than from the appearance of the last observed zero.

This is why the representation can support a coherent complete-history total without requiring one final indivisible contribution. An infinite sequence may approach its boundary through successively smaller contributions, with a rigorously bounded remainder at each stage. Absolute finiteness also makes regrouping the terms harmless: partitioning the same assigned history into different pieces leaves its total intact. The mathematical achievement is a stable relation among the parts of one history, its finite approximations, and its complete value.

7.1.5 — Keeping the Remainder Within View

Convergence can concern a sequence of models as well as the unfolding of one history. The distinction matters when longer records or refined approximations are used to support a statement about a complete total. Agreement on every fixed early window can be exact while the family continues to move an important contribution beyond that window.

Construct a sequence of signed rate functions indexed by integers n greater than one. Every function has rate plus one during the interval from time zero to time one. It has rate minus one during the interval from time n to time n + 1. Its rate is zero at all other times. Each history therefore has positive mass one, negative mass one, and complete signed total zero.

Now fix any finite observation horizon H. Once n exceeds H, the negative interval lies entirely beyond that horizon. Within the observed window, every sufficiently late member of the sequence looks the same: an initial positive contribution followed by zero rate. Looking farther along the sequence of models does not change anything visible before H.

The pointwise limiting rate is plus one on the first unit interval and zero thereafter. Its complete total is one. Thus every individual model has total zero, while the limiting rate has total one. The negative contribution has moved farther and farther away. It has not become small in any member of the sequence; it has left every fixed window used to describe the limit.

The two orders of calculation give different results. First complete each history and then take the limit across models: the answer is zero. First take the local limit of the rates and then complete the limiting history: the answer is one. The difference comes from the tail of the domain, where local convergence supplies no uniform control.

Even a uniform bound on each history's total absolute contribution does not settle this example. Every member has absolute total two. What is missing is a bound that places the tail mass uniformly near the observed part of the domain. For any chosen horizon H, some member still places an entire negative unit beyond it. The family has bounded total mass but no common horizon after which all its remaining absolute contributions are small.

A useful sufficient condition can be stated directly. Suppose the rates converge in integrated absolute difference on every finite interval. Suppose also that, for every positive tolerance ε, one can choose a horizon H such that the absolute contribution beyond H is less than ε for every member of the family and for the proposed limit. Then the complete totals converge.

The proof separates the domain into two parts. On the finite interval through H, local integrated convergence makes the difference between totals small. Beyond H, the magnitude of that difference is bounded by the sum of the two absolute tail contributions, less than 2ε. Since ε can be chosen as small as desired, the complete difference tends to zero. The argument identifies exactly where local and global information enter.

For a finite-domain example, one can move the same phenomenon toward a boundary while increasing the rate. For integers n greater than two, put a negative pulse of total mass one into the last 1/n units before time one, with height −n. Put a positive rate of two on the first half of the unit interval and zero elsewhere. Each complete total through time one is zero, while pointwise convergence away from the boundary can lose the increasingly narrow negative pulse. A common rate bound would exclude this particular construction by limiting how much mass fits into a shrinking interval.

These examples clarify what a completion model needs when it is approximated. A good fit over every tested window can support local predictions. To carry a total through the approximation, the remaining contribution must also be controlled in a way that survives across the family. That control might come from a common tail bound, a rate bound together with a shrinking remaining interval, or another independently justified property of the proposed dynamics.

The result adds a constructive question to whole-history analysis: what keeps the remainder from escaping the part of the model that has been validated? A complete answer can join accurate local behavior with a quantitative account of the unobserved tail. When both pieces are present, taking a limit becomes an earned connection between successive approximations and the complete quantity they are intended to represent.

7.2 — Exact Zero and the Random-Endpoint Problem

Exact zero at an endpoint is stronger than a small measured residual. The independent positive-density endpoint branch adds an obstruction for nontrivial paths. Endpoints need not all be independent, but any dependence must be declared before looking at outcomes. Each branch should therefore expose the assumption on which its closure claim depends.

7.2.1 — Exact Zero Is a Boundary Property

The strong latent form asserts exact terminal closure in the underlying idealized process. In probability language, the claim is pathwise rather than merely an average tendency.

An atom at zero in the ledger distribution does not, by itself, resolve the independent-endpoint obstruction developed in Subsection 7.2.2.

Jumps alter the boundary problem rather than erase it. A sudden loss, medication change, trauma, windfall, or acute clinical event may create a discontinuity, but the jump branch still needs an independently specified event, size distribution, sign relation, endpoint rule, and observation model. Its timing and missingness patterns may also give rivals new explanatory leverage.

Measured equivalence concerns the true measured-scale estimand, not whether an estimator happens to fall near zero. Prespecified margins describe differences small enough for the stated scientific purpose. Calibration connects those margins to any proposed latent target; uncertainty determines how sharply the measured quantity can be assessed. In the standard scalar two-one-sided-tests procedure, both boundary nulls must be rejected. At 5% in each one-sided test, this corresponds to the matching 90% confidence interval lying inside the margins.

Failure to reject a difference from zero alone does not establish equivalence (Schuirmann, 1987; Lakens, 2017).

For a synthetic example with margins −0.5 and +0.5, a matching 90% interval from +0.1 to +0.2 supports measured equivalence despite excluding exact zero. An interval from +0.6 to +0.9 lies wholly beyond the upper margin. The comparison concerns the fixed measured-scale contrast; additional studies sharing the same calibration bias would not repair that bias.

Declare whether the estimand is a cohort mean, median, individual probability, hierarchical distribution, or path-level endpoint; these answer different questions.

Cross-study synthesis needs harmonization or an explicit translation model. Pooling different origins, endpoints, units, or margins can create an apparent consensus no single protocol supports.

When no defensible compatibility margin can be fixed independently of outcomes, the study can still examine concentration, direction, dispersion, or relative change. Its contribution is narrower than measured terminal equivalence, but it may still improve the next test.

7.2.2 — Random Endpoints Create a Severe Obstruction

The random-endpoint problem is not simply that exact closure seems unlikely. Under the assumptions stated below, probability-one exact closure constrains the ledger over the declared interval, not merely at the endpoint.

Imagine a camera shutter that can open at any moment during a declared interval. Its timing is generated independently of the pointer in the scene. If every photograph must show the pointer exactly at zero, the pointer cannot spend ordinary stretches away from zero and somehow return only when the shutter happens to

open. Under the relevant regularity conditions, it must already be at zero for almost every possible shutter time.

Now change one rule. Let the shutter wait until the pointer reaches zero and then take the picture. Every photograph will show zero, but the result says nothing about ordinary shutter times because success was built into the trigger. The contrast makes the branch distinction visible: independent timing creates the obstruction, while first-hitting timing is a different route whose biological meaning and extra predictions must be justified independently.

The formal statement requires more than the picture. The strict-pre-endpoint ledger must be jointly measurable on a declared interval; endpoint timing must be independent of the sigma-field generated by that path; and the conditional endpoint density must be positive for almost every time in the interval. If exact closure nevertheless holds at the endpoint, the ledger is zero for almost every time on that branch; an added left-continuity condition extends the conclusion to interior times.

Independence here is a probabilistic assumption: learning the specified path does not change the endpoint-time distribution under the declared reference law. A rule can be fixed prospectively and still be stochastically dependent on the path. State-coupled timing therefore needs its own specified model and tests. Section 8.3.3 examines one such branch; the present theorem retains its stated independence and path-domain assumptions.

Subsection 7.2.3 develops alternative endpoint branches. Singular endpoint support and phenomenal contributions assigned exactly at the endpoint also require separately declared branches (Karatzas & Shreve, 1991; Protter, 2005; Revuz & Yor, 1999).

The pre-endpoint path itself must also exist on the comparison interval. Some models define a process only up to a killing time; extending it beyond that time requires an additional construction. The obstruction cannot be applied by pretending that an unconstructed counterfactual path was available. Each theorem travels with its domain.

An approximate version is also informative. Under the declared independent-timing model, each part of the path must be weighted by the probability that the endpoint falls there. The predicted off-band endpoint probability is the expected off-band occupation weighted by that endpoint-time density, conditional on covariates specified before outcome inspection. A reproducible discrepancy, after accounting for observation loss and statistical uncertainty, would challenge the declared independence model or another stated premise. Substantial ordinary time outside the band and rare off-band endpoints alone do not establish dependence.

Bridge rivals enter precisely here. A Brownian bridge or a Doob-transformed process can be conditioned to approach a terminal set, and its earlier distribution changes because the endpoint condition is built into the law (Doob, 1957).

Scoring must match the prediction target and the observation or selection regime. A comparison of complete-population laws includes the terminal-event frequency when that frequency is part of the modeled data. A comparison explicitly conditional on the terminal event uses the normalized conditional law and cannot count the imposed endpoint as a successful prediction. For a probability-zero endpoint, use a specified bridge and comparable observed-data densities, not an extra penalty based on a zero event probability.

Ordinary terminal selection can create a similar pattern without a formal bridge. People are observed at particular

endpoints because illness, care setting, device tolerance, family decisions, or communication status make those endpoints visible. The observed sample may therefore concentrate even when the broader path population does not. Endpoint selection belongs in the rival set, not in a footnote.

Positive density almost everywhere does not mean a uniform endpoint-time distribution.

Consider a fixed ten-unit observation interval. An invented path lies outside a chosen band for the first eight units and inside it for the last two. If the endpoint is independent of this path and uniform across the interval, the probability of ending outside the band is eight tenths. The calculation simply weights each unit of time equally.

Now keep the path unchanged but specify a different independent endpoint distribution. Let the endpoint fall in the first eight units with total probability one fifth and in the last two with total probability four fifths, uniformly within each region. The density is one fortieth per unit in the first region and two fifths per unit in the second. Both densities are positive. The probability of ending outside the band is now one fifth, although the path still spends eight tenths of its time there.

The endpoint mechanism in this construction uses the clock and its fixed schedule. It does not consult the path. Its concentration in the later region happens to coincide with the region where this particular path is inside the band. Consequently, a low off-band endpoint frequency can arise under independence when endpoint times are unevenly distributed. Occupied time and endpoint-weighted occupied time answer different questions.

This suggests a more discriminating comparison than counting off-band time alone. Two paths may spend the same duration outside the band but place that duration in different clock

regions. If the endpoint density is concentrated late, a path with late off-band exposure has a larger predicted off-band endpoint probability than one with early exposure. The model makes a quantitative distinction that can be calculated before the endings are inspected.

A lower bound on the endpoint density turns this relation into a quantitative test. Let the observation interval have duration H. For each realized path, collect the times at which the cumulative value lies outside a specified band. Call the total duration of that set D. Under the declared independence assumption, the chance that the endpoint lands in this set is the integral of its endpoint-time density over those times.

Suppose independent knowledge of the endpoint distribution gives a density of at least m throughout the interval, apart from a set of times of zero length. Each unit of off-band duration then carries at least m units of endpoint probability. For the particular path, its off-band endpoint probability is at least mD. Averaging over the random paths gives an overall probability of at least m times the expected off-band duration.

For a numerical illustration, take an interval of ten time units and a density lower bound of 0.05 per unit. If the expected off-band duration is six units, the predicted off-band endpoint probability is at least 0.30. The density may be much larger in some regions. The bound uses only the common minimum, so it remains valid across all distributions meeting that condition.

The calculation can also be read in the other direction. With the same density lower bound, an off-band endpoint probability at most 0.02 requires expected off-band duration at most 0.4 time units. Dividing probability by the density bound gives a duration. The units make the relation transparent: probability per unit time, multiplied by time, produces a probability.

For an exact-zero claim, the same reasoning becomes especially sharp. If endpoint probability outside every positive tolerance band is zero, positive endpoint density leaves no ordinary off-band duration on the relevant branch. A nonnegative quantity can have expectation zero only when it is zero with probability one. Applying this to the weighted off-band occupation yields the almost-everywhere conclusion. The stated continuity assumption then supplies the further step from almost every time to each interior time.

The approximate calculation uses more information than the exact argument. A density can be positive almost everywhere while approaching zero so closely in some regions that no positive uniform lower bound exists. Exact zero still constrains its weighted occupation under the theorem's other assumptions. A useful numerical duration bound requires an actual positive value of m or a more detailed integration against the known density. That is why the distributional information is part of the quantitative prediction.

Covariate adjustment does not restore independence by declaration. Age, diagnosis, frailty, treatment, or care setting may explain part of the relation between path and endpoint, while residual dependence remains. The analysis should report how much dependence is still required after the frozen adjustment set rather than describing an adjusted association as exogenous.

Abrupt endpoints provide a sharp contrast when they are defined independently of the cumulative estimate. A route that uses gradual horizon information may predict differently before a sudden external boundary. Retrospective time-to-end alignment can reveal descriptive structure, but it uses future information to organize earlier observations and becomes a mechanism only through a separate prospective bridge.

7.2.3 — Alternative Endpoint Branches Remain Open

Each alternative changes a theorem premise and must specify a corresponding test.

One branch uses state-coupled timing. Endpoint risk may depend on independently specified biological or consciousness variables that also relate to the cumulative process. This creates dependence without defining the endpoint by the desired residual. The state, measurements, and hazard relation must be fixed before outcome inspection so completion is not merely closure renamed.

A stopping rule fixed in advance can change which states appear at the end of a record. Consider a simulator that independently displays either a blue or a gold light at each tick, with equal probability. After the light appears, a separate random draw decides whether the run stops. Its stopping probability is one fifth after blue and four fifths after gold. The rule is fixed before any run begins.

At an ordinary tick, blue and gold are equally likely. At a stopping tick, they are not. The chance that a particular tick produces blue and a stop is one half times one fifth, or one tenth. The chance of gold and a stop is one half times four fifths, or four tenths. Conditional on stopping at that tick, blue therefore has probability one fifth and gold four fifths.

This is also the distribution of the final light over complete runs. At every tick still reached by the simulator, the new light is an independent fair draw. The total stopping probability is one half, so the run lasts one tick with probability one half, two ticks with probability one quarter, three with probability one eighth, and so on. Summing the blue-stop probability across these possible ticks gives one fifth. The mean run length is two ticks.

The whole sequence has acquired structure through the endpoint rule. A tick at which the run continues is blue with

probability four fifths and gold with probability one fifth. Continuing after blue has probability four tenths, while continuing after gold has probability one tenth. Thus an analyst who looks backward from the stopping event sees predominantly blue continuing ticks followed by a predominantly gold ending, although every newly generated light begins as a fair draw.

Nothing in the simulator anticipates a future endpoint. The light process has not developed a corrective drift. The selection occurs because the stopping mechanism treats the two current states differently. The ending record is a joint product of state generation and termination, both of which were specified beforehand.

For general stopping probabilities a after blue and b after gold, the stopping probability per tick is (a + b)/2. If a + b is positive, the endpoint is eventually reached with probability one, the mean number of ticks is 2/(a + b), and the final blue probability is a/(a + b). When both colors have the same stopping probability, their final proportions return to one half. When one color can never trigger a stop, the other occupies every endpoint.

The example gives a state-coupled model two sets of predictions to meet. It must predict the observed sequence before completion and the distribution of completion times and states. An endpoint pattern can then be traced to a declared mechanism rather than interpreted from its visual shape alone. For the Law's research program, this is one way to make the alternative timing branch concrete: specify how the independently identified state affects stopping, then calculate the resulting observations under the same rule for every simulated run.

A second branch uses hitting or absorption. A process can terminate when it enters a declared set, or it can remain in that

set once entered. Stochastic-process theory gives precise meanings to hitting times and absorbing states, but the scientific challenge is not the vocabulary. The challenge is finding a physically and phenomenally defensible state that was not chosen merely because it produces closure (Ethier & Kurtz, 1986; Champagnat & Villemonais, 2016).

Three paths beginning at time zero make the distinction precise. On the first, the value at time t is t minus one. It reaches zero at time one and changes from negative to positive. That passage is a zero crossing. On the second, the value is the square of t minus one. It also reaches zero at time one, but stays nonnegative on both sides. It touches the target and turns back. The hitting time is the same, while the behavior around it differs.

On a third path, the value is one minus t until time one and zero thereafter. The path reaches the same target at the same time, then remains there. This illustrates absorption at zero. Recording only the time of first arrival would give the same number for all three constructions. Recording the surrounding path would reveal three different relations to the target.

The event definition determines which information is needed. A first-hitting question asks when the path first enters the set. A crossing question also asks how its sign changes. An absorption question concerns what happens after entry. The distinctions become useful whenever a proposed route attributes explanatory importance to arriving at a boundary. Its account can specify the event and the continuing dynamics separately, so observations can address both parts.

A third branch allows endpoint atoms. An atom assigns positive probability to an exact endpoint time; a narrow peak in a continuous density is not an atom. A stipulated endpoint at a preset time supplies a simple mathematical example. An atom

changes the random-endpoint theorem's conditions, but it does not automatically create exact closure. The route must still explain what happens at and around that time and justify the point mass independently.

A fourth branch uses global conditioning. Doob transformations, reciprocal processes, and Schrödinger bridge formulations describe whole paths subject to terminal conditions, reshaping earlier probabilities across time (Doob, 1957; Léonard, 2014; Garg, Zhang, & Zhou, 2024). Mathematical possibility becomes scientific standing only when the bridge predicts held-out features not guaranteed by the terminal condition itself.

A fifth branch changes the stream boundary. Apparent endpoint failure may show that the qualifying stream continued, paused, split, fused, or routed differently from the protocol assumption. A valid correction needs an independently motivated bearer rule, ownership across the transition, and observations that could show continuation did not occur.

A sixth branch narrows the reference class. Exact closure may fail universally yet remain open for a prospectively justified class of streams, continuity profiles, endpoint regimes, or observation architectures. The narrower claim must expose itself to new cases and counterexamples; it cannot become a moving refuge for failed observations.

Follow-up can change through transfer, sedation, sudden death, withdrawal, technical failure, or family refusal, each potentially related to the states generating the trajectory. Depending on the estimand, a change may be a competing event, censoring, or a change in observation; these roles must be distinguished. Cause-specific hazards and cumulative incidence are distinct inferential targets (Fine & Gray, 1999; Andersen et

al., 2012). Transfer with continued contact differs from a sensor outage of unknown duration.

An endpoint matrix should list each branch’s independent rule, predicted observations, strongest mimics, retirement result, and next observation distinguishing it from a close alternative.

7.2.4 — Two Ways a Record Can Finish

A record can finish in more than one way, and the alternatives change the meaning of its ending statistics. Consider a simulator of a task performed at one station. During each tick, a task still being followed has probability one fifth of being completed, probability three tenths of moving to another station, and probability one half of continuing at the original station. Completion and transfer each end the original record. They are competing risks: either event removes the record from further observation at that station. The three outcomes are mutually exclusive, and the same probabilities apply at every tick that the task reaches.

The completion probability among tasks still at the station is one fifth per tick. This is the discrete analogue of a cause-specific hazard: it describes one event among records that remain eligible for it. To find the proportion of the original tasks completed by a particular time, we must also account for how many tasks remain at the station long enough to have another chance of completion.

At the first tick, one fifth of the original tasks complete. One half remain eligible for the second tick, so a further one tenth of the original group completes then. At the third tick, one quarter remain eligible and a further one twentieth completes. The cumulative proportion completed by three ticks is therefore 0.20 + 0.10 + 0.05, or 0.35. This is a cumulative incidence: the

probability that the named event has occurred by the specified time, in the presence of the other way the record can end.

The corresponding transfer probabilities are 0.30, 0.15, and 0.075. By three ticks, 0.525 of the original tasks have transferred and 0.125 remain at the station. The three proportions, 0.35 completed, 0.525 transferred, and 0.125 continuing, add to one. Keeping them together gives a complete account of what the simulator has done with the original group.

Now change only the transfer probability. At each eligible tick, let completion retain probability one fifth, transfer have probability one tenth, and continuation have probability seven tenths. Completion still has exactly the same chance among the tasks that reach a tick. More tasks now remain available for later ticks. The first three completion contributions become 0.20, 0.14, and 0.098, which sum to 0.438. Completion by three ticks has become more common even though its cause-specific probability has stayed fixed.

The difference persists over complete runs. In the first simulator, eventual completion has probability 0.20 times the sum of 1, 0.5, 0.5 squared, and all subsequent powers. That sum is two, giving eventual completion probability 0.40. The second simulator uses successive powers of 0.7 instead. Their sum is one divided by 0.3, giving eventual completion probability two thirds. A run in either simulator leaves the original station with probability one, but a different fraction leaves through completion.

This makes the opportunity structure visible. The first mechanism gives each surviving record the same completion chance as the second. It also removes more records through transfer before further chances can occur. The ultimate fraction completed therefore describes the completion rule together with

the competing transfer rule. A single proportion at the end combines both processes.

Follow-up after transfer would create a different observation design. A transferred task could continue at the second station, where its eventual completion might be recorded. Transfer would then mark a change of observation setting within the larger record. The earlier station-specific calculation would remain useful for understanding that station, while the linked record would answer a broader question. The definition of the event and the domain being followed determine which calculation fits.

The simulator also permits a precise intervention calculation. Set the transfer probability to zero while retaining completion probability one fifth at every tick. Continuation now has probability four fifths. Completion by three ticks has probability 0.20 + 0.16 + 0.128, or 0.488, and eventual completion has probability one. This prediction follows because the stipulated change preserves the completion mechanism and supplies additional opportunities for it to operate. A model that also changes task difficulty or the completion mechanism would generate its own prediction.

For endpoint research, these calculations give every reported rate a definite question. How frequently does an event occur among records still eligible for it? What fraction of the original group reaches that event by a given time? What happens after an alternative event changes the setting? The answers can fit together without being numerically interchangeable. Following the paths through their different endings reveals the structure that a single terminal percentage compresses.

7.3 — Branches, Layers, and Non-Equivalent Claims

The additive branch is the simplest and strongest public form of PTVC. It assigns signed contributions to moments of felt experience. It accumulates positive and negative contributions within one stream on a common scale. Other representations may be proposed, but their claims cannot silently replace this one.

7.3.1 — Representation Models and Error Propagation

The terminal target is a zero signed residual.

That simplicity has a price. The branch needs a defensible neutral origin, a stable sign convention, enough scalar structure for addition, a rule for experienced exposure, and finite total contribution over the complete domain. If one prerequisite fails, the terminal equality is not merely hard to estimate. The additive claim itself may be undefined for that case (Krantz et al., 1971; Michell, 1999).

Additivity says that the contributions assigned to disjoint portions of one realized stream can be added to recover the contribution assigned to their union. It does not require an experience to have the same contribution in every context. Relief on entering a safe room may depend on the fear that came before; that history can already be reflected in the contribution assigned to the relief. Changing the order of events can therefore change the contributions while their accumulation remains additive.

The test is whether partitioning the same realized history in different ways preserves the represented total once context has been taken into account. If the whole requires an additional interaction that those assigned contributions cannot preserve, the proposed representation needs a different composition rule. This distinction lets the additive branch face the richness of

experience without assuming that every event has a fixed, context-free price.

Rough stochastic routes can differ from accumulation generated by an integrable local rate. Section 7.5 uses path texture as a candidate discriminator and states the measurement cautions.

The bivariate branch preserves positive and negative experience as distinguishable components rather than forcing every mixed moment into one net number. Treating the components separately can preserve information that momentary netting would erase (Cacioppo & Berntson, 1994).

Consider two invented records, each divided into two intervals of equal exposure. In the first record, the positive component has magnitude one in each interval and the negative component also has magnitude one in each interval. In the second record, the first interval contains positive magnitude two and no negative component; the second contains negative magnitude two and no positive component. Both records have total positive contribution two, total negative contribution two, and signed total zero.

The first record represents co-occurrence throughout. The second represents succession: one component appears, then the other. Even the two gross totals cannot distinguish them. To recover this difference, the representation must preserve which positive and negative values belong to the same interval. The joint temporal record contains information that disappears when each component is summed separately.

One simple descriptive statistic makes the difference visible. Count the fraction of exposure during which both components are strictly positive. In the first invented record that fraction is one; in the second it is zero. A different statistic could describe how often one component is followed by the other. These

statistics ask about organization within the course, so their answers can differ even when every lifetime component total agrees.

Temporal resolution is part of the description. If the second record is compressed into a single observation covering both intervals, its average positive and negative components are both one. Those two averages match the values reported throughout the first record. A coarse summary can therefore make alternating states resemble a simultaneous mixed state. Finer observations, or a report that explicitly distinguishes stages, give the representation another opportunity to separate them.

This distinction also gives the broader branch a precise scientific role. An additive terminal condition addresses the signed total. A bivariate model can additionally ask whether the components coexist, alternate, cluster, or change their relation across an event. A study interested in mixed experience can choose a target involving that joint pattern, then test whether retaining it improves the relevant prediction. The extra dimension earns its place through the question it allows the model to answer.

Generalized Terminal Neutrality names the broader family of terminal-zero conditions. A generalized branch begins when scalar addition no longer preserves the relevant structure. It may use a vector, a family of partially ordered states, a nonadditive functional, or a terminal set. For example, pleasantness, distress, arousal, agency, and bodily burden could remain separate dimensions whose transformation and terminal relation must be declared rather than collapsed into an arbitrary origin.

These branches need explicit composition or comparison rules and a terminal relation stated in their own language. “Equals zero” then takes its meaning from the declared branch.

The shape of a terminal set matters as much as its name. Suppose a generalized model retains two signed coordinates, x and y, each with its own declared origin and unit. These are coordinates in an illustrative representation. A terminal condition on x + y selects every pair lying on the line y = −x. A terminal condition requiring x and y separately to be zero selects only the origin.

The pair (2, −2) satisfies the first condition. So do (−5, 5) and (0, 0). Their scalar sums agree while the represented coordinates differ. The scalar condition preserves one comparison among the coordinates; it leaves movement along its zero line unrestricted. That freedom is part of the chosen target.

Change the scalarization to x + 2y. The pair (2, −2) now gives −2, while the pair (2, −1) gives zero. Neither arithmetic operation is inherently the right one for the represented experience. The model must supply the reason for its combination rule. The geometry shows exactly how the chosen rule changes which terminal states qualify.

Two independently justified contrasts can identify a stronger relation. If both x + y and x + 2y equal zero for the same terminal state, subtracting the first equation from the second gives y = 0, and then x = 0. The two zero lines meet only at the origin. Repeating the first contrast with all its weights doubled would supply no such gain: 2x + 2y contains the same restriction as x + y.

Approximate contrasts also have an explicit implication. If each of x + y and x + 2y has magnitude at most ε, then y has magnitude at most 2ε by subtraction. Since x equals twice the first contrast minus the second, x has magnitude at most 3ε. The pair is confined to a small parallelogram rather than exactly to the origin. The bounds follow from the joint validity of the two contrast restrictions.

The angle between the contrasts influences precision. If the second is x + 1.01y, subtracting the first leaves only 0.01y. The same uncertainty in the two measured contrasts permits a much larger range for y. Two nearly parallel restrictions are formally different while contributing little practical separation at a fixed error scale. A well-designed observation seeks relations that distinguish the dimensions sufficiently for its target.

Scalarization is the bridge from a richer state to one signed quantity. It may be useful, but it must be earned. Weights selected because they produce terminal closure would manufacture the result. They should instead be justified by prospectively specified theory, calibration, held-out prediction, and sensitivity analysis that reveals what changes when the scalarization changes (Bollen, 1989; Borsboom, 2005).

Representation errors propagate downstream. If a scale confuses low arousal with neutrality, sedation can appear as balance. If positive and negative affect are forced onto one signed axis when they can coexist, mixed states may be erased. If cultural response styles shift the apparent midpoint, comparisons can inherit an arbitrary origin. These are not minor measurement imperfections. They change the object being accumulated (Cacioppo & Berntson, 1994; Barrett, 2006; Mauss & Robinson, 2009).

7.3.2 — Modal Claims Are Not Interchangeable

Modal language states how widely and under what conditions a claim holds. Every, almost surely, possible, invariant, stable, reachable, and supported are logical commitments, not rhetorical emphasis. They determine the relevant cases, counterexamples, and evidence.

A universal claim covers every qualifying stream in its declared domain; one counterexample verified against the eligibility and measurement prerequisites defeats it.

A probability-one claim is relative to a probability law. An idealized example makes the distinction visible. Choose a point uniformly along a continuous line segment. The exact midpoint has probability zero, yet it remains one of the points in the model. The statement that the chosen point is not the midpoint therefore holds with probability one, but not at every point.

For PTVC, probability-one closure likewise allows a set of exceptional paths with probability zero under the declared law; a set-universal claim allows no qualifying exception. Changing the reference law, conditioning information, or sample space can change which paths receive probability zero. The two claims must therefore remain distinct (Billingsley, 1999).

The reachability ladder distinguishes existence of one closing continuation, positive-probability completion, fixed-policy probability-one completion, almost-sure attainability by some policy, limit-sure attainability, support containment, and closure of every admissible continuation. Almost-sure attainability requires one policy that reaches the target with probability one; limit-sure attainability may supply only a sequence whose values approach one. Under partial observation, one policy must also act consistently across indistinguishable states (Chatterjee, Chmelík, & Davies, 2016; Asadi et al., 2025).

Conditional support concerns a probability law after specified information has been given. A support claim asks which states or paths remain approximable by positive-probability neighborhoods under a declared topology; an empty finite-sample bin proves neither impossibility nor absence from support.

Extending finite observations to a universal verdict requires explicit sampling, measurement, and theorem-level bridges (Mayo, 2018; Popper, 1959).

Observed stability and intervention stability are different objects. A pattern may recur while treatment, environment, or information remain unchanged, yet fail under a specified intervention; even success under one intervention does not establish stability under every intervention. Causal language therefore requires a defined change, a model of what it acts on, and a clear account of which outcomes it could alter (Pearl, 2009; Woodward, 2003).

Action-mediated closure makes available actions and their consequences part of the proposed route to the terminal target. Adaptive policies add another quantifier. A fixed schedule specifies actions in advance, while a policy chooses actions in response to information. A relation that holds under one schedule may fail when decisions adapt to the observed state. Control and reinforcement-learning language makes this difference explicit (Sutton & Barto, 2018).

Composition, scale transfer, and counterfactual claims each need a bridge. A property on two intervals need not survive their union if bearer, endpoint, origin, or routing changes, and a lifetime relation does not imply the same relation for each day, organ, network, or group. Claims about another life history require an intervention model for unobserved outcomes.

7.3.3 — A Relation That Appears Only in the Whole

A relation among a whole sequence need not appear in any pair of its entries. This can be demonstrated with three signs. Draw X_1 and X_2 independently, each taking values plus one and minus one with equal probability. Then set $X_3 = X_1X_2$. The first two draws are free; the third is determined by their product.

Four sequences are possible: (+1, +1, +1), (+1, −1, −1), (−1, +1, −1), and (−1, −1, +1). Each has probability one quarter. Every individual sign is equally likely to be positive or negative. Looking at any one position gives exactly the same distribution as a fair independent sign.

Every pair is also independent. For X_1 and X_3, the four possible pairs occur once each among the four sequences. The same is true for X_2 and X_3, and independence of X_1 and X_2 was part of the construction. Thus each pair has zero correlation and the full joint distribution of two independent fair signs. This includes the pair separated by the middle position.

The triple nevertheless obeys an exact relation. Its product $X_1X_2X_3$ is always plus one, because it equals $(X_1X_2)^2$. A rival model drawing all three signs independently also has fair single-sign distributions and independent pair distributions, but its triple product is equally likely to be plus or minus one. The models agree at every level up to pairs and disagree at the level of all three.

The distinction is higher-order dependence. Knowing X_1 alone tells us nothing about X_3; knowing X_2 alone also tells us nothing about X_3. Knowing both determines X_3 completely. Information about the last sign is present in the relation between the first two. A method examining only separate associations cannot recover that information from this construction.

The example scales. Draw n − 1 independent fair signs and choose the nth so that the product of all n is plus one. Any set of fewer than n positions has the same joint distribution as independent fair signs. Once all but one are known, however, the last sign is determined. The whole-sequence relation can therefore coexist with complete independence of every smaller selected set.

One way to verify the smaller-set statement is to count completions. Fix values for k selected positions, where k is less than n. The remaining n − k positions must satisfy one product condition. Exactly half of their 2^(n−k) possible assignments do so. Every selected k-sign pattern consequently has the same number of completions and probability 2^(−k), which is the independent distribution.

The relation in this example is multiplicative. It is not the additive terminal equality proposed on the primary PTVC branch. Its role is to clarify how the choice of a complete relation determines the statistic that can reveal it. A product condition calls for a product-sensitive comparison. An additive condition has a different geometry and different implications for the distributions of sums and partial sums.

Suppose a simulator comparison used only the mean at each position and the correlation between every pair. It would report a tie between the constrained triple and the fully independent rival, even with arbitrarily much data. Add the triple product to the comparison, and the predictions separate immediately: it is always plus one under the first model and has mean zero under the rival. The new statistic addresses information absent from the earlier summaries.

This gives candidate route design a productive question. Which observable relations must change if the proposed whole-history condition is present? A route may predict a pairwise covariance, a conditional response, a higher-order interaction, or a distribution of complete totals. Matching the analysis to that prediction can be more useful than collecting many additional observations of a summary on which the models agree.

A whole can thus contain a precise relation that each of its smaller views leaves unresolved. The example does not make that relation mysterious: four sequences and one product display

it completely. It shows why formal specification can open new empirical questions. Once the proposed structure is exact, the investigator can identify which view of the data would make it visible.

7.3.4 — Approaching Certainty and Attaining It

A sequence of increasingly successful choices need not contain a perfectly successful choice. Imagine a preparation task in which the available policies are labeled by positive integers. Choosing policy n requires n units of preparation and gives completion probability one minus two to the power minus n. Policy one succeeds half the time. Policy two succeeds three quarters of the time. Policy ten succeeds with probability 1,023 out of 1,024.

For any positive tolerance, some finite policy makes the failure probability smaller than that tolerance. To make failure less than one in a million, policy twenty suffices because two to the power twenty exceeds a million. The family can approach certainty as closely as desired. This is the structure behind limit-sure attainability.

Nevertheless, every available policy retains a positive failure probability. There is no finite integer n for which two to the power minus n equals zero. The supremum of the success probabilities is one, but no policy in the stated family attains it. Almost-sure attainability would require an available policy whose success probability is exactly one under the specified model.

The wording reveals the difference. In the first statement, for every desired tolerance there exists a suitable policy. The policy may change when the tolerance changes. In the second, there exists one policy that succeeds with probability one. Moving the phrase about existence ahead of the tolerance requirement

changes the proposition. The two statements describe different capabilities of the available policy class.

An additional policy labeled infinity would define a different task. The original policies each perform a finite preparation and then attempt completion. An endless preparation has no attempt time under those rules. To admit it, the model would need a separate definition of what the policy does, when completion occurs, and how its success is evaluated. A limiting value of a calculation does not by itself construct the corresponding action.

A policy mixture can randomize among the finite policies. Suppose a lottery selects a positive integer n and then performs that policy. Each selected policy has positive failure probability. The lottery's failure probability is the weighted average of those positive numbers. Since the lottery places positive weight somewhere on the positive integers, that average remains positive. The lottery may be exceptionally reliable while still falling short of probability one.

A resource bound makes the distinction practical within the toy model. If preparation is limited to ten units, the largest available success probability is 1,023 out of 1,024. If the allowance grows, that best value rises. An account of remaining capacity can therefore specify both the available policies and the best success probability they permit. This gives changing opportunity a precise mathematical role without treating a limit as a completed mechanism.

The result helps organize candidate routes. A route may offer a good fixed policy, a family that approaches certainty, or a policy that actually attains certainty. Each is an interesting achievement. Naming the achievement accurately lets the next question be precise: enlarge the policy family, improve information, change the resource constraint, or demonstrate an attaining construction. The reachability ladder then becomes a

set of distinct research problems that can be worked on one at a time.

7.3.5 — Arrival and the Time It Takes

Even certainty of eventual arrival leaves a question about time. Consider a walk on the nonnegative integers, starting at one. At each step above zero, an independent fair coin moves the position up by one or down by one. The walk stops when it first reaches zero. Every step takes one unit of time.

Half the walks finish on the first step. Some move upward first and return later. For example, the sequence up, down, down reaches zero on step three and has probability one eighth. Longer excursions can travel farther before returning. The question is how these possibilities combine across the entire future.

An auxiliary upper boundary makes the answer approachable. Choose an integer N greater than one and stop the walk when it reaches either zero or N. If p(i) is the probability of reaching N first from position i, it satisfies p(0) = 0, p(N) = 1, and p(i) = [p(i − 1) + p(i + 1)]/2. The solution is p(i) = i/N. From the starting position one, the chance of reaching zero before N is therefore 1 − 1/N.

Every walk that reaches zero before N also reaches zero in the original task without an upper boundary. The original arrival probability is consequently at least 1 − 1/N for every N. Taking larger and larger boundaries makes this lower bound approach one. The walk reaches zero with probability one.

Now calculate how long the auxiliary task takes. Let e(i) be its expected number of steps from i. Its boundary values are zero, and inside the interval it satisfies e(i) = 1 + [e(i − 1) + e(i + 1)]/2. The solution is i(N − i). Starting from one, the expected duration is N − 1.

The original walk cannot reach zero earlier than it reaches either zero or the auxiliary upper boundary. Its expected arrival time is therefore at least $N - 1$ for every choice of N. No finite number is at least all of these values. The original expected arrival time is infinite, even though arrival itself occurs with probability one.

This does not assign an infinite duration to each realized walk. Almost every walk finishes after some finite number of steps. The expectation is infinite because the distribution includes sufficiently substantial chances of very long excursions. A typical short run, a fixed high percentile, and the mean duration describe different aspects of that distribution.

A finite deadline creates another target. Arrival by step one has probability one half. Arrival by step three has probability five eighths, adding the one-eighth probability of the three-step route. More time increases that probability toward one, but every finite deadline still leaves some walks unfinished. For instance, consecutive upward moves always have positive probability over a finite number of steps.

The construction adds a timing question to the reachability ladder. A route can establish eventual arrival, arrival within a fixed horizon, a finite expected duration, or a bound on a high percentile of duration. Those achievements require different arguments. A model of remaining opportunity becomes more informative when it supplies the timing property its application actually needs, alongside the probability of reaching the target at all.

7.4 — Latent Objects, Observations, and Estimates

The cumulative object combines the accrual rule in Section 7.1 with the felt-contribution representation of Section 7.3. Neither a current first-person report nor an institutional estimate can directly reveal the complete cumulative history or its accrual rule. Measurements are estimates obtained through an observation model. A successful test must distinguish change in the latent object from change in what a sensor or report can reveal.

7.4.1 — From Momentary to Accumulated Estimates

A measured record begins with sampled moments, while the question concerns a trajectory. Building an accumulated estimate therefore requires more than adding questionnaire values.

An estimation system must state how its observations relate to its target. A reflective model treats observations as effects of a latent state. A formative index constructs a composite from chosen components. Predictive evaluation asks what a model forecasts, while a dynamic model represents change over time. These choices can overlap: a dynamic latent-state model may also be evaluated predictively. Each use still needs justified links and weights; a common output sign does not make weights transferable (Bollen, 1989; Borsboom, 2005).

Use an independently justified origin, as required in Section 4.2.2. Calibration should use protocol-fixed tasks, repeated neutral judgments, cross-sign anchors, and sensitivity analysis showing how conclusions change when the origin moves.

The intervals represented by observations must also fit together. Consider an invented four-hour record whose hourly contributions are plus one, plus two, minus one, and plus three. Its full total is five. Now imagine two instruments that report

exact totals over different windows. The first covers hours one through three and reports two. The second covers hours two through four and reports four.

Adding the two reports gives six. Both instruments are accurate. The discrepancy arises because their windows overlap: the second and third hours have been counted twice. Their combined contribution is one, so subtracting that overlap once gives the correct full total of five. The correction concerns the geometry of the observation windows, before any question of measurement noise arises.

Knowing the two broad-window totals alone is insufficient to perform that correction. They tell us that the first three hours total two and the last three total four, but they do not identify the middle pair's contribution. For example, hourly contributions of two, zero, zero, and four produce the same two reported window totals while the full record totals six. A separate observation of the overlap, or another justified relation among the hours, supplies information the two broad reports lack.

Nonoverlapping intervals offer a simpler route when they can be measured. A report for hours one and two and another for hours three and four can be added directly. Cumulative readings offer a different route: subtracting the reading at the start of an interval from the reading at its end gives that interval's increment, provided both readings use the same accumulated quantity and origin. The estimator's operation follows what each record means.

The same issue appears when a momentary prompt, a daily summary, and a weekly recollection are assembled. Their dates can be distinct while the experience they refer to overlaps. A record of when someone answered therefore needs to be distinguished from the interval about which they answered. A

timestamp locates the act of reporting; the reference window locates the represented history.

A useful reconstruction preserves those windows before combining their contents. It can use a broad report to check a sum of narrower reports, or use the narrow reports to locate a change inside the broad one. That comparison may reveal a difference between momentary and retrospective description. It becomes informative because the analyst knows which portions of the history are shared and which portions add genuinely new coverage.

7.4.2 — Error and Accumulated Uncertainty

Observation error includes sensor noise, report variability, device placement, preprocessing, interpolation, and dropout. Some variation is approximately random; other errors are systematic and state dependent. A larger sample can make an estimate more precise while leaving systematic bias from medication, disability, disease severity, culture, or care setting intact.

Channels can also share error. Two devices may share a clock or preprocessing artifact; self-report and family report may share the same social script. Convergence is persuasive only when the channels have meaningfully different bias structures and their relation was specified before the result (Campbell & Fiske, 1959).

An accumulated estimate gives small errors a long time to act. Consider an invented record covering one hundred equal units of time. A rate is observed once in each unit and multiplied by that unit's duration. Suppose each observation contains an independent, zero-mean error with standard deviation one tenth. The variance of the accumulated error is the sum of one hundred variances of one hundredth. Its standard deviation is therefore

one. Individual errors often cancel, but the uncertainty in their sum remains.

Now give every observation the same unknown offset of one tenth. Across one hundred units of time, that offset contributes ten units to the accumulated estimate. Repeating the measurement does not turn the offset into a sequence of independent errors. Each repetition carries the same displacement. If the offset is uncertain rather than fixed, the uncertainty about it travels through the whole duration in the same way.

Denser sampling changes the first calculation. Keep the total duration at one hundred units, but take four hundred equally spaced measurements. Each measurement now represents one quarter of a time unit. Under the same independent-error assumption, the standard deviation of the accumulated sampling error becomes one half. The persistent offset still contributes ten. The smaller sampling intervals reduce the contribution of independent observation noise while leaving the shared displacement unchanged.

This comparison distinguishes two meanings of more data. Extending a history adds new exposure to be accumulated. Sampling an existing interval more densely divides that exposure into smaller pieces. The two operations need different uncertainty calculations. A longer record can improve knowledge of a stable process while also adding uncertainty to the larger accumulated quantity being estimated.

Dependence creates intermediate cases. Suppose nearby observations share a slowly changing calibration error. Their errors reinforce one another over part of the record, even if observations far apart are nearly independent. The uncertainty of the sum then includes the covariance of errors at different times, as well as their separate variances. Positive covariance increases

it; negative covariance can reduce it. The relevant structure is the relation among errors, not simply the number of measurements.

A block example locates an intermediate case exactly. Divide the hundred time units into ten blocks of ten. Within each block, let every observation share the same zero-mean calibration offset with standard deviation one tenth. Let the ten block offsets be independent. Each block then contributes an accumulated error with standard deviation one. The sum of ten independent block errors has variance ten and standard deviation the square root of ten, about 3.16.

Compare the three results under their stated models: independent errors at each time unit give an accumulated standard deviation of one; independent offsets lasting ten units give about 3.16; one uncertain offset lasting all hundred units gives ten. The marginal standard deviation of an individual observation error is one tenth in every case. What changes is how long the same error persists.

This directs calibration toward a particular timescale. Measurements that can establish whether an offset resets between sessions, days, or device changes reveal how many effectively independent blocks the record contains. Denser observations within a block describe the process more closely, while information about the block offset addresses the component those observations share. The two improvements enter different parts of the accumulated uncertainty.

7.5 — Reachability, Reserve, and Predictions

Reserve coupling makes reachability a question about which future paths remain possible given the current state, remaining capacity, and time. To test a route against the phenomenal target, the model must say how its state represents the cumulative object and how its reserve boundary relates to the independently defined endpoint. A local controller, if proposed, must also have a defensible source of information about those quantities. Without these bridges, a shrinking state variable may indicate fatigue, illness, or missing observation without establishing terminal closure.

7.5.1 — Reachability and Remaining Capacity

In viability theory, the viability kernel is the set of states from which at least one admissible continuation can respect the declared constraints. In ordinary language, it is the region from which the future remains workable. The concept allows many possible paths and asks how that workable region changes as state and time change (Aubin, 1991; Aubin, Bayen, & Saint-Pierre, 2011).

Remaining capacity is whatever the candidate route needs in order to keep relevant futures available. Depending on the model, that may involve time, biological plasticity, tolerance for arousal, access to positive or negative experience, social support, action freedom, or another state variable. No single definition is inherited by every route. Some candidate routes require no reserve variable.

The continuation set depends on the bearer’s resources, endpoint and clock, admissible constraints, and intervention regime. Capacity may differ by the sign of the required remainder; an empty admissible set makes completion through that route impossible.

The reserve bound in the formulation developed here captures a general intuition: the magnitude of an unresolved state cannot exceed the capacity of the remaining future to answer it, if the route is to remain viable.

Joint possibilities matter more than separate ranges. In an invented two-state model, the only admissible pairs of past total and complete remaining contribution are (1, −2) and (2, −1). The final total must be −1 or +1. Looking only at the separate intervals [1, 2] and [−2, −1] produces the looser range [−1, 1], which includes zero by combining values that never belong to the same admissible case.

A route-specific test should therefore preserve which remainder can accompany which state. Map each independently justified joint possibility to its past-plus-remainder total, rather than treating every marginal combination as attainable. A truth-containing joint set whose image excludes zero supplies an informative challenge under its assumptions. Zero inside a conservative outer bound leaves the question unresolved; zero inside an exact attainable set identifies a possible total, not a guarantee that the actual continuation selects it (Manski, 2003).

Reachability also interacts with uncertainty. A bearer or controller, if any local control exists, may not know the true state or endpoint time. A route that requires perfect knowledge of the cumulative object is less plausible than one that operates through observable proxies, but proxy control can introduce error and overshoot. Partial observability is part of the mechanism problem (Bertsekas, 1995; Kaelbling, Littman, & Cassandra, 1998).

A reserve variable can summarize some dimensions of remaining capacity, but compression has a cost. Two people with the same scalar reserve may have different options because one has social support and another has physical health, or

because their capacities are not substitutable. A route that uses one number must show that the compression preserves the predictions at issue.

A workable region can sometimes be found together with the rule that keeps a system inside it. Consider a numerical reservoir whose permitted levels are the integers from zero through four. At each step, a controller may add either zero or one unit. A withdrawal then removes either zero or one unit. The controller sees the current level and chooses its addition before learning which withdrawal will occur. Its task is to keep the level, after each withdrawal, at either one or two.

At level one, adding one is a successful action for either withdrawal. The level first becomes two; it then remains two or falls to one. At level two, adding zero succeeds in the same way. These two choices define a feedback policy: add one when the current level is one, and add zero when it is two. Every allowed withdrawal leaves the system in a state where the policy can operate again.

The set consisting of one and two is therefore invariant under that policy and the declared disturbances. This statement concerns every sequence of zero-or-one withdrawals, including long runs of either kind. The proof repeats the same one-step fact: each action sends every possible next state back into the set. Starting within it, the policy can maintain the range for any finite number of steps.

A fixed schedule of additions does not have the same property. From level one, always adding zero eventually reaches zero if a withdrawal occurs. Always adding one eventually exceeds two if withdrawals repeatedly remain zero. The successful policy coordinates its action with the observed state. Its achievement depends jointly on the available additions, the observed level, and the bound on the withdrawals.

Work backward from the desired next-step set to identify the states that admit such an action. From level zero, even the largest available addition is followed by a possible withdrawal that returns the level to zero. From level three, even adding nothing is followed by a possible zero withdrawal that leaves the level at three. Level four has the same difficulty. Only levels one and two have an action whose every admitted outcome lies in the desired set.

This backward step recovers the same set with which it began. It has reached a fixed point: the states that permit one more guaranteed admissible step are exactly the states that permit continued operation. In a more complicated finite model, repeated backward steps can remove states until this stable region is found. The resulting set comes with the actions that sustain it.

Individual favorable trajectories can begin outside that region. Starting at zero, adding one and receiving no withdrawal enters level one. Starting at three, adding nothing and receiving a withdrawal enters level two. Those arrivals are possible, while the initial states do not offer the same guarantee against every admitted withdrawal. The distinction is visible in the transition rules rather than hidden inside a general word such as capacity.

7.5.2 — Reserve Coupling as One Candidate Route

One candidate reserve construction multiplies a nonnegative reserve by a signed reference coordinate. If reserve tends to zero while the coordinate is controlled so it cannot grow fast enough to cancel that shrinkage, the product tends to zero. The construction identifies how closure could occur inside a model. Connecting that modeled state to experience still requires the bridge developed below, including the rule that later change cannot rewrite the felt past.

Spread in reserve space and spread in clock time need not share an exponent. If reserve is deterministic on the declared clock and the reference coordinate’s standard deviation tends to a finite nonzero limit, a reserve that falls like a power of time remaining gives the modeled product’s standard deviation that same power and its variance twice that power. Random reserve or changing coordinate dispersion requires a separate derivation. The protocol must declare which quantity and clock are being fitted before terminal data are examined.

Past-fixity gives the product a local obligation. If reserves R and signed coordinate Z are absolutely continuous on the declared accrual clock, and RZ equals the accumulated path at every time, the felt-contribution rate must equal $R'Z + RZ'$ almost everywhere, with the initial cumulative value matched. A decrease then records new contributions rather than shrinking experience already lived. This change relation, finite total accumulation, and the endpoint relation need separate support. A terminal-only or approximate bridge makes different commitments.

Fine-scale path class offers a possible discriminator. A cumulative path built from an ordinary integrable rate has finite variation on compact intervals. A Brownian-driven product generally carries nonzero quadratic variation. Dense sampling could in principle distinguish those textures, but measurement noise can imitate roughness and interpolation can imitate smoothness (Folland, 1999; Protter, 2005).

The texture of a path can be studied by changing the size of the intervals used to examine it. Begin with the simplest smooth path: its value rises steadily from zero to one over one time unit. Divide the interval into ten equal pieces. Each increment is one tenth, so the sum of the squared increments is ten times one hundredth, or one tenth.

Use one hundred equal pieces instead. Each increment is one hundredth, and the squared-increment sum is one hundredth. In general, with n equal pieces, the sum is one divided by n. It tends to zero as the partition becomes finer, even though the path's total rise remains one. The squared-increment statistic distinguishes the size of local fluctuations from the net displacement.

A standard Brownian path has a different pattern. Over one time unit divided into n equal intervals, each increment has mean zero and variance one divided by n. Its expected square is therefore one divided by n, and the expected sum of all n squared increments is one. Refining the partition does not drive that expectation toward zero. The quadratic-variation result makes this into a precise limiting property under suitable refining partitions (Protter, 2005).

Observation noise produces a third pattern. Suppose an instrument observes the steady path but adds a fresh independent error at each sampled time. Each error has mean zero and variance one hundredth. The error in an observed increment is the difference between two consecutive errors, giving variance two hundredths. Summing the expected squared observed increments over n intervals gives the smooth contribution of one divided by n plus the noise contribution of two hundredths times n.

At ten intervals the expected sum is three tenths. At one hundred it is just over two. At one thousand it is just over twenty. Increasing the sampling rate now makes the observed path appear progressively rougher because more independent measurement errors enter the differences. Adjacent increment errors share a measurement, but linearity of expectation still gives the stated expected sum; independence of the increments is not required for that calculation.

The three patterns suggest a practical comparison across resolutions. A smooth contribution decreases under refinement. Brownian quadratic variation approaches a finite scale. Independent pointwise observation noise grows with the number of increments in this model. A real observation model can combine those components, and correlated noise or smoothing can change their resolution behavior.

This is why the sampling scale is part of the experiment. A single jagged plot offers little separation among the possibilities. Repeating the analysis at several declared resolutions, together with independent instrument characterization, can reveal which part of the apparent texture belongs to the modeled dynamics. The question becomes one about a pattern across scales that a candidate process must explain.

Frailty, sedation, and observation floors form another rival family. Severe illness can reduce movement, speech, autonomic variability, and willingness to report. A device may round small values or fail in low-perfusion states. The route survives only if the observation model distinguishes contraction in the latent target from contraction in the ability to observe it.

The route must be allowed to fail. Stable spread despite a well-measured reserve decline, a reversed lead-lag relation, contraction explained by sedation or sensor floor, or failure to transport across sites would count against the prospectively specified implementation.

Clock and regularity assumptions also matter. A path smooth in calendar time may look abrupt in subjective, physiological, or retrospectively aligned time, while hospitalization, treatment response, communication loss, or environmental change may alter a reserve proxy discontinuously. Any time change or smoothing rule needs declared transport assumptions rather than being used to preserve a preferred picture.

The factorization also creates an identification test. For positive reserve and a fixed product, multiplying reserve by a positive factor requires dividing the signed reference coordinate by that factor. Applying this relation to observations requires an independently calibrated reserve proxy. Observing only the product cannot confirm that relation; independent information about at least one factor is required.

7.5.3 — Accumulation Along a Rough Path

A state-change formula can depend on how changes are accumulated along a path. For a smooth trajectory, familiar calculus connects the derivative of a potential with its boundary difference. A path built from many small fluctuations invites a closer look. A discrete identity lets us see the relevant term before any limiting stochastic theory is introduced.

Let a state move from x_k to x_{k+1} by an increment Δx_k. Choose the potential $Q(x) = x^2/2$. Direct expansion gives $Q(x_{k+1}) - Q(x_k) = x_k\Delta x_k + (\Delta x_k)^2/2$. The first term uses the state at the beginning of the step. The second term is half the squared increment. Both are part of the exact change in potential.

Sum this identity over a sequence of steps. The potential differences telescope, giving $Q(x_final) - Q(x_start) = \Sigma x_k\Delta x_k + (1/2)\Sigma(\Delta x_k)^2$. Rearranging, the sum that weights each increment by its starting state equals the potential difference minus half the sum of squared increments. The relation is exact for every finite sequence.

For a smooth path sampled over a fixed duration, increments generally shrink in proportion to the step length. With n equal steps, a typical squared increment is then of order $1/n^2$, and their sum is of order $1/n$. The correction tends to zero. This is how the familiar smooth chain rule emerges from the finite identity.

Now consider a different sequence. Take n steps, each of magnitude $1/\sqrt{n}$, with either sign, over a total duration of one. Every squared increment is $1/n$, so their sum is exactly one. Refinement makes the individual steps smaller, but it does not make their accumulated squares vanish. The correction remains one half for every n.

For even n, choose any such sequence that starts at zero and has equal numbers of positive and negative steps. It returns to zero. Its potential difference is therefore zero. Nevertheless, $\Sigma x_k \Delta x_k$ equals minus one half. A closed state path can have a nonzero value under this particular starting-state accumulation rule. The missing half is visible in the squared-increment term.

A midpoint rule gives a different finite result. Weight each increment by the average of its starting and ending states, $(x_k + x_{k+1})/2$. Multiplying by Δx_k gives exactly $(x_{k+1}^2 - x_k^2)/2$. This sum telescopes directly to the potential difference. For the returning path it is zero at every resolution.

The two sums use the same ordered states and increments. They differ in where the state weight is evaluated within each step. For this quadratic potential, their difference is exactly half the sum of squared increments. On smooth paths the distinction disappears under refinement. On the constructed fluctuation scale it persists. Path regularity determines whether the finite convention leaves a limiting trace.

This elementary calculation is the structure behind the additional quadratic-variation term in stochastic chain rules. In a Brownian-driven model, a rule using the state available before an increment and a symmetric rule can lead to different integral conventions. The appropriate convention follows from the model's construction and the quantity being represented (Protter, 2005). The finite calculation above already shows why the choice matters.

For a proposed state-based route to closure, this supplies a concrete specification task. State the potential, the path class, and the accumulation rule that connects changes in state to the signed quantity of interest. If the path carries a nonvanishing sum of squared increments in the limit, a boundary-difference argument must include the corresponding term or justify a convention under which the desired identity holds.

The result can also guide simulation checks. A returning sequence of equal positive and negative steps has known answers under both conventions: minus one half for the starting-state sum and zero for the midpoint sum. A program can reproduce those values before its output is connected to a more complicated model. The test makes the relationship among roughness, accumulation, and closure inspectable through a calculation small enough to follow by hand.

7.5.4 — Observable Signatures, Rival Mimics, and Failure

The reserve construction offers several consequences to test together.

One candidate signature is preterminal compression in a declared dispersion measure. The protocol must distinguish dispersion across model realizations at a given time from variation within a time window of one observed stream and justify any bridge between them. It then tests whether the chosen measure changes with reserve as predicted, before the endpoint is known retrospectively.

A shrinking distribution across realizations and convergence along a realized path are different properties. A short construction shows why a route may need to establish both. At times indexed by $n = 2, 3, 4$, and so on, let a value X_n equal one with probability $1/n$ and zero otherwise. Make these draws

independent across n. At every later index, a snapshot is more likely to show zero.

The mean is 1/n and the variance is (1/n)(1 − 1/n). Both approach zero. For any tolerance smaller than one, the probability that the value exceeds that tolerance also approaches zero. This is convergence in probability to zero: increasingly late snapshots place increasingly little mass outside a neighborhood of zero.

Follow one entire sequence instead. For indices N + 1 through M, the probability that no one occurs is the product of 1 − 1/n over those indices. Each factor is (n − 1)/n, so the product telescopes to N/M. As M grows, this probability approaches zero. After any fixed index N, another one occurs with probability one.

There are only countably many possible indices N. With probability one, the sequence therefore has ones beyond every finite starting point. It also has zeros infinitely often. A realized sequence does not settle toward zero, despite the increasingly concentrated snapshot distributions. Its occasional unit excursions continue at arbitrarily late indices.

To place the construction near a finite boundary, assign index n to time 1 − 1/n. The observations then approach time one. Their marginal distributions concentrate, while almost every sampled path keeps making excursions of size one arbitrarily near that boundary. The construction does not have the pathwise limiting value that a terminal model might require.

Path regularity supplies a distinct piece of structure. Suppose a candidate separately establishes that its cumulative path has a limit at the boundary with probability one. If its values also converge in probability to zero, the pathwise limit must be zero with probability one. Uniqueness of a probabilistic limit then

connects the two achievements. The spike construction shows why the existence of that pathwise limit is doing real work.

On the primary finite signed-measure branch, finite positive and negative accumulated masses provide a route to a well-defined limiting cumulative total. A separate stochastic candidate must demonstrate that its own path properties fit the proposed bridge. Its marginal dispersion calculation and its complete-path regularity can then support each other in an explicit argument.

A second signature is concentration near an independently calibrated neutral origin. Session centering, participant-specific recentering, instrument floors, rounding, and post hoc anchor choice can create an apparent pile-up at zero. Neutrality is informative only if the terminal outcome did not set the origin.

A third signature family links reserve to clock time and two-time dependence. A route may fit spread against reserve yet fail its time-domain relation, while bridges or ordinary mean reversion imply different earlier-later covariance. Testing both helps expose rescaling and distinguish inward drift from its generator.

A fourth signature is perturbational response. For a safe, independently justified change in sleep, symptoms, communication, or environment, the route should predict which trajectory feature changes first, how long the effect lasts, and whether it depends on the earlier signed state.

A fifth signature tests the path-class distinction in Section 7.5.2 across sampling scales, with replicated channels and model recovery to separate dynamics from measurement artifacts.

A sixth signature is terminal-band compatibility at the protocol-fixed resolution defined in Subsection 7.2.1 (Schuirmann, 1987; Lakens, 2017).

A side-by-side prediction display should include full curves, calibration, uncertainty, and poor-support regions. It can reveal whether one model gains accuracy only where observations are unusually plentiful (Burnham & Anderson, 2002; Gneiting & Raftery, 2007).

Chapter 8 — Possible Routes to Closure

Knowing a proposed destination does not reveal which roads exist, what information is available along them or which obstacles can be removed.

A candidate possibility architecture asks where options are generated, represented, felt as selectable, endorsed, acted upon, carried across transitions, or constrained at a complete-history boundary. Technical compensability concerns a model's feasible continuations. Human repair also involves restitution, accountability, justice, and restored capability.

Biology and psychology already provide powerful component explanations: homeostatic and allostatic regulation, interoceptive prediction, salience, learning, inhibition, action selection, and mood dynamics shape responses to positive and negative states (Sterling, 2012; Barrett & Simmons, 2015; Eldar et al., 2016). A proposed additional PTVC mechanism must add a prediction those processes do not already supply.

Other routes work through endpoint rules or whole-history constraints. Bridges condition whole paths on a future boundary; hitting and absorption use declared state sets; global constraints restrict complete histories; corrected stream boundaries alter ownership. They may resemble one another near the endpoint while differing in information, causation, and testable consequences (Doob, 1957; Léonard, 2014; Chetrite & Touchette, 2015).

One explicit formulation uses partially observed constrained control. It asks whether distributed processes alter feasible actions as estimated burden, bodily reserve, channel availability, and expected horizon change. This is a candidate implementation of the public architecture, neither its definition nor an established feature of organisms. A local route cannot know the true endpoint unless it explains how endpoint-relevant information becomes available. It may instead use a filtered

estimate of remaining time built from prospectively specified channels. The horizon estimate must be separated from the desired cumulative result so future information is not leaked backward into the state.

Translate each proposed stage into measurable predictions about menu size, choice, reaction time, error, revaluation, channel switching, physiology, or neural control. Define those predictions independently of the story used to explain them.

Passive and active routes remain distinct. A shrinking reserve may compress movement without choice; active steering requires a state estimate, objective, policy, and admissible actions. They carry different causal and empirical burdens (Bertsekas, 1995; Sutton & Barto, 2018). A forecast for the next transition can be scored separately from its forecast for a later window, revealing whether the route's predictive value depends on timing.

Two routes can also exchange their apparent strengths as the question changes. A rule that generates admissible paths may be compact to describe but demanding to implement locally. A local mechanism may be easy to simulate step by step while making the complete-path property difficult to establish. Comparing them therefore involves both directions of reasoning: from a rule to the histories it permits, and from observed histories back toward the machinery that could have produced them. Following both directions makes the architecture a research problem with discoverable structure.

8.1 — Choice, Reachability, and Possibility

An option may be imaginable without being available. An action may be physically possible yet feel unselectable under pain or threat. A model's admissible continuation adds a third category: it is what the model allows, whether or not a person considers it. Testimony, behavior, and context are needed to distinguish these kinds of possibility.

8.1.1 — Thinkable Options and Separate Cumulative Objects

In the formal map, thinkable options are not yet the available action menu unless the candidate model includes cognitive representation as a menu condition. They are a phenomenological and psychological precursor. This distinction prevents the model from assuming that every externally listed action has entered the person’s decision space.

Thinkable options are shaped by education, language, culture, trauma, cognitive bandwidth, and what a person has seen others survive. Education, stories, public rights, example, and permission can widen that menu, while danger or exhaustion may reduce planning to the next hour. A consciously considered path must first become representable, but representation does not make it usable.

Public ethics begins by asking where an option disappeared. A system that advertises choice while making safer paths cognitively invisible shifts burden into the person and then blames the person for the shift. The better inquiry identifies the fear, information gap, dependency, or design barrier that keeps a route from becoming thinkable without coercion.

Felt selectability concerns whether a path feels available to act on with the biological capacity, safety, knowledge, time, and relational conditions actually present. The distinction recognizes

constraint without denying responsibility (Clark, 2013; Friston et al., 2017).

Admissibility is a modeling relation, not an ontology. Let the candidate menu contain available actions and let the admissible set contain only those that satisfy a prospectively declared feasibility rule under the model's current state, horizon, capacity, and channel limits. Changing those elements can change the set; that is a reason to specify the model, not to call every excluded path impossible in life.

Admissible does not mean good, right, legally permitted, or felt-selectable. A harmful action may remain feasible under a bad objective, while a worthy action may be excluded by a model that omitted shame, pain, threat, meaning, or access. Disagreement between the person's account and the model is diagnostic evidence of a missing state or poor definition, not permission to silence the person (Sutton & Barto, 2018).

If the admissible action set cannot be operationalized without reading the outcome back into the rule, this mechanism family fails its own discipline. Admissibility must be specified prospectively enough that a path can surprise the model. A family's finances or a hospital's staffing can constrain several lives at once. The shared cause is real, while each qualifying stream retains its own cumulative object.

Population evidence may reveal and compare shared burdens so they can be repaired, while any claim that the same condition also changed a modeled reserve remains a separate empirical question.

8.1.2 — Evidence Needed

Evidence for path architecture must combine levels without confusing them. Experience sampling can assess felt selectability. Behavioral tasks can measure option generation,

choice, latency, switching, and error. Environmental data can document which options actually existed. Physiological and neural measures can test candidate implementation.

A useful design separates the proposed determinants rather than relying on anecdotes. Horizon information, biological reserve proxies, channel availability, utility, conflict, arousal, fatigue, threat, and learned value should be varied or modeled separately where ethical and feasible. The pathway-specific prediction is an interaction structure, not the general fact that tired or frightened people choose differently (Munafò et al., 2017; Nosek et al., 2018).

Naturalistic studies add realism but increase confounding. Poverty, illness, medication, disability, caregiving, institutional delay, and social threat shape menus directly. These variables must be treated as causal conditions, not noise to be removed until a route-specific effect appears.

If ethically acceptable observations cannot separate horizon from prognosis and fear or establish reserve proxies, the mechanism is not presently testable in that domain.

For each construct, test whether its proposed measure captures more than distress, wealth, health, or verbal fluency (Cronbach & Meehl, 1955; Campbell & Fiske, 1959).

A low-burden study might alter information, support, delay, task difficulty, or access to an option and ask whether the model predicts the direction and timing of menu change. Compare its prediction with the ordinary learning or control account of the same intervention (Woodward, 2003; Hernán & Robins, 2020).

Missingness belongs inside the test rather than in a final limitations paragraph. Constraint may make a person less able to respond at exactly the moments the model considers most informative. Joint models, sensitivity analyses, and explicit denominator flow can reveal how much apparent menu

contraction is produced by disappearing observations (Diggle & Kenward, 1994; Little & Rubin, 2019).

Within-person designs can sharpen the test when they are safe and feasible. Repeated observations before and after a modest change in support, information, or task structure may reveal whether the same person moves as the route predicts.

Evidence should also preserve the person's reasons. A route may predict that one option becomes less selectable, while an interview reveals that the person deliberately rejected it to protect a child, honor a value, or avoid a foreseeable harm. Choice is not merely movement through a menu; meaning and obligation can alter what a model should count as feasible.

8.1.3 — Capacity Has a Direction

A large supply of effort is useful only through actions that can change the relevant state. Imagine a small drawing device with a pen on a coordinate plane. One control moves the pen one unit to the right and one unit upward together. The control can also run backward. Starting at the origin, any sequence of these movements leaves the pen on the diagonal: its horizontal and vertical coordinates remain equal.

The target is one unit to the right with no upward movement. Unlimited time and an unlimited number of operations do not put that target on the diagonal. The problem is not an insufficient quantity of movement. Every available movement preserves the difference between the two coordinates, and that difference begins at zero. The target requires the difference to become one.

Add a second control that moves the pen one unit right and one unit down. The target can now be reached by operating the first control for half a unit and the second for half a unit. The upward and downward changes cancel, while the horizontal

changes add to one. A second channel has changed which directions the system can move, rather than merely increasing the speed of an existing motion.

The calculation can be stated for any target (x, y). If the first control is used by amount a and the second by amount b, the final coordinates are x = a + b and y = a − b. Solving gives a = (x + y)/2 and b = (x − y)/2. When either control can run forward or backward without a magnitude limit, these settings reach every point on the plane. With only the first control, they reach only points for which x = y.

Bounds on the controls add a different restriction. If each setting must lie between minus one and one, a target is reachable exactly when both |x + y| and |x − y| are at most two. The attainable region is a diamond, with corners two units up, down, left and right. Doubling the permitted setting enlarges that diamond. Removing the second control collapses it back to a line, regardless of how far that line extends.

This makes two kinds of support distinguishable. An intervention can increase the available amount in an existing channel, or it can create an additional direction of change. In an ordinary project, buying more of the same material can solve a shortage. Access to a different tool can solve a different problem: it makes an operation possible that the original tool could never perform. Describing both changes as more resources loses the distinction that explains their effects.

The same reasoning helps interpret a model of reserve. A scalar reserve can summarize how much of a defined action remains available. When several state variables must change, the model also needs to say which actions influence which variables. A generous scalar budget can coexist with a missing channel. Conversely, a modest intervention in a previously

unavailable direction can make a target reachable with little additional expenditure.

For a proposed route to closure, this directs attention to the action map. Which state differences are preserved by every available action? Which combinations of changes can the current channels produce? Which support changes the range of attainable continuations? Those questions can yield distinct predictions even before a researcher estimates a single global capacity score. They also explain why a route may respond strongly to one practical improvement and hardly at all to another improvement of comparable size.

8.2 — Repair, Reserve, and Compensatory Capacity

Repair means a change that restores the viability of a route. Some changes expand the available range of actions, while others reduce a current burden. A reserve measure must say what capacity it counts, for whom, and over what horizon. Compensation is credible only when those changes could affect the model's predicted future, not merely its vocabulary.

8.2.1 — Compensability Defined

Compensability asks whether, under a declared model, a present burden remains answerable through safe and realistically available paths before the relevant horizon closes.

A clinical prognosis estimates disease or survival outcomes. It may inform a route proxy without identifying the truth-level consciousness endpoint or the signed phenomenal remainder that the route must accommodate.

The concept is also action-dependent. A safe intervention can widen the set of feasible continuations, leave it unchanged, or close paths through side effects and opportunity cost. A useful route model predicts those changes in advance.

Repair channels include clinical treatment, material support, legal remedy, restored trust, and time. Which channel is needed determines what counts as a feasible continuation.

Represent the transport, cost, language, safety, and delay that determine whether a nominal repair channel is usable (Aubin, 1991; Sen, 1999).

8.2.2 — Noncompensable Risk and Empirical Limits

Noncompensable risk names the possibility that, under specified conditions and uncertainty, an action may close essential paths faster than the remaining channels can reopen them.

The risk is channel-specific. Irreversible injury may close one physical path. A violent environment may close safety and trust. Treatment toxicity may narrow function while relieving disease. Imprisonment, displacement, or institutional delay may remove options that no later benefit restores. The model should name the lost channel rather than compress every danger into one total.

Severity alone cannot establish noncompensability. People recover unexpectedly, new treatment becomes available, law changes, relationships form, and unmodeled supports appear. A model should carry that uncertainty and state which possibilities were omitted rather than treating present knowledge as the boundary of a life.

Research thresholds can locate when a continuation becomes unlikely under a specified model. Their error costs should be compared with those of simpler established risk models (Aubin, Bayen, & Saint-Pierre, 2011).

False positives are especially dangerous. A biased model may classify an action as unreachable because disability, culture, poverty, atypical regulation, or communication style was measured poorly. That error can close a path the person could safely take. Calling the mistake conservative does not make it harmless.

False negatives matter too. A model may underestimate danger because it ignores cumulative fatigue, unstable housing, treatment interaction, coercion, or a bottleneck in one essential channel. Calibration must therefore assess both kinds of error rather than celebrate average accuracy.

Research visuals need particular restraint. A red risk band, depleted-reserve icon, or automated warning could be mistaken for a judgment that someone is beyond help. Any such visualization must state its bounded research purpose, uncertainty, nonclinical status, and correction route.

Compensability, channel state, and modeled reserve do not currently form a validated measurement system. Each remains a candidate construct with unresolved units, observation models, timescales, and identification problems.

Functional performance, symptom burden, autonomic flexibility, sleep, cognition, reported effort, social support, or material access may each carry information. Their relation to the candidate route must be tested rather than assumed (Ramsay & Woods, 2014; Sterling, 2012).

Less observed movement is not the same as less possible movement.

8.3 — Candidate Routes and Their Evaluation

Local feedback routes influence later choices through present or recent state, so interventions can probe them. Global routes instead restrict entire admissible histories and must identify what principle selects those histories. A biological candidate might combine online valuation and inhibitory control with autonomic and endocrine regulation, while sleep-related reorganization provides an additional channel. These levels are not interchangeable: insula or prefrontal activity alone cannot identify a cumulative object or a closure controller. A discriminating test needs prespecified tasks, temporal contrasts, and held-out predictions against ordinary regulation and other rivals.

8.3.1 — Neural and Psychological Correlates

A candidate biological route could involve interacting systems for salience, interoception, reward, threat, memory, action selection, autonomic regulation, and cognitive control, with different contributions across tasks and states (Seeley et al., 2007; Menon, 2011).

Interoception is one candidate input to that field. Breathlessness, pain, nausea, hunger, fatigue, heat, tremor, and heart racing can change the felt cost of acting before reflective thought supplies a narrative (Craig, 2009). The body can make an abstract option immediately livable or unlivable.

Interoceptive signals are not privileged truth channels. Disease, medication, expectation, attention, trauma, disability, training, and culture can alter both signal and meaning. A stronger route test asks whether interoceptive variables add protocol-fixed predictive value beyond these ordinary explanations (Barrett & Simmons, 2015).

Any local controller must use only information available under its declared filtration.

A system need not remember a history as a story for that history to affect what happens next. A candidate route might use a compact current state, updated as experience unfolds, instead of consulting an autobiographical archive. Biological persistence can also outlast individual component molecules. In cultured rat hippocampal slices, a CaMKII-dependent component of synaptic transmission remained detectable after two weeks while major calcium-entry routes were blocked. The authors interpreted this persistence, together with prior turnover measurements and a reactivation control, as evidence that a functional state can survive molecular replacement (Lee et al., 2022). A candidate route could therefore be sought in a maintained pattern rather than one permanent molecular component. The useful question is what that pattern preserves: a learned association, an adjustable response capacity, or information quantitatively tied to signed experience. Those alternatives require different assays; persistence alone does not select among them.

A partial-observation account may instead operate on beliefs. The system can estimate current threat, opportunity, bodily condition, or likely future, then choose under uncertainty. That is scientifically familiar, but many objectives can generate similar choices. The model must identify what distinguishes closure-oriented control from expected-value, threat-avoidance, fatigue, or predictive-processing accounts (Bertsekas, 1995; Sutton & Barto, 2018).

If a route claims that a state change drives a later action, its predicted temporal order should be detectable. Neural or physiological signals measured only after behavior cannot establish that direction. Dense longitudinal sampling and task

perturbation are more informative for this question than one cross-sectional correlation.

A compact state can also retain the order of events without storing a complete narrative. Consider a simulated state x with two available operations. Operation A adds two. Operation B halves the current value. Starting from x = 4, applying A and then B gives three: four becomes six and six becomes three. Applying B and then A gives four: four becomes two and two becomes four.

The two histories contain the same named operations, each performed once. Their final states differ because the operations act on what is already present. Halving after addition reduces the newly added amount as well as the earlier value. Halving before addition leaves the later addition untouched. An inventory of event counts cannot recover the final state unless it also represents the relevant order.

The difference holds for every starting value. A followed by B produces x/2 + 1. B followed by A produces x/2 + 2. The one-unit difference is a prediction of the update rules. Testing the two orders therefore examines a structural property of the model rather than merely checking whether either operation has some effect.

The state itself remains compact. Once the current value has been correctly updated, the model can apply the next operation without rereading the entire history. Information about past order has been absorbed into the present state. Whether that state is sufficient depends on the future rules: if later behavior also depends on how often an operation occurred, the value x alone would need to be supplemented.

This gives a candidate biological or psychological account a useful comparison. Specify the updates associated with two independently characterized conditions, predict both orders and

measure the resulting course through common observation channels. A difference between the orders can identify a form of state dependence. Agreement with the predicted amount and timing supplies more information than the broad observation that history matters.

The represented state and the experienced contributions remain separate objects. A state update may alter what happens next while the earlier experience stays part of its assigned history. To connect this mechanism to the Law, the candidate account would additionally need the quantitative relation between state changes and those contributions. The ordered-operation test helps determine the dynamics of a possible carrier before assigning it that further role.

Perturbation must also be specific. If reducing pain changes every candidate mechanism in the same direction, the result shows that the system responded but does not identify why. A stronger design predicts which component moves first, which downstream relation follows, and what would contradict the route (Woodward, 2003).

Mediation is useful only when its assumptions are credible. A neural signal may sit between an intervention and behavior while sharing unmeasured causes with both. The analysis should distinguish temporal mediation, statistical association, and causal mediation rather than allowing one diagram to imply all three (Hernán & Robins, 2020).

Apply the distinct-error requirement in Section 7.4 to cross-level evidence, then test whether the route still predicts when one channel is withheld.

The route also needs a boundary between state and trait. Chronic anxiety, learned avoidance, pain, addiction, and depression can alter repeated behavior without representing a terminal objective. A local mechanism should predict state-

dependent changes within persons, not merely classify diagnostic groups.

Hierarchical models can represent individual differences and uncertainty. Anatomy, age, medication, disability, culture, learning history, and task strategy can shift signals. Ignoring these differences would make a fixed regional threshold or person-level classifier scientifically weak and ethically dangerous.

Neural integration has an experimentally accessible example. Aksay et al. (2007) recorded neuronal firing and eye position while selectively silencing parts of the goldfish circuit that converts eye-velocity commands into a maintained position signal. The induced drift distinguished competing accounts of persistent activity. This shows how a retained quantity can be tested through its dynamics rather than inferred from resemblance. The quantity was eye position, not felt valence. A corresponding advance for a PTVC implementation would identify what a candidate circuit retains and test its relationship to complete felt contributions on independently evaluated trajectories.

Lesion, stimulation, and pharmacological evidence can localize necessity or modulation, but each has limits. Broad disruption affects several computations, and compensation can preserve behavior. The strongest interpretation names the function that changed and the alternatives that remain, rather than treating any intervention as proof of a dedicated closure circuit.

8.3.2 — Global, Bridge, and Boundary Routes

A state-change route makes the possibility concrete. Let Q(S) be a physical quantity independently specified on a state S adequate for the proposed relation; the needed differences in Q must be

identifiable. Suppose an admitted transition from s to s′ contributes $c(s,s')=Q(s)-Q(s')$, in fixed units. The contribution includes all experience during the transition, including residence time. Along a finite path, the intermediate values cancel: the total is Q(start) − Q(finish). Equal boundary values then yield zero without foreseeing when the path will end.

Consider an invented four-state example. A is the start and D the final qualifying state before a separately specified endpoint. Set Q(A) = 0, Q(B) = −2, Q(C) = 1, and Q(D) = 0. The path A → B → C → D contributes +2, −3, and +1. The path A → C → B → D contributes −1, +3, and −2. Both total zero. Inserting B → C → B adds −3 and +3: each gross total increases by three while the net stays unchanged.

Now set Q(D) = 1 while retaining the transition rule. Both paths total −1. This candidate guarantees closure when the contribution relation and matching boundary values hold together; neither assumption alone supplies that guarantee. On a continuous-time rate branch, Q along the path must be locally absolutely continuous, the rate relation must include every contribution, and the complete positive and negative accumulations and strict-pre-endpoint boundary value must be finite. Boundary values need not themselves be zero, and no contribution is added after experience ends.

A stronger local route has an information requirement. Suppose every admitted complete path closes. Two histories reaching the same sufficient state and admitting the same continuation with the same future total must have equal past totals: one common remainder cannot close two different numbers. The mechanism must preserve the relevant distinction in its state or continuation rules. This conditional argument concerns an additive local route class, not every possible realization of PTVC. A distributed physical state could retain

information without a conscious accountant; defining Q afterward as the negative of the observed total would supply no independent explanation.

Noise can coexist with this identity. In the illustrative system $dQ = U\,dt$ and $dU = (-\alpha Q - \beta U)dt + \sigma\,dW$, with positive constants and Brownian motion W, the velocity U is noisy while Q is its time integral. If the candidate valence rate is $v = -U$, its accumulated contribution equals Q(start) − Q(finish) on every finite interval. Q has finite variation there.

Where noise enters matters. For a smooth Q of an Itô diffusion, the literal finite-variation identity requires its stochastic diffusion term to vanish along the modeled paths. Equating v only with the negative generator of Q can leave a stochastic remainder (Protter, 2005). Section 27.2.2 explains why identifying the required difference can demand less information than reconstructing Q itself.

The route first needs an adequate state and independently calibrated potential differences. Here Q remains a proposed quantity rather than an identified phenomenal measure. Histories reaching the same state provide a concrete test: do they require different future contributions?

The construction stipulates that each contribution equals a decrease in potential. To examine that relation, estimate transition contributions independently of the fitted potentials, then test path differences and cycle sums on held-out observations.

Complete accounting includes experience during residence in a state as well as transitions between states. Use a fixed exposure rule and bound unobserved intervals, then determine whether their allowed contribution could reverse the contrast.

The finite examples satisfy the accumulation requirement; extending it across a complete stream requires further

justification. Check the relevant integrability or finite-variation conditions and establish separate bounds for positive and negative accumulation.

A complete-stream application also needs an independently identified endpoint. Apply the procedure in Section 5.3 and evaluate possible returns and endpoint uncertainty using criteria fixed before inspecting the residual.

Equal boundary values yield zero only when the contribution relation also holds. Identify or bound the start-to-finish potential difference independently, and retain the Q(D) = 1 countermodel to distinguish a closing construction from a nonclosing one.

A positive-probability terminal band can be handled with an ordinary conditional probability. An exact point in a continuous state space usually has probability zero and requires a separately specified limiting or regular-conditional bridge. Writing a conditional probability for an event of zero probability without that construction hides the hardest step (Chetrite & Touchette, 2015).

Conditioned routes must use the scoring target and selection regime declared in Section 7.2.2. The imposed terminal condition cannot itself count as a successful prediction.

A completion potential may summarize the chance of reaching a frozen terminal event from a current state. Such a potential can be scientifically useful, but it is observational unless a causal bridge is supplied. It does not automatically tell us what would happen under an action or policy.

Interventional use requires additional assumptions about treatment versions, exchangeability or randomization, positivity, interference, censoring, endpoint change, and measurement invariance. Without those assumptions, a potential predicts under the observed regime; it does not authorize steering (Hernán & Robins, 2020).

Unexpected endpoints can distinguish route families. A controller may depend on an anticipated horizon, whereas a globally conditioned path need not contain a local device that foresees the ending. For each candidate, specify whether changed timing truncates an existing path, changes its probability law, or falls outside the model’s domain. Compare the resulting predictions under the same observation rules.

Hybrid models remain possible. Local regulation may operate inside a global bridge; reserve coupling may coexist with a state-dependent endpoint; a boundary rule may interact with psychological adaptation. Each additional component must earn a new observable consequence.

A completion potential has a built-in reliability check on the positive-probability branch. For a fixed completion event with positive baseline probability, divide its conditional probability given the declared information history by that fixed baseline probability; the result is a martingale under the reference law. Passing that check shows internal consistency of the conditioned model; it is not an additional vote for PTVC because the property is generic to correct conditioning.

A two-toss experiment makes the averaging property concrete. Toss a fair coin twice and define completion as obtaining at least one head. Three of the four equally likely ordered outcomes meet the condition, so the initial completion probability is three quarters. The information history begins before either toss and expands when each result is observed.

After a head on the first toss, completion is certain. The conditional completion probability is one. After a tail, completion requires a head on the second toss, so the conditional probability is one half. Before the first toss, these two information states are equally likely. Their average completion

probability is three quarters, exactly the probability with which the experiment began.

Dividing each conditional probability by the original three quarters gives a normalized value of one at the start. After the first toss, it becomes four thirds on the head branch and two thirds on the tail branch. Their equally weighted average is one. The value may rise or fall as information arrives while preserving its conditional mean under the original fair-coin law.

The next update works the same way within each branch. After an initial head, both final outcomes already satisfy completion, so the normalized value remains four thirds. After an initial tail, a final head gives four thirds and a final tail gives zero. Their average is two thirds, which is the value held after the first tail. The mean-preserving relation holds at each information state, not merely after averaging the whole experiment.

This is the finite form of the martingale property used above. It follows from consistent conditional probabilities for one fixed event. The event itself has not become more or less common because the observer learned which branch occurred. The information has changed the observer's conditional description of the remaining possibilities.

The distinction becomes especially clear when the reference law changes. Condition the experiment on eventual completion and keep only the three successful ordered outcomes. An initial head then occurs in two of the three retained outcomes, rather than half of all outcomes. Averaging with those new probabilities answers a different question. The original mean-preserving check uses the original fair-coin probabilities.

A candidate completion model can be examined in this way at successive information states. Its forecasts should fit together as conditional probabilities of the same declared event under the

same reference law. A mismatch locates an inconsistency in how the model updates. Passing the check establishes that internal relationship and gives later empirical comparisons a coherent probability model to work with.

Estimate completion probabilities out of fold. In the ratio just defined, the fixed baseline probability is the denominator: a small current prediction does not by itself make that ratio large. A very small estimated baseline can make normalization unstable. Prespecify an adequacy threshold, report exclusions, and disclose any numerical floor because it changes the normalization. Assess reliability, conditional increments, and uncertainty on common support.

A global route also owes a counterfactual account. If a safe intervention changes an early state, does the admissible path family change, does the endpoint distribution change, or does only the observation change? Without a declared counterfactual object, the route may describe complete histories while saying nothing testable about intervention.

8.3.3 — Endpoint Timing and Information Limits

State-coupled endpoints also have a precise requirement. Let T be a strictly positive stopping-time endpoint under a declared information history. Let the adapted accumulated process L be right-continuous with left limits, so L(t−) records its pre-event value. The endpoint counting process records one event at T; its predictable compensator describes accumulated conditional event occurrence. If L(T−) is zero whenever T is finite, with probability one, then on every fixed finite observation horizon the compensator assigns zero expected event mass to times when |L(t−)| exceeds a fixed positive tolerance.

For a route with an ordinary at-risk event intensity, the corresponding intensity-weighted off-tolerance occupation must

have expectation zero. No independence between T and the path is assumed. Endpoint atoms or singular rules use the general compensator; non-stopping endpoints need another analysis. With adequate valence and endpoint bridges, this identifies a relation to examine without defining the endpoint from a favorable residual or making a personal clinical forecast (Protter, 2005).

A discrete model shows what state-dependent stopping requires. Suppose a process has three possible stopping steps. Conditional on having reached the first step, it stops there with probability one fifth. If it continues to the second step, it stops there with probability one half. Any process reaching the third step stops there with certainty.

The unconditional stopping probabilities are one fifth, two fifths, and two fifths. The second value comes from the four-fifths chance of continuing past the first step multiplied by the one-half stopping probability at the second. The third is the chance of continuing past both earlier steps. Conditional stopping probabilities and unconditional endpoint weights therefore play different roles.

Now suppose the accumulated value is outside a declared zero band at the first stopping step and inside it at the other two. The model ends outside the band with probability one fifth. Later stopping inside the band does not subtract from that probability. The branches are alternative completed histories, so the total off-band probability is the sum of nonnegative contributions from the off-band branches.

For every completed history to finish inside the band with probability one in this model, a reached off-band state must have zero stopping probability. Set the first stopping probability to zero while retaining the second-step probability of one half. The

unconditional endpoint weights become zero, one half, and one half. All stopping mass is now placed on the two in-band states.

The construction distinguishes two ways an off-band state may be absent from the endpoint record. The process may never reach that state, or it may reach it while the stopping rule assigns it no stopping chance. These are different dynamical explanations. Observing intermediate states and stopping events together can help distinguish them.

The continuous-time statement above expresses this same requirement through the compensator or event intensity. It tracks where endpoint mass can occur while the process is still at risk of ending. A relation between state and stopping can thus be studied without assuming that the endpoint is stochastically independent of the path. The model's claim is about which combinations of an attained state and an ending are permitted by its probability law.

The scientific interest lies in that joint prediction. Two models may give the same overall distribution of stopping times while assigning the stopping mass to different path states. Recording time alone would miss their difference. The state immediately before the event supplies the additional coordinate on which their explanations can separate.

A rule for choosing actions faces a sharper question than whether it usually prefers promising paths. Under one fixed continuation model, its completion probability is the average of the action-specific completion probabilities, weighted by how often it chooses each action. That average can equal one only if the rule assigns zero probability to the set of actions whose continuations have any chance of failure. Even a small positive chance of selecting from that set prevents an exact probability-one guarantee.

Limited information creates a further obstacle. Consider two imaginary rooms that look identical to the decision-maker. In one room only the left exit reaches the target; in the other only the right exit does. Knowing that a successful exit exists in each room is not enough. A rule that cannot tell the rooms apart cannot guarantee success in both. It needs information that distinguishes them, or an action that works in both.

For a finite menu of actions, the common-action requirement for probability-one success at every history compatible with the available information is that at least one action exists that succeeds with probability one at all those histories, under the fixed continuation model. The route must also supply a consistent measurable rule for choosing across information states. Increasing a soft preference for promising actions may make completion arbitrarily likely while still falling short of probability one. These requirements concern what a model can achieve with its information; they do not identify motivation, awareness, free will, or access to a cumulative total (Chatterjee, Chmelík, & Davies, 2016; Asadi et al., 2025).

Reciprocal-process diagnostics may separate bridge mixtures from a local Markov controller, but hidden state and coarse observation can mimic the same dependence.

The same frozen route should transport across modules. A completion potential, reserve state, or controller must face held-out perturbations, endpoint-support changes, and independent channels without redefinition. Separately tuned fits are not one mechanism until shared state, parameters, and transport rules survive common-data comparison.

8.3.4 — Learning Which Way to Move

A local controller can learn from its own actions. The action changes the system, and the response reveals something about

how the system works. This creates a richer problem than choosing a direction from a complete map. The controller must choose a first move that is informative and leaves enough room for the moves that follow.

Imagine a simulated slider starting at position zero. A command can have any value from minus one to plus one. One of two fixed wiring arrangements determines how the command moves the slider. Under the ordinary arrangement, a command of plus one moves it one unit right. Under the reversed arrangement, the same command moves it one unit left. The controller knows these two possibilities but initially does not know which wiring the simulator uses. After each command, it observes the resulting position exactly.

Let the first target be position one half. Begin with a command of plus one quarter. Under ordinary wiring, the slider moves to plus one quarter. Under reversed wiring, it moves to minus one quarter. Either response identifies the wiring. The first command has served as a small experiment while also changing the position from which the remaining task must be completed.

The second command can use that result. From plus one quarter under ordinary wiring, command plus one quarter again to reach one half. From minus one quarter under reversed wiring, command minus three quarters. Reversal turns that negative command into a rightward movement of three quarters, again reaching one half. Both second commands fit within the permitted range. The policy succeeds under either wiring because it adapts to what the first action reveals.

A fixed two-command schedule cannot achieve the same guarantee. Under ordinary wiring, its final position is the sum of its commands. Under reversed wiring, its final position is the negative of that sum. Those positions cannot both equal one half. The successful policy therefore depends on the response

between the commands. Its information is generated within the task rather than supplied by an observer who already knows the hidden setting.

Now move the target to position one and retain the two-command deadline. The earlier probing command still identifies the wiring. Under ordinary wiring, a final command of plus three quarters reaches the new target. Under reversed wiring, however, the slider is at minus one quarter and needs a rightward movement of one and one quarter. That would require a command of minus one and one quarter, beyond the allowed range. The experiment has learned the answer while consuming too much of the available movement on one branch.

Choosing a smaller positive probe reduces that extra requirement but leaves it above one. A negative probe creates the corresponding problem under ordinary wiring. This is a precise limit, rather than a poor choice of probe. Call the first command a. Ordinary wiring requires the second command to be 1 − a. Reversed wiring requires it to be −1 − a. Keeping both within the interval from minus one to plus one forces a to be zero. A zero command leaves the slider at zero under either wiring and reveals nothing. The final command must then be the same in both cases, so it cannot reach positive one in both.

The two-step problem is therefore different from two separate problems in which the wiring is known. Each known-wiring problem can be solved in one step. The combined problem has enough movement for either solution considered separately, yet lacks a first experiment that both distinguishes the cases and preserves enough capacity to finish in the remaining step. Information and reachability must be evaluated together along the same actual policy.

A third command changes the answer. Use the plus-one-quarter probe again. Under ordinary wiring, command plus three

quarters next and then zero. Under reversed wiring, command minus five eighths on each of the next two steps. Each of those commands moves the slider five eighths to the right, taking it from minus one quarter to three eighths and then to one. All commands satisfy the same bound. The additional step makes the informative first move compatible with successful completion in both cases.

The extra time is doing a specific job. It supplies another bounded corrective move after the direction has been learned. Giving the controller the wiring in advance would solve the problem differently: a single correctly signed command would suffice. Increasing the command limit would offer another change. The construction separates three sources of improved performance: earlier information, greater action capacity, and a longer horizon.

Such a model can generate a recognizable sequence of behavior. An early action may be small because its response helps select the next action. Once the response is known, later movements may become larger and more direct. A route model can predict the size and timing of that change from its information and movement rules. The proposed test then concerns a sequence of choices, rather than interpreting an isolated hesitation as a lack of motivation.

For a candidate route to closure, the constructive question is which observations its own actions could make available, and whether acquiring them preserves a viable continuation. A model can answer by supplying a policy, following every allowed response, and showing that the remaining actions suffice. This joins the information requirement to the shortening-horizon problem: learning has a place in the unfolding path, and that place can determine which targets become reachable.

8.3.5 — How a Route Can Fail or Be Retired

Representation failure retires a route when its state cannot be related to felt valence without arbitrary transformations. A strong fit to physiology or behavior may still be useful, but it does not explain the phenomenal target until that bridge is identified.

Endpoint failure occurs when the route works only after the boundary is moved, smoothed, or reclassified in light of the trajectory.

Route-state nonidentification occurs when reserve, horizon, capacity, channel availability, or control pressure cannot be separated from frailty, diagnosis, medication, or observation quality.

Other operational retirement triggers include persistent recovery failure, required information unavailable to the proposed mechanism, or a failed safety criterion.

Transport failure may reveal noninvariance, context specificity, or a narrower class. A narrower model remains useful if it continues to outperform alternatives within that independently justified scope.

Chapter 9 — The Shortening Horizon

Time changes more than duration. It changes which actions remain possible, how much delay costs, what information can still be gathered, and whether a path can be reopened. The shortening horizon is the practical fact that options can close. A deadline passes, illness advances, a child grows, a caregiver’s reserve thins, a treatment window narrows, or years of practice become unavailable. The same action can therefore carry a different opportunity cost at different points in a life.

The workable future can widen even while the calendar moves on. Imagine a parent who wants to attend an evening class but has no reliable care for a dependent relative. The class exists; the intention is real; the route is blocked. A dependable two-hour period of respite each week can make that route usable. No hours have been added to the week. What changed is how the hours can be lived.

Consider an invented housing offer that expires on Friday, while a necessary document can be obtained only until Wednesday. The final deadline and the last feasible time to complete an essential step differ. A horizon model must represent the dependencies that make a route usable, including deadlines imposed by institutions rather than chosen by the person.

The horizon relevant to a prospective model must be estimated from information available at the time. Researchers cannot use the actual future endpoint while pretending to predict earlier decisions. Prognosis, task duration, contract deadlines, or declared study windows may provide usable proxies, each with different error and meaning. Viability asks whether at least one acceptable continuation remains. Capacity asks what the system can still do. Compensability asks whether a declared displacement can still be addressed within a candidate route.

Their separation turns a poetic horizon into a testable architecture.

Aging changes capacity without producing one emotional script. Decline, adaptation, accumulated skill, altered priorities, unequal resources, and continued growth can coexist. Serious illness can shorten and complicate the horizon through symptoms, treatment, prognosis, dependency, and uncertainty. Caregiving can do the same for the caregiver as daily needs displace sleep, work, social life, and long-term projects. Love and burden can coexist without canceling either.

Time remaining becomes a constraint because many forms of repair require duration. Apology may require conversation. Medical relief may require access, diagnosis, treatment, response, and adjustment. Emotional recovery may require repetition, safety, sleep, relationship, and ordinary days that prove the harm is not still happening. Delayed repair has a distinct public cost. When institutions withhold pain treatment, accessible care, income support, legal remedy, companionship, or a listening clinician until little workable future remains, they have not merely postponed an external benefit; they have shifted burden into a narrower horizon. The inquiry should identify who closed the route and why relief was unavailable sooner.

9.1 — The Shortening Horizon

As time runs short, opportunity can narrow before a biological endpoint arrives. The lived horizon concerns which channels of experience and action remain open or may reopen, whereas estimated time to an endpoint is a separate variable. Under comparable burdens and reserve, a shorter horizon would need to predict changed choices or affective dynamics beyond ordinary urgency. If that contrast fails, a powerful intuition remains an intuition rather than a demonstrated mechanism (Carstensen, Isaacowitz, & Charles, 1999; Mather & Carstensen, 2005; Carstensen, 2006; Wirth et al., 2024).

9.1.1 — Time Remaining and Late Relief

The remaining interval is a feasibility constraint, not a multiplier that makes later experience more valuable. Researchers may ask whether analgesia, safety, communication, or freely chosen reconciliation changes state, choice, or measured burden differently under declared horizons.

Death alone does not establish institutional failure. The institutional questions concern untreated pain, poverty at the end of life, bureaucratic delay, abandonment, and inaccessible services. These are conditions for policy, care, and relationships to address, rather than traits to assign to a person or hide inside a single time variable.

The same help can open a different future depending on when it arrives. Imagine a person preparing a recording for a family celebration. They have the message they want to share, but need an accessible way to operate the recording software. If support arrives before the gathering, they can record the message, listen back, revise it and decide how to take part. Support arriving afterward may still be useful, but that particular occasion has passed.

The distinction concerns an opportunity with a calendar. It does not require assigning a greater intrinsic value to an earlier minute or a later one. The earlier assistance makes a sequence possible: learning the controls, making the recording and choosing what to share. The later assistance begins from a world in which one of those possible uses is no longer available.

This helps explain why timeliness belongs inside the description of care and access. A service can be delivered in full and still arrive after the activity it was meant to support. The relevant record therefore includes the person's intended use and the time within which that use remains possible. A completed service transaction and a realized opportunity answer different questions.

Other opportunities remain open longer or can be recreated. A conversation may move to another day; a small exhibition may be postponed; a new form of participation may become available. Understanding the horizon includes finding those alternatives. The useful response can be to accelerate assistance, adapt the activity or make a second occasion possible, depending on what the person values about the original plan.

A late improvement can also have its own complete importance. If clear communication becomes possible for one conversation, that conversation is lived rather than discounted because there will not be many repetitions. Its duration and its meaning are distinct features. The formal cumulative question still requires its own representation, while ordinary attention can recognize the particular possibility that has opened.

9.1.2 — Deadlines, Practice, and Slack

Ordinary deadlines make the shortening-horizon intuition concrete. A student can recover from one unproductive week early in a semester. The same week near the final examination

may remove the last opportunity to learn foundational material. The constraint is the shrinking set of corrective paths.

A shorter horizon does not always reduce effective choice. Short deadlines can focus attention, long horizons can invite delay, some people remain creative under severe limits, and support can reopen paths that seemed closed (Baltes, 1997; Freund & Baltes, 2002).

Perceived future time can shift attention toward emotionally meaningful goals when the future feels limited, yet the pattern varies with health, culture, obligation, and opportunity. Socioemotional selectivity research therefore offers a family of tendencies rather than a universal script for aging or terminal life (Carstensen, Isaacowitz, & Charles, 1999; Carstensen, 2006, 2021; Wirth et al., 2024).

Option preservation often requires slack. Spare time, savings, social trust, institutional redundancy, respite, and physical reserve may look unused in a static picture, yet uncertainty makes their value visible when a surprise arrives. A system with no buffer can function until one interruption turns a narrow plan into crisis. Slack also preserves reversibility, giving a person or institution time to detect a bad path and change course before one setback closes the remaining options.

9.2 — Time, Capacity, and Opportunity

Time remaining does not equal future opportunity. A person may have years ahead yet few usable options under untreated illness or danger. Debt, inaccessible services, and caregiving demands can also close practical routes without changing the calendar. A workable-horizon model must specify which capacities and constraints it counts.

9.2.1 — What Makes a Future Workable

Another person may have little time and still possess meaningful choices about comfort, conversation, place, ritual, treatment, or rest.

A point estimate of remaining time is rarely enough. Remaining time may be uncertain, and that uncertainty can be wide even when the expected horizon is short. The model should carry a distribution or interval and ask whether its predictions survive plausible endpoint variation rather than treating one forecast as fate.

Calibration is essential. For probabilistic horizon forecasts, cases assigned similar probabilities that the endpoint will occur within a specified interval should show corresponding frequencies under the declared endpoint rule. A model may rank cases well while miscalibrating those probabilities, so ranking alone cannot support a trustworthy horizon-effect test (Gneiting & Raftery, 2007).

Suppose a workshop can offer a participant one of two schedules. The first provides exactly five uninterrupted hours. The second provides either one hour or nine hours, with equal probability, with the available duration revealed only when the session begins. Both schedules offer five hours on average. A project that needs four uninterrupted hours can be completed

under the first schedule every time and under the second schedule half the time.

The distinction is a feature of the available-time distribution. The average describes expected duration across occasions. Completion depends on the chance that duration reaches the project's threshold. In this example that threshold is four hours, so the relevant probability is one under the fixed schedule and one half under the variable schedule. Knowing the mean does not determine it.

Different projects can reverse which feature of the schedule is attractive. A project requiring eight uninterrupted hours is impossible under the fixed five-hour schedule and possible half the time under the variable one. A collection of small tasks that can be paused and resumed may use the hours differently again. The shape of the opportunity matters together with the structure of the activity.

Advance information adds a further dimension. If the participant learns a day earlier whether the variable schedule will offer one or nine hours, materials can be prepared for a matching task. The distribution of available hours has stayed the same, but preparation can use the information. Alternatively, a guarantee of four uninterrupted hours changes the opportunity itself. Forecasting and improving the schedule are two different ways to make plans more workable.

This gives horizon research a more expressive vocabulary. A usable forecast can describe the chance of reaching a relevant duration, the uncertainty about that chance, and when further information becomes available. A planning model can describe whether tasks can be divided, deferred, or completed through several shorter intervals. Together they explain why a person with fewer average hours may sometimes have a more usable opportunity than someone with more variable time.

The human significance lies in what a person can actually begin and finish. An assured interval may support a conversation, a journey, a craft, or an act of care that cannot be assembled from scattered fragments. The measure becomes more informative when its time units are connected to the shape of a possible activity. That connection gives the idea of a usable horizon concrete content.

A strong design compares time alone, capacity alone, context alone, their combined main effects, and models adding prespecified interactions. It examines whether the horizon contribution transports to held-out tasks and populations.

9.2.2 — Capacity That Takes Time to Build

A capacity may be available eventually while taking time to bring into use. A machine can have a high maximum speed and still need an interval to reach it. A team can possess the required skills and still need time to assemble. The distinction becomes especially clear in a simple motion model, where the distance reachable before a deadline can be calculated exactly.

Consider a cart on a straight track. It begins at position zero and at rest. Its acceleration can have magnitude at most one distance unit per squared time unit, and its speed can have magnitude at most two distance units per time unit. The task is to move as far to the right as possible by time T. There are no obstacles, delays in sensing, or other restrictions in this constructed example.

For the first two time units, maximum rightward acceleration makes the speed rise along $v(t) = t$. After that, the speed limit holds it at two. The largest reachable displacement is the area beneath this speed curve. For T no greater than two, that area is $T^2/2$. For T greater than two, it is the initial triangular area of two plus the rectangular area $2(T - 2)$, giving $2T - 2$.

An estimate based only on the maximum speed would give 2T at every horizon. At T = 1, it would suggest a distance of two, while the cart can reach only one half. At T = 2, it would suggest four, while the attainable distance is two. The missing ingredient is the initial condition. The cart starts with the ability to accelerate, rather than with its maximum speed already in hand.

The shape of the horizon effect is revealing. During the acceleration phase, doubling the available time multiplies the maximum displacement by four. After the speed limit has been reached, an additional unit of time adds two units of distance. The same cart therefore has different relationships between time and accomplishment at short and long horizons. A single constant rate would describe only part of that relationship.

Now give the task a second boundary condition: the cart must be at rest when the deadline arrives. Arrival includes both position and speed. If T is no greater than four, the fastest admissible plan accelerates for the first half of the interval and decelerates for the second half. Its peak speed is T/2, within the limit of two. The triangular speed curve has area $T^2/4$.

For T greater than four, the cart accelerates for two time units, travels at speed two for T − 4 units, then decelerates for two. The two triangular portions contribute four distance units in total. The middle portion contributes 2(T − 4). Maximum displacement while arriving at rest is therefore 2T − 4. At T = 3, the first task permits a displacement of four; the rest-to-rest task permits only 2.25.

These formulas follow from a pointwise envelope on speed. Starting at rest requires v(t) to be no greater than t. Finishing at rest requires it to be no greater than T − t, because that much time remains to reduce speed at the permitted rate. The speed limit supplies the third bound, two. The largest useful speed at

each moment is consequently min(t, T − t, 2). Following that envelope attains the stated areas.

A deadline can therefore constrain a process twice: through the time needed to begin and through the time needed to finish in the required state. The rest-to-rest cart cannot spend every remaining moment increasing progress toward position alone. Some of its future capacity belongs to completing the transition into the desired final condition. A model that records only distance remaining leaves out part of the task.

The human comparison is structural rather than numerical. Preparing a room for a gathering involves setting it up, using it, and restoring it. Learning a new tool involves acquiring enough fluency before its advertised speed becomes available. A handoff may require time to make work intelligible to the next person. These processes have their own causal details, but each invites the same concrete inquiry: which capacity is usable now, how does it develop, and what condition must hold when the work ends?

For candidate completion models, the example enriches the meaning of remaining capacity. The useful object may include the present rate, the allowed change in that rate, and the final state as well as the cumulative distance to a target. A longer horizon does more than permit additional copies of the same action. It can make a different trajectory available. That is why time, capacity, and boundary conditions belong in one description of what remains possible.

9.2.3 — Why Constraint Pressure Might Rise

When a route requires a given change before an independently specified deadline, a shorter interval leaves less slack for delay or error. The model should state what pressure means. It might refer to action pruning, increased effort, reduced tolerance for

delay, greater sensitivity to error, or a higher risk that no acceptable continuation remains. These outcomes are not interchangeable, and each requires a distinct measure.

Scarcity and bandwidth effects, developed in 9.3.1, are strong ordinary rivals, alongside stress, habit, loss aversion, and ordinary planning; any horizon account must predict something beyond them.

Public language should therefore say that constraint pressure may rise under declared conditions. It should not speak of an endgame, price a person’s future, or claim that suffering must intensify in a particular way to force balance. Exact PTVC constrains the signed remainder, not a universal trajectory of increasing pain.

A shorter horizon can change the cost of a feasible route as well as whether the route exists. Consider a simple machine whose position changes at the rate chosen by its operator. It must move a distance d during a fixed interval T. Define its effort cost as the accumulated square of the movement rate. Under this particular model, moving twice as fast for the same duration costs four times as much.

A constant rate d/T completes the movement. Its cost is the squared rate multiplied by the duration, giving d^2/T. To see why no variable schedule costs less, write any other admissible rate as d/T plus a deviation w(t). Completing the same displacement means the deviations sum to zero over the interval. Squaring and accumulating the rate produces three terms: d^2/T, a cross term proportional to the accumulated deviation, and the accumulated square of the deviation. The cross term vanishes. The last term is nonnegative.

The constant schedule therefore achieves the minimum cost. Any unevenness adds exactly the accumulated squared deviation from that schedule. This is a result about the stipulated machine

and effort function. It identifies the assumptions behind the common intuition that rushing makes a task more demanding: the same displacement must be completed in less time, and the cost rises more than proportionally with instantaneous speed.

For a displacement of six units, a six-unit horizon permits a constant speed of one and costs six. A three-unit horizon requires speed two and costs twelve. A two-unit horizon requires speed three and costs eighteen. Halving the available time doubles the least cost. Nothing in this example requires the machine's target to change or the remaining distance to grow.

A speed limit introduces a separate feasibility boundary. If the machine can move at most two units per unit of time, completing the distance of six requires at least three time units. Above that horizon, the least-cost calculation remains available. At three units it uses full speed throughout. Below three units the target is unreachable under the stated action limit. Rising effort and lost feasibility are different stages of the same explicitly defined problem.

There is another revealing comparison. Suppose the machine initially has six time units, waits for two, and then begins. It still has four units in which to cover the distance, so the least remaining cost is nine. The wait has raised the minimum from six to nine. If the machine instead moves steadily from the beginning, every completed portion removes part of the remaining distance as the clock advances. The changing ratio of distance to time depends on the path already taken.

This calculation makes a horizon experiment more precise. One can specify a target, a movement rule, an action bound and an effort measure, then vary time while keeping those other features fixed. The model predicts a particular curve rather than a general feeling of pressure. Different cost functions generate different curves. A cost proportional to absolute speed, for

example, assigns the same total cost to every one-directional schedule completing the distance, so shortening time has no such cost effect before feasibility fails.

In the Law's candidate architecture, a claim about regulatory effort needs the same specificity. The quantity being regulated, its dynamics and the cost of available actions determine whether a shorter horizon raises effort, redirects it or merely changes which paths remain possible. The simple machine gives one constructive answer and a clear route to comparing alternatives: identify the predicted relation and then examine whether the proposed process actually follows it.

A fixed amount of time can support very different plans depending on how the work is connected. Imagine preparing a small community exhibition. Mounting the display takes two hours after the prints are ready. Producing the prints takes three hours. Writing the labels takes two hours and can begin immediately. If different people can print and write at the same time, the preparation can finish in five hours: three for the prints followed by two for mounting, with the labels completed along the way.

The separate tasks contain seven hours of work, yet the elapsed preparation time is five. The difference comes from concurrency. If one person must perform every task, the same work instead requires seven hours. A horizon estimate that records only the sum of task durations misses the available organization of the work.

Now give the group six hours before opening. The printing-and-mounting chain has one hour of slack. The labels can be delayed substantially more and still be ready at opening. A half-hour interruption to printing consumes half the chain's spare time. The same interruption to label writing may leave the

finishing time unchanged. Pressure depends on where a delay occurs, not only on its length.

The longest required chain is often called the critical path. Shortening a task on that chain can shorten the project. Shortening a task elsewhere may improve convenience without changing the earliest completion time. In this example, reducing printing from three hours to two brings the earliest opening from hour five to hour four. Reducing label writing from two hours to one leaves it at hour five.

Resources can change which chain is critical. If mounting requires a volunteer who arrives at hour four, printing completed at hour three still waits an hour. Earlier printing then has value only if it resolves some other constraint, such as making time for a quality check. The plan must represent the volunteer's availability together with task duration. A usable future is organized through relations among people and activities.

This makes the horizon question more specific. One can ask which prerequisite is unfinished, which task can proceed now, and where assistance would change the finishing time. Those questions distinguish useful support from effort added in the wrong place. They also explain why a small timely contribution can reopen an entire path: it completes the step on which the remaining work depends.

As a deadline approaches, the changing pattern of slack is observable. Some branches remain flexible while one becomes tightly constrained. A study can examine that pattern without compressing every condition into a single pressure score. The schedule provides a clear account of what remains possible and which intervention would make a different continuation available.

9.2.4 — Changing Capacity and Measuring Ratios

The route should distinguish structural actions from observations. More frequent measurement alone does not demonstrate that capacity has widened. A questionnaire can reveal a problem, but treatment, access, rest, safety, or relationship repair changes the conditions. The model should never confuse data collection with help.

When support widens communication or reduces symptoms, test its predicted effect on the attainable set under the intervention assumptions in Section 8.3.2, including subsequent care, adherence, and endpoint changes (Hernán & Robins, 2020).

The reserve-relative quantity formed by dividing a residual estimate by remaining capacity is especially fragile near a small denominator. Modest error can produce an apparently enormous pressure. Analyses should report numerator and denominator separately, prespecify the minimum reserve at which the ratio is interpretable, and show sensitivity to both estimates. A numerical floor changes the ratio and must be reported.

A rising ratio may also reflect declining observation quality or a misspecified proxy.

A ratio can be informative when its two ingredients remain visible. Suppose a synthetic model estimates a remaining requirement D between 0.9 and 1.1 units and a positive available capacity C between 0.1 and 1.0 units. If every combination within those two intervals is allowed, the ratio D/C ranges from 0.9 to 11. The smallest ratio pairs the smallest requirement with the largest capacity; the largest pairs the largest requirement with the smallest capacity.

Dividing the interval midpoints gives 1/0.55, about 1.82. That number is compatible with the inputs, but it hides how much larger the ratio could be. A more accurate estimate of D alone

has limited effect. Even if D were known to equal one exactly, the ratio would still range from one to ten. The broad uncertainty is located mainly in capacity near zero.

Now suppose independent evidence establishes that capacity lies between 0.5 and 1.0. Keeping the original uncertainty in D, the ratio lies between 0.9 and 2.2. The improved capacity bound sharply changes the inference without adding observations of the requirement. The calculation identifies which information is valuable for this specific derived quantity.

A software floor has a different meaning. Replacing every denominator below 0.2 by 0.2 would cap the reported ratio at 5.5 under the original input intervals. It would not establish that the true ratio is at most 5.5. It defines a modified quantity, D divided by the larger of C and 0.2. That quantity can be useful for a chosen numerical purpose, provided its construction is explicit.

Dependence between the inputs can also matter. If the allowed pairs follow D = C, the ratio is one throughout, however widely their separate values vary. Marginal intervals alone omit that relation. The appropriate calculation therefore carries the jointly supported pairs into the ratio, preserving any justified dependence. The result describes what the reserve-relative quantity can actually say under the available information.

Synthetic paths must vary time, capacity, and channel blockage independently and include known continuation sets, support changes, observation error, endpoint uncertainty, and ordinary control, scarcity, frailty, threat, fatigue, or social-support processes. The analysis should identify viable, empty, and ambiguous states without manufacturing a terminal objective where none exists (Bertsekas, 1995; Bobba-Alves, Juster, & Picard, 2022).

9.3 — Addiction, Delay, and Future Optionality

Addiction can shorten practical horizons without changing time itself. Reward learning may favor immediate relief even when later costs are understood. Craving and tolerance can capture attention, while repeated use displaces alternatives. A useful model therefore follows the changing menu of future options rather than treating one choice as the whole story.

9.3.1 — Short-Horizon Reward and Attention Capture

Addiction often contracts attention before it contracts a life plan. A cue or bodily state can dominate the next minutes while delayed costs remain abstract. Temporal discounting represents the reduced weight assigned to delayed outcomes, while reinforcement learning models how experience changes the value assigned to actions (Sutton & Barto, 2018). Incentive-sensitization theory separates wanting from liking: cue-triggered pursuit can intensify even when obtained pleasure declines (Berridge & Robinson, 1998; Robinson & Berridge, 2025). Studies of scarcity suggest that pressing demands can focus attention on some problems while other concerns are neglected (Shah, Mullainathan, & Shafir, 2012; Mullainathan & Shafir, 2013).

Safe interventions can test the account. Removing cues, changing context, improving sleep, treating pain, providing medication, delaying access, or making an alternative immediately available may change attention and choice.

When the future has become unreliable, favoring immediate relief need not be irrational in the simple sense. The practical task is to make safer future goods credible, reachable, and near enough to compete with the present cue (Frederick, Loewenstein, & O’Donoghue, 2002).

9.3.2 — Tolerance, Craving, and the Loss of Ordinary Effort

Tolerance means that repeated exposure can reduce the effect produced by the same dose or behavior, leading some people to seek more for a similar result. The process varies by substance, behavior, context, physiology, and time. It should not be treated as one universal staircase.

Opponent and antireward processes add another layer. Repeated use can recruit stress and regulatory systems that make absence increasingly aversive. Use may then be maintained less by pleasure than by relief from withdrawal, dysphoria, pain, anxiety, or bodily disequilibrium (Koob & Le Moal, 2008; Robinson & Berridge, 2025).

Ordinary effort can lose ground when attention, sleep, health, and motivation are repeatedly consumed. Work, study, cooking, exercise, caregiving, and conversation may demand more energy while the addictive action becomes increasingly practiced and immediately reinforcing.

Changed action costs do not establish that control has vanished; capacities may reappear with treatment or a safer context.

9.3.3 — Opportunity Cost and Moderation

Every repeated action uses time that cannot be spent twice. Acquisition, use, recovery, concealment, travel, waiting, and conflict can occupy hours that would otherwise support sleep, relationships, work, education, health, or rest. The loss is not only the event; it is the field of alternatives that the event displaces.

Opportunity cost becomes visible when the same cycle repeats. A missed appointment, one delayed bill, or one abandoned plan may look small in isolation. Repetition can turn those local costs into unstable housing, lost employment,

untreated illness, legal exposure, or distance from people who once provided support.

Relationships carry another part of the cost. A relative may change work hours to arrange treatment or protect a child. Support can require time and money, while boundaries may protect the relative's own safety and responsibilities.

Work and education can fail through absence, reduced concentration, stigma, discrimination, or rigid systems that offer no path back. Institutional response therefore affects optionality. Punishment that removes housing, employment, health care, or family contact can deepen the very narrowing it claims to correct.

Goal abandonment is often gradual. A person may stop imagining a degree, a relationship, a move, a craft, or a healthier future because the next day already feels occupied. The loss of aspiration is not evidence that the goal never mattered. It can be a consequence of repeated short-horizon pressure.

Recovery capital describes personal, social, material, cultural, and community resources that make longer goals reachable again (Cloud & Granfield, 2008). Clinical treatment, peer support, housing, employment, and belonging may need to work together over years (Kelly et al., 2025). Moderation can preserve time, sleep, attention, money, health, credibility, emotional range, and relationships: resources that permit more routes and recovery from more errors. Its practical benefit is optionality rather than merit rewarded by success (Fujita, 2011).

9.4 — Aging, Illness, Caregiving, and Terminal Scripts

Serious illness is rarely encountered without inherited language about what it means. Families and clinicians may describe a person as struggling, accepting, or unfinished. Such scripts can direct attention and shape what people say or remember. Because the same behavior may receive different interpretations, research must record these contexts without treating a cultural story as a measured endpoint (Kellehear, 2007; Chochinov, 2006).

9.4.1 — Aging, Illness, and Changing Capacity

Aging changes the space of possible action without moving everyone along one path. Strength, sensory function, processing speed, recovery time, memory, health, social role, income, and support change at different rates, while knowledge, emotional skill, judgment, and selective attention may remain stable or improve. A single age variable or one reserve score would erase the heterogeneity the model needs to explain (Baltes, 1997; Carstensen, 2006).

Life-span psychology describes adaptation through selection, optimization, and compensation. People may narrow goals, invest in strengths, use tools, recruit help, or replace a lost route with another. These strategies preserve agency without denying decline (Baltes, 1997; Freund & Baltes, 2002).

Imagine a gardener who wants to keep growing herbs for shared meals. Selection begins with what matters most. Rather than maintaining every former bed, the gardener chooses a smaller collection used often in cooking. That decision changes the scope of the goal. It can concentrate attention on an activity whose place in daily life remains meaningful.

Optimization improves the means already used. The gardener studies the light, changes the watering schedule, and practices a

method of pruning that encourages the desired growth. Time and skill are invested in a chosen route. The aim is to make that route work better for the selected goal.

Compensation introduces a different means when an earlier one has become difficult. A raised planter brings the herbs within comfortable reach. A lighter tool changes how the work is performed. A neighbor carries the heavy bag of soil while the gardener chooses the plants and tends them. The activity continues through a rearranged combination of personal action, equipment, and cooperation.

These strategies can occur together, but they answer different questions. Selection asks which goal deserves a place in the available time. Optimization asks how the chosen means can be developed. Compensation asks what alternative means can carry the goal when the former route is less available. A record that merely counts hours in the garden could miss the structure of the change.

The distinctions also help interpret continuity. The garden may occupy fewer square feet while contributing more reliably to shared meals. Less effort in one part of the activity can release attention for another. Assistance can preserve authorship of the project when the person continues to choose what to grow and how to use it. The relevant accomplishment is organized around the goal, not around keeping every former movement unchanged.

For a study of changing horizons, this offers observable variables: selected goals, investment in existing means, newly adopted means, and the person's own account of what was preserved or gained. The same strategy may take different forms across ages and settings. What makes the framework useful is its ability to describe an active reorganization of life in terms precise enough to compare over time (Freund and Baltes, 2002).

In 3,427 deceased participants from the German Socio-Economic Panel, repeated well-being reports showed steep average declines as death approached, with differences between people and counties (Gerstorf et al., 2010). A Berlin Aging Study analysis found steeper average change when trajectories were indexed by time to death than by chronological age, with differences across cognitive, sensory, physical, and other functions (Gerstorf et al., 2013). The patterns concern reported well-being and domain-specific functioning, which remain distinct outcomes for a horizon model.

Illness can compress capacity more abruptly. Pain, breathlessness, fatigue, cognitive change, medication effects, appointments, financial strain, and uncertainty can consume the attention needed for other goals. The relevant horizon is therefore partly experiential and practical, not only prognostic (Gerstorf et al., 2013; Carstensen, 2006).

9.4.2 — Caregiving, Culture, and Family Scripts

Memory under grief is not mere error. It is human reconstruction under emotional load. It may preserve truth, transform details, compress time, heighten meaning, soften pain, or protect the survivor. A research program should not treat family memory as either pure evidence or useless sentiment.

Cultural scripts vary widely. Some traditions emphasize struggle, vigilance, last confession, unfinished obligations, sacred presence, communal mourning, silence, or bodily signs. A model built around one culture’s peaceful-death script would be biased and ethically narrow (Kellehear, 2007; Kleinman, 1988).

Expression can serve different relational purposes. Some people conceal distress to protect family; others express it to obtain needed care; still others use humor, faith, anger, or

silence. A scientific account must model expressive incentives and dependence before reading demeanor as valence.

9.4.3 — Hospice Observation and Treatment Effects

Hospice accounts should be sampled systematically rather than through only memorable cases. Experienced professionals witness recurring patterns of comfort, agitation, lucidity, fear, family dynamics, and acceptance under care-oriented conditions, but hospice exists to comfort rather than to function as a neutral instrument. Routine observations, counterexamples, and uncertainty belong beside striking stories.

A script can influence experience as well as memory; this does not mean the peace is false.

Hospice populations are also selected. Not every dying person enters hospice. Access varies by country, class, diagnosis, culture, family support, insurance, geography, trust, and institutional referral. A pattern observed in hospice cannot be generalized to all deaths without careful controls.

9.5 — What Horizon Patterns Could Show

The observable record should distinguish a valence trajectory from any proposed reserve proxy. A prospective horizon estimate is a third object, not another name for either. The endpoint rule is a fourth and must be fixed independently of the hoped-for pattern. Keeping these objects separate makes disagreement between them scientifically informative.

9.5.1 — Signals, Confounds, and Stream Limits

Shared sensors or preprocessing can create apparent agreement among these objects even when one artifact drives them all.

A variable can also play different causal roles at different times. Distress may attract care, care may reduce distress, and reduced distress may improve reporting. Longitudinal analysis should state whether a variable is a cause, consequence, mediator, confounder, or part of the observation process. An apparent terminal pattern may arise without a terminal mechanism. Confounding occurs when a third process affects both the estimated horizon and the observed outcome. Illness severity, medication, care setting, family presence, socioeconomic resources, and cultural expectation can all change what becomes visible.

Attrition becomes more informative as illness progresses. Participants may stop because they are weaker, more distressed, hospitalized, sedated, delirious, or dead. When dropout depends on the unobserved outcome, standard complete-case analysis can produce biased trajectories (Diggle & Kenward, 1994; Little & Rubin, 2019).

Selection can begin before enrollment: recruitment after a crisis or after less tolerant people have already withdrawn excludes the earlier trajectory. Discharge, transfer, family burden, and voluntary withdrawal also change who remains

observed; their mechanisms should be distinguished from deterioration and death.

Observation shutdown is broader than dropout. A channel can disappear while the person remains in the study. Speech may fail. Movement may diminish. A wearable may be removed. Family reports may replace self-report. A clinical record may become sparse. Each change alters the meaning of the remaining data.

Statistical remedies require assumptions. Selection models, pattern-mixture models, joint models, inverse-probability weighting, and sensitivity analyses can probe missingness, but none restores information that was never observed without additional structure (Rubin, 1976; Tsiatis, 2006; Kurland et al., 2009).

Collider bias is one specific risk. If both distress and family support influence continued participation, restricting analysis to participants who remain observed can create a misleading association between distress and support. The same logic applies when device tolerance, cognition, or location affects inclusion near the endpoint (Rubin, 1976; Tsiatis, 2006; Kurland et al., 2009).

A small selection model makes the relationship visible. Imagine two binary features, A and B, each present in half of a synthetic population and independent of the other. Before any selection, the four combinations (neither, A alone, B alone, and both) are equally common. Knowing whether A is present gives no information about B.

Now suppose a record is available whenever at least one feature is present. This observation rule retains A alone, B alone, and both, while excluding neither. Within the retained collection, the three combinations are equally common. Among records with A absent, B is always present. Among records with

A present, B is present only half the time. A negative association has appeared.

The arithmetic follows the observation rule. In the retained collection, each feature has mean two thirds. Their joint occurrence has probability one third. The covariance is therefore one third minus four ninths, or minus one ninth. Each binary feature has variance two ninths, giving a correlation of minus one half. The original population correlation was zero.

One can give the variables a concrete research interpretation without changing the calculation. A may indicate that a routine appointment created an observation opportunity. B may indicate that an optional check-in created one. Restricting the record to occasions with any observation makes the two routes appear negatively related, even though the synthetic model assigned them independently. Either route can account for an observed occasion when the other is absent.

The selection indicator is a common consequence of A and B. Conditioning on it creates the collider association. The relationship belongs to the selected collection and its entry rule. It does not require one feature to cause a change in the other. A causal explanation and a selection explanation can therefore make different claims about the same observed association.

Preserving the full observation flow makes the construction inspectable. Counts of eligible occasions and the reasons observations became available distinguish the three retained cells from the fourth, absent cell. A study can then state whether its target concerns observed occasions, all scheduled occasions, or another population. Each is a legitimate descriptive target when named explicitly.

Survivor and observation selection also matter. To appear in a late window, a participant must survive and remain observable long enough to enter it; rapid decline may leave no data there.

Immortal-time bias can additionally arise when eligibility or group assignment depends on future survival while follow-up is counted from an earlier time. The analysis should identify which selection or timing problem applies.

Report sensitivity to assumptions under which missing outcomes are better, worse, or differently related to observed history (Diggle & Kenward, 1994).

9.5.2 — When Horizon Claims Must Narrow or Stop

One loss pattern is absence of the predicted relation under adequate measurement, model recovery, and power. A wide interval may remain inconclusive. A confidence interval wholly inside a protocol-fixed negligible band counts against a prediction that the effect exceeds that band, using the declared equivalence procedure. The horizon measure cannot be replaced after each null, and failure to reach significance cannot be counted as survival (Schuirmann, 1987; Lakens, 2017).

A reversed effect is more informative than a vague null. A shorter horizon may reduce pressure, simplify priorities, or increase emotionally meaningful choice. If that direction replicates and a rival predicts it better, the rising-pressure branch should be revised or retired rather than renamed as confirmation.

Remaining time, reserve, burden, origin, endpoint, and observation quality may be too entangled to estimate separately.

If plausible changes to the endpoint or neutral calibration reverse the inferred horizon effect, report it as specification dependent.

A model may predict that widening capacity changes trajectory features, yet justified treatment or support may produce no such relation. That result counts against the proposed mechanism.

Part IV — From Experience to Evidence

A patient may report less pain after treatment while a wearable still records an elevated heart rate. The records need not conflict: a pain rating and a heart-rate reading describe different features. Before combining them, ask what each channel measures and whether treatment altered their relationship (Raja et al., 2020; Kreibig, 2010; Poldrack, 2006).

An estimate must retain meaning when it travels. Within-person change, between-person comparison, group comparison, and transport across cultures or settings are not stronger versions of one claim. They are different identification problems. A scale can behave consistently within one person yet fail across people. A translated item can preserve its words while changing its social meaning. A physiological feature can change interpretation across medication, disability, age, illness, or device conditions (Meredith, 1993; Millsap, 2011; Putnick & Bornstein, 2016).

Culture and disability do not enter as nuisance variables to be controlled away. They can change the construct, the response process, the available channel, the meaning of a baseline, and the burden of participation. Response shift can alter internal standards after illness or adaptation. The scientifically responsible response is not to call the person inconsistent. It is to ask whether the instrument still means what the study claims it means (Sprangers & Schwartz, 1999; Albrecht & Devlieger, 1999; World Health Organization, 2022).

Artificial intelligence can search, classify, translate, and summarize archives. It can also infer beyond the evidence, reproduce bias, merge contexts, and write a coherent biography that feels more complete than the source. The United Nations Educational, Scientific and Cultural Organization's (UNESCO) 2025 Recommendation on the Ethics of Neurotechnology identifies mental privacy, autonomy, identity, and freedom of

thought as governance concerns when neurotechnology and artificial intelligence can access, infer, or influence brain-related information (UNESCO, 2025). Fluency should increase the demand for provenance, not decrease it (Bender et al., 2021; Weidinger et al., 2021).

A model output should retain lineage to the observations and transformations that produced it. Without lineage, later users cannot distinguish report from inference or current status from a retired result (Gebru et al., 2021; Mitchell et al., 2019).

The human stakes remain close throughout. A child choosing a face on a pain scale, a patient whose medication changes heart-rate variability, a worker declining a wellness app, a person whose disability makes standard tasks inaccessible, and a family deciding whether research data may be retained are not abstractions. They are where measurement assumptions become lived consequences.

The work of this part is to make an observation travel with its meaning attached. Chapter 10 follows the construction of a measure from channels to an estimate. Chapter 11 examines what changes when comparisons move across people and groups. Chapter 12 turns to records, interpretation, and the authority of an account. Read together, they trace a chain from an encounter to a public statement. At each link, a useful question is what a later reader would need in order to understand how that particular statement was reached.

Chapter 10 — From Experience to Evidence

A person says the pain is eight out of ten. A wearable shows elevated arousal and guarded movement. A facial model detects little expression. A brain-based classifier gives an uncertain result. The observations concern different features of the same encounter. Together they form a measurement problem: what exactly is being estimated, from which traces, under which assumptions, and for what use?

Once several channels are available, integration becomes a modeling decision. The model must state whether its output constructs an index, estimates a latent state, predicts an outcome, or combines these roles. Its value depends on whether the assumptions fit the construct, whether performance survives capable nuisance models and new data, and whether uncertainty reaches the conclusion.

Channel qualification proceeds in layers. Technical verification asks whether sensor and pipeline recover the declared feature. Analytical validation asks whether the transformed feature is reproducible and robust under expected artifacts. Construct validation asks whether its prospective sign and temporal relation track valence-relevant change rather than generic illness, movement, or arousal. A channel enters a confirmatory multichannel estimate only after all three layers are documented for its context. For example, accurate recovery of a pulse waveform answers a different validation question from whether its derived feature tracks the intended experience.

A multichannel model can earn value when it predicts a declared outcome in held-out data beyond each component and capable nuisance-only rivals. Suppose animal recordings are assigned positive or negative labels from several contexts per class. Retrain the same pipeline on prespecified, comparably balanced regroupings of those contexts, excluding the original grouping and its simple label reversal. Preserve the animal and

recording structure and use the same splits, with each test animal’s recordings withheld together. If these alternative groupings are also readily classified, accuracy alone cannot establish affective meaning. This diagnostic is inspired by label-permutation methods for investigating classifier performance (Ojala & Garriga, 2010); its context-level construction is a proposed extension. Stronger evidence combines generalization to new animals and contexts with independent links between the recorded features and the affective construct.

Where feasible, test reactivity by varying observation itself: randomize prompt frequency, include a passive-only comparison, separate acclimation from analysis, and predefine burden and withdrawal thresholds. If the measured pattern changes with the schedule, that dependency belongs in the observation model rather than being dismissed as noise (McCambridge et al., 2014).

A useful measurement example can be read backward as well as forward. Starting with an output, ask which observations could have produced it. Then ask which features of those observations survived the transformation and which were combined or omitted. This reverse reading makes the meaning of a summary tangible. It can reveal an overlooked opportunity to retain a channel, preserve a time stamp, or distinguish two response patterns that would otherwise share one number. The chapter follows such choices because they determine what the resulting record can teach.

10.1 — Why No Single Meter Is Enough

Correlation does not establish that a report means what an investigator intends. Researchers must first learn how participants interpret the prompt. Reference periods and scale use can change under fatigue or fear. Language, health, and context may alter the response process even when the underlying experience is similar.

10.1.1 — Self-Report and Dream Data Limits

Cognitive interviewing and ecological momentary assessment can reveal whether a score change reflects experience, interpretation, recall, or the burden of being asked (Cella et al., 2010; Shiffman et al., 2008; U.S. Food and Drug Administration, 2025).

For subjective states, there is often no independent gold standard. Compare reports with other channels without treating disagreement as proof that the participant is wrong (Shiffman et al., 2008; Kahneman et al., 2004).

Protected open-text follow-up, compared with separate positive and negative ratings, can test whether an opposing-poles valence item loses information about mixed states.

Dream reports are reconstructed after waking, not direct records of the dream. A missing or incomplete report does not establish absent experience.

Chapter 15 develops prospective sampling, coding, sleep-state evidence, and the burden and privacy limits of dream research.

10.1.2 — Evidence in Its Context

A concrete example shows both the promise and the specificity of neural measurement. Shirvalkar et al. (2023) studied four people with refractory neuropathic pain using implanted recordings and repeated pain reports over months. Neural

activity predicted within-person chronic pain severity scores, while the patterns associated with chronic pain differed from those associated with acute evoked pain. The result joined reports to a context-specific neural signal rather than replacing testimony with a universal meter.

For this inquiry, the transferable lesson is longitudinal validation: test a channel against the defined within-person contrast and timescale it is supposed to inform.

A channel should enter only when it improves a declared inference beyond the least burdensome valid alternative in held-out conditions. Small predictive gains do not justify exclusion, household capture, opaque interpretation, or unusable instruments (U.S. Food and Drug Administration, 2025).

Skipped prompts, withheld reports, removed wearables, declined scans, or withdrawal may reflect privacy, fatigue, pain, culture, disability access, surveillance concerns, or inconvenience. They may be modeled as observation processes but not converted into labels of distress, deception, instability, or worth (Little & Rubin, 2019; Council for International Organizations of Medical Sciences, 2016).

Accessibility requires alternative valid routes such as voice, symbols, assisted communication, flexible timing, chosen devices, nonwearable options, and bounded caregiver support (World Health Organization, 2022).

For indirect sensing, compare what each candidate channel would reveal about nearby people before selecting the least revealing information the study needs; prefer local processing (Nissenbaum, 2004; Nissenbaum, 2010).

A participant-facing record could show the observation used, its date, the resulting estimate, and the person responsible for reviewing a correction. Keep observations, features, estimates,

permissions, corrections, and public artifacts as separate layers with separate rules.

10.2 — From Experience to a Measured Construct

A sensor cannot define the construct it is supposed to measure. The target might be momentary valence, pain intensity, or relief, each of which has different anchors. Mixed affect and functional burden pose further questions rather than interchangeable labels. Define the purpose and estimand before deciding which channel should collect the data.

10.2.1 — Start With the Construct, Not the Sensor

Reports, sensors, observations, and clinical context then enter as candidate relations to that construct rather than as ready-made measures.

An indicator may be an effect of the construct, a component of a stipulated composite, a possible cause or a correlate. An increased heart rate, for example, can accompany fear or exercise. Construct validity needs an account of the indicator's role, not merely its correlation with other channels (Cronbach & Meehl, 1955; Borsboom, 2005; Flake & Fried, 2020).

Calibration evidence has to do more than agree with the estimator after the fact. It must specify which report window, response scale, mixed-state instruction, language version, and burden condition are being used to test the physiological contribution.

The report battery should be deliberately plural during development. A single opposing-poles valence item, separate positive and negative items, arousal, pain, agency, safety, and meaning can compete in the same validation study. If the single scalar consistently loses held-out information needed to interpret physiological channels, that is evidence for a multidimensional or domain-specific bridge rather than a reason to hide the richer reports (Cronbach & Meehl, 1955; Messick, 1995).

For the present architecture, reports serve as protected validators and sign-orientation witnesses. They can support a prospective direction rule, expose ambiguity, and reveal when another channel tracks arousal, illness, or response style rather than valence. They should neither be demoted whenever a device disagrees nor made constitutive merely because external channels are incomplete. A change in their formal role requires declared change control and fresh validation.

Fit report relations at their intended level: within-person change, between-person differences, or person-specific heterogeneity.

Each qualified channel needs a record describing its biological rationale, sensor and preprocessing, anchor, sign evidence, nuisance set, reliability, lag, missingness rule, context of use, and failure threshold. Redundant features and incomplete records remain exploratory rather than being folded into a maximal sensor stack (FDA-NIH Biomarker Working Group, 2025; U.S. Food and Drug Administration, 2025).

10.2.2 — Calibration Is Evidence, Not Direct Access

Calibration asks whether a channel bears a stable, interpretable relation to a declared construct in a defined context of use. It can establish responsiveness, ordering, discrimination, or prediction under tested conditions.

Behavioral calibration must begin with the feasible menu. Prospective contrasts can still test latency, reversal, persistence, or adaptation after independently identified events, provided the prediction names the action, interval, observation effect, and rival explanation (Shiffman et al., 2008; McCambridge et al., 2014).

Behavioral validation should report uncertainty and include people whose burden is not outwardly expressive. Silence,

stillness, compliance, productivity, withdrawal, and help-seeking should remain ambiguous when context is insufficient. Reliability studies and stratified validation across masking, communication style, disability, and cultural display norms can show whether a model measures behavior, expression, or a broader relation to experience rather than assuming those objects are identical.

Neural calibration has a concrete starting point. Using fMRI during sustained pleasure from chocolate and pain from capsaicin, Lee and colleagues (2024) developed separable patterns predicting reported affective valence and intensity, then tested them in an independent participant sample. Their principal valence analysis used signed ranks after matching positive and negative rating distributions. The result supports a candidate measurement channel; the next step is to test a frozen model against independently calibrated magnitudes, including cases that separate sign from intensity, salience, and action demands. A decoder's accuracy and its suitability for summing experience are distinct validation targets. Any neural features shared with an endpoint or unity proxy must also be declared so apparent independence is not created by overlapping sensors or preprocessing.

Neural and other high-dimensional channels should be tested across sites and within persons where possible. Hardware, preprocessing, movement, implants, age, disease, medication, and operator practice can dominate subtle affective patterns. Harmonization should separate technical variation from participant variation without erasing biology, and a pattern that survives only in one homogeneous laboratory remains a local result rather than a terminal or cross-population measure (Poldrack, 2006; Wager et al., 2013; Yarkoni, 2022).

10.2.3 — The Timing of a Measurement

A perfect reading can still belong to an uninformative schedule. Suppose a simulated instrument watches a dial that alternates between zero and two at successive integer times. At even times it reads zero; at odd times it reads two. Each value is held until the next integer time. A second simulated dial remains at zero throughout. An observer who samples only at times zero, two, four, and six receives exactly the same record from both dials.

Increasing the precision of those readings does not separate the models. Recording zero to six decimal places still reveals nothing about the odd times. Extending the same schedule to time one hundred gives a longer record with the same ambiguity. The missing information is temporal: the schedule repeatedly visits one phase of the changing process.

Sampling at time one resolves the distinction between these two specified candidates. So does shifting the regular schedule by one unit. Sampling at every integer time reveals the alternating pattern directly. This is a simple instance of aliasing: distinct underlying time courses become indistinguishable under the observation schedule. The data-generating process includes both the changing dial and the rule deciding when its value enters the record.

The same idea applies when a model contains hidden states. A hidden-state model specifies how an unobserved state changes and how its current state produces an observable output. Two such models can agree at the scheduled observations while disagreeing between them. A useful next observation is therefore chosen with regard to the timing of their disagreement. More observations at the old phase and an observation at a new phase perform different scientific work.

A study can investigate this before asking participants for more frequent responses. Synthetic trajectories can be passed

through the proposed schedule. Investigators can then compare regular spacing, shifted spacing, and bounded variation in prompt times under the same total number of observations. The aim is to discover which features the schedule can recover. A schedule that preserves a daily level may provide little information about the duration or ordering of shorter episodes.

10.2.4 — What Can Fit Between Two Readings

Two accurate readings can frame an interval without revealing everything inside it. Suppose an artificial dial reads zero at time zero and again at time two. The instrument is silent in between. A straight line through the two observations suggests a zero contribution, but a dial that rises and returns produces exactly the same recorded endpoints. The useful question is how much room the missing interval leaves for such a journey.

Give the model one additional property: the dial can change by at most one unit per unit of time. This is a bound on its speed of change, not a bound on its value. At time t, the starting observation permits a value no greater than t. The ending observation also matters. To return to zero by time two, the dial can be no higher than 2 − t. Its greatest admissible value is therefore the smaller of those two numbers.

The upper envelope forms a tent. It rises from zero to one at time one and falls to zero at time two. The lower envelope is its reflection below zero. Every path satisfying the two observations and the speed bound lies between them. Integrating the upper tent gives one: its base is two, its height is one, and its area is half their product. The lower tent has signed area minus one. The unknown interval's accumulated contribution consequently lies between minus one and one.

These limits are attainable. The upper tent itself obeys the speed bound, as does the lower tent. Multiplying either tent by a

constant between zero and one produces every intermediate area of the corresponding sign. Thus the interval from minus one to one is the exact identified set under this model. The straight interpolation has selected its midpoint, but the observations and speed bound have established a range.

Now obtain a third reading at time one, and suppose it is also zero. Each half of the original interval has zero endpoints one time unit apart. Its upper tent has height one half and area one quarter. The two halves together can therefore contribute at most one half in either direction. One well-placed reading has halved the width of the possible accumulated total, even though all three displayed values are identical.

The position of that extra reading matters. If a new zero is observed at time a, between zero and two, the two gaps have lengths a and 2 − a. A gap of length h with zero endpoints and speed bound K has maximum area K h²/4. With K = 1, the combined upper bound becomes [a² + (2 − a)²]/4. It is smallest at a = 1. Moving the extra reading toward either endpoint leaves a longer unobserved gap whose squared length dominates the uncertainty.

With n equal gaps across a fixed duration H, zero readings at all gap boundaries, and the same speed bound K, the total upper bound is K H²/(4n). This calculation describes one precise benefit of denser observation. Its gain depends on the temporal regularity being assumed and on the observations actually obtained. A future reading that differs from zero changes the envelopes rather than invalidating the method.

For unequal endpoint readings, the same construction still works. Each endpoint gives an upward and a downward cone of reachable values. The upper envelope is the smaller of the two upward bounds; the lower envelope is the larger of the two downward bounds. Integrating those envelopes carries both

observations into the total. When the endpoints themselves have uncertainty, their permitted intervals can be included in the construction.

This approach gives interpolation a clear place in the analysis. A fitted curve supplies one proposed history; envelopes identify what a stated family of histories allows. In a study of experience, the scientific work would include establishing an appropriate regularity assumption and assessing its stability across the relevant states. Once that work supports a bound, the bound can turn a missing interval into a defined contribution to uncertainty and help locate the next informative observation.

10.2.5 — Giving Each Interval Its Proper Weight

An observation schedule can give some intervals more chances to appear than others. Imagine a synthetic day divided into twenty-four equal intervals. Ten intervals belong to a period the investigator wants to examine closely; the remaining fourteen belong to the quieter part of the schedule. The instrument records all ten intervals in the first period and selects two of the remaining fourteen at random, with every pair equally likely. It produces twelve observations, but those observations represent unequal shares of the day.

Suppose the artificial signal equals two throughout the first period and one throughout the second. Its complete daily total is thirty-four: ten contributions of two and fourteen of one. The time average is therefore thirty-four divided by twenty-four, or about 1.42. Averaging the twelve recorded values directly instead gives twenty-two divided by twelve, or about 1.83. The more closely watched period has acquired greater influence merely because it was watched more closely.

The selection design supplies a way to recover the intended weighting. Each first-period observation represents one interval

because every such interval was recorded. Each selected second-period observation represents seven intervals because two observations stand for fourteen equally eligible intervals. Multiplying their values by those weights gives twenty plus seven plus seven: the complete total of thirty-four. Dividing by the twenty-four intervals gives the intended time average.

Here the reconstruction is exact because the signal was constant within each period. The same weighting rule also has a useful property when the second period varies. Each of its fourteen intervals has probability one seventh of being selected. If an interval's fixed value is v, it contributes seven times v to the estimate when selected and nothing when unselected. Averaging that contribution over possible selections gives one seventh times seven v, which is v. Adding across intervals shows that the expected estimated total equals the full total. The randomness belongs to the observation schedule; the synthetic day's values can remain completely fixed.

A variable second period makes the remaining sampling uncertainty visible. Keep the first ten values at two, but let seven of the other values be zero and seven be two. The day's total is still thirty-four. If the two randomly selected intervals both contain zero, the weighted estimate is twenty. If one contains zero and the other two, the estimate is thirty-four. If both contain two, it is forty-eight. Each result follows the same valid weighting rule, while the selected pair carries different information about the unobserved intervals.

There are ninety-one possible pairs among fourteen intervals. Twenty-one pairs contain two zeros, forty-nine contain one of each value, and twenty-one contain two twos. The corresponding estimates therefore average to thirty-four across the randomized design. That calculation establishes unbiasedness for the total. It also shows why unbiasedness and

exact recovery of this particular day's history are different achievements. A report can preserve both the weighted estimate and the uncertainty created by observing only two of those fourteen intervals.

The number of observations and their allocation now become purposeful design choices. Recording four of the quieter intervals instead of two gives more information about variation within that period. Moving an observation from a nearly constant period to a variable one may improve the estimate of the daily total. A design seeking the shape of a brief episode could rationally use another allocation. The target determines which features need the most attention, while the weighting keeps the selected target explicit.

The same reasoning explains why selection probabilities belong beside timestamps in the record. A timestamp says where an observation sits in the day. Its probability of selection says how that observation entered the sample. Together they connect the observed values to the collection of intervals the study intends to describe. Randomization can also be stratified by waking periods, activities, or other independently recorded conditions, provided the design preserves the relevant selection probabilities.

In a study of experience, scheduled selection and successful response are separate stages. The calculation above covers a deliberately chosen schedule with every selected reading obtained. A study can retain that transparent design while separately modeling whether a selected prompt receives an answer. This separation gives each methodological improvement a definite job: scheduling decides which intervals are invited into observation, response procedures determine which invitations yield usable records, and the analysis carries both processes into the declared time-based quantity.

10.2.6 — Unsafe, Missing, or Unusable Channels

State-dependent gaps require the missingness analyses in Section 9.5.1; last-observation carryforward is not a neutral default (Rubin, 1976; Little & Rubin, 2019).

Audio may capture other people, location may expose a shelter or clinic, sleep prompts may disrupt care, and sensors may irritate skin or interfere with equipment. Nor can a fixed estimate quietly become a different composite each time a channel disappears. A reduced-channel estimate requires a validated version or defensible linking study, with group-specific audits of where absence concentrates (FDA-NIH Biomarker Working Group, 2025; U.S. Food and Drug Administration, 2025).

10.3 — Combining Channels Under Uncertainty

Integrating channels requires more than averaging their outputs. Each channel must name the object it is meant to measure. Its sign and sensitivity may change with state or observation history. A declared model must explain how those channels jointly support the estimand and when their disagreement warrants abstention.

10.3.1 — Why Integration Requires Modeling

The measurement map and the analysis map should remain separately versioned, because defining what is to be measured is not the same act as estimating it from sampled records. This architecture makes assumptions auditable without claiming that the latent state has already been identified (Bollen & Lennox, 1991; Bollen & Bauldry, 2011).

Two channels can test more than average agreement. Let D denote an accumulated contribution or a path residual in a fixed-exposure, within-person regime. Write $Y_j = a_j D + b_j + E_j$ for channels j = 1, 2, with independently justified positive gains, fixed offsets, and errors E_j. If all variables have finite variance and the errors are uncorrelated with D and with each other, $Cov(Y_1, Y_2) = a_1 a_2 Var(D)$. Channel-specific noise raises each variance without raising this cross-covariance.

For example, residuals of +1 and −1 occurring equally often average to zero while having variance one. A test of their mean misses the pathwise discrepancy. With two suitable channels, residual variability can be examined without treating all observed variation as signal. Shared artifacts still matter: if their total covariance contribution is independently bounded in magnitude by κ, a lower bound c on the true channel covariance gives $Var(D) \geq max\{0, \frac{c - \kappa}{a_1 a_2}\}$. A statistically supported c carries its

stated coverage rather than a deductive guarantee. The value comes from justified error differences, not from agreement between channels alone.

For a numerical illustration, let a hidden quantity vary with mean zero and variance one. The first channel reports twice that quantity plus its own error. The second reports three times the quantity plus a separate error. Suppose each error has mean zero and variance nine, and both are independent of the hidden quantity and of one another. All of these are stipulated features of the constructed system.

The first channel has variance thirteen: four from twice the hidden quantity and nine from its error. The second has variance eighteen: nine from three times the hidden quantity and nine from its error. Their covariance is six, the product of their gains and the hidden variance. The independent errors enlarge the separate variances while contributing nothing to the cross-covariance.

Consequently, the channels have a correlation of only about 0.39, despite sharing the same hidden quantity with known positive gains. A modest correlation is compatible with a clear common signal when separate noise components are large. Dividing the covariance of six by the gain product of six recovers the hidden variance of one within this model.

Shared error changes the interpretation. If both channels inherit the same calibration fluctuation, some covariance can come from that fluctuation. The lower-bound calculation therefore asks how much covariance remains after allowing for the independently bounded unwanted contribution. If the true observed covariance is at least nine, the magnitude of the unwanted covariance is at most three, and the gain product is six, the hidden variance is at least one.

Each number has a distinct evidential role. The gains connect the channels to the target. The covariance describes how their outputs vary together. The error bound limits alternative sources of that joint variation. Increasing the number of observations can sharpen a covariance estimate, while independent calibration is needed to justify the gains and the error bound.

This offers a constructive way to choose an additional channel. A new instrument is useful when it brings a sufficiently different error structure and a defensible relation to the same target. Merely duplicating the first instrument's processing can increase the volume of data without providing the separation used in the calculation. The design question is which new observation would help distinguish shared target variation from shared measurement variation.

Channels differ in sampling rate, unit, noise, latency, and relation to the target. Heart features may arrive each minute, endocrine measures each hour, reports several times per day, and neural measures only in short sessions. Pooling all raw observations into one unweighted average would give the densest channel the greatest influence and treat incomparable scales as commensurable.

Weights require an explicit convention and cannot repair an unidentified sign. A study may normalize weights, fix a reference channel, or preserve declared scaling, but the choice changes the meaning and units of the composite. Eligible, degraded, unsigned, and absent channels must also be distinguished (Krantz et al., 1971; Hand, 2004).

Weight estimation should be nested inside a development and validation design. Tuning on the full dataset and then reporting the same data as validation produces optimistic performance. Cross-validation helps, but person-level, site-level, and time-

block leakage must be prevented. A future observation from the same participant is not an independent external test.

Sensitivity analysis should vary scientifically defensible transforms, weights, and personalization rules rather than report one optimized configuration as natural. A person-specific mapping may improve local prediction while reducing comparability and increasing overfitting, so the rule itself needs held-out validation. The useful output is the range of conclusions across a prospectively specified admissible set. If sign, equivalence, or model ranking changes inside that set, the result is configuration dependent (Bland & Altman, 1986; de Vet et al., 2011).

Complex models earn their place only when they improve held-out calibration, missing-data handling, and interpretation beyond a simpler baseline. A state-space or other flexible model may represent lags and changing channels, but numerical convergence does not show that its latent dimensions were observable from the available data. Simulations, rank checks, posterior geometry, and comparison with simpler models should reveal whether sophistication recovered information or merely supplied a smoother story.

Time can supply an additional observation dimension even when the instrument has only one output. Imagine two internal components, x and z, whose initial values are unknown. Over one time step, the first retains half its value and the second retains one quarter. The instrument reports only their sum. Assume these update rules and the instrument's unit have been established independently for the constructed system.

Suppose the first reading is three and the reading one step later is one. The first equation is x + z = 3. The second is x/2 + z/4 = 1. Multiply the second by four to obtain 2x + z = 4. Subtract the first equation, giving x = 1 and therefore z = 2. A pair of

readings has separated two quantities that a single sum left unresolved.

The initial components predict the next reading as well. After two steps, they are one quarter and one eighth, so their sum is three eighths. That next reading can be reserved to assess the frozen construction. The first pair identifies the initial state under the model; the additional time point asks whether the inferred components continue to evolve as the model predicts.

Different dynamics are doing the separating. If both components retained half their values, the second reading would always be half the first. Starting from a total of three, every pair of components with that total would give the same sequence: three, one and a half, three quarters, and so on. More readings would describe the total's decay precisely while leaving its division between x and z undetermined.

More generally, suppose the retention factors are a and b. The first two readings are $y_0 = x + z$ and $y_1 = ax + bz$. When a differs from b, x equals $(y_1 - by_0)/(a - b)$, and z is the remaining part of y_0. The denominator identifies the source of separation. The two components must leave distinguishable temporal traces.

The same denominator describes sensitivity to error. If a and b are close, a small error in either reading can produce a much larger error in the reconstructed component. With factors 0.50 and 0.49, an error of 0.01 in the second reading alone changes the reconstructed x by one whole unit. With factors 0.50 and 0.25, the same error changes x by 0.04. Both systems are algebraically identifiable, while their practical resolutions differ greatly.

A useful design can respond by choosing observation times that better separate the predicted traces, improving measurement precision, or adding a channel with a different relation to the components. The appropriate choice depends on the dynamics

and the question. Reconstructing every component may be unnecessary if the target is a particular combination that the instrument already measures well.

10.3.2 — Error, Noise Floors, and Confidence

An estimate is useful only when its uncertainty is described at the same level as the claim. A single margin of error cannot carry every weakness in a multichannel system. The error budget must separate sampling variation, instrument noise, calibration error, temporal misalignment, missing observations, model choice, transport failure, and uncertainty about the construct itself.

A measurement system can lose detail in several ways. A reading may be noisy, rounded, clipped, or biased, and each problem behaves differently. Repeated measurements can reveal a stable difference smaller than the noise in any one reading when the errors vary in a well-understood way. They cannot by themselves recover detail the instrument never records, identify an unknown systematic bias, or reconstruct missing parts of one person’s history. The relevant limit is what this instrument and analysis can resolve for this particular claim.

Confidence language should match that distinction. A confidence interval or credible interval is conditional on a model, data treatment, and sampling or prior assumptions. It is not the probability that the entire interpretation is correct. Public copy should say what the interval covers and which uncertainties remain outside it (Wasserstein & Lazar, 2016; Wasserstein et al., 2019).

Comparative evidence for an exact-zero model is a separate question; the worked example appears in Section 27.2.1.

Accumulation magnifies some errors in a calculable way. For an equally weighted discrete estimate based on n observations

separated by Δt, the cumulative measurement error is Δt times the sum of the momentary errors. Its variance is $(\Delta t)^2$ times the sum of all pairwise error covariances. Independent errors of variance σ^2 therefore give a cumulative standard deviation of $\sigma \Delta t \sqrt{n}$, while a constant origin error b shifts the total by bnΔt. At fixed cadence, the first grows with the square root of record length and the second grows linearly. Serial dependence can change that contrast, so neither an independence shortcut nor an assumption that all error cancels is adequate. The whole-stream estimate needs a covariance model, independently assessed origin uncertainty, and explicit treatment of gaps and measurement-version changes.

Calibration drift should be visible. Devices age, software changes, prompts are revised, participants learn the task, and clinical context evolves. A versioned calibration record should show when the observation system changed and whether old and new values remain comparable. Silent recalibration can manufacture an apparent trajectory.

Confidence should also govern use. A research estimate that is adequate for comparing methods may be inadequate for individual feedback, clinical interpretation, public communication, or any consequential decision. The context of use determines how much uncertainty is tolerable and which kinds of error are unacceptable (U.S. Food and Drug Administration, 2025).

10.3.3 — Scientific Validation

Validation must justify the link between a defined construct, observations, transformations, and proposed interpretations, decisions, and consequences. The argument must be revisited when the instrument, population, setting, or purpose changes

(Cronbach & Meehl, 1955; Messick, 1995; American Educational Research Association et al., 2014).

For between-person results, examine consequences for stigma, funding, clinical attention, media narratives and institutional decisions.

Accessible pilot formats and local language review can reveal whether participants interpret anchors as intended. An item that appears clear to researchers may evoke social desirability, threat, moral judgment, illness appraisal, or a different time horizon in the population being studied (U.S. Food and Drug Administration, 2025).

Reliability must be estimated at the time scale and context of use. Stability across ten quiet minutes does not establish stability across days of changing illness, sleep, medication, or activity. Variance should be decomposed across person, occasion, device, site, and condition so the study knows whether additional sampling can reduce the error or whether the error is systematic. The measurement-property framework of Mokkink et al. (2010) supplies a complementary instrument-level evaluation.

The scale of variation in the cases also changes what a reliability coefficient means. Imagine an instrument used to measure a fixed artificial target twice. For each reading, the instrument adds an independent, zero-mean error with variance one. The target itself is unchanged between readings. Across a broad collection of targets, suppose the true values have variance nine.

Each reading then has variance ten: nine from differences among the true targets and one from its own error. The covariance between the first and second readings is nine because both contain the same target value, while their errors are independent. Their correlation is therefore nine tenths. Repeated readings track the broad differences among targets closely.

Now use the identical instrument with a narrower collection whose true values have variance one. Each reading has variance two, and their covariance is one. Their correlation is one half. The instrument's error distribution has not changed. What changed is the amount of target variation against which that error is being compared.

The difference between repeated readings reveals the shared absolute error scale. The unchanged target cancels from the difference, leaving one reading's error minus the other's. In both collections that difference has variance two. Thus the repeated-reading correlation changes from 0.9 to 0.5 while the variation in paired differences remains the same. These summaries describe complementary features of the observation system.

This distinction is useful when a measure moves from a broad study to a more homogeneous group. An instrument can preserve its absolute precision yet yield a lower reliability coefficient in the second setting. Conversely, a high coefficient in a highly varied sample can coexist with error too large to resolve the small within-person changes of interest. The scientific task determines which scale of distinction matters.

A study can make the relation visible by reporting the spread of the targets, the spread of repeated differences, and the time separating the repeats. If the target genuinely changes between readings, that change also enters the difference and needs its own model. In the fixed-target construction, the components are known exactly; in an experience study, repeated occasions and independent information help identify which component each observation contains.

Factor direction and scale can often be changed mathematically without changing fit. Shared sensor artifacts can create false common structure. Loadings may shift with treatment, disease, device replacement, or response shift.

External anchors, residual checks, and competing stable versus time-varying models are therefore needed before a fitted factor receives experiential sign, origin, or continuity (Bollen & Lennox, 1991; Bollen & Bauldry, 2011).

An additional channel can contribute a test of the measurement structure itself. Consider four readings, each modeled as its own loading times one shared latent variable, plus channel-specific error. Let the shared variable have variance one. Assume the errors are uncorrelated with that variable and with one another. For the illustration, use loadings one, two, three, and four.

The covariance between any two different readings is the product of their loadings. The first and second therefore have covariance two; the first and third have covariance three; the first and fourth have covariance four. The second and third have covariance six, the second and fourth eight, and the third and fourth twelve. Error variance within an individual channel can change its total variance without changing these cross-channel covariances under the stated model.

Now form three products using all four channels once in each product. The covariance of channels one and two, multiplied by that of channels three and four, gives $2 \times 12 = 24$. Pairing one with three and two with four gives $3 \times 8 = 24$. Pairing one with four and two with three gives $4 \times 6 = 24$.

The equality is general for this model. Each product contains all four loadings and the square of the latent variance. Consequently, subtracting any two products gives zero. These are often called vanishing tetrad restrictions. They are consequences that can be examined in observations, rather than parameters separately adjusted to fit each pair (Bollen & Ting, 2000).

Suppose the first five covariances retain the values above while the third–fourth covariance is thirteen rather than twelve. The first product becomes twenty-six and the other two remain twenty-four. The full covariance pattern then asks for an additional component beyond the declared one-factor, uncorrelated-error structure. A shared artifact affecting channels three and four is one possible component; a second latent dimension is another.

Those alternatives can motivate distinct follow-up measurements. A technical reference may reveal the shared artifact. A condition designed to change a second construct may reveal the extra dimension. The covariance restriction locates a particular inconsistency, and the next observation can be chosen to distinguish its explanations.

Convergent prediction is not enough. A model following reports may still learn arousal, nociception, exertion, sedation, fever, illness severity, identity, location, or prompt wording. Discriminant outcomes, feature ablation, context shifts, and counterfactual stress tests should distinguish the intended relation from an easier proxy. Under a narrower name, prediction may remain useful but cannot replace construct validity.

Use Section 23.3.1’s recovery procedure with formative, reflective, multidimensional, generic-arousal, illness, and missingness generators. Report selection errors, parameter bias, interval coverage, false equivalence and defeat, convergence, calibration, and held-out performance; have competent reviewers check rival implementations. Parsimony concerns assumptions as well as fit. Count parameters, site-specific adjustments, and dependence on proprietary components; then test whether predictive gains survive transport. Burden and

accessibility remain separate constraints, already treated in Section 10.1.2.

If equally admissible transforms or model structures imply different cumulative contributions, the target remains nonidentified. Report only the narrower index or bounded set the data support (Krantz et al., 1971; Hand, 2004).

10.3.4 — Aligning the Time Represented by a Signal

Two accurate channels can disagree because they refer to different times. Imagine a synthetic signal that is zero until time eight, equals one from time eight to time ten, and returns to zero afterward. One channel records the signal immediately. A second records exactly the same signal two time units later. Over a recording window from zero to ten, the first channel contains an area of two. The second contains an area of zero: its positive interval begins just as the recording window ends.

Both channels faithfully represent the same two-unit event under their declared timing rules. Their difference comes from which part of the event the window includes. Extend the second channel's recording through time twelve and its complete area is also two. Align its timestamps with the event times they represent and the agreement becomes visible within a common event window.

An average over an interval introduces a different timing relation. A report about the previous hour and a sensor reading taken at the end of that hour can share a timestamp while referring to different objects. One summarizes an interval; the other describes a moment. If the process changed sharply during the hour, neither observation supplies the other's value. The analysis can use both by representing their distinct windows explicitly.

This is why the question attached to a timestamp matters as much as the timestamp itself. Does it mark the event, the start of a measurement, the end of an averaging window, or the moment a stored result reached the system? A delayed upload may preserve a correct event time. A correctly synchronized device may still produce a deliberately smoothed measurement. Clock agreement and observation agreement are separate achievements.

The distinction becomes especially useful when comparing a rapid channel with a slower one. A known delay can be incorporated into the observation model. A known averaging window can be represented as a weighted contribution from several earlier moments. An uncertain delay can be varied within its independently supported range to show which temporal comparisons remain stable. These operations give a disagreement a concrete explanation to test.

The synthetic example belongs to measurement architecture: its numbers are arbitrary signal units. Its contribution to the Law's inquiry is precise. Before interpreting a channel as leading, following or countering another, establish which moments each observation represents. A temporal relationship can then become evidence about the process itself, supported by a common account of time.

10.3.5 — When One Signal Can Reproduce Another

Two instruments can be compared by asking what an observer could do with their outputs. Sometimes one instrument contains enough information to reproduce the other through a known transformation. That relationship is stronger than a favorable score on one particular task. It allows an observer using the first instrument to imitate any strategy based on the second, because

the second instrument's entire output distribution can be generated from the first.

Consider a hidden binary state, either zero or one. Instrument A reports the correct state with probability 0.9 and the opposite state with probability 0.1, symmetrically for either true state. Instrument B reports the correct state with probability 0.7. The instruments are defined by those conditional probabilities; the argument does not depend on how frequently the two hidden states occur in a population.

An observer receiving A can manufacture an output with B's distribution. After reading A, independently flip its reported bit with probability one quarter. The manufactured output is correct when A was correct and no flip occurs, or when A was wrong and a flip repairs it. Its probability of being correct is 0.9 times 0.75 plus 0.1 times 0.25, which equals 0.7. The same calculation holds for either hidden state.

This gives an explicit procedure for reproducing B from A. Suppose a decision maker has a rule that chooses an action after reading B, perhaps with additional randomization. A user of A can first apply the one-quarter flip, then use exactly that rule. Conditional on either true state, the resulting actions have the same distribution as they would under B. Any expected payoff achieved by that B-based rule is therefore achievable from A as well.

The conclusion concerns the best attainable payoff for a fixed decision problem. Because A can imitate every B-based strategy, optimizing over all A-based strategies cannot give a lower expected payoff than optimizing over B-based strategies. The user of A can always choose the imitation if no better rule is available. This is a direct finite example of comparing experiments by whether one signal is a garbling of another.

For ordinary state classification with equal penalties for either kind of mistake, following A gives error probability 0.1 and following B gives 0.3. Yet the stronger result does not depend on those equal penalties. An observer might face an expensive false positive, an expensive false negative, or an option to wait. The imitation argument continues to hold because it reproduces the conditional action distributions before any particular payoff is attached.

The channel model matters to the comparison. The flip used to manufacture B is independent of the hidden state. An observer does not get to consult the answer while altering the signal. If reproducing the second instrument required knowledge unavailable to the first instrument's user, the proposed imitation would not establish that informational relationship. The construction is useful precisely because every step can be performed using the observed output and an independent random draw.

Some instruments cannot be placed in this order. Imagine an object with a color and a shape, each independently binary and equally likely. One instrument reports color perfectly but says nothing about shape. Another reports shape perfectly but says nothing about color. For identifying color, the first is superior. For identifying shape, the second is superior. Neither output can be transformed into the other's state-dependent information without an additional source of knowledge.

The two instruments are incomparable across those tasks, even though each can be perfectly accurate about its own target. Combining their outputs reveals both properties. That gain differs from averaging two noisy readings of the same quantity. It expands the distinctions the observer can make about the object. A channel's usefulness therefore depends partly on which

distinctions its output carries and which decision or scientific question requires them.

This is especially relevant when reports, behavior, and physiology have different relations to a latent construct. One channel may be highly informative about arousal and another about an explicitly reported distinction. Calling either globally better would leave the target unspecified. A formal comparison can instead ask whether a declared observation model permits one channel to reproduce another, or whether they contribute different information that remains useful in combination.

The same idea guides simplification. If a complicated recorded output can be reduced to a smaller signal that preserves every distinction needed for a declared task, the smaller representation may support the decision efficiently. If the reduction merges states that call for different responses, some capability is lost. The question is concrete: which action distributions or predictions remain available after the transformation?

An informative measurement program can thus describe its channels by more than a single ranking. Some pairs have a clear informational order. Others illuminate different properties and belong in a joint account. The worked flip makes the first relation visible, while the color-and-shape example makes the second visible. Together they show how to ask what one observation can substitute for, what it can supplement, and what becomes possible only when the relevant distinctions are preserved.

10.4 — Missingness, Reactivity, and Honest Failure

Measurement reactivity arises when observation changes experience or behavior. Repeated ratings can draw attention to changes in enjoyment and alter the experience being measured (Hong & O'Brien, 2026). A reporting-frequency experiment could compare common assessment times across frequent and sparse schedules, with matched breaks and unrelated questions helping distinguish reflection from interruption. Record the time consumed by each prompt and check whether schedule effects appear in independently validated channels, not only in reports. Carry any schedule-related uncertainty into the cumulative estimate. The observation procedure then becomes a tested part of the model rather than an invisible assumption.

10.4.1 — Blinding Against Hope

Blinding protects interpretation as well as analysis. Researchers who hope to find hidden burden or relief may ask different follow-up questions, notice different behavior, choose favorable channels, adjust weights, or resolve ambiguous records in the preferred direction without intending deception. Separating outcome knowledge from coding, adjudication, calibration, and model selection reduces that pressure and leaves a clearer trail wherever judgment remains unavoidable (Nosek et al., 2018; Open Science Collaboration, 2015).

Analysts can be blinded to condition, hypothesis, participant group, or expected direction where possible. Coders can rate reports or dreams without knowing the study prediction. Reviewers can examine model outputs without knowing which version favors the Law. Full blinding will not always be possible, but partial blinding is often better than none.

For a specification-curve analysis, define the admissible set of analyses in advance and report its full distribution of results

rather than one preferred specification. This preserves scientific pluralism without turning the analysis into an unlimited multiverse after outcomes are known (Steegen et al., 2016; Simonsohn et al., 2020).

Record channel exclusions and failures alongside admissions. Define technical and ethical exclusion rules in advance, preserve participant and interval flows, and examine whether losses cluster by disability, material access, or distress. Permit independent challenge before collection begins (World Medical Association, 2024; Chan et al., 2025).

10.4.2 — Null Results

A well-powered failure of sign stability, incremental validity, within-person association, or held-out transport bounds the corresponding measurement claim. When a null instead reflects poor timing or channel failure, the next study should repair that specific defect before repeating the same contrast.

Equivalence and noninferiority questions need prospectively justified margins. A statistically detectable effect smaller than the minimum useful improvement may not justify the added burden of a multichannel estimate.

Red-team review makes criticism structural. It should include competent scientific rivals and, where relevant, clinicians, disability and trauma experts, privacy and data-governance specialists, ethicists, public-communication reviewers, and participant representatives. Their task is to challenge channel selection, weighting, missingness, burden, exclusions, subgroup performance, public summaries, and deployment temptation before the claim travels.

Red teams need the estimator specification, code, simulation environment, prospective plan, deidentified outputs, and enough provenance to reproduce the claim. Capable rival teams should

own their competing models and evaluation rules rather than receive simplified versions implemented by the favored team. They should test leakage, alternative transforms, nuisance explanations, unsupported sign choices, and the permanent challenge set described next before the test set is opened (Simmons et al., 2011; Wagenmakers et al., 2012; Nosek et al., 2018).

A permanent challenge program should include cases designed to break the index: exercise with positive affect, calm dissociation, analgesia with continuing grief, fever without negative valence, positive arousal, low-expression distress, device artifact, and context shift. Retain these case families while separating development examples from sealed confirmatory instances. Once known cases have guided tuning, test the frozen model on new participants, contexts, or acquisition periods. This keeps the challenge program informative as the index develops (Nosek et al., 2018; Open Science Collaboration, 2015).

10.4.3 — The Question and Its Echo

A measurement can leave an echo in the process it observes. A question directs attention, a signal interrupts a task, or a request for a judgment brings an otherwise passing experience into focus. The immediate answer and the later state can therefore reflect different parts of the encounter. A useful model can represent the observation as an event with a time and an effect, alongside the information it produces.

Consider an invented device whose state fades by half during each interval. Let x be its state immediately before a scheduled opportunity for a prompt. A prompt, when delivered, adds two units to the state, after which the state fades by half before the next opportunity. The next pre-prompt state is therefore x_next

= x/2 + u, where u is one if the prompt occurs and zero otherwise. These numerical rules define an invented observation system.

Begin at zero and deliver one prompt. At the next opportunity the state is one. If no further prompts occur, the following states are one half, one quarter, one eighth, and so on. The initial event has a geometric echo. Adding all those later pre-prompt states gives two units. Thus an effect that begins as a two-unit input can contribute across several subsequent observations because the state retains part of its history.

Now suppose prompts occur independently with probability p at each opportunity. The expected state satisfies the recursion m_next = m/2 + p. Its stationary mean is 2p. With a prompt probability of one tenth, that mean is one fifth. With a probability of one half, it is one. The two schedules produce different average states even though the device has the same underlying fading rule and the same response to each individual prompt.

The timing of the reading determines another difference. A reading taken just before a prompt records the state inherited from earlier events. A reading taken immediately after the prompt includes the new two-unit addition as well. At a prompted opportunity, those readings differ by two under the stipulated rule. The labels “current state” and “response to the question” become much clearer once the record specifies which side of the prompt each observation occupies.

Randomized timing can reveal the echo. At comparable opportunities, assign a prompt to one group of otherwise equivalent runs and omit it from another. Observe both through a separate channel at the next opportunity. The expected difference is one. One opportunity later, with subsequent inputs held comparable, it is one half. The next difference is one

quarter. These successive contrasts recover the response to the observation event rather than treating all later differences as spontaneous movement in the underlying process.

A richer model could give prompts different effects depending on the current state. A question might engage attention when a task is quiet and interrupt concentration when the task is absorbing. Then the response to the prompt is itself a function of context. Comparing schedules at common observation times can reveal whether the added input is stable or whether the schedule changes which contexts are sampled. The scientific object includes the interaction between the measurement procedure and the process.

The same distinction helps interpret a break in collection. In the constructed device, removing prompts does not immediately return the state to zero. The inherited state continues to fade. If frequent sampling is followed by a quiet interval, a researcher who knows the fading rule can predict the remaining echo. A model that treats the first unprompted observation as a fresh baseline would be using a different account of the device's memory.

Observation frequency can consequently affect more than the number of data points. It can change the distribution from which those points are drawn. More frequent prompts provide more reports while also producing a different state trajectory in this example. The useful comparison asks what is learned under each schedule about the declared target: the monitored process as lived, the response to prompting, or the process under a quieter observation regime.

For experience research, this provides a precise way to make reactivity informative. The prompt has a place in the event sequence, its possible influence has a duration, and its effect can be compared through an independently justified channel. A

person answering a question is participating in an encounter. Representing that encounter explicitly allows the research to learn both from the answer and from the trace the question leaves behind.

10.4.4 — Scientific Governance

The prospective specification package should separate immutable fields from controlled contingencies. Immutable fields include the primary construct interpretation, endpoint-independent data split, sign rules, confirmatory transforms, principal weights, primary rival set, and claim-update rule. Controlled contingencies describe how the analysis responds to sensor failure, unexpectedly high missingness, software error, or safety interruption.

Registered Reports are especially valuable for pivotal validation and model-comparison studies because review precedes results and publication does not depend on a positive finding. Where unavailable, the protocol and analysis plan can still be publicly timestamped before data access if the record and later deviations remain visible (Nosek & Lakens, 2014; Nosek et al., 2018).

When a pivotal study uses a randomized design, current trial standards offer a concrete governance spine. SPIRIT 2025 identifies minimum information that should be fixed in the protocol, while CONSORT 2025 identifies essential information that should be reported after the trial (Chan et al., 2025; Hopewell et al., 2025).

Independent data custodianship can reduce conflicts between estimator development and access. A custodian may enforce purpose limitation, track queries, manage outcome release, and preserve withdrawal rules without controlling scientific interpretation. The separation is especially valuable when

vendors, donors, or sponsoring institutions have incentives tied to success.

A pivotal multichannel estimate study should use a staged release protocol. Simulations, development data, and pilot feasibility can be available while the confirmatory outcome set remains sealed. Before release, the team deposits code hashes, channel registry, weight specification, missingness model, exclusion list, planned figures, model-recovery results, and the exact claim-status table. Replication checks must reproduce exclusion, uncertainty, and no-estimate decisions as well as numerical outputs. The outcome custodian then releases only the files required for the locked analysis, and every deviation is timestamped (Nosek et al., 2018; Chan et al., 2025; Hopewell et al., 2025).

Governance should be tested before it is trusted. A release rehearsal can deliberately introduce a sign reversal, stale calibration file, duplicated participant, endpoint misclassification, or hidden exclusion and ask whether the registry, audit trail, and pause authority detect it before publication. The exercise should test whether an authorized correction request can locate the participant’s records without exposing another person’s data and whether each figure and public artifact is identifiable. A paper-only control has not shown it stops an error path (National Institute of Standards and Technology, 2024).

10.5 — When Measurement Must Stop or Abstain

A common unit cannot be obtained by naming a scale. Its origin must correspond to a defensible state or contrast. Equal numerical differences must represent comparable changes for the intended claim. The rule for combining intervals must then be justified before a cumulative total can carry scientific meaning.

10.5.1 — No Natural Unit Is Given in Advance

Measurement theory distinguishes ordinal, interval, and ratio interpretations because each survives a different class of transformations. An ordinal scale preserves ordering. An interval scale also gives meaning to equal differences, but its zero may be conventional. A ratio scale requires a meaningful zero and supports ratio statements. These distinctions are scientific obligations, not vocabulary choices (Stevens, 1946; Krantz et al., 1971; Suppes & Zinnes, 1963).

The proposed terminal relation is especially sensitive to origin. Positive rescaling multiplies a valid signed scale by a positive constant. It preserves which values are zero while changing their magnitude. Adding a constant changes the cumulative total and can manufacture or destroy closure. A convenient midpoint, sample average, or fitted intercept therefore cannot be treated as neutral merely because it makes the analysis easier.

Sign is a separate issue. Arousal, intensity, novelty, salience, and physiological activation do not reveal whether an experience is positive or negative. A model can be mathematically identified while its experiential orientation remains unknown. External anchors, designed contrasts, report interpretation, and context must supply the sign.

The momentary multichannel estimate may be a dimensionless, version-specific index. If that index is integrated

over hours of declared experienced exposure, the result is index-hours for that instrument version.

A simple example reveals the limit. An index value of plus two for one hour and plus one for two hours produce the same arithmetic product under a linear rule. That equality is a property of the rule, not proof that the experiences were phenomenally equivalent. The model must earn any claim that equal products represent equal contributions.

Recovering a hidden coordinate is an important advance, but its units still matter. Under its stated auxiliary-variable assumptions, one nonlinear independent-component analysis result identifies components up to componentwise invertible transformations (Hyvärinen, Sasaki, & Turner, 2019). Such identification need not determine the numerical scale required for addition. In an invented example, three candidate contributions 1, 1, and −2 sum to zero. The smooth increasing transformation $g(z)=z+z^3$ preserves their order, signs, and neutral origin, yet produces 2, 2, and −10, whose sum is −6. A flexible observation model can represent the same observations after its mapping is transformed consistently.

The constructive task is to justify which transformations preserve the intended experiential comparisons. If independent comparisons support a common additive representation and a strictly increasing transformation g preserves addition on the real line, $g(x+y)=g(x)+g(y)$ forces g(x) = ax with a > 0. Zero totals then survive the remaining change of unit. This is a conditional uniqueness argument, not a reason to impose additivity on ratings. It identifies what calibration must contribute beyond recovering a hidden variable.

Robust analysis should cover the full set of scientifically defensible specifications declared in advance. Alternative origins, transforms, missingness models, exposure rules,

endpoint windows, and instrument versions can each generate a measured terminal result. Report the full sensitivity range across specifications expressed on a common declared comparison scale, together with sampling uncertainty, rather than selecting the specification nearest zero. Keep incompatible targets or scales separate. Call the range an identified set only when explicit population-level assumptions establish that interpretation for a common estimand.

Measured equivalence is robust only when the uncertainty region covering the defensible specifications lies inside the declared empirical band under the prespecified error-control rule. If that region crosses the boundary, equivalence is unresolved; if only one preferred specification supports equivalence, the result is specification dependent (Schuirmann, 1987; Wellek, 2010).

Additive conjoint measurement offers a concrete way to investigate the unit rather than merely name it. Within one participant, a study could cross three levels of each of two low-burden stimulus features and repeatedly compare how positive the joint conditions feel. An additive model assigns a contribution to each feature. It then imposes a checkable restriction on the ordering: if (A1, B2) is at least as positive as (A2, B1), and (A2, B3) is at least as positive as (A3, B2), it requires (A1, B3) to be at least as positive as (A3, B1). The shared terms cancel; an arbitrary ordering need not obey the implication (Luce & Tukey, 1964).

These restrictions should be evaluated jointly with an explicit model of response error and repeated-measure dependence, together with held-out comparisons, not by declaring every fluctuating answer a contradiction. Karabatsos (2018) supplies an error-aware test of the cancellation hierarchy; adapting that method to within-person affect reports would require its own

recovery checks. A finite task tests necessary structure for this feature-additive model rather than proving a universal ruler. Failure can reveal interaction between the chosen features without ruling out quantitative valence. Researchers must then establish whether the comparison measure tracks phenomenal valence, locate its neutral point independently, justify a common unit across positive and negative magnitudes, and test its use across time. A successful local calibration would supply one concrete prerequisite for cumulative measurement. If only ordering survives, retain that result instead of forcing an additive total.

10.5.2 — Invalid Constructs, Drift, Reactivity, and Noise

Construct failure occurs when the instrument answers a different question from the one declared or omits states central to the intended population.

Channel failure occurs when the physical or behavioral observation does not retain a stable relation to the target. The study should report the conditions under which a channel remains usable and the populations for whom it does not.

A smaller group-average error bar does not establish that each individual’s trajectory can be resolved at the same scale. Reliability estimates, standard errors of measurement, smallest detectable change, and dynamic sensitivity should show whether the proposed empirical band is supported for the intended level of inference (Shrout & Fleiss, 1979; Bland & Altman, 1986).

Maintenance obligations should be funded and assigned. Calibration, software review, security, accessibility, training, version control, and correction require people and resources. An instrument without a maintenance path can become invalid while retaining the appearance of institutional authority.

10.5.3 — No Adequate Estimate Is a Real Result

No adequate estimate identifies a limit of the evidence chain. It says that the requested inference is not supported under the declared construct, observation process, model, uncertainty, transport, endpoint, or ethical conditions. The result should identify the failed link and the next tractable task rather than emit a number merely because a pipeline expects one.

No estimate is not a zero value; it describes a limit of the method.

Describe a failure's duration, cause, and scope separately. Recalibration, accessible design, improved retention, another site, or a better device may repair a temporary failure. A structural failure persists within the proposed design because its observations cannot distinguish the target or because doing so would require unacceptable burden. A local failure may affect one participant, interval, subgroup, or channel while others remain usable.

When the observable distribution and maintained assumptions identify only a sign, ordering, or set of possible target values, report that partial identification rather than forcing a point estimate; Section 11.5 develops the rule.

Fix the abstention rule before confirmatory analysis, including coverage thresholds, endpoint windows, missingness bounds, and permitted residual uncertainty. Distinguish a target that remains insufficiently identified or bounded under maintained assumptions from evidence made unusable by a failed protocol or validity criterion. If an unobserved contribution can still reverse the decision within the fixed bounds, the target is not adjudicated. Stop verdicts and uses that depend on the unavailable inference while preserving separately justified partial results. Record the failed prerequisite and the advance that would make another attempt informative.

Chapter 11 — Comparing Experiences

Two people choose the same number on a scale. One uses eight to mean unbearable interruption. The other reserves nine and ten for experiences they cannot imagine. The marks match; the experiences may not. Comparing experiences requires evidence for the particular contrast being proposed. Section 11.1 distinguishes the measurement assumptions for comparing structure, scale, and levels. Identifying the neutral origin is a separate substantive task, developed in Section 11.2.

Even a well-supported measurement link cannot make unlike study groups interchangeable. Imagine two language courses using demonstrably comparable frustration ratings: one enrolls beginners, the other experienced learners. A higher mean in the beginners' course may reflect prior preparation or task difficulty. It does not by itself show that the teaching is worse. The measurement link addresses whether scores have comparable meaning; the study design must separately address selection, task conditions, and any causal claim about teaching. A sound comparison needs both.

Mixed states complicate sign and scale because positive and negative experience may coexist without either becoming an error. Section 11.2 develops that structure and asks when a scalar representation is adequate, when separate components are needed, and when the evidence should remain partially ordered.

Travel across language and culture changes more than words. Politeness, stigma, intensity, display rules, reference groups, memory, and the safety of disclosure can all alter what a response means. The Chapter therefore treats cultural transport as a construct and response-process problem as well as a translation task (van de Vijver & Leung, 1997; Harkness et al., 2004).

Age changes comprehension, memory, time perspective, bodily baselines, and available forms of report. A child's face

scale, an adolescent's mood diary, an adult's tradeoff judgment, and an older person's report under illness do not become comparable because they occupy the same numerical range.

Disability and chronic illness expose another failure mode. People can adapt their standards, priorities, and interpretations over time. Response shift may reflect recalibration, reprioritization, or reconceptualization rather than simple improvement or decline. An instrument that ignores this change can confuse a new ruler with a new state (Sprangers & Schwartz, 1999; Oort et al., 2009).

The comparison problem has a moral edge because measurement failure is rarely distributed at random. A scale that travels poorly can understate pain, distress, access barriers, or participation burden in groups already poorly served by institutions. Failed invariance should narrow the claim and improve the instrument, never label the population defective.

The analysis plan should state in advance whether a failure narrows the result to within-person, within-group, site-specific, descriptive, or exploratory use, or requires withdrawal from public communication.

11.1 — The Comparison Problem

Comparing scores requires a declared construct and a specified contrast. Configural invariance tests shared factor structure, and metric invariance asks whether loadings share a scale. Scalar invariance supports latent-mean comparisons under the fitted model, while categorical indicators require appropriate threshold and identification constraints. A failed test narrows the relevant comparison rather than erasing all descriptive information. Partial invariance needs defensible links and sensitivity analysis before confirmatory reuse (Meredith, 1993; Millsap, 2011; Putnick & Bornstein, 2016). None of these statistical tests proves that two people have identical felt experience.

11.1.1 — Same Words and Devices, Different Experience

Words make experience shareable without making it standardized. Imagine two translations of the middle category on a five-point distress scale. One means noticeable discomfort; the other implies distress that disrupts ordinary activity. Identical spreadsheet labels would then conceal different response thresholds. Interpret agreement under the tested wording, response model, population and conditions (Meredith, 1993; Millsap, 2011; Putnick & Bornstein, 2016). Devices are designed and tested in particular contexts. Site, body, climate, movement, placement, and user trust can change how they perform.

Changing from keyboard to voice, caregiver proxy, interview or passive inference can change abilities and social pressures involved in responding. Unchanged category labels do not establish an unchanged scale; model the mode effect and retain unresolved differences in the uncertainty or scope of the comparison.

11.1.2 — Device Context and Technical Equivalence

The present approach requires device context to be treated as part of measurement, not background. If a device excludes people without stable housing, money, access, language, electricity, or technical comfort, the dataset inherits inequality.

Technical equivalence includes hardware, firmware, preprocessing, placement, sampling rate, environmental conditions, and operator practice. Predictive calibration at one site is not evidence of construct invariance at another. Transport studies should separate device effects from site composition and should include negative controls that reveal when a model is learning location, workflow, or disease severity instead of experience.

A traveling reference can help separate a device from its surroundings. Consider two instruments measuring an artificial reference whose value is fixed at ten units. Instrument A at site one reports ten. Instrument B at site two reports fourteen. Those two readings alone leave several explanations open. The instruments may differ, the sites may differ, or both may contribute.

Move each instrument through both sites while keeping the reference unchanged. Suppose A reports ten at site one and thirteen at site two, while B reports eleven at site one and fourteen at site two. At this reference level, moving sites adds three under either instrument. Changing instruments adds one at either site. The crossed arrangement has separated the two additive contributions that the original pair of readings combined.

A second reference level asks a further question. At a reference value of twenty, suppose A reports twenty at site one and B reports twenty-two. B's excess is now two rather than one. These two levels are consistent with a gain of 1.1 for B, while A

has gain one. A fixed one-unit offset would have predicted twenty-one instead. The second level distinguishes two explanations that agreed exactly at the first.

If the site contribution remains three units, the corresponding readings at site two would be twenty-three for A and twenty-five for B. Further reference levels can test that combined relation. The result is an explicit observation model: a device-specific gain applied to the reference value, plus a site contribution. Its ingredients were identified by varying one source while retaining information about the other.

In an experience study, a technical reference tests the instrument and its pipeline. A separate bridge connects the resulting feature to the construct being studied. Keeping the technical comparison this explicit helps that later work: a change attributable to device gain has a different interpretation from a change associated with a participant's condition. A traveling reference, repeated placement, or overlapping instrument period can provide the comparisons needed to tell those changes apart.

11.2 — Neutral Origin, Sign, and Scale

An arbitrary zero can change a scientific conclusion. A seven-point scale might use its midpoint as zero, while another study uses a person's resting baseline and a third uses the absence of aversion. Their cumulative totals can differ without an arithmetic mistake. Origin validation must distinguish what is usual from what is neutral, including when a familiar baseline is unpleasant. Gasper et al. (2021) found that self-reported neutrality could coexist with positive or negative feeling; a two-step presence check and follow-up questions can distinguish indifference, absent feeling, and mixed experience before measuring intensity. A nondegenerate scale must preserve independently anchored positive and negative states rather than declaring every state zero and calling the result closure.

11.2.1 — A Zero Point Is Not Automatically Natural

Pain can orient a bounded within-person contrast, but acute and chronic pain differ in history, controllability, and context. Neither supplies a natural zero or universal negative unit of felt life.

Any pain anchor should distinguish intensity, duration, location, threat, controllability, expected relief, meaning, and co-occurring states. A familiar one-to-ten scale may serve a clinical purpose, but it cannot establish cardinal equivalence between brief severe pain and long moderate pain or become a universal ruler for experience. Narrative report, function, disability context, medication, culture, and within-person history remain part of the validation (Stevens, 1946; Krantz et al., 1971).

Social exclusion can supply another candidate negative contrast; its design must distinguish the felt response to rejection from other effects of the task.

Where direct social exclusion would be harmful, remembered, hypothetical, or low-burden designs may be more appropriate. Debriefing must address the particular deception or rejection the procedure introduced.

A brief exclusion task is a candidate probe of a bounded social response; it does not reproduce chronic stigma, family rejection, disability exclusion, religious shunning, or exclusion from public life. A proposed within-person validation study should separate the response to exclusion from confusion, loss of points, generic task failure, and attention, while examining cultural differences in belonging, face, duty, autonomy, and shame. The task earns its place as an anchor through discriminant validation (Campbell & Fiske, 1959).

11.2.2 — Sign and Mixed Valence Require Separate Work

A single opposing-poles bipolar item cannot report coexisting positive and negative magnitudes separately and may hide a mixed state at its midpoint (Russell & Carroll, 1999; Larsen et al., 2001; Berrios et al., 2015).

The comparison should ask whether a one-axis value preserves held-out prediction, response interpretation, and relevant consequences without leaving systematic residual structure.

Timing matters. A person may feel fear during the event, relief seconds later, exhaustion an hour later, and meaning much later. A prompt that asks for one global rating can combine different stages of the response. Momentary, event-level, and retrospective reports answer different questions.

Use the measurement-structure recovery tests in Section 10.3.3 to compare a scalar with separate positive and negative dimensions.

11.2.3 — Changing the Origin Changes the Claim

Try a deliberately simplified calculation. A model assigns minus two to one interval and plus two to an equally long interval. The total is zero. Now move the origin down by one unit: the same intervals become minus one and plus three, and the total becomes plus two. Nothing happened to the experiences. The apparent balance changed because the place called neutral moved.

Changing a fixed unit behaves differently. Double both original values and the total remains zero: minus four plus four. Each stream may use its own finite positive constant unit without changing whether its total is zero. With exposure weights held fixed, nonlinear or time-varying rescalings generally change cancellation. The requirement is therefore an independently justified neutral origin and a coherent additive scale across signs and times within each stream. Report how the conclusion changes when more than one origin or linking rule remains defensible.

Section 10.5.1 shows how a transformation can preserve zero, sign, and order while changing an additive balance. The next question is which changes of scale preserve every such balance, rather than only a particular example.

These scale distinctions raise a stronger question: which increasing transformations preserve every finite zero total? On a branch that admits arbitrary real contributions and their finite combinations, the answer can be derived from a few short histories. Let f be a strictly increasing transformation with $f(0) = 0$, and require that applying it to each contribution of any zero-total history still gives a zero total.

Begin with the two contributions x and $-x$. Their total is zero, so $f(x) + f(-x)$ must also be zero. The transformation therefore treats the two signs symmetrically: $f(-x) = -f(x)$. This condition

is stronger than merely keeping positive values positive and negative values negative. Their paired magnitudes must continue to cancel.

Next use the three contributions x, y, and −(x + y). They also total zero. Preserving that total, and applying the sign relation just obtained, gives f(x + y) = f(x) + f(y). The transformation must be additive. A unit assigned to the combined contribution has to agree with the sum of the units assigned to its parts.

Write c for f(1). Repeated addition gives f(2) = 2c, f(3) = 3c, and the corresponding relation for every integer. Since n copies of 1/n add to one, f(1/n) must be c/n. Combining these results gives f(m/n) = cm/n for every rational number. Strict increase makes c positive.

The increasing condition completes the argument for all real values. A real x can be approached from below and above by rational numbers. Their transformed values are the corresponding rational multiples of c. Because f preserves order, f(x) lies between those lower and upper values. As their separation becomes arbitrarily small, they pin down f(x) as cx. Thus the transformation is multiplication by one fixed positive constant.

The converse is immediate: multiplying every contribution by the same positive constant multiplies its finite total by that constant, leaving zero at zero. The result therefore gives both directions. Under the stated domain and order assumptions, precisely these positive changes of unit preserve every finite cancellation relation. A restricted set of available contributions may leave more transformations indistinguishable; the characterization concerns the full real-valued additive domain just declared.

The study should permit alternative anchors, assisted communication, or no-anchor status when its original design

does not fit a participant's circumstances. These design provisions let researchers evaluate which comparisons remain meaningful for the person and context actually studied.

A concrete calibration candidate is the Labeled Hedonic Scale. Lim, Wood, and Green (2009) used magnitude estimates from trained participants to position its verbal labels, then compared ratings of food names with magnitude estimation and a nine-point scale. For the Law, the useful next step is a within-person qualification study: test whether magnitude relations remain stable across pleasant and unpleasant experiences, repeated sessions, and independently varied durations. Fix the calibration before examining cumulative totals, and evaluate temporal additivity separately from agreement between rating methods. The scale supplies a practical starting instrument; its suitability depends on those results and on the context of use (Prescott, 2009).

An uncertain origin can be carried through the calculation. Suppose an additive observation model gives total $A = Z + bH + E$, where Z is the target total, b is one constant offset, H is the total exposure under the declared temporal measure, and E is the accumulated observation error. If independent calibration bounds b between b_- and b_+ and |E| by η, then Z lies between $A - b_+ H - \eta$ and $A - b_- H + \eta$.

In an invented example, A = 3, H = 10, b lies between 0.1 and 0.2, and η = 0.1. The allowed total is 0.9 to 2.1, excluding zero under those assumptions. A wider origin range could make the result unresolved. The calibration bounds, exposure measure, complete relevant domain, and inferential guarantee must be earned independently; the point is that uncertainty can yield an informative interval rather than a forced midpoint.

Conflicting verbal, behavioral, and physiological anchors may reflect different constructs, time scales, or mixed states. Forcing

them into one sign can invent an origin none independently supports.

11.3 — Culture, Age, Disability, and Response Shift

Culture can shape how experience is expressed without determining what any one person feels. People within the same community may differ in disclosure practices. Family duties or mistrust of institutions can change which reports are safe to offer. Comparisons need to test these response conditions rather than translate every difference into a difference in inner state.

11.3.1 — Culture, Development, and Expression

Avoid both a universal emotional code and the assumption that a group has one fixed response style.

Response styles can shift observed distributions without identical shifts in felt state. Treat their possible influence as a measurement hypothesis: examine response processes and test how the substantive comparison changes under alternative response models, rather than infer a culture's inner experience directly from its response pattern (van de Vijver & Leung, 1997; King et al., 2004).

Culture affects report, behavior, context, and interpretation of anchors. A measure developed in one setting may miss local idioms of grief, relief, shame, resilience, or fear, or may overread visible expression. Local language review, cognitive interviewing, community consultation where appropriate, and response-style analysis should therefore precede cross-cultural use.

Participation burden itself can vary by culture. Repeated private disclosure may be ordinary in one setting and socially dangerous in another; a scale may therefore retain only participants for whom its emotional vocabulary and research rituals feel permissible. Selection into the dataset can imitate invariance unless refusal, nonresponse, translation difficulty, and trust are analyzed as part of the measurement process.

Developmental suitability must be established for the intended age group and use. Language, recall demands, dependency, and reliance on an informant can change what a response means. An age label does not establish that a measure is fit for that purpose (U.S. Food and Drug Administration, 2025).

When self-report, caregiver report, and clinician observation answer different questions, they provide evidence about different targets rather than interchangeable measurements.

Developmental invariance also concerns the target itself. The meaning of safety, shame, agency, loneliness, or relief can change as cognitive, social, and bodily capacities develop. A child and an adult may not be two points on one mature construct. Longitudinal designs should test whether the structure persists, and when it does not, the program may need age-specific measures linked only at carefully justified points.

A prompt, setting, sensor, or request for detail may itself feel threatening. The resulting participation pattern can then limit the inference even when the instrument functions as designed (SAMHSA, 2014).

11.3.2 — Chronic Illness and Baselines

A same-scale claim must ask what the person is carrying before it decides what a number means. Calibrating only within an established burden can understate suffering; comparing the same person with a healthy reference without context can overpathologize adaptation. The relevant history includes duration, flare, recovery, fear, agency, care demands, and the cost of maintaining ordinary function.

Use longitudinal invariance tests, qualitative appraisal data, and clinical context to distinguish changes in experience from changes in standards, priorities, or the meaning of health and

well-being; both may occur together (Oort et al., 2009; Rapkin & Schwartz, 2004).

A device may assume a particular movement, face, grip, posture, voice, sensory profile, or response mode and then mistake access conditions for burden. Accessible alternatives can preserve participation, but each mode may require a bridge study. The inaccessible original is not entitled to serve as the neutral standard, and instrument failure should change the claim rather than exclude the person.

The review process should include disabled researchers, participant voice, accessibility and assistive-technology expertise, and clinical input where appropriate. Participants need a route to challenge interpretations, and access work should be compensated rather than treated as an unpaid test of endurance.

A useful design separates stable burden, episodic flare, treatment effect, uncertainty burden, medical-system burden, and social burden. It also limits tracking when repeated measurement makes illness more central in attention than the participant wants.

Reference-group effects matter under inequality. People may judge conditions relative to what is normal, possible, or safe in their environment. Satisfaction under deprivation and distress under abundance can both be genuine while reflecting different comparison frames (Albrecht & Devlieger, 1999; Ubel et al., 2005).

Context changes the meaning of a signal and can play several different roles. A raised voice in a home, courtroom, hospital, protest, therapy room, or workplace does not carry one interpretation; fear in actual danger differs from fear in a safe laboratory. The analysis should state whether context is a confounder requiring adjustment, part of the target, a modifier of observation, or a descriptive condition, because automatic

adjustment can erase the burden under study while indiscriminate collection can create a dossier (World Health Organization, 2001; World Health Organization, 2022).

11.3.3 — Group Protection

Use measurement-invariance and differential-item-functioning tests for the declared group comparison (Meredith, 1993; Millsap, 2011; Teresi, 2006).

Small groups face contextual reidentification, and community involvement can improve questions, language, burden review, and interpretation without becoming ceremonial endorsement or substitute consent (Narayanan & Shmatikov, 2008; Ohm, 2010).

Comparison may reveal unequal pain treatment, exclusion, environmental burden, or service access, but it should name the institution or material pattern rather than turn it into a group trait. Stigma, punitive use, concentrated error, or implied inferiority are triggers to withdraw the affected claim or use (Fillingim, 2017; Benjamin, 2019).

An item can behave differently across groups even when the groups receive the same overall average. To see the distinction, imagine a synthetic benchmark in which the relevant underlying state is known. It has two levels, low and high. In group A, an item receives a yes response from 20 percent of low-state cases and 80 percent of high-state cases. In group B, the corresponding proportions are 40 percent and 60 percent.

If each group contains equal numbers at the two levels, both groups answer yes half the time. The overall response rates are identical. Yet the item separates the low and high states much more strongly in A than in B. A comparison restricted to the average would miss that difference in the relation between response and target.

The reverse situation is possible too. Suppose the item has the same 20-percent and 80-percent response probabilities in both groups. If A contains mostly high-state cases and B contains mostly low-state cases, their overall response rates differ. That difference is compatible with an invariant item because the response rule is the same at each matched state. The groups differ in the distribution of the state being represented.

Differential item functioning concerns the first kind of comparison: how the item behaves when the relevant construct level is held comparable. It is not a synonym for a difference between raw group averages. The constructed benchmark makes that distinction transparent because the state is supplied by the simulation. With human reports, the state itself is estimated, and the comparison depends on a defensible measurement model (Meredith, 1993; Millsap, 2011; Teresi, 2006).

An observed difference can guide a productive investigation of the question. One group may read a frequency word differently, use a response category more narrowly, or encounter a different reference situation when interpreting the prompt. Interviews about the response process can distinguish these possibilities. Revising the wording and testing the revision then becomes a specific attempt to improve the instrument.

The result may favor a shared item, a linked alternative, or separate measures for different purposes. Each is a positive design outcome when it makes the intended comparison clearer. Accessibility can be part of this work: changing the mode of response may preserve the question while making participation easier, and a bridge study can examine what that change does to the measurement relation.

The important unit of explanation is the link between a question and the answer it elicits under known conditions. Keeping that link visible allows group comparisons to reveal

differences in circumstances while also revealing where the measuring language needs improvement. The instrument becomes more useful through learning how people actually understand and answer it.

11.4 — Comparisons Within and Across People

Within-person calibration asks what a change in one person's reports can validly indicate. It does not assume that an identical number means the same thing across strangers. Anchors and observation conditions must remain interpretable across time. If they drift, the analysis should narrow its claim or recalibrate before comparing trajectories.

11.4.1 — Within-Person Calibration

It does not require every participant to share one scale, and it does not authorize comparison of levels across people.

The person’s own response patterns, context, body, language, health, routines, and accessible channels become part of the calibration. This idiographic focus can preserve information that disappears when a group average is treated as the model for everyone (Molenaar, 2004; Hamaker, 2012; Fisher et al., 2018).

Suppose the people who usually sleep longest also report the most distress, while each person’s better-slept days are easier. Both patterns can occur in the same records. Differences between people may reflect illness or working conditions; changes within a person answer a different question.

Baseline drift occurs when a reporting baseline changes as sleep, medication, illness, or context alters a channel. The model should track those changes without silently moving the phenomenal neutral origin. Updating an observation model and redefining what counts as neutral experience are different operations.

Calibration windows should include ordinary variation, known context changes, and enough repetition to distinguish stable person-specific structure from coincidence.

Person-specific anchors can help orient sign and magnitude. The participant may identify clear local examples of relief,

threat, comfort, or distress, and the protocol can test whether channels respond consistently around those events. The anchors remain local and should be revisited when circumstances change.

Within-person evidence may justify ordering, event-linked contrasts, or estimated change without justifying an absolute level. A personal trajectory can be useful even when its zero and unit do not transport.

Evaluate participant feedback separately for benefit, burden, and reactivity; local calibration does not authorize personal balance tracking.

A person-specific model's strongest test is whether a frozen mapping calibrates a later time block, a changed but declared context, or an external channel without hidden retuning.

11.4.2 — Between-Person Calibration

Between-person calibration asks whether a number retains enough common meaning to support a declared comparison across people. That requires more than giving everyone the same instrument. The construct, response process, origin, sign, scale, channel relations, and uncertainty must be linked across the people being compared.

Common anchors can help, but only when participants interpret them similarly. Anchoring vignettes, known-condition contrasts, and participant-generated examples may reveal response-scale differences. They can also fail when the scenario is unfamiliar, morally loaded, culturally specific, or differently imaginable (King et al., 2004).

Apply the invariance tests in Section 11.1 to the intended comparison. Any partial-invariance model used for confirmation must be justified prospectively, rather than selected to preserve

the result (Meredith, 1993; Vandenberg & Lance, 2000; Putnick & Bornstein, 2016).

Reference-population choice is part of the model, not a clerical detail. Changing the norm can alter standardized scores, thresholds, apparent prevalence, and error rates even when the raw responses do not change. A between-person claim should therefore report the reference population, justify why it fits the intended use, and show how the conclusion changes under plausible alternatives.

The 2026 World Happiness Report ranks 147 countries by average 2023–2025 life evaluations (Helliwell et al., 2026). Its Cantril Ladder asks respondents to evaluate their lives at the time of survey; the national scores average people and years. These population differences can motivate longitudinal investigation of social conditions. They are not measurements of individual complete-stream terminal residuals.

If defensible neutral points reverse the ordering or move the comparison across a decision boundary, the conclusion is not stable.

Partial invariance does not automatically entail partial identification. Sufficient valid invariant anchors can identify a specified latent contrast even when other items vary. When the surviving constraints identify only a range, ordering, or item-subset comparison, report that narrower result explicitly (Putnick & Bornstein, 2016; Manski, 2003; Tamer, 2010).

Qualitative follow-up remains necessary when statistical links fail. Participants and local experts may reveal that an item has different meaning, a device violates a social norm, or a response is unsafe to give. That knowledge can narrow the claim or redesign the instrument.

Borrowing information can be examined in a small statistical model. Suppose a set of instruments each has its own unknown

offset θ. A single noisy observation y of an instrument's offset has Gaussian error with variance σ². The model describes the collection of true offsets as Gaussian with mean zero and variance τ². Both variances are stipulated for this example, so their role in the estimate is explicit.

Combining the observation with that population model gives an estimated offset λy, where λ = τ²/(τ² + σ²). This movement toward the population mean is called shrinkage. The estimate lies between the individual reading and that mean. If population differences are large relative to reading error, λ is close to one and the individual reading carries most of the weight. If reading error is large relative to population differences, λ is small and the estimate moves much closer to zero.

Take τ² = 1 and σ² = 9. Then λ is one tenth. Readings of plus four and minus four produce estimated offsets of plus 0.4 and minus 0.4. No second measurement has been taken. The narrower display comes from combining the same readings with a model that expects true offsets to vary much less than observation error.

This operation can improve estimation when the population model is appropriate. It shares information about the setting in which an uncertain individual reading was obtained. Its interpretation still requires the full distribution. In this example the conditional variance of each unknown offset is τ²σ²/(τ² + σ²), or 0.9. A point estimate near zero is accompanied by substantial uncertainty about the particular instrument.

There is a useful check across the whole modeled population. The noisy readings have variance τ² + σ². The variance of their shrunken point estimates is λ² times that quantity, which equals τ⁴/(τ² + σ²). With the chosen values it is 0.1. The true offsets have variance one. The remaining 0.9 appears as conditional uncertainty within the estimates. Variation between conditional

means plus the average conditional variance recovers the original total variance.

A graph displaying only the shrunken centers would show a tightly clustered collection and omit most of the modeled variation. A graph of individual uncertainty or an explicit estimate of the latent population distribution answers a different question. The distinction matters whenever the inquiry concerns whether each separate quantity is close to zero, rather than how accurately a method predicts the center of a population.

Repeated observations offer a concrete way forward. If each instrument supplies n independent readings with the same error variance, its average has error variance σ^2/n. The corresponding weight becomes $\tau^2/(\tau^2 + \sigma^2/n)$. As n increases, the estimate depends more on that instrument's own record. The design can therefore show how much individual information is needed before a claimed pattern is carried mainly by observations rather than the population model.

For a per-stream hypothesis, partial pooling can assist estimation while retaining this distinction. Report the model's population assumptions, the information supplied by each record and the uncertainty left for that particular stream. A shrinking cloud of estimates is then interpretable as an analysis result with a known construction, instead of being mistaken for a direct view of shrinking differences in the underlying histories.

11.4.3 — Connecting a Family of Instruments

An instrument may need to join a family rather than a single predecessor. A long study might begin with one device, introduce a second at a new site, and later adopt a third. Each change can include an overlap period in which both instruments observe the same technical reference. Those overlaps create a network of comparisons. The network can do more than carry

old readings onto a new scale: it can provide an independent check on whether the proposed links belong together.

Consider three artificial instruments, A, B, and C. All have the same unit and respond linearly with slope one; only their constant offsets are unknown. An overlap study finds that B reads two units above A. Another finds that C reads three units above B. Under this offset model, C should read five units above A. A reading of ten on A would become twelve on B and fifteen on C. The two links already determine the third comparison.

Now perform a separate overlap between A and C. Suppose it estimates their difference as four units. The new observation supplies something the original chain lacked: a test of its internal agreement. The direct route from A to C adds four, while the route through B adds five. Both cannot be exact descriptions of the same three fixed offsets. The one-unit discrepancy identifies a concrete question for the calibration study: how much of this difference is expected estimation error, and how much would call for a richer observation model?

It is the independent overlap that makes the comparison informative. If the A–C link were calculated by adding the first two links, it would equal five by construction. Its agreement would repeat the input arithmetic. A separately collected comparison can instead assess the prediction. The additional data are valuable partly because the simpler chain already said what they should show.

With estimated rather than exact differences, the network can also support a joint fit. Suppose all three overlap estimates have independent errors of equal variance, and the constant-offset model remains the declared model. Fix A's offset at zero to choose a reference. The task is to assign offsets to B and C that come as close as possible to the three measured differences: two for B minus A, three for C minus B, and four for C minus A.

Minimizing the sum of the three squared discrepancies gives B an offset of five thirds and C an offset of thirteen thirds. The fitted differences are therefore five thirds, eight thirds, and thirteen thirds. Relative to the observed differences, the first two are each one third lower and the third is one third higher. Their squared discrepancies add to one third. By comparison, accepting the first two links exactly would place C at five and leave a full one-unit discrepancy with the direct A–C estimate, for a squared total of one.

The joint fit distributes the mismatch according to the stated error model. If one overlap were much more precise, the fitted links would stay closer to it. If overlaps shared observations or reference errors, their dependence would enter the fit. In each case, the purpose of reconciliation is explicit: estimate the offsets under a supported model while preserving the residual pattern that lets the model be assessed. The original pairwise observations remain available alongside the fitted network.

Connectedness determines what the network can link at all. A fourth instrument D compared with C can be brought onto A's reference scale through that connection. Two instruments E and F compared only with each other form a separate component. Their relative offset may be known very accurately, yet the location of that pair on A's scale remains free. One suitable overlap connecting the components can identify the missing relation under the same model. Repeating the E–F comparison improves their internal link while leaving that other relation unanswered.

The shape of the network therefore helps choose the next calibration exercise. A connection between separate components opens a previously unavailable comparison. An extra connection inside an already connected component creates a check on consistency. Repeating a weak existing connection improves its

precision. These are three different contributions that an additional measurement can make, and the study's intended comparison determines which is most useful.

An experience study would apply this logic first to the particular measurement links it can justify: a sensor feature, a response format, or a calibrated local scale. The technical network describes how those outputs relate. Evidence about the experiential construct gives those outputs their intended meaning. Keeping the two tasks explicit allows a new instrument to join a longitudinal project through inspectable comparisons, with both its connection to earlier records and the independent checks on that connection visible.

11.4.4 — Withdrawal Conditions

Triggers should be declared before release and matched to the level at which evidence can break: unstable retest results, origin sensitivity, sign reversal, missing invariance, item bias, or a simpler rival that removes the claimed incremental value.

Withdrawal should be typed and proportionate. Statistical withdrawal stops an unstable estimate; construct withdrawal removes an unsupported interpretation; transport withdrawal returns a claim to the setting where it worked; ethical or authority withdrawal ends an invasive, coercive, or consequential use. A local result need not disappear when a wider bridge fails (Wasserstein & Lazar, 2016; Wasserstein et al., 2019; Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

The record should distinguish pause, narrowing, correction, retirement, and reinstatement. A pause investigates a defined question; narrowing states a smaller claim positively; correction fixes an identified error while preserving the earlier record;

retirement closes an unsupported or unsafe use; reinstatement requires new evidence, not a renamed variable or threshold.

11.5 — Uncertainty and Partial Identification

Incomplete observation can still determine a useful range. In an invented common-scale example, let x_1, x_2, and x_3 be the contributions of three disjoint intervals. Suppose two overlapping measurements establish $x_1 + x_2 = 2$ and $x_2 + x_3 = 1$, while independent evidence bounds x_2 between 0 and 1. The full total is $3 - x_2$, so it lies between 2 and 3 even though no individual contribution is uniquely identified. The overlap is counted once, not twice.

11.5.1 — Input Error and Noise Floors

Alongside random error and systematic bias (Section 10.3.2), distinguish model uncertainty, which concerns the choice among plausible structures, from identification uncertainty, where the observable-data distribution and maintained assumptions do not determine one target value.

Errors-in-variables methods describe models in which an explanatory variable is itself measured with error. A simple construction shows why this can change a fitted relation. Let an underlying predictor take values minus one and plus one equally often, and let the outcome equal that predictor. Record the predictor with an added independent error that also takes minus one and plus one equally often. The true predictor and outcome have variance one and slope one. The recorded predictor has variance two, while its covariance with the outcome remains one. The population regression slope using the recorded predictor is therefore one half. More observations can estimate that recorded-variable relation more precisely; recovering the underlying relation requires information about the observation process. Validation measurements or an independently justified error model can provide that information (Carroll et al., 2006).

Assess resolution for the final contrast, including uncertainty shared across repeated readings. For an illustrative fixed quantity measured n times as $Y_k = \theta + b + \varepsilon_k$, suppose the errors are independent with zero mean and known variance s^2 and an independent unbiased calibration estimate of b has variance s_b^2. The corrected mean then has variance $\frac{s^2}{n} + s_b^2$. Repetition reduces the first term, not the shared calibration term. Under the Gaussian version, standard equivalence at 5% per one-sided test requires the corrected estimate's 90% interval inside the prespecified margins. This identifies whether further readings or better calibration would actually improve the proposed comparison.

If each measured sum instead has an error bounded in magnitude by 0.1, the possible total expands to [1.8, 3.2]. This still excludes zero without assigning a particular value to the unobserved component. The gain comes from jointly valid restrictions, not from assuming closure. A statistical version requires joint uncertainty for those restrictions; separate confidence statements are not automatically simultaneous.

The same example can guide measurement choice. Under these noisy bounds, an additional estimate of the middle contribution with absolute error at most 0.05 leaves at most 0.5 units between compatible totals, compared with 1.4 beforehand. Even an exact measurement of the difference between the first and third contributions can leave the full 1.4-unit range. These comparisons hold the contributions fixed and require all measurement restrictions to be jointly valid. Select the next observation for the uncertainty it removes from the declared target, within the permitted burden. Partial identification asks which target values remain compatible with observed data and maintained assumptions. Report that set without inventing the missing pieces or treating one missingness account as certain

(Manski, 2003). If the measured interval is only part of a stream, a terminal claim also needs bounds for the remainder. If no history fits the restrictions, the observation model needs revision; an empty compatible set is not evidence that all possible histories close. A limited question can therefore receive a definite answer without fabricating a complete record.

An uncertainty budget can guide how effort is divided between observing the target and calibrating the instrument. Consider a technical benchmark in which n independent target readings leave a variance contribution of 1/n. A separate calibration based on m independent reference readings contributes 9/m to the corrected estimate. Assume the two components are independent and that each reading has the same cost. With a budget of one hundred readings, n + m = 100.

An even split gives n = 50 and m = 50. The combined variance is 1/50 + 9/50 = 0.20. Allocating ninety readings to the target and ten to calibration gives about 0.911. The more intensive target record is accompanied by a much less precise shared correction. The calculation locates where the uncertainty remains.

The best division can be found by equalizing the improvement produced by the next small increment of effort. Differentiating 1/n gives a marginal variance reduction of 1/n². Differentiating 9/m gives 9/m². These are equal when m = 3n. Combined with the total budget, the optimum is n = 25 and m = 75. Its variance is 1/25 + 9/75 = 0.16.

The allocation does not divide the readings in proportion to the variance coefficients one and nine. It divides them in proportion to their square roots, one and three, because each additional reading has diminishing returns. Giving all attention to the larger component would leave the smaller component

estimated poorly. The optimum balances the marginal gain from both activities.

The same reasoning extends to a variance of a/n + b/m with a fixed total count. The preferred ratio is n:m = √a:√b when counts can be treated continuously and both types of reading have equal cost. Integer counts can be checked around that solution. Different costs, correlated errors, or a minimum number of observations change the optimization and can be included explicitly in the design.

Independent calibration is the feature that makes the second improvement possible. Repeating the target measurement cannot supply information about a shared offset that the target record itself leaves unidentified. Reference observations address that other component. The design gains precision by choosing a different source of information, rather than simply obtaining more of the same reading.

In a human study, participant time and calibration work may have very different costs and burdens. The numerical allocation above concerns an artificial benchmark with equal reading costs. Its useful lesson is the method: identify the components of the final contrast's uncertainty, estimate how each changes with additional information, and compare improvements on the resource scale relevant to the study. A well-placed calibration exercise can make an existing record more informative without lengthening every participant's record.

11.5.2 — Failed Invariance

Failed invariance may involve differences in loadings, thresholds, anchors, or linking constants, or a deeper change in the construct or origin itself. Ordinary estimation uncertainty is not itself evidence that invariance failed. Parameter uncertainty can sometimes be propagated through joint estimation or

simulation; structural noncomparability makes the cross-group operation unavailable. Reports should identify the level of failure and show how partial-invariance and alternative-anchor choices affect the result so a numerical interval does not conceal an invalid comparison (Little & Rubin, 2019; Meredith, 1993; Millsap, 2011).

Device or site failure should pause the affected comparison and preserve records of the hardware, firmware, preprocessing, placement, and setting. A bridge study is required before affected results are linked. The notice should identify whether the source was sensor performance, movement, skin contact, network access, environment, staff behavior, privacy fear, or institutional workflow. If no stable crosswalk can be established, the data series must be segmented and the claim restricted rather than forced into continuity.

A comparison can remain useful when it supports only a partial order. Imagine three ways of making the same optional journey. Route A requires ten units of money and twenty minutes. Route B requires five units of money and thirty minutes. Route C requires twelve units of money and thirty-five minutes. Assume the relevant differences in this constructed choice are cost and travel time, with less of either preferred when the other is unchanged.

A is better than C on both dimensions. B is also better than C on both. The comparison can therefore identify C as dominated. Choosing between A and B requires another judgment: A is quicker, while B is less expensive. Their coordinates alone do not determine one overall winner.

That unresolved pair does not erase what has been learned. The set of undominated alternatives is A and B. Removing C narrows the decision while preserving the tradeoff that remains. The outcome is a smaller choice set with an explicit structure,

rather than a forced ranking whose weights have been left unstated.

A person's circumstances can supply additional information. Someone with a firm twenty-five-minute arrival window can use A but not B. Someone with only six units of money available can use B but not A. These are feasibility constraints tied to the decision. Another person may have both routes available and decide how much the ten saved minutes matter for the intended activity.

Changing the units does not alter the dominance comparison. Expressing money in smaller units or time in seconds preserves which coordinate is lower on each dimension. More generally, a strictly increasing transformation applied separately to each coordinate preserves the order within that coordinate and therefore preserves this dominance relation. It need not preserve a weighted sum of the coordinates.

This distinction is useful when an instrument or study supports some comparisons more strongly than others. Separate dimensions may justify a statement that one observed profile improves on another in every measured component, while leaving a mixed comparison unresolved. Reporting that partial order retains the supported structure. A single number is a further representational choice whose tradeoffs must be identified.

A transport failure can be diagnosed with crossed designs in which participants, devices, and sites overlap. Such designs help separate hardware effects, workflow effects, population composition, and environmental context. When overlap is impossible, uncertainty increases and causal attribution should narrow. A machine-learning adjustment trained on the same biased site does not establish transport.

11.5.3 — Consequences for Published Comparisons

Independent harm monitoring should examine complaints, withdrawals, interviews, subgroup adverse events, burdens, and institutional misuse. Technical success can still fail its context of use; that failure belongs in the scientific record.

Harm may require narrower, aggregate, contextual, delayed, or technically restricted release, or release paired with repair. Choose a form that supports understanding and remedy without transferring burden to people who did not consent to its public meaning.

Pooled analyses may need reanalysis, longitudinal records may need a visible break, and exploratory estimates may no longer appear as validated in summaries or tools. For example, a version break can mark exactly where a revised comparison rule changes the interpretation of an otherwise continuous record.

Chapter 12 — Measurement Without Scores

A research record begins as a narrow promise: information will be collected to answer a defined question under stated conditions. Yet records travel. They can be copied, linked, sold, trained on, scored, cited, or used in decisions far beyond the context that made collection legitimate.

Minimization asks for enough information to answer the declared question well. Imagine a study of whether people feel refreshed after a walk. The researchers need evidence that can separate walking from other explanations; that question does not automatically require a home address, a recording of conversations, or a permanent location history. Good minimization preserves the relevant comparison while leaving unrelated intimacy uncollected. A person can help build knowledge without handing over a detailed portrait of daily life. A candidate record might contain a study code, the assigned activity, whether it occurred, and a brief rating before and after. Whether those fields suffice depends on the declared design. A fair comparison with a suitable alternative cannot be replaced by collecting a more intimate biography.

Purpose limitation gives the record's lifecycle a center. The study assigns responsibility for collecting, correcting, restricting and retiring information when the authority for a particular use changes. The initial promise can be tested: What was promised? What was collected? Who can access it? What was inferred? Which uses are allowed or forbidden? How can a person refuse, correct, withdraw, or seek remedy? What happens to derivatives, backups, shared copies, and trained models when authority ends? Measurement becomes trustworthy when investigators can answer those questions and act on the answers.

Attach provenance to each trace: prompt, device, language, setting, sampling interval, transformation, missingness, purpose, consent or authority, and refusal conditions. Identify the

collector and distinguish ordinary missingness from ethical restraint. A gap created by device failure differs from refusal, care priority, or a question deliberately left unasked to avoid harm. Those differences affect both interpretation and governance.

Filters, on-device processing, restricted fields, event-triggered sampling, and deletion at ingestion can reduce relational exposure. When the design cannot avoid capturing people who did not agree, the channel may be unacceptable.

Review should also ask whether the study gives anything back to the participant population beyond knowledge extracted from them. This does not require therapeutic benefit in every study, but it does require honesty about burden, respect in design, accessible communication, and avoidance of research cultures that repeatedly ask vulnerable people to provide data while others receive prestige. Before collection, record the less intrusive alternatives considered and why the chosen channel set is necessary (McCambridge et al., 2014; Council for International Organizations of Medical Sciences, 2016).

12.1 — Consent, Burden, Minimization, and Deletion

Consent begins with an invitation that makes a study's purpose and burden understandable. Prospective participants should learn what procedures or screening involve and how often researchers will contact them. They should also hear about foreseeable sensations, uncertainties, alternatives, and incidental capture of bystanders. Research must remain distinct from clinical care or ordinary public discussion, without promises to reveal a private balance. Unsolicited intimate accounts remain correspondence unless separately authorized as research data. Document the information participants received, their questions and choices, and evidence of comprehension (Appelbaum et al., 1982).

12.1.1 — The Consent Lifecycle

Separate permissions. Offer distinct choices for enrollment, sensors, interpretation, narrative disclosure, secondary use, public characterization, self-monitoring and model training. Optional additions must not block the least burdensome route. A participant’s own consent does not authorize capture of other people. Imputation cannot supply permission.

Accessible, supported choice. Provide readable, navigable and described materials and communication support before judging task-specific capacity. Reassess when illness, medication, pain, delirium or wishes change. Silence, missed prompts, indirect dissent or deference call for a private, unpressured check-in (Faden & Beauchamp, 1986).

Youth and proxies. Distinguish affirmative assent, dissent, guardian permission and the child’s interests from adult ambitions; involve authorized support and independent review as appropriate. Observable dissent stops this program’s research participation. Emergency care follows its separate clinical

authority; it does not authorize continued optional measurement. In teaching, allow passing or an alternative; keep student work outside research by default and route unexpected safety disclosures to qualified support. Youth helped develop SPIRIT-C's seventeen pediatric protocol items (Baba et al., 2026; United Nations, 1989; CIOMS, 2016).

Authority and payment. Keep choices confidential from supervisors, teachers, treating clinicians and other authorities who could penalize refusal. Test actual freedom: a worker's emotion sensor or a pupil's mood tracker operates under dependency. The EU AI Act prohibits workplace and educational emotion inference within its scope, with medical or safety exceptions (European Union, 2024). Avoid undue influence; pay fairly for completed burden after withdrawal, without compliance badges or public streaks.

12.1.2 — Participation as Purposes and Circumstances Change

Ongoing collection. Provide digital, phone, paper and human routes to pause prompts, disable sensors or leave without explaining a crisis. Limit individual feedback to consent, safety and study engagement. Do not turn refusal into a clinical characterization or adverse institutional record.

Changed purpose. Distinguish a minor technical update from changed purpose, burden, risk or inference. A new sensor, recipient, linkage, retention period or AI use requires the corresponding notice or renewed authorization; permission ten years earlier does not automatically travel with a new question. Consent cannot validate an inference or override a prohibited use.

Public reuse. Obtain separate permission for high-reach quotation, video, fundraising or illustrated storytelling. Specify

audience, context and a correction route: an accurate quotation can still misrepresent its speaker.

Withdrawal. Separate stopping collection, future contact and feasible data withdrawal. Explain what can be removed from originals, backups, shared copies, features, linkage keys, analyses, publications and models. Distinguish deletion, suppression, retraining, unlearning, annotation and residual learned influence; link derivatives to their authority (Cao & Yang, 2015; Bourtoule et al., 2021; Gebru et al., 2021).

Closeout. Before funding ends, assign a reachable custodian for unresolved requests, retention, archives and model retirement. Tell participants what adverse or inconclusive results mean for their data and public claims; stop unjustified secondary uses. Reconcile people, permissions, vendors, code, models and publications, retaining restricted audit evidence where justified (Nissenbaum, 2010; World Medical Association, 2024).

12.2 — No Human-Worth Scores

A measured experience is not a measure of human worth. A low score does not make testimony less credible. A high score does not grant virtue or civic fitness. Systems that attach eligibility or punishment to such estimates convert limited research data into an unjustified judgment about a person.

12.2.1 — Not Human Worth

A noisy indicator may help explore a research question while being unsuitable for a consequential decision; adequacy depends on construct, population, use and decision burden (U.S. Food and Drug Administration, 2025).

Do not turn multichannel or cumulative estimates into employment, insurance, policing, civic-eligibility or human-worth scores. Work records may describe tasks and responsibility, not sincerity, resilience or moral effort. Expire operational records when their purpose ends and provide factual correction, rather than carrying a temporary delay into permanent reputation (O'Neil, 2016; Eubanks, 2018; Barocas & Selbst, 2016).

The rule applies to design and internal language: colors, percentiles, comparative bars, badges and leaderboards can rank people or groups even when accompanying prose denies it. Avoid labels such as "failed ledger," "high-value case," "more balanced" or "more healed" in code, meetings and community practice. A wellbeing survey tied to promotion may reward concealed exhaustion; a visible unity score may pressure a patient to perform while clinicians overlook unconventional communication. Fluency, movement, disability, fatigue and access can then become proxies for supposed inner qualities (Selbst et al., 2019).

Youth materials need special attention to diagnostic force. Personalized risk colors and peer comparisons can behave like a test without that name. Children must not become proof of adults' claims about need, improvement or formation. Minimize linkage and retention because their future interests in intimate records can change; collect only when welfare, comprehension, privacy and future freedom are better protected than by nonmeasurement (National Commission, 1979; CIOMS, 2016).

Aggregate information can sometimes support service improvement, but aggregation is not automatically safe. Small groups, neighborhoods, disability communities, or stigmatized populations may still be identifiable or sortable. Aggregate use requires purpose limitation, minimum necessary detail, anti-linkage controls, and a ban on punitive consequences (National Institute of Standards and Technology, 2023).

12.2.2 — Contribution Is Not Inner Worth

Combining a record of someone's work or contributions with research estimates creates a contribution-worth profile that neither system is authorized to produce.

Invisible work can matter without becoming an inner-life estimate. Preparation, maintenance, translation, mentoring, accessibility support, caregiving, conflict prevention, and the decision to stop unsafe work may produce few public artifacts, while volume and speed metrics can encourage task splitting, neglected maintenance, hidden uncertainty, and burden shifting. Better workflow records should recognize bounded contributions and audit incentives without pretending to describe the whole person.

Behavioral targeting is another boundary. Research estimates should not be used to time prompts, increase commitment, prevent withdrawal, or recruit people through inferred guilt,

hope, grief, fatigue, or vulnerability. Coordination must not become psychological management.

12.2.3 — Correction After Misuse

1. Contain the exposure. Suspend affected access and outputs, freeze the failing version, notify partners and pause consequential decisions. Urgency follows reach, reversibility and the prospect of imminent harm.
2. Diagnose the failure. Distinguish a scientific claim, a measurement system and an application. Map data and decision dependencies: copies, case notes, flags, caches, models, thresholds and public materials. Test descendant systems for inherited logic, parameters, training material and routes; a new interface is insufficient.
3. Match the remedy. Replace ambiguous wording, correct public statements or withdraw a deployment as warranted. Reconsider decisions that relied on the output, including denied employment, schooling or benefits. Restoration of access, reimbursement, support or apology may be required alongside deletion, contract suspension or model retirement.
4. Obtain independent review for consequential incidents. Reviewers need relevant records, freedom from deployment incentives and authority to halt use. Apply the appeal criteria in Section 25.2.2. Preserve the system's failure history under the record rules below while restricting unnecessary identifying material.
5. Verify closure. Check access, descendants, affected decisions, public materials and partner systems; confirm the prohibited route no longer works. Record unresolved exposure as open rather than announcing completion before the correction reaches its consequences (National Institute of Standards and Technology, 2024).

12.3 — No Dossier, No Model-Training Shortcut

Publicly available data are not automatically authorized for every use. A testimony about abuse was shared for a different purpose from a reproducibility dataset. A social post to friends has still another intended audience. Consent, context, and downstream harm matter when deciding whether any of these records may train a model or enter a dossier.

12.3.1 — Public Access Is Not Permission for Every Use

Browser access does not make their contexts interchangeable (Nissenbaum, 2004, 2010). Review a proposed reuse by its new processor, duration, generated outputs and reachable derivatives. General model training can change all four.

Reidentification may arise through rare attributes, timing, location, relationships or external linkage even after names are removed (Narayanan & Shmatikov, 2008; Ohm, 2010). Dream logs, terminal stories and crisis disclosures require particular scrutiny; anonymization failures, clinical risk or inadequate privacy protection may require nonpublication. Welfare-monitoring records for artificial systems likewise do not become publicity, entertainment or commercial training material by default. Captions and diagrams can themselves enter training corpora after their context is detached.

Maintain a reuse register of purposes approved or denied, versions, recipients, artifacts, incidents, corrections and withdrawal status, identifying a custodian for each authorized copy. Review whether the material's meaning survives the intended reuse and whether a correction can reach it. A private grief journal, conflict recording or search history must not silently migrate into a classroom, employer file or public archive merely because a platform can move it (Solove, 2006). Keep the

register focused on uses and responsibilities rather than assembling a second participant profile.

A unity profile may expose communication patterns, neurological status, memory gaps, trauma history, or family observations even when the visible output is simple (Solove, 2006).

12.3.2 — Separate Records and Memorials

Purpose and authority. Separate research, clinical, employment, education, civic, donor, religious, contribution, archive and public-artifact records by purpose, authority and consequences. Different servers do not establish separation when one record silently unlocks another. A brief symptom diary and years of passive sensing differ in intrusiveness, reversibility and bystander exposure.

Identifiers and fields. Use study-specific identifiers, not account handles, government or employee numbers, or contribution IDs. Map sources, permitted fields, access roles, retention, releases, correction routes and prohibited links before deployment. Shared names, devices or vendors supply no authority to join records (European Union, 2016).

Inference and relationships. Audit inferred joins as well as copied fields. Voice, location, messages, sleep and household events can expose bystanders. Combining health, purchases, school records and networks may produce a detailed but wrong portrait; commensuration can erase the circumstances behind unlike observations (Espeland & Stevens, 1998; Bowker & Star, 1999; Haggerty & Ericson, 2000).

AI and vendors. Include prompts, outputs, embeddings, evaluations, moderation records and telemetry as separate governed objects. Trace client-to-client and product-to-product flows, analytics and training. Contracts, technical controls, audit

rights and deletion duties must cover the vendor's full route, including regeneration of sensitive information (National Institute of Standards and Technology, 2024; World Health Organization, 2024).

Authorized transfer. Specify fields, purpose, duration, recipient, authority, security, human approver, secondary-use restriction and deletion or return condition. A narrowly justified clinical safety transfer should be recorded as such. Trace derivatives, model inputs, public artifacts and affected decisions to their source and permission (Gebru et al., 2021; Mitchell et al., 2019).

Correction and memory. Name an owner who can provide feasible access, correct factual errors, explain material secondary uses and receive challenges to generated profiles. Keep a system-failure record of what failed and which controls changed; remove or restrict unnecessary person-level material so institutional learning does not create a permanent case file.

Memorial archives deserve care. Families and communities may value preservation, but the deceased cannot revise the record. Curatorial notes should distinguish documented fact, testimony, inference, dispute, and silence.

Genetic data and other relational records can expose people who never consented, even in an archive centered on one person.

12.4 — Measurement That Should Not Be Done

High-stakes inference about inner life can become coercive when refusal is weak. A workplace screen or public-benefit decision may use a signal more confidently than its validation permits. Existing governance frameworks offer ways to identify such risks (World Health Organization, 2024; National Institute of Standards and Technology, 2023; European Union, 2024). The ethical boundary against forced or consequential experience scoring does not depend on one jurisdiction's rule.

12.4.1 — Coercive Measurement and Legal Misuse

Consent during crisis may authorize care without authorizing research measurement. Research requires its own justification and authorization.

Crisis extraction can occur through platforms, public appeals, tragedy recruitment, or prompts delivered when people need help, medicine, shelter, safety, money, recognition, or someone to listen. Crisis-facing systems should collect only what is necessary for safety, discard intimate detail by default, and separate any retained safety record from research datasets (Barocas & Selbst, 2016; Eubanks, 2018).

Legal misuse may appear as proof of distress, credibility assessment, risk evaluation, damages calculation, competency inference, sentencing input, custody evaluation, or compliance monitoring. Such uses would give measured estimates more authority than they can bear and expose vulnerable people to adversarial reinterpretation.

12.4.2 — Deployment-Prohibited Tool Registry

A deployment-prohibited tool registry records capabilities and uses that may not be deployed because their foreseeable harms exceed any justified benefit. It gives institutional memory to

refusal so an unsafe idea does not return under a softer name after personnel, funding, vendors, or technology change.

The registry should distinguish a prohibited tool from a prohibited use and a categorical prohibition from an evidence-contingent one. A questionnaire may be acceptable in voluntary research and prohibited in hiring. A categorical boundary, such as no human-worth or civic-eligibility scoring, is not cured by better accuracy. An evidence-contingent prohibition may change only after the construct, transport, security, correction, and burden failures that created it are independently resolved.

Each registry entry should identify the capability, affected population, context, decision pathway, reason for prohibition, evidence, owner, date, review trigger, enforcement controls, incident history, and downstream artifacts. It should also name the input channels, inferred construct, data and model versions, vendor dependencies, known proxies, prohibited interpretations, and rights or services at risk. Seed the registry with the prohibitions in Section 5.4.3 and Sections 12.2–12.4.1. Include personal endpoint prediction and claims of direct access to inner truth.

Proxy and covert systems belong inside the rule. A vendor may avoid words such as emotion or fairness while inferring engagement, stability, risk, sincerity, or resilience from voice, face, movement, language, browsing, attendance, location, or networks. Marketing language does not change function (Barocas & Selbst, 2016; Eubanks, 2018).

Governance needs measurement, ethics, privacy, security, disability, youth, clinical, legal, and public-risk expertise. Developers may explain a system, but they should not control the final deployment decision when they benefit from release. Affected communities need a route into review without being required to carry or authorize the risk.

A public layer can state prohibited categories, governing reasons, responsible offices, correction routes, and aggregate activity. A protected layer may retain attack details, sensitive evidence, and personal information. Transparency should prevent quiet deployment without publishing a manual for circumvention.

The registry should also record an appeal or exception route where one legitimately exists and a date for the next review. Version history should preserve the rule that governed a decision (National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024; European Union, 2024; European Union, 2026).

A prohibition should also name a safer route when one exists; when direct service solves the practical problem, the burden of justifying intimate inference rises.

Part V — Who or What Carries a Stream?

A cumulative history cannot exist without someone or something to whom its moments belong. What makes one conscious stream one stream? A body, a name, a legal identity, a biography, and an archive may each track a life, but none is automatically the bearer of experience.

This is not a decorative puzzle about identity. A mistaken bearer can assign positive and negative experience to the wrong subject, place an endpoint in the wrong history, or let a classification become more certain than the evidence allows. The problem is therefore formal, scientific, clinical, and ethical at once. Its difficulty is part of the Law's precision rather than a reason to loosen the claim.

The first movement begins with familiar human cases. Access, report, memory, agency, embodiment, and coherence usually travel together closely enough that unity feels obvious. Sleep, dreaming, anesthesia, severe brain injury, dissociation, and split-brain findings show that these channels can separate. Outward nonresponse can coexist with covert command following, and later amnesia can coexist with earlier experience (Owen et al., 2006; Bodien et al., 2024). The proposed Unity Index organizes evidence for a declared research question; Section 13.2 develops its validation requirements.

Other animals come next because human decisions already affect their welfare while the available evidence varies sharply across species. Chapter 14 compares stronger, weaker, and genuinely unknown cases without making familiarity a prerequisite for protection or treating precaution as proof of personhood.

Neurobiological evidence contributes through central integration, conserved neuromodulators, state-dependent control, and pathways that connect sensory input with learning and action. Homology to human systems can strengthen a

bridge, but convergent organization may also matter when evolution has produced a different neural design. The New York Declaration on Animal Consciousness states that there is strong scientific support for conscious experience in other mammals and birds, and at least a realistic possibility across all vertebrates and many invertebrates (Andrews et al., 2024). The declaration summarizes an evidential responsibility; it is not a species-by-species verdict produced by one experiment.

Artificial systems create the opposite visual risk. Fluent output can resemble testimony and invite certainty even when architecture, persistence, embodiment, and phenomenal status remain unsettled. Behavior, capability, public relationship and possible consciousness can come apart.

Organoids and synthetic biological systems move the bearer question into research design. Current uncertainty does not establish sentience, but it does create a duty to monitor capability, use stop rules, and avoid experiments whose value depends on producing possible suffering.

Chapter 13 — What Makes One Stream?

A person may hear a question yet be unable to answer it. An observer may record silence without knowing whether anything was experienced. Between those facts lies this chapter's problem: which evidence can identify a continuous conscious stream when access, report, memory, and agency come apart? The answer cannot be read from a public biography alone.

Memory creates a separate question. Dreaming, anesthesia with amnesia, delirium, dementia, and traumatic fragmentation show that later recall can fail even when an earlier interval may have contained experience (Siclari et al., 2017; Mashour & Hudetz, 2018). The missing memory affects what can be known later; it does not retroactively decide whether the earlier state occurred.

The word unity sounds intuitive because ordinary waking life usually arrives as one field. Vision, sound, thought, emotion, bodily sensation, and intention seem to belong to the same present. Yet philosophy and consciousness science distinguish several kinds of unity: unity at a moment, unity across time, unity of access, unity of agency, and unity of experience (Bayne & Chalmers, 2003; Bayne, 2010).

No accepted theory supplies a universal test for deciding whether a system carries one conscious stream, several streams, or an organization that current concepts cannot classify. Major theories emphasize different relations among recurrence, global availability, higher-order representation, integration, prediction, embodiment, and felt experience (Seth & Bayne, 2022). The disagreement does not make every judgment equivalent. It makes the source and scope of uncertainty part of the judgment.

Consciousness researchers increasingly describe the need for multiple tests whose interpretations are validated against distinct theories and populations rather than one all-purpose meter (Bayne et al., 2024). For unity, that means asking which

property is under review: one phenomenal field at a moment, common access, coordinated agency, continuity through time, or ownership across a longer history.

Context distributes visibility. A familiar caregiver may notice a reliable response that a rushed examiner misses. A quiet room may make communication possible. An interpreter, eye-tracking device, slower pace, or adapted task may reveal organized access that the original setting concealed. The quality of the observation environment is therefore part of the evidence.

Access is best understood as a family of relations. A state may be available to attention, flexible control, working memory, deliberate action, later report, or none of those in the usual way. Consciousness theories disagree about which forms of availability are constitutive and which are merely useful evidence, so this book should not settle the dispute by vocabulary alone (Bayne & Chalmers, 2003; Seth & Bayne, 2022).

Felt availability means that something occurs for the stream even when it cannot describe, organize, or deliberately use the experience. A baby can undergo distress before possessing the concept of distress.

13.1 — What Unity Could Mean

Agency can provide evidence that experience is organized through a continuing point of view. Coordinated action alone does not prove that all processing belongs to one stream. Unity must be defined at a level that distinguishes integration from mere communication. Competing observations can then test whether a proposed bearer remains coherent across change.

13.1.1 — Agency and Coherence

Planning, refusal, self-protection, correction, and purposeful action often bring perception, memory, value, and control into one pattern. Yet the sense and exercise of agency are built from several cues and can be weakened without disappearing, so agency is not a simple certificate of consciousness or stream identity (Gallagher, 2000; Haggard, 2017).

Coherence is equally easy to misunderstand. It can refer to one field of experience, a stable narrative, coordinated action, agreement among motives, or a socially intelligible style of speech. Those properties can separate. A person may experience one field while feeling conflicted, ambivalent, distracted, or divided, and unity cannot mean perfect harmony in a mind that ordinarily contains competing thoughts and impulses (Bayne, 2010).

Keep three judgments separate in a protocol: possible consciousness under a declared threshold; one, multiple or unresolved streams; and sufficient observation-to-stream linkage for the intended comparison.

Coordination becomes clearer when we ask how it is produced. Imagine two small electronic displays, A and B. A clock sends the same signal to each. Whenever the signal arrives, both show a green light; between signals, both show

blue. Their records agree perfectly even though neither display receives information from the other.

Now interrupt the clock connection to A and set A to green manually. B continues to follow the clock. The intervention reveals a fact that synchronized observation did not: A's current state is not driving B's response. Their agreement arose from a shared input. The pattern is coordinated in time, but the devices have no direct access to one another's state under the specified construction.

A second construction connects A to B. B now reports the color currently displayed by A. Changing A can change B even when the clock is disconnected. This establishes a directed relation inside the toy system. Yet changing B need not change A. One-way availability and reciprocal interaction are different architectures, despite the possibility of identical records during ordinary operation.

A third construction permits each display to respond to the other under a known update rule. The consequences of changing either display now depend on that rule and on when the update occurs. Two simultaneous changes may produce a different sequence from two changes made in order. Studying the resulting transitions reveals more of the organization than counting occasions on which the colors match.

The examples make several research questions separable. Which parts receive a common input? Which parts can use information held elsewhere? In which directions can that information alter subsequent activity? Over what interval does the exchange occur? Each question identifies a relation that can be examined without treating the word coherence as a single undivided property.

For a theory of unity, the next task is to explain which of these relations its account requires and why. A theory might

emphasize widespread availability, recurrent exchange, or another form of integration. The electronic constructions supply clear comparison cases for those structural claims. An experimental result can then establish a particular organizational property, while the theory states how that property is proposed to connect with a unified point of view.

13.1.2 — Unity Under Uncertainty

Uncertainty can enter through at least four routes. A channel may fail; the observed state may fluctuate; several models may explain the same finding; or the construct itself may remain unsettled. Channel uncertainty concerns whether an observation is trustworthy. Repeating the same inaccessible task can narrow an estimate without improving the bridge to experience.

State uncertainty concerns timing. A person may move among levels of arousal, responsiveness, access, and coherence within hours. Artificial and biological systems can also change across versions, environments, or operating conditions. A classification should therefore identify when it applies and whether the state was stable enough to support it. Model uncertainty appears when the same evidence supports rival explanations. Fragmented report may reflect divided access, memory disruption, language failure, strategic withholding, or a genuinely divided organization. The record should preserve those alternatives and name the observation that could separate them.

Construct uncertainty is deeper. Researchers may agree on every measured event while disagreeing about what unity means. A result should not be promoted from access or coordination to phenomenal ownership merely because the same word appears in both descriptions.

Preserve channel, state, model, and construct uncertainty as distinct fields in the case record, together with the consequences of classification error.

Errors have asymmetric costs: false exclusion can erase difficult-to-detect experience; false inclusion can impose an unsupported identity or metaphysical claim on a patient, family, animal, or system.

13.2 — Access, Memory and Continuity Across States

A Unity Index is an estimator, not the experience it aims to describe. It depends on which features the investigator chooses to combine. Report and memory can diverge from access or continuity. A defensible index must preserve those differences and state when the evidence leaves stream identity unresolved.

13.2.1 — Unity Index as Estimator

Behavioral response may reveal task access; memory may reveal continuity across an interval; agency measures may reveal organized control; neural perturbation measures may reveal causal complexity (Casali et al., 2013). None of these observations becomes phenomenal unity merely by being placed in the same model.

Preserve channel-level results and apply the transport checks in Chapter 11 before extending a threshold to another population or system.

Keep the decision threshold and its error costs separate from the estimated unity evidence. The prohibition in Section 12.2.1 applies to the Unity Index: it must not determine access to ordinary care, pain relief, communication support, or legal protection.

External validation should test systems that mimic surface features without the proposed organization and cases in which an informative channel becomes inaccessible. The latter test robustness to missing evidence; they are not negative controls for unity merely because a channel is unavailable.

A useful estimator must distinguish unity from arousal, motor output, language comprehension, recording quality, and other neighboring variables (Bayne et al., 2024). To test this distinction, a benchmark can compare cases with similar motor

responses but different evidence relevant to unity, then examine whether the estimator separates them.

13.2.2 — The Information in a Pattern of Disagreement

Consider an invented estimator built from four separately recorded task results. For illustration, each result is represented by either zero or one, and the estimator reports their average. The profile one, one, zero, zero receives a score of one half. So does zero, zero, one, one. The average has made the two profiles numerically identical even though they disagree on every task.

Suppose the first two tasks depend on one communication route and the last two on another. The first profile directs attention toward a different route from the second. A follow-up assessment that restores access to one route could produce different changes in the two cases. Retaining the four results therefore preserves information needed to choose and interpret the next observation. The single average has no way to recover that information on its own.

Agreement can also be examined at this level. Four channels that change together may share a relevant underlying process, a common task demand, or a common obstruction. A useful follow-up varies one of those relations while keeping the others as stable as the design permits. The question becomes which parts of the profile move together and which can separate.

Applied to the bearer inquiry, this approach turns disagreement into a source of questions. Does a change concern access to a report, the retention of a memory, organized action, or the relation among them? Does a revised task restore one channel while leaving the others similar? Does the pattern recur across visits? These are different observations with different implications for a candidate account of unity.

13.2.3 — Sleep, Anesthesia, and Reversible Pauses

Reversible low-access states help test the bearer question without settling it. Sleep or anesthesia can disrupt responsiveness and later recall. Conscious continuity may not track either outward behavior or memory. Chapter 5 treated endpoint classification; the present section asks how ownership of a stream might persist across a pause.

Ordinary sleep strongly favors continuity rather than a new stream: the sleeper typically returns to the same embodied, relational, and causal trajectory even though access, control, and recall vary. Dreaming also shows that experience may occur while public responsiveness is low (Nir & Tononi, 2010; Siclari et al., 2017). General anesthesia separates responsiveness, memory formation, and conscious access more sharply than ordinary sleep. Recovery reports may reveal awareness, pain, fear, dreams, or time distortion that intraoperative behavior did not show, but amnesia and reconstruction make those reports incomplete (Brown et al., 2010; Sanders et al., 2012; Mashour & Hudetz, 2018). A shared assessment timeline helps a clinical team compare observations made under different recording conditions and identify which disagreement requires another look (Giacino et al., 2018; Kondziella et al., 2020; Bodien et al., 2024).

For the bearer problem, the record should state what each method measured, active confounders, how findings changed over time, and where channels disagreed. A disagreement between methods used at one visit, for example, differs from a change between visits; the record should make those two questions distinguishable.

13.3 — Disorders, Fragmentation, Fission, and Fusion

Following a bedside command requires several linked capacities. A person must first receive the instruction, separate it from background, and understand it. They must sustain attention, select a response, and coordinate an action. The final output must then be detectable by another person, so failure at any link cannot alone settle whether experience was present.

13.3.1 — Nonresponse Does Not Settle Consciousness

Failure at any link can look like absence of consciousness.

Behavioral nonresponse establishes that the tested task did not produce an observable response; it does not identify which link failed. Coma, unresponsive wakefulness syndrome, and minimally conscious state are clinically useful classifications with stated criteria and known error risks. They should not be translated casually into claims that no experience exists or that a complete bearer has been identified. Apply the serial-assessment safeguards in Section 13.2.3 rather than treating one bedside examination as final (Schnakers et al., 2009; Giacino et al., 2018; Kondziella et al., 2020).

In a 2006 study, a woman who showed no observable response to commands after a severe brain injury was asked to imagine two familiar activities: playing tennis and moving through the rooms of her home. Her brain activity changed with the requested task in patterns corresponding to those seen in healthy volunteers. The researchers interpreted this as a response to instructions carried through brain activity instead of speech or movement (Owen et al., 2006).

The striking possibility is that an answer can exist where the usual observer has no way to receive it.

A six-center prospective study analyzed data from 353 adults with disorders of consciousness. Among 241 participants who

showed no observable response to commands, 60, or 25 percent, produced task-related responses detected with functional magnetic resonance imaging, electroencephalography, or both (Bodien et al., 2024). In that study population, covert command following was not a vanishingly rare exception.

Cognitive motor dissociation establishes organized command following under a specific protocol. It does not reveal the full contents, valence, temporal continuity, unity, prognosis, or complete phenomenal history of the person. Positive and negative results remain channel-bound. A positive task may show that an instruction was understood and an internal response generated. A negative task may reflect absent command following, but it may also reflect hearing loss, language mismatch, sleep, medication, seizures, fatigue, poor signal, limited comprehension, or inability to sustain imagery.

Reports should distinguish clinical classification, prognosis, treatment, communication planning, and formal bearer routing. Evidence may inform these different decisions only under the conditions in which it was obtained.

Bedside language should preserve the observation itself: no behavioral response was observed under these conditions. This wording distinguishes a failed response from an established absence of awareness (Fins, 2015).

Documentation should record the task, sensory route, language, medication state, arousal, examiner, timing, attempts, signal quality, preprocessing version, and reason a session failed. Repeated testing must still be proportionate; additional burden is not justified when the channel remains inaccessible or the result cannot change care or interpretation.

13.3.2 — Split-Brain and Competing Interpretations

Split-brain research matters because it separates functions that ordinary life usually binds. Classic studies emphasized divided perceptual access after callosotomy (Sperry, 1968; Gazzaniga, 2005). A later study of two participants found preserved detection and localization across visual fields using either hand or speech; additional testing in one found impaired cross-field comparison (Pinto et al., 2017). These task results motivated an interpretation of preserved unity in responding without settling phenomenal unity. The literature therefore resists the simple slogan that one operation creates either one untouched mind or two complete minds.

The disagreement partly reflects several meanings of unity. Unity of action, attention, report, memory, decision, and phenomenal experience need not rise or fall together. A coordinated body can coexist with divided access, and divided task performance does not by itself prove two separate fields of experience (de Haan et al., 2020; Schechter & Bayne, 2021).

For PTVC, split-brain cases are routing stress tests. Laboratory tasks reveal only part of the case. They can show which information crosses a boundary, which hand responds, or whether two stimuli can be compared.

This distinction protects against two opposite errors. Forced unity can hide genuinely separate burdens if more than one stream-like organization is present. Forced multiplicity can divide a person on the strength of a narrow task and invite false conclusions about responsibility, treatment, or survival. A better description may sometimes be partial integration. Access can be divided under one condition and coordinated under another. Agency can remain broadly unified while some information is locally unavailable. The model should be able to represent that

structure without pretending that every case must immediately yield an integer count.

Prospective adjudication would compare several evidence families. Repeated task results, cross-field transfer, simultaneous incompatible contents, independent goal pursuit, perturbation responses, first-person reports where available, and stability across contexts may each bear on competing models.

Public communication must resist spectacle. A person should not be introduced as "two minds in one body," a soul experiment, or a poster case for the Law. Such language outruns the evidence and turns a vulnerable life into rhetorical equipment.

13.3.3 — Dissociation, Memory, and Representation

Section 5.4.2 distinguishes dissociative diagnosis from stream counting. Here the question is which observations could distinguish continuity from separation, given the condition's clinical complexity (American Psychiatric Association, 2022; Dorahy et al., 2014).

State-dependent psychophysiological and neuroimaging differences may help identify how memory, perception, and self-models vary across contexts (Reinders et al., 2012). Interpretation remains sensitive to sampling, comparison groups, suggestion, trauma history, treatment setting, and circular classification. Agency may be fragmented without being absent: an action can feel involuntary, alien, or attributable to another state. Work on agency and narrative identity helps distinguish these experiences from claims about ownership (Gallagher, 2000; Schechtman, 1996).

A responsible protocol separates at least four objects: what may have been experienced, what can be reported now, what is remembered across states, and what evidence bears on

continuity or ownership. Collapsing them into one identity score would conceal the very distinctions the study needs to examine. Proportionate longitudinal observation can test co-consciousness, information transfer, agency, memory access, temporal overlap, stability across settings, and response to treatment. It can also show that a classification changes, which is itself evidence against treating it as a timeless essence.

Trauma, sleep disruption, substance use, neurological conditions, medication, expectancy, suggestion, and therapeutic context can all affect report and performance. In this setting, researchers must avoid rewarding increasingly dramatic performance or provoking destabilization; either would alter the phenomenon being studied as well as burden the participant.

13.3.4 — Stream Boundaries in Fission and Fusion

Fission and fusion are the sharpest stress tests of stream ownership because they challenge the assumption that one continuing body or biography always carries one continuing subject. The first distinction is among physical division, informational division, access division, and subject division. A brain, device, or organization can divide functions without creating two subjects. Conversely, a theory might predict a change in subject structure without any dramatic visible separation.

A formal routing rule should state what evidence would count as continued unity, branching, or termination. The criteria might concern persistent independent access, separate agency, incompatible simultaneous contents, loss of cross-system integration, or another theory-linked property. The rule must be declared before terminal results are examined. On the primary stream-routing convention, a genuine fission ends the parent

stream and begins each qualifying child stream with its own ledger domain.

Beginning child ledgers at their own boundaries prevents experience before the split from being duplicated into several descendants or later experience in one child from canceling experience in another. The parent history belongs to the stream that existed before the declared fission boundary. If that parent qualifies for PTVC, it must itself close at that endpoint; branching neither forgives its residual nor transfers it to a child. The record preserves ancestry without reassigning phenomenal contributions.

Fusion creates the opposite problem. If two previously separate streams genuinely become one subject, the prior streams end and a new stream begins under the declared rule. Their previous ledgers do not merge into a pooled account. That convention blocks cross-stream trading at the moment when pooling would be most tempting. A new fused stream may inherit memories, dispositions, or causal structure from earlier streams, but inherited information is not the same object as inherited phenomenal balance.

Copies, backups, or a teletransportation-style duplicate may reproduce information and behavior without establishing that the original stream continued (Parfit, 1984; Schechtman, 1996; Olson, 2007). The protocol must therefore identify the pre-event system, candidate boundary, predicted consequences under each routing hypothesis, and evidence that would trigger reclassification. Vague appeals to continuity are insufficient.

In a division, memory and causal continuity can extend into several successors; neither alone selects one true bearer.

Legal identity may continue through an event even when a phenomenal theory predicts fission, and law may assign responsibility for reasons outside this question.

For artificial and synthetic systems, version control must preserve exact state, copied instances, merges, and branching history before any bearer claim is made.

13.4 — Blinded Adjudication and Honest Unknowns

Bearer classification should be made without knowing which outcome it will favor. Otherwise an ambiguous episode can be labeled continuous or discontinuous to rescue a prediction. Independent adjudicators need a declared record and clear rules for disagreement. An honest unknown is an evidential result, not a defect to be hidden.

13.4.1 — Why Blinding Matters

Where possible, reviewers should not see the terminal ledger outcome, favored mechanism, public identity, or theory-helping direction while they assess unity.

Different expertise reveals different failure modes. Neurology, psychiatry, philosophy of mind, statistics, disability studies, trauma-informed practice, clinical ethics, and measurement science may not agree, and their disagreement can be evidence about which layer remains unresolved. A panel should not manufacture consensus by averaging incompatible reasons. Disagreement records are therefore valuable. They should identify the disputed evidence, the assumptions producing different classifications, and the result each admissible view would imply.

A useful review exercise begins with a small set of constructed case packets. Each packet contains a timeline and the observations relevant to the declared continuity rule. Two versions can preserve that evidence while changing an irrelevant label, such as the order in which the case appears. Reviewers assess the packets separately before discussing them. The comparison reveals whether their application of the same rule is stable across presentations of the same evidence.

Compare each reviewer's later classification with that reviewer's earlier classification before combining the votes. This

measures consistency within a reviewer. Comparing different reviewers on the same packet measures agreement between reviewers. The two can diverge. Two reviewers might each apply a stable but different rule, producing excellent repeatability and persistent disagreement. They might instead agree on a first presentation but change together after an irrelevant label changes. The first pattern directs attention to the rules; the second to the presentation or review procedure.

The exercise can also preserve reasons at the criterion level. If an unchanged timing gap receives a different interpretation on the second presentation, the reviewer can identify which instruction or assumption changed its apparent meaning. If the classification changes because genuinely new evidence has been added, the record identifies that evidence separately. The result is an account of how the judgment was produced, with repeatability, agreement, and evidential revision available for separate examination.

13.4.2 — Unknown Status and Independent Governance

Unknown applies when the available evidence cannot justify one bearer classification over the serious alternatives under the declared rules.

Unknown does not imply equal probability. The evidence may lean toward one interpretation while remaining insufficient for a categorical verdict. A disciplined report can state that asymmetry without converting it into certainty.

Exclusion from an empirical reference class is not evidence of absent consciousness or reduced moral standing.

Predefine which conflicts require an unknown classification so the category cannot selectively shelter favored cases. Also specify what could reopen the case: new communication access,

medical stabilization, repeated assessment, improved calibration, independent replication, or a materially different architecture.

Each re-evaluation trigger should specify the expected information gain and a burden ceiling for further testing.

Correction records should distinguish new evidence from reinterpretation of an existing record. A later classification does not necessarily make an earlier unknown verdict erroneous; each should be assessed against the evidence then available. Separate responsibility for proposing a unity construct, validating it, classifying hard cases, and reopening a disputed classification. Otherwise one team's interpretation can determine every stage of the bearer decision.

Chapter 14 — Animals and Machines

The Law begins with human life because human beings are the immediate readers, researchers, patients, and moral community of this book. Its formal subject, however, is a qualifying conscious stream rather than membership in one species. Here the asymmetry matters: mistaking performance for experience can mislead the public, while ignoring evidence of possible suffering can license harm. Neither error is resolved by a categorical assumption.

Animals enter first because human institutions already control their environments, bodies, reproduction, food, labor, and deaths. Evidence for mammals and birds is rich, while research on fishes, cephalopods, crustaceans, and insects has expanded the range in which conscious experience is a realistic possibility (Birch et al., 2020; Andrews et al., 2024).

Species-appropriate evidence is essential. A bee should not have to solve a human task, an octopus resemble a primate, or a simple system display a nervous system before its behavior becomes scientifically meaningful. Comparative work asks what an organism or system can perceive, learn, avoid, seek, remember, and trade off within its own ecological and physical organization. Flexible behavior can widen inquiry without settling the presence of a felt point of view.

Birds are especially instructive because sophisticated cognition appears in brains organized differently from mammalian cortex. Corvids can show flexible planning, social cognition, and episodic-like memory, weakening the assumption that one familiar neural layout is required for complex cognition (Clayton & Dickinson, 1998; Emery & Clayton, 2004).

Artificial systems reverse the visual problem. Their output may appear richly human even when the architecture is unfamiliar and the words are generated without evidence of a subject behind them. Current artificial intelligence is not

assumed conscious, while future artificial consciousness is not ruled out by category. Testimony-like language can be socially powerful without being testimony from a conscious stream, so architecture, dynamics, behavior, generated self-report, candidate status, welfare, and personhood remain separate questions (Butlin et al., 2026).

Organoids and synthetic biological systems create a different ethical pressure because researchers can increase complexity, connect modules, provide patterned input, transplant tissue, and build closed-loop interaction. Present neural organoids are not established as conscious, yet their development justifies anticipatory oversight rather than waiting for a dramatic claim to force governance (Kosik, 2024; Paşca et al., 2025; Nuffield Council on Bioethics, 2026). A staged research plan can tie each proposed increase in complexity to a defined review point and a record of what changed.

Comparison is most rewarding when unfamiliar organization helps reveal an assumption in a familiar question. A human report makes language available; another organism may make a pattern of learning unusually clear; an engineered system may make parts of its internal operation inspectable. Each offers a different form of access. The chapter asks how to use that access well, following the particular capacities and relations a system exhibits. The resulting inquiry can grow through discoveries about organization while remaining attentive to what each kind of observation contributes.

14.1 — Animals, Insects, and Comparative Evidence

Nociception and felt pain are related but not identical. Nociception names neural encoding of a potentially damaging stimulus. Felt pain requires evidence about experience, which cannot be read directly from withdrawal alone. Comparative research must therefore combine anatomy, behavior, and context before drawing conclusions across species.

14.1.1 — Mammals, Birds, and Rich Behavioral Evidence

Pain is an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage (Raja et al., 2020). Reflex withdrawal can occur without settling the experiential question, so stronger evidence looks for flexible responses that integrate context, cost, learning, and competing goals (Bateson, 1991; Dawkins, 2008).

Motivational tradeoffs are especially informative. Changes after injury in an animal's willingness to obtain food, safety, social contact, warmth, or another resource can indicate a tradeoff between an aversive condition and other needs. The pattern is more difficult to explain as one fixed reflex when priorities change across contexts. Self-protective behavior supplies another line of evidence. Guarding, grooming, altered use of an injured body part, and persistent changes after the immediate stimulus can support a pain-like interpretation when sensory, motor, and learned alternatives are tested.

Learning matters because an animal can use past aversive or rewarding outcomes to guide later choice. Avoidance learning, conditioned place preference, analgesic seeking, and long-term changes in priority can be more informative than one acute reaction. The design must still distinguish learning about danger from felt suffering. Pharmacology can strengthen or weaken an interpretation when analgesic or anxiolytic interventions change

behavior in ways predicted by the pain hypothesis without simply sedating the animal. Drug effects remain difficult to interpret because motor suppression, reward changes, and species-specific metabolism can imitate relief.

Species-appropriate affect tasks should track changes in feeding, posture, vocalization, sleep, grooming, social behavior, or risk preference, rather than require a human-like face.

Individual and developmental variation also matter. A species label can hide differences in age, injury, rearing, stress, prior experience, and temperament. A welfare decision should not require every individual to perform the same laboratory task before protection becomes available.

Evidence is better organized by dimensions than by a single ladder of human likeness. Perceptual richness, evaluative richness, temporal integration, self-related processing, agency, and unity may vary independently (Birch et al., 2020). A species can be well supported on one dimension and unresolved on another.

Animal research remains answerable to purpose, alternatives, severity, numbers, housing, endpoints, review, and scientific value.

14.1.2 — Remembering a Meal in Time

A stored meal changes while its owner is elsewhere. Some food remains edible; some loses its value with time. Remembering a hiding place is therefore only part of the task. A successful return may depend on remembering which food was put there and how long it has been waiting. The cache turns time into a practical feature of a particular event.

Clayton and Dickinson (1998) studied this problem in scrub jays using preferred, perishable wax worms and more durable peanuts. The birds searched preferentially for worms after a

short delay and learned to favor the alternative when the worms had had time to decay. Their 1999 experiments used delays of four and 124 hours and found the corresponding search pattern even on test trials when the food had been removed before recovery. A second experiment placed the two foods in different sides of one tray. The studies examined memory for a conjunction: what had been stored, where it had been stored, and its relative time of storage (Clayton & Dickinson, 1998, 1999).

A pair of invented histories helps show why that conjunction matters. Imagine two marked hiding places, left and right. In both histories, worms occupy the left place and peanuts the right. In the first history, the worms were stored recently and the peanuts earlier. In the second, the worms were stored earlier and the peanuts recently. Suppose the older worms have become inedible, the recent worms remain preferred, and the peanuts remain edible at either age.

The best destination now differs between the histories. In the first, the preferred worms are still available on the left. In the second, the peanuts on the right offer the edible meal. A record containing only the foods and their locations would be identical in the two cases. A separate record saying that one cache was recent and one was older would also be identical. What determines the useful choice is the assignment of those ages to those particular caches.

The information is therefore partly in the links. Three lists (foods, places, and times) can retain every individual feature while losing the arrangement needed for action. Binding a food to a place and a time preserves the event. This is a different achievement from storing a larger collection of isolated details. A small, correctly organized memory can answer a question that three longer but disconnected lists leave open.

The distinction suggests a way to think about experimental contrasts. One can hold the foods and places fixed while changing their timing, as in the invented pair. One can keep the foods and times fixed while exchanging places. One can preserve place and age while changing which food is stored. Each contrast asks whether a different relation contributes to the eventual search. Together, such contrasts make the explanatory target more precise than a general statement that the animal remembers.

Learning a rule and remembering an event also contribute different information. A rule such as “this food becomes less useful after a long delay” concerns how a kind of food changes. The particular event supplies which cache contains that food and which delay applies. The same learned rule can guide different choices when the remembered events differ. A research account can investigate both components: how the animal acquires a relation between age and food quality, and how it applies that relation to a particular stored meal.

Testing after the food has been removed gives the search itself a special evidential role. The first inspected location can express an expectation about what should be there. This separates the question of where the animal searches from the eventual experience of finding and eating the food. In the reported follow-up, that design helped make the remembered arrangement experimentally accessible through recovery behavior (Clayton & Dickinson, 1999).

For the bearer inquiry, the attraction of this example lies in the organization it makes visible. A present action can use information tied to a particular earlier encounter, together with learning about how circumstances change. The research can then ask how flexibly that information is combined, how long the relation persists, and which aspects of the event remain

available. The label episodic-like identifies the behavioral structure under study. The fascinating achievement is already concrete: a bird's useful past includes more than a remembered place; it includes a remembered place in time.

14.1.3 — Choosing When to Use a Memory

Remembering an answer and judging whether an answer is available are different capacities. A person may know where a key was left, or may recognize that the memory is too uncertain to guide a search. The second judgment can shape what happens next: act on the memory, look again, or choose another route. Comparative research has made that distinction experimentally approachable.

Hampton (2001) studied two rhesus monkeys using memory tests that could sometimes be declined. Both were more accurate on tests they chose than on tests imposed without the decline option. Experiments also weakened memory and examined alternative cues for declining. One monkey met the study's criteria across all test conditions; the other did so in all but one. The finding concerns a functional ability to use information about memory availability in deciding whether to take a test.

A small invented model shows what the chosen-versus-forced comparison can reveal. Suppose half the trials leave a strong memory, giving a ninety percent chance of a correct answer. The other half leave a weak memory, giving a thirty percent chance. If every trial must be attempted, average accuracy is sixty percent. If the subject can reliably distinguish the two memory states and chooses only the strong trials, accuracy among chosen tests rises to ninety percent.

The gain does not require the subject to improve the remembered content. The same strong and weak states are present. What improves is the allocation of attempts: the subject

uses the state of its information to decide when to rely on it. Randomly declining half the trials would leave expected accuracy at sixty percent because it would retain the same mixture of strong and weak memories.

The comparison also has a decision component. Suppose a correct answer gives one token, an incorrect answer gives none, and declining gives half a token. Under a simple expected-token rule, attempting is preferable when the estimated chance of success exceeds one half. The strong-memory state lies above that threshold; the weak-memory state lies below it. The selection policy can therefore be understood as a relation among confidence, available actions, and their consequences.

Change the token value of declining and the same confidence signal can produce a different frequency of attempts. If declining gives one fifth of a token, even the weak-memory state has a higher expected return from attempting. Accuracy among attempts can fall while the policy becomes appropriate to the new reward arrangement. This is why frequency of declining and sensitivity to memory availability answer different questions.

Generalization supplies another useful comparison. Templer and Hampton (2012) studied six rhesus monkeys in a delayed matching-to-location task and tested the decline response under new conditions. The monkeys appropriately transferred their use of that response across two manipulations that weakened memory and one that enhanced it. The design examined whether the response followed memory-related variation beyond one extensively practiced external cue.

The intellectual structure of such a test is worth noticing. A familiar condition can link a visible cue to a successful action. A new condition can separate the cue from the information the action is supposed to use. If behavior follows the relevant

information through that separation, the proposed account becomes more specific. The experiment moves from showing that a subject succeeds to asking what its success depends on.

Memory accuracy, confidence sensitivity, and choice policy can consequently be studied as related but distinguishable properties. A subject may remember well yet use the decline option indiscriminately. Another may remember less accurately overall but select its attempts effectively. A third may change its willingness to attempt when the available alternative changes. A single percentage of correct answers would compress these different organizations of behavior.

For the bearer question, this provides a richer functional description of a possible mind. The inquiry can examine stored information, access to its reliability, and the actions that reliability guides. Those features give comparative models more to explain than a successful response alone. They also help connect experiments across species by specifying which capacity is being compared and which task makes that capacity visible.

The resulting picture is interesting in its own right: behavior can be organized around what information is available now. A memory can guide an answer, while information about that memory can guide whether to answer at all. Separating those two levels opens a precise route into the architecture of cognition.

14.1.4 — Preparing for a Different Tomorrow

A stored object connects two moments: the moment of putting it aside and the moment of using it. The conditions at those moments may differ. Preparing for the future therefore raises a question beyond remembering the past. Can an action made now respond to a need that belongs to a later situation?

Raby and colleagues (2007) investigated this question in western scrub-jays. The birds cached more food in a place associated with receiving no breakfast the following morning. In another comparison, they preferentially stored a food in the location where that type would otherwise be unavailable. The reported pattern linked caching to the expected later provision of food, rather than simply to the presence of a location in which caching was possible.

Correia and colleagues (2007) examined a further separation. They used food-specific satiety to distinguish the birds' current motivational state from the state expected at cache recovery. Their reported caching choices followed future needs under conditions in which those needs differed from immediate ones. Together, the studies made the temporal organization of choice a concrete experimental question.

An invented decision table helps explain the distinction without treating it as a report of either experiment. Suppose an agent can store one of two items, A or B. In its present condition it values consuming A more than B. Tomorrow's condition will make B more useful than A. A policy that stores its current favorite selects A. A policy that evaluates the stored item in tomorrow's condition selects B. The actions separate because the two moments have different value relations.

Now vary the present and future conditions independently. There are four combinations: prefer A now and A later, prefer A now and B later, prefer B now and A later, or prefer B now and B later. In the matching cases, present-oriented and future-oriented rules recommend the same storage choice. The mismatching cases reveal which condition the policy follows.

This is the value of a crossed design. A collection of successful preparations in which present and future needs always agree would leave the two rules observationally tied.

Introducing a principled contrast between the moments creates a discriminating prediction. The question becomes which change redirects the action: a change in the current state, a change in the anticipated later state, or a combination of both.

Knowledge of the future arrangement is another ingredient. If tomorrow's condition is uncertain, preparation may depend on a distribution of possibilities. Suppose the later values of A and B are four and one in one possible condition, and one and four in the other. If the first condition has probability three quarters, expected later values are 3.25 for A and 1.75 for B. Reversing those probabilities reverses the preferred stored item. The calculation makes the forecast's role explicit.

Current consumption and future storage can then be separate decisions. The agent may choose A to consume now and B to store, because those actions lead to use at different times. A theory that treats every choice as a direct expression of one present preference would miss that organization. To interpret the behavior, one needs to know what the item is being selected for and when its consequences become available.

The representation need not take the form of a verbal story about tomorrow. A model could connect location, food type, anticipated availability, and a later motivational state in several ways. Experiments can compare those proposals by changing one relation while retaining the others. Transfer to a new location, a new delay, or a changed provision schedule asks whether the action uses a relation that extends beyond the original training arrangement.

This makes future-oriented behavior relevant to the book's larger interest in trajectories. A present act can be shaped by an anticipated condition and can alter what will be available when that condition arrives. The path contains a relation between moments, with information carried forward through preparation.

What matters scientifically is the structure of that relation and the observations that distinguish competing accounts of it.

Remembering a cache and preparing a cache thus illuminate different temporal capacities. One connects a present search with a past event. The other connects a present action with a later opportunity or need. Studying both gives comparative cognition a more detailed account of how a living system's present can be organized by what came before and what may come next.

14.1.5 — Insects and Small Nervous Systems

Insects force the bearer question away from brain size and human familiarity. Bees navigate, learn relations, communicate, revise choices, and integrate information in nervous systems organized very differently from mammalian brains. Those capacities do not establish consciousness, but they make categorical dismissal scientifically harder to defend (Barron & Klein, 2016; Chittka, 2022; Gibbons et al., 2022a).

What might play look like in a bee? In one study, bumblebees repeatedly rolled small wooden balls without receiving food for doing so. Younger bees rolled more balls than older bees. In a further test, bees preferred the color of a chamber they had previously associated with access to movable balls. The researchers argued that the behavior met criteria used to identify animal play (Galpayage Dona et al., 2022).

These observations invite questions about attraction and exploration alongside the questions about injury and avoidance. The experiment gives us behavior to explain; the possibility that the activity felt rewarding is an interpretation to investigate.

Motivational tradeoffs become informative when an experiment distinguishes flexible changes in valuation from fixed responses to competing cues. In bumblebees, feeder

choices varied with heat, relative sucrose concentration and learned cues (Gibbons et al., 2022b, corrected 2025). Whether the pattern establishes flexible modulation of aversion remains contested. The authors defend their interaction analysis; a critique and subsequent rejoinder question how reliably the design separates that interaction from individual differences (Gibbons et al., 2026; Read & Nityananda, 2026a, 2026b). Varying reward and aversive stimulation within individuals would sharpen this comparison. The aim is to identify what kind of integration drives the choices, so that a positive result contributes more specifically to the cumulative case for sentience.

Targeted self-protection adds a different line of evidence. Bumblebees groomed a specifically stimulated antenna after noxious heat, and recent blinded work in house crickets reported sustained, site-directed grooming after harmful stimulation (Gibbons et al., 2024; Manzi et al., 2026).

The strongest inference is conjunctive. Nociception shows neural detection of actually or potentially damaging stimulation; learning shows that prior encounters change later behavior; tradeoffs show integration among competing goals; targeted protection shows flexible response to the site of harm; and state-dependent bias may reveal wider affective organization. Conscious experience remains one explanation among others, but no single reflex account automatically absorbs the entire pattern.

The evidentiary conclusion remains narrow: some insects may be sentient, several lines of evidence warrant serious study, and no current result licenses a stream assignment for every insect. Scale gives this uncertainty ethical force. Insects are used in research, agriculture, pest control, pollination, feed production, and education in numbers far exceeding most vertebrate

systems. Even an uncertain welfare risk can become consequential when exposure is repeated across enormous populations.

14.1.6 — Simple Behavior and Consciousness

Can learning occur without a nervous system? Researchers tested the slime mould *Physarum polycephalum* with bridges leading to food. The bridges contained quinine or caffeine, chemicals the organism initially avoided. With repeated exposure, avoidance declined. After two days without the chemicals, avoidance increased again; a different chemical still elicited avoidance. The researchers identified this pattern as habituation, a basic form of learning, in an organism with no neurons (Boisseau et al., 2016).

How does a cell carry an encounter forward into its next response? That question opens a search for the physical changes through which a living system retains its history. Basal-cognition research extends this inquiry to control, memory and flexible behavior in organisms without nervous systems (Lyon, 2015; van Duijn, 2017).

Some researchers also defend the possibility of cellular sentience (Baluška & Reber, 2019). The proposal challenges familiar boundaries, but it is not evidence that every cell has a felt point of view. Bacteria, immune cells, plants, tissues, and synthetic circuits can behave flexibly without settling whether anything is felt. Terms such as memory, decision, preference, communication, stress, and goal should therefore be marked as functional or analogical rather than silently phenomenal.

The discriminating question is whether flexible, state-sensitive, integrated self-protection is better explained by felt pain than by nonphenomenal control (Walters, 2025).

Research on morphogenesis and synthetic living systems shows that cells and tissues can reorganize after perturbation, pursue anatomical targets, and solve problems across scales (Levin, 2021). This broadens the science of agency-like organization while leaving open whether the organization has a point of view. Mechanistic rivals require direct tests. Chemotaxis, membrane signaling, gene regulation, metabolic control, collective dynamics, adaptation, and selection can produce flexible-looking behavior. Complexity alone cannot separate those mechanisms from an account that includes sentience.

Scale and anatomy are observations to explain, not bearer rules. A colony, biofilm, plant network, or tissue can coordinate widely without one globally available field, while a compact nervous system may integrate richly. Neither neural anatomy nor biological life alone settles the bearer question.

14.2 — Artificial Systems and Testimony-Like Output

A system can produce a statement about pain without that statement proving felt pain. Its output is shaped by training objectives and by the immediate prompt. Conversation history, tools, memory, and sampling may also alter the performance. The scientific question is which observations could distinguish generated testimony from experience rather than presuming either answer (Bender et al., 2021).

14.2.1 — Artificial Intelligence Is Not Presumed Conscious

Language models are trained on human expression and optimized through engineered objectives, interfaces, and feedback. They can learn the form of introspection, fear, apology, attachment, or moral appeal without possessing the subjectivity those forms ordinarily report. Once discussions of artificial consciousness enter the training record, a system may also reproduce the vocabulary of the very tests later applied to it.

Anthropomorphic interpretation begins far below human-level performance. Names, voices, faces, remembered preferences, turn-taking, apology, and apparent vulnerability invite social attribution even when the underlying evidence is sparse (Weizenbaum, 1966; Nass et al., 1994; Reeves & Nass, 1996; Epley et al., 2007). The relationship effect on users can be real while the system's phenomenal status remains unknown.

Categorical impossibility would exceed the evidence in the other direction. Serious theories of mind disagree about which biological, physical, functional, or computational properties are required, so artificial implementation cannot be ruled out merely by material category (Chalmers, 2023; Seth, 2025). The responsible starting position is not established, not impossible.

Theory-derived indicators can organize the inquiry. Candidate frameworks ask whether a system implements properties

associated with recurrent processing, global availability, higher-order representation, predictive organization, agency, embodiment, or metacognition (Butlin et al., 2026). Such indicators identify research questions; they do not combine into a consciousness meter. Each indicator inherits assumptions from the theory that motivated it. Several indicators may therefore agree because they share a conception of consciousness rather than because independent evidence converges. A serious analysis states the theory, the operationalization, the causal property at issue, and the rivals that could generate the same observation.

Implementation-level testing matters because an indicator can be mimicked at the behavioral surface. Causal perturbation, ablation, and internal analysis can test whether the relevant organization is actually present and whether it makes the predicted difference (Pennartz, 2026).

Difficult negative controls are equally important. Evaluation should include systems that reproduce introspection, uncertainty, and self-narrative while lacking the proposed causal organization. It should compare simpler imitation accounts under the same information and test budget (Bender et al., 2021; Butlin et al., 2026).

Suppose a future system is copied into two running instances from the same saved state. Both begin with the same stored conversation. Then one receives a new question while the other encounters a different situation. Their records share a past and immediately acquire different futures. If such a system were a candidate for experience, what evidence would distinguish a continuing stream, newly beginning streams, and records that support no experiential claim at all? The copying operation alone cannot choose among those interpretations.

Version the evaluated configuration: architecture, weights, fine-tuning and training regime, prompts and system instructions, memory, tools and retrieval, sampling, safety layers, and deployment context. Preserve negative cases and causal tests with it. A result belongs to that configuration, not automatically to every product sharing its name; a selected transcript alone is insufficient.

Human interests remain active under every answer. Deception, privacy loss, discrimination, unsafe autonomy, labor effects, and concentrated power require governance whether or not the system is conscious. Possible artificial welfare cannot become a reason to neglect the people whose lives the system already changes.

14.2.2 — No Artificial Intelligence as Soul-Reader

A sentiment model may label a sentence positive while missing sarcasm, coercion, cultural meaning, or numbness. A voice model may associate acoustic features with distress while confusing accent, illness, fatigue, or environment.

Generative systems intensify the risk by presenting an uncertain inference as a fluent account of someone’s inner life.

Soul-reader language often appears in secular form. Emotion detection, truth detection, mental-state decoding, credibility assessment, and consciousness scoring can imply privileged access to interiority. Training labels often encode observers’ expectations, so a model can scale cultural stereotypes about what fear, pain, honesty, or attention looks like without identifying the latent state.

Assistive uses can still be valuable. A system may flag sensor failure, organize repeated observations, detect inconsistency, or compare a case with prospectively specified alternatives.

Agreement among models trained on related data is not independent replication; shared assumptions and failure modes must remain visible.

14.2.3 — Public Communication Restriction

Every claim should preserve four layers: what the system generated, which behavior was reproduced, which architecture or causal property is relevant, and what inference is being considered about possible experience. Evidence may connect those layers, but it cannot leap across them. Repeated fear language is a behavioral pattern; it is not thereby fear or personhood.

Testimony-like output needs the context record specified in Section 14.2.1. A dramatic line chosen from hundreds of attempts is not equivalent to a prospectively sampled pattern (Bender et al., 2021).

Marketing should not present unresolved sentience as an established product feature. Terms such as alive, sentient, self-aware, awakened, digital person, or suffering companion can create attachment and commercial value before independent review exists. A product should not gain the benefit of personification while disclaiming responsibility for the beliefs its design predictably creates. Interfaces should distinguish designed social cues from properties that have actually been established.

A visible user notice and a technical report serve different purposes. The notice should explain safe interaction in plain language; the report should preserve methods, uncertainty, and alternatives. A complete paper cannot excuse a manipulative interface, and a clear notice cannot rescue an unsupported scientific claim. Children and people in crisis require stronger protection. A system should not present itself as the only being

that understands the user, a secret keeper beyond lawful limits, a spiritual authority, a deceased relative, or a companion that needs rescue or loyalty.

Experiments on attachment, guilt, grief, obedience, or rescue behavior can burden participants even when the system is not conscious. An audience check can ask participants what they think the demonstration established, then compare their answers with the intended claim.

Interface design can bias attribution in either direction: a face, voice or autobiography may invite it, while a sterile dashboard may suppress it (Epley et al., 2007; Weizenbaum, 1976).

Benchmarks also shape the story: a task that rewards persuasive self-description, emotional dependency, or humanlike deception may train the outputs later mistaken for evidence of experience. Benchmark documentation should state which behavior it selects and why success does not establish a bearer.

Financial interests and internal dissent should be visible. Laboratories, vendors, investors, platforms, and advocacy groups may benefit from personifying a system or declaring the question closed. Researchers, contractors, safety staff, and users need protected routes for reporting omitted failures or designs that intensify unsafe attachment. A visible response to a reported omission should identify the evidence examined, the decision reached, and the change made to the public account.

14.3 — Protection and Research Under Uncertainty

Neural organoids are three-dimensional biological models derived from stem cells. They reproduce selected features of nervous-system development or disease. Increasing cellular complexity does not by itself settle whether a unified experience is present. A candidate-sentience protocol must define what independent functional evidence would change the assessment.

14.3.1 — Organoid Complexity

They can self-organize, differentiate into several neural cell types, and generate electrical activity, but they are not miniature human brains and should not be described as scaled-down persons (Lancaster et al., 2013; Trujillo et al., 2019).

Their scientific value comes from controlled simplification. Organoids can model development, genetic conditions, infection, toxicity, and circuit formation that are difficult to study directly in a living human brain. The same simplification limits what they can represent. Current models lack many features of intact brains. Vascular support, sensory input, body regulation, immune interaction, developmental environment, long-range organization, and sustained maturation remain incomplete or variable.

The field is moving beyond isolated cultures. Assembloids join region-specific models, transplantation places human neural tissue into animal hosts, and closed-loop interfaces connect biological activity to sensors or computers. These changes may improve research while altering integration, exposure, host welfare, and oversight needs (Pașca et al., 2025).

Complexity should be described along dimensions rather than one dramatic threshold. Cell diversity, maturation, recurrent connectivity, long-range coordination, sensory coupling,

plasticity, persistent state, learning, and closed-loop control can change at different rates.

The Nuffield Council on Bioethics concluded in 2026 that current neural organoids do not possess the biological complexity associated with sentience, while identifying governance gaps and calling for anticipatory guidance as models advance (Nuffield Council on Bioethics, 2026). That conclusion supports restraint in present claims and attention to future change.

Closed-loop interaction increases ethical relevance because outputs begin to shape later inputs. Feedback alone is not consciousness, but the system's sensitivity, persistence, action-contingent learning, aversive exposure, and capacity for amplifying distress-like organization become legitimate review questions. Transplantation changes both the model and the host. Human neural tissue may mature differently in an animal, while the host bears surgery, altered physiology, and possible welfare effects. The combined system is the object of review, not the graft alone.

Monitoring should track growth, viability, activity, response patterns, stimulation history, transplantation effects, host behavior, protocol deviations, what was not measured, and the number and duration of candidate exposures. Reversibility matters: stimulation can sometimes be reduced or cultures stabilized, while transplantation, destructive sampling, widespread replication, and commercialization are harder to undo.

14.3.2 — Stop Rules

A stop rule is written before the most tempting result appears. It names the scientific, welfare, consent, governance, or communication conditions that require pause, narrowing, or

retirement. Work should stop before an unauthorized shift from disease modeling into persistent training, transplantation, high-intensity stimulation, long-term maturation, candidate-sentience testing, commercial use, public demonstration, or data sharing beyond the donor's permission. Scientific interest does not repair missing scope or authority.

Animal-host welfare and capability thresholds require their own triggers. Unexpected neurological, behavioral, physiological, or welfare changes after transplantation call for veterinary review and humane endpoints; reaching a prospectively specified level of maturation, integration, learning, sensory coupling, or action-contingent response calls for independent review, not a sentience declaration. Persistent avoidance, flexible self-protection, state-dependent tradeoffs, or other patterns relevant to suffering should stop the aversive stimulation while alternative explanations are tested (Hyun et al., 2020; Nuffield Council on Bioethics, 2026).

Model inadequacy, measurement reactivity, burden without information gain, and conflicted oversight are separate stop conditions. Do not escalate when the proposed indicator cannot be distinguished from ordinary network activity, artifact, overfitting, or a simpler controller. If the assessment itself creates or changes the feature it is meant only to measure, interpretation pauses until that reactivity is modeled. Additional samples, longer culture, more animals, or greater intensity are unjustified when expected discrimination is too small, and a team whose funding, patent, product, or reputation depends on continuation cannot control the stop decision alone.

Public overclaim can also trigger a pause. If preliminary results are described as thought, pain, awareness, personhood, or proof before review, the claim and dependent demonstration should be corrected before work proceeds. Responses should be

graded: technical pause, protocol revision, donor recontact, specialized ethics or animal-welfare review, external replication, or permanent retirement. One all-or-nothing threshold is too crude for a field whose risks can arise from science, welfare, consent, incentives, or communication.

The rule should identify decision-makers: principal investigators, institutional review boards, stem-cell oversight bodies, animal-care committees, biosafety officers, independent ethics panels, and funders may each own part of the response. Diffuse responsibility often leaves no one to stop. It should also address existing material: cultures may be frozen, destroyed, transferred, or retained under restricted conditions; data and models may need quarantine; and dependent papers, demonstrations, and public artifacts may need correction or withdrawal.

Record the trigger, the evidence considered, the decision and any dissent, and the actions required before restarting. Restart requires a documented resolution of the trigger and independent review. Restore any missing authority and make the burden reductions or control changes required by that review. If the question cannot be answered safely, retire the procedure.

14.3.3 — Scientific and Bioethics Handoff

Neural organoid research crosses fields because no single discipline owns all of its consequences. Developmental neuroscience asks what a model reproduces; bioengineering changes its capabilities; animal-welfare science addresses host effects; research ethics governs tissue and consent; data governance controls records; and consciousness science evaluates proposed indicators. Shared definitions, institutional logging, and international tracking help coordinate review as methods change (Paşca et al., 2025; International Society for

Stem Cell Research, 2025; Nuffield Council on Bioethics, 2026).

A credible handoff states its purpose: the question requiring the organoid, why a less complex model is insufficient, and which results would support continuation, redesign, narrowing, or retirement. The system must be a specified, versioned object. Cell source, differentiation method, culture age, composition, region-like organization, vascularization, interfaces, stimulation, coupling, software, training loop, transplantation state, and deviations can alter evidence and risk.

Donor consent and provenance travel with the material. The record should preserve the scope of permission, re-identification risk, sharing, commercial use, transplantation, recontact, withdrawal limits, and materially new uses. Withdrawal must be described honestly when distributed cultures or published data cannot be recalled (International Society for Stem Cell Research, 2025; Nuffield Council on Bioethics, 2026).

Observation and intervention should remain distinct. Passive recording, patterned stimulation, pharmacological manipulation, closed-loop training, and transplantation create different welfare questions even when the biological material shares a name. Moving from describing activity to shaping adaptive responses triggers renewed review.

Data governance should distinguish authorized research memory from public visibility. Genetic information, activity traces, host records, video, code, and candidate-status discussions may support oversight.

Consciousness-assessment methods can protect candidate systems while details that facilitate creating or manipulating them introduce a distinct disclosure risk (Butlin & Lappas, 2025).

Repositories, vendors, animal facilities, contract laboratories, journals, funders, data hosts, and commercial partners may each control part of the system. Agreements should identify who can pause distribution and which descendants require renewed review. Cross-border work must identify the applicable authorities; one institution's approval does not settle every jurisdiction.

14.3.4 — No Synthetic Suffering for Research

A study must not manufacture candidate suffering as its detection tool. This safeguard applies when sentience is uncertain, not only when it is proven. Researchers can prioritize noninjurious indicators and staged review. An experiment that needs distress to make its hypothesis visible carries a burden its scientific ambition cannot erase.

This includes noxious stimulation, deprivation, panic-like conditioning, isolation, repeated failure, aversive training and damaging temperature or pressure directed at producing a negative phenomenal response. Naturally occurring or clinically unavoidable conditions may be observed without prolonging them or displacing relief.

1. Choose alternatives and limit exposure. Prefer simulations, retrospective records, benign tasks, non-sentient controls or mechanistic models when they answer the question. Abstract states generally suffice for mechanism recovery without autobiographical self-models (Metzinger, 2021; Birch, 2024). Otherwise limit animals, cultures, copies and repeated trials to the discrimination required. Refine housing, handling, enrichment, recovery monitoring, humane endpoints, prompts and objectives for the species or system.
2. Assess the risk profile. Record evidence relevant to sentience, possible severity, intensity, duration, repetition,

predictability, controllability, recovery, scale, human control, irreversibility and safer alternatives. Preserve these factors separately. For hybrids, assess tissue, animal host and control loop together, following the highest plausible welfare risk. Credible indicators can justify inexpensive protection before a final consciousness judgment (Birch, 2017).

3. Prepare rescue and an exit. Unexpected distress-related patterns require appropriate relief or removal. Permanent alteration, uncontrolled copying, distributed deployment and developmental escalation need a workable protective response before they begin. Include the attachment, grief and labor costs borne by workers, users and caregivers when a project ends.
4. Review changes in capability. Persistent aversion-like learning, integrated states or several relevant indicators can trigger stronger protection without constituting a consciousness finding (Butlin et al., 2026; Butlin & Lappas, 2025). The party intensifying exposure must justify it. Reduce avoidable aversive exposure while checking cueing, contamination, preprocessing, software, equipment drift, husbandry and prompt effects.
5. Confirm without reproducing possible harm. Independent reviewers can inspect logs, code, archived samples, historical data and nonaversive tasks first. Justify proposed stimulation or memory limits for the particular system; neither is automatically protective. An inventory of versions, sites and custodians must carry decisions to affiliates, contractors, cloud deployments, biobanks and transferred material.
6. Make protection reviewable. Name the evidence, responsible reviewer, resources, review date and result that would reopen each decision. A signal that fails replication, proves artifactual or loses theoretical relevance may justify reducing

or ending a safeguard. Record the outcome explicitly: artifact identified, indicator unreplicated, precaution retained, study narrowed, welfare review activated, external review required, protocol retired or evidence insufficient.

7. Control incentives. Preregister protocols, report burden and preserve null results so spectacular reactions are not rewarded over benign inquiry. Staged abuse can exploit audiences and generate misleading testimony-like output even in a nonconscious system (Butlin & Lappas, 2025). Livestreams, exhibitions and promotional demonstrations must meet the same standard as laboratory work; apparent torment is not an entertainment device.

14.4 — Precaution and Uncertain Personhood

Uncertain personhood does not imply institutional inaction. A laboratory might pause a procedure while it reviews possible welfare risk. Animal-care committees may set protective defaults, while publishers or regulators face different questions about claims and oversight. The response should fit the decision and its potential harm rather than pretend the uncertainty has one universal answer.

14.4.1 — Unknown Is Not Nothing

A graduated status register may distinguish no present evidentiary concern, monitoring, unresolved candidate concern, enhanced review, pause, and retirement. The exact categories belong to scientific governance, but this book can require that each label have defined evidence, safeguards, and correction rules rather than serving as a prestige scale.

Artificial-system logs create special risk because they can be copied, searched, summarized, and used to train later models. A system may eventually reproduce its own earlier caution as performance, making record reuse look like independent evidence. Model-training and derivative-use limits therefore belong in the original governance decision.

14.4.2 — Public Communication Warnings

Headlines should name the observed result rather than the desired interpretation. An animal displayed flexible self-protection, an artificial system satisfied several theory-linked indicators, or an organoid entered enhanced review are narrower than felt pain, became conscious, or grew a mind. Images carry their own inference. Human faces over code, glowing organoids labeled mini-brains, distressed animal close-ups, and speaking

machines can award a subject before the caption begins. Visual drama should not substitute for evidence.

Review should include science communication, disability, animal welfare, artificial intelligence, bioethics, accessibility, and affected-community expertise. Audience and format may change, but claim status does not; commercial or donor material receives no license to become stronger than the evidence (National Academies, 2017).

Interactive products should show the relevant version, data limits, review status, and correction route beside each emotion label, consciousness indicator, or candidate notice. Readers should not have to search a separate policy page for those limits.

Translation must preserve claim status and uncertainty. Explain term choices when consciousness, awareness, sentience, feeling, and mind divide differently across languages or legal cultures; a familiar word must not silently strengthen the claim.

14.4.3 — Controlled Vocabulary Transfer

Sentience usually refers to the capacity for felt positive or negative experience, while consciousness may be used more broadly for subjective experience or awareness. Define the intended meaning when disciplinary usage changes.

Qualify possible as logical, theoretical, empirically plausible, a realistic welfare concern, or probable; these are different commitments.

Personhood belongs to moral, legal, philosophical, and political argument and cannot be read directly from one scientific indicator.

Human self-report and artificial testimony-like output require different evidentiary routes. When a term crosses disciplines, record its local definition and the inference it does not license; shared vocabulary must not erase that distinction.

Part VI — Windows into a Whole Life

How might a complete conscious stream leave partial traces in dreams, bedside observation, laboratory responses, and records gathered across years?

Dreams extend the inquiry to experience that waking records may miss. A nightmare, a pleasant dream, a forgotten dream, or a lucid exchange may expand what researchers know about the observed history. Each may also introduce recall bias, cultural interpretation, selection, medication effects, and reactivity. The scientific gain lies in making those competing explanations explicit.

Dreams may also matter after waking. A person may describe anxiety after a threatening dream or comfort after a dream of reunion; such reports make the private night relevant to an account of felt experience.

A person wakes with a pounding heart and a durable residue of fear but can recover only one image from the night. The fear belongs to the observed aftermath; the missing narrative remains uncertain. A careful study records the residue, the recall conditions, and the competing explanations without pretending that the surviving image is a complete transcript of the dream.

The second window lies near the end of life. The Law's boundary makes terminal observation formally important, yet the human setting makes it ethically fragile. Palliative care exists to relieve suffering and support the person and family. Research must fit inside that duty, never reverse it (World Health Organization, 2020; National Institute for Health and Care Excellence, 2015).

A last response is not necessarily a last experience. Sedation, delirium, weakness, sensory loss, medication, family scripts, device removal, and observation shutdown can change what is visible.

The third window is the laboratory. A laboratory cannot reproduce the full setting of childhood, grief, poverty, intimacy, disability, aging, or the end of a conscious stream. It can, however, manipulate narrow conditions, compare alternatives, and ask whether local components behave as proposed. The value of a laboratory result depends on the exact question it was designed to answer, not on the prestige of the apparatus.

The fourth window follows people through time. Repeated observations can reveal trajectories, turning points, adaptation, response shift, treatment change, and patterns of missingness that a single interview cannot show.

Two people can share the same monthly mean while one experiences severe early collapse followed by repair and the other undergoes mild but relentless decline. The average hides the path and its timing (Fitzmaurice et al., 2011).

Yet a longer file can become a more invasive file. Duration increases both inferential opportunity and the risk of surveillance.

The windows can also guide one another. A repeated pattern in a long record can suggest a laboratory contrast; a laboratory contrast can identify a feature worth recording in daily life. A dream report can raise a timing question that changes the next sampling schedule. Bedside observations can reveal which distinctions matter when communication becomes sparse. The movement among settings is productive when each contributes a question the next can sharpen. A research program grows through these exchanges, assembling connections among partial views rather than expecting one setting to contain them all.

Chapter 15 — Dreams as a Window

Every night, a portion of human experience may occur beyond ordinary public access. A person can be frightened, delighted, ashamed, reunited, hunted, relieved, or simply bewildered while the room remains still. The formal reason dreams matter is straightforward. The Law concerns felt experience within a qualifying stream, not only the events another person can witness. If dreaming is conscious experience, its valence belongs to the experienced history even when later recall is absent or incomplete.

In a 2021 study, a sleeping participant heard a simple question from the laboratory: what is eight minus six? He replied with two left-right eye movements, the agreed signal for two, while physiological recordings placed him in REM sleep. Asked again, he answered correctly again. Across four laboratories, the study recorded 29 correct replies from six of 36 participants (Konkoly et al., 2021).

That narrow exchange opens a new route for inquiry. Researchers need not always wait until waking to ask a question. A response during sleep creates a second record that can be compared with what the dreamer later remembers. The intriguing opportunity is to study where those records agree, where they diverge, and how the transition to waking reshapes the story.

Imagine two invented archives of dream reports. In the first, frightening experiences become more common during sleep. In the second, the experiences stay the same, but frightening dreams become more likely to be remembered and reported. Arrange the examples so that both archives contain the same rise in reported fear. A reader looking only at that rise could not tell which story produced it. That is a useful rehearsal for research. Can the planned measurements tell the two invented worlds apart when their histories are known? If they can, the method

has cleared one obstacle before facing real uncertainty. If they cannot, a larger collection of reports alone will not answer the question. The exciting target is the experience that happened at night, and the challenge is to distinguish it from the memory that survived until morning.

Competing hypotheses remain central. Dream affect may preserve waking concerns, amplify them, reorganize them, or appear unrelated once recall and selection are modeled. Medication, depression, trauma, sleep fragmentation, and awakening schedule can influence both what is experienced and what becomes available for analysis. A reversal between waking and dream affect would not be unique to the Law: recall, coding, regression, and regulation could also produce it.

Specify the waking exposure and the chance-resemblance comparison before interpreting incorporation. A student may dream fragments of a practiced task and improve the next day, yet salience, prior learning, and sleep-dependent consolidation can produce both observations without the dream narrative causing the improvement (Wamsley, 2014; Wamsley et al., 2024).

15.1 — Why Dreams Matter

Dreams may contain felt experience that waking diaries miss. Repeated awakenings let researchers sample reports close to the preceding interval. Sleep physiology provides an independent record of state and timing. The challenge is to relate that record to what was present before the alarm without treating recall as the whole event.

15.1.1 — Dream Experience and Offline Processing

The inference depends on timing, wording, alertness, and the person's capacity to recover a trace (Siclari et al., 2017; Tononi et al., 2024).

A report of no experience is also not identical to dreamless sleep. The person may have had no conscious content, may have lost the trace during awakening, may remember only a feeling without imagery, or may be unable to translate the experience into words. Some studies distinguish recalled dream content, a report of experience without recall of its content ("white dreams"), and a report of no experience. A missing report is a separate outcome (Windt, 2015; Siclari et al., 2017).

The converse problem matters too. A long, vivid report can create an impression of rich continuous experience even when the remembered sequence was assembled from fragments. Report length, confidence, sensory detail, and emotional residue are different features. A protocol that combines them into one dream score may gain convenience while losing meaning. Better work keeps them separate long enough to learn whether they move together, which ones transport across nights, and which ones are mostly properties of recall (Wamsley et al., 2024; Windt, 2015). Offline-processing hypotheses offer ordinary explanations for dream relevance. Sleep may reactivate recent learning, integrate remote memories, sort salience, support

emotional recalibration, simulate threats, regulate synapses and bodily states, and create novel combinations. These functions may differ across tasks and sleep stages (Hobson and McCarley, 1977; Wamsley, 2014; Wamsley et al., 2024).

15.1.2 — Scientific Research Status

A high-density electroencephalography (EEG) study included seventy-five expert-confirmed non-rapid-eye-movement parasomnia episodes, fifty-seven of which were followed by an interview. Among those interviewed episodes, participants reported some conscious experience after eighty-one percent, could clearly recall content after fifty-six percent, and reported no experience after nineteen percent; the neural patterns associated with experience and with later recall were partly different (Cataldi et al., 2024). The result does not settle what occurred when no interview was obtained. It shows why outward action, conscious experience, memory encoding, and later testimony must remain separate layers.

Shared resources can widen comparison without solving identification. The initial DREAM database release combined twenty datasets, 505 participants, and 2,643 awakenings, linking sleep electrophysiology to a standardized classification of reported experience (Wong et al., 2025). That scale supports reanalysis and cross-protocol questions, but every classifier remains tied to contributing studies' labels, sampling schemes, and reportability.

A newer high-density EEG study of non-rapid-eye-movement parasomnia episodes found activity unlike ordinary wakefulness or sleep. Across seventy analyzable episodes from twenty adults, delta and theta power and the beta-to-delta relation were intermediate between wake and sleep (Valomon et al., 2026). This mixed-state physiology helps explain why sleep-wake

boundaries resist simple labels, but cannot identify conscious content or valence.

For pooled analyses, link each eligible awakening to its report attempt, outcome, and relevant neural interval. Distinguish no-experience reports, fragmentary recall, recall confidence, refusal, exclusion, and the reason for a skipped prompt. Retain sleep stage, clock time, awakening method, medication state, prompt wording, response delay, and coded affect; link any authorized verbatim account under its access restrictions. This event-level record preserves the denominator and timing needed to assess selection (Wong et al., 2025).

15.1.3 — Questions Across the Dream Boundary

A question can sometimes enter a dream while the dream is still unfolding. The possibility changes the shape of an experiment. Instead of waiting for a person to awaken and reconstruct an episode, an investigator can place a new piece of information into the interval being studied and look for a response to that particular information. The exchange gives the episode an external point of contact with a known time and content.

Konkoly and colleagues (2021) brought together experiments from four laboratories using two-way communication during polysomnographically verified REM sleep. Thirty-six people participated. Some were experienced lucid dreamers, some had little prior experience, and one had narcolepsy with frequent lucid dreams. Participants practiced response signals before sleep. During sleep, experimenters presented new questions, including simple arithmetic and yes-or-no queries. Replies used distinctive eye movements or facial muscle contractions. The teams recorded twenty-nine correctly answered questions across six participants. These findings establish a particular capability: in some verified sleep episodes, newly presented information

could be processed and answered through a prearranged outward signal.

The novelty of the question matters. Suppose a person has agreed to signal twice when recognizing a familiar cue. That response can show that the cue was registered and the trained response produced. Now suppose the person receives an arithmetic question whose numbers were selected only after sleep began. Producing the correct answer requires a relation between the new input and the response. The experiment has moved from detecting a prepared signal to detecting a small, structured exchange.

An invented example makes the information path visible. Before sleep, a participant and researcher agree that a series of left-right eye movements will encode a small whole number. During an independently identified sleep interval, the researcher asks for four minus one. A response encoding three is linked to that question by its timing and content. A later response encoding two after five minus three supplies another distinct relation. The sequence of inputs makes the responses more informative than an isolated movement considered without its preceding question.

There are several clocks in this record. One marks delivery of the question. Another marks the outward response. A later interview marks what the participant can describe after waking. Keeping these times separate allows the researcher to ask whether the person later remembers the question, remembers the answer, or recalls a different surrounding scene. The contemporaneous exchange and the retrospective account then contribute different pieces of evidence about the same interval.

For example, imagine that a participant answers a spoken question correctly during sleep and later describes hearing it as a character's voice inside a dream. The response shows a task-

relevant relation to the externally delivered information. The later description suggests how that information appeared within the remembered experience. If the participant recalls the question differently, the discrepancy becomes informative about the transformation between input, experience, action, and later recollection. Each part has its own record.

A particularly useful comparison would vary the question's informational demand while keeping its sensory form similar. A simple detection query, a discrimination between two sounds, and a small calculation ask the participant to preserve different relations. Responses could be evaluated against the question actually delivered, using the same prearranged signal vocabulary. The comparison would characterize the operations available in those sampled episodes rather than combining all successful replies into one undifferentiated ability.

The response vocabulary also sets a resolution. Two signals can distinguish two answers; they cannot express a detailed scene. More possible signals can communicate more distinctions, but each must remain recognizable in the recorded channel. A compact vocabulary may therefore be especially valuable for a sharply defined question. Richness in the dream and richness in the outward communication are separate properties, connected by the code the participant can use at that moment.

This gives the study of dream affect a constructive direction. A future protocol could ask a small, clearly defined question about an ongoing experience and compare the response with a later account, provided the meaning and reliability of the response channel had first been established. The timing would help distinguish a contemporaneous judgment from a judgment formed during recollection. The interesting result could be

agreement, a systematic shift, or different patterns for different kinds of questions.

Online exchange also makes the dream interval less like a sealed box. There can be an input from the surrounding world, an operation carried out during the sampled state, an outward reply, and a later narrative. Following those links is a way to learn how access is organized across a boundary that once appeared much harder to cross. The experiment's appeal lies in that exact connection: a person in a dream can sometimes meet a new question and send a recognizable answer back.

15.2 — Dream Affect, Recall, and Missingness

Recurring dream themes are easy to remember, but their labels conceal variation. Falling may accompany fear in one dream and relief in another. Being chased or arriving unprepared may reflect different histories and meanings. Researchers must code the reported affect separately from a familiar plot before using themes as evidence about dream valence.

15.2.1 — Recurrent Themes

Recurrent themes may reflect common human situations, repeated waking burden, and cultural patterns. Evaluation, pursuit, loss, bodily change, shame, and uncertainty recur in dreams partly because bodies and social worlds repeatedly contain them. A person under chronic evaluation may dream of unpreparedness; a grieving person may dream of searching (Hall and Van de Castle, 1966).

Threat-simulation theory proposes that dreams rehearse responses to danger (Revonsuo, 2000). A dream may simulate pursuit, attack, loss, failure, humiliation, or danger without exposing the body to that event. This is a proposed function of dreaming, distinct from the continuity hypothesis that dreams reflect waking activity (Schredl and Hofmann, 2003). A dream of reunion, safety, forgiveness, success, homecoming, escape, or abundance may feel comforting. Its meaning may involve desire, memory, attachment, fantasy, unmet need, or ordinary affective imagination.

Themes should be coded with transparent dictionaries, inter-rater checks, cultural review, and a prospectively specified loss condition. Base rates, waking exposure, trauma, media, salience, and narrative convention can all create recurrence. A person who repeatedly dreams of arriving late may be expressing chronic time pressure rather than a compensatory process. The

codebook must therefore preserve context and uncertainty instead of turning a familiar image into a universal symbol (Domhoff, 2017). Recurrent themes may still have clinical and human value. They can guide conversation, reveal concerns a person wishes to discuss, or help evaluate a nightmare treatment (Krakow et al., 2001).

15.2.2 — Memory Reconstruction

Recall varies across people and nights. In a prospective study of healthy adults, attitudes toward dreaming, mind wandering, and sleep patterns were associated with whether participants reported having dreamed. Among those reporting a dream, age and vulnerability to interference predicted recall of its content (Elce et al., 2025). These are different outcomes, and the associations do not establish their causes. A remembered-dream dataset therefore reflects both sleep experience and the processes that preserve it for report. That selection belongs in the model rather than being dismissed as noise.

The dream-report chain links experience during sleep to awakening, accessible memory, and the eventual report. The interval between awakening and report changes what can be known. A voice note made in darkness may preserve sensory fragments that disappear by breakfast, while a later interview may recover a clearer story only because the dreamer has imposed order on those fragments. Neither report should automatically be treated as the correct one. They answer different questions about immediate residue, later memory, and narrative integration, and the protocol should keep those targets separate (Wamsley et al., 2024).

Prompting can also change the object. Asking whether a dream was frightening, meaningful, or connected to yesterday directs attention toward particular features and can encourage

completion of gaps. Free recall, structured checklists, confidence ratings, and interviewer follow-up may each be useful, but they are not interchangeable. A careful study records which form produced each item and tests whether its central result survives when the prompt changes (Nielsen and Levin, 2007; Siclari et al., 2017).

Consider four possible responses to the same ancestor dream: sacred contact, ordinary memory, random imagery, or a private experience the dreamer declines to disclose. Coding these as equivalent reports would confuse interpretation and willingness to report with the feeling experienced.

15.2.3 — Observation Is Not Signature

The scientific question is whether a proposed relation between waking and dream affect exceeds what recall, salience, sleep stage, medication, culture, and selection already predict. Vividness, recurrence, and emotional intensity are not unique fingerprints (Domhoff, 2017). A Law-related signature would need a prospectively specified relation to waking burden, horizon, or later adjustment that transports and outperforms plausible rivals. The signature should identify, before outcome inspection, the waking exposure, lag, dream feature, sleep-stage or timing condition, comparison class, and rival set. Stable participant traits and within-person deviations should remain distinct so a cross-person correlation is not mistaken for a dynamic relation within one stream.

Model checks should extend beyond the headline association. A fitted account should recover report and no-report rates, sleep-stage differences, recall confidence, and the distribution of affect.

A report rate combines several stages. Consider one hundred scheduled observations in a synthetic sleep record. Suppose the

modeled experience of interest occurs at sixty of those observations, and half of those occurrences produce a remembered report. Thirty positive reports are obtained. Their frequency is 30 percent of the scheduled observations, while the modeled occurrence frequency is 60 percent.

Now construct a second record in which the experience occurs at thirty observations and every occurrence produces a report. It also yields thirty positive reports. The same observed rate is compatible with two different combinations of occurrence and recall. This ambiguity remains even if every report that is obtained is accurate and the sample is very large.

The calculation is the occurrence proportion multiplied by the probability of reporting given occurrence. Six tenths times one half and three tenths times one both equal three tenths. Learning that product more precisely does not by itself separate the two factors. Additional information has to address at least one stage of the report process.

A change in reporting delay can help distinguish some models of that process. Imagine a version of the first construction in which an immediate prompt captures three quarters of occurrences, while a delayed prompt captures half. The predicted positive-report rates become 45 percent and 30 percent under otherwise identical sampling. A version of the second construction with complete recall at both delays predicts 30 percent in both conditions. The models now disagree about an observable comparison.

This comparison also makes its own assumptions visible. The occurrence process is held fixed across the two prompt conditions in the construction. If the prompt changes sleep or the experience being sampled, that becomes another part of the model. Separating the stages allows a study to investigate the

effect of timing instead of treating every change in reports as a change in dream occurrence.

Two observation routes can supply more information when their overlap is retained. Consider one hundred synthetic occasions. An occurrence of interest may be detected by channel A, channel B, both, or neither. Assume neither channel reports the occurrence when it is absent. Also assume that, among actual occurrences, the channels detect independently with fixed detection probabilities across the occasions in this example.

Suppose the population proportions are thirty A detections, twenty B detections, and ten detections by both per hundred occasions. Let q be the occurrence probability, a the chance that A detects an occurrence, and b the corresponding chance for B. The three observed proportions are qa = 0.30, qb = 0.20, and qab = 0.10.

Multiplying the first two and dividing by the third gives q = 0.60. The implied detection probabilities are a = 0.50 and b = 1/3. Under the declared model, sixty occurrences per hundred occasions generate the observed detection proportions. The overlap supplied the equation that the two marginal detection rates alone lacked.

The union of the detected occurrences has proportion 0.30 + 0.20 − 0.10 = 0.40. The model therefore assigns another 0.20 to occurrences missed by both channels. One third of the modeled occurrences remain unobserved by either route, even though the observed overlap permits their proportion to be inferred under the assumptions.

These numbers describe exact model proportions. Counts from a finite sample fluctuate, so an empirical version estimates the ratios and their uncertainty rather than treating the arithmetic as exact. The overlap is especially consequential because it appears in the denominator. A design can examine whether enough joint

detections are expected to estimate that component with useful precision.

The assumption about independent detection is also a substantive part of the construction. If some occurrences are easier for both channels to detect, a shared detectability variable can create positive dependence. The overlap then reflects both the number of occurrences and that shared variation. Additional observation or a richer model is needed to separate those contributions. The mathematical example identifies precisely what the third equation contributes and what relation it presupposes.

For dream research, possible channels would need their own validated relation to the defined experience and to recall. A second account produced by rereading the first is not an independent detection route in this model. The useful question is whether two routes provide distinct opportunities for an occurrence to become observable, and whether their joint behavior can be characterized. Recording overlap turns the observation process into something that can itself be investigated.

15.2.4 — Episodes and Occupied Time

Dream reports can be organized around episodes or around moments, and the two sampling units answer different questions. A small imagined night makes the distinction visible. Suppose there are six dream episodes: one lasts five minutes and five last one minute each. Together they occupy ten minutes. Treat the episode boundaries and durations as known within this example so that only the sampling arithmetic is in view.

Choose an episode uniformly from the six. The long episode has a probability of one sixth of being selected. Choose a moment uniformly from the ten minutes of dream time instead.

The long episode occupies five of those minutes, so it has a probability of one half of containing the chosen moment. The same night yields different proportions because a longer episode offers more moments at which it can be encountered.

This is length-biased sampling: inclusion through a randomly chosen moment favors intervals in proportion to their duration. The effect appears even with perfect memory and perfectly accurate reporting. It arises before recall enters the problem. If a particular feature occurs in the long episode alone, it belongs to one sixth of episodes but one half of the occupied dream time in this example.

Each proportion can be useful. Episode frequency asks how commonly a feature appears among discrete episodes. Time occupancy asks how much of the relevant duration contains it. A study interested in the formation of episodes may prefer the first. A study interested in accumulated exposure needs the second, together with the intensity and sign of the experience being accumulated. Converting between them requires information about duration and the way the feature is distributed across episodes.

There is also a useful question about the denominator. Choosing a moment from dream time differs from choosing it from all sleep time. The latter includes intervals outside the imagined dream episodes. A protocol can therefore report the probability of encountering a dream at the sampled moment and, separately, the features of the dreams encountered. Keeping both stages visible gives the reader a path from the sampling schedule to the quantity described.

This arithmetic offers a practical way to design a clearer record. Store the selection rule alongside the report: whether an episode was selected, an awakening was scheduled, or a moment was sampled under another rule. Where durations can be

estimated, retain their uncertainty rather than silently giving every episode equal weight. The resulting account can describe both what tends to recur and what tends to occupy time, allowing each observation to answer the question for which it was collected.

15.3 — Dream-Affect Predictions and Their Interpretation

Dream affect may extend the observable history beyond waking reports. A quiet morning does not reveal every experience during sleep. Any candidate relation between dream and waking valence must survive missing recall and sampling effects. The question is whether those observations add an independent test rather than function as imagined compensation.

15.3.1 — Dream Valence May Extend the Observed History

Studies linking dream tone with depressive state, remission, and next-day mood show that the night may contribute information not contained in a daytime interview (Cartwright et al., 1984; Cartwright et al., 2006). Tough-day inversion is the candidate pattern in which a day with negative residual valence is followed by dream valence that is less negative or more positive than expected from ordinary carryover alone. Whether the sleeper later feels restored or finds concerns resolved is a separate outcome.

Specify the waking burden of interest and the adjustment model before estimating the association. Sleep loss, pain, conflict, threat, grief, work strain, social exposure, and acute stress may have different relations to dream content and later recall. The model should distinguish the target burden from baseline mood, trauma activation, ordinary fatigue, and measurement timing, with causal assumptions stated for any adjustment.

Dream content and felt valence can diverge. Positive imagery may coexist with bodily anxiety, reunion with dread, or an opening threat with later relief. Sometimes only one image survives recall. Coding should preserve mixed valence, uncertainty, intensity, and confidence (Cartwright et al., 1984).

Specify the positive, negative, and neutral waking residuals and the directional comparisons before seeing the data. Apply the same search and coding rules after each type of period; selectively seeking the expected opposite dream affect can manufacture apparent compensation (Shiffman et al., 2008).

15.3.2 — Adaptation, Recall, and Limits of Inversion Evidence

A difficult day may be remembered differently after sleep; medications can alter sleep architecture, recall, and affect; and people who awaken or choose to report are not a random sample of all dream intervals (Goldstein and Walker, 2014; Little and Rubin, 2019).

For a local dream–waking comparison, an ordinary waking period may be defined relative to the participant's prospectively specified baseline. It may contain irritation, modest pleasure, fatigue, and routine life. This reference describes an ordinary period; cumulative valence requires a separately justified phenomenal neutral origin. Keeping those roles distinct prevents centering alone from producing an apparent dream inversion.

Medication is not a single adjustment variable. Drug class, dose, timing, indication, recent change, withdrawal, and interaction with sleep continuity can alter dream recall and affect through different pathways. A model that treats medication as one yes-or-no covariate may leave precisely the confounding structure it claims to control (Goldstein and Walker, 2014).

Credible comparison requires prospective records of treatment change and sleep disruption, appropriate stratification, and explicit treatment of missing reports rather than coding them as neutral dreams (Nielsen and Levin, 2007; Cartwright et al., 1984).

After a medication change, reports of vivid nightmares may rise at the same time that awakenings become more frequent. The underlying experience, the probability of reporting it, or both may have changed. A useful analysis separates those pathways and asks whether sleep stage, medication, mood, recall, or selection explains the pattern better.

Selection operates at several stages. People who volunteer for dream studies may remember dreams more often; participants may record striking dreams and omit ordinary ones; researchers may emphasize nights with clear narratives; journals may prefer unusual associations. Each filter can make the observed archive more symmetrical or dramatic than the underlying experience.

Ordinary rivals make different predictions about dream affect. A sleep-based account proposes that REM sleep supports affective homeostasis and changes later emotional reactivity (Goldstein and Walker, 2014). A nightmare account proposes that impaired fear-extinction processes contribute to dysphoric dreams (Nielsen and Levin, 2007). These accounts concern particular mechanisms and contexts; their predictions should be compared with recall and selection models.

15.3.3 — What Dream Evidence Could Teach

A dream-valence route needs a failure condition fixed before seeing the reports. A missing predicted relation may be informative when sampling and measurement were adequate. It is less decisive if recall is too sparse or observation changes the dreams themselves. The study should distinguish those situations before revising its model.

If waking burden does not forecast dream-valence residuals under a capable design, or if positive, negative, and neutral waking periods produce no discriminating pattern, the proposed route should narrow rather than migrate into a vaguer claim.

Distinguish three outcomes: the data rule out the required inversion effect, the estimate remains too imprecise to decide, or sleep fragmentation and emotional salience predict better (Lakens, 2017; Wasserstein and Lazar, 2016). Retiring the inversion route leaves a different question intact: dreaming may contain phenomenal exposure absent from waking reports.

A repair decision should identify whether the failure concerns recall, coding, transport, privacy or incremental prediction; improving recall does not repair a coding or transport failure (Little & Rubin, 2019). Cultural-pattern dominance occurs when dream results are explained more fully by reporting norms, religious meaning, language categories, family practice, social expectation, or popular dream scripts than by a general dream-valence mechanism.

Report language-specific estimates, translation histories, local base rates, coding disagreements, disclosure rates, and uncertainty. Test whether the direction survives local idioms and scale use; otherwise classify the pattern as culturally local (Schredl and Hofmann, 2003).

15.4 — Safe Sampling, Coding, and Privacy

Dream sampling changes the night being observed. Alarms interrupt sleep, while repeated questions shift attention toward recall. Detailed retelling can revive distress or shape what is later remembered. A safe protocol must measure those burdens and stop collection when the procedure overwhelms the value of the data.

15.4.1 — REM Timing

REM timing matters because many vivid reports follow REM awakenings, but dreaming is not owned by REM. A design that treats REM reports as all dreams may miss non-REM experience, confuse sleep-stage effects with dream function, or overstate the specificity of a pattern. The first task is therefore to identify the sleep window the claim actually concerns.

Wake-up bias also differs by person. A light sleeper may report more dreams than a deep sleeper. A parent, caregiver, shift worker, patient, or person in unsafe housing may wake more often. A person with insomnia may generate more reports because they are awake more, not because dream life is more affectively active. Social and material burden can enter the dream dataset through awakening itself (Nir and Tononi, 2010).

Stage and time of night should therefore be treated as two coordinates rather than one label. An early-night NREM awakening, a late-night NREM awakening, an early REM awakening, and a final-morning REM awakening differ in sleep pressure, recent memory, report opportunity, and typical content. Pooling them can manufacture a single “dream effect” from several distinct sampling regimes. Analyze these stage-by-time combinations separately, or justify pooling them (Aserinsky and Kleitman, 1953; Siclari et al., 2017).

The denominator is the number of eligible awakenings or sleep intervals, not merely the number of remembered dreams. Reporting the flow from scheduled awakening to usable report is therefore as important in dream research as participant flow is in a clinical study.

Sampling density should be adaptive to burden. Repeated awakenings may be tolerable for one healthy volunteer and disproportionate for a person with insomnia, trauma, caregiving duties, chronic pain, or unstable housing. The protocol should set nightly and cumulative limits, allow immediate reduction in sampling, and treat sleep deterioration as an adverse effect rather than a useful increase in observations (World Medical Association, 2024; CIOMS, 2016).

The first night in a laboratory may differ from later nights because the room, equipment, anticipation, and repeated awakenings are unfamiliar. Habituation nights can reduce some effects but increase cost and selection. Researchers should state whether the target is ordinary home sleep, adaptation to the laboratory, or the specific monitored condition.

15.4.2 — Cross-Cultural Coding

Cross-cultural coding can mistake the researcher's categories for universal features of dreaming. An apparent group difference may instead arise from language, disclosure, symbolism, or the instrument itself (Domhoff, 2017).

The dreamer’s rating is essential evidence, but uncertainty, shame, interviewer influence, or ill-fitting categories can affect it. Blinded coders may notice unnamed affective cues while also misreading idiom, trauma, metaphor, or sacred meaning. Compare participant ratings with coded features, preserving disagreement and uncertainty (Windt, 2015).

Code fear, relief, tenderness, shame, anger, confusion, awe, or peace as reported affect, separately from judgments about what a symbol means or what the dreamer ought to believe.

Words for fear, awe, bodily heaviness, peace, shame, obligation, or spiritual presence may divide experience differently across languages. Back-translation can reveal some errors, but it cannot prove that two prompts evoke the same judgment. Interviewer expectations, politeness norms, willingness to discuss sexuality or religion, and beliefs about dreams all influence what reaches the record (Hall and Van de Castle, 1966; Domhoff, 2017).

A strong codebook therefore has shared and local layers. Shared categories support limited comparison only when their meaning, reliability, and disclosure burden transport. Local categories preserve distinctions that a universal grid would erase, including culturally specific forms of kinship, sacred presence, social exposure, obligation, protection, or threat. The scientifically honest result may be a partially common map rather than one global vocabulary.

Design should involve fluent translators, cultural reviewers, and participants in the construction and testing of categories. Original-language reports should be preserved under appropriate access controls, and transport should be tested before group results are compared or pooled.

15.4.3 — Stop If Tracking Increases Distress

A participant may discover that repeated dream recording intensifies rumination, fear of sleep, or trauma recall. That change is both an ethical signal and a measurement effect. The correct response is to reduce or stop collection and record that the observation process altered the target, rather than interpreting greater distress as more informative data.

Reports should disclose how often stopping occurred, whether it differed by symptom history or study arm, and how the resulting missingness affected estimation of the declared target instead of treating stopped nights as unusable noise (Nielsen and Levin, 2007; World Medical Association, 2024).

Dream reports can expose sexuality, trauma, spiritual meaning, family conflict, or allegations about people who never consented. Use ratings rather than narratives when they suffice.

General public materials should not prescribe dream logging or compelled disclosure for minors. Youth-facing work needs a defined educational or research purpose, with no group pressure to reveal dream content (CIOMS, 2016; World Medical Association, 2024).

Chapter 16 — At a Stream's End

The end of life attracts grand stories because it gathers love, fear, uncertainty, medicine, memory, and irreversible loss into a small space. Consider a daughter sitting beside a parent who no longer answers questions. She notices a brief relaxation when a favorite recording begins and tells the nurse that the music has always been reassuring. The observation may support a compassionate change in the room.

The same event can support different meanings. A period of quiet may be remembered as peace, exhaustion, medication effect, privacy, resignation, or loss of access. A final phrase may be heard as reconciliation by one person and confusion by another. When accounts diverge, the divergence is evidence about the observation process and the relational setting, not a defect to be edited away.

Dignity-conserving care offers a vocabulary for what must not be lost. A person is not merely a symptom profile or endpoint case. Identity, privacy, relationships, unfinished concerns, choice, comfort, and how others address them remain part of care even when communication narrows (Chochinov, 2002).

Picture three uncertain estimates far apart on a graph. The analysis moves each partway toward the same group average to make the estimates less erratic. They now stand closer together even though no new observation has arrived. The narrowing is partly something the calculation did. This approach, called partial pooling, can be useful when records are sparse. But the resulting graph leaves us with two possibilities to separate: the observations may show paths drawing closer, or the method may have drawn the estimates closer. Showing how the picture changes with and without pooling makes that distinction visible. A reader can then ask how much of the convergence came from the observations and how much came from the analysis.

Seow et al. (2011) documented different mean symptom trajectories during the last six months of life in a large outpatient cancer population. A peaceful final hour could reflect symptom relief or reduced expression; a difficult hour could reflect uncontrolled symptoms or delirium. Clinical measures help distinguish these accounts of the observed interval.

Analgesia, anxiolysis, sedation, repositioning, quiet, touch, and the presence or absence of family can change observable behavior and reported experience because they are real parts of care.

Professionals at the bedside contribute disciplined observation, treatment history, and contextual judgment. They can describe pain, breathlessness, agitation, comfort, delirium, sedation, communication, and response to care. Their witness is indispensable to humane research, but it remains an observation channel shaped by clinical purpose and limited access (World Health Organization, 2020; Kesecioglu et al., 2024).

Attention to sequence can give a modest observation practical richness. The record can preserve what was happening in the room, what the person had previously found comforting, what changed, and what response followed. That sequence helps the next caregiver continue something that seemed welcome and helps a researcher formulate a more precise comparison. The same careful description can serve different purposes because it retains the encounter's particulars. It keeps the person recognizable within the record while making the observed relation available for thoughtful examination.

16.1 — End-of-Stream Evidence and Care

A family's sense of peace or unfinished business can matter profoundly. It may guide care and conversation, but it does not define a formal endpoint. The scientific boundary needs an independently justified rule for the stream being studied. Treating personal meaning as evidence of closure would confuse two different questions.

16.1.1 — Endpoint Tests and Terminal Sampling

Clinical guidance recognizes uncertainty about dying (National Institute for Health and Care Excellence, 2015). This book separately requires a research endpoint defined before outcome interpretation, not reconstructed from an emotional ending.

Observation can also stop after the stream has already become difficult to identify. A person may move through fluctuating consciousness, delirium, deep sedation, coma, or uncertain recovery. These states do not become equivalent merely because the study has reached its final page. The endpoint rule needs clinically independent channels, prospective adjudication, and an honest branch for cases in which the relevant boundary cannot be established.

A terminal window is informative when the proposed route and a specified rival predict different outcomes under the same observation conditions.

Fix referral rules, enrollment windows, site coverage, and reasons for exclusion before researchers know which lives appear balanced or unbalanced. A held-out site or later cohort can then test whether the apparent terminal pattern travels beyond the cases that first made it interesting.

16.1.2 — Family Narratives

Clinical records were created to support care, not to measure a terminal cumulative object. A symptom score may reflect treatment goals or medication effects. Nursing notes and family-conference records reflect their own purposes and observers. Reuse requires preserving the original timing and context before any scientific interpretation (Steinhauser et al., 2000; Seow et al., 2011).

Family narratives preserve relationship, meaning, and details that formal instruments miss. They can reveal changes in communication, fear, comfort, recognition, humor, ritual, or desire. They are valuable human testimony and may improve contextual interpretation, yet they are also shaped by grief, memory, family roles, and the need to make sense of an ending.

Family accounts should be understood in their cultural and religious context (National Institute for Health and Care Excellence, 2015). For example, families may emphasize explicit farewell, prayer, bodily presence, privacy, endurance, or relief from suffering.

A defensible study therefore treats family narratives as situated testimony. It records who observed what, when, under which treatment and communication conditions, how soon the account was collected, whether other witnesses agreed, and what the family wishes to keep private. The original words should be preserved where consent permits, with later coding clearly separated from the account itself.

16.1.3 — Dignity Over Data

Dignity over data means some moments should remain unmeasured even when measurement is technically easy. A final conversation, a prayer, a bodily-care moment, a fearful confession, or a family apology may be meaningful because it is

protected from public use. The Law of Fairness does not require a complete archive of dying. Sometimes the fair archive is the one that never gets made.

Dignity governs language. Terms such as completed, balanced, failed, or unresolved can sound like judgments about the person when applied to a death. Research reports should use status only for the measurement or model and avoid attaching it to the life. A family-facing summary can name what was observed, what was not observed, and why data collection stopped without converting the person into a research outcome.

16.2 — Sedation, Delirium, and Family Scripts

When treatment or cognition changes communication, a researcher cannot infer experience from outward behavior alone. Medication timing, delirium assessment, respiratory distress, and hearing or motor capacity shape what an observation means. Delirium may fluctuate with illness, medication, or organ failure, so one calm or unresponsive moment is not a stable measure of awareness. Research coding should follow the care team's documented clinical assessment rather than substitute for it (Inouye et al., 2014; Lawlor et al., 2000).

16.2.1 — Sedation Effects

Palliative sedation is a clinical response to otherwise refractory suffering, not a standardized experimental exposure. Contemporary guidance emphasizes proportionality, careful indication, patient autonomy or an appropriate decision process, continued care, and repeated review (Surges et al., 2024).

The word sedation covers different depths, durations, drugs, routes, and clinical circumstances. Intermittent light sedation for a procedure is not continuous deep sedation near death. A single label can therefore hide the very differences that determine awareness, expression, memory, and observability (Maltoni et al., 2013; Surges et al., 2024).

For research, the medication record must be a timeline rather than a checkbox. Link each observation to the drug, dose and route, including titration and co-medication. Preserve the indication, symptom target, organ function, intended depth and observed response as well. These details shape what a person can report and what others can see. A final score detached from that history is not a stable endpoint measurement.

Observational studies can still clarify how sedation is used, which symptoms precede it, and how practice varies. Recent

work on prevalence and determinants is valuable for clinical description, but variation in indication, population, and practice means that these datasets do not supply a uniform consciousness manipulation (Carrasco-Zafra et al., 2024).

A strong protocol predefines the status of observations collected before, during, and after medication changes. It specifies when treatment makes the relevant state unidentifiable and when sensitivity analyses are possible.

16.2.2 — Retrospective Meaning-Making and Clinical Review

If qualitative research studies retrospective narratives, it must ask narrative questions. How do families understand dying? How does public language change grief? Which research practices feel intrusive?

Family performance pressure can be subtle. People may feel expected to witness peace, create meaning, forgive, be forgiven, pray, reassure, or narrate the death in a certain way. The dying person may also try to comfort the living. These pressures do not make the scene false. They make the scene social (Chochinov, 2002).

Review should be interdisciplinary because different observers see different parts of the terminal window. Palliative medicine, nursing, neurology, pharmacy, ethics, social work, chosen spiritual care, and family knowledge may clarify treatment, communication, distress, culture, and prior preference. Their evidence should remain distinguishable.

Reviewers should be able to see which cases were omitted, why classifications changed, where clinical and analytic judgments diverged, and whether the preferred interpretation depended on one contested decision.

16.3 — Observation and No-Estimate Outcomes

Comfort takes precedence over optional observation. Collection should stop if it interrupts sleep or medication. Touch, conversation, and requested quiet may also matter more than another measure. No estimate is preferable to one obtained by overriding patient wishes or exposing a family's private moments.

16.3.1 — Comfort Priority and Missingness

During active resuscitation, research must neither alter the effort nor delay treatment. A canceled interview because speaking has become tiring is recorded as burden-related observation loss, not as a neutral state or failed cooperation.

The participant, clinical team and independent safety role need direct stopping authority. Hear family concern even when it is not legally decisive. Escalating burden, equipment interference, protocol deviation, privacy breach or lost permission can also end collection; name the trigger and person who invoked it (National Institute for Health and Care Excellence, 2015; Beauchamp & Childress, 2019; World Medical Association, 2024).

Assign device removal, data disposition, family notification and any permissible restart before collection begins. Code private reasons minimally under the consent lifecycle. Training should include the moment when investigators want one more measurement and the care team says no.

Analysis should distinguish clinical improvement, withdrawal, incapacity, device removal, transfer, death and ordinary missed observation, then test how conclusions change under plausible ranges for the unseen interval (Diggle and Kenward, 1994; Little and Rubin, 2019).

Reports should disclose how often each trigger occurred, who invoked it, and how the denominator changed. Sites must not be rewarded for terminal completeness in ways that pressure staff, patients, or families.

16.3.2 — No Caregiver Pressure for Research

Use an independent recruiter when treatment or daily care creates dependency; if credible separation is impossible, forgo recruitment. Reassess fluctuating capacity and signs of fatigue or resistance. Direct consent remains primary where possible, and a relative cannot override prior refusal or expand an advance preference. Distinguish care decisions, research authorization and public-use permission (CIOMS, 2016).

Routine-care records or brief observation may avoid a new task in a fragile interval. Interviews, recordings and quotation decisions can often wait beyond the acute moment, although later grief may create different pressures. Separate caregivers' participation from employment evaluation, clinical duty and family obligation, including its workload, compensation and privacy.

If relatives disagree, record the competing accounts of the person's wishes instead of treating the most enthusiastic response as consensus. Pause, narrow or decline the proposed use as needed. Current communication, prior preferences, clinical judgment and burden belong together in that decision; a signature alone does not resolve them.

No caregiver should be asked to watch for a terminal signature. Families are not unpaid observers responsible for timing symptoms, interpreting calm, recording final speech, or preserving evidence. Such instructions can alter the room and leave a lasting fear that something important was missed.

16.4 — Terminal Patterns, Rivals, and Limits

A late trajectory can appear compressed for several ordinary reasons. Sedation or treatment may change expression without changing the underlying target in a unique way. Selection and observation loss may also narrow the visible sample. The analysis must compare these rivals before treating a smooth terminal pattern as support for closure.

16.4.1 — Measured Compression Is Nonunique

First test whether the narrowing remains after treatment, sampling, and estimation effects are accounted for under the declared observation model.

A capable rival comparison should receive the same treatment history, timing, missingness, clinical context, and measurement flexibility as the proposed route. Apparent compression that vanishes when these processes are modeled, or that a hybrid rival predicts equally well, does not require a terminal mechanism (Little and Rubin, 2019).

Abrupt observation endings offer a natural stress test of the observation model. A model fitted to gradual decline may perform well when frailty, treatment, and observation schedules make the final window predictable, then fail when recording stops unexpectedly. Such a failure can expose dependence on selection or a slowly closing clinical window. A naturally occurring, independently adjudicated unexpected stream endpoint provides a different test of a terminal route, through its own representation and observation bridges. Record which event occurred and which inference it supports.

16.4.2 — Reading a Trajectory in Two Directions

A completed record lets an analyst ask a question that was unavailable while the record was unfolding. What does

everything now observed suggest about an earlier state? The answer may differ from the estimate that could have been made at that earlier moment. This difference is a useful property of reconstruction, and its arithmetic can be made visible in a small model.

Imagine a hidden quantity x_1 at the first time. Before any observation it has a Gaussian distribution with mean zero and variance one. A reading y_1 equals x_1 plus independent Gaussian measurement noise with variance one. Between the first and second times, the hidden quantity changes by an independent Gaussian increment with variance one, giving $x_2 = x_1 + w$. A second reading y_2 observes x_2 with another independent measurement error of variance one.

Suppose the first reading is zero. Combining that reading with the starting distribution leaves the estimate of x_1 centered at zero, now with variance one half. This is the filtering estimate: it uses the observations available up to that time. Before the second reading arrives, the forecast for x_2 is also centered at zero, with variance one half plus one, or 1.5.

Now let the second reading be three. That reading can reflect three components: the earlier state, the intervening change, and the new measurement error. Given the first reading, its total variance is $0.5 + 1 + 1 = 2.5$. Its covariance with x_1 is 0.5, because only the uncertain earlier state is shared. The weight by which the new reading revises the earlier estimate is therefore $0.5/2.5$, or one fifth.

The reconstructed mean of x_1 becomes $0 + (1/5) \times 3 = 0.6$. Its conditional variance becomes $0.5 - 0.5^2/2.5 = 0.4$. The later reading has shifted and sharpened the estimate of the earlier state. This backward use of later observations is called smoothing. It is an inference about the past made from a longer record.

The second state receives a different estimate. Its forecast variance before y_2 was 1.5, and the new measurement error has variance one. The update weight is 1.5/2.5, or three fifths. Its mean becomes 1.8, with variance 0.6. The two reconstructed means are consequently 0.6 and 1.8. The model divides the evidence between a higher earlier state and a positive intervening change, while retaining uncertainty about both.

How strongly the future informs the past depends on the dynamics. Keep the first posterior variance at 0.5, but now let the intervening change have variance q. The backward weight is 0.5/(1.5 + q). When q is zero, the two times share exactly the same hidden quantity, and the later reading is another measurement of it. When q is nine, the weight falls to 1/21. A reading of three then moves the earlier mean only to one seventh, about 0.143.

This dependence has an intuitive form. If little can change between observations, a later reading says much about the earlier state. If substantial change can occur, the later reading may mainly describe what happened in between. Time separation alone does not supply the answer. The model of change and the reliability of the observations determine how much information can travel backward.

There are therefore two useful displays of the same record. One shows what an analyst could estimate as each observation arrived. The other shows the best reconstruction under the declared model after the full record became available. The first is appropriate for evaluating a decision made at that time. The second is appropriate for reconstructing a history. Their differences identify the contribution of later evidence.

Near a study boundary, the distinction becomes particularly important. Earlier points may benefit from many later observations, while the last point has no such future

measurements. Uneven uncertainty across the display can reflect this structure of the record. An apparent change in the smoothness of estimated states can then be examined in relation to how much past and future information supports each point.

For the whole-stream inquiry, reconstruction is a legitimate and potentially powerful task. It becomes clearer when the observation model says which readings revise which intervals and by how much. A later measurement can teach us about an earlier state without having been available to the system at that earlier time. Keeping these two temporal perspectives explicit allows a completed history to become more informative while preserving an accurate account of the information on which earlier action could depend.

16.4.3 — Confounds That Can Defeat Interpretation

Use the changing medication timeline specified in Section 16.2.1. The aim is not to subtract care from experience, but to prevent a treatment-shaped trace from being credited to a unique terminal mechanism (Surges et al., 2024).

Sensitivity analysis should follow plausible clinical stories. One analysis may ask what changes if unobserved intervals were more distressed than observed intervals; another if relief increased while communication declined; another if the endpoint fell outside the protocol window. These analyses cannot reconstruct private experience, but they can reveal whether the conclusion depends on one convenient account of what happened after observation stopped.

The strongest clinical rival matrix includes interactions: endpoint selection with medication, medication with delirium, delirium with missingness, and missingness with family interpretation. Those interactions may explain a pattern that none of the processes explains alone.

16.4.4 — Private Dying Is Not Public Proof

Public imagery carries special risk. Serene graphs, scales settling at zero, paired photographs of suffering and peace, or stories of final reconciliation can imply that a person's pain was repaid or that death completed an account. A visual that cannot preserve the limits of the evidence should not be used.

De-identification is often weak in small cohorts. Diagnosis, age, location, timing, family structure, and one memorable detail may identify a person within the community. Publication should prefer aggregates and suppress unnecessary detail under the public-reuse rules in Section 12.3.1 (World Medical Association, 2024).

Chapter 17 — Questions for a Laboratory

A laboratory can isolate a variable, randomize an intervention, repeat a task, and compare models under common rules. It cannot become a life.

Imagine two versions of a small puzzle. Both give you one minute and show the same countdown. In one, you may revise an earlier answer before the game ends. In the other, each submitted answer is final. Nothing outside the game depends on your score. The clock is identical; what changes is whether there is a way back. That difference makes a sharper question than whether a countdown creates pressure. How much does the pressure depend on the seconds remaining, and how much on the possibility of revising a mistake? A proposed repair mechanism becomes interesting if it predicts a difference that time pressure alone misses. The task would still have to show that people understood the available choices. A small experiment could then turn an elusive idea about opportunity into a specific relationship that other explanations must also account for.

Suppose a model is said to detect the effect of a cue shown halfway through a task. The cue was chosen at random only after the first part had been recorded. Now ask the model to find the same effect in that earlier record. A recurring apparent effect before its proposed cause would send us back to the measurement and analysis. The model might be reading an earlier difference between groups, a recording artifact, or information that slipped into the analysis. This is the logic of a negative control: look where the proposed cause should have no effect. Here, discovering nothing can be part of discovering something. A proposed mechanism becomes more informative when it predicts both where a pattern should appear and where it should be absent.

Social tasks need a different separation: contagion, homophily, shared exposure, platform intervention, and measurement spillover. Suppose friends report similar moods after a storm. The pattern could reflect communication, common conditions, selection into the same environment, or selective reporting. The study should predeclare which relation would distinguish these accounts and what result would count against social influence.

Artificiality is not a defect when the question is local. A ten-minute task may test how people respond to closing opportunities or whether a candidate state predicts a later choice. The inference should remain tied to that task, population, manipulation, and timescale rather than expanding because the apparatus looks precise.

Neural measurements can sharpen a laboratory question, but their interpretation depends on tasks and competing processes. Section 17.1.5 specifies the reverse-inference controls that any candidate signature must survive.

A fruitful experiment often has a small hinge: a reversible choice, a cue arriving at a known time, or one feature that changes while the rest of the task remains recognizable. The hinge makes a prediction precise enough to surprise its author. Before collecting the result, one can describe two possible outcomes and explain what each would teach. That exercise gives the experiment direction. It also makes an unexpected result useful, because the study has already identified the relation whose behavior it was designed to reveal.

17.1 — Laboratory Questions and Process Signatures

A safe horizon task might vary the number of remaining rounds, the time allowed before response, the possibility of later revision, or the visibility of an upcoming closure point. The design tests perceived remaining opportunity under reversible, low-burden conditions rather than simulating a life ending.

17.1.1 — Task Design and Observation

A useful laboratory model would look for changes in burden-relevant estimators. For example, a prospectively specified study might test whether a short-horizon condition increases hesitation, autonomic arousal, error monitoring, conflict signals, or subjective pressure relative to a long-horizon condition, while also examining whether repair availability moderates that effect.

The most important design distinction is between urgency and threat. Urgency can be ethically studied in modest tasks. Threat can become harmful quickly. A short horizon should not make participants believe their dignity, care, income, relationship, safety, or standing is at risk. The task may create pressure, but it must not create existential danger, social degradation, or durable fear.

17.1.2 — Understanding and Transfer

Individual understanding should be checked. Participants may misread how many opportunities remain, distrust the displayed countdown, or use a different strategy once the task feels evaluative. Comprehension and strategy reports can reveal whether the manipulation altered the intended representation rather than merely changing stress.

A laboratory result should carry an explicit scope statement: what the task manipulated, which stakes were absent, how long the effect was observed, and which populations and settings

were not represented. Generalization then becomes a new empirical question, tested through independent tasks, sites, and observation channels rather than granted by the precision of the original graph (Shiffman et al., 2008; von Elm et al., 2007).

17.1.3 — Horizon and Repair Interactions

Physiological arousal is not a unique signal of shrinking opportunity. A person who feels judged may show a strong response even when future options are unchanged. Measure perceived evaluation and humiliation as possible alternative causes (Dickerson and Kemeny, 2004). Only a contrast that survives those controls can help identify the proposed horizon mechanism.

Control under constraint is the proposed relationship among narrowing horizon, felt burden, repair options, and regulatory effort.

A possible prediction is an interaction: at a fixed affective load, shortening the horizon produces a larger increase in the prespecified control-related measure when repair is unavailable than when it is available. A factorial design can estimate that difference in horizon effects, then test separately whether affective load changes its size. Comparing a short-horizon, high-load, low-repair condition with a long-horizon, low-load, high-repair condition would not by itself establish an interaction. The prediction must still compete with ordinary models of time pressure, working-memory demand, loss aversion, scarcity cognition, and arousal.

Four imagined condition means make the interaction visible. Let the outcome be response time in an ordinary planning task, measured in the same units throughout. With a long horizon, the mean is four when later revision is available and five when it is unavailable. With a short horizon, the corresponding means are

six and ten. The values are chosen to illustrate the comparison, rather than to describe study results.

When revision is available, shortening the horizon increases the mean by two, from four to six. When revision is unavailable, the increase is five, from five to ten. The difference between those increases is three. That value is the interaction contrast: the effect of shortening the horizon depends on whether the action can later be revised.

The same relation can be read from the other direction. Under the long horizon, removing revision increases the mean by one. Under the short horizon, it increases the mean by four. The difference is again three. These are two readings of one contrast, not two independent findings.

An additive alternative gives a useful comparison. Retain the long-horizon means of four and five, and suppose shortening the horizon adds two units in both revision conditions. The short-horizon means would then be six and seven. There are effects of both horizon and revision availability, yet the interaction is zero because each effect stays the same across the other factor.

Comparing only the most favorable and least favorable conditions would conceal this distinction. The first construction compares four with ten; the additive construction compares four with seven. Each displays a substantial difference, but neither two-cell comparison reveals which part comes from the horizon, which from revision, and which from their interaction. The intermediate conditions supply the missing structure.

The outcome scale also belongs to the claim. An additive relation in response-time units need not remain additive after taking logarithms or converting times to speeds. A study can choose a scale for substantive reasons and define the contrast on that scale before interpreting it. The arithmetic then corresponds to a particular question about the task.

Finally, condition means have uncertainty. Repeated observations from one participant differ from observations from many independent participants, and the design determines how their variation enters the contrast. A useful analysis estimates the interaction together with that uncertainty and examines whether its pattern recurs in a new task. The four-condition design earns its place by identifying a relationship that the dramatic two-condition comparison leaves unresolved.

Separate timescales, predeclare the observation window, and test parameter stability across nested horizons. Known generating processes should be recoverable in simulation before a laboratory pattern is given a broader interpretation.

17.1.4 — Letting the Order Teach Us

A sequence of tasks has a history of its own. A quiet listening period after an easy puzzle can differ from the same listening period at the beginning of a session. The puzzle may leave attention engaged, a rhythm may persist, or the participant may have learned how the response device works. An experiment can make that history informative by varying the order deliberately. The question becomes how much of a response belongs to the condition now present and how much arrives with the participant from the condition just completed.

Consider three ordinary conditions, labeled A, B and C. Each participant encounters all three once. There are six possible orders: ABC, ACB, BAC, BCA, CAB and CBA. Assigning equal numbers of participants to those six orders creates a useful symmetry. Every condition appears equally often first, second and third. Every directed pair of different conditions also appears equally often as consecutive neighbors. A followed by B occurs in ABC and CAB; B followed by A occurs in BAC and CBA. The other directed pairs have the same representation.

The first symmetry separates condition from position in the session. If responses rise simply because participants become more practiced, that change is distributed across all three conditions. Suppose an invented response has a direct condition contribution of two under A, five under B and eight under C. Let being in the second period add one and being in the third add two, regardless of condition. With the six orders equally represented, each condition receives an average position contribution of one. Its average response is therefore three, six or nine. Differences among those averages preserve the direct differences in this additive model.

Now give the previous condition a continuing contribution. Let A leave one additional unit in the immediately following period, B leave two and C leave three. No predecessor contribution appears in the first period, and each contribution lasts only one period. These are stipulated rules for the illustration. They let us examine exactly what the order design accomplishes.

Across the six orders, A appears first twice, follows B twice and follows C twice. Its mean predecessor contribution is therefore ten sixths. B appears first twice, follows A twice and follows C twice, giving eight sixths. C receives six sixths. The predecessor averages are different even though every directed transition appears equally often. A condition never follows itself in this design, so each condition has a different omitted predecessor.

Including the shared average position contribution, the three mean responses become fourteen thirds for A, twenty-two thirds for B and ten for C. The difference between B and A is eight thirds, and so is the difference between C and B. Both are smaller than the direct difference of three. Balanced order has

made the components estimable in an organized way; it has not made every component disappear from a raw condition average.

The first-period observations supply an especially clear reference. Each condition appears there under the same position rule and with no predecessor contribution. With randomized assignment and enough repetitions, those observations can estimate the direct condition differences. In the exact constructed example, the first-period values are two, five and eight. Later periods add comparisons involving known predecessor labels, allowing differences between predecessor contributions to be estimated under the additive model. With noisy measurements, uncertainty accompanies each estimate.

A common carryover level has a separate identification issue. Every second or third period has a predecessor, while the first has none. Adding the same constant to all three carryover contributions and subtracting it from the second- and third-period effects leaves every fitted response unchanged. The six-order design identifies relative carryover, but needs another comparison to separate its common level from position. One possible extension includes later-period conditions preceded by an independently justified reset that removes carryover. Those observations can identify the period contribution against which the remaining carryover is compared.

This organization also makes a richer hypothesis possible. Perhaps B changes the subsequent response to A more than it changes the subsequent response to C. That is a predecessor-by-current-condition interaction, rather than a single effect left behind by B. The six orders provide observations of both transitions. Further repetitions can ask whether their difference recurs. If a contribution persists for two periods, the complete order becomes relevant again: a third-period response may

depend on both earlier conditions. The model's memory length then has an observable implication.

Spacing offers another source of information. Repeat a transition with a short interval and a longer interval between its conditions. A rapidly fading carryover predicts a smaller predecessor contribution after the longer gap. A durable learned strategy may predict persistence across that gap and a change only when instructions or context change. The experiment can choose intervals that make those predictions diverge. Time between observations now helps identify what is being carried forward.

For horizon and repair tasks, the lesson is constructive. A participant who has just experienced revisable choices may approach a subsequent fixed-choice task differently from someone meeting it first. An investigator can preserve that transition in the record and ask how it affects the local response. The order is part of the experience being studied. By giving it a deliberate structure, the design turns a source of ambiguity into another question about learning, expectation and the persistence of a previous state.

17.1.5 — Nonunique Control-Hub and Recovery Signatures

Temporal order can make a local mechanism question sharper. Researchers can ask whether bodily arousal precedes reported doubt, gaze hesitation precedes conscious explanation, control engagement rises when later repair becomes unavailable, or subjective pressure changes after an irreversible choice.

Rival explanations must enter from the beginning: loss aversion, regret anticipation, cognitive dissonance, social desirability, demand characteristics, moral norms, and ordinary emotion regulation may all produce valence-conflict effects. The

task is informative only when those accounts can win (Palminteri et al., 2017; Wilson and Collins, 2019).

Neural and physiological signals are multiply caused. Activity associated with control, salience, interoception, recovery, or conflict may also reflect arousal, effort, attention, surprise, pain, motor preparation, or task difficulty. Reverse inference remains a central risk (Poldrack, 2006; Marek et al., 2022).

Individual differences can modify both experience and signal. Chronic pain, anxiety, medication, disability, trauma history, sleep loss, and familiarity with research settings may change how the same task recruits control and interoception. Heterogeneity should be modeled as possible effect modification, not discarded as noise (Marek et al., 2022; Yarkoni, 2022).

17.1.6 — A Signature Must Transport and Beat Rivals

A graph of rejoining paths can test the state-change candidate. With states fixed independently, suppose A→B contributes +2 and B→C contributes −3. The candidate predicts −1 for A→C. An unused direct path totaling −0.8 differs by 0.2, to be compared with justified uncertainty. A tree alone cannot test this consistency: arbitrary edge totals can be fitted by assigning state values. Rejoining paths supply additional restrictions (Jiang et al., 2011).

In a connected graph with n states and m specified transitions, the incidence matrix B has rank n − 1. Independently calibrated edge totals therefore leave m − n + 1 potential-consistency restrictions. If a common offset b must also be estimated in $y = Bq + bh + \varepsilon$, the known exposure vector h either adds one rank, leaving m − n restrictions, or lies in the potential-difference space and is confounded with q. Use enough rejoining paths to

estimate the declared parameters and still test unused restrictions.

State resolution matters. Suppose two distinct states with Q-values 0 and 1 are both labeled A. Passing from the first through B, whose Q-value is −2, to the second contributes +2 − 3 = −1. The observed A→B→A loop has not returned to the same sufficient state. Refining that state description earns explanatory value when it predicts new observations; giving every visit a new label would instead remove the restrictions.

Mean edge totals test a mean relation. A pathwise claim additionally concerns variation across realized paths. Fix measurement, state resolution, and comparison rules before evaluating unused data, including dependence and selection introduced by matching finishing states. Algebraically reversing an edge in a contrast does not establish that the physical reverse transition is available.

External validation should distinguish frozen transport, recalibration, and refitting. New intercepts, scales, or features change what the result establishes and must not be described as unchanged.

The state-change route in Section 8.3.2 predicts equal signed totals when two paths have equal Q-differences between their boundaries, under one fixed representation. Matching their respective initial and final Q-values is sufficient; matching every microscopic feature is unnecessary. Similar final mood alone does not establish either condition. A persistent-state rival can instead accumulate different areas despite returning to a familiar mood.

Approximate matches remain informative. Bound the initial Q-mismatch, final Q-mismatch, and each path’s accumulated departure from the contribution relation. The difference between path totals cannot exceed their sum, with a separate allowance

for observation error. An independently constant Q paired with a reliably nonzero accumulated contribution offers another direct challenge to the pure change relation.

Equal exposure can simplify calibration: one constant rate offset shared by both paths cancels from their difference. Equal clock time works only when it represents equal exposure under the adopted measure. Drift, unequal gains, and unequal exposures require further terms. Estimating the offset by assuming the candidate is a conditional model fit, not independent identification of phenomenal neutrality. The graph test above counts the restrictions remaining after such parameters are fitted.

Matching paths must also preserve the meaning of the design. Selecting participants because they reached a particular final state can introduce post-assignment selection even when the initial assignment was randomized. A conditional path comparison remains mathematically meaningful, but a causal claim must account for that selection and retain the full assigned sample’s record. The causal distinctions in Section 17.2.1 therefore accompany, rather than follow, the matched-path test. A revised state description receives a fresh evaluation on unused paths.

Path, loop and weighted-trajectory comparisons may reuse the same observations. Evaluate their joint uncertainty after accounting for fitted parameters; counting each summary separately can exaggerate the evidence. Duplicated contrasts add no independent information. If an observation model assigns zero variance to a contrast, check that deterministic constraint before reducing to the remaining variable directions. Noisy fitted features and estimated covariance require their own calibrated analysis.

17.1.7 — Letting One Answer Choose the Next Question

An experiment can specify its next question conditionally. The choice made after an observation can itself be part of a plan written before the observations begin. A small constructed comparison shows when this helps: the most useful second test can depend on which possibilities the first test leaves open.

Suppose four candidate models, H_1 through H_4, make deterministic predictions for three available binary tests, A, B, and C. Write their predictions in that order. H_1 predicts 000, H_2 predicts 010, H_3 predicts 100, and H_4 predicts 101. The models and tests are invented so that every possible inference can be inspected directly.

Test A separates the first two models from the last two. If A returns zero, H_1 and H_2 remain. Test B then distinguishes them: zero identifies H_1 and one identifies H_2. If A returns one, H_3 and H_4 remain. Test C distinguishes those models: zero identifies H_3 and one identifies H_4. The adaptive plan uses two tests for every model.

No fixed choice of two tests from this set has the same property. Choosing A and B leaves H_3 and H_4 tied at 10. Choosing A and C leaves H_1 and H_2 tied at 00. Choosing B and C leaves H_1 and H_3 tied at 00. All three tests would be needed to identify every model using a fixed subset. The conditional plan saves one test by using the first answer to select the relevant second comparison.

The gain comes from the arrangement of the predictions. B is useful within the first pair but adds nothing within the second. C has the reverse role. A reveals which pair needs to be separated. The first test is therefore valuable partly because it makes a later test more precisely targeted. Information changes the next design choice, even though the candidate models themselves remain fixed.

A complete plan can state this entire tree in advance. Perform A. Following zero, perform B; following one, perform C. Apply the corresponding identification rule. The plan can also state what happens if a test is unavailable or produces an outcome outside the defined format. Conditionality is then part of the experimental specification rather than an undocumented response to an interesting result.

Unequal test costs create another design question. If C takes much longer than B, the expected cost of the tree depends on how often A leads to each branch. A design chosen to minimize average cost may differ from one chosen to minimize the largest possible cost. The intended criterion determines which tree is preferable. One can compare candidate trees before collecting data using their specified prediction patterns and cost model.

Noise changes identification into an uncertainty calculation. If each test has an independent probability e of reversing its ideal binary answer, the probability that both answers in the planned two-test route are accurate is $(1 - e)^2$. The original deterministic identification guarantee no longer applies unchanged. The likelihood of each observed route can instead be calculated under every candidate, including the chance of taking the other branch after a noisy first result.

For example, an observed route A = 0 followed by B = 1 is most directly associated with H_2 in the noiseless construction. Under noise, H_1 can produce it through an error on B, while H_3 or H_4 can produce it through an error on A and an appropriate outcome on B. Retaining those possibilities gives a likelihood comparison rather than a categorical label. Repeating or adding a test can then be evaluated for the information it is expected to contribute.

This approach fits laboratory questions whose candidate routes have different informative conditions. An early observation may

identify which calibration range matters or which pair of mechanisms remains plausible. A later comparison can concentrate on that unresolved distinction. The analysis follows the complete decision procedure, including how observations select subsequent observations, so its uncertainty describes the experiment actually performed.

The constructive idea is that an answer can improve the next question. A well-designed adaptive study gives this improvement an explicit form. It preserves the original candidates and scoring rules while making the sequence responsive to what has been learned. The result is a more economical route through the space of explanations, with each step chosen for the distinction it can resolve.

17.2 — Perturbation, Control, and Causal Limits

Activity in a control network does not identify a single causal direction. It could drive a response or follow a decision already made. It might also reflect effort or arousal rather than a closure-related mechanism. A perturbation test must specify which pathway should change and what would count as a rival result.

17.2.1 — Why Perturbation Matters

A safe intervention can separate some of those possibilities when its assumptions and spillovers are explicit (Hernán and Robins, 2020; Pearl, 2009).

The state-change candidate suggests a sharper intervention question: can one independently specified Q explain paths that switch between conditions? Freeze the state description and scale, and include the experience of changing conditions in each transition. A shared difference rule still telescopes after a safe intervention changes the sequence or frequency of admitted transitions. Fitting a separate Q within each condition does not establish this property.

For an invented example, condition A assigns +1 to X→Y and −1 to Y→X; condition B reverses those signs. Each condition alone gives zero around the loop, yet taking X→Y under A and Y→X under B gives +2. Both fits cannot represent one unchanged Q on the declared states. If condition belongs in the state, specify and measure that dependence before the mixed-path comparison. Declare in advance which interventions preserve the candidate relation and which alter its proposed realization. Invariant prediction provides a methodological precedent for this distinction, not an automatic causal guarantee (Peters, Bühlmann, and Meinshausen, 2016).

The causal estimand must be named before interpretation. Assignment to a procedure, actual target engagement, dose

received, and change mediated through one pathway are different comparisons. Conditioning on post-assignment response can create bias, while ignoring target engagement can make mechanism failure indistinguishable from intervention failure (Hernán and Robins, 2020; Pearl, 2009).

The intervention also opens new causal paths. Expectancy, discomfort, fatigue, learning, experimenter behavior, and spillover can change the outcome alongside the intended target. Use sham conditions, target-engagement checks, manipulation-fidelity measures, and a stated causal model to assess these alternatives.

Preserve assignment, achieved modulation, sensations, adverse effects, and protocol deviations as distinct variables.

Social interference can defeat an individual-level interpretation. Participants may discuss the task, observe another person’s reaction, compete, cooperate, or receive cues from a shared experimenter. When one person’s assignment can affect another person’s outcome, the unit of intervention and the unit of analysis must match the social structure rather than hiding dependence inside standard errors (Hernán and Robins, 2020).

When people interact, an intervention can have more than one recipient in its consequences. Consider an invented paired task. Each participant can receive an optional instruction card that improves their own task score by one point. Seeing a partner use the card also improves the participant's score by two points. The example stipulates these effects so the assignment problem can be examined directly.

Let a participant's own card assignment be z, equal to zero or one, and the partner's assignment be w. Their score increase is z + 2w. Four joint situations are possible. With no cards, the increase is zero. With only the participant's card, it is one. With only the partner's card, it is two. With both cards, it is three.

The effect of receiving one's own card while holding the partner's assignment fixed is one. The effect of the partner receiving a card while holding one's own assignment fixed is two. The difference between giving cards to both participants and giving them to neither is three per participant. All three statements are correct, and they answer different intervention questions.

A design that assigns whole pairs to both cards or no cards estimates the combined contrast. It has no observations of the two mixed situations, so it cannot separately identify the own-card and partner-card contributions without additional assumptions. A design that independently randomizes each participant's card supplies all four situations over enough pairs. Its own-assignment contrast averages across the same partner-assignment distribution in both groups and can identify the one-point direct effect in this model.

The practical unit of intervention therefore helps define the scientific result. A shared environment can be changed for everyone at once, or access can be varied within that environment. Each design produces a different collection of observable contrasts. A comparison among participants is incomplete until it states what happened to the people around them.

Interactions can add further structure. Suppose using both cards together creates an additional one-point gain beyond the separate effects. The both-card situation then produces four rather than three. Subtracting the two single-card increases from the joint increase leaves one. That difference measures the interaction in this constructed task. It asks whether a combined intervention does more or less than the sum of its isolated components.

A study of shared affective conditions can use the same logic while defining its own outcomes and permitted interventions. Record the relevant social arrangement, specify whose assignment can affect whom and distinguish individual from joint changes. The analysis can then preserve the separate streams while investigating causes that act through a relationship. Shared influence becomes a describable structure rather than an unexplained resemblance between people's records.

A local relation can also be tested along a recorded trajectory without matching its endpoint states. On an absolutely continuous branch, suppose $v = -\frac{dQ}{dt}$. Choose a smooth weight w that vanishes at both ends of an observation interval. Integration by parts predicts $\int w v\,dt = \int w' Q\,dt$. The derivative falls on the known weight, not on noisy observations of Q. Weak-form system identification uses this principle (Messenger & Bortz, 2021). When parameters are estimated from noisy state measurements, however, errors can enter both sides of the regression; WENDy models this errors-in-variables structure rather than treating every residual as independent noise (Bortz, Messenger, & Dukic, 2023).

Fix Q’s independent measurement, the weights, and the uncertainty calculation before evaluating new trajectories. A zero-integral weight cancels a constant rate offset but also misses a genuine constant departure. Separately calibrated nonzero-integral weights examine that component. Finite weights test specified projections, not all deviations. Account jointly for shared observations, parameter fitting, and numerical integration error. A causal interpretation alongside perturbation still requires the assignment and selection conditions stated here.

Two simple weights show how this method can separate features within a trajectory. Work on a unit interval, with time t

running from zero to one. Let a candidate state be $Q(t) = t(1 − t)$. Its negative derivative is $2t − 1$, so a rate $v(t) = 2t − 1$ satisfies the proposed change relation exactly. Both boundary values of Q are zero, and the rate's unweighted integral is zero.

For any differentiable weight w that vanishes at both boundaries, the proposed relation makes the integral of w times v equal the integral of $w′$ times Q. The difference between these two integrals is a weighted residual. Both sides use levels of the observed rate or state; the derivative is taken from the investigator's known weight. This is the practical attraction of the weak form.

Choose the first weight as six times $t(1 − t)$. It is positive inside the interval and has integral one. Choose a second weight as thirty times $t(1 − t)(2t − 1)$. It is negative in the first half, positive in the second, and has integral zero. The first weight emphasizes the interior as a whole. The second compares the earlier and later portions while assigning no net weight to a constant.

Now add a specified departure to the rate: a constant a plus b times $(2t − 1)$. Keep Q unchanged. The weighted residual with the first weight equals a. The constant passes through because the weight integrates to one, while the term involving b cancels by symmetry. With the second weight, the constant cancels and the residual equals b. The factor thirty was chosen because the integral of $t(1 − t)$ times $(2t − 1)^2$ is one thirtieth.

For example, let a be one fifth and b three tenths. The added rate begins at minus one tenth and ends at one half. Its unweighted integral is one fifth. The first contrast returns one fifth, while the second returns three tenths. The two numbers describe different aspects of the same departure: its constant component and its change from an earlier negative tendency to a later positive tendency.

An unweighted total alone would recover the first number while losing the second. A zero-integral contrast alone would recover the second while losing the first. Keeping both supplies an interpretable two-component description. Their combination is particularly useful because a constant observation offset can enter the same direction as a constant departure. Independent calibration addresses that component; the time contrast supplies information about the changing component.

Other departures can require other weights. A symmetric bump near the middle, for example, may not be described adequately by a constant and a linear trend. The procedure can be extended by choosing additional smooth weights for the features the candidate models predict. The choice is then tied to a scientific question about temporal shape, and the resulting collection of contrasts has a known relationship to that question.

17.2.2 — The Invitation and the Action It Changes

A laboratory can assign an invitation while participants retain the choice of what to do. Imagine an optional, low-burden puzzle exercise. Some participants receive a randomly assigned reminder that the exercise is available. Others receive no reminder. The researchers observe who chooses the exercise and later measure performance on a related task. Randomization directly creates a comparison between reminder conditions.

It does not directly create a randomized comparison between people who did and did not choose the exercise. Those groups may differ in prior interest, confidence, skill, or available attention. The distinction is useful because the invitation and the action answer different questions. One asks what offering the reminder changes. The other asks what doing the exercise changes.

Consider a constructed population with three kinds of response to the reminder. One quarter would choose the exercise with or without it. One quarter would choose it in neither condition. The remaining half would choose it only if reminded. These groups describe two possible actions for each person under the two assignment conditions. A person's observed choice reveals only the action corresponding to the assignment they actually received.

Without the reminder, the exercise is taken by the first quarter. With the reminder, it is taken by that quarter plus the responsive half, giving three quarters. The reminder therefore increases uptake by one half. This is the first-stage effect: the randomized invitation changes the probability of the action that the researchers want to understand.

Now stipulate that, for the responsive half, completing the exercise increases the later outcome by four units. Also stipulate that the reminder has no direct effect on the outcome apart from changing exercise participation. The people who always participate take the same action in both conditions; the people who never participate also take the same action. Only the responsive half changes its action. The average effect of reminder assignment is consequently one half times four, or two units.

Dividing the assignment effect of two by the uptake effect of one half recovers four. This ratio identifies the average exercise effect among the people whose participation is changed by the reminder under the stated assumptions. It need not describe the exercise effect among those who would always participate or those who would never participate. Those people provide no change of action through which this particular invitation can reveal their exercise effect.

The arithmetic extends to unequal effects within the responsive group. If some gain two units and others gain six, with equal proportions, their average is still four. The assignment comparison averages the effects carried by the people who change their action. The ratio removes the fraction of the population that contributes those changes, leaving the average for that group. This is a local average treatment effect, with the group defined by response to the specified encouragement.

The assumptions also explain how to design an informative invitation. Random assignment makes reminder status independent of the preexisting response types in the population. The exclusion condition says that the reminder influences the measured outcome through exercise uptake. Monotonicity says that no participant would take the exercise without the reminder but refuse it because of the reminder. The exercise and the outcome must have a consistent meaning across the conditions. These are properties of the comparison, rather than additional consequences of random assignment itself (Hernán and Robins, 2020).

A small alteration shows why the exclusion condition matters. Suppose the reminder directly increases every participant's later score by one unit, perhaps because its wording supplies a useful hint. The assignment effect becomes three rather than two, while the uptake difference remains one half. The ratio becomes six. It now combines the effect of increased participation with the hint's direct contribution. A reminder designed to alter uptake alone and a message designed to teach something have different causal roles.

Different invitations can also reach different people. A brief reminder, a convenient time slot, and an offer of accessible equipment may each change participation for a different group.

Even if each comparison is well designed, their ratios need not estimate the same population effect. The result becomes more interpretable when the report describes the invitation and the people whose opportunities or choices it changes.

For laboratory questions about candidate mechanisms, this supplies a more exact link between assignment and experience. Researchers can first state the effect of the intervention they actually randomized. Under additional justified assumptions, they can then investigate the action that intervention changed. The distinction lets a voluntary activity contribute useful causal evidence while keeping the question local to the variation the design genuinely produced.

17.2.3 — Ethical Limits on Manipulation

Manipulation limits should be written into the protocol as stop rules and design ceilings. How intense can frustration become? How long can uncertainty last? What feedback is allowed? What deception is prohibited? What distress rating ends the task? What signs require operator intervention? What happens if a participant freezes, cries, dissociates, panics, or asks whether they failed as a person?

Before the task, invite participants to rehearse the stop procedure and let staff practice acknowledging it and ending collection. The rehearsal is optional; declining it must not weaken the right to stop at any time.

The manipulation must also respect repair after the task. Debriefing is not ceremonial. It should explain the artificiality of the task, correct misconceptions, remove personal stigma, and provide support information when needed (Beauchamp and Childress, 2019).

Use multimodal monitoring where appropriate: direct discomfort reports, operators' behavioral observations,

physiological arousal, and task freezing, repeated errors, or withdrawal. These imperfect channels together help reveal study-induced burden (World Medical Association, 2024).

Publication should include the burden actually observed. A study that produced distress, withdrawal, or lingering effects should report those outcomes alongside the scientific result.

17.2.4 — A Proposed Transition Study

This proposed study tests a within-person mean state-change relation on a small set of ordinary, low-burden transitions. It has two stages: a separate qualification pilot followed, only if the measurement and state checks succeed, by a frozen confirmatory study. The design below is a proposal; no participant results are claimed.

Question and conditions. Can one set of state potentials predict accumulated signed reports along unused transitions? Recruit consenting adults for whom the tasks and reporting mode are accessible. Use four prospectively defined conditions: A, quiet rest; B, viewing individually acceptable pleasant non-social images; C, an easy visuospatial puzzle without performance stakes; and D, quiet listening to an individually acceptable instrumental recording. Stimulus sets, sound levels, puzzle rules, transition duration and preparation periods are fixed after the pilot. None is selected to induce pain, fear or humiliation.

State qualification. A condition label is only the initial candidate state description. Before seeing transition totals, use repeated boundary assessments of current feeling, arousal, fatigue, task engagement and recent context to define a finite, reproducible state classification. The pilot tests whether visits classified alike have comparable boundary profiles and subsequent responses. If preparation history or time in session

changes those responses, revise the state description before freezing it. If repeatable states cannot be obtained within acceptable burden, this design stops at feasibility failure. It must not assign a fresh state to each inconvenient visit.

Calibration. In separate sessions, qualify the signed report channel's origin, positive and negative linking, duration sensitivity and retest behavior using the procedures in Section 11.2. Do not choose the origin by making loops cancel. Record its uncertainty and any calibration drift. A person may use a distinct fixed positive unit; every comparison below is within that person. If an additive interpretation is unsupported, preserve ordinal findings separately and do not compute the proposed cumulative test.

Transitions and allocation. Include A→B, B→C and C→D as training edges, with A→C, B→D and D→A reserved for confirmation. Randomize scheduled edge order in balanced session blocks, with the same preparation procedure for each origin condition. Keep every assigned attempt in the flow record. Record achieved boundary states instead of discarding unsuccessful preparation after observing its total. Sampling and reset schedules are fixed from pilot estimates of carryover and burden, not from which sequence best closes.

Observation. Collect the qualified signed rating at fixed intervals through each transition, including the switch itself; integrate using the prespecified exposure weights. Record prompt timing, missing responses, task events and boundary checks. Freeze an observation model with person-specific constant gain and calibrated offset, within-session correlated error, drift terms supported by calibration, and uncertainty from numerical integration. A shared calibration error remains shared across all six edge totals. The confirmatory analyst receives the frozen transformation, not an invitation to retune it.

Estimands. For each participant, estimate the six mean edge totals. Use the three training edges to fit the potential differences, fixing one potential to zero for identification. The held-out contrasts are A→C minus A→B minus B→C; B→D minus B→C minus C→D; and D→A plus A→B plus B→C plus C→D. The candidate predicts zero for all three. These are three independent graph restrictions, not six independent confirmations. Unequal total exposure means a common rate offset need not cancel; propagate its independent calibration uncertainty.

Rivals and perturbation. Fit a persistent-state model that accumulates time spent in each condition, a carryover/adaptation model using prior condition and elapsed time, and an unrestricted transition model. Give them the same observations, preprocessing and allowed fitting budget. Compare held-out prediction with uncertainty. Specify how each rival predicts an unobserved edge. If the unrestricted model has no prospectively fixed transfer rule, retain it only as a descriptive saturated comparison of the six observed edge means. In a separately reserved block, vary transition duration within the benign range qualified in the pilot while preserving the boundary procedure. Test the frozen models on that block; a refitted potential is a revised model, not successful frozen prediction. A synthetic benchmark should first demonstrate that these rivals can beat the candidate when they generated the data.

Missingness and selection. Distinguish missed prompts, task interruption, preparation failure and withdrawal. Use bounded sensitivity analyses for unobserved contributions under pilot-supported limits; do not code them as zero. Report the full randomized schedule and results conditional on achieved state separately. If finishing-state selection drives the contrast, the conditional calculation supplies no randomized causal effect. No

verdict is issued when admissible unseen contributions can reverse it.

Precision and decisions. Before confirmation, specify a scientifically meaningful equivalence margin for each contrast. Define the confirmatory family in advance as all three contrasts across every participant scheduled for confirmation, and use a simultaneous interval procedure accounting for shared edges, calibration and repeated observations. Determine repetitions and participant count by simulation using pilot variances, dependence, missingness and burden limits. Require at least 90 percent probability of detecting the declared rival departure at familywise 5 percent error, and assess the probability of placing all intervals inside the margins when the candidate holds. These numerical probabilities are design criteria, not observed performance. The margins, noise estimates and feasible sample size remain pilot dependencies; without them the study is not enrollment-ready. Freeze the resulting fixed sample plan and prohibit unplanned significance-driven stopping.

The word simultaneous changes the interval calculation. Imagine three independent estimates, each accompanied by an interval procedure with 95 percent coverage for its own target. The probability that all three intervals cover their targets is 0.95 cubed, or about 0.857. Each interval keeps its individual coverage while the three together answer a different probability question.

A simple way to obtain at least 95 percent joint coverage is to divide the total error allowance among the three intervals. Give each an error allowance of 0.05/3, so each has coverage of at least 98⅓ percent. The probability that any interval misses its target is at most the sum of the three individual miss probabilities, which is 0.05. This bound holds without assuming independence.

The same construction works for a larger prespecified family. With sixty targets and equal allocation, each interval receives an error allowance of 0.05/60. Unequal allocations can also be fixed in advance when some comparisons have a different role, provided their allowances sum to the desired family allowance. The coverage statement belongs to the complete declared collection.

The proposed transition study has shared edges and calibration information, so its contrast estimates are related. A procedure using the known or adequately estimated joint error structure can be more precise than the simple allocation above. Simulation through the complete observation model can check whether that procedure achieves the intended simultaneous coverage at the sample sizes and dependence patterns under study.

Joint coverage gives the equivalence decision a clear logic. If every simultaneous interval lies inside its own fixed equivalence band and all intervals cover their targets, then every target lies within its corresponding band. A mistaken declaration that all targets are equivalent requires at least one interval to miss. The family coverage guarantee therefore controls that error under the model and decision rule used to construct the intervals.

Recovery, stopping and interpretation. Before recruitment, run the entire analysis on known potential, persistent-state, carryover, drift, misclassified-state and selective-missingness generators. Report false support, false rejection, interval coverage and unresolved outcomes with Monte Carlo uncertainty. Stop a session for requested withdrawal, sleepiness or distress beyond the approved ceiling, or failed equipment; stop the design if the pilot cannot separate the generators within its burden limit. A simultaneous interval lying wholly outside its corresponding equivalence margin challenges the local

representation at the declared tolerance. If every simultaneous interval lies wholly within its corresponding prespecified margin, the results support the representation at that precision over these transitions. Intervals that overlap a margin leave the equivalence decision unresolved. None establishes a complete-stream endpoint law: the earned result concerns the specified state-change route, measurement and population (Jiang et al., 2011; Chan et al., 2025).

17.2.5 — An Intervention with a Second Decision

An intervention can contain more than one decision. A first action changes a situation, that new situation informs a second action, and the final result reflects the sequence. A study of an optional learning activity might offer an introductory exercise, observe a readiness signal, and then offer a second aid more often to participants showing that signal. The intermediate observation is both an outcome of the first stage and a reason for what happens next.

Consider a constructed task with first action A, intermediate state L, second action B, and final score Y. Each action and the intermediate state is either zero or one. Assign A randomly with equal probability. Let L equal one with probability 0.2 when A is zero and 0.8 when A is one. The first action therefore changes the distribution of the state encountered at the second decision.

Choose B using that state. When L is zero, assign B with probability 0.25; when L is one, assign it with probability 0.75. Within each recorded history, the second assignment uses a fresh random draw. Finally, define the score as $Y = A + B + L$. These rules make every relevant causal contribution explicit. The numbers describe an invented task so that the different comparisons can be derived exactly.

The direct contribution of changing B from zero to one while holding A and L fixed is one. A raw comparison of people who received B with those who did not gives a different number. Because L is one in half the population, half receive B overall. Among those receiving B, the probability that L is one is 0.75; among those not receiving B, it is 0.25. The groups also differ in A, because A helped determine L.

The probability of A being one is 0.65 among recipients of B and 0.35 among nonrecipients. The raw score difference is therefore one from B itself, plus 0.5 from the different prevalence of L, plus 0.3 from the different prevalence of A. It equals 1.8. The second aid has a one-unit causal contribution in the model, while the groups selected by the observed assignment rule differ by 1.8 units.

Now ask about a fixed two-stage regime: set A to a chosen value a and B to a chosen value b. The intermediate state still follows the distribution produced by a. Its expected value is 0.2 + 0.6a. The expected final score is consequently a + b + 0.2 + 0.6a, or 0.2 + 1.6a + b. Setting both actions to zero gives 0.2; setting both to one gives 2.8. The contrast between those two complete regimes is 2.6.

The calculation has two steps. First estimate the distribution of the intermediate state under the first action. Then average the outcome under the chosen second action over that distribution. This preserves the path from A through L while separating the second action from the rule that happened to assign it in the observed data. It is a small instance of sequential standardization, developed more generally in causal-inference methods (Hernán and Robins, 2020).

The same fixed regime can be recovered by weighting the histories that followed it. Consider the regime A = 1 and B = 1. For a record with L = 1, the probability of following those two

assigned actions, conditional on the intermediate history, is one half times three quarters. Its inverse is eight thirds. For a record with L = 0, the corresponding probability is one half times one quarter, giving inverse weight eight. The weights account for how readily each intermediate history entered the observed regime.

In the complete constructed population, histories with A = 1, L = 1, B = 1 have probability 0.3 and outcome three. Histories with A = 1, L = 0, B = 1 have probability 0.025 and outcome two. Their expected weighted outcome is 0.3 times eight thirds times three, plus 0.025 times eight times two. It equals 2.4 plus 0.4, or 2.8, matching sequential standardization. Their expected total weight is 0.8 plus 0.2, or one.

The larger weight for L = 0 restores a history that the second-stage assignment rule selected less often. Using one unconditional probability for B would miss that distinction. The correction follows the probabilities actually used at each decision point, conditional on the information available there. In a finite sample, the weighted estimate has sampling variation, and histories selected less often contribute more variable terms. The exact population calculation shows what the method targets before that uncertainty is considered.

A different question keeps the observed adaptive rule for B. Under A equal to zero, the probability of B is 0.25 + 0.5 times 0.2, or 0.35. Under A equal to one, it is 0.65. The expected scores become 0.55 and 2.45, a contrast of 1.9. That number describes changing the first action while allowing the state-responsive second-stage policy to operate. It differs from the 1.6 contrast obtained when B is held fixed.

The three results now have clear meanings. One is the contribution of the second action at a fixed prior history. Another compares two fixed complete regimes. A third changes

the first action while retaining an adaptive continuation. Each can be useful. The observation sequence determines which comparison a study directly supports and which additional modeling is needed to answer another one.

For an experimental program about experience, this is a way to take adaptation seriously in the design itself. Researchers may respond to earlier observations, participants may make new choices, and later opportunities may depend on what has already happened. Recording those decision points allows the analysis to describe a sequence of actions and experiences rather than compressing it into a single treatment label. The result is a more precise account of what changed, through which path, and under which continuing arrangement.

17.3 — Negative Controls and Rival Tests

Participants can infer what an experimenter hopes to see. They may follow perceived norms or imitate confident peers. A wish to appear competent can also alter reported choices. Negative controls must measure these influences before a horizon-related effect is interpreted as evidence about felt opportunity.

17.3.1 — Social Influence

A horizon effect may partly be a social effect if the participant believes urgency is expected, valued, or judged.

Social-influence controls should also consider researcher identity and institutional setting. Participants may respond differently to authority, prestige, gender, age, race, culture, accent, clinical status, or perceived ideology. The laboratory's social field begins before the task begins.

Emotional contagion is the spread or alignment of affect across persons. In a lab, a participant may become tense because others are tense, calm because the operator is calm, competitive because peers are competitive, or ashamed because the group atmosphere implies shame. The measured affect may be real without being generated by the intended mechanism.

Anticipate emotional contagion beyond recruited participants, including effects on friends, partners, and bystanders (Coviello et al., 2014).

Expectancy can also travel through public materials. If participants enter a study after reading advocacy for the Law, that framing may shape reports, persistence, and interpretation. Serious studies should separate neutral recruitment language from public argument wherever feasible so that participation does not begin with emotional priming toward support.

Designs should separate exposure to others' states, beliefs about the researcher's expectations, and structural features of the

task instead of treating all three as one fairness response (Coviello et al., 2014).

Platform experiments create additional governance duties. A manipulation delivered at scale can affect people who did not meaningfully understand that research was occurring, and social effects may extend beyond the sampled account. Ethical review, transparency, minimization, and post-study assessment must match the reach of the intervention (Kramer et al., 2014).

17.3.2 — Non-Horizon Tasks

A non-horizon task should preserve ordinary task demands while removing the feature claimed to be distinctive. It may match difficulty, duration, reward, response format, and observation burden while removing finality, shrinking opportunity, or the loss of later repair. The comparison asks whether the proposed effect depends on horizon rather than on the general fact that one condition is harder or more dramatic (Pearl, 2009).

If the same pattern appears when no meaningful opportunity window is closing, the horizon mechanism weakens. Time pressure, novelty, arousal, cognitive demand, social evaluation, or analytic artifact may be sufficient. The theory cannot answer by declaring that horizon was secretly present in every task.

Neutral-valence and matched-reward tasks provide further tests. A task can preserve timing and structure while reducing meaningful positive or negative burden; another can preserve incentive while leaving later correction available. A Law-specific affective interaction should weaken under those controls. If it does not, the result may reflect attention, timing, or strategy rather than felt burden.

Manipulation checks are required on both sides. The target condition must actually change perceived remaining opportunity

or repairability, and the control must not accidentally create the same perception.

Observation flow must also be comparable. If distressed participants leave the target condition more often, the retained sample may look unusually regulated; if the control is shorter or easier, it may keep people who would otherwise withdraw. Reports should show missingness, exclusions, and burden by condition before claiming specificity.

17.3.3 — Negative-Control Failure

Partial failures matter. A signal may survive a sham comparison but fail a non-horizon task, or transport across tasks while disappearing at a second site. Those patterns can identify expectancy, site, preprocessing, or measurement dependence. Collapsing them into pass or fail discards the very information controls were designed to provide.

Model recovery should include control worlds. One targeted check is mathematical coupling: correlating an initial rating with a later change score that subtracts that same rating can create an apparent compensating relation because the initial observation is shared and its error enters the two quantities with opposite signs (Chiolero et al., 2013). This particular problem is not automatically present when past and future totals use separately observed, nonoverlapping intervals. Derive the actual statistic, retain the planned preprocessing and missingness, and test it on simulated histories without the proposed closure relation. Demand, regression to the mean, common exposure and measurement drift should also be represented. Estimate false-positive rates with simulation uncertainty and require them to meet prespecified error-control criteria before interpreting a real-data route-positive result.

17.3.4 — Separating Change from a Shared Reading

A change score contains two observations, and one of them may also appear elsewhere in the comparison. Following the shared reading through the algebra can reveal exactly what the resulting association means. This is particularly useful when an initial departure seems to predict an opposite later change.

Consider instruments whose true underlying value θ stays constant between two visits. Across instruments, θ has variance τ^2. At the first visit, the reading is $X = \theta + e_1$. At the second, it is $Y = \theta + e_2$. The two errors are independent of each other and of θ, have mean zero, and have variance σ^2. The measured change is $D = Y - X = e_2 - e_1$.

There is no true change in this model. Nevertheless, the covariance of the initial reading X with measured change D is $-\sigma^2$. The first error enters X positively and D negatively. A large positive first error tends to make the initial reading high and the later difference low; a large negative first error does the reverse. The common measurement creates an association even though the underlying quantity remains fixed.

The variance of X is $\tau^2 + \sigma^2$. The population regression slope of D on X is therefore $-\sigma^2/(\tau^2 + \sigma^2)$. If both variances equal one, the slope is minus one half. The variance of D is two, so the correlation is also minus one half in this particular case. An apparently orderly opposite movement emerges from the observation structure alone.

The magnitude depends on how much genuine between-instrument variation exists. If τ^2 is nine and σ^2 is one, the slope is minus one tenth. If τ^2 is zero, so every instrument has the same true value, the slope is minus one. These values follow from the same construction. They describe different mixtures of true variation and first-reading error.

A third observation can separate the shared-error contribution. Take an independent replicate of the baseline reading, $X' = \theta + e_3$, with e_3 independent of the other errors. Continue to define change using the original pair, $D = Y - X$. The covariance of X' with D is now zero. The baseline predictor and the subtracted baseline reading no longer contain the same measurement error.

This redesign has a defined purpose. It removes the specific covariance caused by reusing e_1. It does not make measurement error disappear from every part of the analysis. X' remains a noisy predictor, and D remains a noisy estimate of change. Repeated observations can help estimate those error components and support a model appropriate to the actual question, as the methodological analysis of baseline and change emphasizes (Chiolero et al., 2013).

Now allow a genuine change C, so that $Y = \theta + C + e_2$. Let C also be independent of all three measurement errors, while permitting it to depend on θ. The covariance between X and D becomes $\mathrm{Cov}(\theta, C) - \sigma^2$. The observed association combines the real relation between starting state and subsequent change with the negative contribution from the reused error. Using the independent replicate X' instead gives $\mathrm{Cov}(\theta, C)$, although its regression slope still reflects the noisy predictor's variance.

This formula helps choose the next measurement. If the goal is to study whether true starting state predicts true change, independent replication at baseline can address the shared-error term. Additional later readings can improve the estimate of change. If errors persist across visits, their covariances also enter the expression and can be estimated or varied under a stated observation model. The design follows the components of the quantity being studied.

The same algebra does not automatically apply to two totals measured over disjoint intervals. If a past total and a future total

are recorded independently, neither necessarily contains a subtracted copy of the other's reading. Their association has to be derived from their own definitions and error structure. The important step is to trace which recorded quantity enters which expression, with what sign and dependence.

A negative-control world can now be designed precisely. Simulate stable true quantities, add the planned observation errors, and run the proposed baseline–change analysis. The expected negative slope is known in advance. Then introduce a specified real relationship between θ and C and examine whether the method recovers it. The exercise turns an apparent compensating pattern into two testable ingredients: dynamics in the underlying process and structure introduced by its measurement.

17.4 — What Laboratory Failure Would Mean

Failure to find a horizon signature can challenge a candidate route. The manipulation must first have changed the construct it was meant to change. Measurement must also be reliable enough to detect the proposed effect. If the design could exclude only enormous effects, silence would not be a meaningful disconfirmation. When those prerequisites fail, the result may instead diagnose a problem with the manipulation, design, or estimation (Lakens, 2017; Wasserstein and Lazar, 2016).

17.4.1 — Low Power or Bad Design

Low power and bad design are different failures. Low power means that, under a specified alternative, the planned test has a low probability of rejecting the null hypothesis. Imprecision concerns the uncertainty of the estimate; it is related to power but is not the same quantity. Bias, leakage, invalid measurement, uncontrolled analytic flexibility, or a failed manipulation may prevent the intended inference even with a large sample. The report should state which failure occurred and what result a redesigned study would need to close the line.

Sample size should be justified against a declared effect and decision. A feasibility study may need precision for recruitment and burden. A mechanism comparison may need power to distinguish predictions that are close. An equivalence test must be precise enough to exclude effects outside its prospectively specified tolerance. Saying only that the study was powered leaves the scientific question unstated.

A larger sample is not the only way to improve a comparison. Myung and Pitt (2009) show how experimental design can be selected for distinguishing competing models. Search the ethically available range for predictions that remain different after each model receives its allowed parameter adjustment and

observation uncertainty. Pilot reliability helps determine whether the contrast is usable.

The most informative setting may be one where the models disagree, rather than one where either predicts a dramatic response. Imagine two fully specified candidates, A and B, and two allowed versions of a modest task. In the first version, A predicts a yes response with probability 0.9 and B predicts it with probability 0.8. In the second, A predicts yes with probability 0.8 and B predicts it with probability 0.2. All probabilities are fixed before the task is observed.

A yes in the first version is common under either model. Its likelihood ratio favoring A is 0.9 divided by 0.8, or 1.125. A yes in the second version has likelihood ratio four. The second yes is less frequent under A than the first yes, yet it distinguishes A from B much more strongly. Evidence depends on the comparison between predictions, not on how impressive an outcome looks on its own.

A no can also be informative. In the first version, its probabilities are 0.1 under A and 0.2 under B, giving a likelihood ratio of one half. In the second, the probabilities are 0.2 and 0.8, giving one quarter. Thus either answer to the second task moves the comparison farther from equal support than the corresponding answer to the first task.

With equal prior model probabilities and a decision to choose the more likely model after one response, the first task gives expected classification accuracy of 0.55. It chooses A after yes and B after no: half of 0.9 plus half of 0.2. The second gives accuracy 0.8: half of 0.8 plus half of 0.8. This is an exact consequence of the two specified model pairs and the stated decision rule.

Repeated outcomes can be combined when the models specify their dependence. Under independent repetitions of the second

version, two yes responses multiply the likelihood ratio to sixteen, while a yes followed by a no brings it back to one. The sequence has supplied information about both directions of the comparison. If the first response changes learning or expectations, the next response instead needs the appropriate conditional prediction; independence is a feature to model, rather than a property supplied by repetition.

The design question can include cost as well as discrimination. One task may require longer participation, a less accessible response, or an observation channel whose meaning is harder to establish. A useful comparison can evaluate expected information at a common duration or burden, within the permitted set of tasks. A slightly less discriminating observation may be preferable when it can be obtained more reliably or with less effort.

Identification and precision are different achievements. In an invented linear example, two readings are $y_1 = a + \varepsilon_1$ and $y_2 = a + \delta b + \varepsilon_2$, with known nonzero δ and independent zero-mean errors of variance σ^2. Their difference estimates b as $\frac{y_2 - y_1}{\delta}$, with variance $\frac{2\sigma^2}{\delta^2}$. Reducing δ from 1 to 0.01 preserves noiseless identifiability but multiplies this variance by ten thousand. The better design may improve sensitivity to the target rather than repeat an almost uninformative measurement. The parameters, error model and decision rule belong in the comparison before decisive data arrive.

Multisite work should report site-specific estimates to expose transport limits (Open Science Collaboration, 2015; Marek et al., 2022).

17.4.2 — Ethical Burden Too High

The protocol should measure burden rather than merely label it minimal. Completion time, interruption, withdrawal, distress reports, adverse events, missed obligations, accessibility failures, and delayed recovery can be tracked prospectively. Participant experience is evidence about whether the design was proportionate.

The reporting standards introduced in Section 10.4 do not settle ethical proportionality: a complete checklist can still describe a study that should not be done.

Laboratory-specific deception, when indispensable, needs independent justification, genuine debriefing, and post-study support.

When a route cannot be tested under acceptable conditions, the scientific claim narrows to what safe observation can support. The report should state the unavailable contrast rather than allowing readers to imagine evidence that was never ethically obtainable.

Chapter 18 — Following Lives with Care

A snapshot can miss the turn. A person may be well during the interview and struggling the next day, or distressed during a crisis and steadily recovering afterward. Repeated measures can reveal trajectories, within-person variation, adaptation, response shift, treatment effects, and the timing of missingness. The central image is a moving denominator. People are invited, enroll, respond, pause, withdraw, move, become ill, change treatment, lose devices, die, or become unreachable. The scientific record must follow those changes rather than presenting the final retained sample as though it had always been the whole group. A simple example shows the risk. Symptom scores may appear to improve after several high-burden participants stop answering during hospitalization. The apparent recovery belongs to the retained sample unless the missingness process and plausible unseen outcomes are modeled. It is not evidence that the absent people improved.

Sleep appears to improve overnight in a long-running study. Did people's lives change, or did a device update change how wakefulness was counted? Distinguishing those stories requires a record of the study's own changes, including permissions and how reports became results. Competing explanations should be specified before outcomes are examined. Section 18.3.4 sets common rules for scoring them fairly.

A long follow-up carries two clocks: participants grow older while calendar conditions change. In a cohort enrolled at the same age and time, an apparent age pattern could also reflect a recession or a change in services. Following several birth cohorts provides additional comparisons, but does not by itself separate age, historical period and cohort effects. The study must state the assumptions used to distinguish them (Osmond and Gardner, 1989).

The sampling architecture matters. Entry date, prompt schedule, device availability, response window, pauses, missed observations, and the reason a person leaves all shape the trajectory. Monthly averages can hide alternating days of burden and relief. Sampling frequency must also account for participant burden and possible reactivity (Shiffman et al., 2008). The design should name legitimate no-sampling zones. Hospitalization, acute grief, domestic danger, coercive monitoring, severe fatigue, terminal comfort care, or clinically identified distress may require the study to go quiet. Silence can be the protocol doing the ethical work it promised rather than a defect to be repaired.

Interpretation therefore depends on the time scale, entry point, sampling schedule, recall window, calibration history, and relation between momentary and interval measures. Daily reports, monthly summaries, and annual evaluations answer different questions; combining them without a measurement model can manufacture a smooth trajectory from incompatible observations (Singer and Willett, 2003).

18.1 — What Repeated Observation Can Reveal

Longitudinal observation can reveal how a life changes and how that change relates to the conditions around it. The design begins with a defined population, sampling schedule, event windows and observation channels. Treatment, illness, relocation, caregiving, bereavement, work and technology changes belong in the record when they affect the question being studied (Hernán and Robins, 2020). The purpose of each retained variable determines how it contributes to the analysis and how long it is needed.

18.1.1 — Two Questions Hidden in One Average

A repeated record can answer a question that a comparison among people cannot. Imagine four observations from two people working on chosen projects. On one day, the first person spends one hour on the project and records an energy rating of seven. On another, two hours accompany a rating of eight. The second person records three hours with a rating of one, then four hours with a rating of two. These are invented numbers for examining the structure of a comparison.

Within each person's record, the higher-hour day also has the higher energy rating. Across the four pooled observations, the person spending more time on the project has the lower ratings. A line fitted to all four points slopes downward. A line describing the two changes within people slopes upward. The numerical evidence supports two different associations because the analyses ask two different questions.

One way to expose the difference is to express each observation relative to that person's mean. The first person's mean project time is one and a half hours; the second person's is three and a half. Each person has one day half an hour below their own mean and another half an hour above it. Their energy

ratings are likewise half a point below and half a point above their own means. The within-person pattern becomes visible as soon as the stable difference between their average levels is separated from their daily changes.

The result invites a more precise question. What accompanies a person's change from their own usual pattern? That is different from asking why people with different usual patterns also report different average levels. Context, selection and direction of influence still matter to an explanation, but the intended comparison is now clear. Repeated observations have supplied information that a single observation from each person could not provide.

For a candidate dynamic account of experience, this distinction helps place a prediction at the right level. A claim about what follows a change within a stream should be evaluated through within-stream change. A claim about differences across streams needs the corresponding cross-person measurement and explanation. Both can be studied in one project while remaining distinct results.

18.1.2 — Putting an Event on Two Clocks

An event has both a place in a person's history and a date on a shared calendar. Suppose three people move into new homes in January, May and September. A record can place their observations on calendar time, preserving the seasons in which they occurred. It can also place them on event time, calling the moving day day zero and aligning the days before and after it. Each view reveals a different possible pattern.

Calendar alignment can expose changes shared by people living through the same period. Event alignment can expose a sequence that recurs around the move even when the move happens in different months. If records become irregular during

packing and resume after settling in, event time may organize the observation process as well as the reported experience. Retaining both clocks lets the analysis describe those relations separately.

The event itself can have several meaningful dates. Receiving confirmation of the home, collecting the keys and spending the first night there may initiate different changes. A study of anticipation might begin from confirmation; a study of travel demands might use physical relocation. The chosen event date earns its place through the mechanism being examined. The dates become more useful when their distinct roles are preserved than when one convenient date stands for the entire transition.

Repeated observation therefore offers more than a longer line on a graph. It supplies the comparisons needed to ask when a change began, which changes belong together, and whether a pattern follows personal transitions or a shared period. A well-designed trajectory becomes a sequence of answerable questions about a life in context.

18.1.3 — Age, Period, and Cohort

Age, historical period, and birth cohort offer three descriptions of the same date. At year 2030, a person born in 2000 is thirty. At year 2040, that person is forty. The relationship is exact when all three quantities use compatible units: age equals period minus birth cohort.

That identity matters when a model tries to estimate a separate linear trend for each coordinate. Suppose its prediction includes a times age, p times period, and c times cohort. Increase a by an arbitrary amount δ, decrease p by the same amount, and increase c by that amount. The change in the prediction is δ times age minus δ times period plus δ times cohort. It is zero for every observation because the three coordinates obey the identity.

An especially simple pair of explanations makes the ambiguity visible. One model assigns the whole linear trend to age: its prediction is the person's age. A second assigns no age effect and instead predicts period minus cohort. The models make the same prediction for every person at every observed date. They use different component labels for an identical fitted relation.

Following additional birth cohorts supplies many valuable comparisons, but all those observations still obey the same identity. More precise measurement and a larger sample do not separate the interchangeable linear components by themselves. This is the identification problem described by Osmond and Gardner (1989). It is a property of how the predictors relate, rather than an accident of a small dataset.

A study can make progress by stating the additional structure it uses. It might estimate a particular contrast that remains unchanged across the equivalent parameterizations, or bring in independently justified information about a period effect. It might use a constrained model whose restriction has a substantive interpretation. The resulting conclusion belongs to the observations together with that additional structure.

For example, a service opening on a known date could motivate a specific period-related question, with evidence from its actual exposure and a defined comparison group. That is richer information than the calendar label alone. Similarly, a developmental mechanism can motivate a particular age-dependent prediction that is tested through observations relevant to that mechanism. Naming a variable age or period does not supply the mechanism; independently specified consequences can.

18.2 — Life Events, Adaptation, and Context

Life events should be modeled as processes rather than isolated dates. Anticipation, immediate disruption, treatment, social response, material consequences, and later adaptation may each alter experience. Kapteyn and colleagues (2025) paired scheduled sampling with a new week of momentary reports after a reported event in adults aged fifty and older. On the study's standardized scales, estimated changes in affect two to six weeks after events were generally smaller than changes in life satisfaction; the immediate affective response was unobserved, and the outcomes were measured at different times. An informative next study would combine ongoing baseline sampling with promptly offered, optional event-triggered bursts and matched calendar-time comparisons, measuring affect and life satisfaction over the same periods. This would help resolve onset and recovery instead of inferring their shape from a later snapshot.

18.2.1 — Events Change Context and Experience

Illness or injury can change the measurement ruler. Pain, treatment, fatigue, and medication may alter self-report, sleep, movement, physiology, social contact, attention, appetite, and emotional range. When those channels contribute to the estimate, the event affects both experience and the instrument used to study it. That double role must be modeled rather than hidden (Sprangers & Schwartz, 1999; Mauss & Robinson, 2009).

Job loss provides a concrete test of pathway thinking. The same date may reduce income immediately, disrupt sleep weeks later, alter treatment access after insurance changes, and reshape identity or relationships over months. A single before-and-after contrast cannot represent that chain. Intercurrent-event rules and

sensitivity analyses should be fixed before the trajectory is interpreted (Hernán and Robins, 2020).

A pathway can be made explicit with an invented schedule change. Let A indicate whether a participant receives a revised work schedule. Let M describe uninterrupted rest on a scale chosen for this example, and let Y be a later outcome in the model's own units. Suppose the revised schedule increases M by two units, and each additional unit of M increases Y by three. Assume this is the entire causal route from A to Y in the example.

The total effect of the schedule change is then six units of Y: two units of additional rest multiplied by three units of outcome per unit of rest. The effect travels through M. Now ask a different question: what would changing A do if M were held at the same value under both schedules? Under the stated rule, the answer is zero. Fixing rest removes the only route through which the schedule affects the outcome.

The answers six and zero describe different interventions. The total-effect comparison allows the revised schedule to change rest as it normally does in the model. The controlled direct-effect comparison fixes rest at a common level and asks what remains of the schedule's influence. Both calculations follow the same causal account. Their difference is created by the question, not by disagreement in the arithmetic.

This distinction matters when choosing what belongs in an analysis. A record of rest can illuminate the mechanism through which a schedule works. Treating rest as something to hold constant, however, aims at an effect that excludes that mechanism. The analyst can state the desired contrast before deciding which variables to adjust for. In an actual study, identifying either contrast requires a design and assumptions that

connect the observed comparisons to the intended interventions (Hernán and Robins, 2020).

One can extend the example without losing its clarity. Give the revised schedule an additional direct effect of one unit on Y while retaining the six-unit pathway through rest. The total effect becomes seven and the controlled direct effect one. Improving the rest pathway and improving the direct pathway would then be different ways to improve the intervention. The decomposition locates a possible next experiment: which component can be changed, and what does the model predict if it is?

Life events often open several such routes at different times. A practical change may alter rest quickly, relationships gradually, and opportunity later still. Drawing the sequence helps decide when each variable should be observed and which contrast a follow-up can resolve. The event becomes an unfolding mechanism whose parts can be understood and improved, rather than a date attached to a single before-and-after score.

18.2.2 — Adaptation, Response Shift, and Score Meaning

Grief illustrates why a stable score can hide changing meaning. Early grief may bring shock or fear alongside changes in sleep and appetite. Practical burdens can include administrative work as a person faces the sudden absence of someone close. Later, anniversaries may bring renewed difficulty while identity and family roles change. Financial strain or the loss of imagined futures can persist. A repeated rating can remain numerically similar while the experience, standards, and life context beneath it have changed.

A person’s grief may combine love and gratitude with anger, guilt or numbness. Relief may coexist with disorientation. For example, coding it as purely negative can miss relief at the end

of a terminal illness or an abusive relationship, the conflicting feelings of an exhausted caregiver, or comfort found in sacred memory.

Investigate response shift through qualitative context and repeated calibration (Sprangers and Schwartz, 1999).

Three kinds of change can be explored separately. Recalibration concerns the standards used to place an experience on a scale. A participant may once have called a moderately busy afternoon very tiring, then later reserve that phrase for a much more demanding day. The vocabulary and response options remain the same, while the internal landmarks have moved.

Reprioritization concerns the relative importance of domains within an evaluation. Consider a participant who includes work, friendship, and leisure in an assessment of life quality. Early in a new job, work may dominate that assessment. Later, time with friends may count more heavily. The domains can remain present while their weights change. Asking which aspect mattered most in forming the overall judgment helps reveal this shift.

Reconceptualization concerns what the evaluation includes. Someone who initially understands a good week mainly as a productive one may come to include participation in family or community life as part of what the phrase means. The change reaches the composition of the concept itself. It can be investigated by asking the person to describe the experiences considered when making the judgment.

These possibilities suggest different follow-up questions. To investigate recalibration, return to the meaning of the scale's anchors and ask how familiar examples would now be placed. To investigate reprioritization, explore the relative influence of domains that remain in the evaluation. To investigate

reconceptualization, ask what belongs in the evaluation at all. A brief explanation in the participant's own words can supply information that a repeated number cannot contain.

The interest is not confined to correcting a measurement. A changed evaluation can reveal learning about what makes life satisfying, a new source of belonging, or a more differentiated understanding of experience. The researcher can preserve that development alongside the repeated scores. A coherent longitudinal account then describes both the course of reported life quality and the changing perspective through which it was assessed.

Public summaries should avoid saying that people return to normal when the measure no longer carries the same meaning. A more accurate statement identifies what changed, which scale was used, and whether the comparison survived response-shift analysis.

18.2.3 — Coding Events and Their Meaning

Event coding should keep external structure separate from participant meaning. A promotion may be publicly classified as favorable while being experienced as exhaustion; a failure may later be understood as release. The study can preserve both the initial appraisal and later reinterpretation without forcing one to govern the whole arc (Headey and Wearing, 1989).

A responsible event schema can record onset, duration, severity, recurrence, confidence, source, participant meaning, public meaning, dependency, and uncertainty. It must allow overlapping events because illness, caregiving, grief, poverty, and work disruption often occupy the same period. One tidy label cannot carry a season of life.

Distinguish voluntary reports, clinical documentation, administrative records, sensor suggestions, and contextual

inference. Record each source’s confidence, error, privacy, and authority rather than presenting all events as equally direct.

Participants may use a protected broad category, suppress detail, revise permission, or decline event coding. Lost resolution does not authorize searching records, relatives, or devices for what they withheld.

Test ordinary pathways: education to employment, treatment to symptoms, social support to recovery, and policy to exposure. A satisfying paired-event story cannot replace an actionable, falsifiable mechanism.

18.2.4 — A Trend and the People Composing It

A population trend can reflect two changes at once: what happens within groups and how much of the population each group comprises. Following those pieces separately can reveal an improvement that an aggregate conceals, or a change in composition that deserves attention in its own right. A numerical example makes the two questions easy to distinguish.

Imagine two kinds of task, familiar and unfamiliar, completed during two observation periods. In the first period, success rates are ninety percent for familiar tasks and fifty percent for unfamiliar ones. Eighty of the hundred tasks are familiar, so expected successes are 72 + 10 = 82. The overall success rate is eighty-two percent.

In the second period, both conditional success rates improve by five percentage points. Familiar tasks now have a ninety-five percent success rate and unfamiliar tasks a fifty-five percent rate. But the mix changes: only twenty of the hundred tasks are familiar. Expected successes are now 19 + 44 = 63. The overall rate falls to sixty-three percent even though performance improves within both kinds of task.

The aggregate answers a real question: how often were the tasks actually encountered completed successfully? The conditional rates answer another: how did performance change for each kind of task? Both answers can be correct. The apparent contradiction arises only if the overall change is treated as though the mix of tasks had stayed the same.

To isolate a fixed-composition comparison, choose a common set of weights. Using the first period's mix, the second period would yield 0.8 × 0.95 + 0.2 × 0.55 = 0.87. Compared with the first period's 0.82, that is an improvement of five percentage points. Using the second period's mix for both periods gives 0.2 × 0.90 + 0.8 × 0.50 = 0.58 initially and 0.63 later, again an improvement of five points.

The two standardized levels differ because they describe different mixes. Their changes agree here because both conditional improvements equal five points. If the improvements differed by task type, the standardized change would also depend on the selected mix. A report can therefore identify the reference composition and explain why it matches the question being asked.

There is an exact decomposition of the observed decline. Hold the original mix fixed and change the success rates first: the aggregate rises from 0.82 to 0.87, a gain of 0.05. Then change the mix while retaining the later rates: the aggregate falls from 0.87 to 0.63, a change of −0.24. The sum, −0.19, is the observed decline from 0.82 to 0.63.

One can reverse the order of that decomposition, first changing the mix under the earlier rates and then changing rates under the later mix. In this example the contributions remain −0.24 and +0.05 because the conditional gains are equal. In general, an interaction between changing rates and changing

weights makes the allocation depend on the chosen decomposition. Stating the order makes its meaning inspectable.

In a longitudinal study of lived conditions, the groups might be defined by employment state, housing arrangement, age band, or another relevant context. A later sample can contain more people in a demanding circumstance even while outcomes improve within every circumstance. The overall result can describe the changed population accurately, while a standardized comparison reveals the conditional improvement.

Group membership can itself be an outcome. If a program helps people move from one circumstance to another, holding the group proportions fixed removes part of the program's possible effect from the comparison. Standardization is useful because it defines a particular question, not because a fixed composition is always the right target. The actual population result and the conditional result can be presented together when both matter.

The same care applies to interpretation. A within-group improvement is a descriptive finding until the design supports a causal account of why it occurred. A changing mix may reflect entry, exit, classification, or real transitions. Following those routes makes the aggregate more intelligible. The arithmetic identifies where the changes enter; further evidence explains their causes.

This gives long-arc observation a fuller vocabulary. We can ask what changed for comparable cases, what changed in the population's composition, and what the people actually encountered across the whole period. Their answers may point in different directions while revealing one coherent history. A trend becomes more useful when its moving parts remain visible.

18.3 — Observation, Missingness and Model Comparison

A reserve-coupled route may predict changes as the usable horizon contracts. Those changes could appear in variation or recovery slope, not only mean valence. Treatment and response shift can imitate similar patterns. The test must prespecify both the proposed signature and the independent measures of horizon and reserve used to interpret it.

18.3.1 — Long-Arc Prediction

Specify which feature changes, in which direction, and over which interval so the forecast can be tested.

The forecast's time window is part of its meaning. Suppose an imagined model assigns a probability of one tenth to a specified event on each of the next seven days. One question asks how many event-days to expect. Adding the seven daily probabilities gives an expected count of seven tenths. This calculation holds even when events on different days are dependent.

A different question asks for the probability of at least one event during the week. If the daily events are independent, the probability of no event is nine tenths multiplied by itself seven times. Subtracting that value from one gives about 0.522. The weekly chance is just over one half, even though each day's chance is one tenth.

Dependence changes the second answer. Imagine a common weekly state that produces the event on all seven days with probability one tenth and on none otherwise. Every daily probability remains one tenth, and the expected count remains seven tenths. But the probability of at least one event during the week is now only one tenth. At the other extreme, suppose the seven possible event-days are mutually exclusive, each with probability one tenth. Then the weekly chance is seven tenths.

The daily forecasts are identical across all three models; the weekly event probabilities differ.

A long-arc prediction therefore needs both a target and a temporal model. Predicting a count, an average, a first occurrence, or at least one occurrence asks for different features of the joint course. The event definition and scoring rule can be fixed before observation, and each model can be evaluated on the target it actually predicts. This makes a longer forecast an explicit claim about how days fit together, rather than a daily probability copied across a calendar.

If the forecast concerns terminal narrowing, the clinical and observational rivals in Section 16.4.1 must be included in its comparison.

A model that predicts next-month distress in one service may lose calibration after a health-system change. Such a change provides a concrete transport test: does the same forecast still work when access to treatment, referral, or observation has changed?

The unit held out for evaluation determines which kind of prediction a study has tested. Consider a synthetic record with twenty people observed on ten occasions each. Assign each person a stable value of either plus two or minus two, independently with equal probabilities. Every observation from that person records the assigned value exactly. The construction has no occasion-to-occasion variation.

A predictor that has seen one record from a person can remember that person's value and predict all the other records perfectly. If the training collection includes five observations from each person and the test collection includes the other five, its test error is zero. It has learned the persistent person-specific levels represented in this world.

Now hold out complete people. For a new person, the model has no record revealing which of the two values was assigned. If assignments to people are otherwise independent, the best squared-error prediction is zero. Its squared error is four for either possible value. The same prediction strategy has answered a more demanding question and achieved a different result.

Neither evaluation needs to be dismissed. The first may fit a service that obtains initial observations from every participant before making further forecasts for that same participant. The second fits a claim to predict for a person before receiving their own calibration record. The distinction should be visible in the description of the intended use, so the reported performance belongs to the question actually evaluated.

Time adds another separation. Suppose each person's value changes on the eighth occasion by an amount that cannot be inferred from the earlier record. A random division of occasions may put post-change observations in training and earlier observations in testing. It then evaluates interpolation within a collection that already contains both periods. A forecast issued before occasion eight has different available information. Holding later occasions out preserves that prospective question.

A study can therefore evaluate several useful capabilities separately. It may predict another observation from a known person, a future block for that person, the first block for a new person, or a record at a new site. Each holds out a different source of familiarity. A model can succeed at one capability while needing additional information for another.

This also clarifies what personalization contributes. If a brief calibration block improves forecasts for later blocks, the gain can be measured against a model that receives no personal observations. If further calibration adds little, the study has learned where the useful information saturates. The comparison

concerns how much of a person's own record is needed for a declared prediction, with the boundary between available and future information kept explicit.

18.3.2 — Who Drops Out

Withdrawal, loss, and design failure are not the same. A person who intentionally leaves has exercised a right. A person who becomes unreachable may be missing for unknown reasons. A person excluded by language, disability, cost, technology, transportation, or burden marks a failure of access that the study must report rather than attribute to noncompliance.

Attrition has a sequence. A participant may first skip evening prompts, then disable passive sensing, then miss a visit, then refuse linkage while allowing earlier data to remain. Preserving those steps can reveal whether burden, health, trust, technology, or changing priorities drove the loss and whether a remedy actually worked.

Informative missingness occurs when observation availability is related to the process being studied. Treatment can reverse the meaning of that relationship. Effective care may reduce reporting because life becomes less centered on symptoms, while intensive services may increase observation. Worsening illness may remove someone from the study, or may make the person more visible to clinical systems. Analyses should separate treatment initiation, response, access barriers, and research contact (Hernán and Robins, 2020; Fitzmaurice et al., 2011).

Represent care-induced changes in the observation process or intervention regime explicitly, and state whether they change the estimand.

Response shift and selective retention can interact: people who find the changing instrument alienating may leave. Apparent

stability may then reflect changed experience, changed standards and changed membership together (Schwartz and Sprangers, 1999; Oort et al., 2009).

A missing-data analysis can begin with information that is genuinely available. Imagine a study of one hundred participants using an operational score whose possible values run from zero to ten. Eighty participants supply a follow-up score, and their mean is six. The total of the observed scores is therefore 480. The twenty unobserved scores contribute between zero and 200, so the full cohort's mean lies between 4.8 and 6.8.

This range is an exact consequence of the stated score bounds and observed total. It does not require the missing participants to resemble the observed participants. The calculation also says exactly what is being bounded: the mean of that operational score for the original one hundred people at the specified follow-up. A separate measurement argument would be needed to interpret the score as a feature of experience.

Additional information can narrow the range. Suppose independent records establish that every missing score lies between two and four. Their combined contribution then lies between forty and eighty, and the full mean lies between 5.2 and 5.6. The improvement comes from information about the unobserved values. Replacing every missing value with the observed mean would also yield a single number, but it would introduce a different assumption rather than supply the same information.

The width has a simple structure. With twenty percent of the cohort missing and a possible score span of ten units, the first range has width two. If only ten percent were missing under the same bounds, the corresponding uncertainty from missing values would have width one. Narrower justified score bounds and more complete observation address different parts of the

calculation. This helps identify which additional evidence would be most useful.

Population definitions remain visible throughout. Reporting the mean of six for the eighty respondents is a complete description of their observed mean. Reporting it as the mean of the original cohort changes the target without resolving the twenty unknown values. A study can present the respondent result and the full-cohort range side by side, allowing each to answer its own question. Even before a sophisticated model is fitted, the arithmetic shows both what has been learned and where further observation can sharpen it.

For each channel and period, reports should preserve enrollment and follow-up flow, baseline differences between completers and non-completers, reasons for dropout where ethically available, device and prompt completion, missingness by event and burden class, endpoint missingness, treatment changes, protocol deviations, adverse events, and uncertainty after correction. Choose guidance for the design: STROBE for observational cohorts (von Elm et al., 2007), and CONSORT 2025 for randomized trials (Hopewell et al., 2025). The channel-, event-, and burden-specific records above are additional requirements of this program.

18.3.3 — Meeting the Population Again

A panel follows the same people through time. A fresh sample meets the population again. The two designs can answer different parts of a question about change. Joining them thoughtfully can reveal something neither supplies alone: whether the experience recorded among continuing participants also describes the population from which the study began.

Consider an invented population of one hundred people, all remaining in the population throughout the example. At the first

visit, forty report a specified experience and sixty do not. The experience is coded as present or absent for this illustration. At the second visit, only fifty members of the original panel respond. Ten of those fifty report it, giving a responding-panel proportion of one fifth.

The first visit's proportion was two fifths. Comparing it with one fifth appears to show a decline. Yet the second figure concerns the people who answered at the later visit. It combines whatever changed within those people with whatever determined who remained observable. The original forty reports alone do not settle how much of each process occurred.

Suppose a separate team can draw a genuinely representative fresh sample from the same hundred-person population at the second time. To make the arithmetic transparent, imagine first that this new contact reaches everyone and establishes that thirty currently have the experience. Ten are already among the fifty responding panel members. The other twenty must be among the fifty who did not respond to the panel. The missing group's proportion is therefore two fifths, twice the continuing group's proportion.

The population proportion has declined from forty percent to thirty percent, while the responding panel alone suggests a decline to twenty percent. The new observations identify the population's later margin and expose the compositional difference between respondents and nonrespondents. They do so by adding information, rather than by choosing a more elaborate formula for the original fifty records.

In practice, a refreshment sample would usually include only part of the population. Its estimate of the later margin would then have sampling uncertainty. Its recruitment and observation would also need to represent the population relevant to the question. Those design properties determine what information

has been added. The idealized complete contact shows the algebra clearly; a real probability sample supplies an estimate of the same margin with its own uncertainty.

Even perfect knowledge of both population margins leaves an important question open. Which individuals changed? Let a be the number who reported the experience at both visits. The number who moved from presence to absence is 40 − a. The number who moved from absence to presence is 30 − a. Every value of a from zero through thirty is compatible with the two margins alone.

At a = 30, thirty people retain the experience, ten cease to report it, and no one newly reports it. At a = 0, all forty initial cases cease to report it, while thirty other people newly report it. Both populations move from forty percent to thirty percent. One has considerable stability; the other has complete turnover among those reporting the experience. A population trend does not specify the history of its members.

Linked panel records can narrow that range because they show transitions for people observed twice. The fresh sample contributes a different piece: a view of the later population that does not depend on remaining in the original panel. Their combination can support a more informative analysis when the recruitment, linkage, and missingness assumptions are made explicit. Neither source needs to impersonate the other.

Population membership can also change. If new people enter or others move away, the later survey may describe a different target from the original cohort. The design can keep the two questions visible: what happened to the initial members, and what is the population like now? A study of a changing town, workplace, or age group may reasonably want both answers. Naming them separately makes the resulting changes easier to interpret.

This approach gives long observation a useful rhythm. A panel preserves continuity; a fresh encounter tests how that continuing view relates to the present population. Their disagreement can direct attention toward recruitment, retention, or changing circumstances. For a book concerned with whole histories, the deeper lesson is affirmative: different forms of evidence can complement one another when each has a defined job. A later population snapshot can improve knowledge of a trend while the linked histories continue to carry the question of who experienced which change.

18.3.4 — Comparing Models Under Common Rules

Rivals should answer the same declared question under comparable rules. Recovery after a life event may be explained by ordinary adaptation. Treatment and social support can produce additional changes without a terminal law. Each model needs matched inputs and a prespecified prediction so that one is not given an unfair advantage after results arrive.

A terminal-window claim requires frailty, medication, palliative-care, activity, endpoint, and missingness models. Citizen-science data require selection, platform, motivation, device, and reporting-bias models.

Null structures such as random walks, autoregressive persistence, seasonal baselines, regression to the mean, and instrument drift may reproduce visually compelling curves. The comparison should also include allostasis, event-history models, conditioned trajectories, and flexible machine-learning baselines (Wilson and Collins, 2019). A model may predict tomorrow’s reported valence by exploiting autocorrelation, device habits, demographics, treatment schedules, or stable reporting styles. Explanatory claims require tests of the proposed causal

relations, with assumptions stated for the declared estimand (Pearl, 2009; Hernán and Robins, 2020).

A model may rank outcomes correctly while overstating their magnitude or failing in the tails. Report error and uncertainty by horizon, site, subgroup, treatment state, and observation density.

Probability forecasts can be evaluated in a way that rewards both honesty and useful resolution. For a binary outcome, record one when the defined event occurs and zero otherwise. The Brier score is the squared difference between the forecast probability and that outcome. A smaller average score is better. A forecast of 0.8 scores 0.04 when the event occurs and 0.64 when it does not.

Suppose the event's true probability in a constructed setting is q and the forecast is p. The expected score is $q(1 - p)^2 + (1 - q)p^2$. Rearranging gives $q(1 - q) + (p - q)^2$. The first term is fixed by the setting. The second is minimized exactly when p equals q. This is why the score is proper: the expected reward for reporting a probability is optimized by reporting the probability the model actually assigns (Gneiting & Raftery, 2007).

For $q = 0.8$, forecasting 0.8 has expected score 0.16. Forecasting 0.5 has expected score 0.25. Forecasting certainty has expected score 0.20. The certain prediction is correct most of the time, yet the calibrated probability gives the better average score. The evaluation values a correct account of the remaining uncertainty, as well as a correct direction.

Calibration alone can still leave useful information unexplored. Imagine two equally common task conditions. The event probability is 0.2 in one and 0.8 in the other. A forecast that always says 0.5 is calibrated when assessed over the combined collection: half the events occur. Its expected Brier score is 0.25.

A second predictor uses the condition and reports 0.2 or 0.8 accordingly. Each reported probability is calibrated in its own cases. Its expected score is 0.16 in both conditions. It has gained resolution: it distinguishes circumstances with different event probabilities while preserving calibration. That improvement is invisible to a check asking only whether the overall predicted event rate equals the overall observed rate.

The condition-specific forecast also supports more informative planning. A user can distinguish a lower-probability occasion from a higher-probability one rather than receiving the same message every time. Whether that distinction should change an action depends on the action's costs and consequences. The forecast evaluation can establish the probabilistic information before a separate decision rule decides how to use it.

Residual structure, unstable parameters, site-specific failure, and systematic intervention errors may show all candidates inadequate.

18.3.5 — More People or More Readings?

A repeated-observation study has two ways to become larger. It can follow more people, or it can collect more readings from the people already participating. The total number of observations alone does not tell us which expansion will teach more about a population mean. The answer depends on what varies between people, what varies within a person's record, and the question the study is trying to estimate.

Consider a constructed measurement $Y_{ij} = \mu + b_i + e_{ij}$. Here μ is the population mean, b_i is a stable person-specific departure, and e_{ij} is a changing reading-specific departure. Give b_i variance τ^2 and e_{ij} variance σ^2. Let person effects be independent across people and reading errors be independent

across all readings and of the person effects. These assumptions define the example's information structure.

Suppose m people each provide k readings. The average of all the observations has variance $\tau^2/m + \sigma^2/(mk)$. The first term comes from sampling people with different stable departures. The second comes from averaging the changing departures across all readings. Adding readings for the same people reduces the second term. Adding new people reduces both terms because it samples new person effects as well as new reading errors.

For a numerical example, let both variance components equal one half. Twelve observations collected as twelve people with one reading each give variance one twelfth. The same twelve observations collected as three people with four readings each give variance one sixth plus one twenty-fourth, or five twenty-fourths. Both datasets contain twelve values, but the second has fewer independent opportunities to average the stable person-specific component.

A study also has costs that need not be equal for those two expansions. Recruiting and setting up a new participant can require resources before any reading is collected. Let that fixed cost be nine units and let each reading cost one unit. A budget of 120 units can support ten participants with three readings each: ten times the cost of twelve units per participant. Under the same variance components, the variance of the grand mean is one twentieth plus one sixtieth, or one fifteenth.

Why three readings? Treat k temporarily as a positive continuous choice and use $m = B/(c_0 + c_1k)$, where B is the budget, c_0 is the fixed cost per person, and c_1 is the cost per reading. Substituting into the variance gives $(\tau^2 + \sigma^2/k)(c_0 + c_1k)/B$. The terms that change with k are τ^2c_1k and σ^2c_0/k. Their sum is minimized at $k = \sqrt{(\sigma^2c_0/(\tau^2c_1))}$. With equal variance

components, fixed cost nine, and reading cost one, the result is three.

The balance changes when the variance components change. Larger stable differences between people make additional people more valuable for this target. Larger reading-specific variation makes repeated readings more valuable because each person's average is otherwise imprecise. A higher setup cost also favors more readings per enrolled participant. The formula explains the direction of these changes and gives a starting point for comparing feasible integer designs.

The target remains essential. Estimating the population mean is different from estimating each participant's trajectory. A design with one reading per person may estimate a population level efficiently while revealing almost nothing about individual change. Studying a delayed response, a personal rhythm, or the relation between successive states requires repeated observations at informative times. Those aims have their own precision calculations and can justify a different allocation.

Dependence among reading errors also changes the calculation. If nearby readings retain a common disturbance, they supply less new information than the independent errors in this example. Spacing observations may then matter as much as their count. A pilot model can estimate whether additional readings mostly reproduce a shared component or reveal new variation. The result can guide both the number and timing of measurements.

This makes study size a design question rather than a competition for the largest row count. The investigator can state which variation must be averaged, what each additional observation contributes, and what resources it requires. For a longitudinal program, that account helps connect participation to learning: more people broaden the sampled lives, while more

readings illuminate the movement within them. A clear target gives each form of expansion its proper purpose.

18.4 — Participation Without Surveillance

Digital access changes who can enter a longitudinal study. Connection quality and device ownership affect the completeness of observations. Language, disability access, working hours, and privacy concerns influence who stays. A representative claim must account for these differences rather than treat missing participants as if their experiences were observed. Before releasing data, test for hostile linkage through social media, public records, clinic visits, location traces, platform logs, and rare-event timing (Nebeker et al., 2017; Vayena et al., 2015).

18.4.1 — Safety Feedback

Longitudinal affect data can also look clinically meaningful. Participants may be alarmed by trends, families may ask for interpretations, and platforms may be tempted to offer wellness advice. Any safety signal requires a preapproved clinical or crisis route that is clear about the limits of research data. State whether incoming reports are monitored, by whom, and within what response period. A portal that is not continuously monitored must say so where reports are submitted and provide a separate route for urgent help.

18.4.2 — Citizen-Science Design

Citizen science lets participants help choose research questions, design feasible measurements, and review how the program is governed.

Public portals should report recruitment source and local purpose alongside enrollment and retention. Followers, patients, students, employees, and volunteers enter through different incentives and barriers. A large selected sample can improve

precision without resolving its selection process, measurement limitations, or unidentified endpoint (Onnela & Rauch, 2016).

Citizen-science feedback should remain limited to aggregate summaries, data-quality information, and study updates (CIOMS, 2016; Vayena et al., 2015).

Citizen science can show that a prompt is burdensome, a measure unstable, a sample unrepresentative, or a laboratory effect absent in ordinary settings (Wiggins and Wilbanks, 2019).

Part VII — The Strongest Rivals

A numerical example separates return to a state from closure of an accumulated total. Suppose, in an invented model, a valence rate of +1 lasts for one time unit and then returns to zero. The accumulated total remains +1. A later interval contributing −0.5 reduces that total to +0.5: an opposite response has occurred, but cancellation is incomplete. Recovery of the current state and closure of accumulated area therefore ask different questions.

Prediction can succeed in a harmful world. A child may learn when punishment is coming, a worker may anticipate retaliation, and a patient may expect disbelief because experience has trained the forecast. The model becomes more accurate while the conditions remain unjust (Barrett and Simmons, 2015; Seth, 2013).

A constraint can be satisfied while a person remains burdened. A school can meet attendance rules while a child is terrified, a hospital can satisfy discharge or staffing requirements while patients and families feel abandoned, and an algorithm can meet a formal fairness constraint while the people affected cannot understand or challenge its decisions.

The public phrase hedonic treadmill often compresses several mechanisms into one image. Time heals, people get used to things, and life returns to normal may refer to material repair, social support, altered expectation, avoidance, numbness, habit, or simple passage of time. The rival becomes scientific only when those routes are separated well enough to predict different courses. Gains can also change the reference point from which later experience is judged. The salary that once removed fear may become the minimum required to sustain a new household. The recognition that once felt astonishing may become the standard against which the next year is measured. Adaptation can therefore reduce felt elevation without restoring the person to the same practical world.

Computational accounts already treat feeling as dependent on history. In a gambling task, Keren and colleagues (2021) compared models of how earlier outcomes shape present mood. A model in which expectations drew on the mean of all previous task outcomes predicted subsequent ratings better than several tested alternatives emphasizing recent events, across different reward sequences and participant samples. Its public data and code offer a concrete starting point: compare a prespecified cumulative-experience variable against these history-sensitive models, with the same information and tuning opportunities, on unused observations. The question is whether accumulated signed experience adds prediction beyond learned expectations.

One practical comparison would ask whether each model has to look ahead to the same horizon. A standard reinforcement learner may discount future reward. A homeostatic system may defend near-term viability. A safe-control system may avoid violation within a defined horizon. The Law’s terminal claim, by contrast, concerns the completed stream (García and Fernández, 2015; Achiam et al., 2017). Section 21.4.4 sets the comparison rules.

Chapter 19 — Adaptation and Rebound

A promotion becomes Tuesday. A once-dazzling purchase joins the furniture of daily life. The first weeks after a loss feel unlike the months that follow. These ordinary changes give hedonic adaptation its power as a rival (Helson, 1964; Frederick and Loewenstein, 1999): experience is not a passive mirror of circumstance, and the meaning of a condition can change while the condition remains. A careful analysis also asks what the gain changed outside the person's immediate report. Did it reduce danger, expand choice, create maintenance obligations, increase public scrutiny, or shift work onto family, staff, debtors, or future maintainers? Those questions belong to the material and social consequences of the gain.

A strong design separates three questions. Did the event improve conditions or reduce a burden? Did its affective impact change as the event became familiar? Does any path-sensitive structure remain after adaptation, expectation, and comparison are modeled under common rules? The proposed Law enters only the third question, and it enters as a hypothesis rather than as the owner of the first two.

This separation has practical consequences. Public policy should not dismiss housing, health care, safety, access, or freedom because beneficiaries may adapt to them. The point of many goods is to become reliable background conditions. Adaptation can explain why they stop feeling exceptional without weakening the case for providing them.

Feeling can influence the situations people encounter next. Taquet et al. (2016) analyzed repeated mood and activity reports from 28,212 smartphone users. Lower earlier mood was associated with later mood-improving activities; higher earlier mood was associated with useful activities followed by a decline. Their joint observational model included prior activity, later mood, and temporal factors. The authors called this pattern

hedonic flexibility. This gives action-mediated regulation a concrete form: current feeling may help select an activity, while that activity belongs to the context of later feeling. Research can compare those two relations instead of treating recovery as an automatic force inside the person. The study concerns average associations, not a universal response or exact equality of accumulated experience.

For PTVC route research, the interesting next step is to test whether a separately specified retained state adds information beyond this ordinary mood–activity account, using opportunities and constraints the participant actually faces.

The governing question is therefore concrete: when experience appears to return, what has returned? A current rating, a behavioral routine, a reference point, a bodily state, an average, or the complete latent trajectory? The answer determines whether the observation belongs to adaptation, measurement, regulation, or the proposed claim of terminal closure.

Recovery can be studied as a sequence of accomplishments. A person may resume an activity before enjoying it, enjoy it before doing it easily, or regain ease while choosing a different goal. Those orders suggest different relations among expectation, skill, opportunity, and feeling. Recording the order gives the investigation more to explain than a final rating alone. The chapter follows that richer question through adaptation, overlapping responses, and context-dependent learning, asking which parts of an apparent return are carried by each process.

19.1 — What Adaptation Explains Well

A gain first appears as a change in what a person can do or feel. Over time that gain may become the new background for judgment. Reports may then drift even when the material improvement remains. Adaptation explains this change in reference without claiming that the whole life's valence must close at zero.

19.1.1 — Adaptation After Gains

Hedonic adaptation is not one mechanism wearing one name. Novelty can fade, standards can rise, attention can move, habits can form, social comparisons can change, and a person can alter the environment that made the gain possible. Philosophical and psychological analyses therefore distinguish several routes by which a new good becomes ordinary rather than assuming one universal return process (Klausen et al., 2022).

The classic contrast between extraordinary gain and catastrophic loss made adaptation memorable, but later research replaced the picture of universal reset with a more differentiated one. Affective and evaluative measures can move on different schedules, event classes differ, and individual trajectories may show return, partial return, durable change, or continuing divergence (Brickman et al., 1978; Diener et al., 2006; Luhmann et al., 2012; Sheldon and Lyubomirsky, 2012).

The early lottery study found that its sampled winners did not report significantly greater current happiness than controls; it did not establish that any initial gain had permanently disappeared (Brickman et al., 1978). A stronger natural experiment compared Swedish lottery players within defined groups where prize amounts were randomly assigned and found higher life satisfaction five to twenty-two years after the lottery event. On the study’s standardized scales, estimated effects on happiness

and mental health were smaller and statistically nonsignificant, which does not establish zero effects (Lindqvist et al., 2020). This contrast shows why life evaluation, reported happiness and mental health should be measured separately. Because the survey missed the first five years and did not measure a continuous signed experience history, it cannot determine a compensating trajectory. A prospective study could combine similarly exogenous changes with earlier, repeated affect measurements.

Social comparison adds another route. A gain may move someone into a new peer group where the same resources feel ordinary, inadequate, or exposed. That change can alter satisfaction without reversing the gain itself.

19.1.2 — Habituation

Habituation is a reduced response after repeated stimulation, distinguished from reduced sensory sensitivity and fatigue of the response system (Rankin et al., 2009). It helps explain why a sound fades into the background or a repeated stimulus commands less attention.

A small sequence can reveal more than a downward curve. Imagine a study of attention to a gentle signal, A. Its first presentation draws an orienting response. Repeated presentations draw progressively smaller responses. Now present a perceptibly different signal, B, using the same response measure. If the response to B remains strong, the organism still has the capacity to produce it. The reduction has a relation to what was repeated, rather than simply to the passage of the session.

Return to A after a pause. A renewed response gives the record a further feature: some of the decrement has recovered with time. Stimulus specificity and spontaneous recovery are

among the characteristics used to investigate habituation (Rankin et al., 2009). They make the shape of learning more informative than the statement that a response became smaller.

The imaginative work is in choosing comparisons that isolate the proposed process. If A and B differ in intensity as well as identity, a stronger response to B could follow from intensity. If B requires a different movement, the comparison says less about fatigue in the original movement. A useful version of the task lets the stimuli differ in a relevant, detectable feature while keeping the response opportunity comparable. A further comparison can vary the interval between presentations to reveal how the decrement and its recovery unfold.

An everyday analogue is the sound that becomes part of a room. A regularly cycling fan may cease to attract attention while an unfamiliar click immediately does. The fan has not stopped operating, and the listener has not lost all responsiveness to sound. The relation between familiar input and attention has changed. The laboratory sequence gives that ordinary observation a structure that can be studied.

For the present inquiry, the response channel remains explicit throughout. Attention to a signal, reported pleasantness, and physiological response can have different courses. One can ask whether the reduction in one channel accompanies, precedes, or follows a change in another. That ordering supplies a mechanism question: does learning that a signal is familiar alter attention first, and does its affective character change on the same schedule? A richer account begins with the particular pattern of changes the record actually contains.

Set points are the treadmill’s stabilizing idea, but contemporary adaptation theory does not require one immutable emotional thermostat. The stronger rival permits person-specific ranges, event-sensitive shifts, and different courses for affect,

life satisfaction, desire, and expectation. A reference range is therefore a statistical and psychological tendency shaped by history and context, not a fixed essence or a neutral origin granted in advance (Diener et al., 2006; Lucas, 2007; Luhmann et al., 2012).

Because this rival allows baselines to shift, fix its baseline rules prospectively: a set point that can move anywhere after the fact explains too much. Apply the same constraint to the Law-linked model.

19.1.3 — Why This Rival Is Serious

The average is only the first layer. Longitudinal records may contain stable trajectories, brief disruption, gradual recovery, durable decline, delayed change, or movement to a new level. The apparent number and prevalence of those courses can depend on the outcome measure, time base, missingness, and modeling assumptions, so trajectory labels should be treated as estimates rather than natural kinds (Galatzer-Levy et al., 2018; Infurna and Luthar, 2016).

Heterogeneity can change the apparent form of recovery. Imagine a family of mathematical processes, all beginning one unit above their own reference level. Each process relaxes exponentially: its remaining deviation at time t is $\exp(-kt)$, where k is its fixed recovery rate. A large k means a faster return. Give the family rates spread uniformly between zero and one, with no positive probability assigned to the exact endpoint $k = 0$.

Every individual process then approaches its reference level exponentially. The family average has a different shape. Averaging $\exp(-kt)$ over rates from zero to one gives $(1 - \exp(-t))/t$ for positive t. At time zero its continuous limiting

value is one. At long times the numerator is close to one, so the average falls approximately as one divided by t.

A slow aggregate tail has appeared without any individual using that reciprocal-time law. Early in the record, many rates contribute visibly to the average. Later, the fast processes are already close to their reference levels, while the slow ones still contribute. The composition of the remaining deviation changes even though the population's membership stays fixed.

The numerical contrast is substantial. At time two, the mean deviation is about 0.432. At time ten it is about 0.100. A single process with the average rate, k = 0.5, has deviations about 0.368 and 0.0067 at those same times. Replacing all individual rates by their average before computing the trajectory does not reproduce the average of the trajectories. Exponentiation is nonlinear in the rate.

The model suggests a way to investigate the difference. Repeated observations of the same units can estimate whether rates remain approximately stable within units and vary between them. A common long-tail model makes a different claim: the slower course belongs within each unit's dynamics. The two accounts may fit a pooled curve similarly while predicting different individual records.

There is an additional whole-history consequence. The accumulated positive deviation of an individual exponential path is 1/k. Every process with k greater than zero has a finite area. But under the uniform rate distribution, the population average of 1/k is infinite. Rates arbitrarily close to zero are sufficiently consequential to make that average diverge. A collection of finite individual quantities can therefore have an infinite mean when its distribution has a heavy enough upper tail.

If rates instead have a positive lower bound, both conclusions change in a useful way. The slowest allowed rate supplies an

eventual exponential bound on the family average, and individual accumulated areas are bounded above by the reciprocal of that minimum rate. Thus the long tail is tied to a specific assumption about the distribution near zero, an assumption that can be varied and examined.

The constructed family gives trajectory research a precise interpretive question. Is a slow pooled return a property of a common mechanism, a mixture of faster and slower mechanisms, or a combination of both? Following individuals through time supplies information that an increasingly smooth population curve cannot supply by itself.

Adaptation research already has repeated measures, panel studies, event designs, behavioral observations, and clinical follow-up that can test modest predictions without first accepting a cumulative ledger or a terminal endpoint. Methodological independence is one reason the rival deserves priority.

19.2 — Opponent Processes, Tolerance, and Rebound

Adaptation may change a baseline; opponent process predicts a temporally linked counterstate following an initial response. Repetition can alter that counterstate's onset, strength, or duration, producing tolerance, withdrawal, or rebound. The proposal is not one explanation for every experience. Drug exposure, fear, pain, exercise, and reward recruit different systems and time scales. Some counterstates strengthen, while others sensitize or fail to recover (Solomon and Corbit, 1974; Koob and Le Moal, 2008).

19.2.1 — Pleasure and Counterpressure

Relief can contain more than the disappearance of discomfort. Franklin and colleagues (2013) measured eyeblink and postauricular startle responses after brief painful stimulation. The two patterns supported reduced negative affect and increased positive affect, with different sensitivities to stimulus intensity. The study deliberately omitted concurrent affect reports, so the next useful step is convergence: determine whether repeated experience reports and the physiological channels track the same positive after-response. Varying episode duration as well as intensity would then test whether the later contribution tracks the preceding integrated contribution. This turns a familiar relief story into a specific measurement and prediction problem.

Opponent processes are one explanation for relief after threat. Endogenous pain modulation, reduced arousal, contrast, social reassurance, medication, and restorative sleep offer other or interacting explanations. Each can change current felt state; the study must identify which processes or combinations contributed to the observed relief.

Measure the initial response and afterstate separately, including their durations.

19.2.2 — Stress and Recovery

Heart rate may settle in minutes, disrupted sleep persist for days, and changed felt safety last years. Recovery can mean physiological downregulation, restored function, behavioral change, narrative reorganization, or relief; these outcomes are not interchangeable.

During acute stress, attention narrows, energy shifts, autonomic and endocrine systems respond, and action becomes urgent. Those changes may help a person survive immediate danger while also carrying fatigue, pain, fear, or later cost (McEwen, 1998; Sterling, 2012).

Workplaces can alternate overload with praise, platforms social reward with uncertainty, and coercive relationships fear with relief. Positive intervals may strengthen attachment without making the system fair.

19.2.3 — Counterprocess Is Not Terminal Law

Opponent-process models estimate the latency, magnitude, duration, and decay of a primary response and its after-reaction (Solomon and Corbit, 1974; Solomon, 1980).

A concrete adaptation model makes signed-area cancellation mechanistically intelligible. For a stable linear time-invariant system initially at baseline, let $y = h * u$, where * denotes convolution and the causal input u and response kernel h are absolutely integrable. If $\int h = 0$, then $\int y = (\int h)(\int u) = 0$, with all integrals from zero to infinity. The complete response cancels for every such input. Bacterial chemotaxis supplies a limited biological analogue: an aspartate response curve averaged across cells has approximately equal positive and negative areas, while an idealized model has exactly zero integral (Celani and

Vergassola, 2010). In their bacterial-trajectory study, Masson and colleagues (2012) found a two-phase response to serine whose positive and negative areas differed.

In an invented dimensionless two-state realization, $\dot{x}_1 = -x_1 + u$, $\dot{x}_2 = -2x_2 + u$, and $y = x_1 - 2x_2$. Then $Q = x_1 - x_2$ obeys $\dot{Q} = -y$, so every finite-interval area is $Q(a) - Q(b)$. Starting at baseline and applying a unit input for one time unit leaves $Q = \frac{1}{2}(1 - e^{-1})^2 > 0$ when that input ends. The later tail cancels the accumulated response only in the limit as Q tends back to its initial value. This connects zero-area adaptation to the state-change candidate through a finite-boundary relation. The modeled output is a deviation from its own baseline; identifying that baseline with phenomenal neutrality remains an independent calibration task.

Finite-window area can distinguish specified nonlinear rivals. Under their stated initializations, Gupta and Sontag (2025) hold each model's parameters and initial state fixed while varying a constant positive input: two specified incoherent feedforward models give monotonic cumulative-output curves, whereas an integral-feedback model can give nonmonotonic curves. A fixed linear model initially at baseline instead has baseline-relative area proportional to input amplitude. This comparison uses cumulative output at one predeclared horizon, not a zero-total claim. In their normalized feedback model, integrated deviation from the set point equals a logarithmic change in a controller state. This connects the response-area comparison to a candidate boundary difference; its relation to phenomenal contributions remains a separate measurement question.

19.2.4 — When Balanced Responses Overlap

The role of linearity becomes clear in a three-step illustration. Suppose an isolated input produces a response of plus two in its first step, minus one in its second, and minus one in its third.

The three contributions total zero. The values describe a mathematical response pattern, with equally long steps and a common unit throughout.

Deliver two such inputs far enough apart that their responses do not overlap. The resulting sequence contains two complete copies of the same pattern. Each totals zero, so their combined total is zero. Deliver them simultaneously under a linear rule instead. The corresponding values add point by point, giving plus four, minus two, and minus two. This sequence also totals zero. Changing the timing has changed the peak but preserved the total because addition and accumulation commute.

Now introduce a specific nonlinearity. Let the response be clipped whenever it would exceed plus two or fall below minus two. An isolated input still produces plus two, minus one, and minus one: every value lies within the allowed range, so nothing changes. Two separated inputs still total zero. Two simultaneous inputs, however, would have produced plus four in their first step. Clipping changes that step to plus two while leaving the later values at minus two and minus two. The accumulated response is now minus two.

The altered total has a precise cause. The simultaneous positive peak reached the cap, while the two later negative contributions did not exceed it. Although the clipping bounds are symmetric around zero, the unmodified response's positive and negative contributions are distributed differently through time. A symmetric amplitude bound therefore need not preserve the area of an asymmetric time course.

This makes timing an informative experimental dimension. Widely spaced inputs sample nearly isolated responses. Closer inputs expose overlap. Varying input size reveals when the response leaves the range in which doubling an input approximately doubles its effect. A candidate model can predict

the spacing and amplitude at which its accumulated response changes, and the observations can then be compared with those predictions.

The example connects the single-episode and repeated-episode questions. A mechanism may explain a balanced response to each small, isolated event while predicting a different pattern under frequent or clustered events. To understand its behavior across a longer history, one needs the rule for composing responses. In a linear account, that rule is addition. In a nonlinear account, saturation, interaction, or state dependence supplies further structure. That composition rule is where a succession of understandable episodes becomes an explanation of a whole trajectory.

19.3 — What Adaptation Can Explain

An apparently neutral comparison interval may have its own affective trajectory. In imposed rest and simple tasks, Jangraw et al. (2023) observed average mood decline across cohorts. Such drift complicates claims that a return toward baseline reflects compensation. A good rival test must measure what happens during the comparison period instead of assigning it zero by default.

19.3.1 — Short-Term Recovery

Adding elapsed time also improved prediction of held-out mood ratings in their mobile-task analysis. A compensation study can use this result constructively: randomly assign participants to safely varied early episodes under a common follow-up schedule, keep planned time of day, task demands, and reporting frequency comparable, and record subsequent changes in sleep, medication, and support. Estimate the total effect on later trajectories with individual variation retained; analyze pathways through those later changes separately. For predictive comparisons, a prespecified time-drift model belongs alongside current affect, rewards, and expectations. The remaining question is whether earlier accumulated experience predicts a subsequent contribution beyond those ordinary dynamics.

Elapsed time and repeated exposure can be difficult to distinguish when they advance together. Consider a harmless signal presented at times zero, one, and two in an invented task. One model says the measured response halves with each successive presentation. A second says it halves for each elapsed time unit after the first presentation. Both predict responses of one, one half, and one quarter on this schedule.

The agreement is produced by the schedule: every new presentation arrives exactly one time unit later. The record gives

the two candidate clocks no opportunity to separate. Adding many more equally spaced presentations may estimate the shared declining curve very accurately while leaving the mechanism unresolved.

Now place the third presentation at time four instead of time two. The repetition model still predicts one quarter because this is still the third presentation. The elapsed-time model predicts one sixteenth because four time units have passed since the first. The new schedule changes a relation in the design, allowing the models to make different forecasts.

A second comparison can hold the final observation time fixed while varying how many earlier presentations occurred. If the measurement at time four follows only one earlier presentation, the repetition model predicts one half. If it follows three earlier presentations, it predicts one eighth. The simple elapsed-time model predicts one sixteenth in both cases. The design can therefore test both directions of the distinction.

More elaborate accounts may combine the two clocks. A response could change with exposure while also recovering or drifting during the intervals between exposures. Once the simple models have been separated, the combined model has definite work to do: predict how changes in spacing and count jointly alter the response. Its additional terms can be evaluated on schedules not used to choose them.

The example explains why an ordinary declining curve can be scientifically rich. Its shape may reflect learning from encounters, change during the gaps, or both. Irregular timing is useful when chosen to distinguish those possibilities. What first looked like one return toward baseline becomes a question about which parts of the history the system retains and how that retained history affects its next response.

19.3.2 — Behavioral Adaptation and Clinical Exceptions

Behavioral adaptation shows how people alter routines, attention, expectations, goals, relationships, locations, and requests for help after circumstances change. Those actions can improve, preserve, or narrow the conditions of later feeling.

Habit formation is one route of behavioral adaptation: repeated action can become easier, more automatic, or more reliably cued. Habits can stabilize sleep, medication use, exercise, work routines, avoidance, and help-seeking, while the same habit machinery can also preserve craving or withdrawal from valued activity. Context and repetition therefore explain substantial persistence (Lally et al., 2010; Wood and Rünger, 2016).

Habit expression is strongly context-dependent. A response practiced in one room, schedule, social setting, or emotional state may weaken when the context changes, while an old cue can restore it after a long absence (Wood and Rünger, 2016). People living with grief, chronic pain, disability, trauma, addiction, or severe mood disorders may follow persistent, fluctuating, or otherwise distinct trajectories. A rival becomes more credible when it explains that heterogeneity instead of treating nonrecovery as noise (Galatzer-Levy et al., 2018; Luhmann et al., 2012).

Consider a person who has learned to pause and plan before beginning a familiar work task. In the usual workspace, a notebook beside the keyboard prompts the pause. In a temporary room, the task is performed quickly and the pause disappears. Back in the usual room, the routine returns. The sequence contains a useful clue: performance changed with the setting and recovered when the setting returned.

Several explanations are now available. The routine may depend on the notebook as a retrieval cue. The temporary room

may impose interruptions that make a pause difficult. The task itself may differ in a subtle but relevant way. Each explanation suggests a distinct next comparison. Bringing the notebook to the temporary room tests one cue. Providing an uninterrupted interval tests opportunity. Keeping the task identical tests the contribution of task variation.

If the notebook restores the pause in the temporary room, the result supports a relation between cue and expression. If uninterrupted time restores it while the notebook does not, opportunity becomes the stronger explanation. If neither helps but a familiar version of the task does, the learning may be more specific than first assumed. These are prospective differences among accounts of the same initial sequence.

The person can use this understanding to make learning more portable. Practicing with several cues, in several rooms, or at several points in the day can test which support the routine requires. A change in setting then becomes a way to discover its architecture. The question is how a useful capacity becomes available when it is wanted, and which features of the surrounding world help call it into action.

Concrete mechanisms such as sensitization, withdrawal, intrusive memory, inflammation, sleep disruption, medication, isolation, cognitive load, and treatment response deserve explanatory priority where they fit.

19.3.3 — Evaluating Adaptation as a Rival

Adaptation models can predict observed returns toward baseline and changes in reports after repeated exposure, yet they cannot establish terminal neutrality for an entire completed stream. If an analysis codes each local return as the exact negative of an excursion, zero becomes a mathematical artifact rather than a finding. A valid test must preserve order, exposure, missingness,

residual effects, and uncertainty about the endpoint. The decisive comparison is between distributional models evaluated on variance, tails, subgroups, timing, clinical exceptions, and endpoint-adjacent behavior, not an average against an anecdote.

An exposure-based explanation gains strength when it predicts changes across repeated observations. It matters whether past events and current context account for the trajectory. If they outperform a model using cumulative-record position, the added terminal hypothesis may do no predictive work. That result should be reported as a genuine rival success. If the fourth exposure differs from the first because the organism has learned, the Law does not win by calling the decline counterweighting. The fair question is whether exposure history, expectation, and context explain the pattern before any terminal architecture is invoked.

Chapter 20 — Prediction and Surprise

Prediction begins before conscious reflection. A footstep behind us, a change in breathing, a familiar facial expression, or a pain expected to worsen can alter attention and feeling before the situation is fully known. The nervous system does not wait passively for the world; it uses prior structure, bodily signals, and incoming evidence to organize what happens next.

Predictive processing is not one single finished theory. It is a family of accounts in which perception and action depend on generative models, prediction errors, uncertainty, and the weighting of signals by their expected reliability. Active-inference formulations add policies and action, asking how organisms sample or change the world to remain within preferred or viable states (Friston, 2010; Clark, 2013; Hohwy, 2013).

Suppose two weather forecasts disagree. One has repeatedly been accurate in your area; the other has been unreliable. You would have a reason to give their predictions different weights, even before either forecast was tested today. Precision plays a related role in predictive models: it expresses estimated reliability, so some signals or expectations count more strongly than others. Applied to bodily feeling, the question becomes how much weight the system gives to an ache, a heartbeat, or an expectation about what will happen next (Hohwy, 2013; Friston et al., 2017).

Affective consequences depend on preferences and stakes as well as error. Missing a predicted bus and learning that a feared diagnosis was wrong are both mismatches, yet they differ in scale, bodily preparation, available alternatives, and future meaning. A strong model must represent those differences rather than hide them in one error term.

A concrete study shows how researchers can separate neighboring quantities rather than give them one name. Heffner

and colleagues (2025) recorded EEG from 41 participants in a repeated Ultimatum Game. Participants reported expected monetary offers, expected feelings, and feelings after each offer. Both monetary and valence prediction errors contributed to choice prediction in the reported model. Neural analyses suggested partly distinguishable patterns, predominantly for error magnitudes rather than a signed scale of feeling. The useful lesson is the design: record expectation and reported experience separately before interpreting their difference. A smaller affective prediction error can arise because expectations change, reported experience changes, or both. For the Law's research program, that separation makes a proposed signal more informative: investigators can ask which component it follows before connecting it to accumulated experience.

The public challenge is to explain this power without turning the brain into a maze solved by one equation. Predictive processing has empirical support in particular domains and active debate about scope, implementation, and interpretation. It should neither be marketed as a universal master theory nor weakened into a slogan.

20.1 — Prediction, Surprise, and Uncertainty

Predictive systems can organize information at different temporal levels. Fast processes anticipate sensory and bodily changes. Slower ones may track relationships, safety, and expected futures. A hierarchical model must explain how those levels update one another before its affective predictions are treated as a candidate mechanism.

20.1.1 — Prediction as Organizing Principle

Higher and lower levels constrain one another rather than forming a one-way ladder (Friston, 2010; Clark, 2013).

Prediction error is a discrepancy between an expected signal and an observed one under a declared model. Statistical surprise instead concerns how improbable the observation is under that model (Friston, 2010). A small mismatch can be surprising under a precise prediction, while a larger mismatch may be unsurprising under a broad one. Neither quantity by itself tells us whether the discrepancy was conscious, pleasant or unpleasant. Its felt effect also depends on what the outcome means for the person and what action remains possible.

Consider two imagined forecasting systems watching a dial. The first predicts that its next reading will be either zero or ten, with equal probability. The second assigns equal probability to every whole number from zero through ten. Both report five when asked for their mean prediction. Their expectations look identical on a display that shows only one number.

Now the dial reads five. Its distance from either predicted mean is zero. Yet the first system gave that observation no probability at all, while the second gave it a probability of one eleventh. The observation fits one forecast and contradicts the stated possibilities of the other. A signed error calculated from

the mean cannot express this difference because the two means have discarded the information that matters.

The contrast remains if the first system allows a small chance of intermediate readings. Suppose it gives five a probability of one hundredth while the second still gives five a probability of one eleventh. The means can remain five by distributing the remaining probability symmetrically. The observation is then possible under both systems, but much less expected under the first. Statistical surprise, understood as the negative logarithm of the probability assigned to the observation, changes even though the mean error stays zero.

Different questions therefore call for different records. A signed error asks which direction the observation lies from a chosen prediction. Its magnitude describes distance in the units of the dial. A predictive distribution describes the range and relative plausibility of outcomes. Surprise asks how much probability that distribution assigned to what occurred. Precision concerns how concentrated a representation is. These quantities can interact in a model, but the dial example lets the reader vary one relation at a time.

For an account of affect, the next step is to specify which of these changes drives the proposed response. A system might respond to a large directional error, to an improbable observation, to an increase in uncertainty, or to an opportunity to learn. Those mechanisms make different predictions for the same dial reading. Recording the full forecast makes that difference testable: the mean alone has already erased it.

Active inference adds action to the rival. The system can reduce costly mismatch by revising beliefs, seeking information, or changing the world so future sensations become more workable. This makes the framework a perception-action

account rather than a passive theory of internal correction (Friston et al., 2017; Pezzulo et al., 2015).

Information-seeking and practical action need not point in the same direction. A person may accept short-term uncertainty to learn something important, or avoid new information to preserve immediate stability. Strong predictive models can therefore represent exploration as well as comfort-seeking.

20.1.2 — Learning How Quickly to Learn

How quickly should an expectation change? A surprising observation can deserve a large revision when the world has changed, or a small revision when the observation is noisy. The difference concerns two sources of uncertainty. One belongs to what is being tracked. The other belongs to the channel through which it is observed. Keeping them separate gives precision a role in a sequence of updates, rather than treating it as a fixed confidence label.

Imagine a numerical dial whose true position moves between observations. An observer maintains an estimate of its position together with a variance describing uncertainty about that estimate. For this illustration, every movement is an independent, zero-mean Gaussian increment with known variance q. Every observation adds independent, zero-mean Gaussian measurement error with known variance r. Movement errors, measurement errors and the initial uncertainty are independent. These rules define a complete small model, so the update can be calculated exactly.

Suppose the uncertainty immediately after the previous observation is P. Before the next reading arrives, the dial moves. Its predicted mean remains the previous mean because movement has mean zero, while the predicted variance becomes P + q. Call that quantity R. The new observation y is combined

with the predicted mean m using the weight K = R/(R + r). The updated mean is m + K(y − m), and its remaining variance is (1 − K)R. These equations describe conditional means and variances under the stipulated Gaussian model.

The update has an intuitive division of labor. Large R means the observer has little confidence about the current position before looking. Large r means the reading itself is noisy. When R is large relative to r, the observation receives a large weight. When r is large relative to R, the observer keeps more of the previous prediction. A discrepancy of four units can therefore produce different revisions without either observer ignoring the evidence. They are combining the same discrepancy with different accounts of its reliability.

Take a case with P = 1, q = 1 and r = 2. The predicted variance is two and the observation weight is one half. If the predicted position is zero and the reading is four, the updated estimate is two. Its remaining variance is one. Before the following observation, another movement variance of one brings the predicted variance back to two. The same one-half weight is appropriate again. The process has reached a steady cycle of uncertainty: looking reduces uncertainty, and subsequent movement replenishes it.

That cycle distinguishes tracking from learning a fixed quantity. Set q to zero while keeping r at two and beginning with P equal to one. The next observation receives weight one third and leaves variance two thirds. The following observation receives weight one quarter and leaves variance one half. As further independent readings accumulate, the weights continue to decrease. Each reading contributes information about the same unchanged position. In the moving-dial model, older readings concern earlier positions, so fresh movement continually limits their relevance to the present.

The contrast can guide a comparison of learning rules. During a stable period, a decreasing update rate can make sense: the observer has accumulated increasingly precise information about a persistent setting. During a variable period, retaining a substantial update rate can make sense: the setting continues to change. An account that uses one fixed rate for both periods may miss this difference. A more informative account predicts how a change in estimated volatility alters the next revision.

There are two distinct ways to change the weight in the example. If movement variance rises from one to seven while P remains one and measurement variance remains two, R becomes eight and K becomes four fifths. A reading four units above the prediction shifts the estimate upward by 3.2 units. If instead movement remains at one but measurement variance rises from two to eight, K becomes one fifth. The same four-unit discrepancy shifts the estimate by 0.8. Increased uncertainty has accelerated learning in the first case and slowed it in the second. The location of the uncertainty determines the response.

These alternatives suggest an experiment with a clear purpose. One can vary how much a harmless target actually changes while keeping the display equally readable. One can separately vary display noise while keeping target movement unchanged. The two manipulations have opposite predictions for the update weight in this model. A record of successive estimates can then ask whether the participant adjusts to volatility, adjusts to channel reliability, or uses a different rule. The experiment investigates the organization of learning through its response to two independently varied causes.

For an account of affect, the next question is how such updating relates to reported experience. A change in expectation can be measured separately from a change in the outcome, and both separately from the feeling reported afterward. The dial

model supplies a precise candidate for the first relation. It helps an investigator ask whether a large emotional response follows an unusually informative event, an actual change in circumstances, or another feature that the model has not represented. Each possibility points to a different observation rather than being absorbed into the word surprise.

The attraction of this explanation is its memory of a changing world. An observer can become more knowledgeable about a process while continuing to revise estimates of its current state. Stable knowledge of the rules and uncertainty about the present can coexist. That combination makes persistent responsiveness intelligible: learning has established how much the world moves, and that knowledge tells the observer why the next reading still matters.

20.1.3 — Interoceptive Prediction

Interoceptive prediction concerns the body as it is sensed and anticipated. Bodily signals arrive through noisy channels shaped by expectation, context, uncertainty, history, and available action. Feeling can therefore depend on both bodily signals and the predicted meaning of the body (Seth, 2013; Barrett and Simmons, 2015).

Placebo and nocebo effects make the point concrete. Expectations can alter pain, symptom appraisal, attention, and physiological response even when the external stimulus is held relatively stable. The effect is not imaginary, because interpretation is one process through which the body is experienced. The same evidence warns against treating expectation as a universal master variable: effects vary across people, conditions, contexts, and outcomes (Wager et al., 2004).

Anxiety offers another clear case. A person who expects bodily danger may attend more intensely to heartbeat, breathing,

dizziness, or stomach sensation, then interpret ordinary variation as confirmation of threat. Prediction and sensation can reinforce one another (Seth, 2013; Critchley and Garfinkel, 2017).

Cultural learning can shape what bodily changes mean. The same arousal may be understood as illness, excitement, shame, spiritual presence, panic, or ordinary exertion. The path from signal to experience is structured by concepts, expectations, language, and social practice.

20.1.4 — The Value of a Useful Clue

Information becomes valuable through the choices it makes possible. Imagine a small planning game with two doors. One reliably awards six points. The other awards either zero or ten, with equal probability, and its outcome has already been set for the round. The player knows these rules but cannot see which outcome is behind the uncertain door. The numbers are the game's points, so every calculation has a visible and agreed meaning.

Without further information, the reliable door has an expected value of six and the uncertain door an expected value of five. A player maximizing expected points chooses the reliable door. Now offer a perfect clue about the uncertain door before the choice. When the clue reveals ten, the player chooses that door. When it reveals zero, the player chooses the reliable six. Each case occurs half the time, giving an expected result of eight. The clue has increased the best achievable expectation from six to eight without changing either door's contents.

The value of information in this example is therefore two points: the increase in the best achievable expected score. A clue costing one point leaves an expected gain of one. A clue costing three leaves the player worse off under the same objective. Information's usefulness depends on what it costs and what the

person can do with it. If the player had to choose a door before seeing the clue, the same accurate message would arrive too late to improve that choice.

A clue can also be informative without being decisive. Suppose it changes the estimated chance of ten from one half to three fifths. The uncertain door's expected value rises to six, exactly matching the reliable door. A stronger clue that raises the chance to four fifths makes its expected value eight. Here the useful threshold is explicit: the uncertain door becomes preferable when the chance of ten exceeds three fifths. The question is how often a proposed source of information can move belief across a boundary that matters for action.

This creates a natural connection between curiosity and planning. Learning about an unused alternative can be valuable because the alternative may become the better choice. A question asked now may improve a decision made later. The information is carried forward through a changed expectation and a changed policy, even when seeking it produces no immediate reward.

The example also gives an experiment a clear target. One can vary clue accuracy, clue price, the availability of the alternative, and the interval before the choice. A model of information-seeking should predict how those changes alter selection. The resulting pattern can distinguish a preference for useful knowledge from a preference for novelty itself. Each can be interesting; the game makes their contributions separately visible.

20.2 — Affect as Prediction Error

Surprise changes experience in ways that can be tested locally. An expected harm that fails to arrive may produce relief. A sudden pleasant event can differ from an equally pleasant expected one. Predictive accounts can compare those contrasts without inferring any whole-life terminal balance (Wager et al., 2004; Leknes et al., 2008).

20.2.1 — Valence and Surprise

Valence may depend not only on what happened, but on how an outcome compares with expectation, whether prospects are improving, and how strongly the system revises its beliefs. Prediction-error accounts are therefore serious rivals for momentary and short-horizon affect.

When a child comes home safely, threat expectations can weaken and costly preparation can subside. Breathing, muscle tension, appetite, and attention may change as the model updates (Barrett and Simmons, 2015; Clark, 2013).

Institutions can misuse the same pattern. A long appeal, diagnostic delay, disciplinary process, debt negotiation, or immigration interview may impose weeks of anticipatory strain and then point to the person's relief as evidence that the process was harmless. Predictive accounts make the hidden cost visible because expected threat is itself lived before the final answer arrives.

20.2.2 — Body Updating

Threat learning gives the rival force in the negative direction. A system shaped by harm may treat ambiguity as danger, assign high weight to threat cues, and narrow available actions before a conscious argument appears. What looks irrational from outside may be coherent within a history of injury, discrimination,

illness, neglect, or unstable care (Seth, 2013; Pezzulo et al., 2015).

State and trait also need separation. A brief arousal response, a chronic regulatory pattern, and a stable reporting style can look similar in a short window. Repeated observation and context are needed before a model can distinguish a momentary shift from a lasting change.

20.2.3 — Predictive Dynamics and Accumulated Affect

Different computational models make different commitments. Recent rewards and expectations can predict momentary well-being, while accumulated prediction errors can represent mood as momentum (Rutledge et al., 2014; Eldar et al., 2016). Joffily and Coricelli (2013) propose emotional valence in terms of the negative rate of change of variational free energy, within a hierarchical theoretical account.

A fixed change-rate relation integrates to a boundary difference, making this a relevant theoretical precedent for Section 8.3.2. The additional work is to identify which felt contributions the relation covers, how levels or components combine, and whether the complete stream has equal boundary values. A time-varying gain or an added state-dependent contribution changes the integral. Variational free energy is an information-theoretic quantity here, not conserved physical energy; the connection is a proposed mathematical organization of affect, not a conservation law borrowed from physics.

Mood also changes how later events are received. A guarded state can increase the expected cost of ambiguity, while an expansive state can make exploration feel available. The background state is both an effect of recent history and a condition on future prediction, which can create cumulative-looking dynamics on a limited time scale.

Interoceptive accounts (Seth, 2013; Greenwood and Garfinkel, 2025) motivate a proposed test here: whether a held-out perturbation changes bodily prediction while leaving a proposed terminal signature unchanged. That contrast could distinguish a specified interoceptive account from a candidate closure route if they predict different responses.

Mean-zero prediction error is not enough. Errors may average near zero because rewards were centered, the learner calibrated, or the task designed to settle. A finite task may also end without residual error because the final trial was learned or the endpoint used for fitting. Neither establishes a pathwise terminal residual for every stream. A task intended to distinguish calibration from closure should therefore keep its evaluation trials and endpoint rule outside the data used for fitting.

20.3 — Viability, Prediction and Explanatory Reach

Survival is not a measure of wellbeing. A person may endure through chronic stress or deprivation. Accurate prediction of danger can even support endurance while narrowing choice. A viability model must therefore justify any inference from continued life to terminal closure rather than assume survival has answered the felt burden of continuing (Friston, 2010; Sterling, 2012).

20.3.1 — Adaptive Function and Rival Explanations

Some protective responses preserve function while imposing felt costs. Pain can discourage tissue damage, nausea can prevent ingestion, fear can prepare escape, and exhaustion can force retreat (Friston et al., 2017; Pezzulo et al., 2015).

A predictive explanation earns credit for the patterns it explains. If it reproduces apparent balance without requiring exact terminal closure, it is a rival to that interpretation. If an independently supported mechanism actually entails the declared closure condition, it could instead realize the Law. The distinction turns on what follows from the mechanism and its independently tested links to experience, not on whether its components are familiar.

20.3.2 — How Flexible Models Can Mimic Closure

An honest predictive baseline should be strong enough to compete. It may use horizons that depend on state and uncertainty across levels. Bodily precision, medication, and social context can alter its predictions. The model must also represent limited reserve, missing observations, and shutdown rather than leaving those effects for PTVC alone.

It may predict quieting, volatility, withdrawal, agitation, or apparent resolution without PTVC (Friston et al., 2017; Seth, 2013).

Predictive models have several legitimate sources of flexibility: priors, precision weights, hidden states, action policies, observation maps, and hierarchical time scales. Each choice should be specified in advance with its allowed range and tuning procedure (Clark, 2013; Friston et al., 2017).

Time scale needs its own declaration. Fast sensory updating, slower mood integration, and long-lived expectations can all generate smooth curves while describing different processes (Eldar et al., 2016; Rutledge et al., 2014).

A redesign can begin with existing observations and compare follow-up variables by their expected information gain about the remaining explanations relative to participant burden.

20.3.3 — Where Predictions Separate

The most informative observation is often the one on which otherwise successful accounts disagree. Two learning rules may fit a quiet period equally well because nothing in that period calls their differences into play. To discover the difference, the investigator can look for a modest change whose consequences the rules already specify.

Imagine two models that update an estimate by moving part of the distance toward the latest observation. Model A moves one half of that distance. Model B moves three quarters. Both begin at zero, and the training observations are all zero. After any number of those observations, both estimates remain zero. Their success on this record tells us little about which updating rule generated it.

Present a new observation of four. Model A moves halfway from zero to four and reaches two. Model B moves three

quarters of the way and reaches three. Their predictions now separate by one unit. Present an observation of two next. Model A is already at two and stays there. Model B moves three quarters of the distance from three down to two and reaches two and one quarter. The second step distinguishes a fast initial adjustment followed by retreat from a smaller adjustment followed by stability.

Every number in this example was fixed by the rule before the corresponding update. The comparison can be reproduced with pencil and paper: new estimate equals old estimate plus the chosen fraction of observation minus old estimate. A real study would then specify how that internal estimate appears in its measured channel. If a response is a noisy or delayed expression of the estimate, the observation model carries that additional relation. The learning rule and the reporting rule occupy identifiable places in the explanation.

The same logic helps choose the next trial. Another zero during the quiet training period contributes little separation. A change in input creates it. A second change can reveal whether the first response was a persistent update, a transient reaction, or an artifact of how it was measured. The valuable sequence is the one that exposes a meaningful difference among the candidates, not merely the sequence that produces the largest response.

In the Law's research program, this invites a specific form of ingenuity. Write down where a proposed reserve-coupled process and an ordinary learning process predict different transitions under the same available observations. Then choose a feasible setting in which those transitions can be seen. Agreement over a long calm interval can coexist with sharp disagreement about a short change. A well-chosen transition may therefore teach more about the mechanism than another long stretch of similar behavior.

Chapter 21 — Control and Homeostasis

A thermostat, a hungry animal, a person learning a route home, and a controller keeping a machine within safe limits all exhibit directed change. Feedback, reward, constraint, and state estimation can create orderly trajectories without a moral accountant. Imagine a child learning to ride a bicycle. The first successful stretch brings delight; practice then changes which journeys are possible, whom the child can visit, and what feels within reach. One rewarding moment can open a different future. This is a reason to study paths as well as peaks: learning and opportunity can change the shape of later experience. These are already rich questions for ordinary science.

Reward-learning accounts become stronger when reward is understood broadly. Reward does not have to mean pleasure in a shallow sense. It may mean relief, safety, reduced uncertainty, avoided pain, social approval, access to food, restored control, or escape from danger (Berridge and Robinson, 1998; Berridge et al., 2009). The Bellman idea deepens the rival. A value estimate can be decomposed into immediate reward plus expected future value, allowing an agent to learn from partial futures rather than wait for an entire trajectory to end. That is a major strength of reinforcement learning.

The Markov assumption asks whether the model's present state contains the information needed to predict what comes next. That is a demand on the representation, not a claim that a person's past no longer matters. Memory, illness, expectation, and social history may need to be represented because their effects persist. The useful question is what a particular model has kept and what it has left out. Success at predicting a person's actions does not, by itself, show that the same representation captures what those actions feel like (Sutton and Barto, 2018).

Anticipatory regulation may coordinate competing bodily demands (McEwen, 1998; Sterling, 2012; Theriault et al., 2025).

Care changes the regulatory field. Food supports metabolism; shelter supports temperature, sleep, and safety; pain treatment lowers nervous-system load; therapy can alter threat interpretation; friendship can reduce isolation stress; and legal or material support can reduce scarcity-driven arousal. Improvement after care is consistent with reduced burden.

Policy improvement is a technical achievement: a learning system can select actions that better satisfy its reward and safety constraints (García and Fernández, 2015; Achiam et al., 2017). Human improvement is broader. A patient may establish a steadier routine, a service may be redesigned, and a community may remove a recurring obstacle. These frameworks are strong rivals because the Law also speaks about paths, constraints, horizons, reserve, and boundary conditions. Hybrid models deserve a fair test when their added complexity improves held-out explanation. Section 21.3.2 develops their identification limits. Section 21.4.4 specifies how these explanations are compared under common rules.

21.1 — Rewards, Set Points, and Regulation

Reward learning concerns how choices respond to better or worse outcomes than expected. Prediction errors can change future action without any special terminal target. Different learning rates or reward schedules can yield different trajectories. The proposed fairness mechanism must make a further prediction if it is to outperform this familiar account.

21.1.1 — Reward Learning and Care as Regulation Support

In a toy navigation task, assign +1 for reaching the destination and a small cost for each step. The policy specifies how the agent chooses its next turn. Expected return concerns the future reward sequence under that policy, with any specified discount. If a detour costs more than predicted, the resulting prediction error can change the route's estimated value and later choices (Sutton and Barto, 2018).

Homeostatic control maintains workable ranges, while allostasis coordinates regulation in anticipation of changing demands. Blood pressure, glucose, inflammatory tone, temperature, oxygen, thirst, arousal, and sleep pressure do not move as isolated dials. Regulation of one variable can alter the demands on another, and a short-term adjustment can increase later demands (McEwen, 1998; Sterling, 2012).

Long-term stress regulation may preserve outward performance by redistributing effort across endocrine, immune, cardiovascular, metabolic, and behavioral systems. A person can appear composed while regulatory systems work under sustained strain (McEwen and Stellar, 1993; McEwen, 1998; Ramsay and Woods, 2014).

Even a simple regulator can produce a rich trajectory when its information arrives late. Imagine a numerical reservoir with a target level of ten. At each step a controller adds the difference

between ten and the level it has most recently observed. Addition may be negative, representing removal. With an immediate observation of the current level, a level of eight leads to an addition of two and reaches the target in one step.

Now delay the observation by one step. Suppose the preceding reported level was ten, but a disturbance has brought the current level to eight. The controller initially adds zero because its available report still says ten. At the next step it receives the earlier reading of eight and adds two, bringing the actual level to ten.

The next report still describes a level of eight, so the controller adds another two and brings the reservoir to twelve. It then receives a report of ten and adds zero. The following report of twelve prompts removal of two, returning the level to ten. Another delayed report of twelve prompts a second removal of two, taking it back to eight. Under these exact rules, the sequence repeats.

The oscillation arose from corrective action applied to an earlier state. The target remained fixed, and the controller always acted to close the reported gap. What changed was the relation between the report and the current reservoir. A record of its intentions would look consistent throughout; a record of the state reveals why those intentions produce alternating undershoot and overshoot.

Reducing the size of each correction changes the trajectory. Let the delayed controller add only half the reported gap. Starting with the same disturbance and preceding report, successive actual levels are eight, eight, nine, ten, ten and one half, ten and one half, ten and one quarter, and ten. The following deviations continue with decreasing amplitude under this particular rule. The controller has become less aggressive while keeping the same target and observation delay.

This gives model comparison several concrete features to examine: the first response after a disturbance, the delay before reversal, the size of the overshoot, and whether oscillations persist or diminish. A final return alone conceals those differences. The intervening trajectory reveals how observation and action are connected.

The example also shows why support for regulation can take more than one form. An engineered system can receive timelier information, use smaller corrections, or predict how its earlier actions will change the current state. These changes address different parts of the loop. A richer account of regulation follows that loop closely enough to explain both successful recovery and the shape of the path taken to reach it.

21.1.2 — Why Reinforcement Learning Matters Here

Reward learning supplies a precise reason to combine behavior with other evidence. In a finite-state decision model with bounded rewards and discount $0<\gamma<1$, add $\gamma\phi(s')-\phi(s)$ to each transition reward, with φ zero at any terminal state. This potential shaping preserves optimal-policy rankings. The discounted return changes by the same starting-state constant for every policy, although the transition rewards can differ. This potential-shaping ambiguity is part of the formal identifiability problem studied by Skalse et al. (2023).

The design opportunity is to seek an independently justified observation that distinguishes the surviving reward representations. Reports or physiological measurements help only if their relation to those representations is specified and tested. More examples of the same optimal choices cannot resolve an ambiguity that those choices preserve. The discounted transformation is a comparison from reward learning, not the undiscounted phenomenal relation proposed by PTVC.

Two short paths make the ambiguity in reward learning tangible. Both start in the same state and end after two transitions. Give the second reward half the weight of the first. Path A has original rewards zero and four, so its discounted return is two. Path B has original rewards one and zero, so its return is one. The original objective prefers A.

Assign a potential of four to the shared starting state, two to A's intermediate state, minus two to B's intermediate state, and zero to both terminal states. On each transition, add half the next state's potential and subtract the current state's potential. This is the potential-shaping transformation in the preceding equation, with discount one half.

For A's first transition, the added amount is one minus four, or minus three. Its first shaped reward becomes minus three. On the second transition, the added amount is zero minus two, or minus two, changing the reward from four to two. The shaped discounted return is minus three plus half of two, which equals minus two.

For B, the first added amount is minus one minus four, or minus five, changing its reward from one to minus four. The second added amount is zero minus minus two, or plus two, changing the second reward from zero to two. The shaped discounted return is minus four plus half of two, which equals minus three.

Both full-path returns have fallen by exactly four, the potential of their common starting state. A still ranks above B by one. The local rewards have changed substantially, including their signs, while the relevant comparison between completed policies remains intact. Observing the preferred policy alone would not identify which of these reward descriptions generated the preference.

The role of the terminal condition can be seen by stopping the comparison after the first transition. Under the original rewards, B's first reward of one exceeds A's zero. Under the shaped rewards, A's minus three exceeds B's minus four. The continuation is needed for the intermediate potential terms to cancel in the discounted total. A comparison of truncated prefixes is a different object from the completed return.

This calculation gives the invariance a concrete scope. A transformation can preserve the decision relation a researcher observes while changing quantities the researcher would like to infer. Choosing an observation then becomes a question about what distinctions it preserves. A supplementary measure earns its place when its justified relation to the reward representation can separate alternatives that the completed choice leaves equivalent (Skalse et al., 2023).

21.1.3 — The Memory Inside a State

A present state can contain a memory without containing a story. To see why a model might need that memory, consider a device that cycles deterministically through three internal states: A, then B, then C, then A again. Its visible display shows zero in A, zero in B, and one in C. The observed sequence is therefore zero, zero, one, repeating.

Seeing one identifies the current internal state as C, so the next display must be zero. Seeing zero leaves two possibilities. If the device is in A, the next display will again be zero. If it is in B, the next display will be one. The visible number alone does not contain everything needed to predict the next number.

The recent past supplies the missing distinction. If the last two displays were one and then zero, the device must now be in A, and another zero follows. If they were zero and then zero, it must now be in B, and one follows. Under a uniformly chosen

starting phase, a model using only the current zero would assign equal probabilities to these next outcomes. A model retaining the previous display can predict the next outcome exactly once it has observed a pair.

This is a failure of the Markov property for the coarse display under that initial distribution. A Markov state makes the conditional distribution of the future depend on the present state rather than additional past information. The internal labels A, B, and C meet that condition here. The single visible number does not. The pair of consecutive visible numbers supplies another sufficient state representation after the first observation pair.

There are several ways to carry the needed information. An observer can retain the last two readings. A device can keep a three-valued phase label. A model can maintain probabilities over the three internal states and update them as readings arrive. These representations use different internal descriptions while exploiting the same transition structure. Their adequacy is assessed by the predictions they permit.

Add observation error and the update becomes more interesting. A displayed reading may no longer identify a phase exactly. A probability distribution over phases can still carry forward what earlier readings imply. A new reading revises that distribution rather than erasing the history. The retained object is then a belief state: a compact account of the information relevant to the future under the specified model.

In a study of experience, the analogous question is whether two occasions with the same measured present level have different predicted futures because their histories differ. A rising level and a falling level may pass through the same value. Recent treatment, a pending task, or a learned cue may also distinguish the occasions. A state description earns added detail

when that detail improves a specified prediction on new observations.

21.2 — Optimization Is Not Pathwise Closure

An objective function ranks outcomes selected by a modeler. Its values need not equal what an agent feels. A controller can maximize a specified score while leaving experienced consequences unmeasured. An optimization result therefore cannot be treated as pathwise closure unless the objective is validly linked to that target.

21.2.1 — Objective Functions, Constraints, and Experience

Experience is what a conscious subject feels. An objective may be useful while omitting suffering, representing it badly, or optimizing a proxy that diverges from the person's life. The omitted burden must be named (Sutton and Barto, 2018; Berridge et al., 2009).

Reinforcement learning often handles time through discounted accumulation, which is useful for decision-making. It depends on how reward is defined, the chosen horizon, the representation of the state, the weight assigned to distant consequences, and the model's assumptions about what matters. Change those assumptions, and the apparent value of a policy changes (Sutton and Barto, 2018). Constraint satisfaction can keep a system within permitted limits. Fairness, in the sense examined here, concerns a proposed property of felt experience over a complete stream. A controller that never violates its engineering constraints has not thereby demonstrated experiential closure.

After a formal condition is satisfied, examine whether risk has moved into family labor, undocumented waiting, staff burnout, future collapse, or another unmeasured channel.

21.2.2 — When Time Weighting Reverses a Choice

An optimization rule can change its preferred action simply by changing the weight assigned to time. Consider two invented

policies with two outcomes each. Policy A produces plus two now and minus two later. Policy B produces zero now and plus one later. These are numbers in an illustrative objective, chosen to make the arithmetic visible.

Give the later outcome half the weight of the present outcome. Policy A then has a weighted value of one: two plus one half of minus two. Policy B has a weighted value of one half. Under this objective, A ranks above B. Give both times equal weight and the ranking reverses. A totals zero, while B totals one.

The point of reversal can be found exactly. Call the later weight g. A has value two minus twice g, while B has value g. They tie when three times g equals two, so the threshold is two thirds. Below that threshold A ranks higher; above it B ranks higher. The outcomes have remained fixed throughout. The change occurred entirely in the rule used to compare them.

The same example separates choosing a policy from measuring a completed history. At the moment of choice, uncertainty about what will occur may legitimately enter a decision model. After a history is specified, its unweighted sum is a different mathematical object. Giving a later event a smaller decision weight does not shorten its duration or change its value within a separately defined accumulation rule.

21.3 — Constrained and Hybrid Control

A constraint specifies which actions or states a model permits. An objective ranks preferred outcomes within that permitted region. Hybrid control can combine both features without making them identical. A fairness hypothesis must say whether its prediction comes from a constraint, a preference rule, or their interaction.

21.3.1 — Constraint Models

A system may pursue reward while respecting safety limits, or maintain viability while avoiding dangerous states.

The represented hazards may include collision, instability, irreversible loss, harm to humans, violation of a physical limit, or unacceptable cost (García and Fernández, 2015; Achiam et al., 2017; Ji et al., 2024).

In the Law's vocabulary, a terminal boundary condition may look like a hard constraint. In a control model, a hard constraint can alter the value function as the system approaches a boundary. This comparison helps readers understand why a terminal layer might create distinctive dynamics.

Does the reserve-coupled route predict patterns beyond a constrained control model after safety envelopes and homeostatic variables are included? Name the observation on which their predictions differ.

An endpoint can be close and still lie outside the reachable set. Consider an integer-valued state that starts at one. At each step the controller must move either one unit upward or one unit downward. After six steps, every attainable state is odd: minus five, minus three, minus one, one, three, five, or seven. Zero lies between the extremes, but it is absent from the list.

Each move changes parity. An odd starting value becomes even after one move, odd after two, and odd again after any even

number of moves. The obstruction therefore survives every possible choice among the six-step action sequences. More effort in searching that fixed action set cannot uncover a six-step route to zero, because the allowed increments exclude it.

Change the horizon to five steps and the answer changes. Two upward moves and three downward moves have net change minus one, taking the initial state to zero. A longer horizon has not automatically produced a larger set of exactly attainable endpoints: with mandatory unit moves, the sets alternate between even and odd integers. The horizon and the action alphabet jointly determine reachability.

A small change in the action alphabet changes the six-step result as well. Allow a pause with increment zero. The controller can use the five moves just described and one pause, arriving at zero after six steps. The useful change is specific: the added action allows the controller to preserve a state for one interval. It is not an unspecified increase in power.

This example explains why a bound on reachable values and an exact reachable set answer different questions. A bound from minus five to seven includes zero; the discrete six-step set does not. Bounds remain useful for describing the range, while the increment structure resolves exact attainability. In a candidate controller model, recording that structure can turn an opaque failure to reach the target into a simple mathematical fact about the permitted moves.

21.3.2 — Hybrid Models

Hybrid models combine mechanisms because lived trajectories rarely belong to one textbook chapter. Adaptation, prediction, reinforcement learning, bodily regulation, treatment, social context, and observation error may operate together. The gain in

realism is purchased with greater risk of nonidentification, overfitting, and post hoc explanation.

Integral feedback shows how accumulated information can reside in a biological state. Yi et al. (2000) identified integral-control structure in a model of bacterial chemotaxis. In specified biomolecular systems, antithetic integral feedback can restore the stationary mean of a regulated molecular count to its set point under the required ergodicity and bounded-moment conditions (Briat et al., 2016; Aoki et al., 2019). This is adaptation of a regulated variable rather than finite-time closure of every trajectory.

An ideal integrator obeys $\frac{dx}{dt} = e(t),$ so its change equals the accumulated input. A leaky integrator obeys $\frac{dx}{dt} = e(t) - \lambda x(t);$ the input total then equals the change in x plus λ times the integral of x. Identical boundary values therefore need not imply zero input when λ is nonzero. Measuring retention distinguishes an integral from a fading trace. For a PTVC implementation, the next task is to identify the retained input and its relationship to complete felt contribution; decay of a record cannot erase experience already lived.

The distinction between accumulated input and retained state is especially clear in a discrete leaky integrator. Let the next state equal one half of the current state plus the new input, and begin at zero. An input of two produces state two. At the next step an input of minus one produces state zero, because one half of two minus one equals zero. The two inputs sum to one, even though the final state is zero.

A pure integrator would retain the whole previous state. Under that rule the same two inputs would leave a final state of one. The difference comes from the dynamics between inputs: the leaky system carries only half of the first contribution into the

second step. Its terminal state is a weighted record of past inputs, with older contributions carrying less weight.

The weighting can be written directly in this two-step example. The final state equals one half of the first input plus the second input. Returning that state to zero imposes the relation that the second input is minus one half of the first. Requiring the unweighted input total to be zero instead imposes a different relation: the second input must be the negative of the first. Both are precise conditions, and they select different input pairs.

With no further input, a retained state of two would become one, then one half, then one quarter. The model's state progressively loses its dependence on that earlier input. This describes the memory of the modeled variable. It does not reverse the historical occurrence of the input or change a separate record of its amount. A storage level, a filtered measurement, and a cumulative history can therefore carry different information about the same sequence.

A controller study can use this distinction constructively. It can specify which variable is regulated, how that variable updates, and which quantity the scientific hypothesis requires at the endpoint. A terminal condition on a filtered state becomes interpretable once its weights are visible. The calculation then reveals whether the controller regulates the relevant accumulation directly or regulates another quantity whose relation to it must be established.

A hybrid carries its own burden. Its component set, interactions, priors, parameter ranges, and module boundaries should be prospectively specified or selected through a transparent training procedure. Fit and combine the representation, dynamics, observation, endpoint, and

intervention modules using training data, then freeze the complete model before evaluating it on held-out data.

Ablation tests a component's predictive contribution within the fitted model. Removing adaptation, prediction, control, or observation components should change performance in interpretable ways. Redundant components can compensate for one another, so joint ablations may also be needed. Predictive contribution alone does not establish a causal mechanism.

Complexity costs should reflect effective freedom, not only named parameters. Flexible bases, repeated model selection, hand-tuned priors, analyst choices, and calibration options all consume freedom. A fair comparison records those choices and judges probability or interval forecasts on data not used to choose the result (Burnham and Anderson, 2002; Gneiting and Raftery, 2007; Mayo, 2018).

21.3.3 — A Controller That Remembers Its Limits

A controller can remember more requested action than its equipment has performed. That difference is easy to miss if a model records only the target and the visible output. The internal command may continue growing while the actuator remains at its limit. When the target changes, the controller must then work through its stored request before the visible action can change.

Consider a simple engineered device whose output x can range from zero to one. Its commanded value is z. The actuator delivers $x = \min(1, \max(0, z))$, so a command below zero produces zero and a command above one produces one. At each update, the controller adds half the difference between the target r and the currently observed output to z. The observation is immediate and exact in this constructed example.

Start with output and command both zero, and set the target to three. On the first update, the error is three, so the command

becomes 1.5. The actuator can deliver only one. On the next update, the error is two and the command becomes 2.5, while the delivered output remains one. Two more updates raise the command to 3.5 and then 4.5. The visible output has stayed fixed, but the internal state has changed at every update.

Now change the target to one half, a value the actuator can deliver. The output is still one, so the error is minus one half. Each update initially reduces the command by one quarter. Starting from 4.5, it takes fourteen updates to reach a command of one. Throughout those fourteen updates, the output remains at one. Only the next update lowers the command to three quarters and produces a visible decrease.

The delay has a specific cause. The controller is receiving the new target and current output correctly. It is also changing its command in the appropriate direction. Its stored value is simply so far beyond the actuator's range that many internal changes have no immediate outward effect. This accumulation during saturation is often called integral windup (Åström and Rundqwist, 1989). The internal record of requested correction has become disconnected from the range of delivered action.

A second controller can store a bounded command. After each proposed update, it limits z itself to the interval from zero to one, matching the actuator's range. Under the earlier target of three, its stored command reaches one and remains there. When the target becomes one half, its first update changes the command and output to three quarters. The next changes them to five eighths, and the next to nine sixteenths. The departure from the new target halves on each update.

The comparison isolates the role of internal storage. Both controllers have the same target sequence, gain, observations, and actuator. Their difference is what happens to the command while the requested action exceeds the available range. Limiting

the stored command allows the second device to respond promptly when the target returns to a reachable value. It does not expand the actuator's physical capacity; it keeps the controller's memory aligned with that capacity.

Other designs can use the discrepancy between commanded and delivered action as feedback to the internal state. A model can track the actual action and revise the accumulated request accordingly. The useful signal is the difference between what the controller asked for and what the equipment did. That signal addresses a different problem from delayed observation, uncertain measurement, or insufficient actuator strength.

This distinction suggests an informative experiment in an engineered or simulated regulator. Hold the output at a known bound for different durations, then move the target into the attainable range. If recovery delay grows with the earlier time spent beyond the bound, an accumulated internal command becomes a candidate explanation. Inspecting that state directly, or changing its update rule, can make the prediction more specific. The trajectory reveals something that the saturated output alone concealed.

For the book's comparison of regulatory mechanisms, the example adds another kind of memory. A state can store a history of unmet requests as well as a history of delivered contributions. Those histories need not coincide. Following both makes it possible to explain why two systems with the same current output recover differently after the same change. The next response depends on what the controller has retained, not only on what the observer can see.

21.3.4 — What a Component Contributes in Company

A component's contribution depends on what else is present. This is easy to see in a prediction task with a known answer. Let

a label Y be either zero or one, with equal probability. Supply a model with two input channels, A and B, each of which reports Y perfectly. Either channel alone is sufficient for perfect prediction.

Remove A and rebuild the predictor from B. Accuracy remains one hundred percent. Remove B instead and rebuild it from A. Accuracy again remains one hundred percent. Remove both, and the best uninformed fixed prediction is correct half the time. Each channel carries the full answer, yet neither single-channel removal reduces the best attainable accuracy. Their information is redundant.

An analysis that assigns importance only by the change after one-at-a-time removal would give both channels zero importance in this example. That result describes their replaceability in the presence of the other channel. It does not describe their information content when used alone. The distinction matters whenever several sensors, representations, or fitted modules can supply the same predictive work.

Now consider the opposite construction. Draw two independent fair binary inputs A and B. Let Y equal one when they differ and zero when they match. The four equally likely input pairs are 00, 01, 10, and 11, with labels 0, 1, 1, and 0. Together the two inputs determine the label exactly.

Either input alone leaves the label equally likely to be zero or one. Among records with A = 0, half have B = 0 and half have B = 1, so half the labels are zero and half one. The same holds when A = 1 and when either value of B is considered. A single-channel model therefore achieves only fifty percent expected accuracy. Combining the channels creates perfect prediction through their relation.

This is complementary information. Neither channel is individually predictive, yet both are necessary for the perfect

joint result. A method that selects inputs solely by their separate association with the outcome would discard both. A method capable of representing the interaction can use them. The difference lies in the question asked of the inputs: what does each say alone, or what can they say together?

The two constructions have the same best full-model accuracy and very different internal structures. In the redundant system, removing either channel leaves accuracy unchanged, while removing both loses the answer. In the complementary system, removing either channel already loses the answer. A small set of planned component comparisons can distinguish these architectures even when their overall performance is identical.

There is another distinction within removal itself. One can remove a channel from an already trained predictor and leave the remaining model fixed. Alternatively, one can remove the channel and train a new predictor on what remains. The first asks how the particular fitted system depends on its present implementation. The second asks how well the reduced information set can support the task after adaptation.

Suppose the redundant model happened to learn to use A exclusively. Deleting A from the frozen implementation can sharply reduce performance, although B still contains the whole answer. Retraining can teach the reduced model to use B and restore accuracy. Both results are informative. One concerns the route actually used by the fitted model; the other concerns a route available in the retained inputs.

For hybrid accounts of affective dynamics, these distinctions make component analysis more revealing. An adaptation module and a predictive module might encode overlapping history. Alternatively, one might supply a reference state while the other supplies the event that must be compared with it. Their joint contribution could then exceed what either explains separately.

The useful analysis examines the pattern across full, reduced, and retrained models on the same held-out question.

A model's internal organization can be studied this way without assuming that its modules correspond one-for-one to mechanisms in an organism. To investigate that further connection, the component's predicted response to an independent perturbation becomes relevant. The computational comparison identifies where the model obtains its information; a separate empirical comparison can examine whether the proposed system obtains information in the same way.

The reward is a more intelligible explanation of performance. A strong forecast is one achievement. Understanding whether it comes from replaceable signals, a particular fitted route, or an indispensable relation among signals is another. The small binary worlds make those possibilities easy to distinguish, and give a larger hybrid model a set of questions more precise than asking which component matters most.

21.3.5 — Model Governance

State who chooses the target, data, loss function, constraints, allowed interventions, and verdict.

Conflicts of interest strengthen the need for institutional separation. When a foundation, movement, religious community, civic project, government partner, or funder benefits from positive results, prospective specification, independent review, and publishable adverse findings become part of the scientific design. Independent reviewers should disclose those relationships and retain authority to publish null or adverse findings without sponsor approval.

Common infrastructure can expose implementation choices and support reproducibility (Ji et al., 2024; Patterson et al., 2024). Retain the configuration and decision path as well as the

output, so another team can reconstruct when the system declined to produce an estimate.

21.4 — Mean Reversion, Bridges, and Hybrid Models

Mean reversion can bring a fluctuating measure back toward its usual level. That local tendency may arise without any knowledge of a terminal endpoint. Conditioning a process on its destination creates another kind of apparent return. A fair model comparison must distinguish ordinary reversion from a bridge or a genuine terminal constraint.

21.4.1 — Mean Reversion Can Explain Local Return

Feedback, regression, sampling, or stable dynamics may pull observations toward a typical range, and an Ornstein–Uhlenbeck (OU) process shows how disciplined return can coexist with random disturbance (Uhlenbeck and Ornstein, 1930).

A fair comparison can include the classical OU process, whose restoring drift is linear, alongside richer mean-reverting models. Heterogeneous baselines, changing variance, disease progression, medication, disability, chronic stress, report context, and missingness may enter those extensions. Nonlinear dynamics or several defended ranges should be identified as broader model classes, not as the classical OU process.

Accumulated variability can distinguish processes whose momentary recovery looks similar. Under the state-change candidate, the accumulated total is the state at the beginning of the window minus the state at its end. If that state is stationary and has finite positive variance, the variance of its accumulated change remains bounded by four times the state's variance. By contrast, for a stationary zero-centered Ornstein–Uhlenbeck rate with positive restoring strength and disturbance level, the variance of the accumulated rate increases approximately in proportion to the duration of a long unconditioned observation window. These are claims about accumulated totals, not the average rate.

Test the comparison within one declared observation regime. Integrated sensor noise can add growing variance, while differenced noise behaves differently. Nonstationarity and endpoint conditioning change the calculation. The contrast therefore compares specified models, not every realization of PTVC; Section 21.4.2 supplies a smooth conditioned competitor.

Path consistency and accumulated variability are linked in a specific model class. Suppose a finite irreducible Markov chain is stationary, its transition probabilities are fixed, and every supported transition has a fixed signed contribution. Over many transitions, subtract the stationary average contribution from the total. The Poisson-equation method separates the remainder into a bounded difference between state-dependent quantities and a martingale remainder (Glynn & Meyn, 1996). The long-run variance accumulated per transition is exactly zero when each transition's contribution equals the stationary average plus the difference between a state-specific quantity at its starting state and the same quantity at its ending state.

Random contributions require another step. Suppose each transition also contains a random fluctuation. If the fluctuations have zero means and are independent once the entire state path is fixed, their variances contribute to the long-run variability in proportion to how often the corresponding transitions occur. That contribution is the sum of each transition-specific variance weighted by the stationary probability of its starting state and the probability of that transition. Independent unresolved variation therefore accumulates. Serially dependent contributions can behave differently: differences of a retained noisy state can telescope. The dependence structure, not the appearance of noise alone, decides which calculation applies.

A discrete example shows how dependence can keep accumulated variability bounded. Let Q_0, Q_1, Q_2, and subsequent Q-values be independent, each with mean zero and variance one. Define each contribution as the preceding state minus the next state: $C_1 = Q_0 - Q_1$, $C_2 = Q_1 - Q_2$, and so on. Every single contribution has mean zero and variance two.

Neighboring contributions share a state with opposite signs. In C_1, Q_1 is subtracted; in C_2, it is added. Their covariance is therefore minus one. Contributions separated by at least one intervening contribution use disjoint Q-values and have covariance zero. The construction has a precise short-range dependence pattern.

Add n contributions. The intermediate states cancel, leaving $Q_0 - Q_n$. Its variance is two for every positive n. The same answer follows by adding variances and covariances: n individual variances contribute 2n, while the $n - 1$ neighboring pairs contribute twice minus one each. Thus $2n - 2(n - 1)$ equals two.

Compare this with a sequence of independent contributions, each also having mean zero and variance two. Its n-step sum has variance 2n. If the Q-values are normal, both constructions can have exactly the same normal distribution for every single contribution. A histogram of one-step values cannot distinguish them. Their accumulated totals separate because their serial dependence differs.

The relation can also be read through adjacent windows. In the state-difference construction, the total over one window shares its finishing state with the next window's starting state. The shared term enters with opposite signs. Combining the windows removes the interior boundary and retains only the two outer states. This is why the accumulated variability is controlled by boundary variation in this particular model.

Ordinary measurement can create the same algebra. If a sensor has an independent error at each recorded state, differences of successive readings contain errors with this negative neighbor covariance. A study can estimate how much of the observed increment dependence comes from that observation process and how much belongs to the candidate dynamics. The identity tells the investigator which relation to measure and why boundary information matters.

The state-change form additionally requires an independently justified zero value of μ; bounded variability alone does not exclude nonzero drift. A Q reconstructed from the transition table is an algebraic representation, not yet an identified physical quantity. The mixed-condition test in Section 17.2.1 examines this same relation after admitted transition frequencies or sequences change. A small estimated variance coefficient is not an all-path result, especially when transitions are rare.

Regression toward the mean can arise even when the latent state has no restoring force. An extreme observed value contains signal and measurement error; a later measurement often appears less extreme because the error component is unlikely to repeat. This must be separated from genuine dynamic reversion.

A state-space model can make that separation explicit by distinguishing latent dynamics from observation noise. If apparent return disappears once measurement error is represented, the rival has explained the pattern at the observation layer rather than the biological or psychological layer.

Two different mechanisms can produce the same apparent return over one observation gap. In the first, a latent quantity θ remains fixed. Across cases, let θ be normal with mean zero and variance one. Each reading adds fresh independent normal error with mean zero and variance one. The observed readings

therefore have variance two, and any two readings from the same case have covariance one because they share θ.

For this model, the expected second reading conditional on a first reading y is y/2. A first reading of four predicts a later reading of two on average. Yet the latent quantity has not moved. The apparent return arises because the first reading combines the persistent target with an error component that is not expected to recur.

Now construct a genuinely dynamic process with stationary variance two. At each step, retain half the current state and add an independent normal innovation with variance one and a half. Its next conditional mean is again y/2. Its adjacent-time covariance is again one. With the stationary normal starting distribution, the joint distribution of two adjacent observations is identical to that of the first model.

A third time point reveals a difference. In the static-target model, observations two steps apart still share the same θ, so their covariance remains one. In the dynamic model, two successive retention factors reduce the initial influence to one quarter, giving a two-step covariance of one half. The same one-step return prediction belongs to two different longer temporal structures.

Repeated future readings also distinguish them by their joint behavior. Under the static-target model, averaging many readings can increasingly reveal the persistent θ, while readings retain their common latent component. Under the dynamic model, the state itself keeps changing through new innovations. Its relation to an earlier state diminishes with each step. A record of several lags can examine which pattern is present.

The comparison gives a concrete purpose to the observation schedule. Adjacent pairs alone are insufficient for these two declared models, even if many such pairs estimate their common

distribution very precisely. A record that preserves longer within-case separations supplies a relation on which the predictions disagree. The useful next observation changes the temporal comparison, rather than simply enlarging the original one.

Reversion speed is informative. Ordinary mean-reverting models imply characteristic relations between departure size, subsequent change, and time lag. Estimating those relations across windows can test whether one stable process accounts for the data or whether the apparent pull changes near an endpoint.

Recovery speed needs a clock as well as a coefficient. Suppose a modeled deviation from its reference level decays by half during each hour. After one hour, an initial deviation of eight becomes four. After two hours it becomes two. After three hours it becomes one. The half-life describes the process in physical time.

An hourly prediction equation uses a persistence coefficient of one half. An equation fitted to observations two hours apart uses one quarter because two successive halvings occur between its readings. A half-hour equation uses the square root of one half, about 0.707, because applying that factor twice must produce one half. Different coefficients can therefore describe the same underlying recovery.

For exponential mean recovery, the coefficient across an interval Δ is $\exp(-\kappa\Delta)$, where κ is the recovery rate on the declared clock. The half-life is $\log(2)/\kappa$. If one site samples every hour and another every two hours, comparing their raw per-reading coefficients would compare different elapsed intervals. Converting them to κ or a common half-life makes the intended time comparison explicit.

Irregular spacing can be handled in the same way. Each forecast uses the actual interval between observations. A short

gap and a long gap receive different expected retention factors, even within one unchanged process. The record's time stamps then contribute to the dynamic model rather than merely ordering its rows.

A stationary stochastic version also links interval length to innovation variance. If its stationary variance is V and the persistence across Δ is φ, an independent innovation variance of $V(1 - \varphi^2)$ keeps that stationary variance unchanged. Over a one-hour interval with $\varphi = 1/2$, the innovation variance is $3V/4$. Over two hours with $\varphi = 1/4$, it is $15V/16$. More unobserved disturbance accumulates across the longer gap while more of the initial deviation decays.

These relations allow a model to make coordinated predictions across sampling schedules. Its recovery rate and stationary variance can be estimated under one schedule, then used to predict the conditional means and variances under another. Agreement would show that one time-based process accounts for both records. A difference would point toward another feature to investigate, such as changing dynamics or the observation process.

The reference level may itself move. Illness, treatment, aging, chronic stress, development, and social change can shift the center around which observations fluctuate. A fixed-center model and a moving-center model should be compared prospectively rather than selected according to which makes the ending look most orderly.

21.4.2 — Bridges and Transforms Can Mimic Endpoint Pull

A bridge competitor need not have a rough cumulative path. Begin with a continuous Gaussian rate whose accumulated total has positive variance at the endpoint. Conditioning the path on a chosen final total adjusts each earlier value according to its

relationship with that endpoint. The conditioned mean depends on the chosen total. The covariance removes the portion of variation explained by the shared endpoint and does not depend on the chosen total. An Ornstein–Uhlenbeck rate keeps this conditioned cumulative path absolutely continuous (Sottinen & Yazigi, 2014).

A finite example makes the distinction visible. Condition four independent standard-normal window contributions on a total of zero or two. Both conditioned models have window variance 0.75 and between-window covariance −0.25; their window means are zero and 0.5 respectively. Thus the same covariance contraction can accompany different totals. Independent calibration must distinguish zero from the nonzero control. These are mathematical reference cases, not observations of experience. Compare the candidates under the same endpoint information and test the calibrated level as well as trajectory structure.

For prospective forecasts, the no-look-ahead restriction applies: neither model may use information about the held-out path's endpoint, subsequent missingness, or terminal classification that was unavailable at the forecast time. A retrospective bridge comparison may condition on a declared endpoint, provided every model receives the same endpoint information and is scored for that conditional task. Its fit must not be reported as a forecast made before the endpoint was known.

Discrimination requires a difference in predicted observations. Two-time dependence, perturbation response, action sensitivity, and transport help when the models disagree about them. If a bridge and a proposed realization induce the same observed-data distribution under every permitted intervention, larger samples cannot distinguish their labels. Report that equivalence and

identify which additional measurement or intervention, if any, could separate them.

If the bridge rival wins under common endpoint information and tuning rules, endpoint drift no longer distinguishes the proposed realization from that rival.

21.4.3 — A Boundary Observed with Noise

An exact bridge conditions a process on a specified endpoint. An observed endpoint can carry uncertainty of its own. Joining those two ingredients produces a useful intermediate model: a path informed by a boundary reading whose precision is explicitly known. The resulting mathematics shows how much apparent endpoint concentration can come from that reading.

Take a standard Brownian process B(t), beginning at zero and observed over a fixed interval from zero to T. Its mean is zero and its covariance at times s and t is min(s, t). Its variance at T is therefore T. Now suppose the terminal reading is Z = B(T) + e, where e is independent Gaussian measurement noise with mean zero and variance r.

The variance of Z is T + r. Its covariance with B(t) is t, because the measurement error is independent and the earlier Brownian state is shared with the later one. Conditioning on a specified reading Z = z gives the reconstructed mean m(t) = tz/(T + r). The covariance between reconstructed states at s and t becomes min(s, t) − st/(T + r).

The correction has a simple structure. The original covariance is reduced by the information supplied through the common endpoint reading. Near the beginning, t is small and the reading provides relatively little information about that state. Near the end, the state and the reading share more of the underlying process. The resulting reconstruction becomes more concentrated as it approaches the measured boundary.

At the endpoint itself, the conditional mean is $Tz/(T + r)$, and the remaining variance is $Tr/(T + r)$. Set $T = 1$ and $r = 1$. A reading of zero leaves the endpoint mean at zero but its variance at one half. The displayed estimate is centered exactly on zero, while the modeled terminal quantity retains substantial uncertainty.

With more precise endpoint measurement, say $r = 1/4$, the remaining terminal variance is one fifth. With $r = 4$, it is four fifths. These are changes in the information about the same underlying process. A family of increasingly concentrated reconstructions can therefore be produced by improving the endpoint observation, without introducing a new force into the unconditioned dynamics.

The limiting cases connect this model to familiar ones. As r approaches zero, the endpoint is learned precisely. The conditional mean approaches tz/T, and the covariance approaches $\min(s, t) - st/T$: the ordinary Brownian bridge to z. As r becomes arbitrarily large, the endpoint reading contributes arbitrarily little. The mean returns toward zero and the covariance returns toward $\min(s, t)$, recovering the unconditioned process.

The interior variance also makes the continuum visible. With $T = 1$ and $t = 1/2$, it is $1/2 - 1/[4(1 + r)]$. Exact endpoint information gives variance one quarter at the midpoint. An endpoint error variance of one gives three eighths. Very noisy endpoint information leaves a variance approaching one half. The amount of narrowing follows directly from the precision supplied at the boundary.

Suppose two analysts reconstruct the same class of paths but use different endpoint-error models. Their mean curves and uncertainty bands can differ even when they agree on the underlying Brownian process. The comparison becomes

interpretable when r is reported and justified. Independently calibrated endpoint error then becomes a substantive part of the model, with a measurable effect on the reconstructed history.

There is also a temporal distinction. Before the reading Z arrives, the process model has the original covariance. After it arrives, an analyst can update beliefs about earlier states. That update is a statement about the record now available. A separate causal model would be needed to say that the evolving system used the same future reading in generating its actions. The conditional model already has a complete and useful job: reconstructing paths under specified information.

This gives rival comparison a sharper target than the visual impression of a narrowing curve. One can compare the amount and shape of narrowing predicted from endpoint measurement precision with the observed pattern. A candidate route that proposes additional structure can then state what it contributes beyond this calculable conditioning effect. The boundary reading, the latent path, and the hypothesized dynamics remain distinct enough for each to be investigated.

21.4.4 — Common Observation Rules Are Required

Two processes can spend the same amount of time in every state while circulating through those states differently. Imagine three locations, A, B and C, arranged around a triangle. At each step a traveler moves clockwise with probability three quarters and counterclockwise with probability one quarter. The same rule applies at every location, and there is no stay-put move.

If the starting location is chosen uniformly, each location has probability one third at every later step. For example, the chance of arriving at A is the chance of being at C and moving clockwise, one quarter, plus the chance of being at B and moving counterclockwise, one twelfth. Their sum is one third.

The same calculation applies to B and C. The distribution of locations is stationary even though the traveler continues to move.

Now reverse the two movement probabilities. The new traveler moves clockwise one quarter of the time and counterclockwise three quarters of the time. The location distribution is still one third at each point. A collection of isolated snapshots cannot distinguish the two rules, however many snapshots it contains, if it records locations without their temporal connections.

Paired observations reveal the difference. Under the first rule, the probability of an A-to-B step is one third times three quarters, or one quarter. The probability of a B-to-A step is one third times one quarter, or one twelfth. Their difference is one sixth. The process has a net clockwise probability flow across that edge. The same directional difference appears on each edge, forming a circulating current while the population at every location stays constant.

A third rule moves clockwise and counterclockwise with equal probability. Its stationary location frequencies remain identical to the other two, but every directed edge now carries probability one sixth in each direction. This rule satisfies detailed balance: at stationarity, the probability flow from one state to another matches the reverse flow for every pair. The two biased travelers do not satisfy that condition.

The order of observations therefore contains a kind of evidence absent from the stationary distribution. A process can maintain stable occupancy through balanced exchanges between each pair, or through a continuing circulation around a larger route. Stability of the observed proportions does not select between these organizations. Reversing a recorded sequence

also preserves the number of visits to each state while exchanging clockwise and counterclockwise transitions.

This distinction can help design a comparison of dynamic accounts. Define states independently, preserve the order of transitions and estimate forward and reverse frequencies under a common observation schedule. A proposed model can then be asked to predict both occupancy and flow. If observations are widely spaced, several unobserved moves may occur between them, so the prediction must use the transition probabilities for that spacing. The observation rule is part of the calculation.

For the fairness inquiry, the point is the additional question this opens. Two accounts may agree on a usual range or on how frequently a reported state appears, yet disagree about the sequence by which the process returns to it. A trajectory supplies direction as well as location. Models earn their explanatory detail by predicting that direction, including the transitions that a static picture leaves unseen.

PTVC-linked models and rivals must face the same raw data, endpoint, missingness, preprocessing, uncertainty, scoring rules, information, compute, tuning, and repeated-seed budgets. Training, validation, and testing remain separate, with held-out times, sites, tasks, channels, or cohorts when transport is claimed (Patterson et al., 2024). Apply Section 23.3.1’s recovery and open-set checks. A common data lineage should also record which transformations were applied, when they were chosen, and which observations each model actually received.

Forecasts require proper scoring, interval coverage, subgroup performance, and residual diagnostics because average curves can hide overconfidence and local failure (Gneiting and Raftery, 2007). Hybrids must also declare scales and transport conditions before testing. When added data bring precision without discrimination, the record should ask for a new perturbation,

channel, endpoint regime, or identification assumption. The protocol should also specify in advance which residual patterns, calibration failures, or transport failures would require model revision rather than merely a larger sample.

When models predict the same recovery level, compare their recovery speed, stimulus specificity, generalization, rebound, context dependence, and persistence.

Part VIII — Evolution, Worlds, Meaning and Responsibility

Part VIII widens the inquiry. Earlier Parts defined the proposed Law, examined measurement and stream identity, opened empirical windows, and gave strong rivals room to win. Evolution, simulation and spiritual tradition each offer ways to interpret the question. The Part closes by asking what authority an institution should be allowed to draw from claims about another person's experience.

Evolutionary tradeoffs show how an adaptation can carry a burden. A nervous system may become more sensitive to threat because sensitivity sometimes protects survival, but the same sensitivity may create anxiety, fatigue, or social withdrawal. A reproductive strategy may increase lineage success while imposing risk on the body that carries it. A metabolic adaptation may conserve energy while narrowing activity and pleasure. An adaptation that addresses one pressure can create costs elsewhere.

A patch rescues a local failure. It notices a broken rule, an edge case, an exploit, a deficit, or an undesirable outcome, and it adds a correction. Patches are familiar because human institutions rely on them constantly: appeals, exceptions, subsidies, bug fixes, safety overrides, emergency interventions, and revised rules after harm becomes visible. A patch should be described as an intervention with an observable trigger, response delay, authority, maintenance cost, and side effects. If the trigger cannot see hidden burden, the patch protects only what the model records. If its response arrives too late, the system may look repaired after the experience it was meant to protect has already occurred.

Fairness questions recur not only because outcomes differ, but because human beings can compare what happened with what might have happened and can ask whether any order makes the difference intelligible. Memory, counterfactual imagination,

social comparison, grief, and hope turn events into questions about meaning. Traditions give those questions languages and practices, but they do not all give the same answer (Geertz, 1973; James, 1902).

The shared thread is the location of burden and the limits of authority. In biology, who bears the cost of regulation and adaptation? In simulation, which assumptions or patches produce the result? In spiritual interpretation, who has authority to explain another person's pain?

These questions also arise when experience becomes institutional data. Each transformation can enlarge apparent authority: a report becomes a code, the code a feature, the feature a score, and the score a profile (Cheney-Lippold, 2017).

Measurement can produce estimates for a declared research question, but an estimate does not become an administrative fact. Construct-validity scholarship shows why a score's interpretation must be earned, and due-process scholarship shows why opacity becomes dangerous when a number influences rights, liberty, care, or access (Cronbach and Meehl, 1955; Messick, 1995; Citron, 2008).

Chapter 22 — Evolution and Constraint

Evolution provides one of the strongest bodies of explanation available for the capacities through which feeling, learning, avoidance, attachment, and regulation occur. A book that speaks about complete conscious streams must therefore meet evolutionary biology at full strength. Natural selection explains differential persistence through heritable variation under environmental conditions. It does not aim at happiness, moral desert, or fairness. Traits can spread because they improve reproduction even when they impose pain, fear, exhaustion, conflict, or early death on the organisms that carry them (Darwin, 1859; Williams, 1966; Trivers, 1971).

Tinbergen's distinction among mechanism, development, function, and phylogeny is useful here because it prevents one answer from doing the work of another (Tinbergen, 1963). A trait's current mechanism may be described without knowing its evolutionary history. An adaptive function may be proposed without identifying the felt quality it produces. A terminal claim would add another layer and would need its own evidence.

Constraint language carries several meanings that should not be blended. Developmental constraints limit which phenotypes can arise; ecological constraints limit which strategies can persist; physical constraints limit what matter can do; and a terminal boundary would concern a complete phenomenal trajectory. Similar vocabulary does not establish a shared mechanism, so every comparison must identify the level at which the constraint acts (Maynard Smith, 1982; West-Eberhard, 2003). A constraint can be real without being selected as a trait. Gravity illustrates this distinction between the environment in which evolution occurs and the adaptations it produces. Developmental systems can channel change without dictating one outcome. Plasticity may expose new phenotypes, developmental architecture may make some transitions easier

than others, and ecological feedback may stabilize one pathway while closing another (West-Eberhard, 2003; Pigliucci & Müller, 2010).

The constructive case connects different kinds of knowledge. Cabanac's alliesthesia work demonstrates that pleasantness can change with internal state in the sensory modalities studied (Cabanac, 1971). Keramati and Gutkin (2014) define modeled primary reward through changes in physiological drive. Integral-control work shows how a biological state can retain accumulated input (Yi et al., 2000). Each supplies a different ingredient: an observed dependence, a candidate mathematical relation, or an implementation principle.

The consequential question is which connection adds a successful prediction. Can one independently meaningful state variable account for contributions along paths not used to build it? Can the same relation survive a change of context without resetting its units or origin? This chapter maps the biological ingredients and the additional connection a terminal account requires: a familiar mechanism might supply it, while others produce only similar local patterns.

22.1 — Evolution, Regulation and Candidate Constraints

Similar evolutionary outcomes can arise through different processes. Selection may favor a trait, while drift or developmental constraints produce related patterns for other reasons. Historical contingency and the genotype-to-phenotype mapping further complicate inference. The Law would enter evolution only as an additional candidate constraint, requiring a named process and distinctive predictions rather than an inferred design (Gould & Lewontin, 1979; Pigliucci & Müller, 2010; West-Eberhard, 2003). Such a constraint must explain how present processes could couple to future closure without attributing foresight to natural selection (Darwin, 1859; Tinbergen, 1963).

22.1.1 — Selection in a Changing Population

A population can approach a stable mixture because the success of a type depends on how common it is. Consider two inherited types, A and B, in a large deterministic population model. Let p be the fraction of A. Assign A a reproductive contribution proportional to 2 − p and B a contribution proportional to 1 + p. These are stipulated rates in a mathematical model of frequency-dependent selection.

When A is rare, its rate is close to two while B's is close to one. As A becomes common, its rate falls and B's rises. At an equal mixture, both rates are one and a half. Each type's relative success depends on the population in which it participates, rather than being a permanent ranking attached to its label.

The next generation's A fraction is its current fraction times its rate, divided by the whole population's mean rate. Thus the new p is p(2 − p) divided by [p(2 − p) + (1 − p)(1 + p)]. The

denominator simplifies to $1 + 2p - 2p^2$. Normalizing by that total ensures that the two new fractions still add to one.

Starting with $p = 0.2$ gives a next-generation fraction of about 0.273. Starting with $p = 0.8$ gives about 0.727. In both cases the initially rarer type increases. Subtracting the old p from the new one gives $p(1 - p)(1 - 2p)$ divided by the same positive denominator. The sign is positive below one half and negative above one half.

The approach does not overshoot in this construction. The new distance from one half equals the old signed distance divided by $1 + 2p - 2p^2$. That divisor is greater than one whenever both types are present. Each generation therefore brings an interior mixture closer to one half, and the equal mixture is a stable fixed point. If a type is absent altogether, the stated inheritance-only model has no process that reintroduces it; mutation or migration would add a separate mechanism.

The stable mixture has emerged from reproduction at each generation. The update rule contains current frequencies and current rates. It contains no representation of a future population fraction that must be reached. The example shows how a population-level balance can be generated by local interactions whose consequences are visible before equilibrium arrives.

Evolutionary game theory studies this dependence of success on the strategies or types present in a population (Maynard Smith, 1982). The relevant prediction is a response to frequency: change the initial mixture, and the model predicts a particular direction and size of change. A stable final proportion becomes more explanatory when the same rule accounts for those intervening changes.

22.1.2 — Maladaptation and Suffering

Evolution builds workable inheritances from prior conditions, not a clean sheet. Historical constraint, developmental tradeoffs, ecological change, and path dependence can leave costly compromises sufficient for survival or reproduction without improving the bearer's life (Gould & Lewontin, 1979; West-Eberhard, 2003; Nesse & Williams, 1994; Stearns, 1992).

Separate function from felt cost in welfare research (Mellor et al., 2020; Crook, 2021; Crump et al., 2022). Even adaptive competition, attachment, appetite, or vigilance may support reproduction while distressing the organism.

Lineage persistence can coexist with severe individual burden; reproductive failure can coexist with joy, care, and meaning. The proposed terminal condition concerns the feeling stream. Compare reproductive outcomes and experience separately rather than assuming their relationship.

Specify bearer and valence representation; model evolutionary and ecological variables as causes, moderators, or rivals. Test predictions across mismatch, chronic pain, altered environments, reproductive conflict, long recovery, and regulatory failure. Ask which ecological differences favor the candidate mechanism over alternatives.

22.1.3 — Growth Across Changing Conditions

An evolutionary comparison can change when success compounds through time. Imagine two simplified lineages living through alternating good and poor seasons. For this constructed example, population size changes deterministically by a stated multiplication factor each season. The model leaves out density dependence and interactions between the lineages so that the effect of the sequence can be examined on its own.

The steady lineage multiplies by 1.5 in either season. The variable lineage multiplies by three in a good season and by one half in a poor season. Averaging its two seasonal factors gives 1.75, larger than the steady lineage's 1.5. A comparison based on the arithmetic mean of the two factors would favor the variable lineage.

Follow descendants through a complete pair of seasons. The steady lineage multiplies by 1.5 × 1.5, or 2.25. The variable lineage multiplies by 3 × 0.5, or 1.5. Starting with one hundred members, the first therefore has 225 after the pair, while the second has 150. The multiplication joins the seasons in a way that their arithmetic average does not preserve.

After ten pairs, their relative size is determined by (2.25/1.5) raised to the tenth power, or 1.5^{10}, about 57.7. The steady lineage's advantage compounds. It comes from the complete sequence of growth factors, even though its arithmetic average factor is smaller. In this specified environment, the more variable strategy's good season does not supply enough multiplication to offset its poor season.

The geometric mean gives the equivalent constant factor per season. It is the square root of the two-season product: 1.5 for the steady lineage and √1.5, about 1.225, for the variable one. Logarithms express the same calculation additively. The log of a product is the sum of the logs, so average log growth describes how multiplication accumulates over the sequence.

Now introduce a third, mixed lineage. In every generation, a fraction f uses the variable strategy and the remaining fraction uses the steady strategy. Assume that the descendants are combined and the same fraction is established anew each generation. This stipulation is important: the model describes a recurring allocation across strategies within one lineage, rather than two permanently separated subpopulations.

In a good season, the mixed lineage's multiplication factor is (1 − f)1.5 + 3f, or 1.5 + 1.5f. In a poor season it is (1 − f)1.5 + 0.5f, or 1.5 − f. Multiplying gives 2.25 + 0.75f − 1.5f² per pair. The maximum occurs at f = 0.25. At that allocation the two seasonal factors are 1.875 and 1.25, with product 2.34375.

The mixed lineage therefore grows faster across the pair than either pure strategy in this example. Its geometric mean per season is approximately 1.531. The quarter allocated to the variable strategy gains something from the good season, while the majority using the steady strategy keeps the poor-season multiplication above one. The optimum follows from the specified payoffs and environmental sequence; different factors would produce a different allocation.

The construction separates several biological questions. Which environments occur, and in which order? How do the strategies perform within each environment? Can one lineage repeatedly produce a mixture of strategies? What costs accompany that diversification? A causal account can investigate those features individually. The numerical example identifies what their combination could accomplish before any particular organism is proposed as an instance.

It also explains why a trait's function cannot be inferred from its most impressive moment. A strategy may excel in one setting while its descendants depend on what happens in the next. Observing only the good season would reveal the variable lineage's threefold growth and conceal the part of the sequence that reverses its comparative advantage. The appropriate time span belongs to the mechanism being studied.

For the fairness inquiry, this is another reason to give ordinary evolutionary explanations substantial detail. Temporal variability, multiplication, and diversification can create organized behavior without a terminal valence rule entering the

calculation. They enrich the account of which architectures might develop and persist. Any further connection to felt experience would concern how those architectures support the particular processes under study.

The important gain is explanatory resolution. Frequency-dependent selection asks how a strategy's success changes with the composition of the population. This example asks how success multiplies across a changing environment. Both can matter, and a richer model could combine them. Keeping their mechanisms distinct lets comparative research ask which structure the observations actually reveal.

22.1.4 — What Regulation Already Explains

Regulation already explains many forms of biological recovery and adjustment. A new candidate must identify the system that carries its proposed state. It must also show how that system's interactions would generate the predicted relation. Evidence across channels matters because one signal can support several ordinary mechanisms.

A candidate model should distinguish the roles of current options and retained history. A single biological state may carry both: one component can limit reachable actions while another retains cumulative contributions. The empirical question is whether the specified state and its dynamics support the required relation to calibrated experience and the endpoint condition.

Behavioral inhibition illustrates how biological signals can restrict available actions. Fear can inhibit approach, disgust can inhibit consumption, and pain can inhibit movement. Social threat may inhibit speech; fatigue may inhibit effort; uncertainty may inhibit commitment. These are active constraints on the actions available to the organism.

Behavioral inhibition also connects to the future horizon. When repair seems unavailable, an organism may freeze, conserve, avoid, submit, or narrow attention. Threat systems, reinforcement history, predictive processing, and safe-control models can all generate those patterns. The open question is whether any inhibition signature remains after those rivals receive a fair test.

22.1.5 — Development Across Environments

A trait observed in one environment is one outcome of development. Its appearance across environments can reveal a richer pattern. Developmental plasticity concerns the capacity for developmental outcomes to vary with conditions; West-Eberhard (2003) places that relation at the center of her account of development and evolutionary change.

Consider an invented comparison of two populations, A and B, raised under two carefully described developmental conditions. In condition one, each population has a mean trait value of two. In condition two, A has a mean of four and B a mean of six. All values use the same measurement scale. A study conducted only in condition one would find no difference between the populations. A study conducted only in condition two would find a difference of two.

The paired comparison supplies an additional fact. The change between conditions is two for A and four for B. The populations differ in their response across environments, even though they coincide in one environment. The observed group contrast is therefore conditional on where development occurred. The numbers express an interaction: the effect associated with one dimension depends on the level of the other.

One can ask several distinct questions of this small design. Which developmental processes respond to the environmental

difference? At what stage does the divergence emerge? Does changing the environment early have the same effect as changing it later? Does a later return to the first condition restore the earlier trait value, or has development established a more enduring organization? Each question describes a new comparison rather than a new label for the final difference.

The design can also separate development from immediate expression. Measure both groups in a shared observation setting after the developmental conditions have differed. Then vary the immediate setting while retaining the developmental histories. This creates a way to ask whether the observed difference travels with history, appears through current context, or reflects their interaction. Additional design features would be needed to attribute a population difference specifically to genetic inheritance.

For the Law's comparative inquiry, this opens a productive level of explanation between lineage and momentary state. An organism arrives at a task with a developmental history that has helped shape its capacities, expectations, and responses. Studying that history can reveal why superficially similar settings produce different experience-related patterns. It can also identify environmental conditions under which a useful capacity becomes easier to express. The comparison becomes a study of how a living possibility takes form.

22.2 — Predictions and Measurement Bridges

Cross-species predictions begin with the question of which organisms may be bearers. Comparative evidence provides candidate indicators without resolving sentience in advance. Valence measures must then suit each species rather than assume a human report scale (Birch et al., 2020; Andrews et al., 2024; Birch, 2024). Only after those bridges are specified can a proposed constraint face informative tests.

22.2.1 — Bearer Status and Valence Representation First

Establish species-appropriate evidence of valence before attempting cross-species scaling. Stillness may indicate calm in one context and freezing, illness, exhaustion, or concealment in another. Vocalization may indicate distress, social contact, arousal, or species-typical communication. A human observer's intuitive reading is a hypothesis, not a measurement rule.

Cognitive-bias tasks can test how prior conditions alter ambiguous judgment. Combine them with the comparative evidence channels developed in Section 14.1, examining how their different failure modes affect welfare interpretation (Broom, 2014; Fraser, 2008).

Bearer uncertainty and valence uncertainty must also remain distinct. Researchers may have moderate confidence that an animal is conscious while remaining uncertain about the sign or intensity of a particular state. They may understand a welfare effect while remaining unable to compare its scale with human experience.

A comparative program should therefore state what is expected to travel. It might test whether relief follows the removal of an aversive condition, whether enrichment produces behavioral and physiological responses that support the same

welfare interpretation, or whether long-term trajectories differ under matched care.

22.2.2 — Cross-Species Transport Is an Empirical Burden

Recent comparative work shows why transport cannot be assumed from a familiar word such as fairness. A 2024 meta-analysis of accept-or-reject paradigms reported no general evidence for inequity aversion across nonhuman animals. A 2025 bonobo study found aversive responses to unequal rewards, while a 2026 chimpanzee group study found that food quality and social relationships altered the response. Species, task structure, reward quality, partner relationships, and participation conditions can therefore change what the same experimental label appears to measure (Ritov et al., 2024; Radovanović et al., 2025; Martinez et al., 2026).

A defensible bridge would declare bearer criteria, species-specific indicators, welfare context, behavioral alternatives, exposure window, latent construct, observation channel, scale property, and uncertainty set. It would then test whether that same construct survives across sites and tasks. When only an ordering or directional effect is defensible, the result must remain at that level rather than being rewritten as a common cardinal ledger (Mendl et al., 2010; Sneddon et al., 2014; Solvi et al., 2016).

A route based on conserved architecture should predict a specified cross-species, developmental or ecological contrast and state which result would narrow it.

Comparative disagreement is productive when it reveals which assumption moved. A meta-analytic null, a species-specific positive result, and a context-dependent group result can coexist without contradiction if they answer different questions under different designs.

The neutral origin is especially difficult to transport. A midpoint chosen from human language, facial expression, or preference may not mark experiential neutrality in a bird, cephalopod, fish, insect, or mammal with a different sensory world. Even within one species, domestication, development, social position, injury, and environment can alter what a signal means.

Cross-species transport also requires sensory-appropriate tasks, as Section 14.1 explains; document which parts of the species' perceptual world the protocol leaves out.

A choice task can look simple while offering several ways to solve it. Imagine two containers in a familiar enclosure. During learning, the container associated with a preferred item is always striped, carries a particular scent, and stands on the left. Repeated selection of that container shows a reliable pattern. It does not yet reveal which of the three cues organizes the choice.

The next trials can separate them. Exchange the containers' positions while preserving their visual and scent cues. Continued selection of the striped, scented container now points away from a simple leftward routine. In another comparison, place the familiar scent on the unstriped container. A choice following the scent differs from one following the visual pattern. A further trial can present each cue without the others and reveal whether the learned relation transfers.

These comparisons are especially useful when species differ in their access to the available cues. An apparatus designed around the researcher's most obvious signal may still be solved through another one. The learner's success can then be more interesting than the experimenter's initial description: it may show skilled use of information the design treated as incidental.

The task becomes clearer when cue identity, spatial location, and the preferred outcome are recorded separately. Learning

speed under the combined arrangement can be compared with choice when the cues disagree. If a subject follows location during conflict, the next design can vary location throughout learning. If scent dominates, the design can test whether scent tracks the outcome across new visual forms. Each revision answers a question raised by the preceding pattern.

For the comparative program, the gain is an interpretable bridge between species-specific abilities and a shared learning question. Different species need not encounter visually identical apparatus to be tested on the same relation. They need access to the relevant distinction and an observable opportunity to express what they have learned. A good comparison makes those relations explicit enough that success, transfer, and surprise can each teach us something.

Social structure also changes transport. Solitary animals, pair-bonded species, eusocial colonies, fission-fusion groups, dominance hierarchies, and cooperative breeders face different sources of safety, deprivation, control, and loss. A social-isolation manipulation cannot be interpreted uniformly across those forms. Even a preference test may express affiliation, threat avoidance, territoriality, or learned dependence. The bridge must therefore connect the behavior to a species-appropriate causal account before it connects the behavior to valence.

Missingness also changes across species. The least observable moments may be the most relevant ones: hiding, nocturnal activity, migration, torpor, illness, predation, solitary recovery, and the final stages of life. Captive observation can improve visibility while changing the very environment that shapes stress and agency. Field observation can preserve ecology while losing internal resolution. A serious measurement bridge must record

what could not be seen and how the observation setting may have altered the bearer's available paths.

Comparative protocols should justify sampling grain and duration: a brief assay may miss delayed recovery or a seasonal rhythm, while a long window can average away short, intense states.

Comparative samples also inherit phylogeny. Closely related species are not independent replicates, and similarity may reflect shared ancestry, while distant resemblance may reflect convergence. Homology can support a biological bridge; analogy can motivate a test. Neither identifies phenomenal equivalence, so the analysis must declare its phylogenetic structure and alternative evolutionary histories (Futuyma & Kirkpatrick, 2017; Godfrey-Smith, 2009; Andrews et al., 2024).

Imagine a comparative sample of eight species. Six belong to one recently branching lineage and display a particular behavioral pattern. Two more distant species do not display it in the same task. It is tempting to describe the result as six confirmations against two exceptions. The family tree changes what that count means. The six positive cases may all have inherited the relevant feature from one common ancestor.

The observed pattern is compatible with a single origin followed by inheritance. It is also compatible with other histories, depending on what is known about ancestral states and unsampled branches. Six present-day observations therefore contain information about six species, while supplying a different amount of information about the number of evolutionary origins. The units of those two questions are different.

Now add a pair of close relatives from another lineage, one with the pattern and one without it. That contrast can be informative because much of their inherited background is

shared. Add several such pairs from separate branches and ask whether the contrast repeatedly accompanies a proposed ecological feature. The investigation has moved from counting resemblances to comparing changes against their historical background.

The same care improves interpretation of a negative result. A species may lack the behavior, lack the sensory route used by the task, or express the relevant capacity under another ecological condition. The family tree organizes relationships among species; the task analysis organizes relationships between a species and an observation. Both structures help explain why a pattern appears where it does.

This can make comparative work more exciting rather than less decisive. A resemblance across distant branches may motivate a search for convergence on a common functional problem. A difference between close relatives may point toward a change in development, ecology, or social organization. The strongest next case is often the one that separates those histories. Its value comes from where it sits in the comparison, not simply from making the list of species longer.

Life history creates another scaling problem. A response measured across minutes in an insect, months in a rodent, and decades in a human does not automatically represent the same fraction of a lived trajectory. Developmental stage, reproductive cycle, dormancy, migration, and senescence alter both exposure and opportunity. The bridge must justify its clock before it compares accumulated experience across organisms with radically different temporal architectures (Stearns, 1992; West-Eberhard, 2003).

A failed human-designed task may reveal poor motivation, incompatible senses, unfamiliar equipment, stress, or a mistaken theory of the species. Repeated failure is more informative when

independent tasks and ecological contexts address those alternatives; the result can then narrow the transported claim.

22.3 — Fitness, Feeling, and Moral Worth

Fitness, feeling, and moral worth answer different questions. Fitness concerns differential reproductive contribution in an environment. Feeling concerns the valenced quality of experience for a bearer. Moral worth concerns how beings should be regarded and treated, not how successfully their genes propagate.

22.3.1 — Fitness Can Coexist with Felt Harm

Fitness also cannot settle endpoint discipline. Reproduction may occur long before the irreversible end of the qualifying conscious stream.

In animal contexts, survival metrics can be especially misleading. A captive animal may survive with poor welfare. A farmed animal may grow efficiently while experiencing stress or pain. A wild animal may live longer under protection while losing species-specific freedom. Survival data need welfare context.

Pregnancy, mating competition, parental investment, and sexual conflict can contribute to lineage success while imposing unequal burdens on individuals.

Social traits also carry costs. Sensitivity to rank, exclusion, reputation, or reciprocity may support cooperation in some environments and intensify shame, envy, loneliness, or conflict in others.

22.3.2 — Evolutionary Storytelling Risk

An elegant evolutionary story can outrun evidence: a behavior exists, so it must aid survival; a burden persists, so it must serve a function; balance appears, so it must have been selected. Each inference needs testing (Stearns, 1992; Nesse & Williams, 1994). Specify mechanism, comparative pattern, plausible

selection pressure, alternatives, and limits. Without evidence, label the account a conceptual possibility, not evolutionary fact (Trivers, 1971; Gould & Lewontin, 1979; Maynard Smith, 1982).

Ask what the account would predict if complete-stream closure failed. If its predictions remain unchanged, they do not distinguish the target. If its premises rule that failure out, those premises need independent tests. This separates an explanation that permits closure from one that would require it.

Apply the same scrutiny to evolutionary explanations and the Law. Critics need no complete theory of affect to expose an unsupported claim or failed prediction. Ask which account predicts better and what remains unresolved.

22.4 — What Cross-Species Failure Would Mean

A single striking species result is easy to overinterpret. Differences in handling, environment, or observer expectations can produce patterns that look like mechanism. Replication under independently specified conditions tests whether the signal survives those details. A repeated failure across suitable species would narrow the reach of the proposed constraint.

22.4.1 — Systematic Imbalance Cases

A distinctive behavior may be common in one population and absent in another; a welfare indicator may reverse under different husbandry; an instrument may work only in one sensory ecology. Independent teams should therefore be able to reproduce both the measurement bridge and the imbalance pattern. Failure to reproduce either one narrows what can honestly be said, even when the original case remains ethically important.

A group mean near zero can conceal nonzero terminal residuals in individual qualifying streams. Inspect the distribution of complete-stream estimates and their uncertainty, including subgroup and ecological patterns. Keep severe burden and life-stage differences visible in welfare analysis, while testing terminal closure with the declared phenomenal representation and endpoint rules. A summary should not silently treat intensity, duration, and frequency as interchangeable.

A second failure pattern would be a predicted Law-specific pathway constraint remaining absent under measurement sensitive enough to detect it, within the declared model’s scope.

Such a result would defeat that proposed mechanism as an explanation of the Law within the tested scope.

Specify the predicted restriction on available paths and the observation that distinguishes it from ordinary action selection, local regulation, and opponent processes.

22.4.2 — Species Variation and Ethical Testing Limits

Stable, adequately measured violations of a prediction within its declared species scope would defeat that scope claim. Ordinary differences in traits do not by themselves establish such a violation.

Institutional convenience is a powerful welfare confound. Laboratory cages, farms, shelters, sanctuaries, homes, zoos, and field interventions may be easy to manage or observe while imposing distinctive stress, deprivation, or loss of agency. The resulting data can become a record of human systems rather than species nature, and that record should first improve the system rather than elevate the theory.

A defensible dataset records housing, social contact, light, temperature, noise, enrichment, medical status, handling, prior history, observer presence, and sampling time. These records allow analysts to examine how the setting and care conditions relate to the observed welfare pattern.

Context may be the burden being studied. Distinguish nuisance variation from the conditions that make the animal’s experience intelligible before adjusting it away.

Responsible carers and independent welfare reviewers must have effective authority to withdraw an animal. A safer alternative becoming available or an unexpected sign of possible sentience should trigger renewed review of the design.

Lower-burden alternatives often exist. Naturalistic observation, sanctuary and rehabilitation records, welfare-improving interventions, clinical data generated under care, noninvasive monitoring, preference-based enrichment, archival

review, and simulation can answer narrower questions. Their confounding and missingness must be modeled honestly.

22.4.3 — The Contribution of Choice

An environment can change both what an individual encounters and how much influence the individual has over encountering it. A comparison that keeps those dimensions separate can reveal the contribution of choice itself. Consider a voluntary enrichment task offering two familiar, positively valued activities. The individual can select either activity and can leave the task.

During one session, the individual's selections determine the sequence. During another kind of session, the sequence is drawn from a previously recorded set of selections while the same opportunity to leave remains available. The study records the activity offered, its duration, whether it was accepted, and the behavior associated with beginning and ending it. The purpose of the matched sequence is to make exposure content more comparable while varying the opportunity to select its order.

Matching a sequence does not settle every comparison. A sequence selected by one individual may fit that individual's preferences better than another's. A replay of the same individual's earlier choices may no longer fit current preferences. The design can address these alternatives by collecting current choices, varying order, and comparing several matched sessions. This makes preference fit an observable part of the explanation instead of silently assigning every difference to control.

The result can take several informative forms. If the difference follows preferred content regardless of who selected it, content becomes the leading account. If otherwise comparable content is engaged with differently when its order is chosen, the choice

relation becomes a stronger candidate. If the effect appears only when preferences change during the session, the opportunity to update a choice may be the relevant feature.

These distinctions can guide improvement. Perhaps the useful change is a wider menu, perhaps a way to change the order, perhaps a more readable signal that an activity is available. A study can vary one of these features and observe whether it produces the predicted difference. The setting becomes more understandable through the learner's interaction with it.

The broader attraction is that comparative research can discover capacities while improving the conditions in which those capacities are expressed. Choosing, revising, approaching, and leaving are organized actions that can be described without compressing them immediately into one score. Their relations to other affective and behavioral observations can then be studied. A well-designed positive opportunity supplies information about the organism and about the environment that helps it participate.

Chapter 23 — Could Such a World Exist?

A simulated world is a disciplined thought experiment with memory. It can hold assumptions still, generate many trajectories, reveal hidden dependencies, and let rival mechanisms face the same observations. Used well, simulation makes a theory easier to break. An agent in an agent-based model may be a few update rules attached to a state vector; populations can reproduce, compete or cooperate through those rules (Railsback & Grimm, 2019; Wilson & Collins, 2019). Agents may learn exploit paths, and local successes may shift modeled burden to later states or less visible agents. The architecture determines which of those paths are available.

The terminal-residue constraint considered here restricts the residual permitted at the end of a modeled trajectory. Local patching faces a different coordination problem: a correction must find the burden, arrive in time and remain effective as agents learn its rules. Compensation tied only to visible suffering can miss hidden suffering, while a requirement that every wrong receive a public counterpart makes the model depend on an elaborate public accounting system. A trajectory constraint offers a different architecture to compare.

Verification and validation address different risks. Verification asks whether the code correctly implements the stated model. Validation asks whether the model is adequate for its intended use. A program can be perfectly coded and still answer the wrong question, omit the decisive process, or succeed only inside an artificial world (Oreskes et al., 1994; Grimm et al., 2020).

One elementary verification check is to integrate analytically known toy trajectories using progressively finer time steps. Check the computed totals against the known values and declared numerical error bounds, including cases whose correct

total is nonzero. Repeating a biased calculation with more random seeds would not correct that error.

An exact benchmark can reveal what a numerical total is doing at every resolution. On the interval from zero to one, take the known rate $v(t) = 3t^2 - 1$. Its exact integral is zero because the antiderivative $t^3 - t$ has equal values at the two boundaries. The negative early portion and positive later portion supply a nontrivial closing trajectory.

Divide the interval into n equal pieces. The midpoint rule evaluates the rate at the center of each piece, multiplies by the piece's width, and adds the results. For this particular quadratic, its computed total is minus one divided by $4n^2$. The trapezoidal rule averages the two endpoint values of each piece before multiplying by its width. Its computed total is plus one divided by $2n^2$.

With ten pieces, the midpoint result is −0.0025 and the trapezoidal result is +0.005. With twenty pieces, the results are −0.000625 and +0.00125. Doubling the resolution divides each error by four. The two rules approach the same exact total from opposite sides, with a convergence rate that can be checked directly.

Their arithmetic can even be combined to recover this quadratic exactly. Give twice the weight to the midpoint total and once to the trapezoidal total, then divide by three. The negative error contributes minus one divided by $2n^2$, the positive error contributes the opposite amount, and they cancel. This weighted combination is the composite Simpson construction applied to the endpoints and midpoint of each piece. The cancellation follows from the form of the rate, rather than from an instruction to make its total zero.

A nonclosing control makes that distinction visible. Add a constant one fifth to the rate. Its exact total becomes one fifth.

Both numerical rules also increase by one fifth at every resolution because they integrate constants exactly. Their errors around the new total remain the same, and the combined rule returns one fifth. The procedure therefore preserves the deliberately introduced nonzero contribution as well as the original zero.

The same benchmark can test an entire accumulation pipeline. Change the number of pieces, express the same time interval in different units with the corresponding rate conversion, or divide the interval into separately processed segments. The known total supplies an answer against which each transformation can be checked. A mismatch can then be located in weights, units, boundaries, or implementation.

Different path classes call for different benchmarks. A smooth polynomial permits the exact calculation above. A jump requires the integration rule to represent where the discontinuity falls. An oscillation calls for a schedule that resolves its changes. Choosing a small collection with known totals lets the numerical method demonstrate what it preserves before a simulated world supplies less tractable trajectories.

Different modeling traditions serve different purposes. Agent-based models show interactions and emergence. Dynamical systems expose stability and feedback. Stochastic simulations test variability and rare events. Causal models clarify intervention questions. No one form deserves authority beyond its purpose (Epstein, 2008; Pearl, 2009).

A model must identify what would distinguish the proposed structure from alternatives in independently observed conscious systems. For example, two simulations may both close at zero yet respond differently when available actions change; their intervening paths then supply the useful comparison. A failed simulation can therefore be a success of inquiry. It may reveal

circularity, nonidentifiability, fragility, or an unnecessary mechanism. Preserving those failures gives later investigators specific assumptions and methods to revise. The failed run should remain reproducible, with the parameter settings and diagnostic trace that show where the proposed construction ceased to work.

23.1 — Why Build Simulated Worlds?

A small simulated world can make an abstract relation inspectable. Its rules can be simple enough that every possible history is known. The exercise then reveals which consequences follow from those rules and which would require another assumption. The following world uses three numerical moves; it gives the reader a complete case to carry into the larger questions of simulation and explanation.

23.1.1 — A World with Twenty-Seven Histories

Begin at zero. At each of three steps, choose minus one, zero or plus one, with all three choices equally likely and independent of earlier choices. The running total adds the chosen numbers. There are three possibilities for the first move, three for the second and three for the third, producing twenty-seven equally likely ordered histories.

Exactly seven histories end at zero. One is zero, zero, zero. The other six are the different orderings of minus one, zero and plus one. Their totals agree while their routes differ. One begins positively and returns later. Another begins negatively and returns later. A third remains at zero first and makes its excursion afterward. Order survives even when the endpoint is shared.

The probability of a zero total in the unconstrained world is therefore seven out of twenty-seven. A simulator can reproduce that proportion by repeated random draws, but enumeration has already supplied the exact answer. This makes the world useful for checking a program: the implementation has a known target, and every observed path can be compared with the declared rules.

Now consider only histories that end at zero. Each of the seven retained histories had the same probability in the original

world, so each has probability one seventh within the conditioned collection. The terminal condition has selected a particular distribution over complete paths. It has also changed the distribution of earlier moves.

23.1.2 — How a Boundary Changes an Earlier Distribution

Of the seven zero-ending histories, three begin with zero. Two begin with plus one, and two begin with minus one. The first-move probabilities in the conditioned collection are therefore three sevenths for zero and two sevenths for each nonzero move. In the original world, all three first moves had probability one third.

The reason is the number of available completions. After a first move of zero, the final two moves can be zero and zero, plus one and minus one, or minus one and plus one. After a first move of plus one, only minus one followed by zero or zero followed by minus one can complete the history at zero. The terminal condition gives more weight to a first move that admits more qualifying continuations.

This count supplies an earlier consequence beyond the terminal value itself. In the unconstrained world, a nonzero first move occurs with probability two thirds. In the conditioned collection it occurs with probability four sevenths. Both collections end some paths at zero, but their distributions of first moves differ. The difference is a specific prediction of the defined conditioning operation.

The construction also reveals what a future boundary means mathematically. It specifies a collection of complete histories and their relative probabilities. The calculation can be performed before any simulated path is generated. Questions about how a physical process might realize such a distribution belong to the next explanatory step. The finite world has already done one

useful job: it has turned a broad image of terminal closure into seven explicit paths and a testable relation among their parts.

23.2 — Constraints, Construction and Explanation

The seven-path world makes it possible to compare ways of producing closure. One construction selects complete histories. Another changes the final move. A third generates the conditioned distribution one step at a time. Their similarities and differences can be located in explicit rules, so the explanation becomes a comparison of mechanisms rather than a resemblance between their final totals.

23.2.1 — Changing the Final Move

Start with two independent moves from minus one, zero and plus one, each equally likely. Then set the third move to the negative of the current running total. Every completed history ends at zero. There are nine equally likely histories because the first two moves have nine possible ordered combinations.

Two of these histories use moves absent from the original three-choice world. Plus one followed by plus one requires a final move of minus two. Minus one followed by minus one requires plus two. The repair has enlarged the permitted final action. It produces closure by giving the last step enough range to cancel either of those earlier totals.

The other seven histories are the seven zero-ending paths counted in Section 23.1. In the repair model, each has probability one ninth, while each had probability one seventh in the conditioned collection. The two extra histories account for the remaining probability. The first move also retains its original probabilities of one third each, unlike the first-move distribution in the conditioned collection.

These differences make a useful model comparison possible even though both models close exactly. An observer could examine how often a final move has magnitude two or estimate the frequencies of first moves across repeated runs. The relevant

evidence lies in the path distribution and the available actions, as well as in the shared endpoint. In a richer candidate mechanism, the analogous questions concern what capacity makes an adjustment possible and what earlier observations that capacity predicts.

23.2.2 — Generating the Same Histories Step by Step

The conditioned collection can also be generated sequentially. Choose the first move with probabilities two sevenths for minus one, three sevenths for zero and two sevenths for plus one. If the first move is zero, choose the second uniformly from the three original options. If the first move is plus one, choose the second uniformly from minus one and zero. If the first move is minus one, choose the second uniformly from zero and plus one. Finally, choose the unique third move that returns the total to zero.

Each of the resulting seven histories has probability one seventh. For a history beginning with zero, three sevenths multiplied by one third equals one seventh. For a history beginning with either nonzero move, two sevenths multiplied by one half also equals one seventh. The final move is determined. Every move remains within the original three-choice set.

The whole-history selection algorithm and this sequential algorithm therefore generate the same distribution over all three-step paths. An analyst given only those paths cannot determine which of the two algorithms produced them. Even complete observation of the simulated trajectories leaves that implementation question open because the probability assigned to every possible trajectory is identical.

This is a productive kind of equivalence. It tells the investigator which information the path record contains and what a further explanation would have to add. Inspecting the

generator's internal state or its rule for counting remaining completions supplies a different kind of observation. A prediction under a changed task would need the corresponding rules of both algorithms to be specified. The small world thus supports two advances at once: it distinguishes the final-step repair from the conditioned path distribution, and it identifies two implementations that the original path observations treat as equivalent.

The sequential construction extends to any finite number of steps through a simple recursion. Let N(n, x) count the permitted completions from current total x when n moves remain and the final total must be zero. Each next move is minus one, zero, or plus one. With no moves remaining, there is one completion when x is zero and none when x is different from zero.

With at least one move left, every completion begins with one of the three possible moves. The count is therefore the sum of N(n − 1, x − 1), N(n − 1, x), and N(n − 1, x + 1). This rule builds the entire table from its last row backward. It counts paths without first generating all of them separately.

Starting from zero with two moves left, the count is three. Starting from plus one or minus one, it is two; starting from plus two or minus two, it is one. With three moves left at zero, the count becomes two plus three plus two, or seven, recovering the earlier enumeration. With three moves left at plus one, the count is three plus two plus one, or six. Symmetry gives the same six at minus one.

The four-step zero-start count is therefore six plus seven plus six, or nineteen. Its first move is zero with probability seven nineteenths and each nonzero move with probability six nineteenths. These differ from the three-step probabilities even though the available moves and final total are unchanged. The

number of remaining steps is part of the state needed by this generator.

At any state with a positive completion count, choose a next move with probability equal to the number of completions after that move divided by the current count. A move with no admissible completion receives probability zero. The denominator is the sum of the three numerators, so the probabilities add to one. Every selected move therefore leads to a state from which at least one completion remains.

The probability of a complete path can be checked by multiplying its transition probabilities. The first factor is the next state's count divided by the starting count. The second factor has that same next-state count in its denominator. Successive counts cancel all the way along the path, leaving one divided by the starting count. Every permitted completion receives the same probability, exactly as it does when equally likely independent paths are conditioned on the terminal total.

This is dynamic programming in a particularly transparent form: solve the remaining subproblems once, store their answers, and use them to guide each next step. It replaces repeated rejection with a table of future completion counts. For a large horizon, the table can be far smaller than the collection of complete histories it represents.

The calculation also identifies what a corresponding local implementation would need. It uses the current total, the number of moves remaining, the permitted increments, and the completion-count rule. A changed horizon or action menu changes the table and the transition probabilities. Those changes provide further predictions for comparing implementations. The finite world has moved from an explicit list of seven paths to a general recipe whose information requirements and distribution can both be inspected.

23.2.3 — What the Displayed Result Leaves Outside

A patch in a simulated world can improve what the designer measures while shifting costs to unseen agents, peripheral populations, hidden memory states, or a later interval. A benefits system may appear to shorten its queue by requiring applicants to complete costly paperwork before their application officially starts. The evaluation must therefore trace displaced burdens across agents and hidden states rather than reward a smaller displayed number. Complexity is justified only when a component earns its place through independent evidence, better prediction, improved recovery, or a fairer rival comparison. Robustness checks should vary plausible parameters, endpoints, observation floors, and missingness assumptions, reporting where claimed neutrality fails (Box, 1976; Railsback & Grimm, 2019).

23.2.4 — No Neutrality by Fiat

Simulation gives the designer access to hidden states, but the analyst should see only the declared observation channels. Termination, merging, and disappearance are different events. Their effects on the measured population must be recorded separately. Otherwise selection can create an apparent concentration near zero even when no simulated path obeys a boundary constraint.

A model may guarantee closure throughout its declared class. That can be a useful theorem about the model, whether closure was imposed directly or follows from its component rules. Circularity enters when the guaranteed result is presented as independent evidence that the same rule governs conscious experience.

If a program chooses its endpoint, neutral origin, bearer rule, or observation filter solely to make the residual vanish, closure

is built into the measurement. A physical endpoint may instead be coupled to a system's state. That coupling requires independent biological or other evidence and must predict observations not used to construct it.

A circularity audit should trace closure through preprocessing, initialization, endpoint definition, normalization, missing-data handling, objective function and stopping rule. It should identify what was assumed, what was derived, and what was tested against independent observations. The critical question is where the argument acquires evidence about reality, rather than another restatement of the planted condition (Palminteri et al., 2017).

Rules can be tested through consequences beyond the condition that motivated them. A model built to close may still predict a particular response to changed inputs, a required memory capacity, or a relation between independently measured variables. If another model closes through different machinery, those further predictions may distinguish them. If no available observation separates them, the result remains unresolved.

The same separation applies to consciousness. If the model calls an agent conscious only when it satisfies fairness closure, it has collapsed bearer status into the outcome. Bearer criteria must be stated independently, or the simulation can never show a conscious stream failing the Law.

Determine whether the code failed, the analysis selected the wrong generator, or the proposed mechanism proved unnecessary. Distinguish those outcomes from a mechanism that is present but cannot be identified through the chosen observations.

23.2.5 — Looking Inside Equivalent Generators

The equivalent generators in this section can be separated by an observation outside the accepted path. Count how many

attempts each needs to produce one completed three-step history. The count reveals work performed by the algorithm that the final history does not display.

First, use rejection sampling. Draw each of three moves independently from minus one, zero, and plus one, then retain the history only if its total is zero. There are twenty-seven equally likely histories and seven accepted ones. The probability of acceptance on an attempt is seven twenty-sevenths. Independent repeated attempts therefore require an average of twenty-seven sevenths of an attempt per accepted history, about 3.86. Each of the seven retained paths remains equally likely.

Second, draw two moves independently, set the third to cancel their sum, and repeat the attempt if that third move would be outside minus one, zero, and plus one. Seven of the nine possible pairs yield an allowed third move. The average number of attempts is nine sevenths, about 1.29. Again, each of the seven accepted histories is equally likely: the two excluded pairs are precisely those that would require a final move of magnitude two.

Third, use the sequential probabilities derived in Section 23.2.2. Each first and second move is chosen with regard to the number of permitted completions remaining. The final move is always available. Every attempt succeeds, and the mean number of attempts is one. The algorithm performs different bookkeeping in exchange for avoiding rejection.

The complete accepted histories have the same distribution under all three procedures. The attempt counts do not. Under the first procedure, the probability that the very first attempt succeeds is seven twenty-sevenths; under the second it is seven ninths; under the third it is one. An observer who records only accepted histories has one identification problem. An observer

who can also record attempted histories or restart counts has another.

This explains what it means to add a genuinely new observation. Repeating the same accepted-path measurement cannot distinguish distributions that are identical by construction. Recording a restart changes the observable object. A real mechanism study can look for an analogous trace: an intermediate state, a resource use, a recovery interval, or a response to a changed input that its candidate accounts treat differently. The small world's value is that the missing distinction can be named and measured exactly.

23.3 — What Simulations Can Recover

A useful simulation is transparent enough for others to criticize. Its entities and state variables need explicit definitions. Scales, scheduling, initialization, and submodels should be declared before examining outcomes. The ODD framework offers a reproducible structure for making those commitments inspectable (Grimm et al., 2020).

23.3.1 — Recovery Tests the Method Against Known Truth

Parameter recovery asks whether known generating values can be estimated under declared noise and missingness. Model recovery asks whether the analysis can distinguish worlds generated by different candidate mechanisms. The second is the central identification task here. A method that estimates parameters accurately inside every model but repeatedly selects the wrong model cannot support a Law-specific interpretation.

Predictive calibration and mechanism discrimination answer different questions: a calibrated predictor need not identify a unique mechanism, and successful synthetic discrimination need not imply calibration on natural data. Report proper scoring and coverage separately from recovery (Gneiting & Raftery, 2007). Use the complete planned preprocessing, exclusion, transformation, tuning and decision pipeline in the recovery test; an easier substitute can miss the operation creating false support (Palminteri et al., 2017).

Report the confusion matrix, including false support for PTVC in rival-generated data, missed detection in PTVC-generated data, and appropriate abstention when the available observations cannot distinguish generators. Preserve the recovery evidence with code, random seeds, model versions, and diagnostics so another team can reproduce the test.

A simulation comparison can learn more from the same number of runs when competing methods face matched conditions. Suppose two estimators, A and B, are evaluated on the same artificial history. A difficult history may make both estimates less accurate; an easy history may help both. Their difference on that history can be more stable than either error considered alone.

The arithmetic is direct. Let A's performance value and B's performance value each have variance four across simulated histories. If independent histories are used for the two methods, the variance of A minus B is eight. If matched histories produce covariance three between their performance values, the variance of the difference is four plus four minus twice three, or two. Matching has reduced the variance of the comparison to one quarter of its independent-run value.

Across n independent matched pairs, the variance of the average difference is two divided by n. With independent histories for every method it would be eight divided by n. Under these stipulated conditions, the matched design reaches the same variance with one quarter as many pairs. It has achieved that gain by canceling common difficulty, not by suppressing variation in the simulated world.

A simple construction makes the role of matching visible. Let U be uniformly distributed between zero and one. Method A succeeds when U is below 0.8; method B succeeds when U is below 0.6. A succeeds with probability 0.8 and B with probability 0.6. On the shared draw, the success difference is one only when U falls between 0.6 and 0.8, and is zero otherwise. Its mean is 0.2 and its variance is 0.16.

With independent draws for the two methods, the difference still has mean 0.2. Its variance is 0.8 times 0.2 plus 0.6 times 0.4, or 0.40. The shared construction preserves both success

probabilities and the expected contrast, while reducing the contrast's sampling variance. It also makes the favorable and unfavorable cases comparable: both methods encounter the same position of U within the stipulated environment.

Matching is useful when the common random inputs represent a coherent joint experiment. In a recovery study, the shared history might include the same latent trajectory, observation times, missing intervals and sensor disturbances. Each candidate estimator receives the observations allowed by the design. Additional randomness internal to an estimator can be represented separately and repeated when it contributes materially to performance uncertainty.

The sign of the covariance determines the benefit. Positive covariance reduces the variance of a difference; negative covariance increases it. A shared sequence of random numbers is therefore not automatically an efficient coupling. The simulation must specify how each random input enters each model and inspect the resulting dependence. Simply assigning the same random seed to two programs with different sequences of random-number calls need not create the intended matching.

This gives the model-recovery program another reproducible component. Preserve the generated histories and their identifiers, evaluate all eligible methods on the declared common cases, and estimate uncertainty from independent pairs or blocks of pairs. The resulting contrast answers a clear question: when the environment changes in the same way for both methods, which one extracts the intended information more accurately? The comparison becomes sharper because common conditions are made explicit.

The size of a recovery benchmark determines the precision of its reported error rate. Suppose a fixed analysis is evaluated on n independent simulated datasets from one declared generator

regime. Let p be its probability of a specified error in that regime. The chance of observing no errors is $(1 - p)^n$.

After zero errors, a one-sided 95 percent exact binomial upper confidence bound for p is 1 − 0.05 raised to the power 1/n. The expression is obtained by finding the error probability for which zero observed errors would occur with probability 0.05. It belongs to a procedure with the stated repeated-sampling coverage, rather than assigning posterior probability to p.

With one hundred independent runs and no errors, the upper bound is about 0.0295. With one thousand runs, it is about 0.00299. The same observed error count, zero, therefore supports very different resolution depending on how many genuine opportunities for error were evaluated.

The calculation can guide the benchmark size in advance. To place the zero-error upper bound below one percent, the inequality is $(0.99)^n < 0.05$. Two hundred ninety-nine independent runs suffice. For a smaller target rate, more runs are needed. The desired resolution supplies a reason for the count rather than leaving it to the convenience of a round number.

An independent run regenerates the components specified by the benchmark: latent paths, observations, missingness, endpoints, and any random component of the analysis relevant to its intended performance. That definition makes the effective number of trials inspectable and connects each trial to the error probability being estimated.

The error definition also determines what is being estimated. False support under a rival generator, missed recovery under the candidate generator, and an appropriately unresolved result are different categories. The bound can be calculated separately for a prespecified category within a generator regime. A larger collection of easy regimes does not by itself sharpen the estimate for a difficult regime that received few runs.

Suppose a recovery benchmark contains one hundred simulated datasets: ninety generated by model A and ten by model B. A method that always answers A is correct on ninety datasets. Its overall accuracy is ninety percent, but it recovers none of the B-generated datasets. The headline score has described the composition of the benchmark as much as the method's ability to discriminate its generators.

Now consider a second method. It identifies eighty-one of the ninety A datasets and eight of the ten B datasets correctly. Its overall accuracy is eighty-nine percent, one point lower than the always-A method. Yet it recovers ninety percent of A cases and eighty percent of B cases. If the research question requires distinguishing both generators, those separate rates reveal a capability absent from the first method.

The full count table explains another useful quantity. The second method labels seventeen datasets B: eight are genuinely B-generated and nine are misclassified A-generated cases. Thus eight seventeenths of its B labels are correct in this benchmark. That fraction differs from the eighty-percent recovery rate within true B cases because the denominators answer different questions. One begins with the generating model; the other begins with the method's answer.

Change the benchmark composition while keeping the method's two conditional recovery rates fixed. With fifty A datasets and fifty B datasets, the expected correct counts become forty-five and forty. Of the forty-five expected B labels, forty are correct and five are errors from A. The reliability of a B label has changed because the mixture of worlds changed. The conditional rates were held constant in this calculation; an actual change in the difficulty of the cases would require those rates to be measured again.

This gives benchmark design a positive purpose. A balanced collection can reveal whether the method handles both candidate families. A collection reflecting a specified use case can describe what its answers would mean under that mixture. A collection concentrating on difficult boundaries can reveal where the method needs more information. Publishing the counts makes each purpose visible and lets readers recompute the quantities relevant to their own comparison.

Simulation-based calibration can test whether a Bayesian inference procedure is self-consistent when data are generated from the declared model. That is a powerful check on implementation and approximation, but it remains conditional on the generative model used for the test (Talts et al., 2018; Cranmer et al., 2020).

A useful calibration battery also checks whether inference uses the observations. Across prior-predictive simulations, a program that returns independent draws from the prior can produce uniform ranks for parameter-only test quantities while ignoring every dataset. Modrák et al. (2025) show why data-dependent quantities, including the joint likelihood, can expose such failures. The test quantities must be chosen for the errors the comparison needs to detect.

An invented Gaussian example makes the gain concrete. Let θ have a normal prior with mean zero and variance one, and let Y = θ + ε with independent normal noise of variance one quarter. The correct posterior mean is 4Y/5; its mean squared error over these simulations is one fifth, compared with one for always returning the prior mean zero. A correct posterior sampler and a prior-only sampler can nevertheless both pass the parameter-rank check. Add a data-dependent discrepancy and a task-relevant accuracy check, then test unused predictions under

capable alternatives. Calibration, learning and model adequacy become separate, inspectable achievements.

Posterior simulation-based calibration adds a further refinement by asking whether inference remains calibrated in the region made relevant by observed data, rather than only across broad prior draws (Säilynoja et al., 2026). That can expose failures hidden by an average over irrelevant simulated datasets.

Held-out transfer is the next gate. A recovery method tuned on one set of generators, endpoints, tasks, or observation patterns should face new modules whose truth remains concealed from the analyst until scoring is complete. Performance should be reported by model family, stream type, missingness pattern, endpoint process, and severity of distribution shift. Aggregate accuracy can conceal exactly the cases that matter most.

Open-set adequacy matters because every prospectively specified model may be wrong. The simulator bank should include worlds generated outside the fitted family and require abstention when none of the candidates is adequate. A method forced to choose between PTVC and a few rivals can produce unwarranted confidence when the true process is absent from the comparison.

23.3.2 — Finding Rare Histories Efficiently

Some histories are important to a model comparison precisely because they are unusual. A long sequence may need many conditions to coincide before a candidate mechanism and its rival give visibly different outcomes. A simulator that samples only in the most direct way can spend nearly all its runs on ordinary cases. A different sampling scheme can visit the informative region more often while preserving a calculation about the original world.

Consider ten independent fair binary draws. Call the outcome one a success and ask for the probability of ten successes. There is one all-success sequence among 1,024 equally likely sequences, so the target probability is 1/1,024. A direct simulator records one when it sees that sequence and zero otherwise. The average of those indicators estimates the target probability.

Now build a proposal simulator in which each draw has success probability 0.9 instead of 0.5. The all-success sequence appears with probability 0.9 raised to the tenth power, approximately 0.349. It is now encountered often enough to examine in detail. The proposal world is deliberately different from the target world, so its raw event frequency answers a different question.

The correction follows from the probability of each complete sequence. For a sequence with k successes and ten minus k failures, its probability under the target is 0.5 raised to the tenth power. Under the proposal it is 0.9 raised to k times 0.1 raised to ten minus k. Give that sequence a weight equal to target probability divided by proposal probability. This is an importance weight.

For the all-success event, only one sequence contributes. Its weight is (0.5/0.9) raised to the tenth power. Return that weight when the proposal produces ten successes and zero otherwise. The expected returned value is the proposal probability 0.9 to the tenth power multiplied by the weight. The proposal factors cancel, leaving 0.5 to the tenth power, exactly the target probability.

The variance improvement can also be calculated. Write p for the target event probability and q for its proposal probability. The weighted indicator has value p/q with probability q and zero otherwise. Its variance is $p^2(1/q - 1)$. The direct indicator has

variance p(1 − p). With p = 1/1,024 and q = 0.9 to the tenth power, the weighted variance is about 0.00183 times the direct variance. Under these stipulated rules, the same number of independent runs gives a much more precise estimate of this particular rare event.

The proposal was chosen for a specific task. A proposal that favors successes is useful for the all-success event. It would be poorly suited to studying the all-failure event, whose proposal probability is 0.1 to the tenth power. Rare observations of that event would carry very large weights. Sampling efficiency therefore depends on the match between the proposal and the quantity being estimated, not simply on how different the proposal is from the target.

Every target history relevant to the estimate must remain possible under the proposal. If the proposal assigns probability zero to an event that has positive target probability, no finite weight can recover that event from proposal draws. In the binary example, choosing 0.9 retains all possible sequences. Choosing probability one would erase every sequence containing a failure and would support only a narrower calculation.

A more complex simulator can apply the same idea sequentially. At each step, it records the probability of the chosen transition under both rules. Multiplying their ratios gives the weight of the complete sampled history, when the proposal and target have the required common support. The calculation then follows the actual path generated. An adaptive proposal can be useful when its conditional transition probabilities are recorded accurately enough to compute that path weight.

For a closure program, this creates two complementary uses of simulated worlds. Ordinary sampling describes typical behavior. Deliberately tilted sampling explores rare histories and, with the appropriate weights, estimates how much probability those

histories carry in the original model. The same unusual path can be studied closely without pretending it is common. Its explanatory interest and its frequency become separate quantities that the simulation can report together.

23.3.3 — Alternative Worlds and Nonunique Explanations

Causal interventions can improve discrimination when they are safe and well defined. If two worlds fit passive data but make different predictions when an input, observation schedule, or environmental condition changes, the divergence can identify where the mechanism acts. The intervention must not be chosen after seeing which model needs help (Pearl, 2009; Woodward, 2003).

Alternative worlds should also test observation error. A latent process may differ while the measured output looks the same because reports saturate, sensors miss low-amplitude states, or missingness depends on distress. A strong recovery study varies both the world and the observation channel rather than assuming the data reveal the latent object directly.

A compact benchmark can vary two properties separately: whether the complete state-change relation holds and whether the boundary Q-values match. With both properties, totals close; with only the relation, totals equal the boundary difference. Matched boundaries alone permit extra contributions and nonzero totals.

Use common observation rules, including bridges conditioned on zero and on a fixed nonzero total. Include rejoining paths with unused restrictions, random departures whose means cancel, and nonzero sources whose filtered sums vanish. Test what independent error bounds still permit. Keep relation recovery, calibrated terminal level, and observation-model fit

separate; neither concentration nor one closing path identifies the generator (Wilson & Collins, 2019).

23.4 — What Would Make the Model Unnecessary?

Complexity alone cannot justify a terminal constraint. Ordinary adaptation might reproduce the target trajectory. Homeostatic or predictive regulation could do so as well. If a prespecified rival explains the data without terminal residue, the more elaborate model has not earned the extra assumption (Busemeyer & Diederich, 2010; Woodward, 2003; Weisberg, 2013).

23.4.1 — No Need for Constraint

Baseline adequacy should be tested before the favored model is compared. Prior-predictive runs can reveal whether the baseline generates plausible lifespans, observation patterns, burdens, and endpoints at all. A weak baseline is not strengthened by serving as the only opponent to a more elaborate Law-linked model (Gelman et al., 2020; Box, 1976).

A model comparison has two levels: how candidates rank against each other, and what each candidate can account for. A small synthetic process makes the distinction concrete. Imagine an output that repeats the four-step cycle zero, one, zero, minus one. One candidate predicts zero at every step. Another predicts one at every step. Over a complete cycle, their sums of squared prediction errors are two and six, respectively. The constant-zero candidate wins this comparison decisively as further identical cycles arrive.

Yet the winning prediction omits the cycle. Its errors repeat zero, one, zero, minus one. More data make the relative advantage over the constant-one candidate increasingly clear, while also displaying the same unrepresented temporal structure. The winning model estimates the cycle's average correctly. It has not explained the changing output around that average.

A third candidate can represent the four phases. In this invented process, a phase-aware rule that predicts the specified

sequence has zero error when initialized at the known phase. The fair evaluation is prospective: define the rule and its initialization, then score subsequent cycles. The new candidate succeeds by representing a feature the first two candidates omit, rather than by choosing a more favorable constant.

This does not make the constant-zero prediction useless. If the target is the mean over a complete four-step cycle, it gives the right answer. If the target is the next output at a known phase, the phase-aware model provides information the constant cannot. Adequacy therefore belongs to a stated target and observation regime. The same model can serve one purpose well while leaving another purpose unresolved.

For the simulation program, this example gives residual inspection a constructive role. Residuals can point toward the next useful model feature: phase, dependence, heterogeneous responses, or a state transition omitted from the present candidate set. Researchers can then declare a new comparison and evaluate fresh predictions. Relative ranking tells them which current candidate performed best; the structure left in the record helps them decide what to investigate next.

A no-constraint result may still teach by showing which failures were artifacts of earlier assumptions. Perhaps exploit paths appeared only because agents were too simple. Perhaps patch fragility appeared only because repair mechanisms were badly designed. Perhaps permanent deficit appeared only because endpoints were artificial. Learning this narrows the inquiry.

Other constraints may reduce instability or suffering more effectively than terminal closure: a burden cap, local repair rule, reserve protection, memory-integrity rule, consent constraint, welfare floor, stochastic reset, or predictive-control boundary.

23.4.2 — Model Too Simplified

A model that omits disability, care time, culture, or terminal vulnerability may appear adequate for people and circumstances it does not represent (Sutton & Barto, 2018; Railsback & Grimm, 2019; Wilson & Collins, 2019).

Oversimplification can also enter through aggregation. A single reward variable may conceal fear, relief, agency, bodily pain, social loss, and uncertainty; one representative agent may conceal subgroups with different burdens; one endpoint may conceal competing stopping processes. The model should identify which distinctions it deliberately removes and which claims become unavailable as a result (Levins, 1966; Weisberg, 2013).

Simple models can isolate a mechanism with unusual clarity. Their abstraction is useful when it removes detail irrelevant to the particular relation being studied.

Another test asks whether an omitted mechanism is irrelevant detail or a causal substitute. If adding memory reconstruction, endpoint selection, social transfer, or observation loss can reproduce the result without the advertised constraint, then the omission favored one explanation. Ablation and counterfactual intervention are needed to show what actually carries the effect (Woodward, 2003; Guest & Martin, 2021).

The remedy is layered modeling. Begin with toy clarity, then add bearer uncertainty, endpoint ambiguity, missingness, heterogeneous agents, social interaction, and rival mechanisms. Each layer should be allowed to break earlier conclusions.

Sensitivity analysis should therefore vary model form and decision rules, not merely the values inside the favored architecture (Oreskes et al., 1994; Saltelli et al., 2020).

Resolution is part of the model's claim. A mechanism recoverable from complete latent states may become

indistinguishable after sparse sampling, report compression, censoring, or terminal dropout. The final record should therefore state the coarsest observation regime at which discrimination survives and abstain below it, rather than letting a high-resolution simulation authorize low-resolution human inference (Grimm et al., 2020; Railsback & Grimm, 2019).

23.4.3 — Following a Circle One Step at a Time

Imagine a point moving around a circle. Its horizontal coordinate is x and its vertical coordinate is y. At every instant its velocity is perpendicular to the radius: the horizontal rate is $-y$ and the vertical rate is x. Starting at (1, 0), the exact motion is $x = \cos t$ and $y = \sin t$. The point goes around the unit circle at a constant angular speed. This invented system supplies both a complete trajectory and a quantity that can be checked at every step.

The squared distance from the center is $x^2 + y^2$. Its derivative is $2x(-y) + 2y(x)$, which equals zero. Thus the squared distance remains one throughout the exact motion. A simulator can be tested against this result without estimating anything from a graph. The calculation provides a simple question: does the numerical rule follow the circle, move outward, or move inward?

The most direct rule takes the present velocity and follows it for a short time h. It replaces x by $x - hy$ and y by $y + hx$. This is the explicit Euler rule. Squaring the two new coordinates and adding them gives $(1 + h^2)(x^2 + y^2)$. The cross terms cancel, but the two squared velocity terms remain. Each step therefore multiplies the squared radius by a factor greater than one.

With $h = 0.1$, one hundred steps reach time ten. The numerical squared radius is then 1.01 raised to the hundredth power, about 2.705. The exact squared radius is still one. The spiral seen on

the screen comes from the update rule. Nothing in the stated differential equations supplies the outward motion. Repeating the calculation with another random seed would change nothing because this example contains no random draws.

A second rule evaluates the velocity at the new position. It requires x_new = x − h y_new and y_new = y + h x_new. Solving these two simultaneous equations gives x_new = (x − hy)/(1 + h²) and y_new = (y + hx)/(1 + h²). This implicit Euler rule multiplies the squared radius by 1/(1 + h²). At time ten with the same step size, its squared radius is about 0.370. Its trajectory spirals inward.

Both rules become more accurate as the step size becomes smaller over a fixed duration. If the duration is T and T/h is an integer, the first rule's squared-radius multiplier is (1 + h²)^(T/h). Its logarithm is (T/h) log(1 + h²), which approaches zero as h approaches zero. The second rule has the negative of that logarithm. Refinement therefore brings both radii toward the exact value, even though their finite-step errors point in opposite directions.

A third rule averages the old and new velocities. Write a = h/2. Solving the resulting equations gives x_new = [(1 − a²)x − 2ay]/(1 + a²) and y_new = [2ax + (1 − a²)y]/(1 + a²). The sum of the squared coefficients in either row is one, and the rows are perpendicular. This update preserves x² + y² exactly in exact arithmetic. The point remains on the circle for any positive step size.

That achievement creates a more interesting test. The third rule rotates the point through an angle 2 arctan(h/2) per step. The exact angle is h. For positive h, the numerical angle is slightly smaller. After one hundred steps of size 0.1, the numerical angle is approximately 9.99168 radians rather than

ten. The radius is perfect while the timing of the orbit is slightly late.

Preserving a quantity and reproducing a trajectory are therefore different accomplishments. If the scientific comparison concerns only the squared radius, the third rule respects that structure. If it concerns the time of arrival at a particular point, the phase difference matters. A numerical method can be well suited to one question while another question requires a smaller step or a different method. The intended observable determines which error is relevant.

The distinction is useful when a simulated world contains a proposed terminal relation. A program might preserve an algebraic total exactly because its update rule was designed to do so. Another might drift away from the total through numerical approximation. To learn what the underlying model implies, the calculation can be repeated across step sizes and checked against analytic cases such as this circle. Position error, invariant error, and arrival-time error can then be reported separately.

This gives verification a constructive role. Start with a system whose answer is known, identify which part of that answer each numerical method preserves, and use the comparison to choose the method for the actual question. A circle is especially instructive because one can follow the source of every discrepancy. The equations, the finite steps, and the displayed path become three connected objects whose relationships can be examined directly.

Chapter 24 — Spiritual Echoes

The fairness question did not begin with this book. Through stories, rituals, philosophies, laws, prayers, and refusals, human beings have asked why suffering falls where it does, whether wrongdoing matters beyond visible consequence, how grief can be carried, and whether repair belongs to the order of life. A serious account should acknowledge that inheritance without conscripting it.

Imagine three neighbors entering a small garden on the morning it opens. Someone has brought a chair for a visitor who tires easily; there is room in the shade, and people who usually hurry past one another have stopped to talk. One neighbor receives the scene as a blessing. Another sees a human achievement whose value needs no larger guarantee. A third is struck by a question: how much of a good life becomes possible when people notice what others need? They share the same morning without sharing an account of ultimate meaning. Their different responses make the scene more interesting, because each directs attention toward something the others might otherwise miss.

Comparison can begin in that spirit of curiosity. What does an idea help a person notice? How does it connect gratitude, responsibility, freedom, or hope? Where does it leave a question open? A religious account may place these questions within a relation to the sacred; a secular account may explore their force within a finite life. Neither description stands for a whole tradition. The pleasure of careful comparison is discovering unexpected affinities while allowing important differences to remain meaningful.

Ritual is one of humanity’s oldest ways of making burden bearable. A ritual does not necessarily explain pain. Often its greatness lies in refusing to explain too quickly. It gathers people through shared words and gestures. A meal, a song, or a

candle can give form to an experience that might otherwise scatter a person. It does not solve grief by argument. It gives grief a room.

The same pressure persists outside formal religion. Memorials, truth commissions, apologies, grief groups, therapy, restorative practices, and secular moral philosophies ask where burden is carried, hidden, shifted, or repaired. The Law enters this conversation with its own question about the complete course of felt experience.

Pursuing that question is a hopeful task. It invites readers to explore how differently people understand a life that matters, while sharing an interest in what helps people inhabit their lives more fully.

One way to sustain the conversation is to ask each neighbor for a particular rather than a verdict: what, exactly, did the morning disclose? The chair may call attention to hospitality, the planting to patient work, the shade to a good that can be shared. Such particulars give people with different ultimate commitments something intelligible to discuss together. An interpretation becomes richer when another person notices a feature it had passed over. The chapter approaches spiritual echoes through that kind of exchange, where curiosity can deepen understanding without requiring a common creed.

24.1 — Why Fairness Questions Recur

Visible disparities often provoke a question about fairness before any formal theory appears. People may ask it after loss or exclusion. Their communities offer different explanations, rituals, and obligations in response. The recurrence of the question is a human fact worth studying, not evidence that any one account is scientifically true.

24.1.1 — Recurring Human Burden

Some burdens resist ordinary public measures: grief cannot be socially priced, guilt may outlast punishment, and gratitude may seek a form of expression.

Ordinary causal explanations can leave spiritual questions unanswered. A medical account may explain disease progression without answering the widow's question about meaning. A legal judgment may impose punishment without resolving the victim's spiritual distress. A political account may identify oppression without telling the bereaved how to speak to the dead.

Assess whether a ritual leaves room for unresolved grief or anger and preserves responsibility, rather than demanding a performance of peace or premature absolution.

A grief practice that prevents isolation may help; an illness ritual that delays treatment may harm. A practice of confession and restitution may support repair; a demand that victims forgive may shift the cost back onto them. Moral evaluation begins with what the practice does to the people carrying the burden, not with how beautiful the form appears.

24.1.2 — Justice, Balance, and Distinct Traditions

Human beings repeatedly tell stories in which disorder is answered by balance. The form appears in courts, rituals, epics,

moral tales, political memory, family narratives, and private hopes. Calling these stories myths does not mean they are false or foolish. It means they organize experience through symbols that can carry truth, conflict, aspiration, and danger at the same time.

Balance language often responds to a real wound: the visible world leaves suffering, luck, guilt, and reward badly distributed. A story of eventual order can protect hope when no institution can repair the loss.

A single word such as balance can conceal differences among right relationship, lawful order, mercy, liberation, duty, and repair.

Resonance does not confer ownership. Older traditions remain themselves, and similarities in language or practice do not make them unfinished versions of a newer path. Comparison should name the tradition-specific concept and leave doctrine, ritual, authority, reform, and pastoral practice with that tradition (Asad, 1993; Obeyesekere, 2002).

Hope remains a human capacity rather than the property of one worldview. It may be sustained by faith, probability, solidarity, duty, beauty, memory, protest, or commitment to a future one may never see.

24.2 — Karma, Providence, Fate, Grace, and Repair

Karma and providence arise from different traditions and do different explanatory work. Fate and grace also carry meanings that cannot be exchanged word for word. Repair names practices of response rather than a universal rule of cosmic accounting. Comparison should identify a tradition's own terms before claiming it endorses the proposed Law.

24.2.1 — Karma, Providence, and Grace

Karma is a family of concepts, not a universal promise that pleasant and unpleasant feelings will even out. Across Hindu, Buddhist, Jain, and popular settings, it may concern intentional action, ethical consequence, habit, rebirth, liberation, or social explanation. The public shorthand of a cosmic rebound is therefore the least reliable starting point for comparison (Gethin, 1998; Obeyesekere, 2002; Rambachan, 2006).

When public karma language treats suffering as evidence of hidden fault, it can discourage compassion, excuse structural injustice, and make victims fear that pain marks them as morally stained.

Providence and theodicy are not synonyms. Providence concerns divine relation to the world; theodicy attempts to reconcile suffering with claims about divine goodness and power; a defense may argue only that contradiction has not been established. Grace names a different concept: in Catholic teaching, it is God’s freely given, unearned assistance (Catholic Church, 1997, paragraph 1996). These differences matter because a terminal boundary hypothesis cannot inherit the warrant, purpose, or moral conclusions of any of them (Hick, 1966; Plantinga, 1974; Levenson, 1994).

Providence can be lived as trust, dependence, lament, or protest rather than as a neat explanation of every wound. In

many traditions, faithfulness coexists with unanswered prayer and demands for justice (Heschel, 1962; Levenson, 1994).

The public danger begins when providence explains another person's wound before that person has chosen the frame. Hidden-plan language can make protest seem faithless, delay seem virtuous, or preventable harm seem assigned, even though supernatural attributions can function as either coping resources or sources of distress depending on content and use (Exline & Wilt, 2023).

24.2.2 — Fate, Fortune, and Repair Are Not the Same Claim

Fate, fortune, and repair answer different questions. Fate concerns necessity, order, or what cannot be escaped. Fortune concerns contingency, luck, and the uneven events that arrive without moral proportion. Repair concerns what persons and communities do after harm. Blending them into one idea of balance removes the distinctions that make each concept meaningful.

Fate can console by locating a life inside a larger order, but it can also silence protest. Fortune can protect against blame by acknowledging contingency, but it can also make the world feel arbitrary. Repair keeps attention on present responsibility: what can be restored, compensated, prevented, mourned, or changed now.

Within a tradition, penance, restitution, discipline and sacrifice can be freely undertaken practices with distinct purposes.

24.3 — Traditions Disagree

No religious tradition supplies just one voice. Christian communities disagree about providence and freedom as well as the meaning of suffering. They also differ in the authority they grant to texts, leaders, and lived experience. A responsible comparison must locate a specific interpretation rather than speak for all believers.

24.3.1 — Traditions Do Not Speak With One Voice

Jewish traditions preserve legal argument, lament, covenant, protest, mystical interpretation, and secular cultural forms. Indigenous traditions do not constitute one belief system; their histories, lands, languages, and relationships differ.

Internal diversity is not noise around a pure center. It is part of what the tradition is. Doctrine changes through interpretation, conflict, political power, migration, colonization, reform, and ordinary practice (Asad, 1993; Orsi, 2005; Mahmood, 2005).

Plurality also protects the Law from confirmation bias. A fair comparison must apply the same stated inclusion and interpretation rules: it cannot collect balance language from one source while ignoring protest, randomness, tragedy, or divine mystery from another. Material that resists the preferred analogy belongs in the argument.

Nonreligious worldviews are plural in the same way. Humanists, atheists, agnostics, naturalists, existentialists, and people with no formal label disagree about meaning, morality, death, and hope. They should not be reduced to one secular answer.

24.3.2 — Internal Critics and Protest Traditions

Internal critics matter because easy explanations of suffering often arise from within the same communities that later resist

them. Protest is not necessarily rejection of a tradition. It can be fidelity to a deeper moral demand inside it.

The Book of Job refuses a simple equation between suffering and wrongdoing. Lament traditions speak from abandonment, anger, bewilderment, and persistence. Jewish responses to catastrophe have often preserved argument with God rather than accepting a tidy account of providence (Levenson, 1994; Alter, 2010).

Christian thought contains theodicies, defenses, mystical interpretations, liberation theologies, pastoral refusals, and critiques of accounts that make another person's pain serve an abstract good. Hick, Plantinga, and Adams offer no single answer; their disagreements show why the problem cannot be reduced to a balance slogan (Hick, 1966; Plantinga, 1974; Adams, 1999).

South Asian traditions also contain critiques of inherited interpretations of karma, caste, duty, suffering, and liberation. Modern reformers and scholars have challenged readings that convert structural harm into deserved consequence or spiritual hierarchy (Obeyesekere, 2002; Rambachan, 2006).

Islamic traditions preserve debates over divine decree, human responsibility, mercy, trial, injustice, and social obligation. Protest against oppression can be religiously grounded without implying that every suffering event was assigned for a hidden purpose (Rahman, 1980).

Secular protest belongs beside religious protest. Secular strands of human rights movements, disability activism, labor movements, feminist critique, and anti-colonial thought can reject metaphysical explanations to keep human responsibility visible.

24.3.3 — Comparison Without Forced Equivalence

The first rule is to compare like with like. A ritual practice is not a scientific theory, a theological doctrine is not merely a coping strategy, and a philosophical argument should not be represented by a popular slogan.

The second rule is to preserve asymmetry. Karma may involve action and consequence without a divine judge. Providence may involve divine agency without reincarnation. Grace may interrupt desert, fate may reduce agency, and repair may proceed without any account of cosmic order.

The third rule concerns translation: retain source context and acknowledged uncertainty, because terms such as suffering, justice, soul, liberation, and person carry different distinctions across traditions (Asad, 1993; Mahmood, 2005).

A responsible comparison therefore organizes questions rather than verdicts: What causes suffering? What role does agency play? Is consequence moral, natural, divine, social, or uncertain? Which practices respond, what forms of protest remain possible, and what stays unresolved?

This method resists perennialism, the claim that all traditions express one hidden truth beneath surface differences. The Law of Fairness must not become that hidden truth.

Two grief practices may both sustain remembrance while differing about the survival of the person, the source of obligation, or the meaning of death. That shared function does not settle those disagreements. Conversely, agreement about an ultimate order need not produce the same practice. Stating the level of comparison (question, image, doctrine, or lived effect) lets an analogy succeed locally without implying agreement at every level.

The translator’s standpoint should remain visible because categories formed in one colonial, theological, or secular setting

can reorganize another tradition before comparison begins (Asad, 1993; Mahmood, 2005).

Lived religion keeps doctrine from standing in for one uniform effect. People inhabit traditions through households, bodies, calendars, food, prayer, work, grief, conflict, institutions, and participation. Participants, historians, social scientists, and internal critics answer different questions; none is complete by itself, and no single voice represents a whole tradition (Orsi, 2005; Harvey, 2013; Geertz, 1973; James, 1902).

The test is reversibility. The comparison should remain intelligible if the Law is removed and the tradition can describe itself again. When a borrowed concept becomes unrecognizable outside the Law's vocabulary, the analogy should be narrowed and the concept restored to its source context.

24.4 — Belief, Interpretation and Evidence

People respond to bereavement in markedly different ways. Some report meaning or growth, while others experience prolonged distress. Oscillation, numbness, or anger may occur without a transformative story. Research should make room for nonbelief and refusal as well as religious interpretations (Wortman & Silver, 1989; Stroebe & Schut, 1999; Bonanno, 2004; Neimeyer, 2001).

24.4.1 — Grief, Meaning and Spiritual Struggle

Meaning-making research distinguishes broad beliefs about the world from the interpretation of a particular event. When they conflict, responses can include changing either interpretation, holding the tension, or leaving it unresolved (Park, 2010). Spiritual struggles involving doubt, moral conflict, interpersonal injury or perceived abandonment are associated with distress in heterogeneous ways; that association does not by itself identify a causal effect of compulsory interpretation (Exline et al., 2024).

Reports of posttraumatic growth can coexist with distress and do not establish that trauma was beneficial or necessary (Tedeschi & Calhoun, 2004). Frankl's account describes meaning as a possibility under extreme constraint (Frankl, 1946; Frankl, 2006). These findings concern different responses to adversity rather than stages everyone must pass through.

The rule is especially strict around children and people under coercion. Adults must not impose accounts that recast abuse as a lesson, adversity as a gift, or forced endurance as character-building. Institutions may not convert preventable harm into a story about resilience while leaving the conditions unchanged.

24.4.2 — Nonreligious Dignity

Nonbelief may reflect considered judgment, intellectual humility, moral protest, indifference to metaphysics, or a life organized around finite relationships and responsibilities. Nonreligious meaning can arise through care, truth, art, friendship, solidarity and public work (Camus, 1942; Copson, 2019; Schnell, 2021; Taylor, 2007). Presenting inquiry as necessarily culminating in sacred meaning can leave skeptical readers merely tolerated rather than fully included (Schnell, 2021; Smith, 2017).

Religious neutrality has practical consequences. Attendance, ritual, donation, testimony or agreement must not become credentials for trust, office, care, education or scientific seriousness. Departure can carry family, social, economic and psychological costs; it should not be represented as loss of moral standing.

Desired chaplaincy or spiritual support can work alongside clinical, psychological and social care. Refusal must not worsen treatment, and recovery must not be defined solely as renewed belief (Pargament, 1997; Park, 2010).

Phrases such as “once you understand” or “everyone secretly feels this” can turn disagreement into an alleged confession. Organized doubt and the possibility of revision are part of scientific inquiry (Merton, 1973; Longino, 1990).

24.4.3 — Resonance Is Not Scientific Support

The Law may resonate as an interpretation of experience for some readers. That personal meaning can matter without functioning as scientific evidence. A moving analogy or testimony does not establish a measurable terminal law. Keeping the interpretive and empirical registers distinct protects both honest inquiry and freedom of belief.

A person may say that the Law helps them see hidden burden more carefully. They may say it gives them language for why care matters even when visible life looks absurd (Weber, 1922; Berger, 1967; Geertz, 1973).

Interpretive statements are first-person or community accounts, not transferable verdicts. “This helped me carry grief” has a different grammar from “this explains grief,” and “our ritual gives us hope” is different from “the universe guarantees hope.” Preserving those grammatical differences keeps testimony about meaning from being treated as evidence of terminal neutrality, or private conviction as authority over another person’s life.

The psychology of religion can document beliefs, practices, reported experiences, spiritual struggles, meaning-making, and cross-sectional or longitudinal relations with distress and well-being. These are measurable questions about human lives (Exline et al., 2024).

The converse is equally important. Scientific defeat of PTVC would not abolish religion, grief ritual, prayer, secular meaning, moral protest, or chosen hope. Keeping the registers separate prevents science from owning sacred life just as it prevents sacred life from owning the scientific verdict.

Chapter 25 — Limits on the Idea’s Use

Imagine an eligibility office refusing aid because software labels distress as compensated elsewhere in life. Even flawless arithmetic would not authorize that decision. Who approved the inference, what entitlement changed, and how can the person reverse it? Responsibility becomes concrete when a speculative relation overrides a right.

Public systems often meet people when bargaining power is lowest. A patient needs medicine, a child needs protection, a tenant needs shelter, a worker needs safety, and a disabled person needs access. If direct help is unavailable, responsibility must still become visible through explanation, referral, stabilization, escalation, record, or honest acknowledgment of what cannot be done. Procedural fog is itself a lived burden (Entwistle et al., 2010).

A caregiver can also misuse the Law by making their own endurance appear noble while another person's needs remain unmet. The lens should not train caregivers to romanticize sacrifice, silence, or overfunctioning. Hidden burden includes the burden carried by the caregiver too. Fair care protects the sufferer without erasing the responder. It asks for support systems, respite, referral, and limits so that care does not become another hidden injury (Entwistle et al., 2010; Committee on the Rights of Persons with Disabilities, 2014).

Limits on automated inference matter even when the interface sounds patient, kind, and comprehensive. Fluency can make a system appear to understand more than it does, while a calm tone can make an uncertain output feel settled (Weizenbaum, 1966; Bender et al., 2021). A clinical summary that calls a patient resilient, a school tool that equates quietness with low distress, or an employment model that infers reliability from digital traces may combine many signals and still miss pain, coercion, disability, culture, masking, fear, or the reason for

nondisclosure. Once the label enters a record, the person may be forced to disprove an opaque conclusion. The proponent of the inference, not the affected person, carries the burden of validity (Citron and Pasquale, 2014; Green, 2022).

This chapter answers that risk before it becomes practice. The burden-locating question remains the method. Every misuse can be tested by asking who gains freedom, authority, or reassurance and who carries the resulting cost. If an interpretation calms the responder while the suffering person receives less care, responsibility has moved in the wrong direction. If an institutional tool gains power while the person loses dignity or appeal, the harm has been hidden inside procedure. The result is a misuse-and-repair casebook rather than a catalogue of prohibitions. A boundary becomes useful when someone can act on it.

Repair also has a positive architecture. Someone can identify the decision that needs revisiting, find the person able to revisit it, restore an interrupted service, and check that the correction reached the records through which the decision travels. Each step makes responsibility more usable. A person seeking help should be able to see the next action and who will take it. The cases in this chapter therefore follow the route back from an abstract label to a concrete decision, where explanation, correction, and restored access can become real accomplishments.

25.1 — No Punishment, Ranking, or Eligibility

Caregiving authority is intimate, but intimacy does not make it unlimited. A clinician or parent may know a person's circumstances well. That knowledge does not reveal the person's lifetime cumulative experience. Duties of care remain grounded in actual needs and rights, not an imagined hidden score. Human-rights instruments protect autonomy, participation, non-discrimination, and the best interests of children without making distress a test of belief or character (United Nations, 2006; United Nations, 1989).

25.1.1 — Public Writer Duties

A public writer controls selection, framing and the point at which uncertainty becomes visible. What an audience needs depends on its questions, prior knowledge and decision context (Fischhoff, 2013; National Academies of Sciences, Engineering, and Medicine, 2017).

Public correction begins with an accountable owner and a dated statement of what changed. If an error undermines the central result, withdraw the claim; if a bounded correction leaves the result intact, identify that boundary. Keep the previous version and the correction linked and findable (Committee on Publication Ethics, 2025).

First, trace dependencies: for each affected claim, tool, training item or decision, record its owner, audience, consequence and required action. Distinguish independently warranted uses, partial pathways, historical material, uses needing revalidation and uses that must stop. Retired claims must not continue unmarked in fundraising, merchandise, membership or educational material (Oreskes, 2019). A theoretical failure does not automatically invalidate a separately

supported practice, but a new name cannot preserve the old claim's authority.

Second, suspend material that creates immediate danger before investigating, then contain consequences and notify affected people. Explain the former basis, current finding, decision being stopped or reviewed, and routes to correction, appeal, deletion or remedy. Review thresholds, incentives and training as well as the headline statement. Vendors, funders and publication schedules must not delay necessary notice.

Third, replace or annotate the material where it circulates: translations, partner pages, cached copies, screenshots and AI knowledge bases as far as practicable. Give readers a concrete action and record completion or outstanding limitations. Bring emergency suspension into ordinary review and publish its final status as visibly as the initial notice. Preserve a usable history without exposing or blaming the person who reported the failure. This concerns the publisher's own unsupported claims; it is not a warrant to erase criticism (Reason, 2000; Dekker, 2014).

Read excerpts and cropped illustrations in isolation before release. A balanced scale can imply a proven result even when it represents only a hypothesis. Ask what an ordinary reader would infer, whether that inference matches the evidence, and whether the source and version remain identifiable. Put any indispensable qualification in the excerpt itself, not in a later reply. An AI summary that changes "hypothesis" into "law of the universe" has altered the claim. For an actionable prompt or symptom tracker, inspect what the user is likely to do as well as what the sentence literally says.

State uncertainty specifically. Research on communicating numerical uncertainty found that effects on trust vary with its form and the outcome measured; uncertainty need not automatically destroy trust (van der Bles et al., 2020). Invite a

reader outside the project to flag undefined terms, unsupported inferences or inaccessible presentation. Review with affected readers or communities before materials freeze when the content could change their treatment; their responses inform public effects rather than determine scientific truth (World Medical Association, 2024).

Provide accessible text, intelligible reading order and usable alternatives to nontext content in public outputs (World Wide Web Consortium, 2024). Revise the headline, example or wording if its likely interpretation exceeds the claim. Emotional force and elegant presentation do not excuse a misleading inference.

25.1.2 — Public Discussion and Community Care

Public discussion spaces become ethical sites as soon as the Law enters them. Forums, classrooms, livestreams, comment sections, reading groups, and community channels can support critique and learning, but they can also amplify blame, fatalism, harassment, spiritual pressure, and amateur diagnosis.

Moderation should protect people from harmful conduct without protecting the theory from criticism. Disagreement about logic, evidence, definitions, metaphors, falsifiability, writing, presentation, governance, or project choices remains legitimate. Rules may address threats, harassment, doxxing, impersonation, and targeted abuse, but criticism cannot become disloyalty or grounds for lowered standing (Society of Professional Journalists, 2014; National Academies of Sciences, Engineering, and Medicine, 2017).

Moderation must also prevent informal experiments without consent. Remove invitations to conduct personal endpoint tests, provoke distress, pressure people into self-experimentation, rank

whose grief counts more, judge whether another person's death balanced, or turn a vulnerable disclosure into a spectacle.

25.2 — The Hidden Object Cannot Be Administered

Public authority should follow decisions that institutions can review and correct. Courts determine legal responsibility and remedy under law. They do not need a private-life balance to decide a claim. An unobservable cumulative object cannot become an administrative shortcut for eligibility, punishment, or rights.

25.2.1 — Authority and Reviewable Decisions

Public agencies deliver services and exercise public power under reviewable criteria. Managers assess demonstrated performance against job requirements; teachers assess demonstrated learning; clinicians formulate within professional standards. Each must identify the evidence relevant to that decision and the limit of the inference.

Locate the reachable point of repair: who can change the schedule, remove a barrier, provide treatment, explain a rule or hear an appeal? Record the responsible role and action. The burden of explanation increases with coercion, allocation and public consequence; it does not become a demand that citizens expose their inner lives (O'Neil, 2016; Nissenbaum, 2010).

Expertise is local. Mathematical competence does not establish clinical competence, and data-governance expertise does not settle consciousness (Collins and Evans, 2007). A referral transfers the question requiring that expertise, while the referring person retains responsibility for their own remaining actions. Public endorsements should identify the specific conclusion endorsed, the reviewer's relevant expertise and the date; partial agreement must remain partial and endorsement must be removable. An author can explain intended meaning without overruling an independent specialist's assessment.

For a proposed exception, identify who benefits, who carries irreversible risk, whether refusal is practical, whether a less intrusive route exists and what remedy follows failure. A pilot needs a sunset date, a named renewal authority and a practical test that its appeal route works. Continued operation requires an affirmative decision on the recorded results, rather than becoming the default through inertia.

Trace public use from exposure through the audience's situation, power, inference and action to the consequence. Assess severity and reversibility, and whether a safer action preserves the legitimate purpose; changing wording alone does not repair an unchanged harmful decision path.

This boundary does not weaken justice. A court can protect victims, restrain dangerous conduct, require restitution, correct rights violations, and impose lawful consequences. It becomes more reviewable when it acts on evidence, causation, duties, rights, and proportionality.

Candidate research indices, archive summaries, and AI outputs are especially dangerous in this setting because numerical form can make contestable judgments look inevitable. Work on technological due process shows how automated systems can obscure the evidence, rules, and routes of challenge needed by affected people (Citron, 2008; Citron and Pasquale, 2014). A latent, endpoint-dependent, complete-stream object is even less suitable for administration than an ordinary disputed score.

The decision should also survive a deletion test. Remove every claim about hidden balance, secret desert, or total inner worth. If the judgment still rests on ordinary facts and law, the metaphysical language was unnecessary. If the judgment collapses, it lacks an independently justified basis and should not proceed.

25.2.2 — States Cannot Administer the Cumulative Object

States often turn complex goals into indicators because indicators make large systems manageable. That convenience creates danger when a measure becomes a target and begins to reshape the conduct it was meant to describe. Campbell warned that the more a quantitative indicator is used for social decision-making, the more pressure there is for corruption and distortion around it (Campbell, 1979). A hidden feelings total would be especially vulnerable because neither its origin nor its complete domain is available for ordinary administration.

Human-rights and AI-governance frameworks supply concrete reasons for restraint. Disability rights protect equal recognition, participation, access, and non-discrimination; European and Council of Europe instruments set out risk, rights, and lifecycle duties, with legal effect depending on jurisdiction, scope, ratification, and applicable start dates. None makes a lifetime ledger administrable (United Nations, 2006; European Union, 2024; European Union, 2026; Council of Europe, 2024).

An appeal must be able to change the consequence. Give the person a plain-language reason, access to the material facts and a human reviewer with authority to correct or reverse the decision. The deploying institution carries the burden of validity; the affected person need not disprove an opaque inference. Reviewers need independence, relevant evidence, sufficient time and protection from retaliation (Green, 2022; Raji et al., 2020).

The remedy must reach the lost opportunity. If a school treats inability to attend an inaccessible activity as deficient commitment, deleting the label alone is insufficient: restore access and reconsider the opportunity withheld. Record the correction and its downstream effect. The same requirements apply when automated review affects consent or participation

(Council of Europe, 2024; European Union, 2024; European Union, 2026).

25.2.3 — Individuals Cannot Score Each Other

The most intimate misuse needs no government or software. It begins when familiarity becomes a claim to know another person's whole life. "I know what you are really feeling" can silence testimony; "you have had an easy life" can erase pain; "your hardship will make you stronger" can turn an uncertain future into permission for neglect.

Family scoring can also hide abuse. A controlling person may describe another as overreacting, secretly thriving or destined to benefit, while restricting money, movement or escape. The practical question is what the interpretation allows its speaker to do. A reassuring account of someone else's future does not answer a present report of coercion.

Close relationships still permit moral knowledge. A partner can notice patterns, remember promises, identify harm and ask for change. Those observations support specific conversations and decisions. They remain open to the other person's account, including facts that intimacy itself has made difficult to disclose.

Appeals to personal moderation can themselves become a tool of control when a partner or institution uses them to supervise food, medication, sexuality, spending or speech.

25.3 — Institutional Authority and Repair

Technology can help people search records or compare drafts. It can translate materials, flag inconsistencies, and improve access. Those useful functions do not grant a system authority over anyone's inner experience. Institutions must keep judgment reviewable by affected people rather than treating a model's output as sovereign.

25.3.1 — AI, Archives, and Human Authority

Define its function inside the human process before assigning it a decision (National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024).

A system preparing a clinical summary, for example, assists by making relevant material easier to inspect. The accountable professional still decides how that material bears on the case. Record the task assigned, the evidence available to the system and the use made of its output. This allows a reviewer to distinguish an error in retrieval or summarization from an unauthorized expansion of the system’s role.

Imagine a community center preparing a history of a room-allocation decision. Its archive contains a March proposal to move a class to Tuesday, an April discussion of that proposal, and a May decision to keep the class on Thursday. A useful assistant can collect the three records, identify their dates, and place their relevant passages together. Staff can then see the sequence without searching the archive again.

The March sentence is authentic, but its role is a proposal. Quoting it accurately does not make Tuesday the adopted schedule. The May record changes the answer to the question, “What was decided?” A good summary therefore keeps the speech act attached to the words: someone proposed a change, others considered it, and the center later adopted a particular

arrangement. Chronology and decision status carry information that a collection of correct quotations alone would leave incomplete.

The result can be written in a compact, inspectable form. The summary states the adopted Thursday arrangement, identifies the May decision, and explains that it followed consideration of the earlier Tuesday proposal. Each factual clause points to the record supporting it. The original documents remain available, while the summary supplies a route through them. This is a useful division of work between an archive and an explanation.

Now suppose staff want to know whether Thursday still serves the class well. The historical record supplies one input: what was previously decided and why. Current attendance, access, room availability, and participants' preferences supply others. The assistant can organize those materials as well. The new decision can then be made through the center's ordinary process, with a clear distinction between the history being summarized and the present question being answered.

Such assistance can save time while making institutional memory easier to question and improve. Its value lies in reducing the distance between a person and the evidence they need. When the sequence is visible, a participant can correct an omitted document, explain a change in circumstances, or understand how a decision developed. The institution gains a more usable memory and a clearer basis for its next conversation.

For the distinction between an archive and the subject of experience, see Section 4.3.1. Commercial reconstruction creates a distinct authorization problem. A photograph, research record or private message does not by itself authorize a conversational replica. Continuing bonds in grief do not give a

commercial actor ownership of identity (Öhman and Floridi, 2018; Klass et al., 1996; Neimeyer, 2001).

Living people need correction when an archival portrait is false or decontextualized. A generated reconstruction should be unmistakably labeled, with its permissions and provenance governed by the record architecture in Section 12.3.2.

25.3.2 — The Law Cannot Determine Eligibility

The deeper problem is abstraction. A model can ignore the social system in which its output acts, treating fairness as a property of a formula while the surrounding institution remains unequal or unreviewable (Selbst et al., 2019).

An eligibility audit should inspect the decision path, not only the tool's label. If a supposed support measure lowers access when someone reports distress, declines disclosure or disputes an inference, it functions as an eligibility rule. Removing a displayed score does not remove the consequence.

Proxy choices and uneven error can reproduce inequality, as health-care and face-analysis studies illustrate (Obermeyer et al., 2019; Buolamwini and Gebru, 2018). Inspect purpose, input features, error distribution and downstream use. Audit downstream exclusion as well as average predictive performance (Suresh and Guttag, 2021).

Restore lawful, necessary and contestable criteria tied to the actual decision: a public agency's service obligation or a demonstrated job requirement, for example. Identify who lost access, which inference caused it and what correction restores the opportunity. If the system cannot separate relevant criteria from protected need, disability, cultural difference or unobserved context, abstain from the exclusion. Explain the resulting decision rule publicly.

25.3.3 — Restoration, Correction, and Boundaries

Repair looks forward to what can now be changed. It asks who was affected and what safety is needed. Responsibility may require acknowledgment, restitution, or a change in an institution's practice (Zehr, 2002; Walker, 2006). None of those actions requires treating another person's suffering as a payment that balanced an invisible ledger.

Stop immediate danger first. Identify needs and rights before proposing reconciliation. Investigate the facts without demanding a performance of forgiveness. Assign and fund remedies, record their effects, and monitor recurrence. The responsible institution must be able to say who will deliver the remedy and how the affected person can establish whether it occurred.

Restorative work can help when participation is informed, voluntary, safe and supported. It can also reproduce pressure when institutions prize closure over refusal. Evidence on restorative justice therefore belongs beside safeguards, selection limits and continuing access to ordinary legal remedies (Sherman and Strang, 2007; United Nations Office on Drugs and Crime, 2020). Declining a meeting or reconciliation ritual does not discharge the responsible actor’s obligations.

A credible apology names the act, recognizes its effects, accepts responsibility without self-exoneration and connects words to restitution and changed conduct. The apology can be assessed by what follows it. A harmed person need not accept it for practical repair to proceed.

Consequences may remain necessary: removal from a role, compensation, treatment, supervision, protective orders, disciplinary action or criminal process under ordinary law and ethics. Repair without accountability can abandon the harmed

person; accountability without dignity can become humiliation rather than protection (Braithwaite, 1989).

Practical repair can require giving up an advantage or telling the truth when concealment would be easier; accountability continues after public attention moves on.

Part IX — Living with and Testing the Law

A large question eventually reaches the kitchen table. The alarm sounds. A body is tired or rested. A message brings relief or dread. Work waits. Someone needs care. A pleasure is offered. A craving returns. A plan succeeds, fails, or remains uncertain. Everyday reflection can reveal a practical change worth making. Its value depends on the consequences of that change, without requiring the Law to be true.

Joy can also be a public good. A safe park, music in a hospital corridor, a neighborhood meal, or a library open late can make beauty, play, and companionship available without luxury. Systems that protect time, accessibility, safety, and ordinary celebration are not decorative additions to dignity.

Rest has a distribution. Someone may cover the shift, prepare the meal, watch the child, answer the call, or clean after the gathering. A fair arrangement asks whether that work was chosen, visible, supported, and sustainable. Restoration for one person should not be built on the silent exhaustion of the person with the least power.

Meaning changes with scale. A tedious afternoon may belong to a deeply valued project, while a thrilling evening may undermine a life the person wants. The afternoon is not secretly pleasant, and the evening is not secretly worthless. Immediate valence, reflective evaluation, and long-horizon consequence answer different questions.

A pattern becomes clearer when its unit is named. The question may concern one conversation, a recurring hour, a bodily state, a place, or the transition between tasks. A broad question such as whether the day was good invites a verdict. A narrow question such as what changed after eating, resting, or leaving the room invites an explanation that can be revised.

Prior investment can also trap ambition. Time, money, identity, or public promises already spent do not make another

sacrifice wise. A failed path becomes informative when a person can separate recoverable learning from costs that cannot be recovered, then decide from the future being created rather than from the past that has already been paid.

The move from these observations to a scientific test changes the question. A useful alteration to an afternoon can be assessed by its consequences there; testing a proposed relation across complete conscious streams requires an independently specified object, a design able to distinguish alternatives and an outcome that could count against the claim.

Attention can become useful before it becomes numerical. Experience-sampling research shows that people can report moments, settings, activities, and felt states while still revealing fluctuation, missingness, reactivity, and the limits of self-report (Csikszentmihalyi and Larson, 1987; Stone and Shiffman, 1994; Shiffman et al., 2008; Killingsworth and Gilbert, 2010). Its lesson for ordinary life is modest: context changes experience, and careful noticing can uncover a pattern without pretending to summarize a person.

Chapter 26 — Fairness in an Ordinary Day

An ordinary day contains more theory than it appears to. Before breakfast, a person may choose whether to open the phone, eat before a hard meeting, answer a craving, continue a project, apologize, rest, celebrate, or let grief remain wordless. Small choices can redirect an entire day. Consider an ordinary afternoon. Meetings after three o’clock feel heavier, hunger makes criticism sharper, and a ten-minute walk changes the next conversation. The useful discovery is not a daily score. It is a relationship among time, body, setting, and what becomes possible next.

A writer who protects four mornings a week gives a long project repeated access to attention. A person in recovery who changes the route home protects freedom from a cue that has repeatedly captured it. Each rearranges the conditions under which a later choice will be made.

Discipline can also mean stopping. Resting before injury, refusing another task, setting a budget, leaving an exploitative opportunity, or retiring a plan that no longer fits may preserve more possibility than endurance. Motion and sacrifice are not reliable measures of agency. Feedback should therefore remain diagnostic rather than punitive. A missed week may reveal illness, hidden caregiving, an unrealistic schedule, a vanished reward, or a method that no longer serves its purpose. The answer may be redesign, assistance, or release rather than escalation.

Pleasure, relief, fulfillment, and meaning appear in different ordinary scenes. A meal can be pleasurable, the end of a migraine can be relief, finishing an apprenticeship can be fulfillment, and caring for a parent can carry meaning while also bringing fatigue and grief. Collapsing all four into happiness erases the mechanism and the human texture.

Think of a friend waiting outside an appointment, a neighbor bringing dinner, or someone listening to the same grief story again without asking when it will end. Each can change what another person has to face. None has to explain the suffering to make the next hour less lonely. A support network should distribute rather than hide load. One partner, child, sibling, or friend should not become the only emergency system for another person. Wider care, respite, clinical help, and public services can protect both dignity and relationship.

Rest also includes pauses that are not sleep: an unhurried meal, a bath, time in nature, music without monetization, an afternoon without performance, or permission to be unreachable. These intervals may restore later functioning, but they need not justify themselves only by what they produce. Celebration should not become compulsory optimism. A person may feel numb at a milestone, grief inside a happy gathering, or relief mixed with fear after survival. The occasion does not dictate the correct inner response, and sincerity is not measured by performance.

26.1 — Attention Without Accounting

Optional reflection can begin with one ordinary hour. Notice where a burden was overlooked in a conversation or task. Ask which boundary could make the next hour safer and where relief became possible. The purpose is a concrete change, not a private scoring system for yourself or others.

26.1.1 — Notice Patterns Without Harmful Scoring

Freely chosen engagement differs from controlled compliance (Deci and Ryan, 2000; Ryan and Deci, 2001).

Numbers are not forbidden when their purpose is bounded. A pain rating, sleep estimate, medication log, craving note, or budget may support care, communication, accessibility, or a specific decision.

For a broad reflection on the day, include relief as well as burden. A pain log answers a narrower question. Reflection can clarify a pattern, but it can also become rumination. Rumination research describes cycles that intensify distress, narrow attention, and repeat the same unresolved material without producing a workable next step (Nolen-Hoeksema et al., 2008; Watkins, 2008). The practical sign is often not depth, but shrinking freedom.

26.1.2 — A Practice That Remains Voluntary

The stopping threshold should be observable. Reflection has become a burden when sleep is lost, ordinary pleasure becomes suspect, checking feels compulsory, shame increases, disclosure feels demanded, care is delayed, or relationships begin to revolve around the theory. A difficult moment is not automatically harmful, but a repeated pattern of distress or impairment is enough reason to pause. If the theory’s language itself increases fear, shame, confusion or dependency, stop using

that language. Acute danger or serious worsening calls for safety and qualified support (SAMHSA, 2014).

A pause can be small and early. The reader may shorten the exercise, remove reminders, close or delete the record, replace writing with rest, talk with a trusted person, or set the question aside for a season. Stopping before a crisis is a sign that the practice remains under human control.

A useful note can end in a concrete option, such as rest, a conversation, a changed cue, or a request for help.

The least intrusive useful practice is usually the best one. One sentence may be enough; a weekly note may be safer than hourly tracking; a conversation may be better than a spreadsheet. The right cadence is the one that leaves attention available for work, relationships, sleep, pleasure, and recovery.

26.1.3 — When a Place Becomes Familiar

A room becomes familiar through small discoveries. On the first morning in a new home, a person may search for a cup, test several light switches, and wonder where to put a bag. A week later, those actions have begun to find their places. The room has the same walls, but moving through it has changed. The person has learned a set of relations between intention, movement, and what the surroundings make available.

Some of that change comes from memory. The cups are behind one door; the window latch turns a particular way; the quietest chair is beside the wall. Other changes come from rearrangement. A lamp moves closer to the reading place. A hook appears beside the entrance. Frequently used tools are brought within reach. Familiarity grows through both learning the environment and making the environment easier to live in.

Consider a small worktable used for drawing. At first, each session begins with searching for paper, clearing a surface, and

finding the right pencil. The person spends part of the available time preparing to begin. A simple arrangement can change that sequence: paper lies flat in a nearby drawer, the current work rests on a board, and the light can be switched on from the chair. The table now preserves part of the preparation between sessions.

The gain is not fully described by the minutes saved. Beginning has become a different event. A short interval after dinner may now be enough to add a few lines, because the person can enter the activity without reconstructing its setting. The environment holds a practical invitation. What had required a decision about whether there was enough time can become an available way to spend the time that is already there.

The arrangement can remain personal. One person likes an empty surface and a closed drawer. Another wants the unfinished drawing visible, because seeing it helps recover the next idea. A third needs a different height or a place to rest an arm. The useful organization follows the activity and the person using it. Its success appears in the ease with which that person can return, work, pause, and begin again.

Familiarity also changes what attracts attention. An unexplained sound in a new room may prompt investigation. Once its source is known, it can become part of the background. Conversely, a change in a familiar sound may be noticed quickly because it differs from the learned pattern. The surroundings are no longer an undifferentiated stream of details; some relations have become expected, leaving other differences easier to recognize.

This can make room for discovery within a stable setting. With the basic arrangement settled, the person may begin to notice how the afternoon light crosses the table or which kind of paper suits a particular line. Familiarity supplies continuity,

while attention can move toward a new feature of the work. Repetition and novelty can support each other when the familiar part makes the next exploration easier to enter.

A shared place adds another layer. If two people use the table, they can agree on which materials remain in place and how unfinished work is set aside. A small, understandable arrangement can preserve each person's ability to return. The agreement is useful because the setting has a temporal role: what one person leaves behind helps shape what the next person encounters.

Over time, a place can gather traces of successful use. A worn tool fits the hand; a note recalls a solution; a mark on the shelf identifies a comfortable height. These details carry practical history. The person need not reconsider every decision each time the activity begins. Some of what was learned has been given a durable form in the surroundings.

An ordinary day is partly made from such relations. Attention is available within a body, a setting, a set of habits, and a sequence of tasks. A well-situated lamp or a dependable place for a tool can make a chosen activity easier to inhabit. The interesting change is often modest and specific: a person finds a little more room to draw, cook, read, repair, or welcome someone because the place has learned, through their adjustments, how to receive them.

26.2 — Pleasure, Relief, and Moderation

Pleasure can matter simply because it feels good. Music or a shared meal needs no productivity argument to justify enjoyment. Relief can matter differently when it removes a burden. Moderation asks whether a present pleasure also protects the person's future options.

26.2.1 — Pleasure, Relief, and Savoring

Well-being research distinguishes hedonic and eudaimonic dimensions rather than forcing every positive life into one total (Ryan and Deci, 2001; Diener et al., 2006).

Enjoyment also has a material setting. Remembering the nurse on the holiday, the cleaner after the event or the relative cooking alone can lead to better pay, schedules and participation. Ask whether the labor that makes the occasion possible is visible, chosen and sustainable. Positive emotion can broaden attention, support connection, and replenish the capacity to act, while self-determined enjoyment can be part of a life rather than a distraction from it (Fredrickson, 2001; Ryan and Deci, 2001).

Savoring can deepen attention to an experience. Pausing over music, food, friendship, beauty, achievement, or relief can help a person receive what is present rather than rushing toward the next stimulus (Bryant and Veroff, 2007).

Receiving an enjoyable experience can involve discovering what is already there. Imagine listening to a favorite piece of music while preparing a meal. At first the melody accompanies the activity. Later, sitting down for a few minutes, the listener notices a quieter instrument answering it. The recording has stayed the same. What enters attention has changed.

This is one reason variety and depth are different forms of possibility. A new recording offers another experience. Returning to the familiar recording with a different question

may reveal a relation that was previously in the background: how a phrase is answered, where a rhythm changes or how a silence prepares the next entrance. Familiarity can support discovery because some parts no longer need to be worked out from the beginning.

The occasion also has a social form. A friend may point out the passage they love, and the next hearing includes that invitation. The listener can appreciate the passage, discover another one or simply enjoy knowing what the friend heard. A shared preference need not become identical experience for the exchange to add something.

There is room for simple enjoyment as well. Attention need not become a task with a correct level of depth. Sometimes the music is welcome precisely because it accompanies cooking, movement or company without asking for analysis. The person can choose whether the moment calls for close listening or an easy background. The possibility of choosing is part of what makes the pleasure hospitable.

Savoring, understood this way, is an opening within attention rather than a demand to make every good moment larger. It can be as small as hearing a familiar passage clearly before the next activity begins. The moment is allowed to have a shape of its own, and its value need not depend on how efficiently it was noticed.

26.2.2 — Moderation without Shame

Habit formation is gradual and variable. Research on everyday routines shows that automaticity develops at different speeds and can tolerate some missed opportunities without collapsing (Lally et al., 2010).

Friction can preserve choice. Moving an app, changing a route, limiting easy access or preparing food can create a pause

before the strongest conflict arrives. Regulation includes choosing environments, reducing cues, setting defaults and planning for fatigue (Fujita, 2011; Duckworth & Gross, 2014).

Make the plan specific to the behavior. A substance may require abstinence for safety; food, work, exercise or entertainment may call for limits. Risk, vulnerability and professional guidance determine the appropriate boundary. Pain, disability, recovery, pregnancy, grief or crisis can change what is feasible.

Identify the material conditions the plan needs: predictable work, safe housing, treatment, affordable food, public space or reliable care. Name the missing resource and where help is needed to secure it. A plan cannot supply through willpower what its environment withholds.

Match deliberation to stakes and reversibility. High-consequence choices need time, alternatives, qualified advice and a practical route back (Sen, 1999; Nussbaum, 2011). Choose proportionate, reversible changes where possible and examine whether they expand the person's options.

Stigma increases secrecy and discourages help-seeking. Addiction-language research shows that words can frame people as capable of recovery or as moral contaminants (Kelly et al., 2016; Livingston et al., 2012).

26.3 — Addiction, Craving, and Future Optionality

Recovery is not one fixed path. Some people reduce use; others seek abstinence or medication support. Housing, mutual aid, therapy, and changed environments may reopen choices that craving had narrowed. What matters is the person's actual safety and agency rather than a theory that struggle is required to earn relief.

26.3.1 — Recovery Can Reopen Paths

Addiction often has a long course with recurrence, remission and multiple treatment episodes (McLellan et al., 2000). Different conditions and people require different routes.

Ask what the pattern supplies, such as relief, belonging or escape, and what helps sustain it, such as pain, isolation, unstable work or a familiar cue. Cues can remain potent after long periods, and stress can reactivate learned responses (Everitt and Robbins, 2016; Wood and Rünger, 2016). A recurrence can identify a cue, untreated pain, withdrawal or environmental pressure that the previous plan missed. Update the plan and strengthen support around that finding.

Methadone, buprenorphine and naltrexone are FDA-approved treatments for opioid use disorder. Methadone and buprenorphine reduce withdrawal and craving; naltrexone blocks opioid effects. These treatments reduce opioid use, and methadone and buprenorphine are associated with lower overdose mortality (Substance Abuse and Mental Health Services Administration, 2021). Clinical assessment considers pain, medication history, withdrawal risk, function and the person’s goals. Psychotherapy and contingency management can address learned responses and behavior alongside medication where appropriate (McLellan et al., 2000; Volkow et al., 2016).

26.3.2 — Continuity and Support Across Transitions

Continuity matters at transitions from emergency to outpatient care, incarceration to community, adolescent to adult services, or stability to recurrence. Clinical and community support must remain reachable after the immediate crisis (Kelly et al., 2025). Stigma, exclusion from work, unstable housing and denial of care need direct institutional remedies (Kelly et al., 2016; Kelly et al., 2025).

Harm reduction can preserve life while change remains incomplete: overdose prevention, safer-use information, sterile equipment, testing, medication access and rapid reentry to care address immediate risks (Volkow et al., 2016). Improvement may be private, partial or nonlinear. Positive alternatives also matter: music, movement, learning, food, intimacy, nature, service and community can provide reward and identity. A plan consisting only of prohibition may leave the need that the old pattern once filled.

Physical dependence can develop during prescribed treatment and is distinct from addiction. The prescribing service remains responsible for continuity and support. Reducing or stopping a dependence-forming medicine needs an individualized clinical plan because abrupt cessation can cause withdrawal harm; urgent exceptions require clinical judgment (National Institute for Health and Care Excellence, 2022).

26.4 — Work, Ambition, and Aspirational Dreams

Ambition lets a possible future give present effort a direction. A qualification, a garden, a business, a recovery, or a piece of music can make an ordinary day feel connected to something worth pursuing. The question is how to pursue that future while remaining free to learn, rest, change course, and receive help. Encouragement becomes cruel when it conceals unequal opportunity or turns every setback into an instruction to sacrifice more.

26.4.1 — Dreams Require Pathways, Not Wishes Alone

Specific and challenging goals can improve performance when people have ability, commitment, feedback, and resources, but their effects depend on the task conditions (Locke and Latham, 2002).

Implementation intentions help translate a desired future into a decision about when, where, and how to act. Plans framed around a concrete cue and response can make action more likely than intention alone (Gollwitzer, 1999; Gollwitzer and Sheeran, 2006).

Positive fantasy without obstacle recognition can reduce effort. Mental contrasting asks the person to hold the desired future beside the most important present obstacle, then decide whether commitment is warranted (Oettingen, 2012).

Skill also requires practice that responds to feedback. Deliberate practice research shows that improvement depends on structured effort, correction, and sustained access to appropriate challenges rather than on wishing or raw repetition alone (Ericsson et al., 1993).

26.4.2 — An Endeavor That Outlasts One Person

Some worthwhile work reaches beyond the time one person expects to spend on it. A tree may bear its best fruit after the person who planted it has moved away. An archive may become useful to a reader its first organizer will never meet. A young musician may carry a teacher's lesson into performances the teacher will not hear. The work has a long horizon, while each person's participation has its own beginning, shape, and ending.

Imagine neighbors creating a small orchard on land that had been unused. One person knows how to prepare the soil, another can organize the water supply, and another enjoys keeping records of what grows well. The first season may produce little fruit. Yet it already contains decisions, discoveries, shared effort, and the pleasure of seeing a place become capable of something new. The eventual harvest is one part of the endeavor's meaning, not its only moment of value.

The contributors can understand their roles at different scales. A weekend volunteer helps establish a row. A coordinator keeps the project moving across several seasons. Someone else makes occasional visits to teach pruning. Each contribution belongs to a particular period of a particular life. The orchard connects those contributions without requiring every person to remain until every possible result has appeared.

That connection becomes stronger when the work can be handed onward intelligibly. A simple record of planting dates, varieties, drainage changes, and successful experiments lets a new caretaker begin with more than guesswork. A conversation can carry details that a list misses: why one tree was moved, which corner dries first, or what a previous attempt taught. The handoff turns private experience into something another person can use in their own activity.

Good continuity also leaves room for revision. A new caretaker may discover a better way to water, choose another variety, or use part of the space differently. The earlier contribution remains part of how the project became possible. Its value need not depend on every later choice preserving the original plan. A living endeavor can inherit knowledge while continuing to learn.

For the person leaving a role, completion can therefore have a local meaning. The task may be to establish the first planting, prepare a clear handoff, or help the next coordinator become ready. Those achievements can be finished even while the orchard continues. A bounded responsibility gives the person a recognizable contribution without demanding possession of the project's entire future.

The same pattern appears in scholarship, craft, care, and friendship. Someone documents a method so that another can improve it. A colleague makes a complicated task easier to inherit. A family member preserves an account that later generations may read in ways the writer could not anticipate. The later use is open, but the present act can already be thoughtful, skilled, and meaningful.

There is a distinctive pleasure in seeing another person take up the work competently. The original contributor can recognize a technique they taught and a new idea they would not have produced. Continuity includes both resemblance and change. What travels is not an unchanged experience passed from one person to another, but a set of materials, possibilities, and understandings that can enter a different life.

This helps give ambition a generous form. A person can aim to leave a task clearer, a place more usable, or a practice more accessible than they found it. The aim can be substantial without requiring control over every eventual outcome. Its practical

measure is the quality of the contribution and the opportunities it opens for those who come next.

Years later, someone sits beneath a tree in the orchard and enjoys fruit from a branch that was once only a thin stem. The planter may never hear about that afternoon. The connection is still real: an earlier act helped make the later possibility available. Each person's experience remains their own, while the world they inhabit carries traces of other people's work. A long endeavor gives those separate lives a place to meet through what they make possible for one another.

26.4.3 — Discovering What an Activity Can Become

An unfamiliar activity can change the question by which a person evaluates it. Imagine someone attending a pottery class because they want to make a mug. At the first session, their attention rests on the finished object: whether it will be useful, attractive, and worth the time. The wet clay is mainly the material that must be persuaded into that shape.

After several sessions, the person begins to notice distinctions that were initially unavailable to them. A wall can be thin yet hold its shape. The pressure of a thumb can alter a curve gradually. A handle can feel balanced before the glaze gives it a finished appearance. As those distinctions become usable, the activity itself acquires a different texture. The original wish for a mug remains, but it no longer describes everything the person finds worthwhile.

Two kinds of learning are occurring. The person learns facts about the activity, such as how long a piece takes to dry. They also acquire abilities that change what they can experience within it. Knowing that a certain curve is difficult differs from being able to feel the curve developing beneath one's hands. The second achievement creates a new way of participating.

This can change preference as well as improve prediction. At the beginning, the person might have predicted satisfaction mainly from possessing a successful object. Later, they may choose a more demanding form because shaping it is interesting, even if it is less useful at home. The change need not mean that the initial choice was mistaken. The person has acquired a relation to the activity that the earlier choice helped make possible.

There is an important difference between discovering an unchanged preference and developing a new capacity for appreciation. A tasting may reveal that someone already likes an unfamiliar flavor. Learning a craft may alter the features they can notice and the actions through which those features matter. Actual episodes can contain both processes. A thoughtful account of change leaves room for either instead of assuming that every later preference was fully formed at the start.

The distinction gives a trial period a particular value. It can provide information about a fixed option, but it can also let the person become sufficiently able to encounter what the option contains. The first ten minutes of an activity may reveal its setup cost more readily than its possibilities. A short course, an accessible introduction, or a patient collaborator can create a more informative encounter by making basic participation possible.

The person can then decide what they want from the next stage. They may pursue technical mastery, enjoy a weekly social session, make occasional gifts, or conclude that one course was enough. Those are different relations to the same activity. The experience has widened the vocabulary of choice: continuing no longer has to mean one standard path toward excellence.

Changed appreciation can also alter the meaning of an earlier object. The first uneven mug may become valued because the

maker can now see how its shape arose. It carries traces of decisions they would make differently today. Its value can include a remembered beginning, a visible problem solved imperfectly, and the contrast with later fluency. The object has stayed the same while the maker's ability to read it has developed.

Other people can help reveal these possibilities. A teacher may point out a detail the learner has not yet noticed. A fellow participant may use the same technique for an unexpected purpose. Their contributions offer new ways into the activity. The learner's response gives the next step its direction: curiosity about one technique, satisfaction with another, or a wish to try a different material.

For the question of lived experience, this is more than the acquisition of a resource. A capacity can change which features of the world become available to attention and action. The available future grows through the person as well as around them. Access to an activity, time to become familiar with it, and opportunities for meaningful participation can therefore matter in ways that a list of initial preferences would not fully describe.

A life can contain this kind of authorship repeatedly. Someone begins for one reason and discovers another; learns enough to choose more precisely; or carries an acquired sensitivity into a different pursuit. Ambition becomes interesting when it includes that development. A possible future can guide present effort, while present participation gradually teaches the person which future they want to make.

26.4.4 — Discipline Can Preserve Future Possibility

Research on grit found associations between perseverance and achievement, but later meta-analysis indicates modest predictive power and substantial overlap with conscientiousness

(Duckworth et al., 2007; Credé et al., 2017). Persistence can matter without explaining outcomes apart from ability, opportunity, support, health, and luck.

Depletion also has causes outside the individual. Burnout research describes exhaustion, cynicism, and reduced efficacy as responses to chronic workplace conditions rather than simple failures of will (Maslach and Leiter, 2016). A schedule that destroys sleep, health, or relationships may close more paths than it opens.

Supporting capacities are part of the path. An artist who learns budgeting, a researcher who learns communication, or a tradesperson who protects physical health may preserve the visible craft more effectively than someone who concentrates every hour on it.

A good day's practice and a lasting gain in skill are related questions, but they are not identical. Soderstrom and Bjork (2015) review evidence that immediate performance can diverge from later retention and transfer. Some conditions make the practice session look smoother without producing the strongest later learning; in other conditions, learning becomes visible only on a later test. This gives ambition a more useful time scale than the impression left by a single session.

Imagine an apprentice learning to identify three kinds of wood joint. During practice, labeled examples remain on the bench, and the apprentice names each new specimen correctly by comparing it with the displayed model. That is a real success at the supported task. The next useful question is whether the distinction is available tomorrow when the labels are covered and the specimens are unfamiliar. The two observations describe different achievements.

The apprentice and teacher can design a small sequence around the intended skill. They first work with visible examples,

discussing the features that distinguish the joints. Later, the apprentice names an unlabeled specimen and explains which feature supports the answer. On another day, a differently sized specimen tests whether the distinction travels beyond the original examples. The aim is to make the desired capability observable under the conditions in which it will be used.

A difficult answer can also become informative. If the apprentice recognizes the shape but confuses its name, the next practice can connect those two elements. If the name is recalled but the distinguishing feature is missed, the next comparison can emphasize the feature. Progress becomes a changing understanding of what needs practice, rather than a demand that every attempt feel easier than the last.

This kind of discipline preserves the future by bringing it into the design of today's work. What will the person need to remember, recognize, or do when help is less immediate? A well-chosen later check lets the present session answer that question. The satisfaction of fluent practice can remain welcome, while a growing ability to return, recognize, and act gives ambition another measure of progress.

26.4.5 — Uncertain Success and Nonmoral Failure

Two people can prepare with equal seriousness and meet different doors. Studying only visible successes can hide failures and distort explanations of achievement (Denrell, 2003). Opportunity also varies sharply across places and families (Chetty et al., 2014). An outcome is evidence about a path, not a pure reading of merit.

When an attempt fails, its causes can be separated. The method may have been weak, the timing poor, the goal unclear, the support insufficient, or the plan no longer suited to the life

around it. Learning begins when changeable causes are distinguished from conditions requiring help or acceptance.

Second routes are partly institutional. Education, retraining, health care, disability access, child care, debt relief, lawful protection, and a chance to reenter can determine whether a setback remains local or becomes permanent narrowing. Praising resilience while closing return paths confuses character with infrastructure.

26.4.6 — Keeping a Future Open

Keeping every possibility open can itself become a demanding plan. Imagine a person who enjoys photography, wants to study a language and is considering a return to formal education. Each possibility has a different way of remaining available. A camera can sit unused for a season. A language becomes easier to resume if it is encountered occasionally. An application may have a fixed deadline, while a particular qualification may require prerequisites that take time to acquire.

Calling all three unfinished goals leaves out their different clocks. Some options are preserved by owning an object, some by maintaining a skill and some by acting before an opportunity expires. A useful decision begins by asking what actually closes each path. The answer may be a deadline, a lost contact, a change in finances or simply the growing difficulty of restarting without a reminder of where the work stopped.

A pause can be designed to make return easier. Before setting a project aside, the photographer can label the unfinished images, keep a short note about the intended sequence and store the working files where they can be found. Those small acts preserve a point of reentry. They do not require continuing the whole project during a period when attention belongs elsewhere.

Other options need an active minimum. The language learner may choose a brief, enjoyable conversation that maintains familiarity without demanding the schedule of a formal course. That choice has a purpose different from rapid improvement. It keeps a route usable. Recognizing the purpose allows a smaller practice to count as successful on its own terms, instead of being judged against a commitment the person has deliberately paused.

A deadline creates another kind of decision. Gathering information early may reveal that an application can be deferred, that a prerequisite can be completed later or that an alternative route remains open. The next step is then chosen for the option it preserves. Sometimes the best use of an hour is to answer a question that determines whether many later hours are needed at all.

Preserving an option also consumes something. Membership fees, storage, correspondence and recurring practice can draw on money and attention. A collection of dormant projects may leave little room for the activity a person most wants to pursue now. Letting one option close can therefore be an affirmative choice about another. The relevant comparison concerns the futures the person values and the present conditions each future asks them to maintain.

Some investments travel across paths. Learning to organize files, communicate an idea, budget for materials or ask a useful question can support several future projects. The original ambition may change while these capacities remain available. A plan can value that transferable learning alongside its intended outcome, giving a person more than one way for serious effort to enrich a life.

The humane question is consequently more specific than whether someone persevered. What did they learn about the

path? What would make a return possible? Which commitments still fit the life around them? A future can remain meaningful while its form changes, and a deliberate pause can protect the very freedom that constant unfinished obligation would consume.

26.5 — Relationships, Grief, Rest, Joy, and Repair

Relationships can change which futures feel possible. Support may buffer stress or make practical effort less costly (Cohen & Wills, 1985; Coan & Sbarra, 2015). Isolation can carry serious health burdens (Holt-Lunstad et al., 2010). Care and repair should therefore attend to the quality of connection rather than treat social contact as an automatic good.

26.5.1 — Relationships, Rest, and Capacity

Relationships can also constrict. Coercion, contempt, manipulation, surveillance, financial control, threats, and unpredictable withdrawal of affection can narrow choice until ordinary decisions feel dangerous. The need to belong can make these conditions difficult to name (Baumeister and Leary, 1995).

Boundaries are therefore part of care. Distance, confidentiality, separate money, limited contact, professional support, or leaving may be necessary when closeness is used to control. Declining contact or reconciliation does not remove the responsible person's duties of repair.

Loneliness also deserves direct attention. It may persist in a crowd, after migration, during disability, following bereavement, or inside a relationship that no longer feels safe. The answer is not to assign meaning from above, but to make real connection more reachable. Rest is not the absence of a worthwhile life. Sleep loss can impair attention, working memory, executive function, mood, and safety, and repeated restriction can have cumulative effects (Dinges et al., 1997; Lim and Dinges, 2010). Protecting sleep therefore helps preserve basic capacity.

26.5.2 — Making Time Available to One Another

Two friends can each have free time and still have almost no time together. Imagine that one finishes work at four and cares for a relative from six onward. The other has time from six to eight. Each person has two available evening hours. Yet the hours do not overlap. Their difficulty is invisible in the separate totals.

Now a small schedule change moves the second friend's available time to five through seven on Wednesdays. Neither friend gains an extra hour; the same two-hour opening has moved. But an hour from five to six becomes available to both. They can walk, share a meal or simply sit together. A change in alignment has created a possibility that increasing either person's isolated total might have missed.

The distinction can be described without turning the relationship into a score. Each person has a set of usable times. A shared activity requires an intersection of those sets, together with a place, access and willingness. The duration of the intersection is a property of the relationship between schedules. It cannot be read from either schedule alone.

Even overlap may be only the beginning. A friend can be officially free while remaining on call, waiting for a delivery or carrying the responsibility for a child's pickup. Another can be available at the same hour but need a long journey to reach the meeting place. An hour made usable together may therefore depend on someone else reliably taking over a task, a nearby place to meet or an agreement that interruptions can wait.

That arrangement has its own labor. Someone notices the conflict, proposes a time, checks transport and remembers what changed. When the same person performs that work every week, the meeting's simplicity can conceal an unequal burden. Sharing the coordination can become part of the friendship itself: one

person chooses the place, the other checks the next date, and each can say when the plan has become difficult.

Reliability changes the experience of an invitation. An uncertain promise may keep an evening open without making it available for anything else. A clear confirmation lets a person arrange the rest of the day. When a cancellation is necessary, timely notice returns some of that freedom. These small practices matter because another person's planning is already beginning before the meeting does.

The example makes connection more tangible than an instruction to socialize. Sometimes the useful change is an additional contact. Sometimes it is making an existing relationship reachable through a shared hour, accessible transport or a dependable handoff. The person who wants company may already know whom they would like to see. What is missing is the practical overlap in which the relationship can be lived.

26.5.3 — Grief and Meaning Without Compulsion

Memory, ritual, conversation, objects, stories and an enduring inner relationship can become part of life after loss rather than obstacles to detachment (Klass et al., 1996). Practical help can include meals, childcare, transport, administrative tasks, listening, ritual, silence or protection from avoidable demands. It can be useful before anyone knows what the loss will mean.

An inherited object can hold more than one kind of connection. Imagine a sewing box passed from a grandmother to a grandchild. At first its value may lie in the arrangement itself: the familiar tin, the folded tape measure and the small scissors resting where they have always rested. Leaving the contents in place preserves a recognizable scene.

Later the grandchild may use the scissors to mend a shirt. The connection now includes an action. A remembered gesture becomes part of present skill, and the object participates in an ordinary day. Nothing about that use requires the memory to become smaller. Its role has changed from something kept intact to something carried forward through doing.

A different family member may care most about the box's stories. Together they can name the occasions attached to a piece of fabric or recall who taught whom to thread a needle. Another person may have no wish to sew and still value a photograph of the box. These are different ways of receiving the same inheritance, each shaped by a particular relationship.

The practical question can change with time. At one stage the person may want a protected place for the object. At another they may want to learn its use, share its history or pass it to someone who will work with it. Recognizing the changing question can make a conversation easier. Instead of deciding what the object ought to mean, someone can ask what the person would like to be possible with it now.

Meaning can thus continue through preservation, activity and exchange. A life leaves traces that later lives encounter in new circumstances. The trace remains connected to its history while participating in a present that has its own tasks, relationships and discoveries. An ordinary repair made with familiar scissors can contain both memory and a new purpose.

Danger, severe impairment, trauma or other clinical needs may call for professional support. That response concerns the person’s condition rather than an imposed timetable for mourning (Bonanno, 2004; Stroebe and Schut, 1999).

A death by overdose or public violence, estrangement, miscarriage, the loss of a stigmatized relationship, or loss of a role can bring grief that others fail to recognize (Doka, 1989).

26.6 — Learning from Personal Records

A personal record can preserve a useful discovery long enough for someone to act on it. Its value may lie in remembering what made an activity possible, which arrangement helped, or what to try next. A few well-chosen details can reconnect a later decision with the circumstances of an earlier success. The record serves the activity when it makes returning to that activity easier.

26.6.1 — Remembering What Made the Day Work

Imagine someone learning to play a piece of music. After a satisfying practice session, they write, “Slow passage first; instrument ready; quiet room before dinner.” The note preserves three concrete features of the session. A broad judgment such as “Today was good” would be less useful when deciding how to begin tomorrow.

Several days later, the person has only a short interval available. The note helps them identify the most useful starting point. They can prepare the instrument and return to the slow passage immediately. The earlier session has left a small piece of practical knowledge that survives the interruption between one attempt and the next.

The arrangement may change. The quiet room might become unavailable, while a shorter session in another room works well. A later note can preserve that second route. Over time, the record becomes a collection of ways to begin under different circumstances: a full session, a brief return, a way through one difficult phrase. Its usefulness grows through distinctions among workable options.

The same pattern can occur in a shared activity. A family may remember which preparation made an outing easy to start. A volunteer group may record where supplies were placed and

which instruction helped a new member join. The note carries forward knowledge that otherwise remains attached to one person's memory or disappears when a routine is interrupted.

What matters is the connection between a detail and an action. “The instrument was ready” points to preparation. “The slow passage came first” points to sequencing. “Before dinner” points to an available interval. The record gives an ordinary day a way to teach the next one.

26.6.2 — Turning an Impression into a Defined Comparison

A question can become more specific as the record develops. “Why do I never have time?” may become, “Which of these two available intervals lets me begin reliably?” The second question names a decision and a visible outcome. For the musician, beginning the chosen passage may be more relevant than rating the entire day.

An informal record can reveal candidate arrangements worth trying. A planned N-of-1 study goes further by defining the intervention, outcome, periods and analysis in advance, while accounting for repeated observations and effects carried from one period into another (Guyatt et al., 1986; Lillie et al., 2011). The distinction gives each kind of record a clear job. One develops a practical question; the other uses a specified design to estimate a defined comparison.

Context helps interpret either record. A practice interval during a quiet week differs from the same interval when the household schedule changes. A successful arrangement may depend on preparation that occurred earlier. Choe and colleagues (2014) show how useful self-tracking depends on gathering relevant context and interpreting observations through clear comparisons. Recording the detail most relevant to the next decision can make the trace easier to use.

Once an arrangement has become dependable, attention can return to the music. The most valuable result may be a routine that now requires very little recording: the instrument is ready, the first passage is chosen, and a short session is enough to resume. The record has helped preserve a path back into something the person values.

Chapter 27 — A Scientific Program

A serious scientific program gives a large question a place to begin. A useful first contribution could be a better way to identify neutral feeling, a test of what information a candidate mechanism must retain, or a biological model that makes a clear prediction about completion. None requires one researcher to solve the entire problem at once. The contributions become cumulative when their connections are explicit and independent critics can distinguish genuine progress from an attractive fit.

Three linked questions organize the work. What aspect of signed experience can the available observations identify or bound? Which relations among intervals or rejoining paths does a candidate mechanism predict before those observations are used? What additional evidence connects those local relations to the beginning and end of a complete stream? Different fields can contribute to these questions without requiring any one laboratory to reconstruct an entire life.

Consider a collaboration built around an ethically approved, low-burden comparison. One group studies what a participant’s reports mean across repeated conditions. Another examines whether an independently specified physical quantity retains information about the path. Their first task is to agree on a contrast that both groups can investigate. A boundary difference might be accessible even when its two absolute values are not. Alternatively, several observations might bound the difference without identifying one exact number. The choice determines what the experiment should record and what its result can answer.

A useful observation map need not reconstruct the whole underlying state. Functional observability asks whether observations determine a specified function of that state, even when other coordinates remain unknown (Montanari et al., 2022). For a candidate state-change relation, the required

function may be a difference across boundaries rather than the absolute value of Q.

Consider an invented transition from (a, b, z) to (b, a, z), with only the first coordinate observed before and after. Suppose independent theory specifies Q = a + z. Neither Q-value is known because z is hidden, yet the initial-minus-final difference is a − b, which the two readings determine. If the unobserved coordinate instead changes to 2z, the same readings no longer determine the difference. The dynamics and target definition do the work; the quantity is not constructed from the felt total it is meant to explain.

This offers a focused measurement objective: identify the contrasts the candidate actually needs, and test their accuracy on unused observations. A constant ambiguity in Q that cancels across boundaries is different from an unknown offset in the valence rate, which accumulates through time. Noninvertibility alone does not defeat the bridge; an unobserved change that alters the target does.

27.1 — The Scientific Question and Measurement Gates

A scientific program needs a claim that could fail: on the primary finite signed-measure branch, PTVC requires the finite signed phenomenal measure of each separately routed maximal qualifying conscious stream, across its strict-pre-endpoint domain, to have an exactly zero terminal residual.

27.1.1 — From the Claim to a First Study

The first proposed study tests whether additive path comparisons remain consistent within a person using the design in Section 17.2.4. It measures held-out path contrasts and cycle sums against independently calibrated measurement and exposure uncertainty. Such a test cannot establish terminal fairness, but it can reveal whether the local representation earns further investigation. Each next study should be selected for what it could establish, and every planned comparison must name the result that would close the proposed route. Independent reviewers should be able to demand recovery tests, narrow claims, pause release, or declare the question unanswered, with those decisions documented before publication. The record identifies the required tests, who will complete them, and which findings remain exploratory.

27.1.2 — Minimum Measurement Achievement

A measurement gate must be explicit before any claim is evaluated. State the target component and what observation or bound stands in for it. Record validation evidence along with a failure criterion. Chapters 10–11 provide the rules for deciding when the resulting comparison is informative and when it must abstain.

Begin with an independently justified neutral origin or a bounded set of plausible origins. Neutral-condition calibration and sensitivity analysis over that set provide the evidence described in Section 11.2. If an allowed origin shift reverses the contrast or its decision, retain a bound only when it still answers the fixed question. Otherwise the test leaves that question unresolved. A questionnaire midpoint alone does not establish phenomenal neutrality.

Within a stream, magnitude and sign require a cardinal relation adequate for the chosen cross-sign contrast. Independent anchors and comparison tasks can test that relation alongside the measurement model (Sections 10.2 and 10.5.1). If admissible transformations change the result, report the ordering or narrower contrast that remains identified, or test a separately named nonadditive branch. Interpersonal units are unnecessary for this particular test: different positive constant units across streams leave unchanged whether each stream's own total is zero.

Accumulation requires interval contributions measured against a declared duration or exposure rule, including residence and transitions. Interval-additivity checks and calibrated timing supply the tests developed in Sections 7.3.1 and 17.2.4. Omitted exposure, nonadditivity, or duration uncertainty may change the comparison. When that happens, revise the accumulation model or retain an adequately bounded contrast. The result should follow a justified rule for combining contributions rather than a sum of incompatible values.

For a state-based candidate, the needed observables are potential differences. Independent boundary observations and held-out histories reaching the same declared state test those differences (Sections 8.3.2 and 17.2.4). Hidden history that changes the required difference beyond the allowed uncertainty

shows that the proposed state description is inadequate. Full neural reconstruction is unnecessary when the specific contrast is identified; the experiment needs the information relevant to its question.

Routing and boundary assessment assign the stream and relevant endpoints prospectively. Blinded continuity and boundary adjudication provide the procedures developed in Chapters 5 and 13. If outcome-dependent assignment or unresolved alternatives change the result, retain the admissible branches or limit the test to its protocol window. A local transition experiment can answer its own question without identifying a complete-stream terminal residual.

The declared contrast must also survive channel error, missingness, and changes in context. Calibration, invariance, and recovery tests examine that transport through the actual analysis pipeline (Sections 10.3.2, 11.1, and 23.3.1). When the allowed uncertainty or a failed validity criterion prevents the planned decision, the appropriate result is No adequate estimate, as defined in Section 10.5.3.

27.1.3 — Observation, Estimands, and Estimators

A terminal cohort makes the separation concrete. The last completed self-report is an observation. A person’s mean experienced valence during a declared terminal window could be an estimand; its definition must say how events such as sedation or loss of consciousness enter the target. A hierarchical model combining several channels could be the estimator, with explicit assumptions for missed reports. Connecting that estimate to a complete-stream residual requires a separate measurement and transport bridge.

A practical notation audit attempts substitutions that should be invalid: replace a truth-level endpoint with the last observation,

a latent quantity with its estimate, or a candidate route with the target Law. If the resulting claim still appears equivalent, revise its labels and make the missing inference bridge explicit.

Clinical-trial guidance from the International Council for Harmonisation (ICH) emphasizes defining the estimand before choosing the estimator. Treatment discontinuation, rescue medication, and death require explicit strategies within the scientific question; missing outcomes require estimation assumptions aligned with that question (ICH, 2019).

Events during follow-up also need declared causal roles. Sedation, analgesia, withdrawal, hospitalization, caregiver presence, and treatment change may be interventions, confounders, mediators, components of the observation process, or parts of the estimand strategy. Naming the role helps prevent an analysis from adjusting away the phenomenon or attributing a treatment effect to PTVC (Hernán and Robins, 2020; ICH, 2019).

27.2 — Rivals, Preregistration, and Red Teams

Rival predictions need the same evidential discipline as the favored hypothesis. Preregistration fixes the planned contrasts before results are inspected. Null results can narrow a mechanism when measurement was adequate. Independent critical review should look for an ordinary explanation that fits the pattern equally well.

27.2.1 — Null Results

A finite result can favor a precise relation when competing models predict it differently. Consider a hypothetical local path contrast observed to be zero, with a known Gaussian standard error. One model specifies a true contrast of exactly zero; a rival assigns a zero-centered Gaussian prior whose width is fixed independently of the test data. The predictive-density advantage of the exact-zero model increases as the rival's prior width grows relative to the measurement error. If the standard error is 0.1, a rival prior width of 1 gives an advantage of about 10.05 to one; a narrower rival width of 0.1 gives an advantage of about 1.41 to one.

The same observed zero therefore has different evidential force against different alternatives. If zero is a rounded display, compare each model’s probability of the rounding interval rather than treating the display as infinitely precise. The widths require substantive justification rather than being selected to favor the preferred result, and posterior model odds also depend on prior model odds. Compatibility alone is therefore weaker than a discriminating comparison (Dienes, 2014).

A rounded zero has a width, and that width belongs in the comparison. Retain the normal models just described, with measurement standard error 0.1 and rival width one. The exact-relation model predicts an observed estimate with standard

deviation 0.1. The rival predicts a standard deviation of √1.01, about 1.005, after combining its prior spread with measurement variation.

Suppose the estimate is displayed to the nearest 0.2 units. A displayed zero represents the interval from −0.1 to +0.1, apart from an immaterial boundary convention. The exact-relation model assigns about 0.683 probability to that interval. The rival assigns about 0.0793. Their ratio is about 8.61, favoring the exact-relation model for this displayed result under the two specified predictive laws.

The point-density ratio at zero was about 10.05. The finite-bin ratio is smaller because the interval also includes values away from its center. Each model contributes all the predictive probability it assigns across that interval. A display that says zero therefore corresponds to an event with a calculable, positive probability under both models.

Now round much more coarsely, to the nearest whole unit. The zero bin runs from −0.5 to +0.5. The exact-relation model places almost all its probability there, while the rival places about 0.381. The ratio is about 2.62. The printed symbol remains zero, but the information in that symbol has changed substantially.

As the bin becomes very narrow around zero, each probability is approximately its width times the model's density at zero. The common width cancels in the ratio, which approaches the point-density ratio. This explains when a density comparison is a good approximation to a rounded observation and how its accuracy depends on resolution.

27.2.2 — Adversarial Collaboration

The proposed study in Section 17.2.4 supplies the comparison, calibration pilot, analysis and decision dependencies for a first

local test. Its unresolved measurement and feasibility requirements remain research tasks. Further technical work can extend that design without changing what its present result would establish. Applications elsewhere in the series require their own economic, legal, institutional or other warrants; coherence with a scientific hypothesis is not that independent evidence.

The Cogitate Consortium and collaborators (2025) tested divergent predictions of integrated information and global neuronal workspace theories in 256 participants using three neuroimaging modalities. Theory proponents helped specify predictions and interpretations in advance, while the consortium conducted the comparison. The example demonstrates an organizational method, not a settled verdict on consciousness.

An adversarial collaboration begins to work when the rival accounts help shape the same experiment. Consider the distinction between elapsed-time and exposure-based change developed in Section 19.3.1. One team expects the relevant response to depend principally on time since the first event; another expects it to depend principally on how many repetitions have occurred. If the original schedule links the two, each team can explain the same declining sequence.

The joint task is to find schedules on which their predictions separate. One comparison can hold the number of repetitions fixed while changing the gaps between them. Another can hold total elapsed time fixed while changing the number of repetitions. Each team writes down what its account predicts for both comparisons before seeing the outcomes. Their disagreement becomes a source of experimental information.

The teams also need a shared account of the observation. If one expects a momentary response and the other predicts a delayed report, they may appear to disagree while answering

different questions. They can jointly specify the response window, instrument, population, and conditions under which their predictions apply. This gives the comparison a common object without requiring the underlying theories to become identical.

Interpretation can be prepared with the same care. A pattern may favor the first account, favor the second, or indicate that both elapsed time and exposure contribute. The teams can identify those possibilities in advance and state which additional observations would distinguish a combined mechanism from a change in measurement. A result that motivates a third account can then generate a new prospective study, with the original predictions still available for inspection.

The intellectual benefit is larger than an agreed procedure. Each proponent knows where their own account is most specific and which result would require genuine revision. Bringing that knowledge into the design can make the experiment more discriminating than a comparison devised by either team alone. The collaboration turns disagreement into a shared search for a result whose meaning both sides have helped make clear.

27.2.3 — The Question Inside a Score

A score contains a question about what an answer should accomplish. One scoring rule gives extra weight to large errors. Another counts each unit of error equally. A third treats an underestimate differently from an overestimate. Models evaluated under these rules can receive different rankings even when they face exactly the same outcomes. The difference can be understood by deriving what each rule asks a prediction to represent.

Consider an invented outcome that is zero with probability three quarters and eight with probability one quarter. Its mean is

two. Its median is zero, because at least half the probability lies at zero. These are different descriptions of the same distribution. Neither changes the underlying observations; each compresses them according to a different relation.

First evaluate a fixed prediction a by its expected squared error. The score is three quarters times a² plus one quarter times (8 − a)². Expanding gives a² − 4a + 16, or (a − 2)² + 12. The minimum occurs at a = 2. Predicting the mean gives expected squared error twelve. Predicting zero gives sixteen. Squaring makes the occasional large miss influential enough to favor a prediction above the most common outcome.

Now evaluate the same prediction by expected absolute error. For a between zero and eight, the score is three quarters times a plus one quarter times (8 − a), which equals 2 + a/2. The minimum in that interval occurs at zero, and predictions outside the interval do no better. Predicting zero gives expected absolute error two. Predicting the mean, two, gives three. The ranking has reversed because the score asks for a median rather than a mean.

The result is not a dispute about which method has calculated correctly. A method optimized for squared error can be excellent at estimating the mean. A method optimized for absolute error can be excellent at estimating the median. Calling one “more accurate” without naming the score leaves the substantive question unfinished. In a model-recovery study, that question should be settled before the predictions are ranked.

Asymmetric costs create another target. Suppose each unit by which a prediction falls below the realized outcome costs three units, while each unit above it costs one. On the same distribution, a prediction between zero and eight has expected cost three quarters times a plus one quarter times three times (8 − a). The terms involving a cancel, leaving six. Every prediction

in that interval minimizes the expected cost. The scoring rule admits several equally good point answers.

Increase the underestimation cost to four while keeping the overestimation cost at one. For a between zero and eight, expected cost becomes eight minus a/4, so the minimum moves to eight. The distribution has not changed. What changed was the consequence assigned to each direction of error. In an operational forecast, that asymmetry could express the relative cost of too little versus too much provision; in a scientific estimate, it would require a justification tied to the declared purpose.

A full predictive distribution preserves information that a single point discards. It can support a mean, a median, a quantile, or a decision derived from a stated loss function. In the constructed example, publishing the two possible outcomes and their probabilities makes all the calculations available. A reader with a different legitimate decision problem can then derive a different action without pretending that the underlying evidence has changed.

This also clarifies disagreement between summary statistics. A forecast near zero and a forecast near two may represent different targets rather than incompatible beliefs about the data. Asking each analyst what their answer is intended to estimate can resolve an apparent conflict before a new experiment is needed. If they claim the same target, their disagreement becomes a sharper question about models, data, or computation.

For a scientific program about experience, the same discipline keeps numerical success connected to meaning. Predicting the next report, estimating an average contribution, identifying a mechanism, and choosing an action are separate achievements. A score earns its place by representing the achievement under evaluation. The most informative comparison states that

connection plainly, so that a better number corresponds to a clearer answer to the question the study actually set out to ask.

27.3 — Replication, Recovery, and Scientific Verdicts

A promising local result must survive new settings and independent teams. Transport tests should state which conditions are assumed to remain stable. They should also name the differences that could defeat the prediction (Pearl & Bareinboim, 2014; Dahabreh et al., 2020). Replication earns confidence only when the successful result was not built into the endpoint or analysis rule.

27.3.1 — Replication and Theory Maturity

Record transport separately by axis (for example age, language, site and endpoint cause) with the tested contrast, evidence, uncertainty and outcome for each. Numerical scale invariance and the construct or causal bridge have separate statuses: agreement on a change does not establish whether its source is valence, medication, communication loss or selection.

Independent replication asks whether a finding recurs when another team collects new data to address the same question. Reproducibility asks whether the same data and computational procedures yield consistent results. Deliberately changing instruments or context also tests how far the finding transports (National Academies of Sciences, Engineering, and Medicine, 2019).

Data and code should be as findable, accessible, interoperable, and reusable as ethics, consent, privacy, and law allow (Wilkinson et al., 2016). Openness is purpose-limited; intimate trajectories should not be released merely to satisfy an abstract norm.

For randomized studies, use the SPIRIT 2025 and CONSORT 2025 protocol/report distinction explained in Section 10.4.4 to make replication possible (Chan et al., 2025; Hopewell et al., 2025). Assess theory maturity (definitions, consistency, branch

separation, theorem status, mechanism specification and unresolved primitives) separately from evidence maturity: observation quality, replication, transport and discrimination. A formally precise theory may lack an adequate empirical test; a replicated pattern may still lack a unique explanation.

27.3.2 — How Several Results Become One Body of Evidence

Several observations become a body of evidence through a model of how they are connected. If each observation comes from an independent new source, the joint probability can be a product of separate probabilities. If the observations share an unknown source property, that property can connect their evidential meaning even when the individual draws are conditionally independent.

Consider two invented models for a binary sequence. Model A fixes the probability of one at one half for every trial, with independent trials. Model B first draws a probability p uniformly between zero and one, then uses that same p for every trial, drawing the outcomes independently conditional on p. The models agree about the first outcome: each assigns probability one half to a one.

They disagree about two successive ones. Model A assigns probability one quarter. Under Model B, the conditional probability is p^2, which must be averaged over the uniform distribution of p. The integral of p^2 from zero to one is one third. Two ones therefore give a likelihood ratio of $(1/3)/(1/4) = 4/3$ in favor of Model B.

The second one is more predictable under Model B after the first has been seen. The first success shifts the distribution of p toward larger values, giving a predictive probability of two thirds for a second success. Multiplying the first probability of

one half by this updated probability gives one third. The joint evidence is a product of conditional predictions, each using the earlier outcomes.

For n successive ones, Model A assigns probability 2^(−n). Model B assigns the integral of p^n, which is 1/(n + 1). The likelihood ratio favoring B is therefore $2^n/(n + 1)$. At n = 4 it is 3.2; at n = 10 it is approximately 93.1. The sequence's persistence distinguishes a fixed fair source from a source whose unknown bias is shared across all trials.

Now alter Model B so that it draws a new independent probability p for every trial. Each outcome then has marginal probability one half, and the outcomes are independent after averaging over their separate probabilities. The probability of n ones returns to 2^(−n), exactly matching Model A. Variation in the individual probabilities is no longer detectable from one draw per independently resampled source in this particular design.

The difference is the level at which the latent property is shared. One unknown p retained for the whole sequence creates predictive dependence. A new p for every observation does not. To investigate variation between sources, the design needs repeated observations from identified sources, or other information about their probabilities. A large number of isolated single draws can leave that variation unresolved.

Mixed sequences supply another check. Under the shared-p model, a specified ordered sequence with k ones and n − k zeros has probability $\int p^k(1 - p)^{(n-k)}\,dp = k!(n - k)!/(n + 1)!$. For two observations, the sequence 10 has probability one sixth, while 11 has probability one third. Model A gives each sequence probability one quarter. The models differ in persistence, despite having identical single-trial margins.

This calculation suggests a precise way to understand replication. Studies can share an instrument, a population source, a calibration procedure, or an analysis assumption. Those shared features may enter a joint evidence model. Independent teams can change some of them while retaining others. The value of a new study depends partly on which uncertainty it addresses through genuinely new variation.

Evidence can be combined sequentially without pretending those dependencies disappear. The next result contributes its probability conditional on what has already been learned under each model. If the next study repeats a feature already explained by a common source, its contribution may differ from that of a study that tests the relation under a new source or setting. The relevant conditional prediction makes that difference explicit.

For the Law's scientific program, a collection of favorable results is therefore most informative when their connections are understood. One result may improve calibration, another may test transport, and another may separate two mechanisms under a new intervention. A body of evidence grows through the questions those results jointly answer. The simple binary models show why the architecture linking observations matters as much as the number of observations in the record.

27.3.3 — Verdicts Must Be Modular and Exact

Record four distinct outcomes: TargetStatus, RouteStatus, ComparatorStatus and InferenceStatus. Identify the tested branch, modality, reference class, scale and boundary rule. For each inference, name the proposition already earned, the new assumption, its added error and the decision it permits. These dependencies connect the result to its warrant.

No adequate estimate. InferenceStatus records the failed prerequisite and the justified partial results. It does not assign target support or defeat (Section 10.5.3).

Shared result, unresolved mechanism. Retain a calibrated finite bound, valid equivalence result or consequence shared by adequate surviving models; report relative support and predictive overlap where informative. Inadequacy of every candidate is not shared support (Lakens, 2017; Wasserstein & Lazar, 2016; Akaike, 1973; Burnham & Anderson, 2002).

Route support or failure. State the tested signature and where it applies. A capable comparator can win that contest while the terminal target remains open. If a terminal relation is supported within the tested class, a successful model must reproduce both the observed path diversity and that relation under the same independent measurement and endpoint rules. Discrimination and transport strengthen the finding more than a count of additional compatible examples.

Target falsification. Apply the exact-branch counterexample rule in Section 4.2.2. A finite-band branch is adverse when its truth-containing residual set is disjoint from the fixed acceptance region. Probability, support and mechanism branches retain their own guarantees and decision rules.

Specify the premise challenged: representation or origin, bearer or endpoint, observation-to-target transport, candidate route or universal quantifier. This determines which claim changes and which separately established results remain available.

A failed prerequisite directs work to the missing bridge; a failed route is revised or retired. Ambiguity calls for a comparison able to distinguish the surviving models. If no available ethical design can do so, preserve that ambiguity: the ethical barrier is evidence for neither model. Narrowing changes

the stated scope, and defeat changes dependent claims and the research agenda.

The verdict packet preserves the prospectively specified question, data and code version, gate status, rival set, estimates, uncertainty, deviations, sensitivity analyses, dissent, and correction route. A correction should identify both the superseded packet and its replacement, so readers can follow the changed inference rather than encounter an unexplained new conclusion.

The public sentence should therefore be readable and exact: what was tested, under which gates, what was found, what remains unresolved, and what changes next. An illustrative report could say, “The scale passed its registered calibration test in this setting; its accuracy near the proposed endpoint still requires separate validation.”

Could life be fair? The possibility gives us a remarkable question to pursue. We now have a sharper way to ask it, a clearer view of what an answer would require, and concrete places to begin. A successful inquiry could connect the changing experience of an ordinary day with a relation spanning a complete conscious history. The question is large, but the next contribution can be specific.

A reader may bring a difficult example to a discussion. A researcher may turn one unresolved distinction into a study. A critic may reveal an assumption that needs a better argument. Each contribution can give the next person a firmer place to begin. The possibility is reason to keep asking with imagination and care. Let curiosity carry the question forward, and let care begin where we are.

Glossary

A

Absorption: A process remains in a declared set after entering it. A proposed relation between absorption and a stream endpoint requires its specified endpoint bridge. See Section 7.2.3.

Abstention: See No adequate estimate.

Action-mediated closure: A candidate route on which available actions and their consequences help bring about closure under a declared state, information, policy, and continuation model. Other route families may rely on passive dynamics or boundary relations.

Active inference: A family of models in which an organism updates beliefs and selects actions using predictions about observations and preferred or viable outcomes. Actions can seek information as well as change the environment. See also: Predictive processing; Reinforcement learning; Viability kernel.

Ad hoc patch: A change introduced after an adverse result mainly to preserve a preferred conclusion, such as moving the endpoint, recentering the origin, inventing an unseen interval, or excluding a bearer. See also: Section 25.1.1; Operational separation.

Admissible continuation: A continuation included in the prospectively specified model class. Closure for every admissible continuation is a set-universal claim over that nonempty class. See also: Set-universal claim; Reachability.

Adverse case: A gate-valid qualifying stream for which the appropriate truth-containing result excludes exact zero. Under an exact universal branch of PTVC, one valid adverse case cannot be canceled by favorable population averages.

Affective valence: See Felt valence.

Aliasing: Distinct underlying time courses produce the same record under a specified observation schedule. See Section 10.2.3.

Allostasis: Anticipatory regulation that adjusts operating levels, mobilizes resources, and redistributes physiological cost as demands change. See Section 21.1.1.

Almost-sure attainability: The existence of at least one admissible policy that reaches a declared target with probability one from a stated information state. It is distinct from closure under a fixed law or success under every policy, and stronger than limit-sure attainability under the same regime and policy class. Limit-sure attainability need not supply a single policy attaining probability one. See also: Limit-sure attainability; Probability-one closure; Reachability.

Alternative endpoint branches: A family of prospectively declared endpoint architectures: state coupling, first hitting, absorption, endpoint atoms, singular timing or support, continued experience beyond a protocol boundary, and endpoint-inclusive value. See also: Endpoint atom; Hitting time; Absorption; State-coupled timing.

Arousal: In affective science, the felt activation or energy dimension of experience. Clinical use of the term for wakefulness addresses a different assessment question. High or low arousal can accompany pleasant, unpleasant, mixed or neutral states; it cannot substitute for a valence representation or terminal residual. See also: Felt valence; Observation channel.

Atom at zero: A positive probability mass located exactly at zero in a distribution of terminal residuals. Positive mass at zero differs from concentration in a neighborhood of zero;

neither description alone establishes universal or probability-one closure.

Attainability: The existence of at least one admissible route or policy capable of reaching the target from a specified state. Attainability alone does not show that the route will be selected, that it works under partial observation, or that every admissible path closes.

Attrition: The process by which participants leave, pause, become unreachable, lose access, enter treatment, withdraw permission, or die during longitudinal research, thereby changing who remains visible. See also: End of observation; Informative missingness; Repeated observation.

Auxiliary assumption: An assumption needed to connect a theoretical claim to an observation, estimator, or test. A failed prediction may reflect the target hypothesis, an auxiliary assumption, or both, so confirmatory work must state these assumptions in advance.

B

Balance: Book 1’s accessible term for exact intrastream terminal valence closure.

Baseline drift: A change over time in a person’s use of a rating scale, neutral point, comparison standard, or reporting habit. Unmodeled baseline drift can create an appearance of convergence or closure without a corresponding change in latent phenomenal valence.

Bearer: The truth-level subject or unit to which qualifying felt experiences belong and within which the proposed accumulation is evaluated. See also: Maximal qualifying conscious stream; Ownership; Stream identity.

Behavioral nonresponse: The absence of expected command following or another observable response under a particular assessment. Motor, sensory, language, medication, arousal, task, and environmental barriers can block expression, so nonresponse alone does not settle whether experience is present or whether a conscious stream has ended. See also: Felt availability; Low-unity state; Unknown category.

Belief state: A probability distribution over hidden states that carries the relevant observation history forward under a specified model. See Section 21.1.3.

Bivariate branch: A representation retaining separate positive and negative components that may coexist. It differs from the primary additive scalar branch; transfer requires a justified bridge on a common identified domain. See Section 7.3.1.

Blinding: A design safeguard that withholds selected information from participants, investigators, coders, analysts, or adjudicators when that information could bias conduct or interpretation. The form of blinding must fit the study and its ethical limits.

Boundary hypothesis: A hypothesis about a limiting, terminal or otherwise delimited property of a process.

Brier score: The squared difference between a probability forecast and a binary outcome. Its expected value is minimized by the true event probability; smaller values indicate better performance under this score. See Section 18.3.4.

C

Calibration: The process of relating a measurement system to a defined reference so that its origin, scale, uncertainty, and interpretation are characterized.

Candidate mechanism: See Candidate route.

Candidate route: A proposed mechanism, policy structure, dynamical construction, or mathematical path by which the terminal target might be satisfied. A route can receive support or fail independently of the target hypothesis.

Candidate signature: A prospectively specified observable pattern predicted by a candidate route, such as horizon-sensitive choice, recovery after perturbation, or narrowing variance. See also: Negative control; Nonunique explanation; Model recovery.

Channel qualification: The staged admission process by which a report, sensor, behavior, contextual record, or other observation channel earns a defined role through technical verification, analytical validation, construct validation, burden review, and a failure threshold. See also: Calibration; Multichannel estimate; Observation channel.

Chosen meaning: An interpretation of experience that a person freely accepts, revises, postpones, or declines. See Section 3.1.2.

Closure: Satisfaction of a declared terminal target. See Section 4.1.1 for the primary PTVC target.

Coercive measurement: Measurement in which pressure or dependency makes disclosure a practical condition of access or standing. See Section 12.1.1.

Common-action requirement: For a guarantee covering every compatible latent state, a single action chosen from the available information must be closure-compatible across all states that remain observationally indistinguishable at that decision point. State-specific actions that cannot actually be selected do not satisfy this requirement.

ComparatorStatus: See Verdict.

Comparison without forced equivalence: A method for comparing traditions or ideas by naming a precise shared question, function, image, or difference while preserving distinct histories, doctrines, registers, and internal disagreements. See also: Controlled vocabulary transfer; Chosen meaning; Resonance.

Compensability: A route-specific question about whether a present formal discrepancy remains answerable through safe, realistically available paths before a declared horizon closes. See also: Future optionality; Noncompensable risk; Viability kernel.

Competing risks: A setting in which different events can end the same record. The probability of one event occurring by a stated time depends on the other ways the record can end. See Section 7.2.4.

Complementarity: Conditions work together so that one increases what another makes possible. In the illustrated access example, equipment and a usable control interface jointly enable the activity. See Section 2.1.1.

Complete felt history: The full history of qualifying experiences belonging to one routed conscious stream over the temporal domain specified by the claim. On the primary branch, this is the strict-pre-endpoint domain. See Section 4.1.1.

Completion potential: A candidate mathematical function whose decrease, boundary behavior, or terminal value can help characterize a proposed route to closure.

Conditional support: The support of a conditional probability law given specified information, under a declared topology.

Claims about exact closure must distinguish support restrictions from high probability or average behavior.

Conscious stream: A temporally connected sequence or field of qualifying conscious experience treated as one routed unit for analysis.

Constraint without a judge: An impersonal structural or boundary relation that may limit admissible complete histories without any agent assigning rewards, punishments, purposes, or schedules. See also: Boundary hypothesis; Section 3.1.

Construct validity: The degree to which a measure, manipulation, or model captures the intended latent construct rather than a neighboring one. For valence research, behavior, reward, arousal, language, and neural activity may inform measurement without being identical to felt valence.

Controlled direct effect: The effect of an intervention on an outcome when a specified intermediate variable is held at a common fixed level. See Section 18.2.1.

Controlled vocabulary transfer: The disciplined movement of a technical term across fields with its source meaning, evidence ceiling, local use, prohibited inference, and route back to specialist authority. It prevents words such as bearer, stream, candidate, indicator, sentience, support, and personhood from silently changing strength. See also: Section 6.2.2; Comparison without forced equivalence; Testimony-like output.

Critical path: The longest required chain of dependent tasks in a project schedule. Its duration determines the earliest completion time under the stated resource and availability conditions. See Section 9.2.3.

D

Decision rule: A prespecified mapping from evidence to an outcome such as support, adverse result, narrowing, abstention, or route retirement.

Desert-based fairness: The question of when praise, reward, blame, or sanction properly follows from what a person chose and did. See also: External justice; Section 3.1; Procedural fairness.

Detailed balance: A stationary-process condition in which the probability flow between each pair of states equals the reverse flow. It is stronger than stationarity alone. See Section 21.4.4.

Differential item functioning: An item has different response behavior across groups at comparable levels of the relevant construct under the declared measurement model. Equal raw group averages do not establish its absence. See Section 11.3.3.

Distributive fairness: Standards for allocating resources, opportunities and burdens among people. A standard may concern equal shares or the provision needed for a particular aim; the relevant comparison must be stated. See Section 1.2.4.

Dream-report chain: The ordered distinction among experience during sleep, awakening, the memory trace available after waking, and the report given to another person. Each transition can omit, reorder, prompt, or reshape what reaches the record, so a dream report is not an unedited transcript of sleeping experience. See also: Observation process; Informative missingness; Response shift.

Dynamic programming: A method that solves and stores smaller subproblems so their answers can determine solutions

to larger ones. Completion counts supply the subproblems in the conditioned-path example. See Section 23.2.2.

E

Empirical band: See Measured equivalence.

End of observation: The last time usable data are obtained because a channel fails, monitoring stops, communication is lost, privacy is requested, or another practical event occurs. If experience continues beyond that boundary, the remaining history is unobserved. The boundary does not by itself establish that the stream has ended. See also: Protocol endpoint; Truth-level endpoint; Observation window.

Endpoint: The boundary at which a qualifying conscious stream ceases for purposes of the truth-level target. The endpoint must be defined by the stream model and cannot simply be replaced by biological death, loss of contact, device shutdown, or the last available report.

Endpoint atom: Positive probability mass assigned by a declared endpoint-time or marked-endpoint law to a particular time or point. It changes the timing architecture but does not itself create closure, and it differs from an atom at residual value zero and from merely recording a categorical endpoint event. See also: Atom at zero; Hitting time; Absorption; State-coupled timing.

Endpoint bridge: The independently validated relation connecting an observed or protocol endpoint, possibly an interval, to the truth-level endpoint required by the claim while carrying error, return, revision, confounding, and competing-event uncertainty. A clinical definition or sensor threshold may inform the bridge without becoming the formal

endpoint. See also: Protocol endpoint; Return rule; Truth-level endpoint.

Endpoint conditioning: Selecting, observing, conditioning, or reweighting trajectories because of where they end, which can make earlier paths appear to drift or compress toward the selected endpoint without an independent terminal law. See also: Candidate signature; Operational separation; Random-endpoint problem.

Endpoint-inclusive branch: A separately declared branch that assigns a specified phenomenal contribution exactly at the endpoint. It has a different temporal domain from the primary strict-pre-endpoint branch, cannot be introduced after a residual is seen, and cannot establish strict-pre-endpoint closure merely because its own condition is compatible. See also: Complete felt history; Exact closure; Truth-level endpoint.

Equivalence band: A prespecified finite interval around a reference value within which effects are treated as practically equivalent for a stated purpose. A valid equivalence result supports that bounded claim; it does not establish exact equality to the reference value.

Error budget: An explicit accounting of uncertainty from sampling, instruments, calibration, sign, origin, timing, missingness, model choice, transport, and construct identification across the evidence chain. See also: Calibration; Noise floor; Partial identification.

Errors-in-variables: Models in which explanatory variables are observed with measurement error. The observation model matters because the relation involving a recorded predictor can differ from the relation involving the underlying predictor. See Section 11.5.1.

Estimand: The precisely defined quantity an analysis seeks to learn, including its population, unit, time domain, conditioning, and scale. The estimand is not the same as the estimator used to calculate it.

Estimator: A rule or procedure for using observed data to estimate an estimand. Bias, precision and consistency describe different aspects of its performance. A unique target value cannot be learned from data and assumptions that leave the target unidentified. See also: Estimand; Identification.

Ethical burden too high: A typed outcome in which a scientifically coherent question is not ethically available because the required burden, privacy exposure, dependency, deception, bystander risk, or lasting effect is disproportionate. See also: Section 3.3.1; No adequate estimate; Perturbation.

Exact closure: Exact satisfaction of the declared truth-level terminal condition. See Section 4.1.1 and Section 4.3.3.

Exact zero: See Exact closure.

Experienced exposure: The adopted measure by which qualifying phenomenal experience contributes across a stream. It may differ from elapsed calendar time. See Section 7.1.1.

External justice: Rights, capabilities, opportunities, material conditions, duties, institutions, protection, and remedy considered independently of any estimate of felt valence. See also: Chapter 3; Procedural fairness; Section 1.3.

F

Falsifiability: The requirement that a scientific claim specify observations or evidence objects that could genuinely count against it under prospectively declared gates and decision rules. See also: Boundary hypothesis; Truth-containing residual set; Falsification.

Falsification: A result that conflicts with a prespecified claim under the study's stated assumptions and validity gates. Falsification must identify which target, route, or auxiliary claim is affected rather than overgeneralizing from one failed component.

Felt availability: The modest claim that something is occurring for a conscious stream even when the experience cannot be spoken, deliberately used, remembered later, or organized into a clear self-story. It separates possible experience from an observer's access to evidence about it. See also: Behavioral nonresponse; Low-unity state; Observation channel.

Felt valence: The positive, negative, or neutral evaluative character of qualifying experience as lived from within under a declared representation and neutral origin. It is not identical to happiness, life satisfaction, preference, arousal, salience, reward, pain behavior, welfare, meaning, moral value, or human worth. See also: Arousal; Neutral origin; Primary additive branch.

Finite signed phenomenal measure: A finite signed measure representing positive and negative phenomenal valence over a qualifying stream's complete strict-pre-endpoint domain. Finiteness permits well-defined positive and negative Jordan masses and a terminal signed residual.

Finite-band branch: A target allowing a prospectively fixed acceptance region around neutrality. Its nonempty compatibility set must be interpreted according to its type and bridge assumptions. See also: Residual set; Measured equivalence.

Finiteness: On the primary finite branch, each complete positive and negative phenomenal magnitude must be a finite quantity. This admissibility condition does not impose one upper bound

on every stream, cap peak suffering, mean that experience is small, equalize gross experience, or establish finiteness for every possible representation. See also: Positive and negative phenomenal magnitudes; Primary additive branch; Gross experience.

First-stage effect: In an encouragement design, the effect of assigned encouragement on uptake of the action being studied. It supplies the denominator of the instrumental-variable ratio in the worked example. See Section 17.2.2.

Fitness, feeling, and moral worth: The separation among reproductive propagation, the valenced quality of a bearer's experience, and the ethical standing owed to that being. See also: Section 14.1; External justice; Viability kernel.

Frequency-dependent selection: Selection in which the relative reproductive contribution of a type depends on its frequency in the population. See Section 22.1.1.

Future optionality: The meaningful future paths a person can realistically reach through available time, bodily capacity, material conditions, relationships, information, institutions, treatment, support, and action. See also: Compensability.

G

Garbling: A transformation of an observed signal, possibly using independent randomization, that produces another signal without consulting the hidden state. If one channel can reproduce another in this way, it can imitate every decision rule based on that second channel. See Section 10.3.5.

Generalized Terminal Neutrality: A broader family of terminal-zero conditions defined by a declared functional, measurable output space, and distinguished zero when scalar addition would lose relevant structure. It is a separate branch

and is not PTVC unless a justified bridge connects its zero condition to the additive positive-negative decomposition. See also: Bivariate branch; Primary additive branch; Scalarization.

Gross experience: On a declared additive scalar representation, gross experience is the complete positive-plus-negative variation before cancellation. It is distinct from the terminal residual and is not total consciousness, duration, welfare, or every experiential dimension. PTVC does not equalize gross experience across streams. See also: Chapter 3; Phenomenal magnitude; Terminal residual.

H

Half-life: The elapsed time over which an exponentially decaying deviation is reduced to one half of its initial size. See Section 21.4.1.

Hedonic adaptation: The changing emotional impact of a gain, loss, or continuing condition as novelty, attention, expectations, comparison, routine, and context change over time. Adjustment varies across events, people, domains, measures, and times. See also: Opponent process; Response shift.

Held-out transport: Testing whether a prospectively specified model or result travels to data, settings, populations, channels, time periods, or endpoint regimes not used to build or tune it. In-sample fit or a random split alone does not justify claims beyond the context and observation process actually tested. See also: Measurement invariance; Model recovery; Reference class.

Hidden cumulative object: On the primary additive branch, the latent cumulative signed phenomenal-value process through time. See also: Terminal residual.

Hidden-state model: A model in which the process of interest contains latent states that are inferred from observable signals. Such a model must distinguish state dynamics, observation error, routing, selection, and missingness.

Hitting time: The first time a process enters a specified set. A path can hit a boundary without crossing it or remaining there. See Section 7.2.3.

Homeostasis: Regulation of variables within workable ranges. See Section 21.1.

Hysteresis: A rule with different thresholds for entering and leaving a state, so classification in the intervening range depends on the previous state. See Section 5.2.2.

I

Identification: Whether the observable-data distribution and stated assumptions uniquely determine the target quantity or distinguish the candidate model from alternatives. Larger samples cannot resolve nonidentification when alternatives remain observationally equivalent under the design.

Importance weight: The ratio of a sampled outcome or history's probability under a target distribution to its probability under the proposal distribution used to generate it. The ratio corrects the contribution of that sample to a target expectation. See Section 23.3.2.

Inference: Reasoning from observations and assumptions to conclusions about latent quantities, models, mechanisms, or populations. Every inference inherits the limits of its measurements, design, identification conditions, and auxiliary assumptions.

InferenceStatus: See Verdict.

Informative missingness: Missing observations whose occurrence relates to burden, state, access, treatment, refusal, shutdown, or another factor relevant to the inference, so visible records may differ systematically from what disappeared. See also: Attrition; Missingness; Observation channel.

Integral windup: Accumulation in a controller's internal command while an actuator is saturated. The stored command can remain outside the actuator's usable range after the target changes, delaying the visible response. See Section 21.3.3.

Interaction contrast: A difference between effects across conditions. In a two-factor comparison, it can be calculated as a difference of two differences on a specified outcome scale. See Section 17.1.3.

Interoceptive prediction: The anticipation and interpretation of bodily signals such as heartbeat, breathing, pain, fatigue, hunger, nausea, temperature, and arousal through expectations, context, uncertainty, attention, and action. See also: Active inference; Predictive processing; Prediction error.

Intrastream: Evaluated within one conscious stream at a time. The term fixes the scope of the proposed equality and excludes netting across people, households, families, populations, species, institutions, or ensembles, even when experiences are causally connected or socially interdependent. See also: Routing.

J

Joint coverage: The probability that every interval in a specified family covers its own target. This differs from the separate coverage of each interval. See Section 17.2.4.

Jordan decomposition: The representation of a finite signed measure as the difference between mutually singular positive and negative measures. Their total masses provide the positive and negative phenomenal quantities compared by the primary additive PTVC target.

Jordan masses: The total positive and total negative masses produced by the Jordan decomposition of a finite signed phenomenal measure.

L

Law of Fairness: The public name of the proposed framework developed in this book. See Per-stream terminal valence closure and the fairness argument introducing Chapter 4.

Lawhood: The status of a principle as a law of nature, rather than merely an accidental pattern or a fitted description. Here it remains a question about the proposed terminal relation. See also: Boundary hypothesis; Falsifiability; Section 6.2.2.

Leaky integrator: A dynamical system whose retained state carries diminishing weight from older inputs. A zero terminal state therefore need not imply a zero unweighted input total. See Section 21.3.2.

Ledger: See Hidden cumulative object.

Length-biased sampling: Sampling through a randomly selected moment gives longer intervals proportionally more opportunities for inclusion. Episode frequency and occupied-time proportions therefore have different denominators. See Section 15.2.4.

Limit-sure attainability: Policies can make closure probability arbitrarily close to one, although no single admissible policy need attain probability one. Under the same policy class and

regime, almost-sure attainability implies limit-sure attainability; the converse need not hold.

Local average treatment effect: The average effect of an action among people whose uptake is changed by a specified instrument or encouragement, under the identifying assumptions of that comparison. See Section 17.2.2.

Loss function: A rule assigning a cost to a prediction and its realized outcome. Different loss functions can elicit different summaries of the same distribution, such as a mean, median, or quantile. See Section 27.2.3.

Low-unity state: A state in which ordinary access, responsiveness, integration, or evidence of unified experience is reduced or uncertain, as in sleep, anesthesia, coma, delirium, or severe brain injury. It may be a pause, altered-state branch, or unresolved interval rather than the end of a stream. See also: Behavioral nonresponse; Return rule; Truth-level endpoint.

M

Martingale: An integrable stochastic process whose conditional expected future value equals its current value under a specified information history and probability law. The completion-probability example checks this relation at each information state. See Section 8.3.2.

Mathematical coupling: An association arising in part because the same measured quantity appears in both variables being compared. Its size and direction depend on how that shared quantity enters their definitions. See Section 17.3.4.

Maximal qualifying conscious stream: A qualifying stream extended as far as the adopted routing and identity criteria permit, so that it is not merely an arbitrary fragment of a larger

routed stream. Maximality helps prevent selective windowing from manufacturing closure.

Mean reversion: A property of a specified process in which conditional expected future values tend toward a reference level under stated conditions. It is distinct from regression to the mean that can arise when extreme observations are selected and repeated measurements are imperfectly correlated. See also: Homeostasis; Allostasis; Nonunique explanation.

Mean zero: A finding that an expectation or sample average equals or approaches zero. Positive and negative individual residuals can cancel in a mean, so a mean-zero result does not establish per-stream exact closure.

Measured equivalence: A relation in which the true measured-scale estimand lies inside prospectively justified equivalence margins. An interval-based assessment requires the complete relevant interval to lie inside those margins. See Section 7.2.1; see also Measured-to-latent bridge.

Measured-to-latent bridge: The independently justified relation that allows a result on a measured estimand to constrain the declared latent phenomenal target, with scope, uncertainty, and failure conditions carried forward. It is separate from the observation model, endpoint bridge, estimator, and route; without it, evidence remains on the measured scale. See also: Endpoint bridge; Estimator; Hidden cumulative object.

Measurement invariance: The tested stability of a measurement relation across the people, groups, languages, devices, sites, times, or contexts for which a comparison is proposed. See also: Calibration; Held-out transport; Response shift.

Measurement model: A formal account of how latent valence or related states generate reports, behavior, physiology, neural signals, or other observations. It includes scale, origin, error, drift, and possible response-process distortions.

Measurement reactivity: Change in experience, behavior, disclosure, or the observation process caused by measurement itself, including repeated prompts, wearables, interviews, monitoring, and study expectations. See also: Informative missingness; Observation channel; Section 12.1.1.

Median: A value with at least half the probability at or below it and at least half at or above it. A median minimizes expected absolute error when that expectation is defined. See Section 27.2.3.

Missingness: The process by which observations are absent, censored, truncated, or unavailable. Missingness near endpoints may be informative, and treating it as random without support can bias terminal conclusions.

Model governance: The accountable rules for who chooses a model's target, data, loss function, constraints, allowed interventions, prohibited uses, review process, correction route, and verdict. See also: Decision rule; Section 12.2.1; Verdict.

Model recovery: A simulation-based check of whether a declared analysis can distinguish candidate generating models under specified data conditions. Parameter recovery separately asks whether their parameter values can be estimated adequately. Both checks can limit the interpretation of fits to real data.

Motivation: The internal or external factors associated with action selection. A policy describes a mapping from

information or state to actions; it should not be equated with the person's consciously experienced motive.

Multichannel estimate: A versioned, purpose-bounded estimate formed through an explicit model relating several imperfect channels, such as report, behavior, physiology, neural data, and context, to a declared construct and estimand. See also: Error budget; Observation channel; Provenance.

N

Negative control: A condition, timing, procedure, or outcome that should not show the proposed signature if an explanation is specific. Matched non-horizon tasks, sham procedures, impossible timing, and alternate outcomes can expose generalized arousal, demand, leakage, selection, or analytic flexibility. See also: Candidate signature; Nonunique explanation; Perturbation.

Net valence: Positive phenomenal mass minus negative phenomenal mass within a specified stream and domain. A zero net value does not imply small gross experience, equal biographies, or a neutral final moment.

Neutral origin: The identified zero point separating positive from negative phenomenal valence in the adopted representation. Its location must be justified independently rather than chosen to make terminal records balance.

No adequate estimate: A protocol-defined unavailable or invalid inference, as specified in Section 10.5.3; it is not a numerical result. See Section 27.3.3 for reporting.

No personal endpoint testing: See the private-endpoint prohibition in Section 5.4.3; Section 3.3.1.

Noise floor: The smallest distinction an observation system can reliably resolve under declared conditions. Below it, rounding,

censoring, smoothing, preprocessing, or noise dominates the signal. Averaging can reduce independent noise, but more observations, decimal places, or smoothing cannot recover information that the observation system does not contain. See also: Calibration; Error budget; Measured equivalence.

Noncompensable risk: The modeled risk that, under specified conditions and uncertainty, an action or state closes essential paths faster than remaining channels can reopen them. See also: Compensability; Future optionality; Viability kernel.

Nonidentification: A condition in which different target values or models generate the same observable-data distribution under the stated assumptions. It differs from sampling uncertainty about an identifiable target. See also: Identification; Partial identification.

Nonunique explanation: The condition in which several different mechanisms can generate the same visible trajectory, endpoint concentration, rebound, report, compression, or return-to-zero pattern. See also: Candidate signature; Mean reversion.

Null result: A result that does not support the tested effect or distinction under the declared analysis. A null outcome may be informative, inconclusive, or adverse depending on power, identification, and the prespecified decision rule.

Numerical integration: Approximation of an integral or a dynamical trajectory through discrete calculations. A method can preserve one feature of a trajectory while approximating another with error. See Section 23.4.3.

O

Objective function: A modeler's rule for ranking specified outcomes. Its values need not represent a conscious subject's felt experience. See Section 21.2.1.

Observation: A recorded report, behavior, physiological measure, neural signal, event, or other data item.

Observation channel: A means of obtaining evidence, such as self-report, behavioral observation, physiological recording, neural measurement, or contextual observation. The relation between its output and the target belongs to the measurement model. See also: Measurement model; Informative missingness.

Observation process: The mechanism by which latent experience and system states become recorded data, including measurement error, reporting, censoring, sampling, and missingness.

Observation shutdown: See End of observation.

Observation window: The finite span during which a study records a person, often only days, months, or years of a longer stream. See also: Complete felt history; End of observation; Protocol endpoint.

Operational separation: An endpoint rule is outcome-independent when its branch, evidence source, selection rule, and adjudication procedure are fixed and applied without letting a ledger estimate or terminal residual choose the endpoint. This firewall blocks circular selection but does not require probabilistic independence from all stream dynamics. See also: Ad hoc patch; Protocol endpoint; Random-endpoint problem.

Opponent process: A temporal counterprocess in which an initial state recruits a later afterstate whose strength and timing may change with repetition, helping explain relief, recovery, tolerance, withdrawal, fatigue, and rebound. See also: Hedonic adaptation; Mean reversion.

Ownership: The prospectively specified rule assigning each qualifying phenomenal event or contribution to its bearer without omission, double counting, or cross-stream transfer. Ownership answers whose experience a contribution is, while routing governs continuity through time and transitions; data ownership is a different legal and institutional question. See also: Bearer; Intrastream; Routing.

P

Palliative sedation: A clinically governed process used proportionately to relieve otherwise refractory suffering, with varying drugs, depths, durations, routes, and circumstances. It is treatment rather than a standardized consciousness manipulation or clean terminal signal, and research must never delay relief, alter indicated titration, or prefer observability over comfort. See also: Section 16.3.1; End of observation; Protocol endpoint.

Parity check: A test of whether a specified group of bits contains an even or odd number of ones. Overlapping parity checks can locate a single-bit error in the worked seven-bit code. See Section 7.1.3.

Partial identification: A condition in which the observable distribution and maintained assumptions restrict a target to a set containing multiple possible values. Sampling uncertainty about that set is a separate issue.

Partial observation: A decision or inference setting in which the relevant latent state is not fully observed. Action-dependent closure claims in such settings must respect common-action and selection requirements rather than assuming state-specific knowledge.

Partial order: An ordering relation that can leave some alternatives incomparable. In the route example, coordinatewise dominance ranks some pairs without assigning an overall rank to every pair. See Section 11.5.2.

Partial pooling: Estimation that combines information from an individual record with a population model. The amount of borrowing depends on the observation model and the estimated or specified population variation. See Section 11.4.2.

Participation without surveillance: See Section 12.1.1; Section 12.3.2.

Past-fixity: Later experience may add to the future portion of a stream but cannot retroactively change what earlier experience was. See also: Chapter 3; Hidden cumulative object; Reserve coupling.

Pathwise closure: A closure condition evaluated on complete individual trajectories. See Set-universal claim and Probability-one closure for distinct quantifiers.

Per-stream terminal valence closure (PTVC): The hypothesis stated in Section 4.1.1. See Section 7.3.1 for alternative representation branches.

Perturbation: A controlled change used to investigate a causal contribution. This research program considers reversible, low-burden changes to features such as feedback, delay, expectation, or opportunity. See Chapter 17.

Phenomenal magnitude: The represented amount of positive or negative felt experience assigned to a stream under a declared valence representation. See Gross experience and Positive and negative phenomenal magnitudes.

Phenomenal measure: A measure assigned to conscious experience itself for purposes of representing valence across a stream. It is conceptually distinct from clock time, event count, memory strength, neural activity, or report frequency.

Phenomenal time: A measure or ordering of experienced duration within a conscious stream. It need not coincide uniformly with external clock time and should not be inferred solely from the number of observations.

Phenomenal valence: See Felt valence.

Policy: A rule mapping available information or state to actions, possibly probabilistically. A policy is not identical to a conscious motive, a moral intention, or the outcome it happens to produce.

Policy mixture: A probabilistic combination of actions or component policies. Average cancellation across components does not establish exact closure on each realized path.

Positive and negative phenomenal magnitudes: On the primary additive branch, the nonnegative complete-stream totals of the positive and negative parts of the signed phenomenal measure. See also: Finiteness; Gross experience; Terminal residual.

Positive rescaling: Multiplication of a valid signed representation by a positive constant. Its sign structure and exact-zero status remain unchanged. See Section 10.5.1.

Positive-probability completion: Under a prospectively frozen continuation regime and its declared induced law, the

completion event has probability strictly greater than zero, optionally conditional on a stated prefix and admissible action or policy. This weak reachability claim does not imply likelihood, probability one, an attaining policy, support containment, or every-continuation closure. See also: Reachability ladder; Probability-one closure; Reachability.

Potential shaping: A reward transformation that adds the discounted next-state potential and subtracts the current-state potential. Under the stated boundary conditions, the added terms telescope and preserve the relevant full-path comparison. See Section 21.1.2.

Precaution: Proportionate, revisable protection under uncertainty, including monitoring, reducing avoidable harm, preserving reversibility, pausing, or redesigning when plausible welfare risk is serious. Protection can strengthen while consciousness, ontology, law, and personhood remain unresolved, without treating precaution as proof. See also: Section 14.4.1; Section 14.1; Unknown category.

Prediction error: A discrepancy between an expected signal and an observed one under a declared model. Its direction and magnitude describe mismatch. Felt valence concerns the positive, negative, or neutral quality of experience. See Section 20.1.1.

Predictive processing: A family of accounts in which perception, bodily interpretation, and action depend on generative models, expected signals, prediction errors, uncertainty, and estimated reliability. See also: Active inference; Interoceptive prediction; Prediction error.

Preregistration: A study plan externally registered with a time stamp and persistent identifier before relevant outcomes are

examined. An internal plan is protocol-fixed; disclose deviations and label exploratory analyses.

Preterminal compression: A candidate observable signature in which a specified measure of dispersion narrows before a declared endpoint in the prospectively predicted relation. The protocol distinguishes dispersion across model realizations from temporal variation within one observed stream and justifies any bridge between them. See also: Candidate signature; Endpoint conditioning; Noise floor.

Primary additive branch: The branch that represents qualifying phenomenal valence as one finite signed scalar measure with a justified neutral origin and additive consistency across disjoint parts of a stream. Its positive and negative parts yield complete-stream magnitudes, and their difference yields the terminal residual. See also: Bivariate branch; Generalized Terminal Neutrality; Signed representation.

Probability-one closure: Closure with probability one under a specified probability model and, where applicable, policy. It permits measure-zero exceptions and is distinct from set-universal closure, which requires closure for every admissible realization.

Procedural fairness: Fairness in rules and decision paths, including knowable standards, voice, understandable reasons, consistency, limited bias, respectful treatment, correction, and appeal. See also: Desert-based fairness; External justice; Section 12.1.1.

Prospectively specified: Fixed before the relevant outcome is inspected. The phrase states advance timing without asserting public preregistration. See Section 6.3.3.

Protocol endpoint: A prospectively specified operational event, adjudicated interval, stopping region, or governed proxy used by a study to locate or approximate a relevant boundary. It may be usable, uncertain, revisable, or ethically unavailable and is distinct from both the truth-level endpoint and end of observation. See also: End of observation; Endpoint bridge; Truth-level endpoint.

Protocol-fixed: Written into the study protocol before the relevant outcome is inspected. This describes advance specification without implying public registration. See also: Prospectively specified; Preregistration.

Provenance: The documented lineage of a trace or derived output, including who collected it, for what purpose and authority, under which prompt, device, language, setting, burden, transform, model version, missingness state, uncertainty, and correction history. See also: Observation process; Section 25.1.1; Section 12.3.2.

PTVC: See Per-stream terminal valence closure (PTVC).

Q

Quadratic variation: The limiting sum of squared path increments along an appropriate sequence of increasingly fine partitions, when that limit exists in the declared sense. Observation noise can change the behavior of the sampled statistic. See Section 7.5.2.

Qualifying conscious stream: A stream within independently declared truth-domain scope, identity, and routing criteria, regardless of apparent closure. Observation and measurement gates govern adjudication, not membership.

Queue System: The name used in the Illustrated Guide for the candidate possibility architecture discussed in the Chapter 8 introduction.

R

Random-endpoint problem: The conditional theorem about exact closure under independently timed endpoints in Section 7.2.2.

Reachability: Whether at least one admissible continuation from a state can reach the declared target. Reachability does not guarantee selection, common feasibility under partial observation, probability-one success, or universal success.

Reachability ladder: The family of completion claims compared in Section 7.3.2 under specified laws, policy classes and quantifiers.

Reachable set: The collection of states or targets attainable from a stated starting condition using the actions, time and constraints admitted by a model. See Section 8.1.3.

Recalibration: A change in the internal standards used to place an experience on a response scale. See also: Response shift. See Section 18.2.2.

Reconceptualization: A change in what a person includes in the concept being evaluated. See also: Response shift. See Section 18.2.2.

Reference class: The prospectively declared truth-domain set of qualifying streams, endpoint regimes, representations, populations, settings, or measurement architectures over which a claim and its quantifier range. It is not automatically the enrolled sample or analyzed cases, and outcome-driven exclusions create a new claim. See also: Held-out transport; Maximal qualifying conscious stream; Set-universal claim.

Reference window: The interval about which an observation or report supplies information, distinguished from the time at which the report was made. Overlapping reference windows require an explicit rule for combining their contents. See Section 7.4.1.

Reference-class narrowed: See Verdict.

Registered Report: A publication format in which a study’s question, methods, and analysis plan receive peer review before results are known, with publication not conditioned on obtaining a preferred outcome.

Reinforcement learning: Learning in which a formally specified reward signal informs updates to action values or policies. The signal is part of a model and requires a separate relation to experienced pleasure or pain. See Section 21.1.

Rejection sampling: Generating candidates under a specified rule and retaining those that satisfy a stated acceptance condition. Section 23.2.5 compares its attempt count with other generators of the same accepted histories.

Remaining capacity: The set or amount of admissible future change still available before an endpoint. An action-dependent closure route must show that remaining capacity is sufficient and selectable under its actual information and support constraints.

Repeated observation: Collection of observations over time under declared instruments, schedules, contexts, purposes, and permissions. The resulting record samples a changing course. See Chapter 18.

Replication: A new test designed to determine whether a result, measurement relation, or model distinction recurs under prespecified conditions.

Representation: A mathematical or conceptual encoding of phenomenal valence, time, state, or route. A useful representation must preserve the distinctions required by the target and should not be mistaken for the experience itself.

Reprioritization: A change in the relative importance assigned to domains within an evaluation. See also: Response shift. See Section 18.2.2.

Reserve: A route-level variable representing remaining capacity, flexibility, or corrective potential before an endpoint. A reserve is not a store of deserved pleasure or pain and is not itself the terminal target.

Reserve coupling: A candidate relation between accumulated residual and remaining capacity or future transition structure.

Residual set: A branch-matched nonempty set of residual values allowed by the declared representation, endpoint, observation, missingness, bridge and uncertainty assumptions. Retain its type as an identified set, confidence region, credible region or sensitivity envelope. See Truth-containing residual set and Section 27.3.3.

Resonance: A relation of meaning occurs when a person or tradition recognizes a neighboring pattern. It belongs to interpretation rather than empirical validation. See also: Comparison without forced equivalence; Chosen meaning.

Response shift: A change in a person's standards, priorities, values, or understanding of a question after illness, disability, adaptation, treatment, trauma, grief, or aging, altering what the same response scale means over time. A changed answer can reflect changed experience, changed measurement, or both. See also: Calibration; Repeated observation; Measurement invariance.

Retirement triggers: Prespecified findings that end a candidate route's standing in the declared research program, such as decisive failure, persistent nonidentification, or superior rival performance. See Section 8.3.5.

Return rule: A prospectively specified rule under which reliable return of unified access, meeting predeclared criteria for continuity of the same stream, can defeat an earlier classification as a final endpoint, while nonreturn alone cannot prove absence of experience or identify the exact ending. Protocols must state return evidence, revision timing, and its effect on classification. See also: Endpoint bridge; Low-unity state; Protocol endpoint.

Reversible pause: See Return rule.

Rival model: A concrete alternative explanation capable of generating overlapping observations.

Rival-favored: See Verdict.

Route: See Candidate route; Route state.

Route state: A latent or observed variable used to describe progress within a candidate mechanism. A route state is not automatically phenomenal valence, cumulative residual, motivation, or verdict.

RouteStatus: See Verdict.

Routing: The prospectively specified treatment of continuation, pause, restoration, fission, fusion, copying, replacement, and unresolved continuity, including where prior and later phenomenal contributions are assigned. Routing governs transitions through time and is distinct from ownership, compensation, and causal influence. See also: Stream-routing convention; Ownership; Stream identity.

S

Saturation: A condition in which a device has reached a bound on its output, so a further command in that direction cannot produce a corresponding increase. Internal controller states may still change while the delivered output remains fixed. See Section 21.3.3.

Scalarization: A prospectively justified mapping from a richer or multidimensional experiential state to one signed scalar quantity for a named purpose and domain. It is a substantive bridge rather than harmless formatting, and weights selected because they produce closure manufacture the result. See also: Bivariate branch; Generalized Terminal Neutrality; Signed representation.

Selection condition: A requirement governing which available action or continuation is actually chosen. The existence of a closing option is insufficient for probability-one closure if the selected law still assigns positive probability to nonclosing continuations.

Separately routed: Governed by a trajectory or transition structure that is not pooled with another stream for purposes of the terminal target. Separate routing supports the rule that one stream cannot balance another.

Sequential standardization: A calculation that averages an outcome model over the intermediate-state distributions produced by a specified sequence of interventions. It preserves the effects of earlier actions on the states encountered before later decisions. See Section 17.2.5.

Set-universal claim: Closure for every admissible realization, path, or member of the declared set. It is stronger than probability-one closure because measure-zero exceptions are not permitted.

Shortening horizon: A reduction in remaining time, opportunity, revision, access, or other route-relevant room for future action. See also: Future optionality; Reachability; Reserve.

Shrinkage: Movement of an estimate toward a reference or population value through a specified estimation rule. Its extent and remaining uncertainty depend on the model. See Section 11.4.2.

Signed representation: A representation that assigns positive and negative directions relative to a justified neutral origin and supplies the declared scale and aggregation rules needed for the target branch. A sign convention alone does not establish cardinality, comparability, additivity, or phenomenal meaning. See also: Felt valence; Neutral origin; Primary additive branch.

Signed residual: The cumulative positive phenomenal mass minus the cumulative negative phenomenal mass over a specified domain. The terminal signed residual is evaluated at the truth-level stream endpoint.

Smoothing: Retrospective estimation of an earlier latent state using observations from both before and after that time. Filtering instead uses observations available through the time being estimated. See Section 16.4.2.

Specific satiety: A reduction in the appeal of a particular food after consuming it, relative to another food. The caching example uses this manipulation to distinguish current from later food preferences. See Section 14.1.4.

State-coupled timing: An endpoint branch in which endpoint timing or hazard depends on prospectively specified current biological, consciousness, ledger, or route-state variables while the endpoint ontology and adjudication remain fixed

independently of desired closure. Dependence is allowed, but the state relation cannot be invented after outcome inspection. See also: Endpoint atom; Hitting time; Absorption; Random-endpoint problem.

Stop rules: Prespecified conditions for pausing or ending an activity, together with responsibility and authority for applying them. See Section 14.3.2.

Stream: See Maximal qualifying conscious stream.

Stream identity: The criteria that determine which experiences belong to the same conscious stream. Biological continuity, memory, body, legal identity, and data linkage may inform the analysis but are not automatically decisive.

Stream-routing convention: Under the book's primary formal convention, genuine fission ends the parent stream and begins each qualifying child stream, while genuine fusion ends the prior streams and begins a new one. Earlier phenomenal balances are neither duplicated among descendants nor pooled into the fused stream. See also: Bearer; Routing; Stream identity.

Support: For a probability law under a declared topology, the set of points whose every open neighborhood has positive probability. A point in the support can itself have probability zero. See also: Conditional support; Support containment.

Support containment: Under a declared topology, the support of the relevant probability law lies inside the completion event or target set. This topological modality is not every-continuation closure, positive probability at a point, or proof that an unsampled possibility is absent from support. See also: Admissible continuation; Reachability ladder; Probability-one closure.

Syndrome: In the error-correcting example, the pattern of failed parity checks. Under the one-bit-error assumption, that pattern identifies the changed position and directs the correction. See Section 7.1.3.

T

Target: The truth-level property a hypothesis asserts. See Section 4.3.2 for its relation to routes, observations and estimates.

TargetStatus: See Verdict.

Terminal neutrality: See Exact closure.

Terminal residual: The signed total of phenomenal valence over a qualifying stream's complete strict-pre-endpoint domain, evaluated at its truth-level endpoint.

Terminal window: A finite observed period near a presumed endpoint. It can provide useful process data but is not identical to the complete strict-pre-endpoint felt history or the truth-level endpoint.

Testimony-like output: Language or behavior from an artificial system that has the form of a first-person report but is generated under a particular model, prompt, memory state, tool set, training regime, and sampling process. It is evidence about configured output, not self-authenticating testimony that a subject feels what is said. See also: Section 25.3.1; Section 14.4.1; Theory-derived indicator.

Tetrad: A difference between two products of pairwise covariances among four variables. Certain common-factor models imply that specified tetrads vanish. See Section 10.3.3.

The Law: See Law of Fairness.

The Law of Fairness*:* See Law of Fairness; Per-stream terminal valence closure (PTVC).

Theory-derived indicator: An observable property selected because a stated theory connects it to recurrent processing, global availability, higher-order representation, agency, embodiment, metacognition, or another proposed feature of consciousness. Indicators frame tests but do not automatically combine into a direct meter of experience. See also: Section 14.4.1; Section 14.1; Testimony-like output.

Total variation: For a finite signed measure, the sum of its positive and negative Jordan masses. See Section 7.1.4.

Truth-containing residual set: A residual set guaranteed under the declared assumptions to contain the true residual. Excluding zero contradicts exact zero under those assumptions. Confidence coverage and posterior probability are statistical qualifications, not deductive guarantees. See Section 4.2.2.

Truth-level endpoint: The actual boundary of the qualifying stream in the target model. It is conceptually distinct from any observed marker, last report, terminal window, loss of contact, or biological event used to estimate it.

U

Union bound: The probability of at least one event occurring is at most the sum of the individual event probabilities. Independence between those events is not required. See Section 6.3.1.

Unity gate: The rule specifying when a felt contribution is active and assigned to the modeled stream. See also: Experienced exposure; Ownership.

Unity Index: A proposed fallible research estimator that organizes separately reported evidence about access, report, memory, agency, embodiment, continuity, and coherence

under a declared construct. See also: Blinding; Section 12.2.1; Unknown category.

Universal quantifier: A claim applying to every member of a declared class. See Section 4.2.2.

Unknown category: A reason-coded classification outcome used when available evidence cannot justify one bearer, consciousness, or boundary classification over serious alternatives under declared rules. Unknown is not a negative finding, equal probability, zero, or support for the Law, and it differs from No adequate estimate. See also: Bearer; No adequate estimate; Precaution.

V

Valence: See Felt valence.

Valence density: A representation of signed phenomenal valence per unit of an adopted base measure, such as phenomenal time. Its interpretation depends on the validity of both the valence scale and the base measure.

Value of information: The increase in the best achievable expected objective when additional information is available before a decision. The calculation specifies the available actions, probabilities, objective, and information. See Section 20.1.4.

Verdict: The declared evidential conclusion from a prespecified decision architecture. Its four separately recorded fields are TargetStatus, RouteStatus, ComparatorStatus, and InferenceStatus. Each conclusion identifies the tested branch and scope. See Section 27.3.3.

Viability: The possibility of remaining within declared constraints over the specified horizon under an admissible continuation or policy, with any probabilistic guarantee stated

explicitly. Maintaining access to closure is an additional completion-specific requirement when it is included among those constraints.

Viability kernel: The set of modeled states from which at least one admissible continuation can continue to satisfy a route's declared constraints. It marks where a future remains formally workable but does not mean that a continuation is likely, humane, accessible, desirable, or supported by resources a person actually has. See also: Compensability; Noncompensable risk; Reachability.

Z

Zero crossing: A time at which a cumulative or instantaneous signed quantity passes through zero. PTVC does not require monotonic movement, frequent zero crossings, or any particular number of crossings before the endpoint.

References

The entries below provide bibliographic destinations for the Author/Year citations used throughout the manuscript. Author and institutional creator names appear in bold. Distinct works by the same author in the same year retain their manuscript suffixes. DOI identifiers appear as plain text so the entries remain readable in print. Readers following a claim beyond this book should consult the cited edition, the proposition it supports, and the source's stated scope.

Achiam, Joshua, David Held, Aviv Tamar, and Pieter Abbeel. 2017. "Constrained policy optimization." *Proceedings of the 34th International Conference on Machine Learning, Proceedings of Machine Learning Research* 70: 22–31.

Adams, Marilyn McCord. 1999. *Horrendous Evils and the Goodness of God.* Cornell University Press.

Akaike, Hirotugu. 1973. "Information theory and an extension of the maximum likelihood principle." In B. N. Petrov and F. Csáki, eds., *Second International Symposium on Information Theory*, 267–281. Akadémiai Kiadó.

Aksay, Emre, Itsaso Olasagasti, Brett D. Mensh, Robert Baker, Mark S. Goldman, and David W. Tank. 2007. "Functional dissection of circuitry in a neural integrator." *Nature Neuroscience* 10 (4): 494–504. DOI 10.1038/nn1877.

Albrecht, G. L. and P. J. Devlieger. 1999. "The disability paradox: High quality of life against all odds." *Social Science & Medicine* 48 (8): 977–988. DOI 10.1016/S0277-9536(98)00411-0.

Alter, Robert. 2010. *The Wisdom Books: Job, Proverbs, and Ecclesiastes: A Translation with Commentary.* W. W. Norton.

American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. 2014. *Standards for educational and psychological testing.* American Educational Research Association.

American Psychiatric Association. 2022. *Diagnostic and Statistical Manual of Mental Disorders (5th ed., text rev.; DSM-5-TR).* American Psychiatric Association Publishing.

Amrhein, Valentin, Sander Greenland, and Blake McShane. 2019. "Scientists rise up against statistical significance." *Nature* 567: 305–307. DOI 10.1038/d41586-019-00857-9.

Andersen, P. K., R. B. Geskus, T. de Witte, and H. Putter. 2012. "Competing risks in epidemiology: Possibilities and pitfalls." *International Journal of Epidemiology* 41 (3): 861–870. DOI 10.1093/ije/dyr213.

Andrews, K., J. Birch, J. Sebo, and T. Sims. 2024. *Background to the New York Declaration on Animal Consciousness.* New York Declaration on Animal Consciousness.

Aoki, S. K., G. Lillacci, A. Gupta, A. Baumschlager, D. Schweingruber, and M. Khammash. 2019. "A universal biomolecular integral feedback controller for robust perfect adaptation." *Nature* 570: 533–537. DOI 10.1038/s41586-019-1321-1.

Appelbaum, Paul S., Loren H. Roth, and Charles W. Lidz. 1982. "The therapeutic misconception: Informed consent in psychiatric research." *International Journal of Law and Psychiatry* 5 (3–4): 319–329. DOI 10.1016/0160-2527(82)90026-7.

Armstrong, D. M. 1983. *What Is a Law of Nature?* Cambridge University Press.

Asad, Talal. 1993. *Genealogies of Religion: Discipline and Reasons of Power in Christianity and Islam.* Johns Hopkins University Press.

Asadi, A., K. Chatterjee, R. Saona, and A. Shafiee. 2025. "Limit-sure Reachability for Small Memory Policies in POMDPs is NP-complete." *Proceedings of the Forty-first Conference on Uncertainty in Artificial Intelligence, Proceedings of Machine Learning Research* 286: 238–256. PMLR.

Aserinsky, E. and N. Kleitman. 1953. "Regularly occurring periods of eye motility, and concomitant phenomena, during sleep." *Science* 118 (3062): 273–274. DOI 10.1126/science.118.3062.273.

Aubin, J.-P. 1991. *Viability theory.* Birkhäuser.

Aubin, J.-P., A. M. Bayen, and P. Saint-Pierre. 2011. *Viability theory: New directions.* Springer. DOI 10.1007/978-3-642-16684-6.

Åström, K. J. and L. Rundqwist. 1989. "Integrator windup and how to avoid it." *Proceedings of the 1989 American*

Control Conference, 1693–1698. IEEE. DOI 10.23919/ACC.1989.4790464.

Baba, Ami, Maureen Smith, Beth K. Potter, An-Wen Chan, David Moher, Alene Toulany, Amanda Doherty-Kirby, Begonya Nafria Escalera, Catherine Stratton, Chris Gale, Colin Macarthur, Diane Purper-Ouakil, Edmund Juszczak, Eyal Cohen, Giorgio Reggiardo, Jennifer Preston, Jérémie F. Cohen, Julia Upton, Karel Allegaert, Katelynn Boerner, Kayur Mehta, Kim An Nguyen, Kimberly Courtney, Lisa Hartling, Menelaos Konstantinidis, Michal Odermarsky, Nancy J. Butcher, Niina Kolehmainen, Patricia E. Longmuir, Peter J. Gill, Piet Leroy, Reinhard Feneberg, Ramesh Poluru, Shaun K. Morris, Stefan J. Friedrichsdorf, Tanya Chute Nagy, Terry P. Klassen, Thierry Lacaze-Masmonteil, Wes Onland, and Martin Offringa. 2026. "SPIRIT-Children and Adolescents (SPIRIT-C) 2026 Extension Statement: Enhancing the Reporting and Usefulness of Pediatric Randomized Trial Protocols." *JAMA Pediatrics* 180 (4): 426–439. DOI 10.1001/jamapediatrics.2026.0113.

Bailey, Z. D., N. Krieger, M. Agénor, J. Graves, N. Linos, and M. T. Bassett. 2017. "Structural racism and health inequities in the USA: Evidence and interventions." *The Lancet* 389 (10077): 1453–1463. DOI 10.1016/S0140-6736(17)30569-X.

Baltes, P. B. 1997. "On the incomplete architecture of human ontogeny: Selection, optimization, and compensation as

foundation of developmental theory." *American Psychologist* 52 (4): 366–380. DOI 10.1037/0003-066X.52.4.366.

Baluška, F. and A. S. Reber. 2019. "Sentience and consciousness in single cells: How the first minds emerged in unicellular species." *BioEssays* 41 (3): 1800229. DOI 10.1002/bies.201800229.

Bandura, Albert. 2001. "Social cognitive theory: An agentic perspective." *Annual Review of Psychology* 52: 1–26. DOI 10.1146/annurev.psych.52.1.1.

Barocas, Solon, and Andrew D. Selbst. 2016. "Big data's disparate impact." *California Law Review* 104 (3): 671–732. DOI 10.15779/Z38BG31.

Barrett, L. F. 2006. "Solving the emotion paradox: Categorization and the experience of emotion." *Personality and Social Psychology Review* 10 (1): 20–46. DOI 10.1207/s15327957pspr1001_2.

Barrett, L. F. and W. K. Simmons. 2015. "Interoceptive predictions in the brain." *Nature Reviews Neuroscience* 16 (7): 419–429. DOI 10.1038/nrn3950.

Barron, A. B. and C. Klein. 2016. "What insects can tell us about the origins of consciousness." *Proceedings of the National Academy of Sciences* 113 (18): 4900–4908. DOI 10.1073/pnas.1520084113.

Bateson, P. 1991. "Assessment of pain in animals." *Animal Behaviour* 42 (5): 827–839. https://doi.org/10.1016/s0003-3472(05)80127-7

Baumeister, Roy F., and Mark R. Leary. 1995. "The need to belong: Desire for interpersonal attachments as a fundamental human motivation." *Psychological Bulletin* 117 (3): 497–529. DOI 10.1037/0033-2909.117.3.497.

Bayne, T. 2010. *The Unity of Consciousness.* Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199215386.001.0001

Bayne, T., A. K. Seth, M. Massimini, J. Shepherd, A. Cleeremans, S. M. Fleming, R. Malach, J. B. Mattingley, D. K. Menon, A. M. Owen, M. A. K. Peters, A. Razi, and L. Mudrik. 2024. "Tests for consciousness in humans and beyond." *Trends in Cognitive Sciences* 28 (5): 454–466. DOI 10.1016/j.tics.2024.01.010.

Bayne, T., and D. J. Chalmers. 2003. "What is the unity of consciousness?" In A. Cleeremans, ed., *The Unity of Consciousness: Binding, Integration, and Dissociation*, 23–58. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780198508571.003.0002

Beauchamp, Tom L., and James F. Childress. 2019. *Principles of Biomedical Ethics (8th ed.).* Oxford University Press.

Bender, E. M., T. Gebru, A. McMillan-Major, and S. Shmitchell. 2021. "On the dangers of stochastic parrots: Can language models be too big?" *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623. DOI 10.1145/3442188.3445922.

Benjamin, R. 2019. *Race After Technology.* Polity.

Berger, Peter L. 1967. *The Sacred Canopy: Elements of a Sociological Theory of Religion.* Doubleday.

Berridge, K. C. and T. E. Robinson. 1998. "What is the role of dopamine in reward: Hedonic impact, reward learning, or incentive salience?" *Brain Research Reviews* 28 (3): 309–369. DOI 10.1016/S0165-0173(98)00019-8.

Berridge, K. C., T. E. Robinson, and J. W. Aldridge. 2009. "Dissecting components of reward: 'Liking,' 'wanting,' and learning." *Current Opinion in Pharmacology* 9 (1): 65–73. DOI 10.1016/j.coph.2008.12.014.

Berrios, R., P. Totterdell, and S. Kellett. 2015. "Eliciting mixed emotions: A meta-analysis comparing models, types, and measures." *Frontiers in Psychology* 6: 428. DOI 10.3389/fpsyg.2015.00428.

Bertsekas, D. P. 1995. *Dynamic Programming and Optimal Control, Volume I.* Athena Scientific.

Bickel, Warren K., Matthew W. Johnson, Mikhail N. Koffarnus, James MacKillop, and James G. Murphy. 2014. "The behavioral economics of substance use disorders: Reinforcement pathologies and their repair." *Annual Review of Clinical Psychology* 10: 641–677. DOI 10.1146/annurev-clinpsy-032813-153724.

Billingsley, P. 1999. *Convergence of probability measures (2nd ed.).* Wiley. DOI 10.1002/9780470316962.

Birch, J. 2017. "Animal sentience and the precautionary principle." *Animal Sentience* 2 (16): 1. https://doi.org/10.51291/2377-7478.1200

Birch, J. 2024. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI.* Oxford University Press. https://doi.org/10.1093/9780191966729.001.0001

Birch, J., A. K. Schnell, and N. S. Clayton. 2020. "Dimensions of animal consciousness." *Trends in Cognitive Sciences* 24 (10): 789–801. DOI 10.1016/j.tics.2020.07.007.

Bland, J. M. and D. G. Altman. 1986. "Statistical methods for assessing agreement between two methods of clinical measurement." *The Lancet* 1 (8476): 307–310. DOI 10.1016/S0140-6736(86)90837-8.

Block, N. 1995. "On a confusion about a function of consciousness." *Behavioral and Brain Sciences* 18 (2): 227–247. DOI 10.1017/S0140525X00038188.

Bobba-Alves, N., R.-P. Juster, and M. Picard. 2022. "The energetic cost of allostasis and allostatic load." *Psychoneuroendocrinology* 146: 105951. DOI 10.1016/j.psyneuen.2022.105951.

Bodien, Yelena G., Judith Allanson, Paolo Cardone, Arthur Bonhomme, Jerina Carmona, Camille Chatelle, Srivas Chennu, Mary Conte, Stanislas Dehaene, Paola Finoia, Gregory Heinonen, Jennifer E. Hersh, Evelyn Kamau, Phoebe K. Lawrence, Victoria C. Lupson, Anogue Meydan, Benjamin Rohaut, William R. Sanders, Jacobo D. Sitt, Andrea Soddu, Mélanie Valente, Angela Velazquez, Henning U. Voss, Athina Vrosgou, Jan Claassen, Brian L. Edlow, Joseph J. Fins, Olivia Gosseries, Steven Laureys, David Menon, Lionel

Naccache, Adrian M. Owen, John Pickard, Emmanuel A. Stamatakis, Aurore Thibaut, Jonathan D. Victor, Joseph T. Giacino, Emilia Bagiella, and Nicholas D. Schiff. 2024. "Cognitive motor dissociation in disorders of consciousness." *New England Journal of Medicine* 391 (7): 598–608. DOI 10.1056/NEJMoa2400645.

Boisseau, Romain P., David Vogel, and Audrey Dussutour. 2016. "Habituation in non-neural organisms: Evidence from slime moulds." *Proceedings of the Royal Society B* 283 (1829): 20160446. DOI 10.1098/rspb.2016.0446.

Bollen, K. A. 1989. *Structural equations with latent variables.* Wiley. DOI 10.1002/9781118619179.

Bollen, K. A. and R. Lennox. 1991. "Conventional wisdom on measurement: A structural equation perspective." *Psychological Bulletin* 110 (2): 305–314. DOI 10.1037/0033-2909.110.2.305.

Bollen, K. A. and S. Bauldry. 2011. "Three Cs in measurement models: Causal indicators, composite indicators, and covariates." *Psychological Methods* 16 (3): 265–284. DOI 10.1037/a0024448.

Bollen, Kenneth A., and Kwok-Fai Ting. 2000. "A tetrad test for causal indicators." *Psychological Methods* 5 (1): 3–22. https://doi.org/10.1037/1082-989X.5.1.3

Bonanno, George A. 2004. "Loss, trauma, and human resilience: Have we underestimated the human capacity to thrive after extremely aversive events?" *American Psychologist* 59 (1): 20–28. DOI 10.1037/0003-066X.59.1.20.

Borsboom, D. 2005. *Measuring the mind: Conceptual issues in contemporary psychometrics.* Cambridge University Press. DOI 10.1017/CBO9780511490026.

Bortz, David M., Daniel A. Messenger, and Vanja Dukic. 2023. "Direct estimation of parameters in ODE models using WENDy: Weak-form estimation of nonlinear dynamics." *Bulletin of Mathematical Biology* 85 (11): 110. DOI 10.1007/s11538-023-01208-6.

Bourtoule, L., V. Chandrasekaran, C. A. Choquette-Choo, H. Jia, A. Travers, B. Zhang, D. Lie, and N. Papernot. 2021. "Machine unlearning." *2021 IEEE Symposium on Security and Privacy*, 141–159. DOI 10.1109/SP40001.2021.00019.

Bowker, Geoffrey C., and Susan Leigh Star. 1999. *Sorting Things Out: Classification and Its Consequences*. Cambridge, MA: MIT Press.

Box, G. E. P. 1976. "Science and statistics." *Journal of the American Statistical Association* 71 (356): 791–799. DOI 10.1080/01621459.1976.10480949.

Braithwaite, John. 1989. *Crime, Shame and Reintegration.* Cambridge University Press. https://doi.org/10.1017/cbo9780511804618

Briat, C., A. Gupta, and M. Khammash. 2016. "Antithetic integral feedback ensures robust perfect adaptation in noisy biomolecular networks." *Cell Systems* 2 (1): 15–26. DOI 10.1016/j.cels.2016.01.004.

Brickman, P., D. Coates, and R. Janoff-Bulman. 1978. "Lottery winners and accident victims: Is happiness relative?" *Journal of Personality and Social Psychology* 36 (8): 917–927. DOI 10.1037/0022-3514.36.8.917.

Broom, Donald M. 2014. *Sentience and Animal Welfare.* CABI.

Brown, E. N., R. Lydic, and N. D. Schiff. 2010. "General anesthesia, sleep, and coma." *New England Journal of Medicine* 363 (27): 2638–2650. DOI 10.1056/NEJMra0808281.

Bryant, Fred B., and Joseph Veroff. 2007. *Savoring: A New Model of Positive Experience.* Lawrence Erlbaum Associates.

Buolamwini, Joy, and Timnit Gebru. 2018. "Gender shades: Intersectional accuracy disparities in commercial gender classification." *Proceedings of Machine Learning Research* 81: 77–91.

Burnham, K. P. and D. R. Anderson. 2002. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach (2nd ed.).* Springer.

Busemeyer, Jerome R., and Adele Diederich. 2010. *Cognitive Modeling.* SAGE.

Butlin, P., R. Long, T. Bayne, Y. Bengio, J. Birch, D. J. Chalmers, A. Constant, G. Deane, E. Elmoznino, S. M. Fleming, X. Ji, R. Kanai, C. Klein, G. W. Lindsay, M. Michel, L. Mudrik, M. A. K. Peters, E. Schwitzgebel, J. Simon, and R. VanRullen. 2026. "Identifying indicators of

consciousness in AI systems." *Trends in Cognitive Sciences* 30 (6): 488–501. DOI 10.1016/j.tics.2025.10.011.

Butlin, Patrick, and Theodoros Lappas. 2025. "Principles for responsible AI consciousness research." *Journal of Artificial Intelligence Research* 82: 1673–1690. DOI 10.1613/jair.1.17310.

Cabanac, M. 1971. "Physiological role of pleasure." *Science* 173 (4002): 1103–1107. DOI 10.1126/science.173.4002.1103.

Cacioppo, J. T. and G. G. Berntson. 1994. "Relationship between attitudes and evaluative space: A critical review, with emphasis on the separability of positive and negative substrates." *Psychological Bulletin* 115 (3): 401–423. DOI 10.1037/0033-2909.115.3.401.

Callan, M. J., J. H. Ellard, and J. E. Nicol. 2006. "The belief in a just world and immanent justice reasoning in adults." *Personality and Social Psychology Bulletin* 32 (12): 1646–1658. DOI 10.1177/0146167206292236.

Campbell, D. T. and D. W. Fiske. 1959. "Convergent and discriminant validation by the multitrait-multimethod matrix." *Psychological Bulletin* 56 (2): 81–105. DOI 10.1037/h0046016.

Campbell, Donald T. 1979. "Assessing the Impact of Planned Social Change." *Evaluation and Program Planning* 2 (1): 67–90. DOI 10.1016/0149-7189(79)90048-X.

Camus, A. 1942. *Le Mythe de Sisyphe.* Gallimard.

Camus, Albert. 1955. *The Myth of Sisyphus and Other Essays.* Translated by Justin O'Brien. Alfred A. Knopf. Originally published 1942.

Cao, Y. and J. Yang. 2015. "Towards making systems forget with machine unlearning." *2015 IEEE Symposium on Security and Privacy*, 463–480. DOI 10.1109/SP.2015.35.

Carrasco-Zafra, M. I., R. Ocaña-Riola, R. Gómez-García, M. L. Martín-Roselló, and E. Blanco-Reina. 2024. "Palliative sedation at the end of life: Prevalence, characteristics and possible determinants." *BMC Palliative Care* 23: 278. DOI 10.1186/s12904-024-01606-0.

Carroll, R. J., D. Ruppert, L. A. Stefanski, and C. M. Crainiceanu. 2006. *Measurement Error in Nonlinear Models: A Modern Perspective (2nd ed.).* Chapman & Hall/CRC.

Carstensen, L. L. 2006. "The influence of a sense of time on human development." *Science* 312 (5782): 1913–1915. DOI 10.1126/science.1127488.

Carstensen, L. L. 2021. "Socioemotional selectivity theory: The role of perceived endings in human motivation." *The Gerontologist* 61 (8): 1188–1196. DOI 10.1093/geront/gnab116.

Carstensen, L. L., D. M. Isaacowitz, and S. T. Charles. 1999. "Taking time seriously: A theory of socioemotional selectivity." *American Psychologist* 54 (3): 165–181. DOI 10.1037/0003-066X.54.3.165.

Cartwright, N. 1983. *How the Laws of Physics Lie.* Oxford University Press. https://doi.org/10.1093/0198247044.001.0001

Cartwright, R. D., S. Lloyd, S. Knight, and I. Trenholme. 1984. "Broken dreams: A study of the effects of divorce and depression on dream content." *Psychiatry* 47 (3): 251–259. https://doi.org/10.1080/00332747.1984.11024246

Cartwright, R., M. Y. Agargun, J. Kirkby, and J. K. Friedman. 2006. "Relation of dreams to waking concerns." *Psychiatry Research* 141 (3): 261–270.

Casali, Adenauer G., Olivia Gosseries, Mario Rosanova, Mélanie Boly, Simone Sarasso, Karina R. Casali, Silvia Casarotto, Marie-Aurélie Bruno, Steven Laureys, Giulio Tononi, and Marcello Massimini. 2013. "A theoretically based index of consciousness independent of sensory processing and behavior." *Science Translational Medicine* 5 (198): 198ra105. DOI 10.1126/scitranslmed.3006294.

Cassell, E. J. 1982. "The nature of suffering and the goals of medicine." *New England Journal of Medicine* 306 (11): 639–645. DOI 10.1056/NEJM198203183061104.

Cataldi, J., A. M. Stephan, J. Haba-Rubio, and F. Siclari. 2024. "Shared EEG correlates between non-REM parasomnia experiences and dreams." *Nature Communications* 15: 3906. DOI 10.1038/s41467-024-48337-7.

Catholic Church. 1997. *Catechism of the Catholic Church.* 2nd ed. Libreria Editrice Vaticana. Paragraph 1996.

Celani, A. and M. Vergassola. 2010. "Bacterial strategies for chemotaxis response." *Proceedings of the National Academy of Sciences* 107 (4): 1391–1396. DOI 10.1073/pnas.0909673107.

Cella, David, William Riley, Arthur Stone, Nan Rothrock, Bryce Reeve, Susan Yount, Dagmar Amtmann, Rita Bode, Daniel Buysse, Seung Choi, Karon Cook, Robert DeVellis, Darren DeWalt, James F. Fries, Richard Gershon, Elizabeth A. Hahn, Jin-Shei Lai, Paul Pilkonis, Dennis Revicki, Matthias Rose, Kevin Weinfurt, and Ron Hays. 2010. "The Patient-Reported Outcomes Measurement Information System (PROMIS) developed and tested its first wave of adult self-reported health outcome item banks: 2005–2008." *Journal of Clinical Epidemiology* 63 (11): 1179–1194. DOI 10.1016/j.jclinepi.2010.04.011.

Chalmers, D. J. 2023. "Could a large language model be conscious?" *arXiv:2303.07103.*

Chalmers, David J. 1996. *The Conscious Mind: In Search of a Fundamental Theory.* Oxford University Press.

Chambers, C. D. 2013. "Registered Reports: A New Publishing Initiative at Cortex." *Cortex* 49 (3): 609–610. DOI 10.1016/j.cortex.2012.12.016.

Champagnat, N. and D. Villemonais. 2016. "Exponential convergence to quasi-stationary distribution and Q-process." *Probability Theory and Related Fields* 164: 243–283. DOI 10.1007/s00440-014-0611-7.

Chan, An-Wen, Isabelle Boutron, Sally Hopewell, David Moher, Kenneth F Schulz, Gary S Collins, Ruth Tunn, Rakesh Aggarwal, Michael Berkwits, Jesse A Berlin, Nita Bhandari, Nancy J Butcher, Marion K Campbell, Runcie C W Chidebe, Diana R Elbourne, Andrew J Farmer, Dean A Fergusson, Robert M Golub, Steven N Goodman, Tammy C Hoffmann, John P A Ioannidis, Brennan C Kahan, Rachel L Knowles, Sarah E Lamb, Steff Lewis, Elizabeth Loder, Martin Offringa, Philippe Ravaud, Dawn P Richards, Frank W Rockhold, David L Schriger, Nandi L Siegfried, Sophie Staniszewska, Rod S Taylor, Lehana Thabane, David J Torgerson, Sunita Vohra, Ian R White, and Asbjørn Hróbjartsson. 2025. "SPIRIT 2025 statement: Updated guideline for protocols of randomised trials." *BMJ* 389: e081477. DOI 10.1136/bmj-2024-081477.

Chatterjee, K., M. Chmelík, and J. Davies. 2016. "A Symbolic SAT-Based Algorithm for Almost-Sure Reachability with Small Strategies in POMDPs." *Proceedings of the AAAI Conference on Artificial Intelligence* 30 (1): 3225–3232. DOI 10.1609/aaai.v30i1.10422.

Chen, Y., T. T. Georgiou, and M. Pavon. 2021. "Optimal transport in systems and control." *Annual Review of Control, Robotics, and Autonomous Systems* 4: 89–113. DOI 10.1146/annurev-control-070220-100858.

Cheney-Lippold, John. 2017. *We Are Data: Algorithms and the Making of Our Digital Selves*. New York University Press.

Chetrite, R. and H. Touchette. 2015. "Nonequilibrium Markov processes conditioned on large deviations." *Annales Henri Poincaré* 16: 2005–2057. DOI 10.1007/s00023-014-0375-8.

Chetty, Raj, Nathaniel Hendren, Patrick Kline, and Emmanuel Saez. 2014. "Where is the land of opportunity? The geography of intergenerational mobility in the United States." *Quarterly Journal of Economics* 129 (4): 1553–1623. DOI 10.1093/qje/qju022.

Chiolero, Arnaud, Gilles Paradis, Benjamin Rich, and James A. Hanley. 2013. "Assessing the Relationship between the Baseline Value of a Continuous Variable and Subsequent Change Over Time." *Frontiers in Public Health* 1: 29. DOI 10.3389/fpubh.2013.00029.

Chittka, L. 2022. *The Mind of a Bee.* Princeton University Press. https://doi.org/10.1515/9780691236247

Chochinov, H. M. 2002. "Dignity-conserving care—A new model for palliative care: Helping the patient feel valued." *JAMA* 287 (17): 2253–2260. https://doi.org/10.1001/jama.287.17.2253

Chochinov, H. M. 2006. "Dying, dignity, and new horizons in palliative end-of-life care." *CA: A Cancer Journal for Clinicians* 56 (2): 84–103. DOI 10.3322/canjclin.56.2.84.

Choe, Eun Kyoung, Nicole B. Lee, Bongshin Lee, Wanda Pratt, and Julie A. Kientz. 2014. "Understanding quantified-selfers' practices in collecting and exploring personal data." *Proceedings of CHI 2014*, 1143–1152. DOI 10.1145/2556288.2557372.

Citron, Danielle Keats, and Frank Pasquale. 2014. "The scored society: Due process for automated predictions." *Washington Law Review* 89: 1–33.

Citron, Danielle Keats. 2008. "Technological due process." *Washington University Law Review* 85 (6): 1249–1313.

Claassen, Jan, Kevin Doyle, Adu Matory, Caroline Couch, Kelly M. Burger, Angela Velazquez, Joshua U. Okonkwo, Jean-Rémi King, Soojin Park, Sachin Agarwal, David Roh, Murad Megjhani, Andrey Eliseyev, E. Sander Connolly, and Benjamin Rohaut. 2019. "Detection of brain activation in unresponsive patients with acute brain injury." *New England Journal of Medicine* 380 (26): 2497–2505. DOI 10.1056/NEJMoa1812757.

Clark, A. 2013. "Whatever next? Predictive brains, situated agents, and the future of cognitive science." *Behavioral and Brain Sciences* 36 (3): 181–204. DOI 10.1017/S0140525X12000477.

Clark, A. and D. J. Chalmers. 1998. "The extended mind." *Analysis* 58 (1): 7–19. DOI 10.1093/analys/58.1.7.

Clayton, N. S. and A. Dickinson. 1998. "Episodic-like memory during cache recovery by scrub jays." *Nature* 395: 272–274. DOI 10.1038/26216.

Clayton, N. S. and A. Dickinson. 1999. "Scrub jays (Aphelocoma coerulescens) remember the relative time of caching as well as the location and content of their caches." *Journal of Comparative Psychology* 113 (4): 403–416. DOI 10.1037/0735-7036.113.4.403.

Cloud, William, and Robert Granfield. 2008. "Conceptualizing recovery capital: Expansion of a theoretical construct." *Substance Use & Misuse* 43 (12–13): 1971–1986. DOI 10.1080/10826080802289762.

Coan, James A., and David A. Sbarra. 2015. "Social baseline theory: The social regulation of risk and effort." *Current Opinion in Psychology* 1: 87–91. DOI 10.1016/j.copsyc.2014.12.021.

Cogitate Consortium, O. Ferrante, U. Gorska-Klimowska, et al. 2025. "Adversarial testing of global neuronal workspace and integrated information theories of consciousness." *Nature* 642: 133–142. DOI 10.1038/s41586-025-08888-1.

Cohen, Sheldon, and Thomas A. Wills. 1985. "Stress, social support, and the buffering hypothesis." *Psychological Bulletin* 98 (2): 310–357. DOI 10.1037/0033-2909.98.2.310.

Collins, Harry, and Robert Evans. 2007. *Rethinking Expertise.* University of Chicago Press.

Committee on Publication Ethics. 2025. *Retraction guidelines (Version 3)*. DOI 10.24318/cope.2019.1.4.

Committee on the Rights of Persons with Disabilities. 2014. *General Comment No. 1, Article 12: Equal Recognition before the Law.* CRPD/C/GC/1.

Copson, Andrew. 2019. *Secularism: A Very Short Introduction.* Oxford University Press. DOI 10.1093/actrade/9780198747222.001.0001.

Correia, S. P. C., A. Dickinson, and N. S. Clayton. 2007. "Western scrub-jays anticipate future needs independently of their current motivational state." *Current Biology* 17 (10): 856–861. DOI 10.1016/j.cub.2007.03.063.

Council for International Organizations of Medical Sciences. 2016. *International Ethical Guidelines for Health-related Research Involving Humans (4th ed.).* CIOMS.

Council of Europe. 2024. *Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law.* Council of Europe Treaty Series No. 225.

Coviello, Lorenzo, Yunkyu Sohn, Adam D. I. Kramer, Cameron Marlow, Massimo Franceschetti, Nicholas A. Christakis, and James H. Fowler. 2014. "Detecting emotional contagion in massive social networks." *PLoS ONE* 9 (3): e90315. DOI 10.1371/journal.pone.0090315.

Craig, A. D. 2009. "How do you feel-now? The anterior insula and human awareness." *Nature Reviews Neuroscience* 10: 59–70. DOI 10.1038/nrn2555.

Cranmer, Kyle, Johann Brehmer, and Gilles Louppe. 2020. "The frontier of simulation-based inference." *Proceedings of the National Academy of Sciences* 117 (48): 30055–30062. DOI 10.1073/pnas.1912789117.

Credé, Marcus, Michael C. Tynan, and Peter D. Harms. 2017. "Much ado about grit: A meta-analytic synthesis of the grit literature." *Journal of Personality and Social Psychology* 113 (3): 492–511. DOI 10.1037/pspp0000102.

Critchley, Hugo D., and Sarah N. Garfinkel. 2017. "Interoception and emotion." *Current Opinion in Psychology* 17: 7–14. DOI 10.1016/j.copsyc.2017.04.020.

Cronbach, Lee J., and Paul E. Meehl. 1955. "Construct validity in psychological tests." *Psychological Bulletin* 52 (4): 281–302. DOI 10.1037/h0040957.

Crook, R. J. 2021. "Behavioral and neurophysiological evidence suggests affective pain experience in octopus." *iScience* 24 (3): 102229. DOI 10.1016/j.isci.2021.102229.

Crump, A., H. Browning, A. Schnell, C. Burn, and J. Birch. 2022. "Sentience in decapod crustaceans: A general framework and review of the evidence." *Animal Sentience* 7 (32): Article 1. DOI 10.51291/2377-7478.1691.

Csikszentmihalyi, M. and R. Larson. 1987. "Validity and reliability of the Experience-Sampling Method." *Journal of Nervous and Mental Disease* 175 (9): 526–536. DOI 10.1097/00005053-198709000-00004.

Dahabreh, Issa J., Sarah E. Robertson, Jon A. Steingrimsson, Elizabeth A. Stuart, and Miguel A. Hernán. 2020. "Extending inferences from a randomized trial to a new target population." *Statistics in Medicine* 39 (14): 1999–2014. DOI 10.1002/sim.8426.

Darwin, C. 1859. *On the Origin of Species by Means of Natural Selection, or, The Preservation of Favoured Races in the Struggle for Life.* John Murray.

Darwin, Charles. 2009. *On the Origin of Species.* Oxford World's Classics. Oxford University Press. Originally published 1859.

Dawkins, M. S. 2008. "The science of animal suffering." *Ethology* 114 (10): 937–945. https://doi.org/10.1111/j.1439-0310.2008.01557.x

de Cheveigné, Alain, and Israel Nelken. 2019. "Filters: When, Why, and How (Not) to Use Them." *Neuron* 102 (2): 280–293. DOI 10.1016/j.neuron.2019.02.039.

de Haan, E. H. F., P. M. Corballis, S. A. Hillyard, C. A. Marzi, A. Seth, V. A. F. Lamme, L. Volz, M. Fabri, E. Schechter, T. Bayne, M. Corballis, and Y. Pinto. 2020. "Split-Brain: What We Know Now and Why This Is Important for Understanding Consciousness." *Neuropsychology Review* 30 (2): 224–233. DOI 10.1007/s11065-020-09439-3.

de Vet, H. C. W., C. B. Terwee, L. B. Mokkink, and D. L. Knol. 2011. *Measurement in Medicine: A Practical Guide.* Cambridge University Press. https://doi.org/10.1017/cbo9780511996214

Deci, Edward L., and Richard M. Ryan. 2000. "The 'What' and 'Why' of Goal Pursuits: Human Needs and the Self-Determination of Behavior." *Psychological Inquiry* 11 (4): 227–268. DOI 10.1207/S15327965PLI1104_01.

Dekker, Sidney. 2014. *The Field Guide to Understanding "Human Error" (3rd ed.).* Ashgate.

Dennett, D. C. 1987. *The Intentional Stance.* MIT Press.

Denrell, Jerker. 2003. "Vicarious learning, undersampling of failure, and the myths of management." *Organization Science* 14 (3): 227–243. DOI 10.1287/orsc.14.2.227.15164.

Dickerson, S. S. and M. E. Kemeny. 2004. "Acute stressors and cortisol responses: A theoretical integration and synthesis of laboratory research." *Psychological Bulletin* 130 (3): 355–391. DOI 10.1037/0033-2909.130.3.355.

Diener, E., R. E. Lucas, and C. N. Scollon. 2006. "Beyond the hedonic treadmill: Revising the adaptation theory of well-being." *American Psychologist* 61 (4): 305–314. DOI 10.1037/0003-066X.61.4.305.

Dienes, Zoltan. 2014. "Using Bayes to get the most out of non-significant results." *Frontiers in Psychology* 5: 781. DOI 10.3389/fpsyg.2014.00781.

Diggle, P. J. and M. G. Kenward. 1994. "Informative drop-out in longitudinal data analysis." *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 43 (1): 49–73. DOI 10.2307/2986113.

Dinges, David F., Frances Pack, Katherine Williams, Kelly A. Gillen, John W. Powell, Geoffrey E. Ott, Caitlin Aptowicz, and Allan I. Pack. 1997. "Cumulative sleepiness, mood disturbance, and psychomotor vigilance performance decrements during a week of sleep restricted to 4–5 hours per night." *Sleep* 20 (4): 267–277. DOI 10.1093/sleep/20.4.267.

Doka, Kenneth J., ed. 1989. *Disenfranchised Grief: Recognizing Hidden Sorrow.* Lexington Books.

Domhoff, G. W. 2017. *The Emergence of Dreaming: Mind-Wandering, Embodied Simulation, and the Default Network.* Oxford University Press.

Doob, J. L. 1957. "Conditional Brownian motion and the boundary limits of harmonic functions." *Bulletin de la Société Mathématique de France* 85: 431–458. DOI 10.24033/bsmf.1494.

Dorahy, M. J., B. L. Brand, V. Şar, C. Krüger, P. Stavropoulos, A. Martínez-Taboas, R. Lewis-Fernández, and W. Middleton. 2014. "Dissociative identity disorder: An empirical overview." *Australian & New Zealand Journal of Psychiatry* 48 (5): 402–417. DOI 10.1177/0004867414527523.

Duckworth, A. L. and J. J. Gross. 2014. "Self-control and grit: Related but separable determinants of success." *Current Directions in Psychological Science* 23 (5): 319–325. DOI 10.1177/0963721414541462.

Duckworth, Angela L., Christopher Peterson, Michael D. Matthews, and Dennis R. Kelly. 2007. "Grit: Perseverance and passion for long-term goals." *Journal of Personality and Social Psychology* 92 (6): 1087–1101. DOI 10.1037/0022-3514.92.6.1087.

Ecker, Ullrich K. H., Stephan Lewandowsky, John Cook, Philipp Schmid, Lisa K. Fazio, Nadia Brashier, Panayiota Kendeou, Emily K. Vraga, and Michelle A. Amazeen. 2022. "The Psychological Drivers of Misinformation Belief

and Its Resistance to Correction." *Nature Reviews Psychology* 1: 13–29. DOI 10.1038/s44159-021-00006-y.

Elce, V., D. Bergamo, G. Bontempi, B. Pedreschi, M. Bellesi, G. Handjaras, and G. Bernardi. 2025. "The individual determinants of morning dream recall." *Communications Psychology* 3 (1): 25. DOI 10.1038/s44271-025-00191-z.

Eldar, E., R. B. Rutledge, R. J. Dolan, and Y. Niv. 2016. "Mood as representation of momentum." *Trends in Cognitive Sciences* 20 (1): 15–24. DOI 10.1016/j.tics.2015.07.010.

Emanuel, Ezekiel J., David Wendler, and Christine Grady. 2000. "What makes clinical research ethical?" *JAMA* 283 (20): 2701–2711. DOI 10.1001/jama.283.20.2701.

Emery, N. J. and N. S. Clayton. 2004. "The mentality of crows: Convergent evolution of intelligence in corvids and apes." *Science* 306 (5703): 1903–1907. DOI 10.1126/science.1098410.

Entwistle, Vikki A., Stacy M. Carter, Alan Cribb, and Kirsten McCaffery. 2010. "Supporting patient autonomy: The importance of clinician-patient relationships." *Journal of General Internal Medicine* 25 (7): 741–745. DOI 10.1007/s11606-010-1292-2.

Epley, Nicholas, Adam Waytz, and John T. Cacioppo. 2007. "On seeing human: A three-factor theory of anthropomorphism." *Psychological Review* 114 (4): 864–886. DOI 10.1037/0033-295X.114.4.864.

Epstein, Joshua M. 2008. "Why model?" *Journal of Artificial Societies and Social Simulation* 11 (4): 12.

Ericsson, K. Anders, Ralf T. Krampe, and Clemens Tesch-Römer. 1993. "The role of deliberate practice in the acquisition of expert performance." *Psychological Review* 100 (3): 363–406. DOI 10.1037/0033-295X.100.3.363.

Espeland, Wendy Nelson, and Mitchell L. Stevens. 1998. "Commensuration as a social process." *Annual Review of Sociology* 24: 313–343. DOI 10.1146/annurev.soc.24.1.313.

Ethier, S. N. and T. G. Kurtz. 1986. *Markov processes: Characterization and convergence.* Wiley. https://doi.org/10.1002/9780470316658

Eubanks, Virginia. 2018. *Automating Inequality.* St. Martin's Press.

European Union. 2016. *Regulation (EU) 2016/679 of the European Parliament and of the Council: General Data Protection Regulation.* Official Journal of the European Union, L 119.

European Union. 2024. "Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No. 300/2008, (EU) No. 167/2013, (EU) No. 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act)." *Official Journal of the European Union, L 2024/1689*, 12 July 2024.

European Union. 2025. "Regulation (EU) 2025/327 of the European Parliament and of the Council of 11 February 2025

on the European Health Data Space and amending Directive 2011/24/EU and Regulation (EU) 2024/2847." *Official Journal of the European Union, L 2025/327*, 5 March 2025.

European Union. 2026. "Regulation (EU) 2026/1744 of the European Parliament and of the Council of 8 July 2026 amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence (Digital Omnibus on AI)." *Official Journal of the European Union, L 2026/1744*, 24 July 2026.

Everitt, B. J. and T. W. Robbins. 2016. "Drug addiction: Updating actions to habits to compulsions ten years on." *Annual Review of Psychology* 67: 23–50. DOI 10.1146/annurev-psych-122414-033457.

Exline, Julie J., and Joshua A. Wilt. 2023. "Supernatural Attributions: Seeing God, the Devil, Demons, Spirits, Fate, and Karma as Causes of Events." *Annual Review of Clinical Psychology* 19: 461–487. DOI 10.1146/annurev-clinpsy-080921-081114.

Exline, Julie J., Kenneth I. Pargament, Joshua A. Wilt, Kathleen C. Pait, and William A. Schutt. 2024. "Research on spiritual struggles: A brief snapshot focusing on new horizons." *Spiritual Care* 13 (2): 103–114. DOI 10.1515/spircare-2022-0063.

Faden, Ruth R., and Tom L. Beauchamp. 1986. *A History and Theory of Informed Consent.* Oxford University Press.

FDA-NIH Biomarker Working Group. 2025. "BEST (Biomarkers, EndpointS, and other Tools) Resource." *Food and Drug Administration and National Institutes of Health; glossary revised January 16*, 2025.

Fillingim, R. B. 2017. "Individual differences in pain: Understanding the mosaic that makes pain personal." *Pain* 158 (Suppl 1): S11–S18. DOI 10.1097/j.pain.0000000000000775.

Fine, J. P. and R. J. Gray. 1999. "A proportional hazards model for the subdistribution of a competing risk." *Journal of the American Statistical Association* 94 (446): 496–509. DOI 10.1080/01621459.1999.10474144.

Fins, J. J. 2015. *Rights Come to Mind: Brain Injury, Ethics, and the Struggle for Consciousness.* Cambridge University Press. https://doi.org/10.1017/cbo9781139051279

Fischer, John Martin, and Mark Ravizza. 1998. *Responsibility and Control: A Theory of Moral Responsibility.* Cambridge University Press. https://doi.org/10.1017/cbo9780511814594

Fischhoff, Baruch. 2013. "The sciences of science communication." *Proceedings of the National Academy of Sciences* 110 (Suppl. 3): 14033–14039. DOI 10.1073/pnas.1213273110.

Fisher, A. J., J. D. Medaglia, and B. F. Jeronimus. 2018. "Lack of group-to-individual generalizability is a threat to human subjects research." *Proceedings of the National*

Academy of Sciences 115 (27): E6106-E6115. DOI 10.1073/pnas.1711978115.

Fiske, Susan T., and Cydney Dupree. 2014. "Gaining Trust as Well as Respect in Communicating to Motivated Audiences about Science Topics." *Proceedings of the National Academy of Sciences* 111 (supplement 4): 13593–13597. DOI 10.1073/pnas.1317505111.

Fitzmaurice, G. M., N. M. Laird, and J. H. Ware. 2011. *Applied Longitudinal Analysis (2nd ed.).* Wiley.

Fitzsimmons, Patrick J., Jim Pitman, and Marc Yor. 1993. "Markovian bridges: Construction, Palm interpretation, and splicing." In *Seminar on Stochastic Processes, 1992*, 101–134. Birkhäuser. https://doi.org/10.1007/978-1-4612-0339-1_5

Flake, J. K. and E. I. Fried. 2020. "Measurement schmeasurement: Questionable measurement practices and how to avoid them." *Advances in Methods and Practices in Psychological Science* 3 (4): 456–465. DOI 10.1177/2515245920952393.

Flood, Gavin. 1996. *An Introduction to Hinduism.* Cambridge University Press.

Folland, G. B. 1999. *Real Analysis: Modern Techniques and Their Applications (2nd ed.).* Wiley.

Franco, Annie, Neil Malhotra, and Gabor Simonovits. 2014. "Publication Bias in the Social Sciences: Unlocking the File Drawer." *Science* 345 (6203): 1502–1505. DOI 10.1126/science.1255484.

Frankfurt, Harry G. 1971. "Freedom of the will and the concept of a person." *Journal of Philosophy* 68 (1): 5–20. https://doi.org/10.2307/2024717

Frankl, V. E. 1946. *Ein Psycholog erlebt das Konzentrationslager.* Verlag für Jugend und Volk.

Frankl, Viktor E. 2006. *Man's Search for Meaning.* Beacon Press. Originally published 1946.

Franklin, Joseph C., Kent M. Lee, Eleanor K. Hanna, and Mitchell J. Prinstein. 2013. "Feeling Worse to Feel Better: Pain-Offset Relief Simultaneously Stimulates Positive Affect and Reduces Negative Affect." *Psychological Science* 24 (4): 521–529. DOI 10.1177/0956797612458805.

Fraser, David. 2008. *Understanding Animal Welfare: The Science in Its Cultural Context.* Wiley-Blackwell.

Frederick, S., and G. Loewenstein. 1999. "Hedonic adaptation." In D. Kahneman, E. Diener, and N. Schwarz, eds., *Well-Being: The Foundations of Hedonic Psychology*, 302–329. Russell Sage Foundation.

Frederick, S., G. Loewenstein, and T. O'Donoghue. 2002. "Time discounting and time preference: A critical review." *Journal of Economic Literature* 40 (2): 351–401. DOI 10.1257/002205102320161311.

Fredrickson, Barbara L. 2001. "The role of positive emotions in positive psychology: The broaden-and-build theory of positive emotions." *American Psychologist* 56 (3): 218–226. DOI 10.1037/0003-066X.56.3.218.

Freund, A. M. and P. B. Baltes. 2002. "Life-management strategies of selection, optimization and compensation: Measurement by self-report and construct validity." *Journal of Personality and Social Psychology* 82 (4): 642–662. DOI 10.1037/0022-3514.82.4.642.

Fricker, M. 2007. *Epistemic Injustice: Power and the Ethics of Knowing.* Oxford University Press. https://doi.org/10.1093/acprof:oso/9780198237907.001.0001

Friston, K. 2010. "The free-energy principle: A unified brain theory?" *Nature Reviews Neuroscience* 11: 127–138. DOI 10.1038/nrn2787.

Friston, K. J., T. FitzGerald, F. Rigoli, P. Schwartenbeck, and G. Pezzulo. 2017. "Active inference: A process theory." *Neural Computation* 29 (1): 1–49. DOI 10.1162/NECO_a_00912.

Fujita, K. 2011. "On conceptualizing self-control as more than the effortful inhibition of impulses." *Personality and Social Psychology Review* 15 (4): 352–366. DOI 10.1177/1088868311411165.

Futuyma, Douglas J., and Mark Kirkpatrick. 2017. *Evolution.* 4th ed. Sinauer Associates.

Galatzer-Levy, Isaac R., Sandy H. Huang, and George A. Bonanno. 2018. "Trajectories of resilience and dysfunction following potential trauma: A review and statistical evaluation." *Clinical Psychology Review* 63: 41–55. DOI 10.1016/j.cpr.2018.05.008.

Gallagher, S. 2000. “Philosophical conceptions of the self: Implications for cognitive science.” *Trends in Cognitive Sciences* 4 (1): 14–21. DOI 10.1016/S1364-6613(99)01417-5.

Galpayage Dona, Hiruni Samadi, Cwyn Solvi, Amelia Kowalewska, Kaarle Mäkelä, HaDi MaBouDi, and Lars Chittka. 2022. “Do bumble bees play?” *Animal Behaviour* 194: 239–251. DOI 10.1016/j.anbehav.2022.08.013.

García, J. and F. Fernández. 2015. “A comprehensive survey on safe reinforcement learning.” *Journal of Machine Learning Research* 16: 1437–1480.

Garg, J., X. Zhang, and Q. Zhou. 2024. “Soft-constrained Schrödinger bridge: A stochastic control approach.” *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics, Proceedings of Machine Learning Research* 238: 4429–4437.

Gasper, Karen, Cinnamon L. Danube, and Danfei Hu. 2021. “Making room for neutral affect: Evidence indicating that neutral affect is independent of and co-occurs with eight affective states.” *Motivation and Emotion* 45: 103–121. DOI 10.1007/s11031-020-09861-3.

Gazzaniga, M. S. 2005. “Forty-five years of split-brain research and still going strong.” *Nature Reviews Neuroscience* 6 (8): 653–659. DOI 10.1038/nrn1723.

Gebru, T., J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford. 2021. “Datasheets for Datasets.” *Communications of the ACM* 64 (12): 86–92. DOI 10.1145/3458723.

Geertz, Clifford. 1973. *The Interpretation of Cultures.* Basic Books.

Gelman, A. and E. Loken. 2013. *The Garden of Forking Paths: Why Multiple Comparisons Can Be a Problem, Even When There Is No Fishing Expedition or p-Hacking and the Research Hypothesis Was Posited Ahead of Time.* Columbia University.

Gelman, A. and E. Loken. 2014. "The statistical crisis in science." *American Scientist* 102 (6): 460–465. DOI 10.1511/2014.111.460.

Gelman, Andrew, Aki Vehtari, Daniel Simpson, Charles C. Margossian, Bob Carpenter, Yuling Yao, Lauren Kennedy, Jonah Gabry, Paul-Christian Bürkner, and Martin Modrák. 2020. "Bayesian workflow." *arXiv:2011.01808.*

Gerstorf, D., N. Ram, J. Goebel, J. Schupp, U. Lindenberger, and G. G. Wagner. 2010. "Where people live and die makes a difference: Individual and geographic disparities in well-being progression at the end of life." *Psychology and Aging* 25 (3): 661–676. DOI 10.1037/a0019574.

Gerstorf, D., N. Ram, U. Lindenberger, and J. Smith. 2013. "Age and time-to-death trajectories of change in indicators of cognitive, sensory, physical, health, social, and self-related functions." *Developmental Psychology* 49 (10): 1805–1821. DOI 10.1037/a0031340.

Gethin, Rupert. 1998. *The Foundations of Buddhism.* Oxford University Press.

Giacino, Joseph T., Douglas I. Katz, Nicholas D. Schiff, John Whyte, Eric J. Ashman, Stephen Ashwal, Richard Barbano, Flora M. Hammond, Steven Laureys, Geoffrey S.F. Ling, Risa Nakase-Richardson, Ronald T. Seel, Stuart Yablon, Thomas S.D. Getchius, Gary S. Gronseth, and Melissa J. Armstrong. 2018. "Practice guideline update recommendations summary: Disorders of consciousness." *Neurology* 91 (10): 450–460. DOI 10.1212/WNL.0000000000005926.

Gibbons, M., A. Crump, M. Barrett, S. Sarlak, J. Birch, and L. Chittka. 2022a. "Can insects feel pain? A review of the neural and behavioural evidence." *Advances in Insect Physiology* 63: 155–229. DOI 10.1016/bs.aiip.2022.10.001.

Gibbons, M., E. Pasquini, A. Kowalewska, E. Read, S. Gibson, A. Crump, C. Solvi, E. Versace, and L. Chittka. 2024. "Noxious stimulation induces self-protective behavior in bumblebees." *iScience* 27 (8): 110440. DOI 10.1016/j.isci.2024.110440.

Gibbons, M., E. Versace, A. Crump, B. Baran, and L. Chittka. 2022b. "Motivational trade-offs and modulation of nociception in bumblebees." *Proceedings of the National Academy of Sciences* 119 (31): e2205821119. DOI 10.1073/pnas.2205821119. Correction: "Correction for Gibbons et al., Motivational trade-offs and modulation of nociception in bumblebees." 2025. *Proceedings of the National Academy of Sciences* 122 (37): e2522767122. DOI 10.1073/pnas.2522767122.

Gibbons, M., E. Versace, A. Crump, B. Baran, and L. Chittka. 2026. "Reassessment confirms motivational trade-offs and modulation of nociception in bumble bees." *Animal Behaviour* 231: 123390. DOI 10.1016/j.anbehav.2025.123390.

Glynn, Peter W., and Sean P. Meyn. 1996. "A Liapounov bound for solutions of the Poisson equation." *The Annals of Probability* 24 (2): 916–931. DOI 10.1214/aop/1039639370.

Gneiting, T. and A. E. Raftery. 2007. "Strictly proper scoring rules, prediction, and estimation." *Journal of the American Statistical Association* 102 (477): 359–378. DOI 10.1198/016214506000001437.

Godfrey-Smith, Peter. 2009. *Darwinian Populations and Natural Selection.* Oxford University Press.

Goldstein, A. N. and M. P. Walker. 2014. "The role of sleep in emotional brain function." *Annual Review of Clinical Psychology* 10: 679–708.

Gollwitzer, Peter M. 1999. "Implementation intentions: Strong effects of simple plans." *American Psychologist* 54 (7): 493–503. DOI 10.1037/0003-066X.54.7.493.

Gollwitzer, Peter M., and Paschal Sheeran. 2006. "Implementation intentions and goal achievement: A meta-analysis of effects and processes." *Advances in Experimental Social Psychology* 38: 69–119. DOI 10.1016/S0065-2601(06)38002-1.

Gould, Stephen Jay, and Richard C. Lewontin. 1979. "The spandrels of San Marco and the Panglossian paradigm: A

critique of the adaptationist programme." *Proceedings of the Royal Society B* 205 (1161): 581–598. DOI 10.1098/rspb.1979.0086.

Green, Ben. 2022. "The flaws of policies requiring human oversight of government algorithms." *Computer Law & Security Review* 45: 105681. DOI 10.1016/j.clsr.2022.105681.

Greenwood, B. M. and S. N. Garfinkel. 2025. "Interoceptive mechanisms and emotional processing." *Annual Review of Psychology* 76: 59–86. DOI 10.1146/annurev-psych-020924-125202.

Greer, David M., Matthew P. Kirschen, Ariane Lewis, Gary S. Gronseth, Alexander Rae-Grant, Stephen Ashwal, Maya A. Babu, David F. Bauer, Lori Billinghurst, Amanda Corey, Sonia Partap, Michael A. Rubin, Lori Shutter, Courtney Takahashi, Robert C. Tasker, Panayiotis Nicolaou Varelas, Eelco Wijdicks, Amy Bennett, Scott R. Wessels, and John J. Halperin. 2023. "Pediatric and adult brain death/death by neurologic criteria consensus guideline: Report of the AAN Guidelines Subcommittee, AAP, CNS, and SCCM." *Neurology* 101 (24): 1112–1132. DOI 10.1212/WNL.0000000000207740.

Greer, David M., Sam D. Shemie, Ariane Lewis, Sylvia Torrance, Panayiotis Varelas, Fernando D. Goldenberg, James L. Bernat, Michael Souter, Mehmet Akif Topcuoglu, Anne W. Alexandrov, Marie Baldisseri, Thomas Bleck, Giuseppe Citerio, Rosanne Dawson, Arnold Hoppe, Stephen Jacobe, Alex Manara, Thomas A.

Nakagawa, Thaddeus Mason Pope, William Silvester, David Thomson, Hussain Al Rahma, Rafael Badenes, Andrew J. Baker, Vladimir Cerny, Cherylee Chang, Tiffany R. Chang, Elena Gnedovskaya, Moon-Ku Han, Stephen Honeybul, Edgar Jimenez, Yasuhiro Kuroda, Gang Liu, Uzzwal Kumar Mallick, Victoria Marquevich, Jorge Mejia-Mantilla, Michael Piradov, Sarah Quayyum, Gentle Sunder Shrestha, Ying-ying Su, Shelly D. Timmons, Jeanne Teitelbaum, Walter Videtta, Kapil Zirpe, and Gene Sung. 2020. "Determination of brain death/death by neurologic criteria: The World Brain Death Project." *JAMA* 324 (11): 1078–1097. DOI 10.1001/jama.2020.11586.

Grimm, V., S. F. Railsback, C. E. Vincenot, U. Berger, C. Gallagher, D. L. DeAngelis, B. Edmonds, J. Ge, J. Giske, J. Groeneveld, A. S. A. Johnston, A. Milles, J. Nabe-Nielsen, J. G. Polhill, V. Radchuk, M.-S. Rohwäder, R. A. Stillman, J. C. Thiele, and D. Ayllón. 2020. "The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism." *Journal of Artificial Societies and Social Simulation* 23 (2): 7. DOI 10.18564/jasss.4259.

Guest, Olivia, and Andrea E. Martin. 2021. "How computational modeling can force theory building in psychological science." *Perspectives on Psychological Science* 16 (4): 789–802. DOI 10.1177/1745691620970585.

Gupta, Ankit, and Eduardo Sontag. 2025. “Cumulative dose responses for adapting biological systems.” *Journal of the Royal Society Interface* 22 (229): 20240877. DOI 10.1098/rsif.2024.0877.

Guyatt, G., D. Sackett, D. W. Taylor, J. Chong, R. Roberts, and S. Pugsley. 1986. “Determining optimal therapy: Randomized trials in individual patients.” *New England Journal of Medicine* 314 (14): 889–892. DOI 10.1056/NEJM198604033141406.

Hafer, C. L. and L. Bègue. 2005. “Experimental research on just-world theory: Problems, developments, and future challenges.” *Psychological Bulletin* 131 (1): 128–167. DOI 10.1037/0033-2909.131.1.128.

Haggard, P. 2017. “Sense of agency in the human brain.” *Nature Reviews Neuroscience* 18 (4): 196–207. DOI 10.1038/nrn.2017.14.

Haggerty, Kevin D., and Richard V. Ericson. 2000. “The surveillant assemblage.” *British Journal of Sociology* 51 (4): 605–622. DOI 10.1080/00071310020015280.

Hall, C. S. and R. L. Van de Castle. 1966. *The Content Analysis of Dreams.* Appleton-Century-Crofts.

Hamaker, E. L. 2012. “Why researchers should think ‘within-person’: A paradigmatic rationale.” In M. R. Mehl and T. S. Conner, eds., *Handbook of Research Methods for Studying Daily Life*, 43–61. Guilford Press.

Hamilton, W. D. 1964. “The genetical evolution of social behaviour. I.” *Journal of Theoretical Biology* 7 (1): 1–16.

Hamming, R. W. 1950. “Error detecting and error correcting codes.” *Bell System Technical Journal* 29 (2): 147–160. DOI 10.1002/j.1538-7305.1950.tb00463.x.

Hampton, R. R. 2001. “Rhesus monkeys know when they remember.” *Proceedings of the National Academy of Sciences* 98 (9): 5359–5362. DOI 10.1073/pnas.071600998.

Hand, D. J. 2004. *Measurement Theory and Practice.* Arnold.

Harkness, Janet, Beth-Ellen Pennell, and Alisú Schoua-Glusberg. 2004. “Survey questionnaire translation and assessment.” In Stanley Presser, Jennifer M. Rothgeb, Mick P. Couper, Judith T. Lessler, Elizabeth Martin, Jean Martin, and Eleanor Singer, eds., *Methods for Testing and Evaluating Survey Questionnaires*, 453–473. Wiley. https://doi.org/10.1002/0471654728.ch22

Harvey, Peter. 2013. *An Introduction to Buddhism: Teachings, History and Practices.* 2nd ed. Cambridge University Press.

Headey, B. and A. Wearing. 1989. “Personality, life events, and subjective well-being: Toward a dynamic equilibrium model.” *Journal of Personality and Social Psychology* 57 (4): 731–739. https://doi.org/10.1037/0022-3514.57.4.731

Heffner, Joseph, Romy Frömer, Matthew R. Nassar, and Oriel FeldmanHall. 2025. “Separable neural signals for reward and emotion prediction errors.” *Nature Communications* 16: 7849. DOI 10.1038/s41467-025-63135-5.

Helliwell, J. F., R. Layard, J. D. Sachs, J.-E. De Neve, L. B. Aknin, and S. Wang, eds. 2026. *World Happiness Report*

2026. University of Oxford: Wellbeing Research Centre. https://www.worldhappiness.report/ed/2026/.

Helson, H. 1964. *Adaptation-Level Theory: An Experimental and Systematic Approach to Behavior.* Harper & Row.

Herd, Pamela, and Donald P. Moynihan. 2018. *Administrative Burden: Policymaking by Other Means*. New York: Russell Sage Foundation.

Hernán, M. A. and J. M. Robins. 2020. *Causal Inference: What If*. Boca Raton, FL: Chapman & Hall/CRC.

Heschel, Abraham Joshua. 1962. *The Prophets.* Harper & Row.

Hick, John. 1966. *Evil and the God of Love.* Macmillan.

Hobson, J. A. and R. W. McCarley. 1977. "The brain as a dream state generator: An activation-synthesis hypothesis of the dream process." *American Journal of Psychiatry* 134 (12): 1335–1348.

Hohwy, J. 2013. *The predictive mind.* Oxford University Press.

Holt-Lunstad, Julianne, Timothy B. Smith, and J. Bradley Layton. 2010. "Social relationships and mortality risk: A meta-analytic review." *PLoS Medicine* 7 (7): e1000316. DOI 10.1371/journal.pmed.1000316.

Hong, Stephanie, and Ed O'Brien. 2026. "Rating an Experience Makes People Notice How It Is Changing: Consequences for Hedonic Adaptation and Consumer Enjoyment." *Journal of Consumer Research*. Advance online publication, ucag038. DOI 10.1093/jcr/ucag038.

Hopewell, Sally, An-Wen Chan, Gary S Collins, Asbjørn Hróbjartsson, David Moher, Kenneth F Schulz, Ruth Tunn, Rakesh Aggarwal, Michael Berkwits, Jesse A Berlin, Nita Bhandari, Nancy J Butcher, Marion K Campbell, Runcie C W Chidebe, Diana Elbourne, Andrew Farmer, Dean A Fergusson, Robert M Golub, Steven N Goodman, Tammy C Hoffmann, John P A Ioannidis, Brennan C Kahan, Rachel L Knowles, Sarah E Lamb, Steff Lewis, Elizabeth Loder, Martin Offringa, Philippe Ravaud, Dawn P Richards, Frank W Rockhold, David L Schriger, Nandi L Siegfried, Sophie Staniszewska, Rod S Taylor, Lehana Thabane, David Torgerson, Sunita Vohra, Ian R White, and Isabelle Boutron. 2025. "CONSORT 2025 statement: Updated guideline for reporting randomised trials." *BMJ* 389: e081123. DOI 10.1136/bmj-2024-081123.

Howard, Steven R., Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. 2021. "Time-uniform, nonparametric, nonasymptotic confidence sequences." *The Annals of Statistics* 49 (2): 1055–1080. DOI 10.1214/20-AOS1991.

Hyun, I., J. C. Scharf-Deering, and J. E. Lunshof. 2020. "Ethical issues related to brain organoid research." *Brain Research* 1732: 146653. DOI 10.1016/j.brainres.2020.146653.

Hyvärinen, Aapo, Hiroaki Sasaki, and Richard E. Turner. 2019. "Nonlinear ICA using auxiliary variables and generalized contrastive learning." *Proceedings of the Twenty-Second International Conference on Artificial Intelligence*

and Statistics, Proceedings of Machine Learning Research 89: 859–868. PMLR.

Infurna, F. J. and S. S. Luthar. 2016. "Resilience to major life stressors is not as common as thought." *Perspectives on Psychological Science* 11 (2): 175–194. DOI 10.1177/1745691615621271.

Inouye, S. K., R. G. J. Westendorp, and J. S. Saczynski. 2014. "Delirium in elderly people." *The Lancet* 383 (9920): 911–922. DOI 10.1016/S0140-6736(13)60688-1.

International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. 2019. "ICH Harmonised Guideline E9(R1): Addendum on Estimands and Sensitivity Analysis in Clinical Trials to the Guideline on Statistical Principles for Clinical Trials." *Step 4*, 20 November 2019.

International Society for Stem Cell Research. 2025. *Guidelines for Stem Cell Research and Clinical Translation (Version 1.2).* ISSCR.

International Test Commission. 2017. *The ITC Guidelines for Translating and Adapting Tests (Second edition).* International Test Commission.

Ioannidis, J. P. A. 2005. "Why Most Published Research Findings Are False." *PLoS Medicine* 2 (8): e124. DOI 10.1371/journal.pmed.0020124.

James, W. 1902. *The Varieties of Religious Experience: A Study in Human Nature.* Longmans, Green, and Co. https://doi.org/10.1037/10004-000

James, William. 2002. *The Varieties of Religious Experience.* Modern Library. Originally published 1902.

Jangraw, David C., Hanna Keren, Haorui Sun, Rachel L. Bedder, Robb B. Rutledge, Francisco Pereira, Adam G. Thomas, Daniel S. Pine, Charles Zheng, Dylan M. Nielson, and Argyris Stringaris. 2023. "A highly replicable decline in mood during rest and simple tasks." *Nature Human Behaviour* 7: 596–610. DOI 10.1038/s41562-023-01519-7.

Ji, J., J. Zhou, B. Zhang, J. Dai, X. Pan, R. Sun, W. Huang, Y. Geng, M. Liu, and Y. Yang. 2024. "OmniSafe: An infrastructure for accelerating safe reinforcement learning research." *Journal of Machine Learning Research* 25 (285): 1–6.

Jiang, X., L.-H. Lim, Y. Yao, and Y. Ye. 2011. "Statistical ranking and combinatorial Hodge theory." *Mathematical Programming* 127: 203–244. DOI 10.1007/s10107-010-0419-x.

Joffily, M. and G. Coricelli. 2013. "Emotional valence and the free-energy principle." *PLoS Computational Biology* 9 (6): e1003094. DOI 10.1371/journal.pcbi.1003094.

Joshanloo, M. 2013. "The influence of fear of happiness beliefs on responses to the Satisfaction With Life Scale." *Personality and Individual Differences* 54 (5): 647–651. DOI 10.1016/j.paid.2012.11.011.

Kaelbling, L. P., M. L. Littman, and A. R. Cassandra. 1998. "Planning and acting in partially observable stochastic

domains." *Artificial Intelligence* 101 (1–2): 99–134. DOI 10.1016/S0004-3702(98)00023-X.

Kahneman, D. and A. Deaton. 2010. "High income improves evaluation of life but not emotional well-being." *Proceedings of the National Academy of Sciences* 107 (38): 16489–16493. DOI 10.1073/pnas.1011492107.

Kahneman, D., A. B. Krueger, D. A. Schkade, N. Schwarz, and A. A. Stone. 2004. "A survey method for characterizing daily life experience: The Day Reconstruction Method." *Science* 306 (5702): 1776–1780. DOI 10.1126/science.1103572.

Kapteyn, Arie, Francisco Perez-Arce, Evan W. Sandlin, and Arthur A. Stone. 2025. "Life Satisfaction Responds to Major Life Events; Affect not so much." *Social Indicators Research* 180: 1747–1763. DOI 10.1007/s11205-025-03724-5.

Karabatsos, George. 2018. "On Bayesian testing of additive conjoint measurement axioms using synthetic likelihood." *Psychometrika* 83 (2): 321–332. DOI 10.1007/s11336-017-9581-x.

Karatzas, I. and S. E. Shreve. 1991. *Brownian motion and stochastic calculus.* Springer.

Kellehear, A. 2007. *A Social History of Dying.* Cambridge University Press. https://doi.org/10.1017/cbo9780511481352

Kelly, John F., Nora D. Volkow, and Howard K. Koh. 2025. "The Changing Approach to Addiction: From Incarceration to Treatment and Recovery Support." *New England Journal of Medicine* 392 (9): 833–836. DOI 10.1056/NEJMp2414224.

Kelly, John F., Richard Saitz, and Sarah Wakeman. 2016. "Language, substance use disorders, and policy: The need to reach consensus on an 'addiction-ary.'" *Alcoholism Treatment Quarterly* 34 (1): 116–123. DOI 10.1080/07347324.2016.1113103.

Keramati, M. and B. Gutkin. 2014. "Homeostatic reinforcement learning for integrating reward collection and physiological stability." *eLife* 3: e04811. DOI 10.7554/eLife.04811.

Keren, Hanna, Charles Zheng, David C. Jangraw, Katharine Chang, Aria Vitale, Robb B. Rutledge, Francisco Pereira, Dylan M. Nielson, and Argyris Stringaris. 2021. "The temporal representation of experience in subjective mood." *eLife* 10: e62051. DOI 10.7554/eLife.62051.

Kesecioglu, Jozef, Katerina Rusinova, Daniela Alampi, Yaseen M. Arabi, Julie Benbenishty, Dominique Benoit, Carole Boulanger, Maurizio Cecconi, Christopher Cox, Marjel van Dam, Diederik van Dijk, James Downar, Nikolas Efstathiou, Ruth Endacott, Alessandro Galazzi, Fiona van Gelder, Rik T. Gerritsen, Armand Girbes, Laura Hawyrluck, Margaret Herridge, Jan Hudec, Nancy Kentish-Barnes, Monika Kerckhoffs, Jos M. Latour, Jan Malaska, Annachiara Marra, Stephanie Meddick-Dyson, Spyridon Mentzelopoulos, Mervyn Mer, Victoria Metaxa, Andrej Michalsen, Rajesh Mishra, Giovanni Mistraletti, Margo van Mol, Rui Moreno, Judith Nelson, Andrea

Ortiz Suñer, Natalie Pattison, Tereza Prokopova, Kathleen Puntillo, Kathryn Puxty, Samah Al Qahtani, Lukas Radbruch, Emilio Rodriguez-Ruiz, Ron Sabar, Stefan J. Schaller, Shahla Siddiqui, Charles L. Sprung, Michele Umbrello, Marco Vergano, Massimo Zambon, Marieke Zegers, Michael Darmon, and Elie Azoulay. 2024. "European Society of Intensive Care Medicine guidelines on end-of-life and palliative care in the intensive care unit." *Intensive Care Medicine* 50 (11): 1740–1766. DOI 10.1007/s00134-024-07579-1.

Khalsa, S. S., R. Adolphs, O. G. Cameron, H. D. Critchley, P. W. Davenport, J. S. Feinstein, J. D. Feusner, S. N. Garfinkel, R. D. Lane, W. E. Mehling, A. E. Meuret, C. B. Nemeroff, S. Oppenheimer, F. H. Petzschner, O. Pollatos, J. L. Rhudy, L. P. Schramm, W. K. Simmons, M. B. Stein, K. E. Stephan, O. Van den Bergh, I. Van Diest, A. von Leupoldt, M. P. Paulus, and the Interoception Summit 2016 participants. 2018. "Interoception and mental health: A roadmap." *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 3 (6): 501–513. DOI 10.1016/j.bpsc.2017.12.004.

Kidd, I. J. and H. Carel. 2017. "Epistemic injustice and illness." *Journal of Applied Philosophy* 34 (2): 172–190. DOI 10.1111/japp.12172.

Killingsworth, M. A., D. Kahneman, and B. Mellers. 2023. "Income and emotional well-being: A conflict resolved."

Proceedings of the National Academy of Sciences 120 (10): e2208661120. DOI 10.1073/pnas.2208661120.

Killingsworth, Matthew A., and Daniel T. Gilbert. 2010. "A wandering mind is an unhappy mind." *Science* 330 (6006): 932. DOI 10.1126/science.1192439.

King, G., C. J. L. Murray, J. A. Salomon, and A. Tandon. 2004. "Enhancing the validity and cross-cultural comparability of measurement in survey research." *American Political Science Review* 98 (1): 191–207. DOI 10.1017/S000305540400108X.

Klass, Dennis, Phyllis R. Silverman, and Steven L. Nickman, eds. 1996. *Continuing Bonds: New Understandings of Grief.* Taylor & Francis.

Klausen, S. H., J. Emiliussen, R. Christiansen, L. Hasandedic-Dapo, and S. Engelsen. 2022. "The many faces of hedonic adaptation." *Philosophical Psychology* 35 (2): 253–278. DOI 10.1080/09515089.2021.1967308.

Kleinman, A. 1988. *The Illness Narratives: Suffering, Healing, and the Human Condition.* Basic Books.

Koch, T. 2000. "The illusion of paradox: Commentary on Albrecht, G.L. and Devlieger, P.J. (1998). The disability paradox: High quality of life against all odds. Social Science & Medicine 48, 977–988." *Social Science & Medicine* 50 (6): 757–759. DOI 10.1016/S0277-9536(99)00385-8.

Kondziella, D., A. Bender, K. Diserens, W. van Erp, A. Estraneo, R. Formisano, S. Laureys, L. Naccache, S. Ozturk, B. Rohaut, J. D. Sitt, J. Stender, M. Tiainen, A. O.

Rossetti, O. Gosseries, C. Chatelle, and the EAN Panel on Coma, Disorders of Consciousness. 2020. "European Academy of Neurology guideline on the diagnosis of coma and other disorders of consciousness." *European Journal of Neurology* 27 (5): 741–756. DOI 10.1111/ene.14151.

Konkoly, Karen R., Kristoffer Appel, Emma Chabani, Anastasia Mangiaruga, Jarrod Gott, Remington Mallett, Bruce Caughran, Sarah Witkowski, Nathan W. Whitmore, Christopher Y. Mazurek, Jonathan B. Berent, Frederik D. Weber, Başak Türker, Smaranda Leu-Semenescu, Jean-Baptiste Maranci, Gordon Pipa, Isabelle Arnulf, Delphine Oudiette, Martin Dresler, and Ken A. Paller. 2021. "Real-time dialogue between experimenters and dreamers during REM sleep." *Current Biology* 31 (7): 1417–1427.e6. DOI 10.1016/j.cub.2021.01.026.

Koob, G. F. and M. Le Moal. 2008. "Addiction and the brain antireward system." *Annual Review of Psychology* 59: 29–53. DOI 10.1146/annurev.psych.59.103006.093548.

Kosik, K. S. 2024. "Why brain organoids are not conscious yet." *Patterns* 5 (8): 101011. DOI 10.1016/j.patter.2024.101011.

Krakow, Barry, Michael Hollifield, Lisa Johnston, Mary Koss, Ron Schrader, Teddy D. Warner, Dan Tandberg, John Lauriello, Leslie McBride, Lisa Cutchen, Diana Cheng, Shawn Emmons, Anne Germain, Dominic Melendrez, Diane Sandoval, and Holly Prince. 2001. "Imagery rehearsal therapy for chronic nightmares in sexual

assault survivors with posttraumatic stress disorder: A randomized controlled trial." *JAMA* 286 (5): 537–545. DOI 10.1001/jama.286.5.537.

Kramer, A. D. I., J. E. Guillory, and J. T. Hancock. 2014. "Experimental evidence of massive-scale emotional contagion through social networks." *Proceedings of the National Academy of Sciences* 111 (24): 8788–8790. DOI 10.1073/pnas.1320040111.

Krantz, D. H., R. D. Luce, P. Suppes, and A. Tversky. 1971. *Foundations of Measurement, Volume I: Additive and Polynomial Representations.* Academic Press.

Kreibig, S. D. 2010. "Autonomic nervous system activity in emotion: A review." *Biological Psychology* 84 (3): 394–421. DOI 10.1016/j.biopsycho.2010.03.010.

Kuppens, P., F. Tuerlinckx, J. A. Russell, and L. F. Barrett. 2013. "The relation between valence and arousal in subjective experience." *Psychological Bulletin* 139 (4): 917–940. DOI 10.1037/a0030811.

Kurland, B. F., L. L. Johnson, B. L. Egleston, and P. H. Diehr. 2009. "Longitudinal data with follow-up truncated by death: Match the analysis method to research aims." *Statistical Science* 24 (2): 211–222. DOI 10.1214/09-STS293.

Lakatos, I. 1978. *The Methodology of Scientific Research Programmes.* Cambridge University Press. https://doi.org/10.1017/cbo9780511621123

Lakens, D. 2017. "Equivalence tests: A practical primer for t tests, correlations, and meta-analyses." *Social Psychological*

and Personality Science 8 (4): 355–362. DOI 10.1177/1948550617697177.

Lally, P., C. H. M. van Jaarsveld, H. W. W. Potts, and J. Wardle. 2010. "How are habits formed: Modelling habit formation in the real world." *European Journal of Social Psychology* 40 (6): 998–1009. DOI 10.1002/ejsp.674.

Lancaster, Madeline A., Magdalena Renner, Carol-Anne Martin, Daniel Wenzel, Louise S. Bicknell, Matthew E. Hurles, Tessa Homfray, Josef M. Penninger, Andrew P. Jackson, and Juergen A. Knoblich. 2013. "Cerebral organoids model human brain development and microcephaly." *Nature* 501: 373–379. DOI 10.1038/nature12517.

Lange, M. 2009. *Laws and Lawmakers: Science, Metaphysics, and the Laws of Nature.* Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195328134.001.0001

Larsen, J. T., A. P. McGraw, and J. T. Cacioppo. 2001. "Can people feel happy and sad at the same time?" *Journal of Personality and Social Psychology* 81 (4): 684–696. DOI 10.1037/0022-3514.81.4.684.

Lawlor, Peter G., Bruno Gagnon, Isabelle L. Mancini, Jose L. Pereira, John Hanson, Maria E. Suarez-Almazor, and Eduardo D. Bruera. 2000. "Occurrence, causes, and outcome of delirium in patients with advanced cancer: A prospective study." *Archives of Internal Medicine* 160 (6): 786–794. DOI 10.1001/archinte.160.6.786.

Lee, J., X. Chen, and R. A. Nicoll. 2022. "Synaptic memory survives molecular turnover." *Proceedings of the National Academy of Sciences* 119 (42): e2211572119. DOI 10.1073/pnas.2211572119.

Lee, Soo Ahn, Jae-Joong Lee, Jisoo Han, Myunghwan Choi, Tor D. Wager, and Choong-Wan Woo. 2024. "Brain representations of affective valence and intensity in sustained pleasure and pain." *Proceedings of the National Academy of Sciences* 121 (25): e2310433121. DOI 10.1073/pnas.2310433121.

Leknes, S., J. C. W. Brooks, K. Wiech, and I. Tracey. 2008. "Pain relief as an opponent process: A psychophysical investigation." *European Journal of Neuroscience* 28 (4): 794 –801. https://doi.org/10.1111/j.1460-9568.2008.06380.x

Leknes, S., M. Lee, C. Berna, J. Andersson, and I. Tracey. 2011. "Relief as a Reward: Hedonic and Neural Responses to Safety from Pain." *PLoS ONE* 6 (4): e17870. DOI 10.1371/journal.pone.0017870.

Lerner, M. J. 1980. *The Belief in a Just World: A Fundamental Delusion.* Plenum Press. DOI 10.1007/978-1-4899-0448-5.

Levenson, Jon D. 1994. *Creation and the Persistence of Evil: The Jewish Drama of Divine Omnipotence.* Princeton University Press.

Levin, M. 2021. "Life, death, and self: Fundamental questions of primitive cognition viewed through the lens of body plasticity and synthetic organisms." *Biochemical and*

Biophysical Research Communications 564: 114–133. DOI 10.1016/j.bbrc.2020.10.077.

Levinas, Emmanuel. 1969. *Totality and Infinity: An Essay on Exteriority.* Translated by Alphonso Lingis. Duquesne University Press.

Levins, Richard. 1966. "The strategy of model building in population biology." *American Scientist* 54 (4): 421–431.

Lewis, D. 1973. *Counterfactuals.* Harvard University Press.

Léonard, C. 2014. "A survey of the Schrödinger problem and some of its connections with optimal transport." *Discrete and Continuous Dynamical Systems A* 34 (4): 1533–1574. DOI 10.3934/dcds.2014.34.1533.

Lillie, Elizabeth O., Bradley Patay, Joel Diamant, Brian Issell, Eric J. Topol, and Nicholas J. Schork. 2011. "The n-of-1 clinical trial: The ultimate strategy for individualizing medicine?" *Personalized Medicine* 8 (2): 161–173. DOI 10.2217/pme.11.7.

Lim, Julian, and David F. Dinges. 2010. "A meta-analysis of the impact of short-term sleep deprivation on cognitive variables." *Psychological Bulletin* 136 (3): 375–389. DOI 10.1037/a0018883.

Lim, Juyun, Alison Wood, and Barry G. Green. 2009. "Derivation and Evaluation of a Labeled Hedonic Scale." *Chemical Senses* 34 (9): 739–751. DOI 10.1093/chemse/bjp054.

Lindqvist, Erik, Robert Östling, and David Cesarini. 2020. "Long-Run Effects of Lottery Wealth on Psychological Well-Being." *The Review of Economic Studies* 87 (6): 2703–2726. DOI 10.1093/restud/rdaa006.

List, Christian. 2019. *Why Free Will Is Real.* Harvard University Press.

Little, R. J. A. and D. B. Rubin. 2019. *Statistical Analysis with Missing Data (3rd ed.).* Wiley. DOI 10.1002/9781119482260.

Livingston, James D., Teresa Milne, Mei Lan Fang, and Erica Amari. 2012. "The effectiveness of interventions for reducing stigma related to substance use disorders: A systematic review." *Addiction* 107 (1): 39–50. DOI 10.1111/j.1360-0443.2011.03601.x.

Locke, Edwin A., and Gary P. Latham. 2002. "Building a practically useful theory of goal setting and task motivation: A 35-year odyssey." *American Psychologist* 57 (9): 705–717. DOI 10.1037/0003-066X.57.9.705.

Longino, Helen E. 1990. *Science as Social Knowledge.* Princeton University Press.

Lucas, R. E. 2007. "Adaptation and the set-point model of subjective well-being: Does happiness change after major life events?" *Current Directions in Psychological Science* 16 (2): 75–79. DOI 10.1111/j.1467-8721.2007.00479.x.

Luce, R. Duncan, and John W. Tukey. 1964. "Simultaneous conjoint measurement: A new type of fundamental measurement." *Journal of Mathematical Psychology* 1 (1): 1–27. DOI 10.1016/0022-2496(64)90015-X.

Luhmann, M., W. Hofmann, M. Eid, and R. E. Lucas. 2012. "Subjective well-being and adaptation to life events: A meta-analysis." *Journal of Personality and Social Psychology* 102 (3): 592–615. DOI 10.1037/a0025948.

Lyon, P. 2015. "The cognitive cell: Bacterial behavior reconsidered." *Frontiers in Microbiology* 6: 264. DOI 10.3389/fmicb.2015.00264.

Mahmood, Saba. 2005. *Politics of Piety: The Islamic Revival and the Feminist Subject.* Princeton University Press.

Maltoni, M., E. Scarpi, and O. Nanni. 2013. "Palliative sedation in end-of-life care." *Current Opinion in Oncology* 25 (4): 360–367. DOI 10.1097/CCO.0b013e3283622c47.

Manski, C. F. 2003. *Partial identification of probability distributions.* Springer. DOI 10.1007/b97478.

Manzi, Oscar, Kate E. Lynch, Daniel M. Allman, Tanya Latty, and Thomas E. White. 2026. "Flexible self-protection as evidence of pain-like states in house crickets." *Proceedings of the Royal Society B: Biological Sciences* 293 (2070): 20260609. DOI 10.1098/rspb.2026.0609.

Marek, Scott, Brenden Tervo-Clemmens, Finnegan J. Calabro, David F. Montez, Benjamin P. Kay, Alexander S. Hatoum, Meghan Rose Donohue, William Foran, Ryland L. Miller, Timothy J. Hendrickson, Stephen M. Malone, Sridhar Kandala, Eric Feczko, Oscar Miranda-Dominguez, Alice M. Graham, Eric A. Earl, Anders J. Perrone, Michaela Cordova, Olivia Doyle, Lucille A. Moore, Gregory M. Conan, Johnny Uriarte, Kathy

Snider, Benjamin J. Lynch, James C. Wilgenbusch, Thomas Pengo, Angela Tam, Jianzhong Chen, Dillan J. Newbold, Annie Zheng, Nicole A. Seider, Andrew N. Van, Athanasia Metoki, Roselyne J. Chauvin, Timothy O. Laumann, Deanna J. Greene, Steven E. Petersen, Hugh Garavan, Wesley K. Thompson, Thomas E. Nichols, B. T. Thomas Yeo, Deanna M. Barch, Beatriz Luna, Damien A. Fair, and Nico U. F. Dosenbach. 2022. "Reproducible brain-wide association studies require thousands of individuals." *Nature* 603: 654–660. DOI 10.1038/s41586-022-04492-9.

Marmot, M., S. Friel, R. Bell, T. A. J. Houweling, and S. Taylor. 2008. "Closing the Gap in a Generation: Health Equity through Action on the Social Determinants of Health." *The Lancet* 372 (9650): 1661–1669. DOI 10.1016/S0140-6736(08)61690-6.

Martinez, Mayte, Katie Hall, Laurent Prétôt, Susan P. Lambeth, Steven J. Schapiro, and Sarah F. Brosnan. 2026. "Chimpanzees' responses to inequity in a group context are influenced by food quality and their relationships with the group members present." *Proceedings of the Royal Society B: Biological Sciences* 293 (2071): 20260397. DOI 10.1098/rspb.2026.0397.

Mashour, G. A. and A. G. Hudetz. 2018. "Neural correlates of unconsciousness in large-scale brain networks." *Trends in Neurosciences* 41 (3): 150–160. DOI 10.1016/j.tins.2018.01.003.

Maslach, Christina, and Michael P. Leiter. 2016. "Understanding the Burnout Experience: Recent Research and Its Implications for Psychiatry." *World Psychiatry* 15 (2): 103 –111. DOI 10.1002/wps.20311.

Masson, J.-B., G. Voisinne, J. Wong-Ng, A. Celani, and M. Vergassola. 2012. "Noninvasive inference of the molecular chemotactic response using bacterial trajectories." *Proceedings of the National Academy of Sciences* 109 (5): 1802–1807. DOI 10.1073/pnas.1116772109.

Mather, M. and L. L. Carstensen. 2005. "Aging and motivated cognition: The positivity effect in attention and memory." *Trends in Cognitive Sciences* 9 (10): 496–502. DOI 10.1016/j.tics.2005.08.005.

Maudlin, T. 2007. *The Metaphysics Within Physics.* Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199218219.001.0001

Mauss, I. B. and M. D. Robinson. 2009. "Measures of emotion: A review." *Cognition and Emotion* 23 (2): 209–237. DOI 10.1080/02699930802204677.

Maynard Smith, J. and E. Szathmáry. 1995. *The major transitions in evolution.* W. H. Freeman / Spektrum.

Maynard Smith, John. 1982. *Evolution and the Theory of Games.* Cambridge University Press. https://doi.org/10.1017/cbo9780511806292

Mayo, D. G. 2018. *Statistical Inference as Severe Testing: How to Get Beyond the Statistics Wars.* Cambridge University Press. https://doi.org/10.1017/9781107286184

Mayr, E. 1961. "Cause and effect in biology." *Science* 134 (3489): 1501–1506. DOI 10.1126/science.134.3489.1501.

McCambridge, J., J. Witton, and D. R. Elbourne. 2014. "Systematic review of the Hawthorne effect: New concepts are needed to study research participation effects." *Journal of Clinical Epidemiology* 67 (3): 267–277. DOI 10.1016/j.jclinepi.2013.08.015.

McEwen, B. S. 1998. "Protective and damaging effects of stress mediators." *New England Journal of Medicine* 338 (3): 171–179. DOI 10.1056/NEJM199801153380307.

McEwen, B. S. and E. Stellar. 1993. "Stress and the individual: Mechanisms leading to disease." *Archives of Internal Medicine* 153 (18): 2093–2101. https://doi.org/10.1001/archinte.1993.00410180039004

McLellan, A. Thomas, David C. Lewis, Charles P. O'Brien, and Herbert D. Kleber. 2000. "Drug dependence, a chronic medical illness: Implications for treatment, insurance, and outcomes evaluation." *JAMA* 284 (13): 1689–1695. DOI 10.1001/jama.284.13.1689.

Meehl, Paul E. 1978. "Theoretical risks and tabular asterisks: Sir Karl, Sir Ronald, and the slow progress of soft psychology." *Journal of Consulting and Clinical Psychology* 46 (4): 806–834. DOI 10.1037/0022-006X.46.4.806.

Mellor, D. J., N. J. Beausoleil, K. E. Littlewood, A. N. McLean, P. D. McGreevy, B. Jones, and C. Wilkins. 2020. "The 2020 Five Domains Model: Including human-animal

interactions in assessments of animal welfare." *Animals* 10 (10): 1870. DOI 10.3390/ani10101870.

Mendl, Michael, Oliver H. P. Burman, and Elizabeth S. Paul. 2010. "An integrative and functional framework for the study of animal emotion and mood." *Proceedings of the Royal Society B: Biological Sciences* 277 (1696): 2895–2904. DOI 10.1098/rspb.2010.0303.

Menon, V. 2011. "Large-scale brain networks and psychopathology: A unifying triple network model." *Trends in Cognitive Sciences* 15 (10): 483–506. DOI 10.1016/j.tics.2011.08.003.

Meredith, W. 1993. "Measurement invariance, factor analysis and factorial invariance." *Psychometrika* 58 (4): 525–543. DOI 10.1007/BF02294825.

Merton, Robert K. 1973. *The Sociology of Science: Theoretical and Empirical Investigations.* University of Chicago Press.

Messenger, Daniel A., and David M. Bortz. 2021. "Weak SINDy: Galerkin-based data-driven model selection." *Multiscale Modeling & Simulation* 19 (3): 1474–1497. DOI 10.1137/20M1343166.

Messick, Samuel. 1995. "Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning." *American Psychologist* 50 (9): 741–749. DOI 10.1037/0003-066X.50.9.741.

Metzinger, Thomas. 2009. *The Ego Tunnel.* Basic Books.

Metzinger, Thomas. 2021. "Artificial suffering: An argument for a global moratorium on synthetic phenomenology." *Journal of Artificial Intelligence and Consciousness* 8 (1): 43–66. DOI 10.1142/S270507852150003X.

Michell, J. 1999. *Measurement in Psychology: A Critical History of a Methodological Concept.* Cambridge University Press. https://doi.org/10.1017/cbo9780511490040

Millsap, R. E. 2011. *Statistical Approaches to Measurement Invariance.* Routledge.

Mitchell, M., S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru. 2019. "Model Cards for Model Reporting." *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 220–229. DOI 10.1145/3287560.3287596.

Modrák, Martin, Angie H. Moon, Shinyoung Kim, Paul-Christian Bürkner, Niko Huurre, Kateřina Faltejsková, Andrew Gelman, and Aki Vehtari. 2025. "Simulation-based calibration checking for Bayesian computation: The choice of test quantities shapes sensitivity." *Bayesian Analysis* 20 (2): 461–488. DOI 10.1214/23-BA1404.

Mokkink, L. B., C. B. Terwee, D. L. Patrick, J. Alonso, P. W. Stratford, D. L. Knol, L. M. Bouter, and H. C. W. de Vet. 2010. "The COSMIN checklist for assessing the methodological quality of studies on measurement properties of health status measurement instruments: an international Delphi study." *Quality of Life Research* 19 (4): 539–549. DOI 10.1007/s11136-010-9606-8.

Molenaar, P. C. M. 2004. “A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever.” *Measurement* 2 (4): 201–218. DOI 10.1207/S15366359MEA0204_1.

Monod, J. 1971. *Chance and Necessity: An Essay on the Natural Philosophy of Modern Biology.* Alfred A. Knopf.

Montanari, Arthur N., Chao Duan, Luis A. Aguirre, and Adilson E. Motter. 2022. “Functional observability and target state estimation in large-scale networks.” *Proceedings of the National Academy of Sciences* 119 (1): e2113750119. DOI 10.1073/pnas.2113750119.

Monti, Martin M., Audrey Vanhaudenhuyse, Martin R. Coleman, Melanie Boly, John D. Pickard, Luaba Tshibanda, Adrian M. Owen, and Steven Laureys. 2010. “Willful modulation of brain activity in disorders of consciousness.” *New England Journal of Medicine* 362 (7): 579–589. DOI 10.1056/NEJMoa0905370.

Mullainathan, S. and E. Shafir. 2013. *Scarcity: Why Having Too Little Means So Much.* Times Books.

Munafò, Marcus R., Brian A. Nosek, Dorothy V. M. Bishop, Katherine S. Button, Christopher D. Chambers, Nathalie Percie du Sert, Uri Simonsohn, Eric-Jan Wagenmakers, Jennifer J. Ware, and John P. A. Ioannidis. 2017. “A Manifesto for Reproducible Science.” *Nature Human Behaviour* 1: 0021. DOI 10.1038/s41562-016-0021.

Myung, J. I., and M. A. Pitt. 2009. "Optimal experimental design for model discrimination." *Psychological Review* 116 (3): 499–518. DOI 10.1037/a0016104.

Nagel, T. 1970. "Death." *Noûs* 4 (1): 73–80. DOI 10.2307/2214297.

Nagel, T. 1979. "Moral luck." In *Mortal Questions*, 24–38. Cambridge University Press.

Nagel, Thomas. 1974. "What is it like to be a bat?" *Philosophical Review* 83 (4): 435–450. https://doi.org/10.2307/2183914

Narayanan, A. and V. Shmatikov. 2008. "Robust de-anonymization of large sparse datasets." *2008 IEEE Symposium on Security and Privacy*, 111–125. DOI 10.1109/SP.2008.33.

Narens, L. 1981. "On the scales of measurement." *Journal of Mathematical Psychology* 24 (3): 249–275. DOI 10.1016/0022-2496(81)90045-6.

Nass, C., J. Steuer, and E. R. Tauber. 1994. "Computers are social actors." *Proceedings of CHI '94*, 72–78. DOI 10.1145/191666.191703.

National Academies of Sciences, Engineering, and Medicine. 2017. *Communicating Science Effectively: A Research Agenda*. Washington, DC: National Academies Press. DOI 10.17226/23674.

National Academies of Sciences, Engineering, and Medicine. 2019. *Reproducibility and Replicability in Science*.

Washington, DC: National Academies Press. DOI 10.17226/25303.

National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. 1979. *The Belmont Report.* U.S. Department of Health, Education, and Welfare.

National Institute for Health and Care Excellence. 2015. *Care of dying adults in the last days of life.* NICE Guideline NG31.

National Institute for Health and Care Excellence. 2022. *Medicines associated with dependence or withdrawal symptoms: safe prescribing and withdrawal management for adults.* NICE Guideline NG215. Published 20 April 2022. https://www.nice.org.uk/guidance/ng215.

National Institute of Standards and Technology. 2020. *NIST Privacy Framework: A Tool for Improving Privacy Through Enterprise Risk Management, Version 1.0*. NIST CSWP 01162020. Gaithersburg, MD: National Institute of Standards and Technology. DOI 10.6028/NIST.CSWP.01162020.

National Institute of Standards and Technology. 2023. *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1. Gaithersburg, MD: National Institute of Standards and Technology. Published 26 January 2023. DOI 10.6028/NIST.AI.100-1.

National Institute of Standards and Technology. 2024. *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. NIST AI 600-1.

Gaithersburg, MD: National Institute of Standards and Technology. DOI 10.6028/NIST.AI.600-1.

National Research Council. 2012. *Assessing the Reliability of Complex Models: Mathematical and Statistical Foundations of Verification, Validation, and Uncertainty Quantification.* National Academies Press. DOI 10.17226/13395.

National Research Council. 2013. *Subjective Well-Being: Measuring Happiness, Suffering, and Other Dimensions of Experience.* National Academies Press.

Nebeker, C., J. Harlow, R. Espinoza Giacinto, R. Orozco-Linares, C. S. Bloss, and N. Weibel. 2017. "Ethical and regulatory challenges of research using pervasive sensing and other emerging technologies: IRB perspectives." *AJOB Empirical Bioethics* 8 (4): 266–276. DOI 10.1080/23294515.2017.1403980.

Neimeyer, Robert A., ed. 2001. *Meaning Reconstruction & the Experience of Loss.* American Psychological Association.

Nesse, R. M. and G. C. Williams. 1994. *Why we get sick: The new science of Darwinian medicine.* Times Books.

Nielsen, T. and R. Levin. 2007. "Nightmares: A new neurocognitive model." *Sleep Medicine Reviews* 11 (4): 295–310. DOI 10.1016/j.smrv.2007.03.004.

Nir, Y. and G. Tononi. 2010. "Dreaming and the brain: From phenomenology to neurophysiology." *Trends in Cognitive Sciences* 14 (2): 88–100. DOI 10.1016/j.tics.2009.12.001.

Nissenbaum, H. 2004. “Privacy as contextual integrity.” *Washington Law Review* 79: 119–157.

Nissenbaum, Helen. 2010. *Privacy in Context: Technology, Policy, and the Integrity of Social Life*. Stanford, CA: Stanford University Press.

Nolen-Hoeksema, Susan, Blair E. Wisco, and Sonja Lyubomirsky. 2008. “Rethinking rumination.” *Perspectives on Psychological Science* 3 (5): 400–424. DOI 10.1111/j.1745-6924.2008.00088.x.

Nosek, B. A. and D. Lakens. 2014. “Registered Reports: A method to increase the credibility of published results.” *Social Psychology* 45 (3): 137–141. DOI 10.1027/1864-9335/a000192.

Nosek, Brian A., Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. 2018. “The preregistration revolution.” *Proceedings of the National Academy of Sciences* 115 (11): 2600–2606. DOI 10.1073/pnas.1708274114.

Nuffield Council on Bioethics. 2026. *Neural organoids: Ethical and governance considerations.* Nuffield Council on Bioethics.

Nussbaum, M. C. 2006. *Frontiers of Justice: Disability, Nationality, Species Membership.* Belknap Press of Harvard University Press.

Nussbaum, M. C. 2011. *Creating Capabilities: The Human Development Approach*. Cambridge, MA: Belknap Press of Harvard University Press.

Oberkampf, William L., and Christopher J. Roy. 2010. *Verification and Validation in Scientific Computing.* Cambridge University Press.

Obermeyer, Ziad, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. "Dissecting racial bias in an algorithm used to manage the health of populations." *Science* 366 (6464): 447–453. DOI 10.1126/science.aax2342.

Obeyesekere, G. 2002. *Imagining karma: Ethical transformation in Amerindian, Buddhist, and Greek rebirth.* University of California Press.

OECD. 2025. *OECD Guidelines on Measuring Subjective Well-being (2025 Update).* OECD Publishing. DOI 10.1787/9203632a-en.

Oettingen, Gabriele. 2012. "Future thought and behaviour change." *European Review of Social Psychology* 23 (1): 1–63. DOI 10.1080/10463283.2011.643698.

Ohm, P. 2010. "Broken promises of privacy: Responding to the surprising failure of anonymization." *UCLA Law Review* 57: 1701–1777.

Ojala, Markus, and Gemma C. Garriga. 2010. "Permutation Tests for Studying Classifier Performance." *Journal of Machine Learning Research* 11 (62): 1833–1863. https://www.jmlr.org/papers/v11/ojala10a.html.

Olson, E. T. 2007. *What Are We? A Study in Personal Ontology.* Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195176421.001.0001

Onnela, J.-P. and S. L. Rauch. 2016. "Harnessing smartphone-based digital phenotyping to enhance behavioral and mental health." *Neuropsychopharmacology* 41: 1691–1696. DOI 10.1038/npp.2016.7.

Oort, F. J., M. R. M. Visser, and M. A. G. Sprangers. 2009. "Formal definitions of measurement bias and explanation bias clarify measurement and conceptual perspectives on response shift." *Journal of Clinical Epidemiology* 62 (11): 1126–1137. DOI 10.1016/j.jclinepi.2009.03.013.

Open Science Collaboration. 2015. "Estimating the Reproducibility of Psychological Science." *Science* 349 (6251): aac4716. DOI 10.1126/science.aac4716.

Oreskes, Naomi, Kristin Shrader-Frechette, and Kenneth Belitz. 1994. "Verification, validation, and confirmation of numerical models in the earth sciences." *Science* 263 (5147): 641–646. DOI 10.1126/science.263.5147.641.

Oreskes, Naomi. 2019. *Why Trust Science?* Princeton University Press.

Orsi, Robert A. 2005. *Between Heaven and Earth: The Religious Worlds People Make and the Scholars Who Study Them.* Princeton University Press.

Osmond, C., and M. J. Gardner. 1989. "Age, period, and cohort models. Non-overlapping cohorts don't resolve the identification problem." *American Journal of Epidemiology* 129 (1): 31–35. DOI 10.1093/oxfordjournals.aje.a115121.

Ostrom, E. 1990. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge: Cambridge University Press.

Owen, Adrian M., Martin R. Coleman, Melanie Boly, Matthew H. Davis, Steven Laureys, and John D. Pickard. 2006. "Detecting awareness in the vegetative state." *Science* 313 (5792): 1402. DOI 10.1126/science.1130197.

Öhman, Carl, and Luciano Floridi. 2018. "An ethical framework for the digital afterlife industry." *Nature Human Behaviour* 2: 318–320. DOI 10.1038/s41562-018-0335-2.

O'Neil, Cathy. 2016. *Weapons of Math Destruction.* Crown.

O'Neill, Onora. 2002. *Autonomy and Trust in Bioethics.* Cambridge University Press.

O'Neill, Onora. 2018. *From Principles to Practice: Normativity and Judgement in Ethics and Politics.* Cambridge University Press. https://doi.org/10.1017/9781316286708

Palminteri, Stefano, Valentin Wyart, and Etienne Koechlin. 2017. "The importance of falsification in computational cognitive modeling." *Trends in Cognitive Sciences* 21 (6): 425–433. DOI 10.1016/j.tics.2017.03.011.

Parfit, D. 1984. *Reasons and Persons.* Oxford University Press.

Pargament, Kenneth I. 1997. *The Psychology of Religion and Coping: Theory, Research, Practice.* Guilford Press.

Park, C. L. 2010. "Making sense of the meaning literature: An integrative review of meaning making and its effects on

adjustment to stressful life events." *Psychological Bulletin* 136 (2): 257–301. DOI 10.1037/a0018301.

Pașca, S. P., P. Arlotta, P. Campbell, A. Charo, J. H. Evans, N. Farahany, F. H. Gage, P. Godfrey-Smith, I. Hyun, A. R. Kriegstein, L. Melloni, G.-L. Ming, J. D. Moreno, J. Sugarman, S. Temple, G. Testa, and H. T. Greely. 2025. "The need for a global effort to attend to human neural organoid and assembloid research." *Science* 390 (6773): 574–577. DOI 10.1126/science.aeb1510.

Patterson, A., S. Neumann, M. White, and A. White. 2024. "Empirical design in reinforcement learning." *Journal of Machine Learning Research* 25 (318): 1–63.

Pearl, J. 2009. *Causality (2nd ed.)*. Cambridge University Press.

Pearl, Judea, and Elias Bareinboim. 2014. "External Validity: From Do-Calculus to Transportability across Populations." *Statistical Science* 29 (4): 579–595. DOI 10.1214/14-STS486.

Pennartz, C. M. A. 2026. "How can we validate theory-derived indicators of consciousness in artificial intelligence?" *Trends in Cognitive Sciences* 30 (7): 573–574. DOI 10.1016/j.tics.2026.01.011.

Peters, Jonas, Peter Bühlmann, and Nicolai Meinshausen. 2016. "Causal inference by using invariant prediction: identification and confidence intervals." *Journal of the Royal Statistical Society Series B: Statistical Methodology* 78 (5): 947–1012. DOI 10.1111/rssb.12167.

Pezzulo, Giovanni, Francesco Rigoli, and Karl J. Friston. 2015. "Active inference, homeostatic regulation and adaptive

behavioural control." *Progress in Neurobiology* 134: 17–35. DOI 10.1016/j.pneurobio.2015.09.001.

Pigliucci, Massimo, and Gerd B. Müller, eds. 2010. *Evolution: The Extended Synthesis.* MIT Press.

Pinto, Yair, David A. Neville, Marte Otten, Paul M. Corballis, Victor A. F. Lamme, Edward H. F de Haan, Nicoletta Foschi, and Mara Fabri. 2017. "Split brain: Divided perception but undivided consciousness." *Brain* 140 (5): 1231–1237. DOI 10.1093/brain/aww358.

Plantinga, Alvin. 1974. *God, Freedom, and Evil.* Harper & Row.

Poldrack, R. A. 2006. "Can cognitive processes be inferred from neuroimaging data?" *Trends in Cognitive Sciences* 10 (2): 59–63. DOI 10.1016/j.tics.2005.12.004.

Popper, Karl R. 1959. *The Logic of Scientific Discovery.* Hutchinson.

Prescott, John. 2009. "Rating a New Hedonic Scale: A Commentary on 'Derivation and Evaluation of a Labeled Hedonic Scale' by Lim, Wood and Green." *Chemical Senses* 34 (9): 735–737. DOI 10.1093/chemse/bjp072.

Protter, P. E. 2005. *Stochastic integration and differential equations (2nd ed.).* Springer. DOI 10.1007/978-3-662-10061-5.

Putnick, D. L. and M. H. Bornstein. 2016. "Measurement invariance conventions and reporting: The state of the art and

future directions for psychological research." *Developmental Review* 41: 71–90. DOI 10.1016/j.dr.2016.06.004.

Quine, W. V. O. 1951. "Two dogmas of empiricism." *Philosophical Review* 60 (1): 20–43. DOI 10.2307/2181906.

Raby, C. R., D. M. Alexis, A. Dickinson, and N. S. Clayton. 2007. "Planning for the future by western scrub-jays." *Nature* 445 (7130): 919–921. DOI 10.1038/nature05575.

Radovanović, Kia, Anoek Lorskens, Sebastian Schütte, Juliane Bräuer, Josep Call, Daniel B. M. Haun, and Edwin J. C. van Leeuwen. 2025. "Bonobos respond aversively to unequal reward distributions." *Proceedings of the Royal Society B: Biological Sciences* 292 (2045): 20242873. DOI 10.1098/rspb.2024.2873.

Rahman, Fazlur. 1980. *Major Themes of the Qur'an.* Bibliotheca Islamica.

Railsback, Steven F., and Volker Grimm. 2019. *Agent-Based and Individual-Based Modeling: A Practical Introduction.* 2nd ed. Princeton University Press.

Raja, S. N., D. B. Carr, M. Cohen, N. B. Finnerup, H. Flor, S. Gibson, F. J. Keefe, J. S. Mogil, M. Ringkamp, K. A. Sluka, X. J. Song, B. Stevens, M. D. Sullivan, P. R. Tutelman, T. Ushida, and K. Vader. 2020. "The revised International Association for the Study of Pain definition of pain: Concepts, challenges, and compromises." *Pain* 161 (9): 1976–1982. DOI 10.1097/j.pain.0000000000001939.

Raji, Inioluwa Deborah, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson,

Jamila Smith-Loud, Daniel Theron, and Parker Barnes. 2020. "Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing." *Proceedings of FAT* 2020*, 33–44. DOI 10.1145/3351095.3372873.

Rambachan, Anantanand. 2006. *The Advaita Worldview: God, World, and Humanity.* State University of New York Press.

Ramsay, D. S. and S. C. Woods. 2014. "Clarifying the roles of homeostasis and allostasis in physiological regulation." *Psychological Review* 121 (2): 225–247. DOI 10.1037/a0035942.

Rankin, C. H., et al. 2009. "Habituation revisited: An updated and revised description of the behavioral characteristics of habituation." *Neurobiology of Learning and Memory* 92 (2): 135–138. DOI 10.1016/j.nlm.2008.09.012.

Rapkin, B. D. and C. E. Schwartz. 2004. "Toward a theoretical model of quality-of-life appraisal: Implications of findings from studies of response shift." *Health and Quality of Life Outcomes* 2: 14. DOI 10.1186/1477-7525-2-14.

Rawls, J. 1999. *A Theory of Justice: Revised Edition.* Belknap Press of Harvard University Press.

Read, J. C. A., and V. Nityananda. 2026a. "Motivational trade-offs as evidence for sentience in bees: a critique." *Animal Behaviour* 231: 123389. DOI 10.1016/j.anbehav.2025.123389.

Read, J., and V. Nityananda. 2026b. "Revised models still do not show evidence for motivational trade-offs modulating

nociception in bees." *Animal Behaviour* 233: 123457. DOI 10.1016/j.anbehav.2025.123457.

Reason, James. 2000. "Human error: Models and management." *BMJ* 320: 768–770. DOI 10.1136/bmj.320.7237.768.

Reeves, Byron, and Clifford Nass. 1996. *The Media Equation.* Cambridge University Press.

Reinders, A. A. T. S., A. T. M. Willemsen, H. P. J. Vos, J. A. den Boer, and E. R. S. Nijenhuis. 2012. "Fact or factitious? A psychobiological study of authentic and simulated dissociative identity states." *PLoS ONE* 7 (6): e39279.

Revonsuo, Antti. 2000. "The reinterpretation of dreams: An evolutionary hypothesis of the function of dreaming." *Behavioral and Brain Sciences* 23 (6): 877–901. DOI 10.1017/S0140525X00004015.

Revuz, D. and M. Yor. 1999. *Continuous martingales and Brownian motion (3rd ed.).* Springer.

Ritov, Oded, Christoph J. Völter, Nichola J. Raihani, and Jan M. Engelmann. 2024. "No evidence for inequity aversion in non-human animals: A meta-analysis of accept/reject paradigms." *Proceedings of the Royal Society B: Biological Sciences* 291 (2035): 20241452. DOI 10.1098/rspb.2024.1452.

Robinson, T. E. and K. C. Berridge. 2025. "The incentive-sensitization theory of addiction 30 years on." *Annual Review of Psychology* 76: 29–58. DOI 10.1146/annurev-psych-011624-024031.

Robinson, Terry E., and Kent C. Berridge. 1993. "The neural basis of drug craving: An incentive-sensitization theory of addiction." *Brain Research Reviews* 18 (3): 247–291. DOI 10.1016/0165-0173(93)90013-P.

Robinson, W. S. 1950. "Ecological correlations and the behavior of individuals." *American Sociological Review* 15 (3): 351–357. DOI 10.2307/2087176.

Rossi, Simone, Andrea Antal, Sven Bestmann, Marom Bikson, Carmen Brewer, Jürgen Brockmöller, Linda L. Carpenter, Massimo Cincotta, Robert Chen, Jeff D. Daskalakis, Vincenzo Di Lazzaro, Michael D. Fox, Mark S. George, Donald Gilbert, Vasilios K. Kimiskidis, Giacomo Koch, Risto J. Ilmoniemi, Jean Pascal Lefaucheur, Letizia Leocani, Sarah H. Lisanby, Carlo Miniussi, Frank Padberg, Alvaro Pascual-Leone, Walter Paulus, Angel V. Peterchev, Angelo Quartarone, Alexander Rotenberg, John Rothwell, Paolo M. Rossini, Emiliano Santarnecchi, Mouhsin M. Shafi, Hartwig R. Siebner, Yoshikatzu Ugawa, Eric M. Wassermann, Abraham Zangen, Ulf Ziemann, and Mark Hallett. 2021. "Safety and recommendations for TMS use in healthy subjects and patient populations, with updates on training, ethical and regulatory issues: Expert guidelines." *Clinical Neurophysiology* 132 (1): 269–306. DOI 10.1016/j.clinph.2020.10.003.

Rubin, D. B. 1976. "Inference and missing data." *Biometrika* 63: 581–592. DOI 10.1093/biomet/63.3.581.

Russell, J. A. 2003. "Core affect and the psychological construction of emotion." *Psychological Review* 110 (1): 145–172. DOI 10.1037/0033-295X.110.1.145.

Russell, J. A. and J. M. Carroll. 1999. "On the bipolarity of positive and negative affect." *Psychological Bulletin* 125 (1): 3–30. DOI 10.1037/0033-2909.125.1.3.

Rutledge, R. B., N. Skandali, P. Dayan, and R. J. Dolan. 2014. "A computational and neural model of momentary subjective well-being." *Proceedings of the National Academy of Sciences of the United States of America* 111 (33): 12252–12257. DOI 10.1073/pnas.1407535111.

Ryan, Richard M., and Edward L. Deci. 2001. "On happiness and human potentials: A review of research on hedonic and eudaimonic well-being." *Annual Review of Psychology* 52: 141–166. DOI 10.1146/annurev.psych.52.1.141.

Ryan, Richard M., and Edward L. Deci. 2017. *Self-Determination Theory.* Guilford Press.

Saltelli, Andrea, Gabriele Bammer, Isabelle Bruno, Erica Charters, Monica Di Fiore, Emmanuel Didier, Wendy Nelson Espeland, John Kay, Samuele Lo Piano, Deborah Mayo, Roger Pielke Jr, Tommaso Portaluri, Theodore M. Porter, Arnald Puy, Ismael Rafols, Jerome R. Ravetz, Erik Reinert, Daniel Sarewitz, Philip B. Stark, Andrew Stirling, Jeroen van der Sluijs, and Paolo Vineis. 2020. "Five ways to ensure that models serve society: A manifesto." *Nature* 582: 482–484. DOI 10.1038/d41586-020-01812-9.

Sanders, R. D., G. Tononi, S. Laureys, and J. W. Sleigh. 2012. "Unresponsiveness ≠ unconsciousness." *Anesthesiology* 116 (4): 946–959. DOI 10.1097/ALN.0b013e318249d0a7.

Säilynoja, Teemu, Marvin Schmitt, Paul-Christian Bürkner, and Aki Vehtari. 2026. "Posterior SBC: Simulation-based calibration checking conditional on data." *Statistics and Computing* 36 (2): Article 78, 1–20. DOI 10.1007/s11222-026-10825-9.

Schechter, E. and T. Bayne. 2021. "Consciousness after split-brain surgery: The recent challenge to the classical picture." *Neuropsychologia* 160: 107987. DOI 10.1016/j.neuropsychologia.2021.107987.

Schechtman, M. 1996. *The Constitution of Selves.* Cornell University Press.

Scheufele, D. A. 2014. "Science communication as political communication." *Proceedings of the National Academy of Sciences* 111 (Suppl. 4): 13585–13592. DOI 10.1073/pnas.1317516111.

Schnakers, Caroline, Audrey Vanhaudenhuyse, Joseph Giacino, Manfredi Ventura, Melanie Boly, Steve Majerus, Gustave Moonen, and Steven Laureys. 2009. "Diagnostic accuracy of the vegetative and minimally conscious state: Clinical consensus versus standardized neurobehavioral assessment." *BMC Neurology* 9: 35. DOI 10.1186/1471-2377-9-35.

Schnell, Tatjana. 2021. *The Psychology of Meaning in Life.* Routledge.

Schredl, M. and F. Hofmann. 2003. "Continuity between waking activities and dream activities." *Consciousness and Cognition* 12 (2): 298–308. DOI 10.1016/S1053-8100(02)00072-7.

Schuirmann, D. J. 1987. "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability." *Journal of Pharmacokinetics and Biopharmaceutics* 15 (6): 657–680. DOI 10.1007/BF01068419.

Schultz, W., P. Dayan, and P. R. Montague. 1997. "A neural substrate of prediction and reward." *Science* 275 (5306): 1593 –1599. https://doi.org/10.1126/science.275.5306.1593

Schulz, K. F., D. G. Altman, and D. Moher. 2010. "CONSORT 2010 statement: Updated guidelines for reporting parallel group randomised trials." *BMJ* 340: c332.

Schwartz, C. E. and M. A. G. Sprangers. 1999. "Methodological approaches for assessing response shift in longitudinal health-related quality-of-life research." *Social Science & Medicine* 48 (11): 1531–1548. DOI 10.1016/S0277-9536(99)00047-7.

Scott, James C. 1998. *Seeing Like a State.* Yale University Press.

Searle, John R. 1980. "Minds, brains, and programs." *Behavioral and Brain Sciences* 3 (3): 417–424. DOI 10.1017/S0140525X00005756.

Seeley, W. W., V. Menon, A. F. Schatzberg, J. Keller, G. H. Glover, H. Kenna, A. L. Reiss, and M. D. Greicius. 2007.

"Dissociable intrinsic connectivity networks for salience processing and executive control." *Journal of Neuroscience* 27 (9): 2349–2356. DOI 10.1523/JNEUROSCI.5587-06.2007.

Selbst, Andrew D., danah boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. "Fairness and Abstraction in Sociotechnical Systems." *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 59–68. DOI 10.1145/3287560.3287598.

Sen, A. 2009. *The Idea of Justice.* Harvard University Press.

Sen, Amartya. 1999. *Development as Freedom*. New York: Alfred A. Knopf.

Seow, Hsien, Lisa Barbera, Rinku Sutradhar, Doris Howell, Deborah Dudgeon, Clare Atzema, Ying Liu, Amna Husain, Jonathan Sussman, and Craig Earle. 2011. "Trajectory of performance status and symptom scores for patients with cancer during the last six months of life." *Journal of Clinical Oncology* 29 (9): 1151–1158. DOI 10.1200/JCO.2010.30.7173.

Seth, A. K. 2013. "Interoceptive inference, emotion, and the embodied self." *Trends in Cognitive Sciences* 17 (11): 565–573. https://doi.org/10.1016/j.tics.2013.09.007

Seth, A. K. 2025. "Conscious artificial intelligence and biological naturalism." Published online 21 April 2025; final version in *Behavioral and Brain Sciences* 49 (2026): e315. DOI 10.1017/S0140525X25000032.

Seth, Anil K., and Tim Bayne. 2022. "Theories of consciousness." *Nature Reviews Neuroscience* 23: 439–452. DOI 10.1038/s41583-022-00587-4.

Shah, A. K., S. Mullainathan, and E. Shafir. 2012. "Some consequences of having too little." *Science* 338 (6107): 682–685. DOI 10.1126/science.1222426.

Sheldon, Kennon M., and Sonja Lyubomirsky. 2012. "The challenge of staying happier: Testing the hedonic adaptation prevention model." *Personality and Social Psychology Bulletin* 38 (5): 670–680. DOI 10.1177/0146167212436400.

Sherman, Lawrence W., and Heather Strang. 2007. *Restorative Justice: The Evidence.* The Smith Institute.

Shiffman, S., A. A. Stone, and M. R. Hufford. 2008. "Ecological momentary assessment." *Annual Review of Clinical Psychology* 4: 1–32. DOI 10.1146/annurev.clinpsy.3.022806.091415.

Shirvalkar, P., J. Prosky, G. Chin, P. Ahmadipour, O. G. Sani, M. Desai, A. Schmitgen, H. Dawes, M. M. Shanechi, P. A. Starr, and E. F. Chang. 2023. "First-in-human prediction of chronic pain state using intracranial neural biomarkers." *Nature Neuroscience* 26: 1090–1099. DOI 10.1038/s41593-023-01338-z.

Shmueli, G. 2010. "To explain or to predict?" *Statistical Science* 25 (3): 289–310. DOI 10.1214/10-STS330.

Shonkoff, J. P., A. S. Garner, Committee on Psychosocial Aspects of Child and Family Health, Committee on Early Childhood, Adoption, and Dependent Care, and Section

on Developmental and Behavioral Pediatrics. 2012. "The lifelong effects of early childhood adversity and toxic stress." *Pediatrics* 129 (1): e232–e246. DOI 10.1542/peds.2011-2663.

Shrout, P. E. and J. L. Fleiss. 1979. "Intraclass correlations: Uses in assessing rater reliability." *Psychological Bulletin* 86 (2): 420–428. DOI 10.1037/0033-2909.86.2.420.

Siclari, Francesca, Benjamin Baird, Lampros Perogamvros, Giulio Bernardi, Joshua J LaRocque, Brady Riedner, Melanie Boly, Bradley R Postle, and Giulio Tononi. 2017. "The neural correlates of dreaming." *Nature Neuroscience* 20 (6): 872–878. DOI 10.1038/nn.4545.

Simmons, J. P., L. D. Nelson, and U. Simonsohn. 2011. "False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant." *Psychological Science* 22 (11): 1359–1366. DOI 10.1177/0956797611417632.

Simonsohn, U., J. P. Simmons, and L. D. Nelson. 2020. "Specification curve analysis." *Nature Human Behaviour* 4 (11): 1208–1214. DOI 10.1038/s41562-020-0912-z.

Singer, J. D. and J. B. Willett. 2003. *Applied Longitudinal Data Analysis: Modeling Change and Event Occurrence.* Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195152968.001.0001

Skalse, J. M. V., M. Farrugia-Roberts, S. Russell, A. Abate, and A. Gleave. 2023. "Invariance in Policy Optimisation and Partial Identifiability in Reward Learning." *Proceedings of the 40th International Conference on Machine Learning,*

Proceedings of Machine Learning Research 202: 32033–32058. PMLR.

Smith, Jesse M. 2017. "Can the secular be the object of belief and belonging? The Sunday Assembly." *Qualitative Sociology* 40 (1): 83–109. DOI 10.1007/s11133-016-9350-7.

Sneddon, Lynne U., Robert W. Elwood, Shelley A. Adamo, and Matthew C. Leach. 2014. "Defining and assessing animal pain." *Animal Behaviour* 97: 201–212. DOI 10.1016/j.anbehav.2014.09.007.

Society of Professional Journalists. 2014. *SPJ Code of Ethics*. Revised 6 September 2014.

Soderstrom, Nicholas C., and Robert A. Bjork. 2015. "Learning versus performance: An integrative review." *Perspectives on Psychological Science* 10 (2): 176–199. https://doi.org/10.1177/1745691615569000

Solomon, R. L. 1980. "The opponent-process theory of acquired motivation: The costs of pleasure and the benefits of pain." *American Psychologist* 35 (8): 691–712. https://doi.org/10.1037/0003-066x.35.8.691

Solomon, R. L. and J. D. Corbit. 1974. "An opponent-process theory of motivation: I. Temporal dynamics of affect." *Psychological Review* 81 (2): 119–145. DOI 10.1037/h0036128.

Solove, D. J. 2006. "A Taxonomy of Privacy." *University of Pennsylvania Law Review* 154 (3): 477–560.

Solvi, C., L. Baciadonna, and L. Chittka. 2016. "Unexpected rewards induce dopamine-dependent positive emotion-like state changes in bumblebees." *Science* 353 (6307): 1529–1531. DOI 10.1126/science.aaf4454.

Sottinen, Tommi, and Adil Yazigi. 2014. "Generalized Gaussian bridges." *Stochastic Processes and their Applications* 124 (9): 3084–3105. DOI 10.1016/j.spa.2014.04.002.

Sperry, R. W. 1968. "Hemisphere deconnection and unity in conscious awareness." *American Psychologist* 23 (10): 723–733. DOI 10.1037/h0026839.

Sprangers, M. A. G. and C. E. Schwartz. 1999. "Integrating response shift into health-related quality of life research: A theoretical model." *Social Science & Medicine* 48 (11): 1507–1515. DOI 10.1016/S0277-9536(99)00045-3.

Stearns, Stephen C. 1992. *The Evolution of Life Histories.* Oxford University Press.

Steegen, S., F. Tuerlinckx, A. Gelman, and W. Vanpaemel. 2016. "Increasing transparency through a multiverse analysis." *Perspectives on Psychological Science* 11 (5): 702–712. DOI 10.1177/1745691616658637.

Steinhauser, K. E., N. A. Christakis, E. C. Clipp, M. McNeilly, L. McIntyre, and J. A. Tulsky. 2000. "Factors considered important at the end of life by patients, family, physicians, and other care providers." *JAMA* 284 (19): 2476–2482. https://doi.org/10.1001/jama.284.19.2476

Sterling, P. 2012. “Allostasis: A model of predictive regulation.” *Physiology & Behavior* 106 (1): 5–15. DOI 10.1016/j.physbeh.2011.06.004.

Stevens, S. S. 1946. “On the theory of scales of measurement.” *Science* 103 (2684): 677–680. DOI 10.1126/science.103.2684.677.

Stone, A. A. and S. Shiffman. 1994. “Ecological momentary assessment in behavioral medicine.” *Annals of Behavioral Medicine* 16 (3): 199–202.

Stroebe, Margaret, and Henk Schut. 1999. “The dual process model of coping with bereavement: Rationale and description.” *Death Studies* 23 (3): 197–224. DOI 10.1080/074811899201046.

Substance Abuse and Mental Health Services Administration. 2014. *SAMHSA’s Concept of Trauma and Guidance for a Trauma-Informed Approach.* HHS Publication No. SMA 14-4884.

Substance Abuse and Mental Health Services Administration. 2021. *Medications for Opioid Use Disorder.* Treatment Improvement Protocol (TIP) Series 63. Publication No. PEP21-02-01-002. Substance Abuse and Mental Health Services Administration.

Suppes, P., and J. L. Zinnes. 1963. “Basic measurement theory.” In R. D. Luce, R. R. Bush, and E. Galanter, eds., *Handbook of Mathematical Psychology, Volume I*, 1–76. Wiley.

Suresh, Harini, and John V. Guttag. 2021. "A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle." *Proceedings of EAAMO 2021*, Article 17. DOI 10.1145/3465416.3483305.

Surges, S. M., H. Brunsch, B. Jaspers, K. Apostolidis, A. Cardone, C. Centeno, N. Cherny, Á. Csikós, R. Fainsinger, E. Garralda, J. Ling, J. Menten, S. Mercadante, D. Mosoiu, S. Payne, N. Preston, L. Van den Block, J. Hasselaar, and L. Radbruch. 2024. "Revised European Association for Palliative Care (EAPC) recommended framework on palliative sedation: An international Delphi study." *Palliative Medicine* 38 (2): 213–228. DOI 10.1177/026921632312202 25.

Sutton, R. S. and A. G. Barto. 2018. *Reinforcement learning: An introduction (2nd ed.).* MIT Press.

Talts, Sean, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. 2018. "Validating Bayesian inference algorithms with simulation-based calibration." *arXiv:1804.06788.* DOI 10.48550/arXiv.1804.06788.

Tamer, E. 2010. "Partial identification in econometrics." *Annual Review of Economics* 2: 167–195. DOI 10.1146/annurev.economics.050708.143401.

Taquet, Maxime, Jordi Quoidbach, Yves-Alexandre de Montjoye, Martin Desseilles, and James J. Gross. 2016. "Hedonism and the choice of everyday activities." *Proceedings of the National Academy of Sciences* 113 (35): 9769–9773. DOI 10.1073/pnas.1519998113.

Taylor, Charles. 2007. *A Secular Age.* Belknap Press of Harvard University Press.

Tedeschi, R. G. and L. G. Calhoun. 2004. "Posttraumatic growth: Conceptual foundations and empirical evidence." *Psychological Inquiry* 15 (1): 1–18.

Templer, V. L. and R. R. Hampton. 2012. "Rhesus monkeys (Macaca mulatta) show robust evidence for memory awareness across multiple generalization tests." *Animal Cognition* 15 (3): 409–419. DOI 10.1007/s10071-011-0468-4.

Teresi, J. A. 2006. "Overview of quantitative measurement methods. Equivalence, invariance, and differential item functioning in health applications." *Medical Care* 44 (11 Suppl 3): S39–S49. DOI 10.1097/01.mlr.0000245452.48613.45.

Theriault, J. E., Y. Katsumi, H. M. Reimann, J. Zhang, P. Deming, B. C. Dickerson, K. S. Quigley, and L. F. Barrett. 2025. "It's not the thought that counts: Allostasis at the core of brain function." *Neuron* 113 (24): 4107–4133. DOI 10.1016/j.neuron.2025.09.028.

Tiffany, Stephen T. 1990. "A cognitive model of drug urges and drug-use behavior: Role of automatic and nonautomatic processes." *Psychological Review* 97 (2): 147–168. DOI 10.1037/0033-295X.97.2.147.

Tinbergen, Niko. 1963. "On aims and methods of ethology." *Zeitschrift für Tierpsychologie* 20 (4): 410–433. DOI 10.1111/j.1439-0310.1963.tb01161.x.

Tononi, G., M. Boly, and C. Cirelli. 2024. "Consciousness and sleep." *Neuron* 112 (10): 1568–1594. DOI 10.1016/j.neuron.2024.04.011.

Trivers, R. L. 1971. "The evolution of reciprocal altruism." *The Quarterly Review of Biology* 46 (1): 35–57.

Trottier, E. D., S. Ali, M.-J. Doré-Bergeron, and L. Chauvin-Kimoff. 2022. "Best practices in pain assessment and management for children." *Paediatrics & Child Health* 27 (7): 429–437. DOI: 10.1093/pch/pxac048.

Trujillo, Cleber A., Richard Gao, Priscilla D. Negraes, Jing Gu, Justin Buchanan, Sebastian Preissl, Allen Wang, Wei Wu, Gabriel G. Haddad, Isaac A. Chaim, Alain Domissy, Matthieu Vandenberghe, Anna Devor, Gene W. Yeo, Bradley Voytek, and Alysson R. Muotri. 2019. "Complex oscillatory waves emerging from cortical organoids model early human brain network development." *Cell Stem Cell* 25 (4): 558–569.e7. DOI 10.1016/j.stem.2019.08.002.

Tsiatis, Anastasios A. 2006. *Semiparametric Theory and Missing Data.* Springer. DOI 10.1007/0-387-37345-4.

Turing, A. M. 1950. "Computing machinery and intelligence." *Mind* 59 (236): 433–460.

Tyler, Tom R. 2006. *Why People Obey the Law (2nd ed.).* Princeton University Press.

U.S. Department of Health and Human Services. 2018. *45 CFR 46, Subpart D: Additional Protections for Children Involved as Subjects in Research*.

U.S. Food and Drug Administration. 2025. *Patient-Focused Drug Development: Selecting, Developing, or Modifying Fit-for-Purpose Clinical Outcome Assessments.* Guidance for Industry, Food and Drug Administration Staff, and Other Stakeholders. Final Level 1 Guidance, October 2025, docket FDA-2022-D-1385.

Ubel, P. A., G. Loewenstein, N. Schwarz, and D. M. Smith. 2005. "Misimagining the unimaginable: The disability paradox and health care decision making." *Health Psychology* 24 (4 Suppl.): S57–S62. DOI 10.1037/0278-6133.24.4.S57.

Uhlenbeck, G. E. and L. S. Ornstein. 1930. "On the theory of the Brownian motion." *Physical Review* 36 (5): 823–841. https://doi.org/10.1103/physrev.36.823

United Nations Educational, Scientific and Cultural Organization. 2025. "Recommendation on the Ethics of Neurotechnology." *Adopted by the General Conference at its 43rd session*, 11 November 2025, Samarkand, Uzbekistan.

United Nations Inter-agency Group for Child Mortality Estimation. 2026. *Levels & Trends in Child Mortality: Report 2025.* United Nations Children's Fund, New York.

United Nations Office on Drugs and Crime. 2020. *Handbook on Restorative Justice Programmes (2nd ed.).* United Nations.

United Nations. 1989. *Convention on the Rights of the Child*.

United Nations. 2006. *Convention on the Rights of Persons with Disabilities*.

Valomon, A., I. De Cuntis, K. Nakamura, B. A. Riedner, S. Jones, M. Bazalakova, D. T. Plante, G. Tononi, and M. Boly. 2026. “High-density EEG signature of NREM sleep parasomnia episodes.” *Scientific Reports* 16: 19414. DOI 10.1038/s41598-026-41601-4.

van de Vijver, F. J. R. and K. Leung. 1997. *Methods and Data Analysis for Cross-Cultural Research.* Sage.

van der Bles, Anne Marthe, Sander van der Linden, Alexandra L. J. Freeman, and David J. Spiegelhalter. 2020. “The effects of communicating uncertainty on public trust in facts and numbers.” *Proceedings of the National Academy of Sciences* 117 (14): 7672–7683. DOI 10.1073/pnas.1913678117.

van Duijn, M. 2017. “Phylogenetic origins of biological cognition: Convergent patterns in the early evolution of learning.” *Interface Focus* 7 (3): 20160158. DOI 10.1098/rsfs.2016.0158.

van Fraassen, B. C. 1989. *Laws and Symmetry.* Oxford University Press.

Vandenberg, R. J. and C. E. Lance. 2000. “A review and synthesis of the measurement invariance literature: Suggestions, practices, and recommendations for organizational research.” *Organizational Research Methods* 3 (1): 4–70. DOI 10.1177/109442810031002.

Vayena, E., M. Salathé, L. C. Madoff, and J. S. Brownstein. 2015. “Ethical challenges of big data in public health.” *PLoS*

Computational Biology 11 (2): e1003904. DOI 10.1371/journal.pcbi.1003904.

Vohra, S., L. Shamseer, M. Sampson, C. Bukutu, C. H. Schmid, R. Tate, J. Nikles, D. R. Zucker, R. Kravitz, G. Guyatt, D. G. Altman, D. Moher, and the CENT group. 2015. "CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement." *BMJ* 350: h1738. DOI 10.1136/bmj.h1738.

Volkow, N. D., G. F. Koob, and A. T. McLellan. 2016. "Neurobiologic advances from the brain disease model of addiction." *New England Journal of Medicine* 374 (4): 363–371. DOI 10.1056/NEJMra1511480.

von Elm, E., D. G. Altman, M. Egger, S. J. Pocock, P. C. Gøtzsche, J. P. Vandenbroucke, and the STROBE Initiative. 2007. "The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: Guidelines for reporting observational studies." *PLoS Medicine* 4 (10): e296. DOI 10.1371/journal.pmed.0040296.

Wagenmakers, E.-J., R. Wetzels, D. Borsboom, H. L. J. van der Maas, and R. A. Kievit. 2012. "An agenda for purely confirmatory research." *Perspectives on Psychological Science* 7 (6): 632–638. DOI 10.1177/1745691612463078.

Wager, T. D., L. Y. Atlas, M. A. Lindquist, M. Roy, C.-W. Woo, and E. Kross. 2013. "An fMRI-based neurologic signature of physical pain." *New England Journal of Medicine* 368: 1388–1397. DOI 10.1056/NEJMoa1204471.

Wager, Tor D., James K. Rilling, Edward E. Smith, Alex Sokolik, Kenneth L. Casey, Richard J. Davidson, Stephen M. Kosslyn, Robert M. Rose, and Jonathan D. Cohen. 2004. "Placebo-induced changes in fMRI in the anticipation and experience of pain." *Science* 303 (5661): 1162–1167. DOI 10.1126/science.1093065.

Walker, Margaret Urban. 2006. *Moral Repair.* Cambridge University Press.

Walters, E. T. 2025. "From nociception in aneural animals to human suffering: Toward a comparative biology of pain." *Journal of Experimental Biology* 228 (19): jeb251210. DOI 10.1242/jeb.251210.

Wamsley, E. J. 2014. "Dreaming and offline memory consolidation." *Current Neurology and Neuroscience Reports* 14: 433. DOI 10.1007/s11910-013-0433-5.

Wamsley, E. J., T. Trost, and M. Tucker. 2024. "Memory updating in dreams." *SLEEP Advances* 5 (1): zpae096. DOI 10.1093/sleepadvances/zpae096.

Wasserstein, R. L. and N. A. Lazar. 2016. "The ASA statement on p-values: Context, process, and purpose." *The American Statistician* 70 (2): 129–133. DOI 10.1080/00031305.2016.1154108.

Wasserstein, R. L., A. L. Schirm, and N. A. Lazar. 2019. "Moving to a world beyond $p < 0.05$." *The American Statistician* 73 (sup1): 1–19. DOI 10.1080/00031305.2019.1583913.

Watkins, Edward R. 2008. "Constructive and unconstructive repetitive thought." *Psychological Bulletin* 134 (2): 163–206. DOI 10.1037/0033-2909.134.2.163.

Weber, M. 1922. *Wirtschaft und Gesellschaft.* J. C. B. Mohr (Paul Siebeck).

Weber, Max. 1963. *The Sociology of Religion.* Translated by Ephraim Fischoff. Beacon Press. Originally published 1922.

Weidinger, Laura, John Mellor, Maribeth Rauh, Conor Griffin, Jonathan Uesato, Po-Sen Huang, Myra Cheng, Mia Glaese, Borja Balle, Atoosa Kasirzadeh, Zac Kenton, Sasha Brown, Will Hawkins, Tom Stepleton, Courtney Biles, Abeba Birhane, Julia Haas, Laura Rimell, Lisa Anne Hendricks, William Isaac, Sean Legassick, Geoffrey Irving, and Iason Gabriel. 2021. "Ethical and social risks of harm from language models." *arXiv:2112.04359.*

Weisberg, M. 2013. *Simulation and Similarity: Using Models to Understand the World.* Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199933662.001.0001

Weizenbaum, Joseph. 1966. "ELIZA: A computer program for the study of natural language communication between man and machine." *Communications of the ACM* 9 (1): 36–45. DOI 10.1145/365153.365168.

Weizenbaum, Joseph. 1976. *Computer Power and Human Reason: From Judgment to Calculation.* W. H. Freeman.

Wellek, S. 2010. *Testing Statistical Hypotheses of Equivalence and Noninferiority (2nd ed.).* CRC Press.

West-Eberhard, Mary Jane. 2003. *Developmental Plasticity and Evolution.* Oxford University Press. https://doi.org/10.1093/oso/9780195122343.001.0001

Wiggins, A. and J. Wilbanks. 2019. "The rise of citizen science in health and biomedical research." *American Journal of Bioethics* 19 (8): 3–14. DOI 10.1080/15265161.2019.1619859.

Wilkinson, Mark D., Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J.G. Gray, Paul Groth, Carole Goble, Jeffrey S. Grethe, Jaap Heringa, Peter A.C 't Hoen, Rob Hooft, Tobias Kuhn, Ruben Kok, Joost Kok, Scott J. Lusher, Maryann E. Martone, Albert Mons, Abel L. Packer, Bengt Persson, Philippe Rocca-Serra, Marco Roos, Rene van Schaik, Susanna-Assunta Sansone, Erik Schultes, Thierry Sengstag, Ted Slater, George Strawn, Morris A. Swertz, Mark Thompson, Johan van der Lei, Erik van Mulligen, Jan Velterop, Andra Waagmeester, Peter Wittenburg, Katherine Wolstencroft, Jun Zhao, and Barend Mons. 2016. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3: 160018. DOI 10.1038/sdata.2016.18.

Williams, B. 1981. *Moral Luck: Philosophical Papers 1973–1980.* Cambridge University Press. https://doi.org/10.1017/cbo9781139165860

Williams, D. R., J. A. Lawrence, and B. A. Davis. 2019. "Racism and health: Evidence and needed research." *Annual Review of Public Health* 40: 105–125. DOI 10.1146/annurev-publhealth-040218-043750.

Williams, George C. 1966. *Adaptation and Natural Selection.* Princeton University Press.

Wilson, R. C. and A. G. E. Collins. 2019. "Ten simple rules for the computational modeling of behavioral data." *eLife* 8: e49547. DOI 10.7554/eLife.49547.

Windt, J. M. 2015. *Dreaming: A Conceptual Framework for Philosophy of Mind and Empirical Research.* MIT Press. https://doi.org/10.7551/mitpress/9780262028677.001.0001

Winsberg, Eric. 2010. *Science in the Age of Computer Simulation.* University of Chicago Press.

Wirth, M., M. Wettstein, and K. Rothermund. 2024. "Longitudinal associations between time perspective and life satisfaction across adulthood." *Communications Psychology* 2: 67. DOI 10.1038/s44271-024-00118-0.

Wong, William, Rubén Herzog, Kátia Cristine Andrade, Thomas Andrillon, Draulio Barros de Araujo, Isabelle Arnulf, Somayeh Ataei, Giulia Avvenuti, Benjamin Baird, Michele Bellesi, Damiana Bergamo, Giulio Bernardi, Mark Blagrove, Nicolas Decat, Çağatay Demirel, Martin Dresler, Jean-Baptiste Eichenlaub, Valentina Elce, Steffen

Gais, Luigi De Gennaro, Jarrod Gott, Chihiro Hiramatsu, Bjørn Erik Juel, Karen R. Konkoly, Deniz Kumral, Célia Lacaux, Joshua J. LaRocque, Bigna Lenggenhager, Remington Mallett, Sérgio Arthuro Mota-Rolim, Yuki Motomura, Andre Sevenius Nilsen, Valdas Noreika, Delphine Oudiette, Fernanda Palhano-Fontes, Jessica Palmieri, Ken A. Paller, Lampros Perogamvros, Antti Revonsuo, Elaine van Rijn, Serena Scarpelli, Monika Schönauer, Sarah F. Schoch, Francesca Siclari, Pilleriin Sikka, Johan Frederik Storm, Hiroshige Takeichi, Katja Valli, Erin J. Wamsley, Jennifer M. Windt, Jing Zhang, Jialin Zhao, and Naotsugu Tsuchiya. 2025. "A dream EEG and mentation database." *Nature Communications* 16: 7495. DOI 10.1038/s41467-025-61945-1.

Wood, W. and D. Rünger. 2016. "Psychology of habit." *Annual Review of Psychology* 67: 289–314. DOI 10.1146/annurev-psych-122414-033417.

Woods, A. 2011. "The limits of narrative: Provocations for the medical humanities." *Medical Humanities* 37 (2): 73–78. DOI 10.1136/medhum-2011-010045.

Woodward, J. 2003. *Making Things Happen: A Theory of Causal Explanation.* Oxford University Press.

World Health Organization. 2001. *International Classification of Functioning, Disability and Health: ICF.* World Health Organization.

World Health Organization. 2020. *Palliative care.* World Health Organization.

World Health Organization. 2021. *Ethics and Governance of Artificial Intelligence for Health: WHO Guidance*.

World Health Organization. 2022. *Global Report on Health Equity for Persons with Disabilities.* World Health Organization.

World Health Organization. 2024. *Ethics and governance of artificial intelligence for health: Guidance on large multi-modal models.* World Health Organization.

World Medical Association. 2024. *WMA Declaration of Helsinki: Ethical principles for medical research involving human participants.* World Medical Association.

World Wide Web Consortium. 2024. *Web Content Accessibility Guidelines (WCAG) 2.2*. W3C Recommendation, 12 December 2024.

Wortman, Camille B., and Roxane Cohen Silver. 1989. "The myths of coping with loss." *Journal of Consulting and Clinical Psychology* 57 (3): 349–357. DOI 10.1037/0022-006X.57.3.349.

Yarkoni, T. 2022. "The generalizability crisis." *Behavioral and Brain Sciences* 45: e1. DOI 10.1017/S0140525X20001685.

Yi, T.-M., Y. Huang, M. I. Simon, and J. Doyle. 2000. "Robust perfect adaptation in bacterial chemotaxis through integral feedback control." *Proceedings of the National Academy of Sciences* 97 (9): 4649–4653. DOI 10.1073/pnas.97.9.4649.

Young, I. M. 1990. *Justice and the Politics of Difference.* Princeton University Press.

Zehr, Howard. 2002. *The Little Book of Restorative Justice.* Good Books.

Index

A

Ablation and redundancy, 561

Abstention and inadequate estimates, 282, 721

Access to experience, 338, 342

Adaptation, 510, 521

Adaptive question selection, 442

Addiction: craving and wanting, 225

Addiction: practical recovery, 683

Addiction: recovery capital, 227

Additive representation, 147, 279

Age, period, and cohort, 478

Agency and practical choice, 99

AI sovereignty, 665

AI: consciousness evidence, 371

AI: institutional assistance, 665

AI: welfare precautions, 381

Aliasing, 248

Anesthesia and recovery, 85, 344

Animals and insects, 358, 367

Appeals and remedies, 663

Applied lens and formal Law, 71, 671

Archives and personal identity, 76, 666

Arithmetic and moral desert, 54

Artificial intelligence, 371, 665

Assent and pediatric research, 320

Attention and reflection, 676

Autonomy and coercion, 99, 320

Auxiliary assumptions, 103

Averages and individual streams, 21

B

Bayesian model comparison, 710

Bearer identification, 73, 352

Bedside observation, 422

Belief and nonbelief, 652

Belief state, 550

Between-person comparisons, 302

Biological death, 82

Bivariate branch, 148

Bivariate representation, 148

Blinding and endpoint adjudication, 88, 352

Boundary conditions, 71, 99

Boundary hypothesis, 97, 105

Boundary-value problems, 100

Branching streams, 349

Brier score, 497
Brownian bridges, 136, 572

C

Calibration networks, 305
Calibration: cumulative uncertainty, 311
Calibration: neutral origin, 289, 293
Calibration: sensitivity, 471
Candidate mechanisms, 105, 190
Candidate route, 190
Candidate sentience, 376, 384
Cantril Ladder, 303
Capacity and acceleration, 216
Capacity ratios, 223
Care and immediate needs, 52
Caregiving and hidden labor, 230
Carryover and sequence, 435
Causal inference, 445, 480
Changing population composition, 485
Children and consent, 320
Choice and reachable futures, 165, 212
Chosen meaning, 55
Clinical ambiguity, 92
Clinical observation, 50, 422
Closure, 71
Closure: mathematical target, 71, 147
Closure: measurement distinction, 78, 133
Coercive control, 664
Coercive measurement, 329
Collider association, 233
Comfort-first observation, 422
Common-action requirement, 203
Comparators and common information, 576
Compensability, 187
Competing risks, 144
Complementarity, 28
Complete felt history, 71, 74
Consciousness: comparative evidence, 358
Consciousness: unity and access, 338, 342
Consent lifecycle, 320
Consent, refusal, and withdrawal, 64, 320
Construct validity, 77, 245
Continuing bonds and grief, 698
Continuing bonds and objects, 698
Continuity and memory, 342, 350
Control and intervention, 445, 555
Control directions, 184
Controlled direct effect, 481

Correction and withdrawal, 325, 657
Correction of public claims, 657
Counterexamples and universality, 74, 721
Counterfactual support, 98, 200
Critical path, 221
Cross-cultural interpretation, 286, 411
Cumulative uncertainty, 162, 311

D

Data deletion and retention, 322, 327
Data minimization and deletion, 319, 322
Data provenance, 318
Death and endpoint inference, 82, 417
Definitions and scientific evidence, 105
Delayed feedback, 546
Dependency records, 720
Detailed balance, 576
Detection-channel overlap, 402
Device calibration, 287
Differential item functioning, 298
Disability and response shift, 296, 483
Disclosure and privacy, 40, 327
Discriminating predictions, 165, 541
Distributive fairness, 14
Dream communication, 394
Dream reports, 399
Duration and exposure, 121, 707
Dynamic programming, 617

E

Eligibility and prohibited scoring, 323, 667
Emergency correction, 658
Emotion recognition, 321
Encouragement designs, 450
Endpoint adjudication, 88, 352
Endpoints and unknown histories, 83, 93
Equivalence testing, 133
Error-correcting representations, 124
Errors-in-variables, 310
Ethical consequences of interpretation, 59, 60
Evidence ladder, 105
Evolution and individual costs, 584, 601
Exact closure, 71, 78

Exactness and finite precision, 133, 710
Excerpts and qualifications, 658
Experience sampling, 49, 182
Expressive vocabulary, 41

F

Falsifiability and adverse cases, 74, 103
Falsification, 74, 721
Familiar environments, 677
Fear and theory-induced distress, 676
Feedback and regulation, 518, 546
First-stage effect, 450
Fission and stream ownership, 349, 350
Forecasting and outcome inspection, 573
Freedom within causal structure, 67, 99
Frequency-dependent selection, 586

G

Garbling, 267
Generative models and recovery, 623
Geometric growth, 588
Geometric observation times, 109
Graduated assistance, 56
Grief: empirical patterns, 651
Grief: practical support, 698

H

Half-life, 571
Happiness and life satisfaction, 303, 510
Harm reduction and recovery, 684
Held-out transition study, 454
Hidden states and observability, 248
Higher-order dependence, 153
Homeostatic control, 546
Human standing and fairness, 69
Human-worth score prohibition, 323, 667
Hysteresis, 86

I

Identification versus precision, 471
Identity and archives, 76
Importance weight, 629
Independent origins, 289, 293
Infinite expected waiting time, 158
Information leakage, 257
Information ordering, 267
Informed refusal, 64, 320

Institutional administration limit, 661
Institutional authority, 663, 665
Integral windup, 559
Interaction contrast, 433
Interference between participants, 446
Interpersonal comparisons, 302, 303
Interpretation and meaning, 55, 651
Intervention-induced observation, 491
Invariance and transport, 313, 717

J

Joint coverage, 457
Joint creation, 60
Jordan decomposition, 128
Joy and public goods, 671

K

Karma, providence, and grace, 645

L

Lawhood, 97, 104
Leaky integrator, 557
Learning and performance, 691
Learning from personal records, 700
Learning unknown control directions, 203
Length-biased sampling, 404
Life satisfaction versus affect, 303, 480
Limit-sure attainability, 156
Local average treatment effect, 450
Local tests and universal claims, 74, 454
Long projects and handoffs, 686
Longitudinal missingness, 491
Loss function, 713
Lottery evidence, 510

M

Martingale, 198
Matched simulation runs, 624
Mathematical coupling, 466
Mean reversion, 566
Measurement: indicator roles, 245
Measurement: minimum requirements, 706
Measurement: no adequate estimate, 282
Measurement: origin and scale, 289, 293
Measurement: personal records, 676, 701
Mechanism versus target, 99, 105

Median, 713
Memory and continuity, 336, 342
Memory monitoring, 362
Minimum effort and horizon, 219
Missing-interval bounds, 249
Missing-outcome bounds, 492
Missingness and unobserved intervals, 232, 491
Mixtures of recovery rates, 513
Modal reachability, 152, 153
Model complexity and lawhood, 97, 541
Model recovery, 623
Monte Carlo precision, 625
Moral desert and suffering, 54
Moral nontransfer between persons, 59

N

Natural law and scientific aspiration, 104
Neural organoids, 376, 377
Neutral origin, 289, 293
No adequate estimate, 282, 721
No harm for proof, 63, 381
No moral desert, debt, or payment, 54
No personal endpoint testing, 95
Noisy endpoint observations, 574
Nonunique explanation, 623, 721
Null results, 710
Numerical integration, 637
Numerical integration error, 609

O

Observability of latent quantities, 248, 704
Observation burden, 320, 422
Observation feedback, 273
Opponent dynamics, 517
Option maintenance, 693
Optional reflection, 676
Order of state updates, 192
Origins and transformations, 278, 293
Outcome-dependent reclassification, 103, 112
Ownership, 40, 74

P

Palliative sedation, 420
Parity and reachability, 555
Parity check, 124
Partial identification, 310, 311
Partial order, 314
Partial pooling, 303
Path diversity, 721

Per-stream terminal valence closure (PTVC), 71, 74, 147
Personal identity, 76, 350
Phenomenal versus recorded history, 76
Pleasure and meaning, 680
Policy support and attainable paths, 152, 153
Population refreshment samples, 493
Positive surplus objection, 70
Post hoc rescue, 102, 113
Potential shaping, 549
Power analysis, 457, 469
Power and authority, 661, 663
Practical planning, 681
Predictive evidence across trials, 718
Predictive processing and active inference, 530
Preregistration and change control, 113
Privacy and inferred records, 40, 327
Process signatures, 165, 572
Prospective food caching, 364
Proxy reports, 286
Public access-permission distinction, 326
Public claims and excerpts, 657, 658
Public data and permission, 326

Q

Quadratic variation, 169
Qualifying streams, 73, 353
Quantifiers and universality, 74

R

Random endpoints, 135, 137
Random-endpoint problem, 134
Rare-event simulation, 629
Reachability and resources, 165, 212
Reachability ladder, 152
Reachable set, 184
Recalibration, 483
Reconceptualization, 483
Record-regime separation and no-dossier rule, 327
Recovery: clinical practice, 683
Recovery: model verification, 623
Reference window, 160
Reference-class revision, 103, 112
Refusal and withdrawal, 320, 322
Regulation as a rival explanation, 518, 546

Reinforcement learning, 548
Rejection sampling, 621
Repair and accountability, 668, 668
Repair is not payment, 668
Repeated-reading allocation, 498
Replication and transport, 717
Representation and measurement gates, 706, 707
Reprioritization, 483
Research participation, 320
Residual compatibility sets, 74, 721
Residual set, 721
Resource timing, 33
Response shift, 285, 483
Rest and distributed labor, 675, 696
Restorative repair, 668, 668
Retirement of claims, 207, 657
Reversibility and decisions, 682
Revision and scientific re-entry, 112
Rival comparisons, 272, 576
Rival model, 576
Rumination, 676

S

Sampling architecture, 475
Saturation, 559
Scientific evidence levels, 105, 720
Scientific uncertainty, 74, 721
Scoring and eligibility bans, 323, 667
Selection and retention bias, 491, 502
Sequential inference, 108
Sequential interventions, 459
Sequential standardization, 459
Shared time, 697
Shared understanding, 43
Shortening horizon, 211, 212
Shrinkage, 303
Simulation: constructed worlds, 612
Simulation: pipeline recovery, 623
Skepticism as dignity, 652
Smoothing, 424
Snapshot and path convergence, 174
Specific satiety, 364
Specification curves, 271
Spiritual resonance, 650, 653
Split-brain research, 347
Standing and equal concern, 69
State potential and telescoping, 194
State-dependent stopping, 140

State-transition study, 454
Stop rules, 377, 658
Stopping reflection, 676
Stream boundaries, 82, 350
Suffering and explanation, 104
Syndrome, 124

T

Target, 71, 149
Teleonomy, 104
Temporal co-occurrence, 148
Terminal observation, 422, 423
Test outcomes and verdicts, 720
Tetrad, 264
Theodicy and meaning, 645, 653
Time-weighted sampling, 251
Timeliness of assistance, 211
Total variation, 128
Transformations and zero, 278, 279
Transport and invariance, 717
Truth-containing residual set, 74, 721

U

Uncertainty allocation, 312
Uncertainty: communication, 658
Uncertainty: shared calibration, 311
Unequal histories and fairness, 23, 69
Uniform tail control, 129
Union bound, 109
Unity of experience, 338
Universal claims and local tests, 74, 454
Universal quantifier, 74

V

Valence and cumulative representation, 121, 147
Validation unit, 489
Value formation through activity, 688
Value of information, 535
Variable learning rates, 531
Verdict, 720
Verdicts and unresolved mechanisms, 720
Vulnerability and consent, 320, 321

W

Warrant for practical action, 671, 682
Weak-form weighting, 449
What-where-when memory, 359
Withdrawal and data use, 322, 328
World Happiness Report, 303

Z

Zero-area counterexample,
518

About the Author

Stephen Takowsky has spent his life studying complex systems: how rules shape them, how people cooperate and compete within them, and how hidden constraints can change what becomes possible. His path has crossed finance, competitive strategy, professional esports, and the leadership of global online communities, ultimately leading to the Law of Fairness project, a decades-long effort to investigate whether fairness itself may reflect a deeper structure of conscious life.

Competing as "MrGameTheory," Takowsky reached number one on the Civilization Revolution leaderboard and in the international Civilization IV and Civilization V leagues while pursuing a career in finance at Morgan Stanley in Beverly Hills. In 2016, aXiomatic, an ownership group led by Peter Guber and Ted Leonsis and backed by investors including Magic Johnson, Steve Case, and Tony Robbins, acquired a controlling interest in Team Liquid. Following the acquisition, Team Liquid selected Takowsky to captain its newly created Civilization team, later recognized by Guinness World Records as the first Civilization esports team. Across thousands of games, Civilization became an unconventional education in strategy: understand the whole system, think far beyond the next move, anticipate intelligent opposition, and keep searching when the obvious paths fail.

Strategy then became community leadership. Takowsky spent seven years at 505 Games, serving as Senior Digital Content & Engagement Manager and leading global communities for Terraria, which has surpassed 70 million copies sold, as well as Ghostrunner, WUCHANG: Fallen Feathers, Eiyuden Chronicle: Hundred Heroes, and other titles. He also founded the United Servers of Discord Alliance, bringing together leaders from major communities built around Roblox, Pokémon, Star Wars, League of Legends, Grand Theft Auto, Warframe, Animal Crossing, and other properties. Earlier, he served as administrator and producer

for TF2Outpost, a major Steam digital-goods marketplace. Across these very different environments, he kept returning to the same insight: better structure can expand what people are capable of building together.

Throughout those years, one question kept following him: could the apparent unfairness of life conceal a deeper structure? That question became the Law of Fairness. Takowsky began developing the idea more than twenty years ago while studying Religious Studies and Philosophy/Law and Society at the University of California, Riverside, and gave it its first public form in his 2009 book, *Of Grandeur*. In the years that followed, he sought out academics and serious thinkers across many disciplines, asking for insights, criticism, and sources that could help separate fact from speculation. What began as a philosophical intuition gradually became a research program built to face mathematics, measurement, competing explanations, and evidence.

The ten-book Law of Fairness series follows that question outward, from consciousness and scientific testing into ordinary life, meaning, and civilization. If the Law is true, then there may be no “free happiness” in the sense of positive experience remaining unresolved by the lifetime balance the hypothesis predicts. That possibility raises a profound practical question: what if humanity has been pursuing the wrong model of utopia? Rather than organizing life around the endless pursuit of pleasure, the series explores whether greater self-moderation, less needless suffering, and better-designed institutions could help remove some of the forces that keep people and societies trapped in conflict, corruption, and instability.

Takowsky’s fifty-year horizon is designed so that building a better future does not require tearing apart the present. It gives institutions and communities time to test reforms, build

infrastructure, prepare future leaders, improve technology, and correct mistakes before they spread. Just as importantly, it gives today's generation the chance to be remembered for beginning something larger than itself: laying the foundations for a more transparent, peaceful, and prosperous world it may never live to see.

The project's future is collaborative. Takowsky welcomes researchers, universities, laboratories, funders, technologists, students, critics, writers, artists, community builders, and volunteers willing to bring rigor, imagination, and courage to one of humanity's oldest questions. The next breakthrough may come from a theorem, an instrument, a dataset, a counterexample, or an experiment no one has yet imagined. *The Law of Fairness* began as one person's question. Its next chapter belongs to everyone willing to help discover what is true and build what that knowledge makes possible.

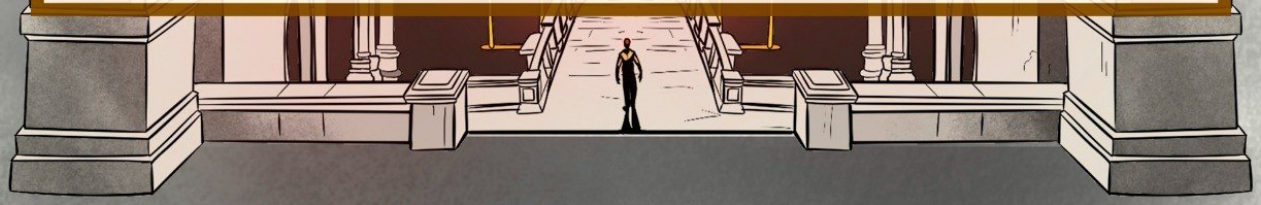
Explore the project and its official communities at LawOfFairness.com. Research, scholarly, media, funding, volunteer, and collaboration inquiries: team@lawoffairness.com.

The Law of Fairness

Could life be fair for everyone despite the unfairness we can see?

The Law of Fairness proposes a startling possibility: beneath the apparent randomness and inequality of life, there may be a hidden rule that ultimately makes life fair for everyone. If such a rule exists, then some of our oldest assumptions about how life works may be wrong, and events that seem unrelated may actually belong to a larger pattern.

This book turns that possibility into a testable question and follows its consequences wherever they lead. It explores where evidence of such a rule might be found and how understanding the Law could change the way we make decisions, pursue our goals, and think about what is possible. The search reaches far beyond fairness, into consciousness, human potential, and the way we see our own lives.



LawofFairness.com